II श्री गणेश प्रसन्न II

# Book Core

*The high-yield core of revision books 2–17 · key points, algorithms, MCQ one-liners and every line with a number to memorise*

| BOOK | TITLE | ITEMS |
|---|---|---|
| 2 | Oncology, DSD & Ageing | 390 |
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| 13 | Kidney & UTUC | 319 |
| 14 | Adrenal | 139 |
| 15 | Renal & Transplant | 157 |
| 16 | BPH | 251 |
| 17 | Prostate Cancer | 570 |
| | **Total** | **5773** |

*Colour key:* ● ***key point*** · ● ***algorithm*** · ● ***MCQ*** · ● ***table row*** · ***number to memorise***

# 2 Oncology, DSD & Ageing

390 core items · page numbers refer to the full book PDF

## Ch 24 · Basic Principles of Immunology and Immunotherapy in Urologic Oncology

### 1 · INNATE AND ADAPTIVE IMMUNITY

- ALGORITHM
  Danger signal (TNF-α, TLR ligand) → DC stops antigen uptake → upregulates MHC and costimulatory B7-1/B7-2 → migrates via afferent lymphatics (CCL21 sensed by CCR7) → activates CD4, CD8, B cells in node
  p.2

### 2 · IMMUNE EDITING AND ESCAPE

- KEY POINT PD-L1 is largely induced by the immune attack itself — a marker of an ongoing but suppressed response, not a fixed tumour trait. p.2

### 4 · BCG IN BLADDER CANCER

- ALGORITHM
  Instil 1–5 × $10^8$ bacilli → fraction adheres via **fibronectin attachment protein** → TLR2/TLR4 on urothelium → IL-8 + TNF-α → neutrophil wave (hours) → DC activation → CD4 Th1 response in nodes (weeks) → memory
  p.3

- Memory implied by ~**50%** of non-muscle-invasive bladder cancer (NMIBC) in durable remission. p.3

### 5 · CANCER VACCINES

- **Sipuleucel-T**: autologous peripheral blood mononuclear cells incubated with PA2024 (prostatic acid phosphatase [PAP]–GM-CSF fusion), intravenous every 2 weeks × 3; asymptomatic/minimally symptomatic metastatic castration-resistant prostate cancer (mCRPC). Overall survival (OS) **+4.1 months** but **no effect on time to progression**; first FDA-approved solid-tumour vaccine. p.3

### 6 · CHECKPOINT BLOCKADE

- Single-agent first-line pembrolizumab (KEYNOTE-052) is restricted to cisplatin-ineligible with combined positive score (CPS) **≥10**, or any-platinum-ineligible — CPS <10 did worse than carboplatin chemotherapy. p.4

- KEY POINT Every successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, immunogenic cell death, VEGF inhibition) or removes a brake (PD-1, CTLA-4). Failures cluster where the brake is not limiting — PD-L1-negative, Treg-dominated prostate cancer. p.4

### 8 · MCQ ONE-LINERS

- MCQ BCG PAMPs signal through TLR2 (and TLR4). p.5
- MCQ BCG adheres via fibronectin attachment protein. p.5
- MCQ Th1 (IL-2, IFN-γ, TNF-α) = BCG success; Th2 = failure. p.5
- MCQ IL-12 drives Th1; IL-4 drives Th2. p.5
- MCQ Treg: CD4 cell marked by FoxP3. p.5
- MCQ CD8 recognises class I MHC, ~9-amino-acid peptides. p.5
- MCQ Immune editing: elimination, equilibrium, escape. p.5
- MCQ PD-L1 is induced by IFN-γ from infiltrating CD8 cells. p.5
- MCQ Renal paradox: high CD8 density predicts worse RCC outcome. p.5

- MCQ Sipuleucel-T antigen: prostatic acid phosphatase. p.5
- MCQ Sipuleucel-T prolongs OS without delaying progression. p.5
- MCQ Nadofaragene: adenoviral IFN-α2b with Syn3; first gene therapy for NMIBC. p.5
- MCQ CTLA-4 is mainly on Tregs; PD-1 on CD8 TILs. p.5
- MCQ Avelumab: PD-L1 inhibitor, maintenance after platinum. p.5
- MCQ First-line pembrolizumab monotherapy needs CPS ≥10 if cisplatin-ineligible. p.5
- MCQ Enfortumab vedotin targets nectin-4. p.5
- MCQ First tumour-agnostic biomarker approval: pembrolizumab for MSI-H/dMMR. p.5
- MCQ Anti-PD-1 fails in most prostate cancer: tumour PD-L1-negative. p.5
- MCQ Nivolumab + ipilimumab: first line intermediate/poor-risk RCC. p.5
- MCQ Highest ORR among TKI combinations: pembrolizumab + lenvatinib. p.5

## Ch 25 · Molecular Genetics and Cancer Biology

### 3 · CELL CYCLE

- ALGORITHM
  Hypophosphorylated RB1 binds E2F + histone deacetylases (HDACs) → S-phase genes silent → cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1 → E2F released → S-phase genes transcribed → restriction point passed (irreversible commitment)
  p.7
- KEY POINT Cancer must break both RB1 (entry) and TP53 (integrity) — which is why these two pathways are lost in almost every tumour. p.7

### 4 · DNA DAMAGE AND REPAIR

- KEY POINT Repair defects are therapeutic targets: HR loss → poly(ADP-ribose) polymerase (PARP) inhibitor (synthetic lethality); MMR loss → immunotherapy; ERCC2 loss → cisplatin sensitivity. p.8

### 6 · SENESCENCE, TELOMERES, AUTOPHAGY

- Telomerase (reverse transcriptase) is high in **85–90%** of tumours; normally only in testis, endometrium, gut, activated lymphocytes. Alternative lengthening of telomeres (ALT) is telomerase-independent (ATRX/DAXX loss). p.8

### 8 · HYPOXIA, VHL AND ANGIOGENESIS

- ALGORITHM
  Normoxia: hypoxia-inducible factor-α (HIF-α) prolyl-hydroxylated → bound by von Hippel-Lindau protein (VHL, E3 ubiquitin ligase) → proteasomal degradation. Hypoxia or VHL loss: HIF-α accumulates → dimerises with HIF-1β → vascular endothelial growth factor (VEGF), angiopoietin and glycolytic genes
  p.9
- KEY POINT VHL loss = permanent pseudohypoxia; angiogenesis, Warburg metabolism, invasion and immunosuppression all follow — why clear cell RCC responds to VEGF, HIF-2α and immune blockade. p.9

### 10 · GU TUMOUR GENOMICS

- KEY POINT The germline syndrome names the somatic driver (VHL → clear cell, MET → papillary, FH → type 2, BRCA → prostate) and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss). p.11

### 13 · MCQ ONE-LINERS

- MCQ Feature unique to malignancy: local invasion. p.11
- MCQ Knudson two-hit: retinoblastoma; familial cases often bilateral. p.12

- MCQ Cyclin D–CDK4/6 phosphorylates RB1, releasing E2F. p.12
- MCQ INK4 inhibitors act only on CDK4/6. p.12
- MCQ MDM2 is the negative regulator of TP53. p.12
- MCQ HR needs sister chromatid — only S/G2. p.12
- MCQ ERCC2 mutation (NER) → cisplatin sensitivity in bladder cancer. p.12
- MCQ MMR defect raises mutation rate ~100-fold. p.12
- MCQ Extrinsic apoptosis is TP53-independent. p.12
- MCQ Telomeres lose ~100 bp per division. p.12
- MCQ GSTP1 hypermethylation: commonest somatic change in prostate cancer. p.12
- MCQ 5-methylcytosine deamination gives C→T transitions. p.12
- MCQ EZH2 writes H3K27me3. p.12
- MCQ siRNA silences one mRNA; miRNA silences many. p.12
- MCQ VHL is an E3 ubiquitin ligase targeting HIF-α. p.12
- MCQ Commonest prostate fusion: TMPRSS2–ERG. p.12
- MCQ SPOP: commonest mutated suppressor in primary prostate cancer. p.12
- MCQ PBRM1 mutation: better prognosis; BAP1: worse. p.12
- MCQ Largest consensus bladder subtype: basal/squamous. p.12
- MCQ Testis: 12p gain; TP53 mutation rare. p.12
- MCQ HLRCC gene: fumarate hydratase. p.12
- MCQ VAF ~50% without germline data: suspect germline variant. p.12

## Ch 55 · Management of Abnormalities of the External Genitalia in Males

### 1 · NORMAL DEVELOPMENT AND PENILE SIZE

- **Stretched penile length (SPL)**: pubic symphysis (fat pad fully compressed) to glans tip; correlates with erect length better than flaccid length. Term neonate **3.5 ± 0.7 cm** (diameter 1.1 cm); adult 13.3 ± 1.6 cm. p.13
- Micropenis cut-off (mean − 2.5 SD): **1.9 cm** at term–5 months; 1.5 cm at 30 weeks, 2.0 cm at 34 weeks; ≈ 3.7–3.9 cm at 5–11 years; adult 9.3 cm. Always compare with the cut-off column, not the mean. p.13

### 2 · PREPUCE: PHIMOSIS AND PARAPHIMOSIS

- **Physiological phimosis** at birth (glanular adhesions ± preputial ring). Smegma accumulation (first 3–4 years) and intermittent erections separate the prepuce: ≈ **90% retractable by 3 years**, < 1% phimotic at 17 years. p.13

### 3 · CIRCUMCISION

- TABLE Amputated glans · Re-attach within **8 h** (often without microscope) p.14
- KEY POINT Late sequelae are problems of skin quantity (too much → trapped penis, chordee, torsion; too little → redundancy, adhesions) or of the newly exposed meatus (stenosis). p.14

### 4 · INCONSPICUOUS PENIS VERSUS MICROPENIS

- KEY POINT **Stretched length is the single discriminator.** Inconspicuous penis (buried, webbed, trapped) = normal SPL and diameter hidden by skin, fat or scar → anatomical treatment. Micropenis = small but normally formed organ → endocrine work-up. p.15
- Normally formed penis ≥ 2.5 SD below mean SPL (< **1.9 cm** at term); length-to-girth ratio normal. Testes often small/cryptorchid; scrotum fused, small. p.15

- ALGORITHM

  History, examination, **karyotype**, SPL, corpora, cryptorchidism → paediatric endocrinologist → testosterone before/after hCG stimulation (± gonadotrophin-releasing hormone [GnRH] test) → anterior pituitary screen (serial glucose, Na/K, cortisol, thyroid) → magnetic resonance imaging (MRI) brain (hypothalamus, pituitary, midline) → **test end-organ response: intramuscular testosterone enanthate for 3 months** (short courses do not affect height; prolonged courses advance bone age); repeat at puberty.

  p.16

## 9 · SCROTAL ANOMALIES

- **Penoscrotal transposition**: failed inferomedial migration of labioscrotal swellings; partial forms — shawl, doughnut, prepenile scrotum. Associated with perineal/scrotal hypospadias + chordee, caudal regression, sex chromosome anomalies, Aarskog–Scott; with complete transposition, **up to 75%** have significant upper-tract anomalies → renal US + VCUG. p.18

## 10 · INGUINAL HERNIA AND HYDROCELE

- ALGORITHM

  Skin-crease incision → Scarpa's fascia → open external oblique along fibres sparing ilioinguinal nerve (in young infants rings coincide — Mitchell-Banks, external oblique not opened) → split cremaster → find sac anteromedially → separate atraumatically from vas and vessels, divide → dissect to internal ring and **high-ligate** (open sac first if contents suspected) → tighten a wide ring medially → excise redundant tunica, remove appendages → **confirm testis is scrotal**.

  p.19

- Contralateral patency ≈ 40% infants → 20% after 10 years, but metachronous hernia only ≈ **7%** — argument against routine contralateral exploration; laparoscopy through the sac sees the other ring (> 99% accurate). p.20

## 11 · ACUTE SCROTUM

- TABLE Orchidectomy risk · **≈ 5%** · 20% · 40% · 60% · 80% · 90% p.20

- ALGORITHM

  Suspected torsion → scrotal (hemiscrotal or raphe) exploration → affected side first: open tunica, note colour, twists, anatomy → detorse, warm moist packs → **meanwhile fix contralateral testis with non-absorbable suture** → reassess (subjective; Doppler or tunica albuginea incision to see bleeding unvalidated): viable → dartos pouch or direct dartos fixation (non-absorbable); non-viable → orchidectomy (cord ligated in segments); prosthesis after healing or in puberty.

  p.21

## 12 · ADOLESCENT VARICOCELE

- Prevalence ≈ **15%** (as in adults); appears after 10 years, peaks **Tanner 3**; tall thin boys; familial (brothers). p.21

- KEY POINT **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain (rare; relieved in most), abnormal semen analysis (reliable at Tanner 5 / ≥ 18 y), intratesticular varicocele with asymmetry. Total testicular volume < 30 mL and low total motile count predict poor semen. p.22

## 14 · MCQ ONE-LINERS

- MCQ Micropenis in a term neonate: stretched length < 1.9 cm (−2.5 SD). p.23
- MCQ Commonest cause of micropenis: hypogonadotrophic hypogonadism. p.23
- MCQ First therapeutic step in micropenis: intramuscular testosterone enanthate for 3 months to test response. p.23
- MCQ Buried penis has normal stretched length — distinguishes it from micropenis. p.23
- MCQ Phimosis topical therapy: betamethasone 0.05%; ≈ 1 in 6 fail. p.23

- MCQ Commonest circumcision complication: bleeding, usually frenular. p.23
- MCQ Circumcision degloving: heal by secondary intention, no immediate graft. p.23
- MCQ Meatal stenosis occurs almost exclusively after circumcision. p.23
- MCQ Best neonatal circumcision analgesia: dorsal penile nerve block (better than EMLA). p.23
- MCQ Amputated glans: reattach within 8 hours. p.23
- MCQ Penile torsion is usually counter-clockwise; corpora at base normal. p.23
- MCQ Lateral curvature: plicate the corpus opposite the maximal curvature. p.23
- MCQ Urethral duplication: ventral urethra is functional; Effmann classification. p.23
- MCQ Megameatus intact prepuce: hypospadias variant hidden by a normal foreskin. p.23
- MCQ Fusiform megalourethra: spongiosum and cavernosa deficient; prune-belly association. p.23
- MCQ Anterior urethral valve: commonest bulbar; often distal lip of a diverticulum. p.23
- MCQ Aphallia: commonest type has postsphincteric opening at anal verge, best survival. p.23
- MCQ Complete penoscrotal transposition: up to 75% have upper-tract anomalies. p.23
- MCQ Suprainguinal ectopic scrotum: ipsilateral renal agenesis — image upper tracts. p.23
- MCQ Commonest cause of paediatric priapism: sickle cell disease (ischaemic). p.23
- MCQ High-flow priapism follows straddle injury; embolise if persistent. p.23
- MCQ Neonatal priapism: no treatment, resolves in 2–6 days. p.23
- MCQ Commonest acute scrotum cause overall (prepubertal): torsion of appendix testis. p.23
- MCQ Present cremasteric reflex does not exclude testicular torsion. p.23
- MCQ Whirlpool/snail sign on high-resolution US = twisted spermatic cord. p.23
- MCQ Torsion < 6 hours: orchidectomy ≈ 5%; > 24 hours ≈ 80%. p.23
- MCQ Always fix the contralateral testis with non-absorbable suture in torsion. p.23
- MCQ Perinatal torsion is extravaginal; bilateral in 5–22%. p.23
- MCQ Paediatric hernia recurrence after open repair: 0.5–1%. p.23
- MCQ Abdominoscrotal hydrocele: processus closed; no ring tightening needed. p.23
- MCQ Varicocele peaks at Tanner 3; repair if > 20% volume discrepancy. p.23
- MCQ Lowest hydrocele + recurrence: microscopic subinguinal varicocelectomy. p.23
- MCQ Isolated right varicocele → image abdomen for mass. p.23
- MCQ Bilateral absent vas = CFTR mutation; unilateral absent vas = renal anomaly. p.23

## Ch 58 · Management of Abnormalities of the Genitalia in Girls

### 2 · EVALUATION AND CLASSIFICATION

- KEY POINT Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery. p.24

### 4 · VESTIBULE: THE INTERLABIAL MASS

- KEY POINT Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Urethral- or vaginal-origin cyst → **always renal US**. p.25

### 5 · VAGINA: OBSTRUCTION AND ABSENCE

- Renal anomaly ≈ **one-third** (agenesis/ectopia); skeletal 10–20%; aberrant pelvic vessels. No lumbosacral dysraphism. p.26

### 6 · VAGINAL REPLACEMENT

- KEY POINT **Cervical atresia**: perineum-to-uterus anastomosis loses the endocervical barrier → life-threatening ascending infection. Hysterectomy + neovagina is often best; otherwise endometrial suppression and in vitro fertilisation with transmyometrial transfer. p.26

### 8 · ACQUIRED DISORDERS

- **Vaginal rhabdomyosarcoma**: grapelike bleeding mass, <2 years; best prognosis of genital primaries (embryonal). Biopsy → computed tomography (CT) chest/abdomen + marrow → chemotherapy ± local resection. Intergroup Rhabdomyosarcoma Study (IRS) trials cut surgery from **100% to 13%**. p.27

### 9 · MCQ ONE-LINERS

- MCQ Urogenital diaphragm and external anal sphincter share pudendal innervation. p.27
- MCQ Müllerian anomalies travel with ipsilateral renal anomalies. p.27
- MCQ Caudal regression: weeks 4–5 insult; maternal diabetes. p.27
- MCQ MURCS: müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia. p.27
- MCQ MRI is gold standard for müllerian anatomy. p.27
- MCQ Commonest cause of clitoromegaly: CAH. p.27
- MCQ Diminutive clitoris suggests CAIS. p.27
- MCQ Urethral prolapse: doughnut with central meatus; excise circumferentially. p.27
- MCQ Skene cyst: displaced meatus, eccentric stream. p.27
- MCQ Ectopic ureter below sphincter: daytime wetting despite normal voiding. p.27
- MCQ Any introital cyst: image the kidneys. p.27
- MCQ Imperforate hymen: transverse incision, never simple aspiration. p.27
- MCQ Commonest transverse septum level: upper/middle third. p.27
- MCQ Commonest septum-repair complication: stenosis. p.27
- MCQ Hollow mould when uterus present. p.27
- MCQ MRKH: second commonest cause of primary amenorrhoea. p.27
- MCQ MRKH: ovaries always functional. p.27
- MCQ MRKH renal anomalies ≈ one-third, type B. p.27
- MCQ Dilation is first-line neovagina therapy. p.27
- MCQ Ileal neovagina stenoses more than sigmoid. p.27
- MCQ Bowel neovagina: adenocarcinoma; skin: squamous carcinoma. p.27
- MCQ Cervical agenesis: avoid uterovaginal anastomosis. p.27
- MCQ HWWS: ipsilateral renal agenesis; no amenorrhoea. p.27
- MCQ Vaginal rhabdomyosarcoma: grapelike mass, under 2 years. p.27
- MCQ Labial adhesions: 80% resolve within a year. p.27
- MCQ WHO type III FGM = infibulation. p.27
- MCQ Round ligament absent in genetic males. p.27

## Ch 59 · Differences of Sexual Development

### 2 · NORMAL SEX DEVELOPMENT

- KEY POINT The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. Any DSD phenotype can be reasoned from which hormone was missing, where and when — e.g. a uterus in 46,XY means Sertoli (AMH) failure, i.e. dysgenesis, not a synthesis or receptor defect. p.28

### 4 · 46,XX DSD

- CAH = commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) = **95%** of CAH; autosomal recessive, ≈ 1 in 15,000. p.29
- Diagnosis: ↑**17-hydroxyprogesterone (17-OHP)**, measured after **36 h** (physiological neonatal surge). Also ↑androstenedione, testosterone, renin; ↓cortisol; cosyntropin test. p.29
- **Salt-wasting crisis**: hyponatraemia, hyperkalaemia, acidosis, hypotension, shock — usually not before **day 5–7**, so normal early electrolytes do not exclude CAH. Treatment: glucocorticoid + mineralocorticoid + salt — lifesaving († hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone and sodium chloride supplements in infancy; double-to-triple hydrocortisone during illness or surgery). p.29

### 5 · 46,XY DSD

- TABLE **5α-reductase 2 deficiency** · Autosomal recessive; atypical/female at birth, blind vaginal pouch; marked **virilisation at puberty**, fertility reported; testosterone:DHT **>20:1**; Dominican "guevedoces" (Imperato-McGinley 1979) p.30

### 7 · EVALUATION OF THE NEWBORN WITH ATYPICAL GENITALIA

- KEY POINT **Emergency**: atypical genitalia with bilateral impalpable gonads = salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP; watch for crisis from the end of week 1. Everything else (designation, surgery) is not an emergency. p.31
- Multidisciplinary review within **24–48 h**; congratulate, say "your baby", never "it"; gender-neutral terms. A urologist without a team should transfer early. p.31
- ALGORITHM

  Karyotype (≥45 cells; fluorescence in situ hybridisation for X/Y in hours) + electrolytes + 17-OHP at ≈ 36 h → pelvic ultrasound (US) for uterus (best neonatally) and gonads → at ≈ 1 week: LH, follicle-stimulating hormone (FSH), testosterone, oestrogen, androstenedione, DHT, AMH ± hCG stimulation → genetic panel / microarray / exome → if unclear: MRI, genitography, endoscopy, laparoscopy ± gonadal biopsy (**no gonadectomy during diagnosis**)

  p.32

### 12 · MCQ ONE-LINERS

- MCQ "True hermaphrodite" is now ovotesticular DSD. p.33
- MCQ "XY sex reversal" = 46,XY complete gonadal dysgenesis. p.33
- MCQ SRY upregulates SOX9 to drive Sertoli differentiation. p.33
- MCQ Sertoli cells secrete AMH from 7–8 weeks. p.33
- MCQ DHT is the active androgen for external genitalia and prostate. p.33
- MCQ Commonest cause of neonatal atypical genitalia: CAH (21-hydroxylase). p.33
- MCQ Second commonest: mixed gonadal dysgenesis (45,X/46,XY). p.33
- MCQ Measure 17-OHP after 36 hours of age. p.33
- MCQ Salt-wasting crisis usually occurs after day 5–7. p.33
- MCQ 11β-hydroxylase deficiency causes hypertension and hypokalaemia. p.33
- MCQ 17α-hydroxylase deficiency: hypokalaemia, not hyperkalaemia. p.34

- MCQ 3β-HSD deficiency undervirilises 46,XY and virilises 46,XX. p.34
- MCQ Uterus in 46,XY points to gonadal dysgenesis. p.34
- MCQ Testosterone:DHT >20:1 suggests 5α-reductase deficiency. p.34
- MCQ Guevedoces: 5α-reductase deficiency virilising at puberty. p.34
- MCQ CAIS gonads can stay until after breast development (risk 2%). p.34
- MCQ 1–2% of girls with inguinal hernia have CAIS. p.34
- MCQ Frasier syndrome: gonadoblastoma 60%, no Wilms predisposition. p.34
- MCQ Denys–Drash: Wilms 90%, congenital nephrotic syndrome. p.34
- MCQ PMDS: AMH or AMH receptor mutation; hernia uteri inguinale. p.34
- MCQ PMDS underlies 30–50% of transverse testicular ectopia. p.34
- MCQ 46,XX testicular DSD: SRY translocation; azoospermic. p.34
- MCQ Palpable gonad = testicular tissue; ovaries do not descend. p.34
- MCQ Klinefelter: commonest sex chromosome aneuploidy, 1 in 600 males. p.34
- MCQ All 45,X/46,XY patients need Turner screening. p.34
- MCQ Vas always accompanies a testis in ovotesticular DSD. p.34
- MCQ Precise diagnosis reached in <50% of 46,XY DSD. p.34
- MCQ No gonadectomy during the diagnostic phase. p.34
- MCQ Micropenis: stretched length <2.5 SD below mean. p.34
- MCQ Highest dysphoria: 5α-reductase deficiency raised female. p.34
- MCQ MRKH uterine remnant must never be connected to a neovagina. p.34
- MCQ Testicular adrenal rest tumours regress with increased glucocorticoid. p.34
- MCQ Pubertal gonadal biopsy: ≥30 tubules with OCT3/4 staining. p.34

## Ch 60 · Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

### 2 · UROGENITAL SINUS AND CLOACA — CLASSIFICATION

- CAH: commonest cause of newborn atypical genitalia — half of all and **90%** of 46,XX cases; salt wasting in **67–75%**; karyotype within **48 h**. p.35
- KEY POINT **Governing principle**: **true urethral length** (bladder neck → confluence) decides the operation far more than common-channel length — in sinus and cloaca alike. Confluence level is a continuum, not simply "high" (suprasphincteric) vs "low". p.35
- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants); virilisation lengthens the **vagina-to-meatus** distance, so "high" CAH confluence is an illusion of virilisation, reinforced by a male-like sphincter and verumontanum-like confluence. p.35

### 3 · CLINICAL EVALUATION

- **Clitoris**: **Huffman index** (glans width × length) normal **<3.5 mm²**, concern **>10 mm²**. Isolated clitoral hypertrophy in a preterm infant is not a DSD. p.35
- TABLE Uterovaginal · Duplication/septation · **53–67%** p.36
- KEY POINT Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in. p.36

## 5 · DSD RECONSTRUCTION — TIMING AND CONSENT

- TABLE **After puberty** · Infant stenosis; avoids dilation; consent; delay the small vagina (**<3 cm**) p.37

- KEY POINT Oestrogen is invoked by both camps (maternal vs pubertal). With no comparative data, the decision is ethical rather than evidential; it currently rests with parents and the team. The Endocrine Society CAH guideline (2018) leaves it to families with experienced surgeons; the 2006 consensus limits surgery to Prader III–IV. p.37

## 6 · OPERATIVE PRINCIPLES

- KEY POINT Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error. p.38

## 7 · LOW CONFLUENCE REPAIR

- ALGORITHM
  Metabolic stability → **stress-dose steroids in CAH** → antibiotics → endoscopy → **Fogarty balloon in vagina** + Foley in bladder → total lower-body preparation (allows supine–prone rotation; lithotomy blocks conversion)
  p.38

- ALGORITHM
  Glans traction suture → incision around glans, inner prepuce kept for hood (**second only to glans in sensitivity**) → strips either side of the ventral mucosal plate → omega flap → Y-incisions on labia majora → deglove to bifurcation, ventral plate intact
  p.38

- ALGORITHM
  Flap vaginoplasty: omega flap with fat → posterior sinus dissected off rectum (**hardest step**) → posterior sinus opened in midline into posterior vaginal wall → **continue to normal-calibre vagina** → flap inset
  p.38

- KEY POINT Two fixed points: never touch the dorsal neurovascular bundle, and always open back to normal-calibre vagina. p.39

## 8 · HIGH CONFLUENCE: PULL-THROUGH

- Only ≈ **5%** of CAH; commoner in pure sinus. p.39

- ALGORITHM
  Prone → midline dissection between sinus and rectum → sinus divided to normal vagina → retractor inside vagina → anterior wall dissected off urethra **under direct vision** → sinus closed as urethra in 2–3 layers → vagina mobilised → supine: flaps + labioplasty
  p.39

## 10 · OUTCOMES OF SINUS/DSD SURGERY

- Infant vaginoplasty usually needs a **secondary introitoplasty** after puberty (**25–100%**) — the orifice does not grow because nothing passes through it; revision is easy. p.39

- **Azziz (1986)** MCQ: poor coitus rates, but almost all had **cut-back** — not applicable to modern repairs. p.39

- **Elmasri meta-analysis (2018)**: observational only; mean age at surgery 2.7 years; female gender identity ≈ 89%; vaginal stenosis **27%**; only half comfortable intercourse; no early-vs-late comparison. p.40

## 11 · CLOACA — INITIAL MANAGEMENT

- ALGORITHM
  (1) Decompress bowel → (2) decompress genitourinary tract → (3) correct nephron-destructive anomalies → (4) definitive repair
  p.40

- **Kidneys**: structural anomalies **60–83%**; chronic kidney disease (CKD) **75%**, end-stage renal disease (ESRD) **17%**; treat obstruction promptly. Urodynamics abnormal in **90%** (overactivity, atonia, dyssynergia, high leak point pressure). Reflux is managed medically if UTI is prevented. p.40

### 12 · CLOACA — DEFINITIVE REPAIR

- KEY POINT Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags it below the continence mechanism → leakage. p.41

- ALGORITHM

  **Posterior sagittal anorectovaginourethroplasty (PSARVUP, Peña)**: coloured catheters in each channel → **prone jackknife** → muscle stimulator marks anal site → midline incision → rectal fistula opened, rectum separated and mobilised to reach skin → TUM or vaginal separation → urethra closed in 2–3 layers + **fat flap** → perineal body → rectum centred in sphincter mass

  p.41

### 13 · CLOACA — RESULTS AND FOLLOW-UP

- **Urinary**: about half void spontaneously; Peña: CIC **28%** (channel <3 cm) vs **78%** (>3 cm). Lifelong yearly urological review. p.41

- **Gynaecological**: introital stenosis, retained septum, persistent fistula, deficient perineal body (often from an anteriorly placed rectum); about half of outside repairs need redo; vaginal stenosis/obstructed menstruation **36–41%**; obstructive haematometra; pregnancy possible, **caesarean preferred**. Leave atretic genitalia — they may develop. p.41

- ALGORITHM

  US **6–9 months after thelarche** → repeat every 6–9 months until menses are normal

  p.41

### 14 · MCQ ONE-LINERS

- MCQ True urethral length, not channel length, decides the operation. p.41
- MCQ CAH confluence: 1.5–2 cm from bladder neck. p.41
- MCQ Huffman index normal <3.5 mm². p.41
- MCQ Both gonads descended: 46,XX unlikely. p.41
- MCQ Hydrometrocolpos: often the only early sign of pure sinus. p.42
- MCQ Cerebriform adrenals suggest CAH. p.42
- MCQ Rectal fistula commonest in the vaginal septum. p.42
- MCQ Schmid (1961): first nerve-sparing clitoroplasty. p.42
- MCQ Flap vaginoplasty for high confluence: classic error. p.42
- MCQ Pull-through: Hendren and Crawford (1969). p.42
- MCQ PUM stops at the pubourethral ligament. p.42
- MCQ Hendren's rule: never a separate rectal pull-through. p.42
- MCQ CIC is first-line decompression in cloaca. p.42
- MCQ Cloaca: CKD 75%, ESRD 17%. p.42
- MCQ TUM in cloaca: channel <3 cm and urethra >1.5 cm. p.42
- MCQ Short spinal cord most impairs continence. p.42
- MCQ Four sinus entities: genital atypia, pure sinus, cloaca, female exstrophy. p.42
- MCQ Hypertension in a virilised infant: 11β-hydroxylase deficiency. p.42
- MCQ Cervical impression on genitogram indicates female internal organs. p.42

- MCQ Ovotestis biopsy: deep longitudinal incision. p.42
- MCQ Fetal ascites in cloaca: urine refluxing through uterus and tubes. p.42
- MCQ Tethered cord in ≈ one-third of cloacas, mostly without skin stigmata. p.42
- MCQ Clitoral erectile tissue removed through ventral incisions; dorsal bundle untouched. p.42
- MCQ Labia minora from split clitoral skin; labia majora moved by YV-plasty. p.42
- MCQ ASTRA divides the anterior rectal wall; no colostomy needed. p.42
- MCQ Mobilised sinus split ventrally = vestibule; dorsally = anterior vaginal wall. p.42
- MCQ Azziz series: poor results reflect cut-back vaginoplasty. p.42
- MCQ Cloaca colostomy of choice: divided descending colostomy. p.42
- MCQ Preserve the appendix at colostomy for later MACE. p.42
- MCQ Transabdominal TUM never: channel ischaemia. p.42
- MCQ Ultrasound for menstrual obstruction 6–9 months after thelarche. p.42

## Ch 65 · Adolescent and Transitional Urology

### 1 · DEFINITIONS

- KEY POINT Evidence for transition looks weak because the wrong endpoint is measured. Renal function and glycated haemoglobin (HbA1c) are expected to worsen, so **stability and retention in follow-up** are success; fewer patients are lost when actively transitioned. p.43

### 3 · THE PROCESS

- Begin ≈ **12 years** (American Academy of Pediatrics, AAP). Most important skill: finding valid information. Complete by late teens–early 20s; **26** = latest (US parental insurance limit). If independence is unachievable, plan custodianship (e.g. sibling). p.43
- KEY POINT Transfer is not the endpoint: transition ends only when integration into adult care is confirmed. p.44

### 5 · DEVELOPMENTAL CONSIDERATIONS AND CONSENT

- Genital surgery windows: both developmental theory and the 1975 AAP panel favour **<6 months**; second window differs (2–3 years vs 4th–5th year). Elective timing is driven by school, parental anxiety and surgeon availability; semi-elective (exstrophy closure) → as soon as safe. p.44

### 7 · MCQ ONE-LINERS

- MCQ Transition: planning, transfer, integration. p.45
- MCQ Stability, not improvement, is success in progressive disease. p.45
- MCQ Vesicoureteric reflux: may be discharged after early-adult review. p.45
- MCQ Pelvic rhabdomyosarcoma survivors: surveillance for treatment late effects. p.45
- MCQ Paediatric provider initiates transition. p.45
- MCQ Avoid transition during a medical crisis. p.45
- MCQ AAP: begin transition at about 12 years. p.45
- MCQ Age 26: latest commonly cited completion age. p.45
- MCQ Transfer of care: 18–21 years. p.45
- MCQ Transition complete by 23, after integration confirmed. p.45
- MCQ New adolescent problem is an ideal route. p.45
- MCQ Emergency post-discharge referral: most challenging route. p.45
- MCQ Model 1b combines continuity with adult expertise. p.45

- MCQ Most complex MDT needs: spina bifida. p.45
- MCQ Nursing is the backbone of transitional care. p.45
- MCQ Avoid genital surgery at 3–6 years. p.45
- MCQ Both schemes agree genital surgery before 6 months. p.45
- MCQ Spina bifida: deficits in working memory, executive function. p.45
- MCQ Confidentiality discussed with family, then alone. p.45
- MCQ Biggest adult-provider barrier: unfamiliarity with congenital conditions. p.45

## Ch 120 · Aging and Geriatric Urology

### 1 · CONCEPTS AND BIOLOGY

- "Geriatrics" first used as a clinical term by Nascher (1909) MCQ. p.46
- **Cellular senescence** (irreversible replication arrest) is the fundamental driver of ageing. Replicative senescence follows telomere shortening — the **Hayflick limit** (1961) MCQ; stress-induced senescence follows reactive oxygen species, radiation or DNA damage. p.46

### 2 · CLINICAL EVALUATION

- ALGORITHM
  Frailty screen before elective major surgery (Society for Perioperative Assessment and Quality Improvement, SPAQI) → positive → diagnostic frailty assessment → comprehensive geriatric assessment (CGA)
  p.46

### 3 · GERIATRIC SYNDROMES

- TABLE Cut-off · Continuous · **≥3 frail; 1–2 pre-frail** p.47
- KEY POINT Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance. p.47

### 4 · INCONTINENCE MANAGEMENT

- UI affects three-quarters of long-stay residents and drives admission. Catheter colonisation reaches **98%**, so treating asymptomatic bacteriuria (ASB) causes harm; criteria guide antibiotics: p.48
- KEY POINT A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used. p.48

### 6 · CANCER, END OF LIFE

- **Prostate**: USPSTF grade D — no prostate-specific antigen (PSA) screening >70; life expectancy **<10 years** → non-curative plan. p.48
- KEY POINT Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide. p.48

### 8 · MCQ ONE-LINERS

- MCQ "Geriatrics" coined clinically by Nascher, 1909. p.49
- MCQ Hayflick limit: telomere-driven replicative arrest. p.49
- MCQ Homeostenosis: narrowing of physiological reserve. p.49
- MCQ First BADL lost: bathing; last: feeding. p.49
- MCQ Surgical risk: operative time, ASA, urgency — not age. p.49
- MCQ Fried: ≥3 of 5 frail; 1–2 pre-frail. p.49
- MCQ Deficit-count index does not predict postoperative outcome. p.49

- MCQ NSQIP/AGS checklist: 13 items, 2 organ-system. p.49
- MCQ Trospium: quaternary amine. p.49
- MCQ High-dose tolterodine: QT prolongation. p.49
- MCQ Mirabegron interacts with metoprolol. p.49
- MCQ Azithromycin: Beers-exempt macrolide. p.49
- MCQ Catheter colonisation: up to 98%. p.49
- MCQ Loeb catheter criteria include mental status change. p.49
- MCQ ASB treated only in pregnancy or before instrumentation. p.49
- MCQ USPSTF grade D: ASB screening and PSA >70. p.49
- MCQ Life expectancy <10 years: non-curative prostate plan. p.49
- MCQ Urethral caruncle: topical oestrogen. p.49
- MCQ ASPREE: aspirin increased mortality in healthy elderly. p.49
- MCQ 4 Ms: Mobility, Medication, Mentation, Matters. p.49
- MCQ SASP spreads senescence to neighbouring tissue. p.49
- MCQ GSM causes urgency and urgency incontinence, not only atrophy. p.49
- MCQ Bacillus Calmette–Guérin (BCG) less effective in elderly: immunosenescence. p.49
- MCQ Confusion alone does not diagnose UTI in older adults. p.49

## Ch 126 · Management and Considerations in Transgender and Nonbinary Individuals

### 2 · DIAGNOSIS AND ASSESSMENT

- TABLE Criteria · Marked incongruence ≥**6 months**, clinically significant distress or impairment, not solely due to prejudice · Marked, persistent incongruence; desire for characteristics and treatment of experienced gender p.50
- KEY POINT SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling remain firm because their consequences are irreversible. p.51

### 3 · GENDER-AFFIRMING HORMONE THERAPY

- KEY POINT **Why the asymmetry?** Physiological-dose oestrogen alone does not suppress testosterone into the female range, so transfeminine therapy needs two components; testosterone suppresses the ovarian axis unaided, so transmasculine therapy needs one. p.51
- **Ethinyl oestradiol is not recommended** — ~20-fold thromboembolic risk and cannot be monitored on assays. Commonest: oral 17β-oestradiol, target **100–200 pg/mL**. **Transdermal if venous thromboembolism (VTE) risk**: age >45 or previous VTE. p.51
- Monitoring: levels 3-monthly in year 1, then 1–2×/year; target testosterone **<50 ng/dL**; electrolytes/creatinine on spironolactone; prolactin (prolactinoma risk); bone density if risk factors; stop smoking. p.51
- **After gonadectomy: reduce oestrogen by 50–75%** (titrate to luteinising hormone [LH]/follicle-stimulating hormone [FSH], keep bone density) and **stop the antiandrogen**. p.52
- Target testosterone **300–1000 ng/dL**. Persistent menses → progestin or endometrial ablation; GnRH analogues possible but costly. p.52

### 5 · FERTILITY PRESERVATION

- TABLE Standard · **Sperm cryopreservation** — simple, cheap; hold GAHT ≥**3 months** before collection (GAHT degrades semen) · **Oocyte or embryo cryopreservation** after controlled ovarian hyperstimulation; oocytes avoid committing to a sperm source p.53

### 6 · CANCER SCREENING — SCREEN THE ORGAN, NOT THE GENDER

- **Prostate (transfeminine)**: rare but may be aggressive, mostly when GAHT began after 50. No dedicated guideline — shared decision-making on prostate-specific antigen (PSA) at 55–69. **On GAHT, upper limit 1 ng/mL or rise ≥0.3 ng/mL from nadir**. †Androgen suppression lowers PSA, hence the lower threshold. p.53

### 8 · MCQ ONE-LINERS

- MCQ ICD-11 moved gender incongruence to Conditions Related to Sexual Health. p.53
- MCQ DSM-5 gender dysphoria: ≥6 months' incongruence plus distress. p.53
- MCQ WPATH SOC-8 (2022): single professional opinion suffices. p.53
- MCQ Fertility counselling precedes any treatment, not only surgery. p.53
- MCQ Ethinyl oestradiol: not recommended — thrombosis. p.53
- MCQ Transfeminine targets: oestradiol 100–200 pg/mL, testosterone <50 ng/dL. p.53
- MCQ After gonadectomy: oestrogen cut 50–75%, antiandrogen stopped. p.53
- MCQ After gonadectomy: testosterone dose unchanged. p.53
- MCQ Transmasculine target testosterone: 300–1000 ng/dL. p.53
- MCQ Monitor haematocrit on testosterone — erythrocytosis. p.53
- MCQ Irreversible on testosterone: clitoral growth, voice, facial hair, baldness. p.54
- MCQ Post-phalloplasty dribble: pooling at native urethra–pars fixa junction. p.54
- MCQ Hold GAHT ≥3 months before sperm banking. p.54
- MCQ Ovarian tissue freezing needs no GAHT pause. p.54
- MCQ PSA on GAHT: abnormal >1 ng/mL. p.54
- MCQ Prostate after vaginoplasty: palpated via anterior neovaginal wall. p.54
- MCQ SAMHSA Four Rs: Realise, Recognise, Respond, Resist. p.54
- MCQ Testis cancer in transfeminine people: self-examination only. p.54
- MCQ Transdermal oestradiol if age >45 or previous VTE. p.54
- MCQ Neovaginal LUTS: nocturia commonest. p.54

## Ch 127 · Male-to-Female Gender-Affirming Surgery: Vaginoplasty

### 1 · AIM AND PREPARATION

- TABLE Hormone therapy before gonadectomy · **6 months** of hormone therapy suffices (patient experiences testosterone blockade before an irreversible change) p.55
- KEY POINT SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory. p.55
- Stay within **1 hour's drive** of the hospital for the first **3 weeks**. p.55
- BMI is the main factor in wound healing and closure tension. No proven cut-off; many require **≤35 kg/m²**; higher may be acceptable for vulvoplasty. p.55
- KEY POINT Plan every orchiectomy as though vaginoplasty will follow — this dictates the midline scrotal incision and leaving peri-cord fat for labial bulk. p.56

## 2 · PENILE INVERSION VAGINOPLASTY (PIV) — PERINEAL APPROACH

- ALGORITHM
  Foley, drain, plug → measure; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion) → harvest scrotal graft + small posterior rhomboid flap (relieves posterior introital tension) → defat, fulgurate follicles, tubularise over dilator → orchiectomy, leave cord fat → penile disassembly → corporal excision + clitoroplasty → urethroplasty → canal dissection → line canal → labiaplasty → pack
  p.56

- Closure: labia majora from lateral scrotal skin with cord fat for bulk; keep incision **≥1 cm** from groin crease (dehiscence). Antibiotic-soaked pack + negative-pressure dressing. p.57

## 3 · ABDOMINOPERINEAL AND INTESTINAL OPTIONS

- **Peritoneal (Davydov-type) vaginoplasty**: external steps identical; proximal canal made robotically. Rationale: historical rectal injury **~7%** (poor exposure), reduced to ~2.3% (urethral 1.1%). p.57

- KEY POINT Lining ladder: penile/scrotal skin → peritoneum (more depth or safer exposure) → bowel (no skin left). Each step adds depth and lubrication but imports donor-organ pathology; none removes the duty to dilate. p.57

## 4 · COMPLICATIONS AND MANAGEMENT

- TABLE **Rectovaginal fistula** (most devastating) · Rectal injury; **0.8%** primary, 6.3% revision · Intra-op: watertight closure + bulbospongiosus flap ± faecal diversion. Fistula: stop dilating, let canal stenose, revise later (may reopen) **or** delay closure until canal stable p.57

- KEY POINT Three failure modes: a violated plane (fistulas), tissue inadequately excised (bulge, engorgement), or a graft not taken/maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient — hence dilation is part of the operation. p.58

## 5 · OUTCOMES, POSTOPERATIVE CARE, FOLLOW-UP

- Stay ~**5 days**; **day 5: remove pack and catheter**, teach dilation and douching. Retention → recatheterise, remove at 1 week. p.58

- KEY POINT The body treats the canal as a defect to close — lifelong stenosis risk. Any narrowing → increase dilation, contact surgeon, physiotherapy. p.58

## 6 · MCQ ONE-LINERS

- MCQ SOC8: one assessment; 6 months' hormones before gonadectomy. p.58
- MCQ SOC8 removed 12-month real-life experience requirement. p.58
- MCQ Dysphoria-driven depression is not a reason to defer surgery. p.58
- MCQ Main psychosocial surgical risk: failure to dilate. p.58
- MCQ Oestrogen continued perioperatively — no added VTE risk. p.58
- MCQ Orchiectomy incision: vertical midline scrotal. p.58
- MCQ Tranexamic acid 1 g intravenously at start: front-loaded bleeding. p.58
- MCQ Clitoral position landmark: adductor longus tendons. p.58
- MCQ Short penile shaft shortens labia minora, not canal. p.58
- MCQ Residual erectile tissue causes arousal engorgement and canal obstruction. p.58
- MCQ Bulbospongiosus kept until canal done — covers rectal injury. p.58
- MCQ Prior radical prostatectomy favours vulvoplasty. p.58
- MCQ Robotic peritoneal approach: rectum may mimic seminal vesicles. p.58
- MCQ Rectal injury ~7% historically, ~2.3% abdominoperineal. p.58

- MCQ Intestinal vaginoplasty: penoscrotal hypoplasia or revision stenosis. p.58
- MCQ Commonest minor complication: introital wound dehiscence. p.58
- MCQ Rectovaginal fistula: 0.8% primary, 6.3% revision. p.58
- MCQ Bladder-neck urethrovaginal fistula → incontinence; distal → vaginal voiding. p.58
- MCQ Stenosis: pelvic floor physiotherapy first. p.58
- MCQ Neovaginal prolapse reflects poor graft take. p.58
- MCQ Median time to orgasm ≈ 180 days. p.58
- MCQ Dilation 4 × daily for first 6 weeks. p.58

## Ch 128 · Special Urologic Considerations in Transmasculine Individuals

### 2 · PREOPERATIVE CARE AND CONSENT

- WPATH (Standards of Care version 7): ≥**12 months** continuous testosterone before genital GCS († SOC8, 2022, lowered this to 6 months) — mainly for **clitoral hypertrophy** (growth plateaus ≈2 years). p.59

### 3 · REMOVAL OF FEMALE STRUCTURES

- ALGORITHM
  Hydrodissect urethrovaginal plane with dilute lidocaine → circumferential introital incision, 4 clamps → blunt rectovaginal plane, gauze protects rectum → **sharp urethrovaginal dissection, no diathermy** (sphincter) → bladder plane (submucosal); bladder injury → close, cystoscopy if ureteric orifices in doubt → lateral walls: free levators, cautery/3-0 sutures for paravaginal plexus → approximate levators (prevents prolapse) → bulbospongiosus over fixed urethra → layered closure
  p.60

### 4 · GENITAL RECONSTRUCTION

- TABLE Phallus · Own clitoris, **3–8 cm** · Flap p.60

### 5 · POSTOPERATIVE CARE AND CATHETERISATION

- Suprapubic catheter (SPC) + neourethral catheter; remove neourethral at **14–21 days**; clamp SPC, open after each void to record PVR; keep until voiding normal (oedema, post-vaginectomy neurogenic dysfunction). p.60

- ALGORITHM
  One gentle Coudé attempt → stylet-guided attempt → any difficulty: stop → flexible cystoscopy, guidewire, open-tip catheter. Emergency: **suprapubic catheter**
  p.61

### 6 · URETHRAL COMPLICATIONS

- **Fistula**: commonest at phallic base. Keep diversion until wound heals; small ones close spontaneously; persisting >**3 months** → excise, layered closure with fat interposition (no dartos), catheter 10–12 days. p.61
- **Stricture**: commonest at the **fixed-phallic anastomosis** (poorly perfused flap end); also meatus. Plateau uroflow → retrograde urethrography (RUG). Treat only after **≥6 months**; retention → SPC or perineal urethrostomy. p.61
- KEY POINT The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure. p.61

### 7 · PROSTHESES

- Erectile: ≥**1 year** after phalloplasty, once **protective sensation** present and urethra stable. One cylinder; no tunica → anchor to pubic periosteum (rear tip extender, Dacron sleeve or dedicated device with pubic base plate). Malleable is simpler. p.61

### 8 · MCQ ONE-LINERS

- MCQ Testosterone duration before genital GCS: ≥12 months (SOC7); † 6 months (SOC8). p.61
- MCQ Testosterone-induced clitoral growth plateaus at ~2 years. p.61
- MCQ Testosterone helps metoidioplasty but hinders vaginectomy. p.61
- MCQ Puberty blockade reduces scrotoplasty tissue. p.61
- MCQ No vaginectomy → no urethral lengthening. p.61
- MCQ Commonest vaginectomy complication: haemorrhage. p.61
- MCQ Urethrovaginal dissection without diathermy protects sphincter. p.61
- MCQ Neourethra: native, fixed, phallic parts. p.61
- MCQ Commonest stricture site: fixed–phallic anastomosis. p.61
- MCQ Commonest fistula site: phallic base. p.61
- MCQ Fistula >3 months: permanent, repair. p.61
- MCQ Defer stricture treatment ≥6 months post-phalloplasty. p.61
- MCQ Standard for long phallic strictures: two-stage urethroplasty. p.61
- MCQ Post-void dribbling: near-universal. p.61
- MCQ Erectile prosthesis ≥1 year, needs protective sensation. p.61
- MCQ Emergency retention after GCS: suprapubic catheter. p.61
- MCQ Metoidioplasty phallus 3–8 cm, spontaneous erection. p.62
- MCQ Scrotoplasty flaps: external pudendal artery, rotated 180°. p.62
- MCQ Repeated DVIU for anastomotic stricture: futile. p.62
- MCQ Catheterisation sequence: Coudé, stylet, then cystoscopy-guided. p.62
- MCQ Erectile cylinder anchored to pubic periosteum. p.62

## ★ Putting it together: genitalia, DSD, gender and special populations

### A · NEWBORN WITH ATYPICAL GENITALIA

- ALGORITHM

  **1. Treat as an emergency for the adrenal, not the genitalia:** congenital adrenal hyperplasia (CAH) is the commonest cause (half of all, 90% of 46,XX cases). Salt-wasting crisis (low sodium, high potassium, shock) usually appears at day 5–7, so normal early electrolytes do not exclude it (→ Ch 59).

  **2. Tests:** karyotype (with SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound for a uterus; then hormonal and genetic work-up guided by karyotype.

  **3. Team:** † defer sex assignment until evaluated by a multidisciplinary DSD team (endocrinology, urology, genetics, psychology) with the family.

  **4. Gonads at risk:** dysgenetic gonads with Y material (e.g. complete gonadal dysgenesis, 30–45% tumour risk) → prophylactic gonadectomy.

  p.63

### B · UROGENITAL SINUS AND CLOACA: WHAT DECIDES THE OPERATION

- KEY POINT True urethral length (bladder neck to confluence) matters more than common-channel length. Low confluence → flap vaginoplasty; high confluence → pull-through or total/partial urogenital mobilisation. Clitoroplasty only for severe hypertrophy (Prader III–IV), sparing the dorsal neurovascular bundle. In cloaca: early colostomy, decompress hydrocolpos and protect the kidneys, then image (cloacogram, MRI) and endoscope before definitive posterior sagittal repair; look for VACTERL associations and tethered cord (→ Ch 60). p.63

### D · IMMUNOTHERAPY AND HEREDITARY CANCER

- ALGORITHM

**Checkpoints:** tumour interferon-γ induces programmed death ligand-1 (PD-L1), which switches off PD-1-positive T cells; cytotoxic T-lymphocyte antigen-4 (CTLA-4) inhibitors act earlier, on priming. † Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathy): hold the drug and give steroids for grade ≥2; endocrinopathies need lifelong replacement (→ Ch 24).

**Hereditary syndromes to recognise:** von Hippel–Lindau (VHL, clear cell renal cancer; belzutifan), hereditary leiomyomatosis and renal cell cancer (fumarate hydratase; aggressive papillary), Birt–Hogg–Dubé (FLCN; hybrid oncocytic tumours), Lynch (mismatch repair; upper-tract urothelial cancer). BRCA/homologous-repair defects in prostate cancer → PARP inhibitor; FGFR alteration in urothelial cancer → erdafitinib (→ Ch 25).

p.63

### E · OLDER PATIENT BEFORE UROLOGICAL SURGERY

- KEY POINT Function, not age, predicts outcome. Screen for frailty, then do a comprehensive geriatric assessment if positive: cognition and capacity, delirium risk, nutrition, falls, polypharmacy. Avoid anticholinergic burden (prefer a β3-agonist for overactive bladder). For new incontinence, first look for transient causes († DIAPPERS: delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess urine output, restricted mobility, stool impaction) (→ Ch 120). p.64

### F · TRANSGENDER AND NONBINARY PATIENTS: UROLOGICAL POINTS

- ALGORITHM

Use the patient's name and pronouns; organ inventory guides screening — retained prostate after vaginoplasty (examine through the anterior neovaginal wall; † PSA runs lower on oestrogen), retained cervix after testosterone (→ Ch 126).

Vaginoplasty complications: canal stenosis, rectovaginal fistula (most devastating), urethrovaginal fistula (→ Ch 127). Phalloplasty: urethral lengthening brings fistula and stricture; skipping vaginectomy raises fistula risk (→ Ch 128).

p.64

# 3 Infections

321 core items · page numbers refer to the full book PDF

## Ch 26 · Infections of the Urinary Tract

### 1 · DEFINITIONS AND CLASSIFICATION

- **Factors making a UTI complicated**: anatomical or functional abnormality (benign prostatic hyperplasia [BPH], diverticula, ureteric obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO], concentrating defects), **male sex**, pregnancy, elderly, diabetes mellitus (DM), immunosuppression, childhood UTI, recent antibiotics, catheter, instrumentation, hospital-acquired, symptoms >7 days. Each item either impairs host defence or selects a resistant organism. MCQ Male sex alone makes UTI complicated. p.2

- TABLE Reinfection · New event; bacteria reintroduced (~**70%** of recurrences, often different organism) · Cleared, then new inoculation p.2

- **Catheter-associated UTI (CAUTI)**: commonest cause of secondary bacteraemia; ~**80%** of nosocomial UTIs. p.2

### 2 · PATHOGENESIS

- ***Escherichia coli***: ~**85%** community, ~50% hospital UTIs; share falls from ~three-quarters in uncomplicated cystitis to under half in complicated UTI, pyelonephritis and urosepsis. p.3

- MCQ *Enterococcus* is a Gram-positive coccus (streptococcal group), not an Enterobacteriaceae. p.3

- TABLE Tropism · Bladder → cystitis · Kidney → **pyelonephritis** (~**90%** of pyelonephritis strains) p.3

- ALGORITHM
  FimH binds uroplakin → invasion of umbrella cell (15 min–1 h) → intracellular replication → **intracellular bacterial community (IBC)**, biofilm-like (4–16 h) → fluxing and filamentation, escape and reinvasion (16–24 h) → exfoliation exposes deeper cells → **quiescent intracellular reservoirs (QIRs)** (from ~2 weeks)
  p.4

### 3 · EVALUATION AND DIAGNOSIS

- Dysuria + frequency without vaginal discharge → ~**90%** probability of UTI. Haematuria + positive nitrite → ≥90%. Dysuria is the most reliable single symptom; vaginal discharge lowers probability. p.5

- Microscopy: centrifuge 6–8 mL; pyuria = ≥**5–10 WBC/high-power field (HPF)**; squamous cells mark contamination. No pyuria → question the diagnosis. Culture-positive without pyuria → repeat with catheterised sample. p.5

- TABLE ≥**$10^2$ CFU/mL** of a known pathogen · Symptomatic women with cystitis; catheterised specimens p.5

### 4 · ANTIMICROBIAL PRINCIPLES

- TABLE Nitrofurantoin · Urine-only levels → useless for pyelonephritis, prostate, bacteraemia. Spares flora, minimal resistance. Toxicity: pulmonary reactions/fibrosis (long-term), neuropathy, haemolysis in glucose-6-phosphate dehydrogenase (G6PD) deficiency. Renal cut-off: Beers 2015 creatinine clearance (CrCl) <**30 mL/min** (older 50–60). Beers advises against long-term suppression in the elderly. p.8

- KEY POINT Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis. Broad, tissue-penetrating agents (quinolones, cephalosporins, carbapenems) are kept for invasive infection. Resistance is made mainly in the bowel, so an antibiotic's breadth, duration and repetition matter more than its MIC. p.8

## 5 · ASYMPTOMATIC BACTERIURIA

- TABLE Either sex · One catheterised · ≥**$10^2$** CFU/mL p.8

- KEY POINT ASB is treated only where bacteria are about to gain a route into blood or tissue: pregnancy (hydroureter → pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective. p.9

## 6 · CYSTITIS

- TABLE First-line · **Nitrofurantoin** 100 mg twice daily × **5 d** · Not if pyelonephritis suspected p.9

- TABLE First-line · **TMP-SMX** 160/800 mg twice daily × **3 d** · Avoid if local resistance >**20%** or used for UTI in past 3 months p.9

- ~**90%** medical: broad Gram-negative cover (add ampicillin if Gram-positive cocci), catheter drainage, glucose control. Debridement or partial/total cystectomy is rarely needed (necrosis, non-response). p.10

## 7 · RECURRENT UTI

- ALGORITHM
  Confirm true recurrent UTI (symptoms + positive culture) → same organism at short intervals? → yes: evaluate for persistence (upper tract imaging + cystoscopy) and remove nidus → reinfection: behavioural measures (≥1.5 L water, frequent and postcoital voiding, stop spermicide) → cranberry ≥36 mg proanthocyanidin (PAC) → intercourse-related: **postcoital antibiotic**; postmenopausal: **vaginal oestrogen**; neither: **methenamine hippurate** (if renal function normal) → continuous antibiotic prophylaxis as last resort
  p.11

- KEY POINT Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigation is reserved for features of persistence or complication. Prevention targets flora, pH, adhesion and washout rather than repeatedly sterilising urine. p.11

## 8 · KIDNEY INFECTIONS

- TABLE Outpatient, no vomiting · Ciprofloxacin 500 mg twice daily if community quinolone resistance ≤**10%**; levofloxacin 750 mg daily; TMP-SMX DS twice daily only if known sensitive. ± one initial IV dose of ceftriaxone 1 g or aminoglycoside. · Cipro 7; levo 5; TMP-SMX 14 p.12

- ALGORITHM
  Pyelonephritis → no sepsis/vomiting: urine culture ± US → oral outpatient 7–10 d | sepsis: blood + urine cultures, imaging → IV inpatient → improved by **72 h**: oral step-down | not improved: repeat cultures + **CT** → drain obstruction or abscess
  p.12

- KEY POINT Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication. p.12

- KEY POINT Fluid is a good sign and streaky gas a bad one. A liquefied, contained collection can be drained. Diffuse streaky gas without fluid means parenchyma is necrosing faster than the host can wall it off. p.13

- TABLE ≤**3 cm** · IV antibiotics alone, even with abnormal tract if stable p.13

- TABLE >**5 cm** · Percutaneous drainage first line; may need several drains, open drainage or nephrectomy p.13

- KEY POINT Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics. Pyonephrosis and large or obstructed abscesses need drainage. When the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy. p.15

### 9 · FOURNIER GANGRENE AND PERIURETHRAL ABSCESS

- **Fournier gangrene**: necrotising fasciitis of genitalia, perineum and/or perianal region. Fournier's original idiopathic disease of young men has given way to contemporary disease: a source is found in ~**95%**, age 50–79, male:female ~10:1. p.15

- TABLE **Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)** · CRP, WBC, haemoglobin, sodium, creatinine, glucose · ≥**6** suspicious; ≥**8** strongly predictive (diagnostic adjunct) p.15

- ALGORITHM
  Suspicion → resuscitation + electrolytes + broad IV antibiotics (piperacillin–tazobactam; or carbapenem/vancomycin + clindamycin or metronidazole) → **immediate radical debridement** to bleeding normal fascia, leave open → **re-look at 24–48 h** → suprapubic diversion (urethral source) / diverting colostomy (rectal source) → hyperbaric oxygen (adjunct) → reconstruction (grafts, flaps)
  p.15

### 10 · UROSEPSIS †

- TABLE Sepsis · Life-threatening organ dysfunction from dysregulated host response to infection: acute rise in Sequential Organ Failure Assessment (SOFA) score ≥**2** p.16

- ALGORITHM
  Hour-1 bundle: lactate → blood + urine cultures → broad-spectrum IV antibiotic within 1 h → 30 mL/kg crystalloid if hypotension or lactate ≥4 → vasopressor (noradrenaline) for MAP ≥65 → **source control**: decompress obstruction (stent or nephrostomy; drain only, no definitive stone surgery) → de-escalate by culture
  p.16

### 11 · SPECIAL POPULATIONS

- Changes: kidney +1 cm; ureteric dilatation (right > left) from progesterone and uterine compression, resolving by 6 weeks postpartum; GFR +30–50% → creatinine falls (0.7 → **0.5 mg/dL**), so a "normal" 0.8–0.9 may mean renal impairment. Proteinuria is abnormal >300 mg/24 h. p.16

- **Screen by culture** at the first visit or **12–16 weeks** (dipstick misses cases). Treat with a standard-length course (4–7 days beats single dose). p.16

- Definition: catheter in place >2 days + symptom (fever, suprapubic/CVA pain) + $\geq 10^5$ CFU/mL; in catheterised patients $\geq 10^2$ is significant. Bacteriuria is inevitable: **5–10% per catheter-day** (intermittent catheterisation 1–3% per catheterisation); 100% long term. p.17

- Catheter >**10 years** → bladder cancer risk (often squamous †) → surveillance. p.17

- Transplant: do not screen or treat ASB >1–2 months after transplant. Early UTI: uncomplicated 5–10 days (not 3); pyelonephritis 14–21 days. Stents raise UTI by about half → remove within **4 weeks**. Recurrent UTI → PVR, imaging → VCUG (graft VUR) → UDS. p.18

### 13 · PERIOPERATIVE ANTIMICROBIAL PROPHYLAXIS

- TABLE Timing · Within **60 min** of incision; vancomycin and quinolones within 120 min p.18

- TABLE Duration · **Single dose** default; ≤**24 h** only for PCNL, prostheses, special cases p.18

- KEY POINT Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit. Otherwise give one well-timed, flora-matched dose; anything longer is treatment. p.19

### 15 · MCQ ONE-LINERS

- MCQ *S. saprophyticus*: ~10% of cystitis in young sexually active women. p.20

- MCQ P pili bind glycolipid receptors in the kidney and are found in ~90% of pyelonephritis strains. p.20

- MCQ Type 1 pili adhesin FimH binds uroplakin Ia; mannose inhibits it. p.20
- MCQ Most abundant urinary protein blocking type 1 pili: uromodulin (Tamm–Horsfall). p.20
- MCQ Quiescent intracellular reservoirs explain same-strain recurrence without a urological focus. p.20
- MCQ Nitrite-negative uropathogens: enterococci, staphylococci, *Pseudomonas*. p.20
- MCQ Cystitis culture threshold in symptomatic women: $10^2$ CFU/mL. p.20
- MCQ ASB in women needs two consecutive voided ≥$10^5$ specimens; catheter specimen ≥$10^2$. p.20
- MCQ Only screen-and-treat ASB groups: pregnancy and mucosa-breaching urological procedures. p.20
- MCQ TMP-SMX empirical for cystitis only if local resistance ≤20%. p.20
- MCQ Quinolone for outpatient pyelonephritis only if community resistance ≤10%. p.20
- MCQ Cranberry acts via type A proanthocyanidin blocking P-fimbrial adhesion; dose ≥36 mg. p.20
- MCQ Methenamine requires urine pH <6; ineffective against urease producers. p.20
- MCQ Oral oestrogen does not prevent recurrent UTI; vaginal oestrogen does. p.20
- MCQ Enterococci are never covered by cephalosporins. p.20
- MCQ Ceftazidime covers *Pseudomonas*; ceftriaxone does not. p.20
- MCQ Perinephric abscess: symptoms >5 days or fever ≥5 days on antibiotics. p.20
- MCQ Renal abscess ≤3 cm: antibiotics alone; >5 cm: percutaneous drainage. p.20
- MCQ XGP commonest organism: *Proteus*; pelvis contracted around a staghorn. p.20
- MCQ Michaelis–Gutmann bodies: pathognomonic of malakoplakia. p.20
- MCQ EPN without fluid, streaky gas: worst prognosis. p.20
- MCQ Fournier hypocalcaemia: lipase-released fatty acids chelate calcium. p.20
- MCQ LRINEC ≥6 suggests necrotising fasciitis. p.20
- MCQ Testes are spared in Fournier gangrene because of independent blood supply. p.20
- MCQ CAUTI: bacteriuria 5–10% per catheter-day; change catheter before treating. p.20
- MCQ SCI bladder management with fewest long-term complications: clean intermittent catheterisation. p.20
- MCQ Surgical prophylaxis: within 60 min; single dose; ≤24 h for PCNL and prostheses. p.20
- MCQ Pins, plates and screws never require urological antibiotic prophylaxis. p.20

## Ch 27 · Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

### 1 · PROSTATITIS: DEFINITIONS AND CLASSIFICATION

- **Drach (1978)** gave the first systematic classification, built on the segmented cultures of **Meares and Stamey (1968)**: acute bacterial, chronic bacterial, nonbacterial (inflammation, negative culture), prostatodynia (neither). MCQ p.21
- Categories I + II = **5–10%**; category III = **90–95%**. Old nonbacterial prostatitis + prostatodynia were merged into III. p.21
- **Category III definition (NIH Chronic Prostatitis Collaborative Research Network, CPCRN)**: pelvic pain/discomfort for **≥3 of the previous 6 months**. Exclusions: uropathogens on standard culture, urogenital cancer, prior radiation/chemotherapy, urethral stricture, neurological bladder disease. p.21

- Histological inflammation is near-universal (autopsy: chronic inflammation in **>70%**) and incidental in benign prostatic hyperplasia (BPH) chips and biopsies, so on its own it cannot define a clinical disease. p.21

## 2 · CATEGORY I: ACUTE BACTERIAL PROSTATITIS

- KEY POINT **Why young healthy men get ABP:** the organism, not the host, explains it — virulent uropathogenic *E. coli* that forms **biofilm** (bacteria in a polysaccharide matrix) more often than cystitis/pyelonephritis strains; biofilm blocks antibiotic penetration, which is why prostatic infection needs longer, well-penetrating therapy. p.22
- In febrile male UTI, **>90%** show rise in PSA and/or prostate volume even if the picture suggests pyelonephritis — prostatic involvement is the rule. p.22
- **PSA not recommended** acutely; if raised, follow to normal (up to **6 months**) before attributing to cancer. p.22
- **European Association of Urology (EAU)**, systemic illness: amoxicillin + aminoglycoside; 2nd-generation cephalosporin + aminoglycoside; or 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%**, fully oral course, not hospitalised, or β-lactam anaphylaxis. p.22
- **Duration**: traditionally 4 weeks; **2 weeks** ciprofloxacin gives equivalent cure. Relapse lower with FQ/parenteral β-lactams than co-trimoxazole/oral β-lactams. TB prostatitis: ≥**6 months** anti-TB therapy. p.23
- ALGORITHM
  Abscess <1–2 cm → antibiotics (drainage may shorten course) → progression mandates intervention → peripheral/localised: US-guided percutaneous drainage (recurrence 15–33%) → large or failed: **transurethral unroofing** (intra-op US to locate; massage after unroofing) → extraprostatic extension: open drainage (rare)
  p.23

## 3 · CATEGORY II: CHRONIC BACTERIAL PROSTATITIS

- TABLE **2-glass pre/post-massage** (current standard) · Midstream pre-massage + first 10 mL post-massage · Predicts 4-glass result in **>95%**; simpler p.23
- Exclude outlet obstruction/retention: PVR **>180 mL** raises infection risk. p.23
- Men **<45** with first UTI: imaging normal — but assess for **urethral stricture** (up to 41%). Image if diabetes, stones, voiding or neurological problems, poor response, persistent haematuria. p.24
- KEY POINT **Why fluoroquinolones?** No active transport into the prostate; entry is by passive diffusion, governed by lipid solubility, ionisation, protein binding and size. FQs are lipid-soluble **zwitterions** and concentrate in the alkaline CBP fluid; they also cover Gram-negatives (incl. *Pseudomonas*), Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate well. p.24
- TABLE First line · FQ (ciprofloxacin = levofloxacin) **4–6 weeks** p.24
- TABLE FQ-resistant, TMP-SMX-sensitive · TMP-SMX **3 months** (cure 30–50% vs 60–80% for FQ) p.24

## 4 · CATEGORY III: CP/CPPS — AETIOLOGY

- TABLE Pelvic floor · Muscle tenderness and **failure to relax** (not weakness); electromyography: high resting tone, poor endurance. Tender in **81%** on standardised examination. p.24

## 5 · CP/CPPS — CLINICAL FEATURES AND PHENOTYPING

- **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)**: pain, urinary, quality of life (QOL) domains; pain and QOL subscales are responsive (trial endpoints). †Total 0–43 (pain 0–21, urinary 0–10, QOL 0–12). A fall of **≥6 points** = clinically meaningful response. Limitation: no bladder-pain items → **Genitourinary Pain Index (GUPI)** adds pain on bladder filling/emptying. p.24

### 7 · CP/CPPS — TREATMENT

- KEY POINT **Governing fact:** no monotherapy has a positive definitive trial — every failure tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team (urology, physiotherapy, psychology, pain, gastroenterology). p.26

- TABLE **Antibiotics** · Placebo-controlled ciprofloxacin and levofloxacin trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. Benefit may be anti-inflammatory (↓ interleukin-6, tumour necrosis factor-α) or due to VB3-only bacteria. **Repeated courses without positive culture not accepted**. p.26

### 9 · ACUTE EPIDIDYMITIS

- Pain, swelling, inflammation of the epididymis for **<6 weeks**. Route: **ascending from urethra** (~80% urine–epididymis concordance). p.27

- TABLE Chlamydia/gonorrhoea · Ceftriaxone **500 mg IM** single dose (1 g if ≥150 kg) + doxycycline 100 mg bd **10 days** p.27

### 10 · CHRONIC ORCHITIS AND EPIDIDYMITIS

- Four-drug anti-TB therapy ≥**6 months**. Surgery (drain abscess, epididymectomy/orchidectomy) **only while on anti-TB drugs**; histology decides further duration. p.27

### 11 · CHRONIC SCROTAL CONTENT PAIN (CSCP)

- Peak 20–30 years. Post-procedural: **inguinal hernia repair** (commonest; ↑ with mesh/recurrent repair), vasectomy (long-term pain ~**1 in 1000** — consent figure), scrotal surgery. p.28

- **Spermatic cord block (SCB)**: **20 mL 0.25% bupivacaine** into the cord at the **pubic tubercle**. **>50% temporary relief** = cord nerves involved and independently predicts success of microsurgical denervation of the spermatic cord (MDSC). p.28

- ALGORITHM
  Identifiable lesion → directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy for intermittent torsion, **microsurgical varicocelectomy**) → post-vasectomy pain → **vasectomy reversal** (improvement 82–93%; relief may take ≥3 months) → no lesion + positive SCB → **MDSC** → localised epididymal pathology → epididymectomy → last resort: **inguinal orchidectomy** p.28

### 13 · MCQ ONE-LINERS

- MCQ NIH category IV prostatitis: no pain; found on biopsy or semen analysis. p.29
- MCQ IIIA vs IIIB: leucocytes in EPS/VB3/semen; no therapeutic relevance. p.29
- MCQ CP/CPPS definition: pelvic pain ≥3 of previous 6 months. p.29
- MCQ Commonest organism in ABP: *E. coli* (65–80%). p.29
- MCQ Febrile prostatitis after transrectal biopsy: carbapenem; avoid FQ and TMP-SMX. p.29
- MCQ Retention in ABP needing long-term drainage: suprapubic catheter. p.29
- MCQ No response to ABP therapy at 48 h: image for prostatic abscess. p.29
- MCQ CBP hallmark: recurrent UTI, same organism, asymptomatic between episodes. p.29
- MCQ Current localisation standard: 2-glass pre/post-massage test (predicts 4-glass >95%). p.29
- MCQ FQs penetrate prostate because they are lipid-soluble zwitterions. p.29
- MCQ CBP first line: FQ 4–6 weeks. p.29
- MCQ Macrolides help in CBP because they penetrate biofilm. p.29
- MCQ BCG-induced granulomatous prostatitis: transient benign PSA rise. p.29
- MCQ PSA in CP/CPPS: not indicated; raised value must not be ascribed to it. p.29

- MCQ Pseudodyssynergia: sphincter activity during voiding without straining → physiotherapy. p.29
- MCQ Clinically meaningful NIH-CPSI response: ≥6-point fall. p.29
- MCQ UPOINT: Urinary, Psychosocial, Organ, Infection, Neurologic/systemic, Tenderness. p.29
- MCQ Best combination in CP/CPPS meta-analyses: antibiotic plus α-blocker. p.29
- MCQ Commonest viral orchitis: mumps; atrophy in about half. p.29
- MCQ Epididymitis <35 y: chlamydia/gonorrhoea; >35 y: *E. coli*. p.29
- MCQ STI epididymitis: ceftriaxone 500 mg IM + doxycycline 10 days. p.29
- MCQ Drug causing chronic epididymitis: amiodarone. p.29
- MCQ Michaelis–Gutmann bodies: malakoplakia. p.29
- MCQ TB epididymal mass enlarging after 3 weeks of therapy: inguinal exploration. p.29
- MCQ Spermatic cord block: 20 mL 0.25% bupivacaine at pubic tubercle. p.29
- MCQ >50% relief after cord block predicts MDSC success. p.29
- MCQ Trifecta nerve complex: cremaster, perivasal sheath, posterior lipomatous tissue. p.29
- MCQ Orchidectomy for scrotal pain: inguinal approach better than scrotal. p.29

## Ch 28 · Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

### 1 · DEFINITION AND NOMENCLATURE

- **American Urological Association (AUA) definition** (clinical): "an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with lower urinary tract symptoms (LUTS) of **>6 weeks**' duration, in the absence of infection or other identifiable causes." Pressure and discomfort are included deliberately because patients describe them so. p.30
- † **European Society for the Study of Interstitial Cystitis (ESSIC) definition (2008)**: chronic (**>6 months**) pelvic pain, pressure or discomfort perceived to be related to the bladder, with ≥1 other urinary symptom (persistent urge or frequency), confusable diseases excluded. † ESSIC types by cystoscopy with hydrodistension (1 normal, 2 glomerulations, 3 HL; X not done) and biopsy (A normal, B inconclusive, C inflammatory infiltrate/detrusor mastocytosis/granulation tissue/fibrosis) — e.g. **type 3C** = classic HL disease. p.30
- **National Institutes of Health/National Institute of Diabetes and Digestive and Kidney Diseases (NIH/NIDDK) criteria (1987)**: a *research* definition for homogeneous trial populations; applied clinically it missed **~60%** of patients → definitions broadened. p.30

### 2 · EPIDEMIOLOGY (HEADLINE)

- RAND Interstitial Cystitis Epidemiology (RICE) study: **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms. Mean onset 30–40 years. p.30
- Female:male **5:1** in clinic series but **2–3:1** in population studies → men are under-diagnosed. Many men labelled CP/CPPS report painful bladder filling/painful urgency → **ask men about bladder pain** and reconsider IC/BPS when CP/CPPS treatment fails. p.30
- **HL prevalence**: working figure **~20%** (range 3.5–56%, method- and centre-dependent). Rises with age: uncommon <30, significantly higher ≥50, >half of patients over 60–70. Same rate in men. p.30
- KEY POINT Why uncontrolled series "work" but randomised controlled trials (RCTs) fail: regression to the mean, placebo effect and more attention in early follow-up. Expect improvement in the first 3–4 weeks of any treatment. p.30

### 3 · AETIOLOGY AND PATHOPHYSIOLOGY

- KEY POINT Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical. p.31

### 5 · ASSOCIATED CONDITIONS

- **Regional**: **high-tone pelvic floor dysfunction (HTPFD)** in **~80%**; vulvodynia (vulvar pain ≥3 months); endometriosis. p.32

- KEY POINT Commonest ≠ most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Associated disorders are phenotype markers — widespread pain predicts failure of bladder-directed therapy. p.32

### 7 · DIAGNOSIS — EXCLUSION PLUS LOCALISATION

- **Potassium sensitivity test (PST)**: intravesical **0.4 M KCl** vs sodium (water); provocation of pain/urgency = positive, thought to reflect epithelial permeability (GAG defect) or hypersensitivity. **Not recommended**: painful, and neither sensitive nor specific — negative in up to ¼ of NIDDK patients; positive in ~⅓ of normal men, ¼ of OAB, most CP/CPPS, virtually all radiation/acute bacterial cystitis. Does not predict PPS response. p.34

- Under anaesthesia: fill to **80 cm $H_2O$ for 1-3 min** → drain (terminal effluent blood-tinged) → refill → **glomerulations** (pinpoint submucosal petechiae); mucosal tears, often near bladder neck. p.34

### 8 · PHENOTYPE-DIRECTED MANAGEMENT

- KEY POINT MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain** only; amitriptyline helped **widespread pain** only. A trial negative in all-comers can hide a positive result in the right phenotype. p.34

### 9 · TREATMENT — AUA 2022 PRINCIPLES

- ALGORITHM

  Symptoms ≥6 weeks → history, examination (pelvic floor, Q-tip, DRE), PVR, urinalysis, culture ± cytology → exclude confusable disease → cystoscopy if HL suspected (>50) → HL: fulguration/triamcinolone → refractory: CyA
  No HL → match to phenotype: pelvic floor → physiotherapy; bladder-centric (lidocaine responder) → intravesical; widespread → systemic/CBT → behavioural + oral + intravesical as needed → BTX-A or SNM → (rare, bladder-centric, all else failed) diversion ± cystectomy

  p.35

### 10 · TREATMENT OPTIONS (AUA 2022)

- Landmark RCT: 10 sessions myofascial therapy vs global massage → **59% vs 26%** improved. p.35

- **Amitriptyline** (tricyclic antidepressant, TCA): anticholinergic, serotonin/noradrenaline reuptake block, potent H1 antagonism, sedation, ?β-adrenergic detrusor relaxation. Dose: **10 mg at night, +10 mg weekly to ~50 mg** (25–75 mg). NIH RCT: 55% vs 45% (EBMP alone), not significant overall, but **77% vs 53%** in those tolerating ≥50 mg. Efficacy independent of HL status. Side-effects: sedation, anticholinergic effects, weight gain; caution with incomplete emptying. **Contraindications**: long QT, conduction block, recent myocardial infarction, heart failure, ventricular arrhythmia. p.36

- **PPS**: synthetic sulfated polysaccharide resembling GAGs; **100 mg three times daily**; only **3-6%** excreted in urine; effect takes **3-6 months**. Only FDA-approved oral drug for IC. Trials conflicting; phase IV RCT negative (≈40% response in both arms); probably less effective in HL disease. AUA Option. Adverse effects (<4%): reversible alopecia, diarrhoea, nausea, rash, rare bleeding; caution in high breast-cancer risk. p.36

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone; biopsy if never biopsied (exclude CIS). **75-90%** improve; re-treatment needed at mean 7–12 months (triamcinolone) or 11–29 months (fulguration) — counsel about periodic repeat. p.37

- **ALGORITHM**
  General anaesthesia → on entry, map HL site/number/size before distension bleeding → mild distension (lesion centres bleed) → biopsy suspicious and any trigonal lesion → fulgurate at **low volume** (Bugbee) from the **outer margin inwards** → triamcinolone 40 mg/mL (or diluted 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally → fulgurate bleeding sites → empty, instil lidocaine–bupivacaine
  p.37

- **KEY POINT** Why fulgurate from the margin inward? Cautery-induced erythema and oedema keep spreading peripherally, so the true lesion edge is lost if you start centrally. p.37

- Anaesthetic capacity **<400 mL** marks a bladder-centric phenotype (worse pain, more inflammation, often HL). p.37

## 11 · WHAT NOT TO DO (AUA)

- **KEY POINT** Nothing is worse than irreversible surgery followed by continued pain and a surgeon with nothing more to offer — every predictor of surgical success marks bladder-generated pain. p.38

## 13 · MCQ ONE-LINERS

- **MCQ** AUA IC/BPS definition requires symptoms for >6 weeks; ESSIC requires >6 months. p.39
- **MCQ** NIDDK criteria were research criteria and missed ~60% of clinically diagnosed patients. p.39
- **MCQ** ESSIC type 3C = Hunner lesion with inflammatory infiltrate on biopsy. p.39
- **MCQ** About 20% of IC/BPS patients have HLs; prevalence rises with age (>50). p.39
- **MCQ** Commonest associated condition: IBS; largest relative excess: autoimmune disease (IBD, SLE). p.39
- **MCQ** Pain and urgency incontinence distinguish IC/BPS from OAB. p.39
- **MCQ** Detrusor overactivity on UDS (15–19%) does not exclude IC/BPS. p.39
- **MCQ** AUA 2022: office cystoscopy when HL suspected, especially age >50. p.39
- **MCQ** Glomerulations are non-specific and do not correlate with severity. p.39
- **MCQ** Potassium sensitivity test uses 0.4 M KCl; no longer recommended. p.39
- **MCQ** Local anaesthetic challenge localises pain to the bladder and selects intravesical therapy. p.39
- **MCQ** Sterile pyuria in suspected IC/BPS: culture for tuberculosis. p.39
- **MCQ** Young patient, inflamed bladder, hydronephrosis: think ketamine cystitis. p.39
- **MCQ** Myofascial PT beat global massage 59% vs 26% — Guideline Standard. p.39
- **MCQ** Amitriptyline: start 10 mg nocte, increase 10 mg weekly to ~50 mg. p.39
- **MCQ** PPS: only FDA-approved oral IC drug; 3–6% reaches urine; 3–6 months to act. p.39
- **MCQ** PPS causes cumulative-dose pigmentary maculopathy; retinal examination within 6 months of starting. p.39
- **MCQ** Ciclosporin A 1–3 mg/kg/day is for refractory HL disease. p.39
- **MCQ** DMSO 50%: FDA-approved intravesical agent; dwell 15–20 min; garlic breath. p.39
- **MCQ** HL treatment: fulguration/triamcinolone — 75–90% respond; fulgurate from margin inward. p.39
- **MCQ** Therapeutic hydrodistension: 60–80 cm $H_2O$ for 1–3 min; benefit gone by 6 months. p.39
- **MCQ** Intradetrusor BTX-A 100 U into trigone; patient must accept CIC. p.39
- **MCQ** IC/BPS has the highest SNM explantation rate among indications. p.39

- MCQ AUA: no long-term antibiotics, BCG, long high-pressure hydrodistension, or chronic oral steroids. p.39
- MCQ Pain persisting after spinal block to T10 predicts failure of bladder surgery. p.39

## Ch 29 · Sexually Transmitted Diseases

### 1 · SCREENING, PREVENTION AND PARTNER NOTIFICATION

- **Nucleic acid amplification test (NAAT)** is the standard screen. Sites follow anatomy and exposure: men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia. p.40
- **Partner notification**: partners from the previous **60 days** for urethritis, chlamydia and *Mycoplasma genitalium*. All partners for trichomoniasis; the previous month for scabies and lice. Abstain for 7 days and until partners are treated. p.40

### 2 · URETHRITIS

- Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**; p.40
- ALGORITHM
  Persistent NGU → check adherence and untreated partner → NAAT for *M. genitalium* ± *T. vaginalis* → negative → think chronic pelvic pain syndrome
  p.41
- KEY POINT Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease (PID) or infertility. Treating the man protects his partner. Chlamydia can also cause reactive arthritis. p.41

### 3 · EPIDIDYMITIS BY AGE

- TABLE Group · Organisms · Empiric regimen (**10 days**) p.41

### 4 · GENITAL ULCER DISEASE

- TABLE Lymphogranuloma venereum (LGV) (*C. trachomatis* L1–L3) · Small, painless, transient · Painful, matted, with fistulae; †groove sign · Chlamydia NAAT · Doxycycline bd **21 days** p.42

### 5 · SYPHILIS STAGING AND SEROLOGY

- A significant change is **fourfold (two dilutions)**. Use the same NTT every time. p.43
- ALGORITHM
  **Reverse algorithm**: TT positive → NTT → NTT negative → second, different TT → positive without past treatment = treat as late latent
  p.43

### 8 · HIV FOR UROLOGISTS

- **PEP**: start within **72 h** after a substantial-risk exposure, i.e. mucosal, broken-skin or percutaneous contact with blood or genital fluid from an HIV-positive source. Do not give PEP after 72 h or for urine, saliva or tears. p.44
- **Needlestick**: risk about **0.3%**. Give three-drug PEP even if the source viral load is low. p.45
- **Circumcision** cuts men's heterosexual HIV acquisition by **50–60%**. The thin inner foreskin is rich in Langerhans cells. It does not protect partners. p.45

### 9 · MCQ ONE-LINERS

- MCQ Gram-stain urethritis threshold: ≥2 WBC per oil-immersion field. p.45
- MCQ Gonorrhoea: ceftriaxone 500 mg IM once. p.45
- MCQ Gonorrhoea test of cure: pharyngeal infection only. p.45
- MCQ Recurrent NGU: *M. genitalium*. p.45
- MCQ Resistant *M. genitalium*: doxycycline, then moxifloxacin. p.45

- MCQ Chlamydia: doxycycline 7 days. p.45
- MCQ Epididymitis in a man >35: levofloxacin 10 days. p.45
- MCQ Commonest genital ulcer: herpes. p.45
- MCQ Painful undermined ulcer with bubo: chancroid. p.45
- MCQ Donovan bodies: granuloma inguinale. p.45
- MCQ LGV: doxycycline 21 days. p.45
- MCQ HSV latency: sacral ganglia. p.45
- MCQ Palmar rash and condylomata lata: secondary syphilis. p.45
- MCQ Late latent syphilis: benzathine weekly × 3. p.45
- MCQ Neurosyphilis: IV aqueous penicillin. p.45
- MCQ Fever after syphilis treatment: Jarisch–Herxheimer reaction. p.45
- MCQ Significant NTT change: fourfold. p.45
- MCQ HPV-6: giant condyloma. p.45
- MCQ Podofilox: contraindicated in pregnancy. p.45
- MCQ HPV vaccine before 15: two doses. p.45
- MCQ Scabies: permethrin 5%. p.45
- MCQ PEP: within 72 h, for 28 days. p.45
- MCQ Needlestick HIV risk: 0.3%. p.45
- MCQ Indinavir stones: radiolucent. p.45
- MCQ Circumcision: 50–60% less HIV. p.46
- MCQ Male trichomonas: metronidazole 2 g once. p.46
- MCQ Chancroid diagnosis: by exclusion of syphilis and HSV. p.46
- MCQ Commonest LGV presentation now: proctocolitis. p.46
- MCQ Molluscum: umbilicated papules; HAART first in HIV. p.46
- MCQ HIV ED: low-dose sildenafil with protease inhibitors. p.46
- MCQ Salmonella UTI in HIV: lifelong prophylaxis. p.46

## Ch 30 · Cutaneous Diseases of the External Genitalia

### 1 · APPROACH TO GENITAL DERMATOSES

- TABLE Papule · Solid, raised, **≤0.5 cm** · Erosion · Epidermal loss only → **no scar** p.47

### 3 · ECZEMATOUS DERMATOSES

- Chronic relapsing, flexural, intensely pruritic; ill-defined border (papulosquamous lesions are sharp). **90%** present before age 5; asthma/rhinitis association. p.48
- TABLE Share · ~**80%** · Minority; genital rare (2.4%) p.48
- Possible response to *Malassezia furfur*. Up to **83%** of acquired immunodeficiency syndrome (AIDS) patients; also Parkinson disease → **extensive SD should prompt human immunodeficiency virus (HIV) testing**. p.48

### 4 · PAPULOSQUAMOUS DISORDERS

- TABLE First-line · Low-potency steroid (hydrocortisone butyrate 0.1%; tar in 1% hydrocortisone) — **≤2 weeks** continuous; triamcinolone 0.1% for thick plaque p.49

### 5 · LICHEN SCLEROSUS (LS) / BALANITIS XEROTICA OBLITERANS (BXO)

- Chronic inflammatory **scarring** disease, cause unknown, familial tendency; female 6–10×; in males usually >60 years. **15–20%** extragenital. p.50

- **ALGORITHM**
  Biopsy → **potent steroid (clobetasol 0.05%) for 3 months** → taper → twice-weekly maintenance (or mometasone 0.1% / calcineurin inhibitor) → lifelong surveillance of non-healing fissures, ulcers, nodules
  p.50

- Deliberately inverts the short-course rule because the disease scars and transforms. Evidence strongest in women and children; recurrence **40–50%** → proactive maintenance. p.50

- **KEY POINT** Steroid strength tracks the scarring and malignant stakes of the disease; LS is the single exception to the short-course rule. p.50

### 6 · VESICULOBULLOUS AND DRUG ERUPTIONS

- **TABLE** Lesion · Symmetric **target lesions**; palms, soles, mouth · Prodrome → red macules → blisters → necrosis; genital ~**60%** · Solitary violaceous plaque with halo, often glans; may blister p.50

- **TABLE** Course · Self-limited, recurs · 4–6 weeks; mortality to ~**30%** (TEN overlap) · **Recurs at the same site within 24 h** of re-exposure p.50

- **ALGORITHM**
  SJS: stop offending drug → burn-type supportive care → intensive care / burn unit (lowers mortality) → multispecialty wound care; options with limited data: cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis, intravenous immunoglobulin (IVIG); systemic steroids controversial
  p.51

- **KEY POINT** The same drug class can cause a trivial FDE or a lethal SJS: extent and systemic illness, not the culprit, separate them. Genital FDE mimics herpes — the "same spot each time" history is the clue. p.51

### 8 · INFECTIONS

- **ALGORITHM**
  Remove irritants → hygiene → topical antibiotic/antifungal → short low-potency steroid → **failure: biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV) → recurrent with phimosis: **circumcision**
  p.52

- **KEY POINT** Fournier gangrene is a fascial disease — skin shows a fraction of the necrosis; testes are spared (own blood supply within tunica). Imaging (CT most specific for gas) must never delay surgery. Debride to bleeding tissue, second look at 24–48 h. Details → Ch 26. p.52

- **TABLE** Treat · Topical imidazole **≥2 weeks**; fluconazole 150 mg occasionally; test for diabetes · Topical azole/allylamine; deep follicular → terbinafine 250 mg/day 1 week; no steroids · Antibacterial soap, clindamycin 1%, miconazole; oral erythromycin if severe · Low-potency steroid p.53

- **KEY POINT** Antifungal spectrum: azoles cover yeast and dermatophyte (use when unsure); nystatin misses tinea; allylamines are weaker on *Candida*. A negative KOH from an untreated lesion → rethink the diagnosis. p.53

### 9 · PREMALIGNANT AND MALIGNANT LESIONS

- **TABLE** SCCis — accessible (scrotum) · Excision with **5-mm margin** (favoured) p.54

- **KEY POINT** HPV mapping: 6/11 → condyloma and verrucous carcinoma (low-grade, locally destructive, rarely metastasises); 16 → bowenoid papulosis and SCCis. Invasive penile SCC → penile cancer chapter. p.54

- TABLE Extramammary Paget disease (EPD) · Sharp red plaque at apocrine sites; Paget cells (CEA, EMA +); underlying carcinoma **10–30%** (urethra, prostate, bladder, rectum) · **Mandatory search for internal malignancy**; excision/Mohs; imiquimod, 5-FU, radiotherapy p.54

### 10 · BENIGN MALE GENITAL LESIONS

- KEY POINT Zoon balanitis vs erythroplasia of Queyrat: both are red glans plaques in uncircumcised men — one benign and cured by circumcision, the other SCCis. Hence a persistent red glans plaque always needs biopsy. p.55

### 12 · MCQ ONE-LINERS

- MCQ Size cut-off separating papule from nodule and vesicle from bulla: 0.5 cm. p.55
- MCQ Erosion heals without scar; ulcer (involves dermis) scars. p.55
- MCQ Prepuce enhances topical steroid penetration as an occlusive dressing. p.55
- MCQ Key defect in atopic dermatitis: filaggrin. p.55
- MCQ Commonest allergic contact allergen: nickel. p.55
- MCQ Severe allergic contact dermatitis: 3-week prednisone taper. p.55
- MCQ Most recurrent erythema multiforme minor is triggered by HSV. p.55
- MCQ SJS commonest cause: drugs; genital involvement ~60%. p.55
- MCQ FDE recurs at the same site, typically glans, within 24 h of re-exposure. p.55
- MCQ Psoriasis histology: "sandwich sign". p.55
- MCQ Biologic with best data for genital psoriasis: ixekizumab (IL-17). p.55
- MCQ PUVA contraindicated in genital psoriasis: genital SCC risk. p.55
- MCQ Balanitis circinata: HLA-B27 reactive arthritis; histology like psoriasis. p.55
- MCQ LP primary lesion: polygonal violaceous flat-topped papule. p.55
- MCQ LS is associated with HPV-independent penile SCC. p.55
- MCQ LS: clobetasol for 3 months then maintenance; recurrence 40–50%. p.55
- MCQ Late stage of male LS with stricture: balanitis xerotica obliterans. p.55
- MCQ Extensive seborrhoeic dermatitis: test for HIV. p.55
- MCQ Behçet: HLA-B51; genital ulcer most specific sign. p.55
- MCQ Behçet 2014 criteria threshold: ≥4 points. p.55
- MCQ PG biopsy taken from the ulcer edge; avoid debridement. p.55
- MCQ Fournier gangrene: testes spared; second look at 24–48 h. p.55
- MCQ HS: avoid repeated incision and drainage; adalimumab is effective biologic. p.55
- MCQ Lithium exacerbates hidradenitis suppurativa. p.56
- MCQ Erythrasma fluoresces coral-red under Wood's lamp. p.56
- MCQ Scrotal involvement favours Candida over tinea cruris. p.56
- MCQ Nystatin does not treat dermatophytes. p.56
- MCQ Genital bite organism: Eikenella corrodens; treat with amoxicillin–clavulanate. p.56
- MCQ Ecthyma gangrenosum: Pseudomonas bacteraemia, black eschar. p.56
- MCQ Permethrin is repeated at 1 week to kill newly hatched lice/mites. p.56
- MCQ Erythroplasia of Queyrat: SCCis of glans in uncircumcised men. p.56
- MCQ Bowenoid papulosis: HPV-16, young adults, may regress. p.56

- MCQ Buschke–Lowenstein: HPV 6/11; radiotherapy relatively contraindicated. p.56
- MCQ EPD: search for underlying urogenital or rectal carcinoma. p.56
- MCQ Zoon balanitis: plasma cell balanitis, cured by circumcision. p.56

## Ch 31 · Tuberculosis and Parasitic Infections of the Genitourinary Tract

### 1 · UROGENITAL TUBERCULOSIS — ORGANISM AND PATHOGENESIS

- UG TB = **15–20%** of pulmonary TB in developing countries; second commonest extrapulmonary site after lymph nodes. p.57
- **Latent** (90–95%): calcified granuloma, not infectious; lifetime reactivation **5–15%**. **Active**: caseation → bacillary release. Reactivation triggers: old age, renal failure, diabetes, malnutrition, human immunodeficiency virus (HIV), immunosuppression. p.57
- TABLE Ascending · Uncommon · Intravesical BCG: UG TB in **~1%** (pyelonephritis, renal abscess, ureteric obstruction, cystitis, prostatitis, epididymo-orchitis) p.57
- KEY POINT UG TB is a blood-borne disease landing in the two best-perfused UG organs — kidney and epididymis — and spreading from there. Haematogenous seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction. p.57

### 2 · CLINICAL FEATURES AND PATHOLOGY BY SITE

- Constitutional symptoms in **< 20%** — absence does not exclude TB. **Lower urinary tract symptoms dominate** (dysuria, storage symptoms); haematuria and flank pain ~one-third; colic < 10% (sloughed papillae, clot, caseous debris). p.57
- TABLE Kidney · **80%**, haematogenous · Caseating cavities → autonephrectomy (up to one-third) p.57

### 3 · DIAGNOSIS

- ALGORITHM
  Hierarchy: culture or MTBC DNA positive → diagnosis. Negative → caseating granulomas on tissue in right context. Negative → compatible picture + exposure + response to empirical treatment. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.
  p.58
- Sensitivity 10–90% (intermittent shedding). Ziehl–Neelsen smear of urine for acid-fast bacilli (AFB): sensitivity **< 50%**. p.59
- TST (Mantoux) read at 48–72 h; cut-offs by risk: **≥ 5 mm** contacts, immunosuppressed; **≥ 10 mm** immigrants, high-risk workers, drug injectors; **≥ 15 mm** no risk factors. False positive with BCG; false negative 10–25%. p.59
- KEY POINT Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). Hence obstruction is the commonest finding and must be watched during treatment. p.59

### 4 · MEDICAL TREATMENT

- ALGORITHM
  **Standard 6-month short course (2HRZE / 4HR)**: Intensive 2 months — daily INH (H) + rifampicin (R) + pyrazinamide (Z) + ethambutol (E); stop E once fully susceptible → kills rapidly multiplying bacilli. Continuation 4 months — HR daily (or thrice weekly by DOT; twice weekly no longer recommended) → eradicates persisters. Pyridoxine throughout.
  p.60

### 5 · SURGICAL TREATMENT

- **Timing: 4-6 weeks after starting chemotherapy** — inflammation subsides, bacillary load falls, lesions stabilise (emergencies excepted). p.60

- ALGORITHM
  Uraemia or sepsis → emergency decompression → retrograde double-J stent (preferred; succeeds in < half, 41%) → fails → antegrade stent → fails → percutaneous nephrostomy (PCN; loculated systems may need more than one) → definitive repair; unsalvageable kidney → nephrectomy.
  p.61

- TABLE Moderate gap · Psoas hitch · Up to **5 cm**; needs good capacity; avoid genitofemoral/femoral nerves p.61

- TABLE Long gap · Boari flap ± hitch · **10–15 cm**; poor in contracted bladder p.61

- KEY POINT Trap: in TB the bladder itself may be contracted, ruling out bladder-based bridges (psoas hitch, Boari flap) — assess capacity before choosing; ileal ureter ± augmentation then becomes the option. p.61

- Indication: intolerable frequency, urgency, pain, haematuria, typically capacity **< 100 mL**. Operate after chemotherapy has controlled active disease. p.61

### 7 · RELAPSE, SPECIAL SITUATIONS

- Relapse after UG TB **6.3–22%** (vs 2–6% pulmonary), mean time ~5 years — a diseased kidney holds many hard-to-sterilise foci. Follow-up: some advise **10 years**. Culture any recurrence for susceptibility. p.62

### 8 · SCHISTOSOMIASIS (BILHARZIA)

- ALGORITHM
  Fork-tailed cercariae penetrate intact skin in fresh water → schistosomulae → lungs → liver sinusoids → paired adults (female in male's gynaecophoric canal) migrate to vesical plexus (80–110 days; live 3–5 years) → eggs (~140 µm, **terminal spine**) cross bladder wall into urine → in fresh water hatch to miracidia → *Bulinus* snail → asexual multiplication → cercariae.
  p.63

- KEY POINT Trap: a negative urine egg count does not exclude severe disease — the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients. p.64

- **Praziquantel (PZQ)** — only WHO drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart. Cure/egg reduction 60–90%; even partial response cuts worm burden. Mild transient side effects. Pregnancy: United States Food and Drug Administration (FDA) category B; many programmes exclude, though adverse effects are rarely reported. Katayama fever: add steroids. p.64

### 11 · MCQ ONE-LINERS

- MCQ Commonest site of UG TB: kidney (80%); commonest genital site: epididymis. p.66

- MCQ Epididymal TB starts in the globus minor (tail). p.66

- MCQ Hallmark urine finding in UG TB: sterile pyuria. p.66

- MCQ Urine for AFB culture: early-morning samples on 3–5 consecutive days. p.66

- MCQ GeneXpert detects MTBC plus rifampicin resistance (MDR surrogate). p.66

- MCQ NAAT cannot monitor treatment — dead bacilli shed DNA. p.66

- MCQ IGRA has no BCG false positives; neither IGRA nor TST separates latent from active. p.66

- MCQ Pathognomonic end-stage renal TB on KUB: lobar calcified "putty" kidney. p.66

- MCQ Phantom calyx = stenosed infundibulum, non-opacified calyx. p.66

- MCQ TB ureteric calcification intraluminal; schistosomal mural. p.66

- MCQ Commonest site of TB ureteric stricture: VUJ. p.66

- MCQ Thimble bladder: capacity < 20 mL → orthotopic substitution. p.66

- MCQ Golf-hole ureteric orifice: retracted orifice from ureteric fibrosis. p.66

- MCQ Only non-hepatotoxic first-line anti-TB drug: ethambutol. p.66
- MCQ *M. bovis* (and BCG) is innately pyrazinamide-resistant. p.66
- MCQ Elective surgery: 4–6 weeks after starting anti-TB therapy. p.66
- MCQ Stricture not improving after 6 weeks of chemotherapy → open surgery. p.66
- MCQ TB nephrectomy: ligate renal artery and vein separately (AV fistula). p.66
- MCQ UG TB relapse up to 22%; follow up ~10 years. p.66
- MCQ HIV-positive UG TB: less fibrosis, fewer strictures, higher mortality. p.66
- MCQ *S. haematobium* snail host: *Bulinus*; egg has terminal spine. p.66
- MCQ Schistosomal bladder cancer: mainly SCC, younger patients. p.67
- MCQ "Fetal head in the pelvis": calcified schistosomal bladder. p.67
- MCQ Praziquantel 40 mg/kg single dose for *S. haematobium*. p.67
- MCQ *Salmonella* sheltering in the schistosome tegument causes chronic bacteriuria — treat both. p.67
- MCQ Genital filariasis is caused by *W. bancrofti* only. p.67
- MCQ Filarial dance sign: live adult worms on scrotal US. p.67
- MCQ DEC contraindicated with onchocerciasis or *Loa loa*. p.67
- MCQ Persistent filarial chyluria: pelvic instillation sclerotherapy, then hilar lymphatic disconnection. p.67
- MCQ Renal hydatid: albendazole ± surgery; PAIR not used for kidney. p.67

## ★ Putting it together: infection decisions

### B · WHEN DOES A UTI NEED MORE THAN ANTIBIOTICS?

- ALGORITHM

**Image** (US or CT) if: no response by 72 hours, sepsis, suspected obstruction or stone, gas-forming infection, men with febrile UTI, recurrent pyelonephritis.
**Drain** any obstructed infected kidney (stent or nephrostomy) ; renal abscess ≤3 cm → antibiotics alone, and abscesses >3–5 cm; treat the stone later.gt;5 cm → percutaneous drainage. Treat the stone later.
**Emphysematous pyelonephritis:** CT class guides care. Gas confined to the collecting system or kidney → antibiotics + drainage; extension beyond the kidney with poor prognostic features → nephrectomy (→ Ch 26).
**Fournier gangrene:** resuscitation, broad-spectrum antibiotics and **urgent radical debridement**, repeated until healthy tissue; scores (LRINEC, FGSI) help, but surgery must not wait for them.

p.68

### C · RECURRENT UTI IN WOMEN: PREVENTION LADDER

- ALGORITHM

Confirm with culture; exclude anatomical causes in atypical cases → behavioural measures and hydration → vaginal oestrogen (postmenopausal) → non-antibiotic options (methenamine hippurate, cranberry, D-mannose, immunostimulants) → self-start or post-coital antibiotic → continuous low-dose prophylaxis.

p.68

ALGORITHM

**Suspect:** sterile pyuria, recurrent "UTI" with negative cultures, contracted bladder, beaded vas or epididymal mass.

**Confirm:** three early-morning urine samples for acid-fast bacilli, culture and nucleic-acid amplification (GeneXpert).

**Treat:** 2 months isoniazid, rifampicin, pyrazinamide, ethambutol, then 4 months isoniazid + rifampicin; watch for ureteric stricture during treatment.

**Operate** after at least 4–6 weeks of drugs: nephrectomy for a non-functioning kidney, reimplantation for stricture, augmentation for a contracted bladder (→ Ch 31).

p.69

# 4 Paediatric Urology 1

459 core items · page numbers refer to the full book PDF

## Ch 41 · Management Strategies for Vesicoureteral Reflux

### 2 · EMBRYOLOGY AND ANTIREFLUX MECHANISM

- **Passive flap-valve**: intramural ureter compressed against detrusor backing during filling. **Paquin**: tunnel length : ureteric diameter **5:1** in non-refluxing vs 1.4:1 in refluxing UVJs; orifice configuration matters as much. p.2

### 3 · PRIMARY VS SECONDARY VUR

- Spina bifida: VUR in up to **30%** → early clean intermittent catheterisation (CIC), anticholinergic or β3-agonist, serial imaging. Look for occult dysraphism (sacral dimple, hairy patch, gluteal cleft anomaly, refractory constipation). p.2

- KEY POINT Primary and secondary are not exclusive: "primary" reflux often has a treatable secondary component (overactivity, bladder and bowel dysfunction). Removing it leaves only the true primary VUR to manage. p.2

### 4 · EPIDEMIOLOGY (HEADLINE)

- VUR in **~30%** of children with UTI; ~17% of those imaged for hydronephrosis. Younger child with UTI → likelier VUR (**38.6%** under 2 y vs 7.6% at 12–21 y). p.3

### 6 · DIAGNOSIS

- **Top-down approach**: dimercaptosuccinic acid (DMSA) scan first; VCUG only if abnormal. Normal DMSA → **<1%** chance of high-grade VUR; misses ~1/3 of (mostly low-grade) VUR. p.3

### 7 · CONGENITAL DYSPLASIA VS ACQUIRED SCAR

- Long-term: hypertension in **34–38%** of scarred children. **Correcting VUR does not lower blood pressure** (parenchymal mechanism); nephrectomy/segmental resection of a small kidney may. p.4

### 8 · BLADDER AND BOWEL DYSFUNCTION (BBD)

- **Phimosis**: in boys with VUR number needed to treat (NNT) by circumcision to prevent one UTI falls from ~111 to **~4**; topical steroid is equivalent. p.4

### 10 · MEDICAL MANAGEMENT — CONTINUOUS ANTIBIOTIC PROPHYLAXIS (CAP)

- 2-year BUTI on CAP: **8.6% low, 26% intermediate, 62.5% high**. About two-thirds of children are low risk. p.5

### 11 · PRACTICAL ALGORITHM (AUA-BASED, RISK-STRATIFIED)

- ALGORITHM

  History of number/severity of true febrile UTIs + RBUS → VCUG if multiple febrile UTIs, admissions or upper-tract findings (afebrile, doubtful UTI with normal US → conservative)
  All: treat BBD (constipation first), hygiene, timed voiding
  Low risk (grade I–II, no BBD, male, older) → observation ± CAP; treat febrile UTI promptly
  High risk (grade ≥III, girl, BBD, febrile presentation, scars, infant) → CAP
  Breakthrough febrile UTI, new scars, high grade with poor function, persistent VUR with low tolerance for infection → correction (endoscopic or reimplant)

  p.5

### 12 · SURGICAL CORRECTION

- Open correction succeeds in **>95%**; robotic correction is probably less morbid. No superiority of surgery over medical therapy for scarring or renal growth (International Reflux Study in Children (IRSC), Birmingham). p.6

## 14 · SIBLING SCREENING

- Sibling prevalence ~**30%** (up to 50%), higher in younger siblings; identical twins near-complete concordance. p.6

## 15 · FOLLOW-UP AND ADULTHOOD

- **Pregnancy**: VUR → pyelonephritis in ~1/3; renal scars → ~3× hypertension, **~8× pre-eclampsia**. **Scarring, not VUR presence, drives pregnancy morbidity.** Reimplanted women still at UTI risk but no excess miscarriage. p.6

## 16 · LANDMARK TRIALS

- TABLE Swedish Reflux Trial · 1–2 y, grade III–IV; CAP vs endoscopic vs surveillance · Girls febrile UTI: 19% CAP, 23% endoscopic, **57%** surveillance; new scars mostly on surveillance, none on CAP; boys no difference · Treat girls with dilating VUR p.7
- TABLE RIVUR · 607 children, grade I–IV after first/second UTI; TMP-SMX vs placebo · Recurrent UTI **12.9% vs 23.6%**; scarring unchanged; benefit greatest with febrile index UTI/BBD · CAP prevents UTI, not scars p.7
- KEY POINT Why do the trials disagree? Population risk. RIVUR was diluted by low-risk children (only ~8% grade IV); reanalysis gives NNT 5 in high-risk vs 18 in low-risk. Prophylaxis reduces infection in proportion to baseline risk; neither CAP nor surgery prevents scarring better than the other. p.7

## 17 · MCQ ONE-LINERS

- MCQ Early ureteric budding → lateral, high orifice → short tunnel → VUR. p.7
- MCQ Paquin ratio of tunnel length to ureteric diameter in non-refluxing UVJ: 5:1. p.7
- MCQ VUR prevalence in children with UTI: ~30%. p.7
- MCQ Grade V VUR: loss of papillary impressions. p.7
- MCQ First dilating grade on international classification: III. p.7
- MCQ Direct RNC misses grade I VUR. p.7
- MCQ Cyclic VCUG detects 12–20% more VUR. p.7
- MCQ Reflux during filling phase predicts poor spontaneous resolution. p.7
- MCQ Normal DMSA: <1% chance of high-grade VUR. p.7
- MCQ Compound papillae at poles allow intrarenal reflux and polar scars. p.7
- MCQ Ransley–Risdon pig model: scarring needs infected urine. p.7
- MCQ Congenital reflux nephropathy: male infant, grade IV–V, no UTI. p.7
- MCQ VUR correction does not cure reflux-related hypertension. p.7
- MCQ BBD lowers endoscopic but not open reimplant success. p.7
- MCQ Circumcision NNT to prevent one UTI in boys with VUR: ~4. p.7
- MCQ RIVUR agent: trimethoprim-sulfamethoxazole; halved recurrent UTI, no effect on scarring. p.7
- MCQ Swedish trial: surveillance girls had 57% febrile UTI recurrence. p.7
- MCQ Avoid sulfonamides under 2 months: bilirubin displacement → neonatal jaundice. p.7
- MCQ Breakthrough febrile UTI on CAP: indication for surgical correction. p.7
- MCQ Hidas high-risk group: 62.5% BUTI at 2 years. p.7
- MCQ Duplex VUR usually affects the lower pole. p.7
- MCQ Ureter entering diverticulum: VUR will not resolve. p.7
- MCQ UPJO with VUR: correct obstruction first. p.7
- MCQ Megacystis–megaureter: treat by ureteric reimplantation, not outlet surgery. p.7

- MCQ Calcified Dx/HA mound may mimic distal ureteric stone. p.7
- MCQ Pregnancy morbidity is driven by renal scarring, not VUR. p.7

## Ch 49 · Anomalies of the Kidney and Collecting System

### 1 · EMBRYOLOGY THAT EXPLAINS EVERYTHING

- KEY POINT Every anomaly of number is a failure of UB–MM induction: timing and extent set the phenotype. An early insult drags the WD/MD with it (genital anomalies). Arrested ascent leaves an unrotated kidney with anterior pelvis and fetal blood supply; fusion freezes kidneys where they meet. p.8

### 2 · BILATERAL RENAL AGENESIS (BRA)

- **≈1 in 5000** births; **75% male**. First-degree relatives ≈1000× population risk → **ultrasound (US) parents and siblings** of any infant with unilateral renal agenesis (URA), BRA or dysgenesis. p.8
- KEY POINT BRA is lethal through the lungs, not the kidneys. Amnioinfusion replaces only the amniotic step, so it converts pulmonary deaths into neonatal end-stage renal disease (ESRD). p.8

### 3 · UNILATERAL RENAL AGENESIS AND THE SOLITARY FUNCTIONING KIDNEY (SFK)

- **1 in 1200**; male 1.8:1; left slightly more. Many apparent URAs are **involuted multicystic dysplastic kidneys (MCDK)** — clues: curvilinear calcification on computed tomography (CT); splenic or hepatic flexure in its normal place (a kidney once occupied the fossa). Renal aplasia (rudimentary parenchyma + ureter) may be the commonest cause of a congenital solitary kidney. **Pitfall at the 20-week scan**: a flattened adrenal or the spleen mistaken for a kidney. p.8
- **Contralateral tract**: vesicoureteric reflux (VUR) **≈30%**, pelviureteric junction (PUJ) obstruction ≈11%, vesicoureteric junction obstruction ≈7%. p.9

### 5 · ECTOPIA (ANOMALIES OF ASCENT)

- **1 in 900** (autopsy); left slightly more; bilateral ≈10%. Types: **pelvic (commonest)**, lumbar/iliac, abdominal, cephalad, thoracic. "Vascular barrier" theory rejected — aberrant vessels are a consequence, not cause. p.10

### 6 · FUSION ANOMALIES

- **Commonest fusion anomaly**: **1 in 400** (0.25%); male 2:1. **Lower-pole fusion in 95%**, before rotation. p.11
- KEY POINT † PUJ obstruction in a horseshoe kidney: dismembered pyeloplasty with a dependent, anterior anastomosis is standard; division of the isthmus (symphysiotomy) is not routinely needed. p.12

### 7 · MALROTATION AND VASCULAR ANOMALIES

- **Renal artery aneurysm** (≈0.1%; Abeshouse: saccular ≈93%, fusiform at branch points — congenital, dissecting, arteriovenous); ring ("wreath") calcification ≈30%; mostly silent; hypertension renin-mediated. Rupture only >3 cm. **Treat** (endovascular/open): rupture, rapid growth, pregnancy (or planned), **>2 cm**, symptoms. Otherwise observe. p.12

### 8 · COLLECTING SYSTEM ANOMALIES

- KEY POINT Collecting-system anomalies split on one question — is there obstruction? Dilatation without obstruction (bifid, extrarenal pelvis, megacalycosis) needs nothing; stasis (diverticulum, hydrocalycosis) is treated for symptoms; infundibulopelvic stenosis threatens function through bilateral dysplasia. p.13

### 10 · MCQ ONE-LINERS

- MCQ Flattened adrenal lying along the spine on US ("lying down" sign) = absent ipsilateral kidney. p.13
- MCQ Potter facies are deformations from oligohydramnios, not primary malformations. p.13

- MCQ URA incidence 1 in 1200; ipsilateral ureter absent in ≈60%. p.13
- MCQ Commonest associated urological anomaly in URA: contralateral VUR (≈30%). p.13
- MCQ 79% of men with absent vas have ipsilateral renal agenesis. p.13
- MCQ MRKH type II carries renal anomalies in 40–60%. p.13
- MCQ Maternal diabetes triples the risk of renal agenesis. p.13
- MCQ Supernumerary kidney requires its own capsule and separate blood supply. p.13
- MCQ Thoracic kidney lies in the foramen of Bochdalek, outside the pleural space. p.13
- MCQ Cephalad ectopia is classically seen with omphalocele. p.13
- MCQ 90% of crossed ectopic kidneys are fused; left-to-right is 3× commoner. p.13
- MCQ Commonest crossed fused form: inferior ectopia. p.13
- MCQ Anterior-facing pelvis in fused kidneys indicates fusion before rotation. p.13
- MCQ Horseshoe kidney: 1 in 400; isthmus lies below the IMA at L3/L4. p.13
- MCQ About 60% of Turner syndrome patients have a horseshoe kidney. p.13
- MCQ Horseshoe kidney raises Wilms tumour risk, not RCC risk. p.13
- MCQ PUJ obstruction in horseshoe kidney: 13–34%. p.13
- MCQ Commonest malrotation: ventral (no rotation). p.13
- MCQ Direction of malrotation is told by the course of the renal vessels. p.13
- MCQ Treat renal artery aneurysm >2 cm, enlarging, symptomatic or with pregnancy. p.13
- MCQ Congenital renal AVF may cause high-output cardiac failure. p.13
- MCQ Stone in a crossed ectopic ureter causes pain on the anephric side. p.13
- MCQ Zinner syndrome: seminal vesicle cyst with ipsilateral renal agenesis. p.13
- MCQ Bifid pelvis occurs in ≈10% of normal kidneys. p.13
- MCQ Calyceal diverticulum: stones in ≈40%; delayed contrast pooling confirms. p.13
- MCQ Megacalycosis is male-predominant (6:1); bilateral disease occurs almost only in males. p.13
- MCQ Megacalycosis: increased calyces, normal pelvis, normal diuretic renogram — no surgery. p.13
- MCQ Infundibulopelvic stenosis is usually bilateral and can progress to ESRD. p.13

## Ch 50 · Renal Dysgenesis and Cystic Disease of the Kidney

### 2 · DYSGENESIS: DYSPLASIA VS HYPOPLASIA

- KEY POINT The ureteric orifice predicts the kidney: a bud from the wrong place on the wolffian duct meets the wrong blastema, so ectopia, obstruction and dysplasia travel together. p.14

### 5 · ADPKD — PAEDIATRIC FOCUS

- KEY POINT ADPKD is a volume disease: cyst growth stretches vessels (hypertension), compresses nephrons and hides stones and tumours. Every therapy is judged by whether it slows total kidney volume. p.16

### 8 · MULTICYSTIC DYSPLASTIC KIDNEY (MCDK)

- ALGORITHM
  Prenatal MCDK → US day 2–3 and at 1 month (contralateral kidney, bladder) → technetium-99m dimercaptosuccinic acid (DMSA) or mercaptoacetyltriglycine (MAG3) scan if doubt → voiding cystourethrogram (VCUG) only if UTI or abnormal contralateral kidney/bladder → primary care: BP, mass, UTI
  p.18

- **KEY POINT** MCDK removes itself; management looks away from it to the solitary contralateral kidney and its reflux or obstruction. Mistaking a non-functioning PUJ kidney for MCDK costs little. p.18

## 13 · MCQ ONE-LINERS

- **MCQ** Sine qua non of renal dysplasia: primitive ducts with fibromuscular collars. p.20
- **MCQ** Leading cause of childhood ESRD: renal dysplasia. p.20
- **MCQ** Degree of hypodysplasia correlates with degree of ureteric orifice ectopia. p.20
- **MCQ** Oligomeganephronia: fewer, hypertrophied nephrons; proteinuria first sign. p.20
- **MCQ** Ask-Upmark kidney: segmental hypoplasia, severe hypertension, cured by nephrectomy. p.20
- **MCQ** ARPKD gene: PKHD1 on chromosome 6 (fibrocystin). p.20
- **MCQ** ARPKD is always accompanied by congenital hepatic fibrosis. p.20
- **MCQ** Neonatal ARPKD US: large homogeneously hyperechoic kidneys. p.20
- **MCQ** Neonatal ADPKD may mimic ARPKD — scan the parents. p.20
- **MCQ** PKD1 (chr 16) ≈85%; PKD2 (chr 4) milder, later ESRD. p.20
- **MCQ** PKD1–TSC2 contiguous deletion gives TSC with polycystic kidneys. p.20
- **MCQ** Commonest extrarenal ADPKD manifestation: hepatic cysts. p.20
- **MCQ** Berry aneurysm in ADPKD: 10–30%. p.20
- **MCQ** Tolvaptan is a vasopressin V2 antagonist lowering renal cAMP. p.20
- **MCQ** Infected ADPKD cyst: use lipophilic antibiotics. p.20
- **MCQ** NPH: salt wasting, no hypertension, ESRD around 13 years. p.20
- **MCQ** Senior–Løken syndrome: nephronophthisis plus retinitis pigmentosa. p.20
- **MCQ** TSC AML >3 cm and growing: everolimus first line. p.20
- **MCQ** Fat-poor AML is missed on US in 25–30%; MRI preferred. p.20
- **MCQ** VHL gene: chromosome 3p25; RCC is clear cell type. p.20
- **MCQ** Commonest renal cystic disease in newborns: MCDK. p.20
- **MCQ** Contralateral VUR in MCDK: 18–43%. p.20
- **MCQ** About 40% of MCDKs involute spontaneously. p.20
- **MCQ** MCDK nephrectomy indication: increasing solid tissue or mass effect. p.20
- **MCQ** Multilocular cyst: boys <4 y and women >30 y. p.20
- **MCQ** Bosniak IIF: follow-up; III and IV: excision. p.20
- **MCQ** Paediatric simple cyst <15 mm: repeat US at 2 years, then discharge. p.20
- **MCQ** Medullary sponge kidney stone prevention: potassium citrate first. p.20
- **MCQ** Acquired cystic disease is caused by uraemia, not dialysis. p.20
- **MCQ** Ask-Upmark hypertension is not cured by reflux correction. p.20
- **MCQ** Simple cyst calcification is peripheral; tumour calcification is central. p.20
- **MCQ** VHL: nephron-sparing surgery aims at cancer control, not cure. p.20
- **MCQ** Nephronophthisis allografts do not develop recurrent disease. p.20

## Ch 51 · Pathophysiology of Urinary Tract Obstruction

### 1 · DEFINITION AND CLASSIFICATION

- Incidence ≈ **1 in 500** live births. High-grade obstruction: lifetime end-stage renal disease (ESRD) risk up to **30%**. p.21

- KEY POINT Trap: **detrusor hypertrophy** is a lower-tract cause of upper-tract (UVJ) obstruction — the thickened wall narrows the intramural ureter. Neurogenic bladder obstructs without an anatomical lesion. p.21

### 2 · HAEMODYNAMIC CHANGES

- **Hypertension**: ↑ intrarenal pressure → ↓ perfusion → renin–angiotensin–aldosterone system activation → vasoconstriction + Na retention. Bilateral ≈ **77%** hypertensive; unilateral ≈ **20%** (renin lateralises to affected kidney). p.21

### 6 · POST-OBSTRUCTIVE DIURESIS (POD)

- Polyuric phase after relief; unilateral ≈ **2%**, bilateral ≈ **50%** (no normal kidney to regulate volume). p.22

- ALGORITHM
  Monitor intake/output, serum and urine electrolytes → replace losses, correct electrolytes → **do not over-replace** (perpetuates POD) → usually resolves in **24 h**, seldom > 48 h
  p.23

### 7 · FETUS AND CHILD

- Urinary tract dilation classification; anteroposterior diameter (APD) most reliable: obstruction considered at **≥ 7 mm** (16–27 weeks), **≥ 10 mm** (≥ 28 weeks). p.23

- LUTO: raised pressure impairs glomerular development; anhydramnios → pulmonary hypoplasia. Second-trimester oligo/anhydramnios → mortality **60–80%**. p.23

- Postnatal: ultrasound (US) first; mercaptoacetyltriglycine (MAG3) for drainage; dimercaptosuccinic acid (DMSA) for scar; voiding cystourethrogram (VCUG) for outlet and vesicoureteral reflux (VUR). Whitaker test: obstructed if pelvic pressure > **12 cm $H_2O$** or continuous rise. p.23

### 9 · MCQ ONE-LINERS

- MCQ Commonest cause of LUTO: posterior urethral valves. p.23
- MCQ Commonest upper tract obstruction: UPJO; male:female 2:1. p.23
- MCQ Ischaemic injury starts within 2 h of complete obstruction. p.23
- MCQ Unilateral obstruction late phase: RBF ↓ and ureteric pressure ↓. p.23
- MCQ Bilateral obstruction: ureteric pressure stays high. p.23
- MCQ Characteristic acidification defect: distal renal tubular acidosis. p.23
- MCQ Potassium: wasted early, retained in prolonged obstruction. p.23
- MCQ Aquaporins downregulated in obstruction: AQP1–4. p.23
- MCQ Key fibrotic signalling: TGF-β → Smad. p.23
- MCQ MicroRNA linked to obstructive fibrosis: miR-34a. p.23
- MCQ Compensatory growth = glomerulomegaly, not nephrogenesis. p.23
- MCQ POD: unilateral ≈ 2%, bilateral ≈ 50%. p.24
- MCQ POD usually resolves within 24 h; over-replacement perpetuates it. p.24
- MCQ Preterm infants: low AQP2 → POD risk after valve ablation. p.24
- MCQ Hypertension: bilateral ≈ 77%, unilateral ≈ 20%. p.24

- MCQ Dangerous bladder storage pressure: > 40 cm $H_2O$. p.24
- MCQ Prenatal APD threshold: ≥ 7 mm (16–27 wk), ≥ 10 mm (≥ 28 wk). p.24
- MCQ Favourable fetal urine Na < 100 mEq/L; β2-microglobulin < 6 mg/L. p.24
- MCQ Fetal bladder not seen by 15 weeks: abnormal. p.24
- MCQ Whitaker test positive: pelvic pressure > 12 cm $H_2O$. p.24
- MCQ Megacystis: fetal bladder failing to empty over 40 min at ≥ 18 weeks. p.24
- MCQ FENa < 1% in acute obstruction; > 1% with tubular injury. p.24

## Ch 52 · Ectopic Ureter, Ureterocele and Ureteral Anomalies

### 3 · ECTOPIC URETER — SITES AND SEX ASYMMETRY

- KEY POINT Why does only the girl leak? Everything follows from where the ureter opens relative to the external sphincter. The female ureter may open distal to it and leaks continuously; the male ureter, confined to wolffian derivatives above it, cannot leak and presents as infection or obstruction. p.26

### 6 · MANAGEMENT PRINCIPLES

- ALGORITHM
  Prenatal hydronephrosis → postnatal ultrasound → antibiotic prophylaxis if significant hydroureteronephrosis → VCUG (optional if it will not change decompression) → renal scan after 2 months
  Small ureterocele, minimal hydronephrosis, no obstruction → observe
  Significant obstruction, parenchyma at risk → TUI (or cutaneous distal ureterostomy if endoscopy impossible, until old enough for reimplant)
  Sepsis not responding → TUI (ureterocele) / nephrostomy or end ureterostomy (ectopic ureter); after stabilisation an elective drainage procedure is still needed, as the system has proven infection-prone
  p.29

### 7 · OPERATIVE OPTIONS

- Simplest decompression; relieves obstruction in **78–97%**. Procedure of choice in **sepsis** (nephrostomy an alternative) and in the **infant**. p.29
- Aftercare: uninfected well child — no catheter, outpatient. Ultrasound and VCUG at **4–6 weeks** (decompression, lower pole and new ureterocele reflux). p.30
- New reflux after TUI **0–50%**. **Intravesical**: decompression without reflux in 70–80%. **Extravesical**: secondary surgery up to 70%. Reoperation relative risk 1.74 with lower pole reflux, 2.78 if extravesical (not additive). Single-system ureteroceles do better than duplex. p.30
- KEY POINT Why offer TUI first even if half need more surgery? It is an outpatient procedure that may be definitive; the second operation can be deferred to an older child; and decompressing the ureter often avoids tapering at later reimplantation. Less appealing in the older child with a massive non-functioning upper pole, in whom definitive surgery at diagnosis is preferable. p.30
- Outcomes: ectopic ureter — usually curative (stump rarely problematic). Ureterocele — pre-existing lower pole reflux resolves in up to 20%; new reflux **15–50%**; secondary surgery **40–50%** (depends on whether any reflux or only clinical reflux triggers surgery). p.30
- Persisting reflux **5–10%**, more with tapering. p.31

## 8 · DECISION SUMMARY

- **ALGORITHM**
  Keep upper pole? → Yes + lower pole reflux → common sheath reimplant, or lower pole reimplant + distal ureteroureterostomy
  Yes + no lower pole reflux → proximal or distal ureteroureterostomy
  Function unclear / massive dilation / sepsis → temporary end ureterostomy → DMSA → reconstruct or remove
  Remove (marked dilation, preference to excise dysplastic tissue) → upper pole nephrectomy with ureterectomy
  p.31

- **KEY POINT** VUR is the pivot: no VUR → upper tract solution; VUR → bladder-level solution — except in the infant, where TUI is chosen precisely to defer bladder surgery. Upper pole function then chooses nephrectomy vs ureteroureterostomy (though ureteroureterostomy is now also used for non-functioning poles). TUI is not a lesser version of the others: its failure costs little and may simplify the next operation. p.32

## 10 · OTHER URETERAL ANOMALIES

- ~**1 in 125** (0.8%) at autopsy; F:M 1.6:1; unilateral 6× bilateral. **Look for contralateral duplication** — an occult second ectopic ureter explains persistent wetting. p.32
- Smith: type 1 three separate ureters (**35%**, commonest); 2 incomplete, two orifices (21%); 3 trifid, one orifice (31%); 4 two ureters, three orifices (13%). Presents with infection or wetting; associated ureterocele, ectopia, UPJ obstruction, VUR. Weigert–Meyer usually holds; violated in quadruplication. p.33
- Flank pain, haematuria, hydronephrosis; commonest at **UPJ**; **89% male, 78% left**; only 22% seen preoperatively (filling defect, mistaken for clot). A filling defect on retrograde study before pyeloplasty → look for a polyp. p.33

## 11 · MCQ ONE-LINERS

- **MCQ** Ectopic ureter in a duplex system always drains the upper pole. p.33
- **MCQ** Weigert–Meyer: upper pole orifice lies caudal and medial to the lower pole orifice. p.33
- **MCQ** Female ectopic ureter: continuous dribbling with normal voiding. p.33
- **MCQ** Male ectopic ureter never causes incontinence; always above external sphincter. p.33
- **MCQ** Commonest male presentation of ectopic ureter: epididymitis or UTI. p.33
- **MCQ** Zinner syndrome: ectopic ureter, ipsilateral renal agenesis, seminal vesicle cyst. p.33
- **MCQ** Female ectopic ureter reaches vagina via the Gartner duct. p.33
- **MCQ** Most useful ureterocele classification: intravesical versus ectopic (extravesical). p.33
- **MCQ** Cecoureterocele: intravesical orifice, submucosal cavity extending into urethra. p.34
- **MCQ** Chwalle membrane persistence: classic theory of ureterocele formation. p.34
- **MCQ** Smooth interlabial mass: prolapsed ureterocele; grape-like: sarcoma botryoides. p.34
- **MCQ** "Drooping lily": lower pole displaced by dilated upper pole. p.34
- **MCQ** Pseudoureterocele: bilaminar thick wall, lumen outside bladder. p.34
- **MCQ** Eversion on VCUG: weak trigone, likely needs open repair. p.34
- **MCQ** Negative urine culture does not exclude infected ectopic system. p.34
- **MCQ** Ectopic perineal orifice lies between urethral meatus and vagina. p.34
- **MCQ** Preoperative upper pole function does not predict post-decompression function. p.34
- **MCQ** TUI incision: low, transverse, close to bladder floor. p.34

- MCQ Watering-can puncture: equal decompression, less reflux. p.34
- MCQ Intravesical ureterocele after TUI: 70–80% decompressed without reflux. p.34
- MCQ Extravesical ureterocele: secondary surgery after TUI up to 70%. p.34
- MCQ Secondary surgery after upper pole nephrectomy for ureterocele: 40–50%. p.34
- MCQ Major heminephrectomy risk: loss of lower pole function. p.34
- MCQ Never separate duplex ureters intravesically: shared blood supply. p.34
- MCQ Distal ureteroureterostomy: stent lower pole ureter first. p.34
- MCQ End ureterostomy for ectopic ureter: do not trim redundant ureter. p.34
- MCQ Post-repair voiding dysfunction: inadequate trigonal support; lateral voiding views. p.34
- MCQ Duplication incidence: ~1 in 125 (0.8%). p.34
- MCQ Lower pole: UPJ obstruction and VUR; upper pole: ectopia and ureterocele. p.34
- MCQ Commonest Smith triplex type: type 1. p.34
- MCQ Fibroepithelial polyp: male, left, at UPJ. p.34
- MCQ Retrocaval ureter: persistent right subcardinal vein; fishhook deformity at L3–L4. p.34
- MCQ Retroiliac ureter obstructs at L5–S1. p.34

## Ch 53 · Surgery of the Ureter in Children

### 1 · PELVIURETERIC JUNCTION (PUJ) OBSTRUCTION — DEFINITION AND AETIOLOGY

- Paediatric hydronephrosis: renal pelvic anteroposterior diameter (APD) **12 mm**; hydronephrosis is **not synonymous with obstruction**. p.35
- Male and left predominance; bilateral in **10–40%**. p.35
- **Lower-moiety PUJ obstruction (duplex)**: incomplete duplication → upper-pole ureter to lower-pole pelvis (ureteropyelostomy); complete → standard pyeloplasty; moiety ≤ **5–10%** function → partial nephrectomy. p.35

### 2 · DIAGNOSIS

- Neonatal washout curves are confounded by renal immaturity, hydration, diuretic timing, posture and pelvic volume/compliance — false positives up to **44%**, so decide on DRF trend and dilatation, not T½ alone. p.35

### 4 · ANDERSON–HYNES DISMEMBERED PYELOPLASTY

- ALGORITHM

  Open Gerota fascia → expose PUJ → free lower pole (do not miss an accessory vessel) → sparing bipolar diathermy (protect pelvic/ureteric blood supply) → stay sutures on pelvis and ureter → decompress tense pelvis with 21-G needle → dismember ± reduce pelvis → **spatulate laterally** well into healthy ureter, stenotic segment kept as handle ("no-touch") → apex of spatulation to inferior pelvis → tension-free, watertight 6-0/5-0 absorbable anastomosis → excise stenotic segment just before completion → irrigate clot → cover with perinephric fat

  p.36
- KEY POINT Every variant answers two questions: is there a crossing vessel (dismember), and is the gap too long for tension-free anastomosis (flap, mobilisation)? p.36
- KEY POINT Repair has one fixed principle and many routes: dismember, excise, spatulate, anastomose without tension. Access, stent and robot are choices of convenience; failure is managed by repeating the same principle. p.36

## 8 · PRIMARY MEGAURETER

- Descriptive: dilated ureter **≥ 7 mm** (normal ≤ 5–6.5 mm to 16 years). Primary (intrinsic) vs secondary (neurogenic bladder, outlet obstruction). p.37

- KEY POINT Why not rely on the renogram? Neonatal washout depends on immature function, hydration and the capacity of a huge ureter — so serial DRF, diameter and infection history drive surgery. Megaureter is the PUJ problem moved to the bottom of the ureter: an immature adynamic segment that usually matures on its own. p.38

- ALGORITHM
  Excise stenotic distal segment → straighten → taper → non-refluxing reimplant at **5:1** tunnel length:diameter → stent (feeding tube 7 days or double-J 4 weeks)
  p.38

- Success > **90%**; POM better than refluxing megaureter (stiffer collagen-rich UVJ). Worse with valves/neurogenic bladder. p.38

## 9 · SURGERY FOR VESICOURETERAL REFLUX

- KEY POINT **Principles**: exclude or treat secondary (bladder) causes first · handle the ureter gently (blood supply) and the bladder gently (haematuria, spasm) · mobilise enough ureter for a tension-free repair · tunnel length:diameter **5:1** (Paquin) · keep the course straight at both empty and full bladder (kinking). p.38

- TABLE **Cohen** · Intravesical; transhiatal cross-trigonal · Commonest; success > **98%**; thick trabeculated bladder; with bladder-neck reconstruction · Difficult later retrograde catheterisation p.39

- TABLE **Lich-Gregoir** · Extravesical detrusorotomy, ≈ 5 cm · No haematuria or spasm; short catheter time · Transient voiding dysfunction/retention up to **10%**, especially bilateral p.39

- ALGORITHM
  Open bladder vertically to 2 cm from neck → pack dome, ring retractor; stay sutures on lateral walls → 5-Fr feeding tube in orifice as handle → circumscribe orifice 1–2 mm with fine diathermy → free ureter in correct plane (preserve distal blood supply) until it reaches the contralateral wall without tension → submucosal tunnel across trigone → new orifice just above contralateral orifice → mucosa-to-mucosa 6-0, detrusor hitch at 6 o'clock
  p.39

- ALGORITHM
  Bladder one-third full → divide obliterated umbilical artery → free ureter **hugging it**, only down to the UVJ → mark 5-cm tunnel along natural course with bladder moderately distended → detrusorotomy to intact mucosa (uniform bulge) → raise flaps wide enough not to constrict → lay ureter in → close detrusor caudocranially, catching ureteric adventitia at apex (prevents slippage) → Foley 12–48 h
  p.39

- KEY POINT Why voiding dysfunction? Pelvic plexus lies ≈ 1.5–2 cm dorsomedial to the UVJ, with branches along the medial ureter. Stay on the ureter, avoid the dorsomedial UVJ, stop at the UVJ. The Zaontz modification (detrusorotomy around the UVJ) encroaches on these nerves and is now avoided. p.39

- Open benchmark: American Urological Association (AUA) meta-analysis — success **95.9%**. Minimally invasive results converging with experience. p.39

- KEY POINT Early complications are tissue reactions — drain and wait. Late ones are design errors (blood supply, hiatus position, tunnel ratio) or an untreated bladder. Reimplanting into an abnormal bladder only worsens dilatation. p.40

- Dextranomer/hyaluronic acid (Dx/HA, Deflux) is the standard agent; ≈ **90%** initial resolution of low-grade reflux after one injection. Teflon (STING, subureteric Teflon injection) abandoned — particle migration, never approved by the US Food and Drug Administration. p.40

- Technique: cystoscopy first (abandon if inflamed) → needle bevel up at 6 o'clock → mound should appear within **0.1–0.2 mL**, else reposition → aim for a **"volcano"** mound with slit/crescent orifice. Bleeding: press with scope, **do not cauterise** (extrusion). Day case, no catheter. p.40
- **Late obstruction** reported (Dx/HA, especially polyacrylate polyalcohol [Vantris]) → long-term US. Deflux loses ≈ 23% volume; Swedish Reflux Trial: **20%** recurrence at 2 years. p.40

### 11 · MCQ ONE-LINERS

- MCQ Gold-standard PUJ repair: Anderson–Hynes dismembered pyeloplasty (≈ 95%). p.41
- MCQ Anderson–Hynes was originally devised for retrocaval ureter. p.41
- MCQ Commonest associated anomaly with PUJ obstruction: contralateral PUJ obstruction. p.41
- MCQ Crossing vessel PUJ obstruction typically presents in older children with intermittent pain. p.41
- MCQ Foley Y-V-plasty is contraindicated with a crossing vessel. p.41
- MCQ Spiral flap length:width ratio ≤ 3:1. p.41
- MCQ Vascular hitch needs no intrinsic stenosis and visible peristalsis. p.41
- MCQ Posterior lumbotomy: infants and toddlers only. p.41
- MCQ Retroperitoneoscopic positioning: table not flexed. p.41
- MCQ Emergency drainage before pyeloplasty: nephrostomy, not stent. p.41
- MCQ Stent duration 1–4 weeks does not affect pyeloplasty outcome. p.41
- MCQ Redo pyeloplasty: transperitoneal route preferred. p.41
- MCQ Endopyelotomy niche: failed primary pyeloplasty. p.41
- MCQ Diuretic renogram false-positive rate in neonates up to 44%. p.41
- MCQ Megaureter: diameter ≥ 7 mm. p.41
- MCQ Ureteric muscle matures craniocaudally; juxtavesical segment last. p.41
- MCQ Megaureter < 10 mm virtually never needs surgery. p.41
- MCQ Megaureter surgery if DRF < 40% or falls ≥ 5%. p.41
- MCQ Tunnel length:diameter for reimplant: 5:1 (Paquin). p.41
- MCQ Folding/plication limit: ureter < 1.75 cm. p.41
- MCQ Hendren excisional tapering: risk of ischaemic stricture. p.41
- MCQ POM outcomes better than refluxing megaureter. p.41
- MCQ Balloon dilatation of POM: double-J 2–6 months, complications up to one-third. p.41
- MCQ Duplex lower-moiety with ≤ 5–10% function: partial nephrectomy. p.41
- MCQ Fibroepithelial polyp: upper ureter, left, males. p.41
- MCQ Psoas hitch: unilateral only; second short ureter → transureteroureterostomy. p.41
- MCQ Externalised nephro-ureteric stent: removed at 7–14 days without anaesthesia. p.41
- MCQ Secondary PUJ obstruction from high-grade VUR: two-stage repair, not simultaneous. p.41
- MCQ Commonest reason for conversion of minimally invasive pyeloplasty: bleeding. p.41
- MCQ Commonest intravesical reimplant: Cohen cross-trigonal; drawback is difficult retrograde access. p.41
- MCQ Glenn–Anderson infrahiatal tunnel: least kinking, length limited by bladder neck. p.41
- MCQ Lich–Gregoir voiding dysfunction: pelvic plexus injury dorsomedial to the UVJ. p.41
- MCQ Open reimplant success (AUA meta-analysis): 95.9%. p.41

- MCQ Endoscopic injection: mound within 0.1–0.2 mL; volcano shape predicts success. p.41
- MCQ Injection-site bleeding: compress, never cauterise. p.41

## Ch 56 · Hypospadias

### 1 · DEFINITION AND DIAGNOSIS

- Incidence ≈ **1 in 150–300** male births; one of the commonest congenital anomalies. Diagnosis is clinical. p.42

### 2 · EMBRYOLOGY

- Critical window **weeks 8–16**; urethral closure proceeds from the anus towards the glans tip. p.42
- **Endodermal "two-zipper" theory** (Kurzrock, current): plate continuous through the glans; "opening zipper" canalises plate into a groove (both sexes), then androgen-dependent "closing zipper" (male only). Older **ectodermal ingrowth theory** (Glenister): glanular urethra from ectodermal ingrowth fusing at the corona. MCQ p.42
- Prepuce fuses dorsally first; ventral fusion awaits urethral fold fusion (≈ 20 weeks), so ectopic meatus → incomplete foreskin in **95%**. p.42
- KEY POINT Hypospadias is an arrest of the androgen-dependent closing zipper: the meatus stops where closure stopped, the prepuce cannot close ventrally over an open urethra, and unfused dysplastic ventral tissue tethers the shaft into curvature. p.42

### 4 · CLASSIFICATION AND SEVERITY

- By meatal position: distal (glanular, coronal, distal shaft) **70–85%**; proximal (proximal shaft, penoscrotal, scrotal, perineal) 10–25%. p.42

### 5 · ASSOCIATED ANOMALIES AND DSD WORK-UP

- **Disorder of sex development (DSD) work-up** when: UDT (uni- or bilateral) + proximal hypospadias; perineal hypospadias; atypical genitalia. Since 2006 severe hypospadias counts as 46,XY DSD. DSD risk ≈ **50%** with non-palpable vs 15% palpable testis; 64% proximal vs 7% distal. p.43
- KEY POINT Hypospadias + cryptorchidism signals a systemic androgen problem, not a local one — hence karyotype before surgery. p.43

### 6 · PRE-OPERATIVE ASSESSMENT

- **Timing**: genital awareness ≈ 18 months. American Academy of Pediatrics (AAP, 1996): 6–12 months; authors start at 6 months within **6–18 months**. Systematic review (>16,000) → fewer complications when younger; repair by 18 months. One short anaesthetic appears low-risk for neurodevelopment. p.43
- **Androgen stimulation**: glans width **<14 mm** → more complications. Testosterone, DHT or human chorionic gonadotrophin (hCG) enlarge glans, reducing glansplasty tension. Authors: testosterone cypionate **2 mg/kg intramuscular (IM) at 5 and 2 weeks** pre-op if glans <15 mm (≈ 4 mm gain). Two randomised trials → fewer complications; meta-analyses inconclusive. Side effects (aggression, erections, pigmentation) resolve within 6 months; concerns: bleeding, healing; severe cases may be partly androgen-insensitive. p.43

### 8 · CURVATURE: ASSESSMENT AND CORRECTION

- ALGORITHM

  Circumscribe → deglove to penoscrotal junction, excise dysplastic dartos → **artificial erection** (Gittes–McLaughlin tourniquet + saline; or corporal compression; or prostaglandin) → straight → urethroplasty · residual <30° → dorsal plication · >30° → divide urethral plate → still >30° → ventral corporal lengthening (usually staged)

  p.44

### 9 · URETHROPLASTY TECHNIQUES

- KEY POINT Why proximal repairs fail — Poiseuille: resistance ∝ length / radius⁴. A long, rigid, spongiosum-free skin tube ending in a fixed non-distensible glans raises proximal pressure → fistula, diverticulum, stricture. 25–75% need extra procedures after planned two-stage repair. p.45

### 10 · DRESSING, STENTS AND POST-OPERATIVE CARE

- Compressive dressing (circumferential Tegaderm including glans, or "sandwich" dressing) for ≈ **72 h**; then petroleum jelly. p.45
- Diversion **3–21 days** by complexity; soft dripping stent; suprapubic tube for older/redo. Short distal repairs can be unstented. p.45

### 11 · COMPLICATIONS AND MANAGEMENT

- TABLE Meatal stenosis · Obstructive symptoms + calibration **<8 Fr** prepubertal / <12 Fr postpubertal p.45
- TABLE Recurrent curvature · **>30°** on erection p.45
- **Urethrocutaneous fistula** — commonest (just under 10%), usually at original meatus or coronal margin. Causes: ischaemia, haematoma, infection, overlapping suture lines and especially **distal obstruction**. Rarely closes spontaneously. Repair ≥ **6–12 months** after surgery: calibrate (bougie à boule) ± urethroscopy, stress test with saline/methylene blue → excise tract, close in offset layers with dartos/tunica vaginalis interposition. Coronal fistula → redo urethroplasty and glans closure, not simple excision. Recurrence ≈ 20%. p.46
- **Balanitis xerotica obliterans (BXO)/lichen sclerosus**: white scarring of meatus/neourethra. Topical clobetasol/betamethasone usually fails → excise all involved tissue (histology), staged buccal or inner preputial graft. Recurrence up to **40%**; BXO can invade buccal grafts. p.46
- KEY POINT Nearly every urethroplasty complication traces to distal obstruction, ischaemia or tension — so fistula/diverticulum repair begins by calibrating the distal urethra. p.46

### 13 · MCQ ONE-LINERS

- MCQ Hypospadias triad: ventral ectopic meatus, ventral curvature, dorsal hooded foreskin. p.47
- MCQ Critical window of urethral development: weeks 8–16; DHT at 8–12 weeks is key. p.47
- MCQ Current embryological model: endodermal "two-zipper" theory (Kurzrock). p.47
- MCQ Megameatus intact prepuce: ≈ 5%; found on foreskin retraction. p.47
- MCQ Distal hypospadias forms 70–85% of cases. p.47
- MCQ GMS score ranges 3–12; higher predicts complications. p.47
- MCQ Hypospadias + undescended testis → karyotype; DSD ≈ 50% if testis non-palpable. p.47
- MCQ Glans width <14 mm increases complication risk. p.47
- MCQ Pre-op testosterone cypionate 2 mg/kg at 5 and 2 weeks before surgery. p.47
- MCQ Recommended repair age: 6–18 months. p.47
- MCQ Residual curvature <30° after degloving → dorsal plication. p.47
- MCQ Curvature >30° after plate division → ventral corporotomy ± graft, usually staged. p.47
- MCQ MAGPI suits a glanular meatus with no curvature. p.47
- MCQ TIP: midline plate incision; commonest late problem meatal stenosis. p.47
- MCQ Mathieu flip-flap: fewer meatal stenoses than TIP, similar fistula rate. p.47
- MCQ Onlay island flap complication typical: diverticulum. p.47
- MCQ Duckett tube = transverse preputial island flap tube. p.47
- MCQ STAG has the lowest reported proximal complication rate (≈ 23%). p.47

- MCQ Byars two-stage flaps: highest complication rate among staged options. p.47
- MCQ Poiseuille: resistance ∝ length / radius⁴ — explains proximal failures. p.47
- MCQ Commonest complication: urethrocutaneous fistula. p.47
- MCQ Coronal fistula → redo urethroplasty, not simple closure. p.47
- MCQ Meatal stenosis: symptoms + <8 Fr prepubertal. p.47
- MCQ Recurrent curvature defined as >30°. p.47
- MCQ BXO recurrence after excision and buccal graft: up to 40%. p.47
- MCQ Graft of choice for redo: buccal mucosa; lip mucosa within glans. p.47
- MCQ Only half of complications appear within 1 year. p.47

## Ch 57 · Etiology, Diagnosis and Management of the Undescended Testis

### 2 · EMBRYOLOGY OF DESCENT

- TABLE Inguinoscrotal · **20–28 weeks** · Gubernacular swelling + cremaster development widen the canal, processus vaginalis forms; distal gubernaculum released → rapid transinguinal passage (rarely before 22 weeks) → gubernaculum regresses; most scrotal after 27 weeks · Androgens acting via **genitofemoral nerve (GFN)**, released **calcitonin gene-related peptide (CGRP)**; cremaster myogenesis (Wnt pathways) p.49
- KEY POINT Outgrowth creates the canal; regression lets the testis follow — so a defect in either produces the same phenotype. INSL3 and androgen act on one target (gubernacular/cremaster myogenesis), hence one clinical picture whichever fails. p.49

### 3 · EPIDEMIOLOGY, AETIOLOGY AND ASSOCIATIONS

- Prevalence **2–4% at birth** (term) → **1–2% at 3 months**; up to 45% in preterm. Spontaneous descent mainly in first 3–6 months; rarely after 6 months (none in referred boys presenting after 6 months). High scrotal testes descend far more often (≈ 88%) than extrascrotal ones (≈ half). p.49

### 5 · MALIGNANCY

- Testicular germ cell tumour (TGCT) risk **2–5×** (pooled odds ratio ≈ 4); ≈ 1% of cryptorchid men; 5–10% of TGCT patients have a history of UDT. Contralateral testis: odds ratio ≈ 1.5. p.51
- **Swedish population study**: cryptorchidism odds ratio 3.2 — ≈ 2.8 if diagnosed before puberty vs ≈ 5.0 after. † Swedish orchidopexy cohort (Pettersson 2007): relative risk 2.23 if operated before 13 years vs 5.40 at ≥ 13 years. Meta-analysis: orchidopexy after 10–11 years → relative risk **5.8**. p.51

### 6 · DIAGNOSIS

- ALGORITHM
  Warm room, warm hands, distraction; supine (then cross-legged, squatting or standing if needed); thigh abduction inhibits cremaster; lubricant/liquid soap on fingers → sweep from anterior superior iliac spine along the canal towards the scrotum with one hand → palpate with the other for the lowest attainable position → while compressing the canal, invaginate scrotum to feel SIP/distal testis → sustained gentle traction fatigues cremaster: **stays in scrotum = retractile; springs back = undescended**.
  p.52
- ≈ **80% palpable**; 60–70% unilateral. Non-palpable at surgery: abdominal/peeping 25–50%, vanished 15–40%, extra-abdominal missed 10–30%. p.52

## 7 · MANAGEMENT PRINCIPLES AND TIMING

- **Timing**: observe to **6 months** (corrected age) → refer → orchidopexy **6–18 months**, preferably by 12 months. Rationale: spontaneous descent unlikely after 6 months; better testicular growth after surgery at 9 months than 3 years; abdominal orchidopexy easier soon after minipuberty. Preterm: observe to 6 months beyond due date (1 year if position marginal). p.53

- KEY POINT Hormones fail on both claims for one reason: position, not endocrinology, is the modifiable variable — descent is not androgen-limited after fetal life, and germ cells track age at scrotal placement. p.53

## 8 · SURGERY FOR THE PALPABLE TESTIS

- ALGORITHM

  EUA confirms palpability and lowest position → skin-crease incision superolateral to pubic tubercle → look for testis in SIP during subcutaneous dissection → open external oblique from external ring, sparing ilioinguinal nerve → free testis and divide gubernaculum distal to the sac (protect a long-looping vas) → incise internal spermatic fascia, **separate the patent processus from vas and vessels without skeletonising them and high-ligate at internal ring** (the main length-gaining step) → if needed: divide lateral spermatic fascial bands → incise transversalis fascia for retroperitoneal dissection → **Prentiss manoeuvre** (route medial to inferior epigastric vessels) → transverse scrotal incision, **subdartos pouch**, tunnel anterior to pubis → pass testis without twisting the cord → excise appendages, inspect epididymis, record volumes → close dartos around cord incorporating tunica vaginalis edge (transalbugineal sutures unnecessary — 98.7% success without) → local infiltration, ilioinguinal or caudal block.

  p.53

## 9 · THE NON-PALPABLE TESTIS

- ALGORITHM

  EUA — testis now palpable? → inguinal/scrotal orchidopexy. **Still non-palpable → diagnostic laparoscopy** — inspect internal ring patency, vessels, vas, testis size and position:
  • **Blind-ending spermatic vessels** (± vas) above a closed ring = vanished testis → stop (only unequivocal endpoint).
  • **Vessels + vas entering a closed or open ring** → inguinal/scrotal exploration; excise the nubbin (viable tubules in 5–15%).
  • **Intra-abdominal testis** → laparoscopic orchidopexy: primary (vessels long enough) or Fowler–Stephens (one or two stage); orchidectomy if postpubertal or dysgenetic.
  • Peeping testis at ring → laparoscopic or inguinal orchidopexy.
  • Neither vas nor vessels seen → search perivesical area and retroperitoneum to the kidney (agenesis extremely rare; perirenal testis may accompany renal dysplasia/agenesis).

  p.54

- ALGORITHM

  Empty bladder and stomach → open (Hasson) umbilical 5-mm port, carbon dioxide ($CO_2$) 8–12 mmHg → inspect both rings → 2–3-mm working ports triangulated to the ring → divide gubernaculum distal to the ring (use it as a handle, minimal cautery) → incise peritoneum lateral to vessels and distal to vas → mobilise vessels proximally **preserving the peritoneal triangle between vas and vessels** (keeps Fowler–Stephens possible) → new hiatus medial to inferior epigastric vessels and lateral to medial umbilical ligament (or via internal ring) → trans-scrotal clamp/port brings testis to subdartos pouch; avoid tension that avulses vessels.

  p.54

- **Single-stage**: clip/coagulate and bring down at once. **Two-stage**: clip vessels high (1.5–3 cm above testis; as proximal as possible) → wait ≈ **6 months** for collaterals → mobilise on a wide peritoneal pedicle. Two-stage slightly favoured (better position, similar atrophy). p.55

- **KEY POINT** Every abdominal technique answers one question: how to gain length without losing blood supply. FS buys length by sacrificing the testicular artery; traction buys it over time while keeping it — so vessel division is the last resort. p.55

## 11 · MCQ ONE-LINERS

- **MCQ** Cryptorchidism prevalence: 2–4% at term birth, 1–2% at 3 months. p.56
- **MCQ** Commonest ectopic site: superficial inguinal pouch. p.56
- **MCQ** Retractile testis stays in scrotum after cremaster fatigue; observe yearly. p.56
- **MCQ** INSL3 (Leydig cell) acts on gubernacular RXFP2 receptor. p.56
- **MCQ** Inguinoscrotal descent: 20–28 weeks, androgen via genitofemoral nerve and CGRP. p.56
- **MCQ** Gubernaculum attaches to the epididymis, not the testis. p.56
- **MCQ** Gonocyte → Ad spermatogonia transformation occurs during minipuberty (1–3 months). p.56
- **MCQ** Kollin randomised trial: orchidopexy at 9 months beats 3 years for testicular growth. p.56
- **MCQ** About 80% of UDTs are palpable. p.56
- **MCQ** Patent processus vaginalis in 87% of unilateral UDT. p.56
- **MCQ** Imaging (US/MRI) not recommended for UDT (AUA 2014). p.56
- **MCQ** Bilateral non-palpable testes in newborn: exclude CAH urgently. p.56
- **MCQ** Anorchia: high FSH/LH, undetectable AMH and INHB, no hCG response. p.56
- **MCQ** Contralateral testis > 2 mL suggests monorchia but exploration still needed. p.56
- **MCQ** Orchidopexy timing: 6–18 months, ideally by 12 months. p.56
- **MCQ** Hormonal therapy for descent: not recommended; LHRH success ≈ 20%. p.56
- **MCQ** Laparoscopy blind-ending vessels: stop, vanished testis confirmed. p.56
- **MCQ** Vessels entering internal ring: inguinal exploration and nubbin excision. p.56
- **MCQ** Fowler–Stephens testis survives on deferential artery collaterals. p.56
- **MCQ** Second stage of Fowler–Stephens at about 6 months. p.56
- **MCQ** Preserve peritoneum between vas and vessels to keep Fowler–Stephens possible. p.56
- **MCQ** Prentiss manoeuvre: pass cord medial to inferior epigastric vessels. p.56
- **MCQ** Shehata technique: staged traction preserving testicular vessels. p.56
- **MCQ** Gastroschisis-associated UDT: observe; 50–55% descend. p.56
- **MCQ** Congenital diaphragmatic hernia-associated UDT is always ipsilateral. p.56
- **MCQ** Paternity after unilateral UDT ≈ controls; bilateral markedly reduced. p.56
- **MCQ** TGCT risk 2–5×; uncorrected UDT tumour is usually seminoma. p.56
- **MCQ** Orchidopexy after puberty: TGCT risk roughly double that of prepubertal surgery. p.56
- **MCQ** Postpubertal UDT with normal contralateral testis: orchidectomy. p.56
- **MCQ** Microlithiasis is not an independent TGCT risk; no routine US surveillance. p.56
- **MCQ** PMDS: AMH or AMHR2 mutation; transverse testicular ectopia. p.56
- **MCQ** Agenesis is associated with ipsilateral Müllerian duct persistence. p.56

## Ch 61 · Urologic Considerations in Pediatric Renal Transplantation

### 1 · CAUSES OF END-STAGE KIDNEY DISEASE (ESKD) IN CHILDREN

- Children needing kidney replacement therapy (KRT) more often have **congenital anomalies of the kidney and urinary tract (CAKUT)**: hypoplasia/aplasia/dysplasia, then obstructive uropathy; acquired: focal segmental glomerulosclerosis. Commonest age at transplant **13–17 years**. p.57

- KEY POINT **Five goals**: unimpeded drainage · capacious low-pressure reservoir with social continence · volitional emptying · no infection · fewest operations. Chronic rejection is unavoidable, but every added insult accelerates it. p.57

### 2 · PRE-TRANSPLANT UROLOGICAL EVALUATION

- ALGORITHM
  Positive history → US + VCUG → abnormal dynamics → urodynamics → drugs + clean intermittent catheterisation (CIC) ± bladder cycling → reassess → still unsafe → augmentation → clear for transplant
  p.57

### 3 · VALVE BLADDER AND BLADDER MANAGEMENT

- **Target**: expected capacity for age at detrusor pressure **< 40 cm $H_2O$**, socially continent. p.57

- Indication: cannot reach > **75%** expected capacity at < 40 cm $H_2O$ with CIC no more often than every **3–4 h**, despite maximal medical therapy. No compelling evidence it increases transplant complications. p.58

- **Timing**: major reconstruction **before** transplant, allowing **6–12 weeks** healing. Exception: transplant before 2 years of age, before continence is expected. p.58

### 5 · TRANSPLANT INTO THE ABNORMAL BLADDER

- Graft retroperitoneal via Gibson incision; child < **15 kg** may need aorta/vena cava anastomosis and transperitoneal approach. Know augment pedicle position. p.58

- Stent if difficult reimplant or poor ureter: short double-J out at **4–6 weeks**; early removal at 5–7 days may cut infection. Use the largest urethral catheter tolerated. p.58

### 6 · UROLOGICAL COMPLICATIONS

- TABLE Ureteric stenosis · **6%** · Most within 100 days; usually distal (ischaemic donor ureter); focal → balloon + stent 4–6 weeks; otherwise open repair with native ureter or bladder flap — do not delay p.59

- TABLE Urine leak · **3%** · Early; drain-fluid creatinine; check catheter → US → mercaptoacetyltriglycine (MAG3) scan or cystogram → small: observe; large: drain/explore; short necrosis → reimplant; long → native ureter, psoas hitch, flap p.59

- ALGORITHM
  Indwelling catheter 1–2 weeks → creatinine falls = bladder cause → if not, stent + catheter; signs of rejection → biopsy; failed retrograde stent → nephrostomy + antegrade study
  p.59

### 8 · MCQ ONE-LINERS

- MCQ Commonest cause of paediatric ESKD: CAKUT (hypoplasia/dysplasia), then obstructive uropathy. p.59

- MCQ Commonest obstructive cause in boys: posterior urethral valves. p.59

- MCQ Routine pre-transplant VCUG: not required. p.59

- MCQ Safe storage pressure target: < 40 cm $H_2O$. p.59

- MCQ Augment if < 75% expected capacity despite CIC every 3–4 h and drugs. p.59

- MCQ Valve bladder end-stage: myogenic failure. p.59
- MCQ PUV bladder: postpone augmentation; neuropathic: earlier. p.59
- MCQ Gastrocystoplasty problem: haematuria–dysuria syndrome. p.59
- MCQ Preferred augment material if available: dilated ureter. p.59
- MCQ Never use the allograft to test a bladder. p.59
- MCQ Reconstruction heals 6–12 weeks before transplant. p.59
- MCQ Catheterisable stoma sited medial to transplant incision. p.59
- MCQ Native kidneys: keep by default. p.59
- MCQ Child < 15 kg: graft to aorta/vena cava. p.59
- MCQ Preferred ureteric anastomosis: extravesical non-refluxing ureteroneocystostomy. p.60
- MCQ Salvage for graft ureteric stenosis: ureteroureterostomy to native ureter. p.60
- MCQ Commonest urological complication: ureteric stenosis (≈ 6%). p.60
- MCQ Most obstructions occur within 100 days. p.60
- MCQ Leak fluid identified by high creatinine. p.60
- MCQ First step in creatinine rise + hydronephrosis: indwelling catheter. p.60
- MCQ Allograft mass: PTLD until proven otherwise. p.60
- MCQ Endoscopic injection for graft VUR: ≈ one-third success. p.60
- MCQ Asymptomatic graft VUR: observe. p.60

## Ch 63 · Paediatric Urologic Oncology: Renal and Adrenal

### 1 · WILMS TUMOUR (WT): EPIDEMIOLOGY AND GENETICS

- Nephroblastoma: **6–7%** of childhood cancers, **95%** of kidney cancers under 15 y; embryonal tumour from immature renal remnants. Median age **3.5 y**, >80% before 5 y; younger in bilateral and syndromic disease. p.61
- TABLE High (>20%) · WAGR, DDS (**50%**); Fanconi D1 (BRCA2); Perlman (DIS3L2); mosaic variegated aneuploidy p.61
- KEY POINT Trap: BWS is the **commonest** predisposing syndrome but only moderate risk; WAGR/DDS are rarer and highest. p.61
- ALGORITHM
  WT risk >1% → renal ultrasound (US) every **3–4 months** until **7 y** → suspicious lesion → computed tomography (CT)/magnetic resonance imaging (MRI)
  p.61

### 2 · PATHOLOGY

- **Anaplasia** (3 criteria): nuclei ≥**3×** neighbours, hyperchromasia, abnormal mitoses. ~2% under 2 y, ~13% over 5 y. Chemoresistant; **diffuse** worse than focal. Acquired (later age, absent from rests); TP53-driven. p.62

### 4 · STAGING AND COG VS SIOP

- KEY POINT Because COG stage is surgical-pathological, the operating surgeon — not the scan — decides chemotherapy intensity and irradiation: one spill converts stage I/II to III. p.63

### 5 · SURGICAL PRINCIPLES

- **Lymph node sampling mandatory** (formal dissection not needed): hilar, paracaval/para-aortic, interaortocaval (skip metastases). COG requires **≥5 nodes**. Omission is the commonest protocol violation (10–20%) → under-staging. p.63

- ALGORITHM
  Synchronous bilateral (5–7%) → **no upfront nephrectomy, no biopsy** → VCR + AMD + DOX 6 weeks → imaging → NSS if feasible → else more chemo but **surgery by week 12** → non-responder: **bilateral open biopsy**
  p.63

- Unilateral NSS indications: syndromic/predisposed or solitary kidney, bilateral risk; SIOP UMBRELLA criteria: **<300 mL**, substantial remnant, negative nodes. Use frozen section and a margin of normal kidney. p.63

### 6 · RELAPSE AND LATE EFFECTS

- Late effects track **radiotherapy and anthracycline**: heart failure 4.4% after DOX (rare below **150 mg/m²**); second cancers 1% at 10 y (in-field); gonadal failure (alkylators, pelvic/whole-abdominal RT). Renal failure after unilateral nephrectomy only **0.6%** at 20 y — higher in WT1 syndromes. p.64

### 7 · OTHER RENAL TUMOURS

- KEY POINT Age points to diagnosis: neonate → CMN; toddler → WT/rhabdoid; adolescent → RCC. Metastatic site sets staging tests: lung (WT), bone (CCSK), brain (rhabdoid/CCSK). p.64

### 8 · NEUROBLASTOMA

- Urinary **VMA** (vanillylmandelic acid) and **HVA** (homovanillic acid) raised in **90–95%**; used to monitor response/relapse. p.65

- TABLE MS · Age **<18 months**; metastases to skin, liver, marrow (<10%) only; no bone (old INSS 4S: <12 months) p.65

- KEY POINT Trap: in neuroblastoma diploid is bad and hyperdiploid good. Stage 4S/MS: OS 80–88% with frequent spontaneous regression; deaths occur in infants <2 months from massive hepatomegaly with respiratory compromise or coagulopathy. p.65

- KEY POINT Surgical value runs inversely with biology: favourable tumours are cured by little or no resection; unfavourable ones are not rescued by radical resection — so vital structures are never traded for margin. p.65

### 9 · ADRENOCORTICAL TUMOURS

- KEY POINT Why the open approach? Cure depends on complete removal of an intact tumour; capsular rupture seeds the retroperitoneum, with poor salvage. p.66

### 11 · MCQ ONE-LINERS

- MCQ WT1 is at 11p13; WT2 putative locus at 11p15 (IGF2). p.66
- MCQ Denys–Drash: dominant-negative WT1 point mutation, 46,XY DSD, mesangial sclerosis. p.66
- MCQ WAGR and Denys–Drash carry ~50% WT risk; BWS 5–10%. p.66
- MCQ Screening: renal US every 3–4 months until age 7 if risk >1%. p.66
- MCQ Anaplasia: nuclei ≥3× normal, hyperchromasia, atypical mitoses; TP53 mutated. p.67
- MCQ Nephrogenic rests in >90% of bilateral WT. p.67
- MCQ Any tumour spill upstages WT to COG stage III. p.67
- MCQ Minimum lymph nodes sampled at Wilms nephrectomy (COG): five. p.67
- MCQ Acquired von Willebrand disease in up to 8% of WT. p.67
- MCQ Bilateral WT: definitive surgery by week 12 of chemotherapy. p.67
- MCQ Thrombus above hepatic veins → preoperative chemotherapy. p.67
- MCQ Doxorubicin cardiotoxicity rare below 150 mg/m² cumulative. p.67
- MCQ Adrenocortical tumour: open adrenalectomy, avoid rupture; test TP53. p.67

- MCQ Commonest neonatal renal tumour: congenital mesoblastic nephroma (ETV6–NTRK3). p.67
- MCQ Clear cell sarcoma metastasises to bone; rhabdoid tumour associated with brain tumours. p.67
- MCQ Rhabdoid tumour: SMARCB1/INI1 loss, most lethal paediatric renal tumour. p.67
- MCQ Commonest paediatric RCC: Xp11.2 translocation (TFE3). p.67
- MCQ Renal medullary carcinoma: sickle cell trait. p.67
- MCQ Neuroblastoma: urinary VMA/HVA raised in 90–95%. p.67
- MCQ 123I-MIBG diagnostic; 131I-MIBG therapeutic. p.67
- MCQ MYCN amplification: >10 copies at 2p24, high risk. p.67
- MCQ Hyperdiploidy favourable in neuroblastoma; diploidy adverse. p.67
- MCQ INRG MS: age <18 months; skin, liver, marrow <10%; no bone. p.67
- MCQ Opsoclonus–myoclonus–ataxia: half have neuroblastoma. p.67
- MCQ Calcification and vessel encasement favour neuroblastoma over Wilms tumour. p.67

## Ch 64 · Paediatric Urological Oncology: Pelvis, Bladder, Testis and Genitalia

### 1 · RHABDOMYOSARCOMA (RMS): BIOLOGY

- GU sites: bladder/prostate (BP), paratesticular (PT), vulvovaginal (VV). GU-RMS 5-y survival **>80%**. p.68
- KEY POINT Fusion status predicts outcome better than histology, so current trials assign risk by **fusion status**, not microscopy. MYOD1 mutation (fusion-negative) and circulating tumour DNA at diagnosis are emerging adverse markers. p.68

### 2 · PRESENTATION AND WORK-UP

- ALGORITHM
  Ultrasound (US) → CT or MRI of primary + nodal basin (BP: pelvic and retroperitoneal to renal vessels; PT: retroperitoneal; VV: inguinal, pelvic, retroperitoneal) → **chest CT** (lung commonest metastatic site) → bone scan + bilateral marrow (omit in fusion-negative ERMS without nodal/lung disease)
  p.68

### 3 · STAGE, GROUP AND RISK

- KEY POINT Organ-preserving **biopsy** deliberately places BP-RMS in group III → **intermediate risk**; PT-RMS (resected by orchidectomy) is almost always low risk. Age <1 y or >10 y is independently adverse. Trap: resected nodes with most distal node positive = IIc, not IIb. p.69

### 4 · TREATMENT PRINCIPLES

- Primary exenteration fell from almost all (IRS-I) to **13%** (IRS-IV) with unchanged OS (78–83%). IRS = Intergroup Rhabdomyosarcoma Study; OS = overall survival. p.69
- **Ipsilateral nerve-sparing retroperitoneal lymph node dissection (RPLND)** before chemotherapy if **age ≥10 y** or radiological N1 at any age; ≥7–13 nodes, up to renal vessels. Open, laparoscopic or robotic — none proven superior. p.70
- KEY POINT Why age, not CT? Older boys have far more retroperitoneal disease and CT has a high false-negative rate for nodes. p.70

### 5 · OTHER BLADDER TUMOURS

- **Spina bifida/exstrophy**: baseline lifetime bladder cancer risk **2–4%**, aggressive, **not increased by intestinal augmentation**; screening asymptomatic patients is neither effective nor cost-effective (cancers arise within months of normal surveillance). Evaluate symptoms (haematuria) promptly instead. p.70

- TABLE Half-life · **5 days** · **24 h** p.71

- ALGORITHM
  Inguinal approach → markers normal + benign-looking US → **testis-sparing partial orchidectomy** with frozen section → benign: close; germ cell malignancy or raised AFP → radical inguinal orchidectomy
  p.71

- **YST**: radical inguinal orchidectomy; **90%** stage I. CT chest/abdomen/pelvis; follow AFP decline by half-life. Stage I → **surveillance, no RPLND or chemotherapy**. Scrotal violation → stage II (chemo, no hemiscrotectomy). Lymphadenopathy with normal markers → node biopsy to confirm; nodes + raised markers = stage III; stage III–IV → cisplatin, etoposide, bleomycin. Persistent mass → biopsy/RPLND. p.71

- KEY POINT Trap: **complete** androgen insensitivity is intermediate (not high) risk; partial is high. p.71



- MCQ Commonest paediatric soft-tissue sarcoma: rhabdomyosarcoma. p.72
- MCQ Commonest GU-RMS histology: embryonal; botryoid variant in bladder/vagina. p.72
- MCQ Alveolar RMS fusion: PAX3/7–FOXO1 (chromosome 13). p.72
- MCQ Risk now assigned by fusion status, not histology. p.72
- MCQ Paratesticular and vulvovaginal are favourable sites; bladder/prostate unfavourable. p.72
- MCQ BP-RMS is at least stage 2. p.72
- MCQ Biopsy-only BP-RMS = group III → intermediate risk. p.72
- MCQ Resected nodes, most distal positive = group IIc. p.72
- MCQ Pretreatment re-excision improves GU-RMS survival; delayed excision does not. p.72
- MCQ First drainage in obstructing BP-RMS: urethral catheter. p.72
- MCQ Stable end-of-therapy bladder mass: observe. p.72
- MCQ Paratesticular RMS RPLND: age ≥10 years or N1. p.72
- MCQ Paratesticular RMS: radical inguinal orchidectomy. p.72
- MCQ Exenteration fell to 13% by IRS-IV without survival loss. p.72
- MCQ Recurrent paediatric urothelial tumour: consider Lynch or other syndrome. p.72
- MCQ Neurogenic bladder cancer risk 2–4%, not raised by augmentation. p.72
- MCQ Commonest prepubertal testicular tumour: teratoma; malignant: yolk sac tumour. p.72
- MCQ AFP half-life 5 days; β-hCG 24 hours. p.72
- MCQ AFP reaches adult levels by 6–9 months of age. p.72
- MCQ Onion-skin appearance on US: epidermoid cyst. p.72
- MCQ Stage I prepubertal YST: surveillance after orchidectomy. p.72
- MCQ Reinke crystals: Leydig cell tumour. p.72
- MCQ Testicular adrenal rests: optimise steroid replacement first. p.72
- MCQ Complete androgen insensitivity: intermediate gonadal tumour risk. p.72
- MCQ Sertoli–Leydig ovarian tumour: test for DICER1. p.72
- MCQ Li–Fraumeni surveillance: annual whole-body MRI plus 3–4-monthly abdominal US. p.72
- MCQ Gonadoblastoma progresses to dysgerminoma. p.72

- **MCQ** Currarino triad: sacral mass, sacral defect, anorectal malformation. p.72

## ★ Putting it together: common paediatric decisions

### A · VESICOURETERAL REFLUX (VUR) AFTER A FEBRILE URINARY TRACT INFECTION (UTI)

- **ALGORITHM**

**1.** Renal and bladder ultrasound (US) → voiding cystourethrogram (VCUG) if febrile UTIs are multiple, need admission, or US is abnormal.
**2.** Everyone: treat bladder and bowel dysfunction (BBD), starting with constipation.
**3.** Low risk (grade I–II, no BBD, boy, older) → observation ± continuous antibiotic prophylaxis (CAP). High risk (grade ≥III, girl, BBD, scars, infant) → CAP.
**4.** Breakthrough febrile UTI, new scars, or persistent high grade with poor function → correction: endoscopic injection or ureteric reimplantation (→ Ch 41, 53).

p.73

### C · UNDESCENDED TESTIS (UDT)

- **ALGORITHM**

**Palpable** → orchidopexy at **6–18 months** (ideally by 12). Imaging is not recommended.
**Non-palpable** → examination under anaesthesia → still impalpable → diagnostic laparoscopy: blind-ending vessels = vanished testis → stop · vessels and vas entering the ring → groin exploration, excise the nubbin · intra-abdominal testis → laparoscopic orchidopexy (primary or Fowler–Stephens).
**Bilateral non-palpable or UDT + hypospadias** → work up for a difference of sexual development (DSD) first (→ Ch 59, in the Oncology, Genitalia & DSD book).

p.73

### E · NEWBORN WITH ATYPICAL GENITALIA

- **KEY POINT** Bilateral impalpable gonads + atypical genitalia = salt-wasting congenital adrenal hyperplasia (CAH) until proven otherwise. Check electrolytes and 17-hydroxyprogesterone (after 36 h) and karyotype; the salt-wasting crisis usually comes at days 5–7. Sex designation and surgery are not emergencies and need a multidisciplinary team. p.74

# 5 Paediatric Urology 2

647 core items · page numbers refer to the full book PDF

## Ch 32 · Embryology of the Human Genitourinary Tract

### 1 · OVERVIEW

- **Indifferent stage** ≈ **8–9 wk** (Carnegie collection); thereafter androgens drive the sexes apart. p.2

### 2 · KIDNEY AND URETER

- KEY POINT Why one defect gives both dysplasia and an abnormal orifice: bud position on the Wolffian duct sets where it meets the blastema (kidney quality) and where its orifice lands in the bladder (see §6). p.2

### 4 · GONADS

- **SRY** (sex-determining region Y, short arm of Y) → **Sertoli cells** enclose PGCs in cords (≈ **6 wk**). Without SRY protein → ovary. p.3
- TABLE Effect · Regression **8–10 wk** (sensitivity window 9–13 wk) · Wk 8–12 → epididymis, vas, seminal vesicle p.3
- KEY POINT Two independent testicular signals act on two independent duct systems: MIS decides Müllerian, testosterone decides Wolffian and external genitalia. Every disorder of sex development (DSD) below is a combination of the two. p.3

### 6 · GENITAL DUCTS

- **Prostate**: buds at **10 wk** from FOXA1+ endodermal UGS epithelium below the ureteric insertion; six stages (bud → branch → canalise → differentiate). Most buds in gutters lateral to the **verumontanum** (Wolffian ducts + utricle open on it; mesoderm–endoderm interface). p.4

### 7 · EXTERNAL GENITALIA

- Timeline: coronal sulcus 11–12 wk; shaft fused **16 wk**; complete 18 wk; fetal ventral curvature resolves 20 wk. p.4

### 9 · CLINICAL CORRELATIONS

- KEY POINT Homologue map predicts remnants: Littre ↔ Skene, Cowper ↔ Bartholin, spongiosum ↔ vestibular bulb, gubernaculum ↔ ovarian + round ligaments, labioscrotal swellings ↔ labia majora. Traps: vagina is UGS-derived; appendix testis is Müllerian, appendix epididymis Wolffian. p.5

### 10 · MCQ ONE-LINERS

- MCQ Indifferent stage of external genitalia: 8–9 weeks. p.6
- MCQ First sign of masculinisation: increased anogenital distance. p.6
- MCQ Entire male urethral epithelium: urogenital sinus endoderm. p.6
- MCQ Canalisation of urethral plate is androgen-independent; fold fusion is androgen-dependent. p.6
- MCQ Commonest hypospadiac meatal site: shaft–glans junction. p.6
- MCQ Penile nerve-free zone: 12 o'clock — dorsal plication site. p.6
- MCQ Prostatic buds appear at 10 weeks from FOXA1+ UGS epithelium. p.6
- MCQ 5αR2 deficiency: rudimentary prostate and severe hypospadias. p.6
- MCQ DHT has about tenfold higher AR affinity than testosterone. p.6
- MCQ SRY lies on the short arm of Y; acts in pre-Sertoli cells. p.6
- MCQ MIS from Sertoli cells; Müllerian regression 8–10 weeks. p.6
- MCQ Appendix testis: Müllerian; appendix epididymis: Wolffian. p.6

- MCQ INSL3 knockout: isolated intra-abdominal cryptorchidism. p.6
- MCQ Gubernacular swelling is androgen-independent. p.6
- MCQ Inguinoscrotal descent: androgen via genitofemoral nerve CGRP. p.6
- MCQ Acquired ascending testis: failed involution of processus vaginalis. p.6
- MCQ Entire vaginal epithelium and hymen derive from UGS (FOXA1). p.6
- MCQ Commonest cause of vaginal obstruction: imperforate hymen. p.6
- MCQ Transverse vaginal septum: junction of middle and upper thirds. p.6
- MCQ Uterus didelphys: failed fusion of caudal Müllerian ducts. p.6
- MCQ Urorectal septum divides cloaca at weeks 5–6. p.6
- MCQ Trigone muscle: mainly bladder smooth muscle (lineage tracing). p.6
- MCQ Upper-pole duplex ureter: caudal, medial, ectopic; lower pole: lateral, refluxing. p.6
- MCQ CAH clues in "male": impalpable gonads, hyperpigmentation. p.6
- MCQ 46,XY streak gonads: remove (gonadoblastoma). p.6
- MCQ PMDS: usually defective MIS receptor; hernia uteri inguinale. p.6
- MCQ †GDNF (mesenchyme) binds RET (ureteric bud) to drive branching. p.6
- MCQ †Horseshoe kidney ascent arrested by inferior mesenteric artery. p.6

## Ch 33 · Urologic Aspects of Pediatric Nephrology

### 1 · RENAL FUNCTION IN THE CHILD

- Glomerular filtration rate (GFR): doubles in the first 2 weeks, adult level by **2 years**; term infant ≈40 mL/min/1.73 m². p.7
- Creatinine: first 48 h reflects **maternal** function; by day 7 the neonate's own (**<0.5 mg/dL** at term). p.7
- KEY POINT The neonatal kidney is short and "hormonally deaf": maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate and poor concentration all reflect immaturity, not disease. p.7

### 2 · HAEMATURIA

- Microscopic: **>5 red blood cells (RBCs)/high-power field** on fresh spun urine; ≈1% of school children, mostly resolves within 1 year. p.7

### 3 · PROTEINURIA

- Normal urine protein-to-creatinine ratio (UPCR): **<0.2 mg/mg** (<0.7 newborn); timed <4 mg/m²/h; adolescents <150 mg/24 h. p.7

### 4 · GLOMERULAR DISEASE

- Proteinuria **>40 mg/m²/h or UPCR >2.0** + hypoalbuminaemia + hyperlipidaemia; oedema of eyelids, scrotum, labia. p.8
- **Post-streptococcal GN**: weeks after pharyngitis/pyoderma; C3 <50% normalises by **8–12 weeks**; persistently low C3 → biopsy. p.8

### 7 · HYPERTENSION

- Repeated BP **>95th percentile** for age, sex, height; adolescents normal <120/80. Up to a quarter white-coat. p.9
- Lifestyle first (stage 1, no end-organ damage); drugs for stage 2, symptoms, end-organ damage or secondary cause. Target <90th percentile; chronic kidney disease (CKD) **<50th**. p.9

### 8 · ACUTE KIDNEY INJURY

- Fractional excretion of sodium (FeNa) **<1%** = prerenal (tubule intact); >3% = intrinsic; on diuretics use urea <35%. p.9

### 9 · CHRONIC KIDNEY DISEASE

- Abnormality >3 months; stages (children >2 years) by GFR: I ≥90, II 60–89, III 30–59, IV 15–29, V **<15** or dialysis. p.9

### 10 · DIALYSIS AND TRANSPLANTATION

- Peritoneal dialysis first if **<9 years / <20 kg** (venous access fails); prior vesicostomy acceptable; contraindicated by omphalocele, gastroschisis, diaphragmatic hernia, bladder exstrophy. p.9
- Transplant is the goal, often pre-emptive living-related; many wait for **≥10 kg and 65 cm** (graft thrombosis). p.10

### 11 · MCQ ONE-LINERS

- MCQ Commonest cause of paediatric ESKD: CAKUT. p.10
- MCQ GFR reaches adult levels by 2 years. p.10
- MCQ Neonatal creatinine in first 48 h: maternal. p.10
- MCQ Microscopic haematuria: >5 RBCs/high-power field. p.10
- MCQ Commonest cause of gross haematuria in children: UTI. p.10
- MCQ Heme-positive dipstick, no RBCs: haemoglobinuria or myoglobinuria. p.10
- MCQ Normal UPCR in older children: <0.2 mg/mg. p.10
- MCQ Nephrotic-range proteinuria: >40 mg/m²/h or UPCR >2.0. p.10
- MCQ Commonest nephrotic lesion in children: minimal change. p.10
- MCQ Steroid-resistant nephrotic syndrome: usually FSGS. p.10
- MCQ Low C3 and C4: lupus nephritis or MPGN. p.10
- MCQ C3 normalises after post-streptococcal GN by 8–12 weeks. p.10
- MCQ Commonest primary GN worldwide: IgA nephropathy. p.10
- MCQ Alport: COL4A5, X-linked, sensorineural deafness. p.10
- MCQ Atypical HUS treatment: eculizumab. p.10
- MCQ Distal RTA: hypokalaemia, nephrocalcinosis. p.10
- MCQ Hyperkalaemic RTA (type 4) may be caused by obstructive uropathy. p.10
- MCQ ARPKD gene: PKHD1; with congenital hepatic fibrosis. p.10
- MCQ Hypertension in school-age child: renal parenchymal, including reflux nephropathy. p.10
- MCQ Prerenal AKI: FeNa <1%. p.10
- MCQ CKD stage V: GFR <15 mL/min/1.73 m². p.10
- MCQ Growth failure in CKD: growth hormone resistance, not deficiency. p.10
- MCQ Only proven measure slowing paediatric CKD: strict BP control. p.10
- MCQ Peritoneal dialysis preferred <9 years and <20 kg. p.10
- MCQ Transplant size threshold: ≥10 kg and 65 cm. p.10

## Ch 34 · Perinatal Urology

### 1 · FETAL RENAL DEVELOPMENT AND PRENATAL IMAGING

- Metanephric kidney appears at **day 28**; urine production begins in earnest at **14 weeks**. After **~16 weeks** amniotic fluid is mainly fetal urine — the basis of oligohydramnios in urinary obstruction and of pulmonary hypoplasia, since lungs need fluid to develop. p.11
- Screening ultrasound (US) at 18–20 weeks. Amniotic fluid index (AFI) = sum of four largest pockets: normal 8–18 cm; oligohydramnios **<5–6 cm**. p.11
- KEY POINT The value of prenatal detection is triage and counselling (differential, postnatal plan, natural history); there is no proof it improves outcome. p.11

### 2 · GRADING SYSTEMS

- Medullary pyramidal thickness (PT) **≤3 mm** predicts pyeloplasty and reduced function; reliable between raters. p.11

### 3 · PRENATAL RISK STRATIFICATION AND FOLLOW-UP

- ALGORITHM
  A1 before 28 weeks → repeat US after 32 weeks → postnatal US after 48 h
  A2–3 → US every 4 weeks to delivery (lesion may progress and change delivery plan; second anomaly likelier) → postnatal US after 48 h
  p.12

### 4 · FETAL LOWER URINARY TRACT OBSTRUCTION (LUTO) AND OLIGOHYDRAMNIOS

- Poor prognosis: oligohydramnios **before 27 weeks**, renal cortical cysts, increased echogenicity. Oligohydramnios after 27 weeks → favourable lungs. p.12
- Early delivery (lungs mature), vesicoamniotic shunt (VAS), fetal cystoscopy, serial amnioinfusion. Window **18–24 weeks**; need ≈1 in 60,000. p.12
- Complications ≈**44%**: incomplete drainage/shunt failure (commonest), preterm labour, membrane rupture, urinary ascites, chorioamnionitis, bowel herniation. Rodeck shunt lies flush to limit migration. p.13
- KEY POINT Counsel three facts: intervention improves perinatal survival (lungs); complications exceed 40%; it does not reliably prevent long-term renal replacement, because dysplasia is often established before the needle goes in. p.13

### 5 · POSTNATAL IMAGING

- **Technetium-99m mercaptoacetyltriglycine (MAG3)** diuretic renogram: preferred over DTPA (diethylenetriaminepentaacetic acid) for better extraction; "well-tempered" protocol standardises hydration, bladder catheter, diuretic dose/timing. Surgery considered if **DRF <40%**, T½ >20 min, or retention on delayed upright images. p.13

### 6 · POSTNATAL MANAGEMENT, UTI AND PROPHYLAXIS

- CAP without VUR in P3 continues until pyeloplasty or **1 year**, as ~90% of UTIs occur in year 1. p.14
- Circumcision protects: number needed to treat (NNT) = **10**; odds ratio ≈0.3. p.14
- TABLE Nitrofurantoin; trimethoprim · **≥8 weeks** (immature hepatic metabolism) p.14

- **ALGORITHM**
  Prenatal UTD → US after 48 h → normal (<10 mm, all parameters normal) → transient, insignificant
  P1 → US 3–6 months, no CAP/VCUG → P2 → CAP if ureter >7 mm; VCUG/MAG3 at discretion → P3 → CAP + VCUG + US ~1 month → persistent → MAG3
  Bilateral severe dilation ± oligohydramnios → treat as outlet obstruction: decompress → CAP → urgent evaluation
  p.14

## 7 · SPECIFIC ANTENATAL DIAGNOSES

- Commonest **surgical** cause. Third-trimester APD **15 mm**: sensitivity and specificity ≈80% for pyeloplasty; postnatal APD predicts better than prenatal. Even severe UTD: only ~a fifth need surgery. p.15
- Most resolve. Surgery predicted by ureter **>13 mm**, DRF <30%. p.15
- British Association of Paediatric Urologists: prenatal ureter **>7 mm after 30 weeks** → postnatal US, CAP, VCUG for all; consider MAG3. p.15
- **KEY POINT** Across PUJ, VUR and megaureter, dilation alone is not an indication: operate for loss of function, infection or progression; grade sets surveillance intensity. p.15
- **KEY POINT** Bilateral dilation with normal fluid = urine still leaves the fetus (reflux, megacystis); with low fluid = true outlet obstruction. p.15
- **ALGORITHM**
  Team at delivery → (pulmonary resuscitation if needed) → 5 Fr feeding tube, finger in rectum to guide over high bladder neck → confirm on kidney–ureter–bladder (KUB) film → fails → US-guided suprapubic catheter (Seldinger)
  p.16
- **CAH**: 21-hydroxylase (chromosome 6p), autosomal recessive; classic salt-wasting 75%. Only **1 in 8** at-risk pregnancies yields an affected girl, so prenatal treatment is controversial. Postnatal: electrolytes first; 17-hydroxyprogesterone + genetics; decompress if LUTO. p.17

## 8 · NEONATAL UROLOGICAL EMERGENCIES

- Anorectal malformation: genitourinary/spinal anomalies up to **75%** → renal + spinal US (tethered cord); CAP until videourodynamics at 6 weeks. p.17

## 10 · MCQ ONE-LINERS

- **MCQ** Second commonest prenatal anomaly: urinary tract dilation. p.18
- **MCQ** Amniotic fluid mainly fetal urine after ~16 weeks. p.18
- **MCQ** Oligohydramnios: AFI <5–6 cm. p.18
- **MCQ** Normal APD: <4 mm before 28 weeks, <7 mm after, <10 mm postnatal. p.18
- **MCQ** Severe APD: >10 mm second trimester, >15 mm third trimester. p.18
- **MCQ** Prenatal APD grade does not predict VUR. p.18
- **MCQ** SFU grade 4: parenchymal thinning. p.18
- **MCQ** Pyramidal thickness ≤3 mm predicts pyeloplasty. p.18
- **MCQ** First postnatal US after 48 hours. p.18
- **MCQ** P3 at any APD if parenchyma abnormal. p.18
- **MCQ** P1: no prophylaxis, no VCUG. p.18
- **MCQ** P2 with ureter >7 mm: prophylaxis. p.18
- **MCQ** Circumcision NNT to prevent one UTI: 10. p.18

- MCQ Nitrofurantoin/trimethoprim avoided before 8 weeks. p.18
- MCQ Commonest surgical cause of prenatal UTD: PUJ obstruction. p.18
- MCQ Third-trimester APD 15 mm predicts pyeloplasty. p.18
- MCQ MAG3 surgical threshold: DRF <40%, T½ >20 min. p.18
- MCQ Megaureter in infancy needing surgery: temporising diversion; reimplant after 1 year. p.18
- MCQ Poor fetal LUTO prognosis: oligohydramnios before 27 weeks. p.18
- MCQ Ruano stage II: intervention indicated. p.18
- MCQ Fetal cystoscopy specific complication: urethrocutaneous/rectourethral fistula. p.19
- MCQ PLUTO: survival benefit, no renal benefit. p.19
- MCQ Avoid balloon catheter in PUV neonate. p.19
- MCQ Commonest cause of neonatal urinary ascites: PUV. p.19
- MCQ MCDK: no increased Wilms tumour risk. p.19
- MCQ Duplex: upper pole obstructs, lower pole refluxes (Meyer–Weigert). p.19
- MCQ Omphalocele distinguishes cloacal from bladder exstrophy. p.19
- MCQ Commonest solid renal tumour <6 months: mesoblastic nephroma. p.19
- MCQ Renal vein thrombosis triad: haematuria, flank mass, thrombocytopenia. p.19
- MCQ Renal artery thrombosis: umbilical artery catheter. p.19
- MCQ Anorectal malformation: genitourinary anomalies up to 75%. p.19
- MCQ Prenatal torsion: urgent contralateral orchidopexy; salvage nearly nil. p.19

## Ch 35 · Urologic Evaluation of the Child

### 1 · TRIAGE BY ACUITY

- KEY POINT Context shifts the tier: hernia emergent if incarcerated, semiurgent if reducible; hydronephrosis urgent if bilateral/solitary. p.20

### 3 · EXAMINATION

- Heart and respiratory rate fall, blood pressure (BP) rises with age. Minimum systolic BP: **≥60** neonate; ≥70 at 1 mo–1 yr; **70 + 2×age** at 1–10 yr; ≥90 above 10 yr. p.20
- **Micropenis**: stretched length (pubis to glans tip) < mean − 2.5 SD — term newborn **1.9 cm**, adult 9.3 cm. Length barely changes 5–11 yr; preterms need gestation-specific values. p.21

### 4 · URINE AND BLOOD TESTS

- American Academy of Pediatrics (AAP) definition: pyuria and/or bacteriuria + **≥50,000 colony-forming units (CFU)/mL** of a uropathogen (catheter/suprapubic), in a symptomatic child. p.21
- TABLE Microscopic haematuria · **>5 red cells (RBC)/high-power field** (adult ≥3), >2 occasions p.21

### 6 · EMERGENT PRESENTATIONS

- TABLE Absent cremasteric reflex; nausea/vomiting; high-riding testis · 1 each · **≥6** → explore without US p.22
- **Ambiguous genitalia**: stabilise, diagnose, assign sex of rearing (multidisciplinary). Exclude **salt-wasting congenital adrenal hyperplasia (CAH)** (21-hydroxylase ≈ 95%): crisis day **4–15** → do not discharge early; check **17-hydroxyprogesterone + electrolytes**. p.22
- **Retention**: expected capacity (mL) = **(age + 2) × 30**. Try analgesia, sitz bath, disimpaction before catheter; no cause → neurological evaluation. p.22

### 7 · COMMON REFERRAL PRESENTATIONS

- Scan after **48 h** (earlier underestimates dilation, neonatal oliguria) unless outlet obstruction suspected. Creatinine reflects maternal value for 48 h → follow trend. p.22
- **Varicocele** (≈15% adolescents): measure volume (Lambert: L×W×H×**0.71**). Non-decompressing supine or right-sided → retroperitoneal US for tumour. p.23
- KEY POINT Red flags: absent cremasteric reflex with acute pain; newborn ambiguous genitalia with impalpable gonads; bilateral hydronephrosis in a boy; fever with obstruction; hypotension (late shock); abdominal mass; meatal blood; lumbosacral lipoma or deviated cleft over a mass. p.23

### 9 · MCQ ONE-LINERS

- MCQ Commonest cause of paediatric acute scrotum: torsion of testicular appendage. p.23
- MCQ TWIST ≥6: explore without ultrasound. p.23
- MCQ Commonest cause of newborn ambiguous genitalia: CAH (21-hydroxylase). p.23
- MCQ Salt-wasting crisis: days 4–15 of life. p.23
- MCQ Expected bladder capacity (mL) = (age + 2) × 30. p.24
- MCQ Paediatric microhaematuria: >5 RBC/hpf on more than two occasions. p.24
- MCQ AAP UTI culture threshold: ≥50,000 CFU/mL by catheter or aspirate. p.24
- MCQ Bag urine specimens: high false-positive rate. p.24
- MCQ Postnatal scan ideally after 48 hours. p.24
- MCQ Commonest renal tumour under 15: Wilms, peak 1–5 years. p.24
- MCQ Commonest prepubertal malignant testis tumour: yolk sac tumour. p.24
- MCQ Commonest cause of ischaemic priapism in children: sickle cell disease. p.24
- MCQ Neonatal priapism: benign, resolves in about 4–5 days. p.24
- MCQ Renal vein thrombosis triad: haematuria, flank mass, thrombocytopenia. p.24
- MCQ Micropenis: stretched length below mean − 2.5 SD (term ≈1.9 cm). p.24
- MCQ First male pubertal sign: testicular enlargement (Tanner II). p.24
- MCQ Minimum systolic BP 1–10 years: 70 + 2 × age. p.24
- MCQ Ultrasound not recommended to locate undescended testis. p.24
- MCQ Simple sacral dimple: no MRI needed. p.24
- MCQ MAG3 answers obstruction; DMSA answers scarring. p.24

## Ch 36 · Pediatric Urogenital Imaging

### 1 · PRINCIPLES — ALARA AND THE THREE HARMS

- KEY POINT Every choice trades **radiation, sedation and contrast** against yield; the child's long lifespan tips it towards US, and the question is which test fills US's functional gap (obstruction, reflux, pyelonephritis) most cheaply. p.25

### 2 · ULTRASONOGRAPHY

- Fetal urine production begins at **10–12 weeks**; by **20 weeks** US detects most urological anomalies. p.25
- Meta-analysis: area under curve (AUC) **0.97**, sensitivity ~92%, specificity ~91–94%, first- and second-generation agents equal → recommended for VUR detection. p.26

### 3 · VOIDING CYSTOURETHROGRAPHY

- **Trauma**: blood at meatus → **retrograde urethrogram before any catheter** → then fill bladder by gravity to capacity. Expected capacity (mL) = **(age + 2) × 30**, max 400. "Pie-in-the-sky" bladder = pelvic fracture with extraperitoneal injury. p.27

### 4 · NUCLEAR MEDICINE

- TABLE Timing · Scan within days; resolves over ~5 months · Lesion at **≥6 months** = scar p.27
- TABLE Furosemide · **1 mg/kg IV** at maximal filling · Too early → drainage measured before system full p.27

### 5 · MR UROGRAPHY

- Protocol: IV hydration + catheter → **furosemide 1 mg/kg (max 20 mg)** → T2 fast spin-echo (3D fat-saturated) → slow gadolinium → dynamic T1 until ureters fill; **Patlak** analysis for function. Hydration + diuretic give better distension and fewer artefacts. p.28

### 8 · MCQ ONE-LINERS

- MCQ SFU grade 4 = grade 3 plus parenchymal thinning. p.29
- MCQ Large bright kidneys bilaterally without discrete cysts → ARPKD. p.29
- MCQ Neonatal solid renal mass → congenital mesoblastic nephroma. p.29
- MCQ Neonate with renal enlargement, AKI and hypertension → renal vein thrombosis. p.29
- MCQ Twinkling artefact on colour Doppler → calculus. p.29
- MCQ Keyhole sign → posterior urethral valves. p.29
- MCQ Whirlpool sign → testicular torsion; raises sensitivity to ~96%. p.29
- MCQ Fountain sign → acute idiopathic scrotal oedema. p.29
- MCQ Onion-ring avascular testicular lesion → epidermoid cyst. p.29
- MCQ CeVUS agent → sulfur hexafluoride microbubbles; AUC 0.97. p.29
- MCQ Expected bladder capacity = (age + 2) × 30 mL. p.29
- MCQ Blood at meatus → retrograde urethrogram before catheterisation. p.29
- MCQ "Pie-in-the-sky" bladder → pelvic fracture, extraperitoneal injury. p.29
- MCQ VCUG during acute febrile UTI may overgrade VUR. p.29
- MCQ DMSA lesion persisting at 6 months = scar. p.29
- MCQ Diuretic renogram furosemide dose → 1 mg/kg IV at maximal filling. p.29
- MCQ T½ is timed from maximal filling, not injection. p.29
- MCQ MRU furosemide → 1 mg/kg, maximum 20 mg. p.29
- MCQ Arterial-phase CT alone misses collecting-system injury. p.29
- MCQ Genitogram classification in DSD → Shopfner. p.29

## Ch 37 · Infection and Inflammation of the Paediatric Genitourinary Tract

### 1 · OVERVIEW AND THE PARADIGM SHIFT

- Urinary tract infection (UTI) affects ~**1% of boys, 3–5% of girls** in childhood; 30–50% recur. Commonest bacterial infection in febrile children (~7% of febrile infants). p.30

### 2 · PATHOGENESIS AND VIRULENCE

- KEY POINT Why does the same organism recur? IBCs and biofilms shield bacteria from neutrophils and antibiotics; a biofilm survives 1000–1500× normal killing concentrations because of poor penetration, slow growth and altered targets. p.30

### 3 · HOST RISK FACTORS

- **Circumcision**: reduces UTI in the first 6 months **~10-fold** (preputial colonisation peaks in infancy). Classic US Army cohort: circumcised 0.11%, uncircumcised 1.12%, girls 0.57% MCQ. Number needed to circumcise: **111** normal boys, 11 recurrent UTI, **4** high-grade VUR. Protective in antenatal hydronephrosis. p.31
- **VUR**: 1–2% of newborns, **25–40%** after first UTI; most pyelonephritis occurs without VUR, but grade III–IV doubles pyelonephritic change → Ch 41. p.31
- KEY POINT Every host factor either raises inoculum and dwell time (prepuce, faecal flora, stasis, constipation, catheter) or lowers defence (immature immunity, receptor-bearing blood groups); prevention = reverse whichever applies. p.31

### 4 · CLASSIFICATION

- **Febrile UTI (presumed pyelonephritis)**: high fever, vomiting, flank pain, lethargy. **Cystitis**: afebrile, dysuria, frequency, urgency, suprapubic pain. Clinical localisation is poor: only **50–66%** of febrile children show acute changes on dimercaptosuccinic acid (DMSA) scan (gold standard); fever does not exclude cystitis. p.31
- Kass: two consecutive cultures **≥$10^5$ colony-forming units (CFU)/mL**, same organism, no symptoms. Low-virulence strains. p.31

### 5 · DIAGNOSIS

- Pyuria threshold **≥10 WBC/HPF** more reliable under 2 years. p.32
- TABLE AAP 2011 (2–24 months) · Catheter/SPA · **≥50,000 + pyuria** p.32

### 6 · IMAGING AFTER UTI

- DMSA (± single-photon emission computed tomography): proximal tubular uptake, ~1 millisievert (mSv). APN sensitivity maximal within **1 week**; scar assessment at **≥6 months**; ~15% of acute lesions scar. Scar is indistinguishable from congenital dysplasia. p.33
- **NICE "atypical" UTI**: seriously ill, poor urine flow, abdominal/bladder mass, raised creatinine, septicaemia, no response within **48 h**, non-*E. coli* organism. NICE late DMSA at 4–6 months. p.33

### 7 · TREATMENT

- TABLE Febrile UTI · **7–14 days** (shorter courses inferior) p.34
- TABLE Afebrile cystitis · **2–4 days** (better than single-dose; equals 7–14 days) p.34

### 8 · RENAL SCARRING AND SEQUELAE

- Long term: childhood UTI accounts for **<1% of CKD**; end-stage risk after childhood UTI ~1 in 10,000. Bilateral scars lower GFR over decades; scarring raises risk of hypertension and pre-eclampsia. p.34

### 9 · PREVENTION AND PROPHYLAXIS

- Recurrence **10–30%**, highest in first 3–6 months; higher after index UTI >1 year and in girls (up to 40%). p.35

### 12 · MCQ ONE-LINERS

- MCQ Boys outnumber girls with UTI only in the first year. p.36
- MCQ Circumcision reduces infant UTI about tenfold. p.36
- MCQ Number needed to circumcise in high-grade VUR → 4. p.36
- MCQ P fimbriae bind Galα1–4Galβ → pyelonephritis. p.36
- MCQ Type 1 fimbriae are mannose-sensitive; FimH binds uroplakins. p.36
- MCQ Aerobactin is a siderophore. p.36
- MCQ Quiescent intracellular reservoir explains same-strain recurrence. p.36

- MCQ Bag urine is reliable only when negative. p.36
- MCQ Suprapubic aspiration contamination → <1%. p.36
- MCQ Most specific dipstick test → nitrite (~98%). p.36
- MCQ Gram-positive organisms give false-negative nitrite. p.36
- MCQ AAP catheter threshold → ≥50,000 CFU/mL plus pyuria. p.36
- MCQ Kass ASB → two cultures ≥$10^5$ CFU/mL, same organism. p.36
- MCQ ASB in schoolgirls without VUR → do not treat. p.36
- MCQ Gold standard for acute pyelonephritis → DMSA scan. p.36
- MCQ DMSA scar assessment → ≥6 months after infection. p.36
- MCQ Procalcitonin ≥0.5 ng/mL predicts APN and scarring. p.36
- MCQ NICE atypical UTI includes non-*E. coli* organism. p.36
- MCQ Top-down approach → DMSA first, VCUG if abnormal. p.36
- MCQ Febrile UTI course → 7–14 days; cystitis → 2–4 days. p.36
- MCQ Oral antibiotics equal IV in non-toxic children >2 months. p.36
- MCQ Neonatal empirical regimen → ampicillin plus gentamicin or cefotaxime. p.36
- MCQ Nitrofurantoin contraindicated in febrile UTI, <3 months, G6PD deficiency. p.36
- MCQ TMP-SMX avoided <6 weeks → kernicterus. p.36
- MCQ Prophylaxis dose → one-quarter therapeutic, at bedtime. p.36
- MCQ RIVUR → fewer recurrences, no scar reduction. p.36
- MCQ Scar risk doubles if antibiotics start after 72 h. p.36
- MCQ Childhood UTI causes <1% of CKD. p.36
- MCQ Renal abscess urine culture negative in up to 20%. p.36
- MCQ XGP organisms → Proteus or *E. coli*; diffuse → nephrectomy. p.36
- MCQ Adenovirus 11 → haemorrhagic cystitis. p.36
- MCQ BK virus cystitis → cidofovir. p.36

## Ch 38 · Core Principles of Perioperative Management in Children

### 2 · PREOPERATIVE ASSESSMENT

- **Investigations are indication-driven**: no routine bloods, chest radiograph or electrocardiogram (ECG) in healthy children; a murmur alone is not an ECG indication. Urine human chorionic gonadotrophin (hCG) in females of menstrual age — MCQ false positive after augmentation cystoplasty. p.37
- **URTI**: airway hyperreactive for **6–8 weeks** → laryngospasm, bronchospasm, desaturation; laryngeal mask safer than endotracheal tube. Postpone elective surgery ≥ 4, ideally 6 weeks. p.37
- **Sepsis**: antibiotics within **1 h** of septic shock, fluids ± vasopressors; avoid neuraxial blocks (hypotension, coagulopathy). **Trauma**: always a full stomach → rapid-sequence induction. p.37

### 3 · FASTING (NIL PER OS, NPO)

- TABLE **2 h** · Clear liquids, carbohydrate drinks (European and Canadian 2022: 1 h) p.37
- TABLE **4 h** · Breast milk p.37
- TABLE **6 h** · Infant formula, non-human milk, light meal, tube feeds p.37
- TABLE **8 h** · Regular meal p.37

### 4 · ANAESTHESIA CONSIDERATIONS

- **Postoperative apnoea** peaks in the first 12–23 h and may be fatal → admit for 23-h cardiorespiratory monitoring if postconceptional age is **< 50–60 weeks** (ex-preterm) or **< 44 weeks** (term). Hence elective surgery is deferred beyond these windows. p.38
- US Food and Drug Administration (FDA) warning (2016): repeated or lengthy (**> 3 h**) anaesthesia or sedation at **< 3 years** or in third-trimester pregnancy may affect brain development — based on animal neuroapoptosis. p.38
- **Autonomic dysreflexia** (lesion ≥ **T6**): bladder distension or noxious stimulus → severe hypertension, bradycardia, stroke. † Stop stimulus, drain bladder, sit up, rapid antihypertensive. p.38

### 5 · FLUID AND ELECTROLYTE MANAGEMENT

- **Holliday–Segar 4-2-1 rule**: **4 mL/kg/h** for first 10 kg + **2 mL/kg/h** for next 10 kg + **1 mL/kg/h** per kg above 20. E.g. 22 kg → 40 + 20 + 2 = 62 mL/h. p.38
- KEY POINT **Why isotonic?** Surgical stress, pain, nausea and hypoxia release antidiuretic hormone (ADH) independent of osmolality → aquaporin-2 water retention. Hypotonic fluid then supplies free water → hyponatraemia (< 135 mmol/L). † The child's large brain-to-skull ratio leaves little room for cerebral oedema, so hyponatraemic encephalopathy can be fatal. p.39

### 6 · BLOOD LOSS AND TRANSFUSION †

- Transfuse red cells at haemoglobin **< 7 g/dL** in stable children (higher in neonates, cyanotic heart disease, active bleeding); 10–15 mL/kg raises haemoglobin ≈ 2–3 g/dL. p.39

### 8 · ANTIBIOTIC PROPHYLAXIS AND INTRAOPERATIVE CRISES

- **Timing**: within **60 min** of incision; vancomycin and fluoroquinolones within **120 min** (long infusion). p.39
- Doses: cefazolin **30 mg/kg** (redose 4 h); ampicillin 50 mg/kg (2 h); gentamicin 2.5 mg/kg, vancomycin 15 mg/kg, metronidazole 15 mg/kg, ciprofloxacin 10 mg/kg (not redosed — long half-life). p.39

### 12 · MCQ ONE-LINERS

- MCQ Fasting: clear 2 h, breast milk 4 h, formula 6 h, regular meal 8 h. p.40
- MCQ Non-human milk is fasted as a light meal. p.40
- MCQ FDA warning: anaesthesia > 3 h or repeated, age < 3 years. p.40
- MCQ GAS trial: no neurodevelopmental difference at 5 years. p.40
- MCQ Apnoea monitoring: ex-preterm < 50–60 weeks, term < 44 weeks postconceptional age. p.40
- MCQ 4-2-1 rule: 22 kg → 62 mL/h. p.40
- MCQ Maintenance fluid: isotonic with dextrose, not 0.45% saline. p.40
- MCQ Postoperative hyponatraemia: non-osmotic ADH release. p.40
- MCQ Avoid lactated Ringer and propofol in mitochondrial disorders. p.40
- MCQ Post-URTI airway hyperreactivity lasts 6–8 weeks. p.40
- MCQ Prophylaxis within 60 min; vancomycin within 120 min. p.40
- MCQ Cefazolin 30 mg/kg, redosed 4-hourly. p.40
- MCQ False-positive urine hCG after augmentation cystoplasty. p.40
- MCQ Earliest MH sign: unexplained hypercarbia; treat with dantrolene. p.40
- MCQ Nitrous oxide does not trigger MH. p.40
- MCQ Autonomic dysreflexia: lesions at or above T6. p.40
- MCQ Spina bifida: latex-free care. p.40

- MCQ Phaeochromocytoma: α-blockade before β-blockade. p.40
- MCQ Sickle cell: avoiding dehydration matters most; acute chest syndrome kills. p.40
- MCQ Codeine contraindicated — ultrarapid metabolisers. p.40
- MCQ Ketorolac limited to 72–96 h; NSAIDs not under 6 months. p.40
- MCQ Methadone requires ECG for QT. p.40
- MCQ Emergence delirium: child does not recognise parents. p.40
- MCQ Metformin stopped 24 h preoperatively. p.40
- MCQ Avoid surgery within 4–6 h after haemodialysis. p.40

## Ch 39 · Principles of Laparoscopic and Robotic Surgery in Children

### 1 · EVOLUTION AND INDICATIONS

- Sequence: diagnostic laparoscopy for non-palpable testis (**1976**, first paediatric urological laparoscopy) → nephrectomy (1994) → pyeloplasty, first reconstructive minimally invasive surgery (MIS) (1995) → first paediatric robotic case (2002). p.41

### 2 · PHYSIOLOGY OF PNEUMOPERITONEUM

- TABLE Maximum IAP · **≤ 6** · **≤ 8** · **≤ 10** · **≤ 12 mm Hg** p.41

### 6 · ROBOTICS IN CHILDREN

- **Working space** is the infant problem: ≈ **1 L at 1 year vs 6 L in adults** → arm collisions, reduced range. Aids: **port spacing** as wide as the small abdomen allows, linear placement, "burping" (lifting ports against the abdominal wall); minimise inhalational anaesthetic and use preoperative enema and rectal tube to reduce bowel distension. p.43
- **Limitations**: no 5-mm robotic ports (8-mm minimum); da Vinci single-port needs a 25-mm cannula and ≈ **10 cm** from cannula to target — prohibitive in small children; no haptic feedback; cost; harder to teach; often sited in adult theatres away from paediatric equipment. p.43

### 7 · COMPLICATIONS AND PREVENTION

- KEY POINT **Ergonomics**: laparoscopy strains hands, thumb and shoulders (fulcrum effect, wrist deviation) → table at 70–80% elbow height, monitor at eye level. The robot offloads the upper limb but fixes the neck; master the clutch to keep elbows on the armrest. p.43

### 8 · MCQ ONE-LINERS

- MCQ First paediatric urological laparoscopy: non-palpable testis, 1976. p.43
- MCQ First paediatric reconstructive MIS: laparoscopic pyeloplasty, 1995. p.43
- MCQ Maximum IAP: neonate 6, infant 8, 1–2 years 10, 2–8 years 12 mm Hg. p.43
- MCQ Only strict contraindication: inability to tolerate pneumoperitoneum. p.43
- MCQ Oliguria during pneumoperitoneum: do not give fluid boluses. p.43
- MCQ Insufflation may provoke vagal bradycardia — desufflate. p.43
- MCQ Commonest major vessel injured at Veress entry: right common iliac. p.43
- MCQ Commonest minor vascular injury: inferior epigastric. p.43
- MCQ Palmer's point: left midclavicular line, 3 fingerbreadths below costal margin. p.43
- MCQ No outcome difference between Hasson and Veress access. p.43
- MCQ Smallest multiport robotic port: 8 mm. p.44
- MCQ Single-port robot needs ≈ 10 cm working distance. p.44
- MCQ Infant working space ≈ 1 L vs 6 L adult. p.44

- MCQ Laparoscopic instruments 4 DOF; robotic 7 DOF. p.44
- MCQ Retroperitoneal approach preferred in peritoneal dialysis. p.44
- MCQ Vesicoscopic approach: cross-trigonal reimplantation. p.44
- MCQ Port-site hernia risk highest near 1 year — close fascia. p.44
- MCQ HIdES places ports below Pfannenstiel line. p.44
- MCQ da Vinci lacks haptic feedback. p.44
- MCQ $CO_2$ embolism: falling end-tidal $CO_2$, mill-wheel murmur. p.44

## Ch 40 · Clinical and Urodynamic Evaluation of Lower Urinary Tract Dysfunction in Children

### 1 · PRINCIPLES AND TERMINOLOGY

- **Expected bladder capacity (EBC)**: infant ≤1 y = **weight (kg) × 7 mL**; older child = **(age in years + 2) × 30 mL**. Small capacity <65% EBC; large >150% (some use >130%). Functional capacity (largest diary void, or voided volume + post-void residual [PVR]) is normal at ≈70–115% EBC. p.45

### 2 · HISTORY

- **Storage symptoms**: urgency, frequency, incontinence (continuous vs intermittent, day vs night, how often). **Secondary enuresis** = new wetting after **≥6 months** dry → look for stressors. Caffeine is a bladder irritant. p.45

- ALGORITHM
  Bowel and bladder dysfunction (BBD): Rome IV → 7-day diary + urinalysis + pelvic ultrasound (US) → **treat bowel first** → LUTS persist → uroflow + electromyography (EMG) → urotherapy → anticholinergic → β3-agonist → botulinum toxin / biofeedback / neuromodulation.
  p.45

### 5 · NON-INVASIVE URODYNAMICS

- Key variables: **voided volume, maximum flow rate (Qmax), pattern** (also average flow, Qavg). Qmax counts only if the peak lasts **≥2 s** (spikes are artefact). p.46
- **Valid only if**: toilet-trained child; voided volume **≥50% EBC** (pattern needs ≈50–100 mL); >1 curve; read as emptying data only. p.46
- KEY POINT The curve reflects detrusor vs outlet. Too much detrusor → tower; too much outlet → plateau; outlet activity that comes and goes → staccato; detrusor effort that comes and goes → interrupted. Patch EMG tells which side is at fault. p.46
- **Lag time** (pelvic-floor relaxation → onset of flow): **<2 s** = DO; **2–6 s** normal; **>6 s** = primary bladder-neck dysfunction MCQ. p.47
- Bladder wall thickness abnormal **>3 mm full / >5 mm empty** (increased detrusor workload; DO, pelvic-floor overactivity). p.47
- Rectal diameter **>30 mm** without urge = faecal loading. p.47
- **PVR** (repeat — very variable): abnormal at 4–6 y if **>30 mL or >21%** of capacity; 7–12 y **>20 mL or >15%**; repeated >20 mL is abnormal. Valid only if measured **<5 min** after void and bladder 50–115% EBC — a late or over-distended PVR is invalid, not abnormal. p.47

### 6 · INVASIVE (FORMAL) URODYNAMICS

- Routine pre-study culture/prophylaxis not needed — post-study UTI ≈ **1.4%**; target high-risk children (CIC, recent UTI, immunosuppression, symptoms). p.47
- **Stop filling** at strong urge, discomfort, voiding, leak > fill rate, detrusor pressure (Pdet) persistently **>40 $cmH_2O$**, or volume **>150% EBC**. p.47

- **Compliance = ΔV/ΔPdet** (mL/cm$H_2O$), calculate in every study: **>20 normal; 10–20 moderately; <10 severely impaired**. In small contracted bladders absolute Pdet (<30–40) is the better safety guide. An **early** pressure rise means the child spends most of the day at high pressure. p.47
- TABLE Threshold · **>40 cm$H_2O$** (or no leak at high pressure) = danger · No paediatric norms (adult <60 intrinsic sphincter deficiency) p.48
- KEY POINT **McGuire (1981)**, myelodysplasia: leak pressure >40 cm$H_2O$ → hydronephrosis or secondary VUR in **85%**. Above 40 cm$H_2O$, bladder pressure is transmitted up the ureters to the kidneys. Aim for the lowest achievable pressure. p.48
- Normal Pdet at Qmax ≈ boys 66, girls 57 cm$H_2O$; infant boys up to 118 cm$H_2O$ (physiological) MCQ. p.48
- KEY POINT **Findings threatening the upper tracts**: compliance <20 (esp. <10); DLPP >40 cm$H_2O$; sustained high filling Pdet; DSD with poor emptying; high voiding pressure with poor flow. Interobserver agreement is poor. p.48

### 7 · EVALUATION IN SPECIFIC CONDITIONS

- TABLE Spinal dysraphism · UMPIRE risk by DLPP/end-fill Pdet: **<25 low; 25–39 or DO intermediate; ≥40 high** · Baseline, serial, pre-surgery p.48
- KEY POINT UMPIRE dropped DSD from its high-risk group because reviewers could not agree who had it — poor agreement, not proof that DSD is benign. Goals in every neurogenic bladder: normal capacity, low storage pressure, complete emptying, upper-tract protection. p.48

### 9 · MCQ ONE-LINERS

- MCQ EBC in children >1 year: (age + 2) × 30 mL. p.49
- MCQ EBC in infants: weight (kg) × 7 mL. p.49
- MCQ ICCS small capacity <65% EBC; large >150%. p.49
- MCQ Secondary enuresis: wetting after ≥6 months of dryness. p.49
- MCQ Rome IV functional constipation needs ≥2 criteria for ≥1 month, age ≥4. p.49
- MCQ Bell-shaped flow = normal. p.49
- MCQ Tower flow = overactive bladder. p.49
- MCQ Staccato flow never reaches zero; interrupted flow does. p.49
- MCQ Qmax is valid only if sustained ≥2 seconds. p.49
- MCQ Flow interpretable only if voided volume ≥50% EBC. p.49
- MCQ EMG lag time >6 s suggests primary bladder-neck dysfunction. p.49
- MCQ Abnormal PVR at 7–12 y: >20 mL or >15% capacity. p.49
- MCQ PVR must be measured within 5 minutes of voiding. p.49
- MCQ Filling rate in paediatric cystometry: 5–10% EBC/minute. p.49
- MCQ Post-urodynamic UTI rate ≈1.4%; routine prophylaxis unnecessary. p.49
- MCQ Compliance <10 mL/cm$H_2O$ = severely impaired. p.49
- MCQ DLPP >40 cm$H_2O$ → 85% upper-tract deterioration (McGuire). p.49
- MCQ Anatomical obstruction: high Pdet, low flow, quiet EMG. p.49
- MCQ Normal Pdet at Qmax: boys ≈66, girls ≈57 cm$H_2O$. p.49
- MCQ UMPIRE high risk: DLPP/end-fill pressure ≥40 cm$H_2O$. p.49
- MCQ Continuous incontinence in a girl: search for ectopic ureter. p.49
- MCQ SCI: baseline video-urodynamics 3–6 months after injury. p.49

## Ch 42 · Bladder Anomalies in Children

### 2 · FETAL BLADDER ON ULTRASOUND (US)

- Empties every **15–20 min** → if not seen, **repeat scan in the same sitting**. Amniotic fluid is mainly fetal urine only after **16 weeks**. p.50

### 3 · MEGACYSTIS

- TABLE 10–14 weeks · Sagittal length ≤6 mm; <10% of crown–rump length · **≥8 mm**; ≥13% of crown–rump length p.50
- Euploid 7–15 mm → most resolve; **>15 mm** → none (all obstructive). p.51
- KEY POINT Female fetus with megacystis cannot have PUV → think MMIHS, cloacal anomaly, urethral atresia. p.51

### 4 · BLADDER DUPLICATION

- ALGORITHM
  relieve obstruction, prevent infection → later: both sphincters competent → join distal urethras; one incompetent → close that bladder neck, excise its urethra; vaginas joined + vulvoplasty
  p.51

### 6 · URACHAL ANOMALIES

- ALGORITHM
  infected → drain + antibiotics first → balloon catheter fills bladder → small transverse infraumbilical incision → feeding tube in tract → circumscribe at umbilicus (preserve umbilicus) → extraperitoneal dissection to dome → excise with **bladder cuff** → two-layer closure → catheter overnight
  p.52

### 7 · ACQUIRED LESIONS

- TABLE Nephrogenic adenoma · Female 5:1; after infection, stones, surgery, augmentation; distal tubular origin · Transurethral resection; **80% recur** → surveillance cystoscopy p.52

### 8 · MCQ ONE-LINERS

- MCQ Urachus becomes the median umbilical ligament. p.52
- MCQ Persistent megacystis with oligohydramnios: LUTO. p.52
- MCQ Fetal bladder empties every 15–20 min — repeat scan before calling it absent. p.52
- MCQ Normal amniotic fluid + absent bladder filling → exstrophy. p.52
- MCQ Congenital megacystis: massive VUR is cause, not consequence, of dilation. p.52
- MCQ First-trimester megacystis ≥8 mm; >15 mm never resolves. p.52
- MCQ Commonest cause of fetal megacystis: PUV (~50%). p.52
- MCQ ~1/5 megacystis fetuses aneuploid (trisomy 13/18). p.52
- MCQ MMIHS gene: ACTG2; female predominance; normal amniotic fluid. p.52
- MCQ Leading cause of death in MMIHS: TPN complications. p.52
- MCQ Commonest bladder duplication: complete sagittal. p.52
- MCQ Hutch diverticulum: paraureteral. p.52
- MCQ Multiple giant diverticula: Ehlers–Danlos, Williams, Menkes. p.52
- MCQ Secondary diverticulum: treat the outlet first. p.53
- MCQ Commonest organism in infected urachal cyst: *S. aureus*. p.53
- MCQ Urachal excision must include a bladder cuff. p.53
- MCQ Urachal adenocarcinoma: not reported in children. p.53

- MCQ Nephrogenic adenoma recurs in 80% of children. p.53
- MCQ Eosinophilic cystitis: biopsy mandatory. p.53
- MCQ Cyclophosphamide cystitis: mesna prevents. p.53

## Ch 43 · Exstrophy–Epispadias Complex

### 1 · SPECTRUM AND EPIDEMIOLOGY

- CBE ≈ **1 in 46,000**; male:female ≈ 2.3:1. CBE is >50% of the complex. p.54
- Recurrence in a family ~1 in 100; offspring of an affected parent **1 in 70** (~500-fold). Inconsistent twin concordance → multifactorial. p.54

### 3 · ANATOMY OF CLASSIC BLADDER EXSTROPHY

- Not simply a wide symphysis: three-dimensional computed tomography (3D CT) shows **external rotation of the posterior pelvis (~12°) and anterior segment (~18°)**, acetabular retroversion, **anterior pubic segment 30% short**; posterior segment normal; mean diastasis **4.8 cm**. p.54

### 4 · VARIANTS

- KEY POINT Superior vesical fissure inflated the belief that variants do better. All variants except superior vesical fissure need formal closure and follow-up like CBE. p.55

### 6 · NEWBORN CARE AND SELECTION FOR CLOSURE

- ALGORITHM
  tie cord with 2-0 silk (no clamp — traumatises mucosa) → cover with non-adherent plastic wrap → irrigate with saline and replace at each nappy change → cardiopulmonary assessment, renal ultrasound → transfer to exstrophy centre; counselling with multidisciplinary team
  p.56
- **Assess**: template size and quality (≥**3 × 3 cm** suitable), pubic diastasis and pelvic malleability under anaesthesia, urethral plate, penile length, associated anomalies. p.56

### 7 · SURGICAL PHILOSOPHIES

- KEY POINT Erlangen, Kelly and CPRE claim to avoid BNR, yet almost all patients still need an outlet procedure. Compressing reconstruction adds ischaemic and outlet risk without removing that need. p.56

### 8 · PRIMARY CLOSURE (MSRE STAGE 1)

- ALGORITHM
  outline 2-cm mucosal strip from distal trigone past verumontanum to mid-shaft (male) / vaginal orifice (female) → incise round bladder–skin junction → separate rectus fascia from bladder; ligate umbilical vessels; strip peritoneum from dome → **divide urogenital diaphragm fibres sharply down to levator hiatus** → free corpora from inferior rami → close bladder, neck and posterior urethra onto the shaft; **taper over 12-Fr sound** → suprapubic Malecot (4 weeks) + ureteric stents; **urethra unstented** → pressure on greater trochanters → **No. 2 nylon horizontal mattress** pubic suture, knot away from neourethra → neoumbilicus (tubes exit)
  p.57

### 9 · OSTEOTOMY AND IMMOBILISATION

- Osteotomy if: closure **>72 h**, diastasis **>4 cm**, stiff pelvis, large template, **failed closure**, **cloacal exstrophy**. Diastasis >6 cm → staged pelvic closure with fixator. p.57
- TABLE Spica cast · 47% · 30% · **66%** p.58
- TABLE External fixation + traction · **7.5%** · 8% · — (few) p.58

### 10 · AFTER PRIMARY CLOSURE

- Capacity **<30 mL at 1–2 years** → warn that a continence procedure may not be feasible. p.58

### 13 · BLADDER NECK RECONSTRUCTION (STAGE 3)

- Capacity **≥100 mL** (under anaesthesia), stable urodynamics, motivated child (usually 5–7 years), voiding programme started ≥6 months earlier, psychologist assessment. p.59

- **ALGORITHM**
  transverse bladder-neck incision with vertical extension → **cephalotrigonal / Cohen cross-trigonal reimplant** (moves orifices off the trigone) → posterior mucosal strip **15 × 30 mm** from mid-trigone to posterior urethra → denude lateral muscle triangles → small edge incisions, no cephalad transverse cut (avoids denervation, ischaemia) → tube over strip → **double-breast muscle flaps** → 8-Fr guide removed → suspend to rectus fascia
  p.59

- MSRE: ~**70–75%** dry by day (>3 h) in both sexes; CPRE followed by YDL lowest (~56%). p.60

### 17 · EPISPADIAS

- **ALGORITHM**
  vertical incision from mons base through urethral wall at 9 and 3 o'clock → excise urethral roof towards bladder neck (dorsal wedge taper) → inverting closure over 10-Fr Foley → denude medial hemiclitoris and labia minora → mons fat over suture line → unite clitoral halves and labia minora → clitoral hood → catheter 5–7 days → BNR at 4–5 years
  p.62

- Old procedures (plication, Marshall–Marchetti suspension, muscle transplants) failed — raised resistance without correcting anatomy. Urethral closure increases capacity. Continence after BNR ~**87%**; epispadias girls do as well as CBE girls. p.62

### 18 · CLOACAL EXSTROPHY

- **TABLE** Spinal defects · Uncommon · **85–100%** (lipomeningocele commonest) p.62

- **ALGORITHM**
  Stage 1 (newborn): excise/close omphalocele → separate caecal plate from hemibladders → join hemibladders (= bladder exstrophy) → colostomy → Stage 2: bladder closure + **combined osteotomy** + fixation 6–8 weeks → 5–7 years: YDL only if >100 mL (rare), usually augmentation + continent stoma
  p.63

### 19 · MCQ ONE-LINERS

- **MCQ** Core embryological defect: failed mesodermal reinforcement of the cloacal membrane. p.63
- **MCQ** Offspring risk of an affected parent: 1 in 70. p.63
- **MCQ** Commonest genetic variant in BEEC: 22q11.2 duplication. p.63
- **MCQ** Anterior pubic segment 30% short; posterior segment normal. p.63
- **MCQ** Anterior corpora 50% shorter — penile lengthening is apparent. p.63
- **MCQ** Intrasymphyseal band contains no striated sphincter. p.63
- **MCQ** VUR in 100% of closed exstrophy bladders. p.63
- **MCQ** Rectal prolapse after closure → suspect outlet obstruction. p.63
- **MCQ** Cervix lies in anterior vaginal wall in female exstrophy. p.63
- **MCQ** Uterine prolapse is predicted by diastasis width, not prevented by osteotomy. p.63
- **MCQ** Newborn exstrophy bladder: immature, not neuropathic. p.63
- **MCQ** Commonest exstrophy variant: skin-covered (split symphysis). p.63
- **MCQ** Superior vesical fissure: sphincter intact, no BNR needed. p.63
- **MCQ** Template ≥3 × 3 cm suitable for newborn closure. p.63
- **MCQ** Cord tied with silk, not clamped. p.63

- MCQ Osteotomy if closure >72 hours or diastasis >4 cm. p.63
- MCQ Preferred osteotomy: combined anterior innominate + vertical iliac. p.63
- MCQ Spica without osteotomy fails in about two-thirds. p.63
- MCQ Commonest closure error: inadequate urogenital diaphragm division. p.63
- MCQ Urethra not stented after closure; ureters stented. p.63
- MCQ Recurrent UTI after closure: look for intrapubic stitch erosion. p.63
- MCQ Testosterone before urethroplasty: 2 mg/kg at 5 and 2 weeks. p.63
- MCQ Epispadias repair increases bladder capacity. p.63
- MCQ Complete disassembly leaves most boys hypospadiac. p.63
- MCQ BNR capacity threshold: 100 mL. p.63
- MCQ Most important determinant of continence: capacity at BNR. p.64
- MCQ YDL mucosal strip: 15 × 30 mm. p.64
- MCQ BNR failure: no 3-hour dry interval by 2 years. p.64
- MCQ Ureterosigmoidostomy cancer: adenocarcinoma ~30 years later; yearly colonoscopy. p.64
- MCQ Exstrophy pregnancy: planned caesarean with urologist present. p.64
- MCQ Male epispadias: ~70% penopubic. p.64
- MCQ Female epispadias classification: Davis. p.64
- MCQ OEIS = omphalocele, exstrophy, imperforate anus, spinal defects. p.64
- MCQ Cloacal exstrophy "elephant trunk" sign: prolapsed terminal ileum. p.64
- MCQ Cloacal exstrophy gender: assign by karyotype. p.64

## Ch 44 · Prune-Belly Syndrome

### 1 · DEFINITION AND TRIAD

- Incidence ~**1 in 29,000–44,000** live births (≈ bladder exstrophy); **95% male**; more in twins, African ancestry, young mothers. Name "prune-belly" given by **Osler** MCQ. p.65

### 2 · GENETICS AND AETIOLOGY

- No inheritance pattern established. Multiplex families: variants in **smooth muscle genes** — **MYOCD (myocardin)** and **FLNA (filamin A)** MCQ. p.65
- KEY POINT Weakness of the obstruction theory: most patients have **no demonstrable obstruction postnatally**. Whatever the cause, the tract is **dilated, collagen-rich and largely non-obstructed** — that observation drives management. p.65

### 3 · GENITOURINARY PATHOLOGY

- Dysplasia in ~**50%**, often asymmetric. Worst with outlet obstruction **without urachal decompression**. US: echogenic parenchyma with cortical cysts. p.65
- Vesicoureteric reflux (VUR) ~**75%**; obstruction (pelviureteric junction (PUJ) / vesicoureteric junction (VUJ)) uncommon. p.65
- Massive, **thick-walled but smooth** (no trabeculation — unlike an obstructed bladder), collagen excess; **urachal pseudodiverticulum**; urachus patent at birth in **25–30%**. p.65

### 4 · ASSOCIATED (EXTRAGENITOURINARY) ANOMALIES

- KEY POINT Two mechanisms explain most non-urinary morbidity: **oligohydramnios** (lung hypoplasia, limb deformity, knee dimples) and **absent abdominal wall pressure** (weak cough, atelectasis, constipation). p.66

### 5 · PRESENTATION AND SPECTRUM

- **RUBACE score** (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism + Extragenitourinary): severity score for survivors (categories II–III) to compare outcomes; renal is the heaviest domain MCQ. p.67

### 6 · EVALUATION

- ALGORITHM

  Team (neonatology, nephrology, urology, cardiology, orthopaedics) → **cardiorespiratory first**: chest X-ray (pneumothorax, hypoplasia) → creatinine/blood urea nitrogen (BUN)/electrolytes (trend over days–weeks; day-1 value is maternal) → renal–bladder ultrasound (US) → antibiotic prophylaxis + circumcision → selective imaging

  p.67

### 7 · MANAGEMENT PHILOSOPHY

- Delay major reconstruction to **≥3–6 months** (pulmonary maturity). Comprehensive single-stage surgery (urinary reconstruction + orchidopexy + abdominoplasty + circumcision) does not add morbidity. p.67

### 10 · ABDOMINOPLASTY

- ALGORITHM

  **Monfort sequence**: excise skin ellipse, umbilicus spared → incise central plate at lateral rectus borders → genitourinary surgery → score peritoneum over lateral muscle → suture plate edges to scored line → overlap lateral flaps over drains

  p.69

### 11 · LONG-TERM OUTCOME AND FOLLOW-UP

- **Renal predictor**: nadir/baseline creatinine **<0.7 mg/dL at 1 year** → stable function through childhood unless pyelonephritis or obstruction supervenes. p.69
- Up to **30%** develop chronic renal failure in childhood/adolescence (mainly those with poor initial function). Drivers: **recurrent pyelonephritis, dysplasia, polyuria**. p.69
- KEY POINT Long-term care is a contest against **stasis**: whatever lowers residual urine and infection protects the kidneys; dysplasia and polyuria set the ceiling surgery cannot raise. p.69

### 12 · MCQ ONE-LINERS

- MCQ PBS triad: abdominal wall deficiency, bilateral intra-abdominal testes, urinary tract anomaly. p.69
- MCQ Main determinant of survival: renal dysplasia (plus pulmonary hypoplasia), not abdominal wall. p.69
- MCQ Genes implicated in multiplex families: MYOCD and FLNA (smooth muscle). p.69
- MCQ Degree of hydronephrosis does not correlate with dysplasia. p.69
- MCQ Ureter: distal worse than proximal; preserve proximal in reconstruction. p.69
- MCQ VUR present in ~75%. p.69
- MCQ Bladder: large, smooth, normal compliance, urachal pseudodiverticulum. p.69
- MCQ Megalourethra more common in PBS than any other syndrome. p.69
- MCQ Fusiform megalourethra lacks corpora cavernosa and spongiosum — more severe than scaphoid. p.69
- MCQ Urethral atresia is lethal unless a patent urachus decompresses the bladder. p.69
- MCQ Commonest orthopaedic anomaly: talipes equinovarus. p.69
- MCQ Lateral knee dimples: sign of oligohydramnios. p.69
- MCQ Woodard category I: pulmonary hypoplasia + dysplasia — supportive care. p.69

- MCQ Controversy over management centres on Woodard category II. p.69
- MCQ Pseudo-prune: delayed diagnosis, higher late renal failure. p.69
- MCQ Neonatal VCUG only with renal insufficiency or suspected outlet obstruction, under prophylaxis. p.69
- MCQ DMSA for differential function after 4–6 weeks. p.69
- MCQ Temporary diversion of choice: Blocksom vesicostomy with a wide stoma. p.69
- MCQ Reduction cystoplasty: capacity recurs; CIC empties better. p.69
- MCQ Orchidopexy of choice: bilateral transabdominal at ~6 months. p.69
- MCQ Staged Fowler–Stephens more successful than single-stage. p.69
- MCQ Monfort abdominoplasty preserves umbilicus in situ and allows transperitoneal surgery. p.69
- MCQ Randolph: transverse incision excising lower full-thickness wall. p.69
- MCQ Creatinine <0.7 mg/dL at 1 year predicts stable childhood renal function. p.69
- MCQ No spontaneous paternity reported; ICSI has achieved paternity. p.69

## Ch 45 · Posterior Urethral Valves

### 1 · DEFINITION AND CLASSIFICATION

- Term coined by **Hugh Hampton Young** (1919); first endoscopic resection by Randall MCQ. p.70
- Genetics: mostly sporadic; chromosome 11 duplications in siblings; **copy number variant burden** predicts higher nadir creatinine and progression to chronic kidney disease (CKD) stage 5 — links genotype to renal prognosis, not to presence of the valve. Whole-genome loci: **TBX5, PTK7** MCQ. p.70

### 2 · PATHOPHYSIOLOGY

- KEY POINT Every pop-off is a false reassurance: a normal creatinine in a growing infant hides an abnormal GFR. Follow VURD as closely as any PUV child. p.71

### 3 · DIAGNOSIS

- **US** first: bladder wall thick if **>3 mm distended / >5 mm empty**; perirenal fluid = urinoma. p.71

### 4 · FETAL INTERVENTION

- TABLE Sodium / chloride · **<100 / <90 mEq/L** p.72
- TABLE Osmolality · **<200 mOsm/L** p.72
- TABLE β2-microglobulin · **<6 mg/L** (most sensitive) p.72
- Both carry **40–50%** pregnancy loss — mandatory in counselling. Deliver at a centre with neonatal surgery. p.72
- KEY POINT Fetal intervention can rescue the lung, not the kidney: amniotic fluid deficiency is still in progress, but dysplasia is already established. p.72

### 5 · NEONATAL PRESENTATION AND INITIAL MANAGEMENT

- ALGORITHM
  Stabilise airway/ventilation → **5–7 Fr feeding tube** per urethra (coudé, stylet-curved tip, or rectal finger guiding it anteriorly if it curls) → confirm position on US → minimal balloon fill (spasm, ureteric orifice occlusion) → creatinine, electrolytes, fluid balance → US, then VCUG → antibiotic prophylaxis → ablation when stable
  p.72

### 7 · VUR IN PUV

- KEY POINT VUR in valves is a pressure gauge, not a lesion: fix the pressure, not the reading. p.74

## 8 · BLADDER DYSFUNCTION AND VALVE BLADDER SYNDROME

- **ALGORITHM**
  Compensated high-pressure voiding → fibrosis, weaker contraction → rising PVR + growing urine volume (nephrogenic polyuria) → upper tracts store urine and refill the bladder → bladder never empty → pressure back-transmitted → renal decline → more polyuria
  p.74

## 9 · PROGNOSIS FOR RENAL FUNCTION

- Lifetime end-stage renal disease (ESRD) **20–50%**. Risk factors: age at diagnosis, dysplasia ± VUR, **nadir creatinine in year 1**, recurrent UTI, bladder dysfunction. p.75
- **TABLE** <0.4 / 0.4–0.69 / 0.7–0.99 / >1.0 mg/dL · **0% / 2% / 27% / 100%** p.75
- **0.7 mg/dL is the watershed** — risk leaps rather than grades. 1-year value beats 1-month value; creatinine velocity over first 30 days also predicts. p.75

## 14 · MCQ ONE-LINERS

- **MCQ** Commonest cause of infant male bladder outlet obstruction: PUV. p.76
- **MCQ** Young type I (95%): leaflets from verumontanum fusing anteriorly. p.76
- **MCQ** Type II is non-obstructing and probably not a true entity. p.76
- **MCQ** Type III: annular membrane, persistent urogenital membrane. p.76
- **MCQ** Antenatal US sign of dilated posterior urethra: keyhole sign. p.76
- **MCQ** Commonest cause of early death in PUV: pulmonary hypoplasia. p.76
- **MCQ** Bladder wall thick: >3 mm full, >5 mm empty. p.76
- **MCQ** Definitive diagnosis: VCUG — withdraw catheter during voiding. p.76
- **MCQ** Renogram timing: MAG3 at 2–3 months. p.76
- **MCQ** Neonatal creatinine reflects mother for first 48 hours. p.76
- **MCQ** Favourable fetal urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6 mg/L. p.76
- **MCQ** Most sensitive fetal urine marker: β2-microglobulin. p.76
- **MCQ** Vesicoamniotic shunting: possible survival/lung benefit, no renal benefit. p.76
- **MCQ** Pregnancy loss with fetal intervention: 40–50%. p.76
- **MCQ** Neonatal hyponatraemia: transient pseudohypoaldosteronism. p.76
- **MCQ** Ablation incisions: 5 and 7 o'clock ± 12 o'clock. p.76
- **MCQ** Hot-loop resection risks urethral stricture. p.76
- **MCQ** Repeat VCUG within 1 month of ablation. p.76
- **MCQ** PADUA: progressive urethral dilation 5 → 8–10 Fr over 3–5 days. p.76
- **MCQ** Vesicostomy: bring the dome to skin to prevent prolapse. p.76
- **MCQ** Vesicostomy does not defunctionalise the bladder. p.76
- **MCQ** High diversion: no renal benefit; biopsies show dysplasia. p.76
- **MCQ** Circumcision reduces UTI risk by over 80% in PUV. p.76
- **MCQ** VURD does not protect long-term renal function. p.76
- **MCQ** VUR in PUV: treat bladder, rarely reimplant. p.76
- **MCQ** Valve bladder syndrome coined by Mitchell. p.76
- **MCQ** Overnight catheter drainage: 7–10 hours nightly. p.76
- **MCQ** Stop oxybutynin if residual and capacity rise. p.76

- MCQ Preferred augmentation in PUV: ureterocystoplasty. p.77
- MCQ Nadir creatinine <0.8 mg/dL at 1 year: minimal ESRD risk. p.77
- MCQ Lifetime ESRD in PUV: 20–50%. p.77
- MCQ Every transplant candidate: videourodynamics first. p.77
- MCQ Duplication: ventral urethra is functional. p.77
- MCQ Anterior urethral valves: often with urethral diverticulum. p.77
- MCQ Urethrorrhagia: conservative; up to 92% resolve. p.77
- MCQ Upper tract dilation persisting after ablation is expected, not residual obstruction. p.77

## Ch 46 · Neuromuscular Dysfunction of the Lower Urinary Tract in Children

### 1 · DEFINITION AND CAUSES

- Commonest cause: **neural tube defect (NTD)**, used almost synonymously with spina bifida (SB). With survival now ≈90% to 10 years, **renal failure** becomes a leading cause of late morbidity/death; **10%** of SB deaths are urological. p.78
- Pivotal advance: **clean intermittent catheterisation (CIC)**, Lapides 1971 MCQ. p.78

### 2 · MYELODYSPLASIA (NTD)

- Neural tube closes by **day 35** (from day 18). Prenatal detection: maternal α-fetoprotein + ultrasound (US). p.78
- **Folic acid** cuts NTDs 50–70%: **400 µg/day** for all women of child-bearing age (tube closes before pregnancy is recognised), from ≥4 weeks before conception through the first month; **4–5 mg/day** for women with SB. p.78
- Recurrence **5%** after one affected child; offspring risk 3.7% if one parent has SB (≈15% if both). p.78
- KEY POINT **Management of Myelomeningocele Study (MOMS)**: prenatal repair (<26 weeks) reduced death or shunt need (relative risk 0.70) and improved motor/cognitive scores at 30 months, at the cost of prematurity (34 vs 37 weeks), uterine-scar thinning/dehiscence (⅓) and more re-tethering. Rationale: "two-hit" in-utero injury. p.78
- KEY POINT Every pattern = detrusor (over/under) × outlet (competent/incompetent/dyssynergic). The dangerous combination is **overactive or poorly compliant detrusor against a closed or dyssynergic outlet**: high storage pressure → vesicoureteric reflux (VUR) and hydroureteronephrosis. An incompetent outlet "pops off" — wet but kidney-safe. p.78
- ALGORITHM
  Birth → renal–bladder US before discharge + post-void residual, PVR (decides need for CIC) → serum creatinine after first week → urodynamics once stable, **safe from ≈3 months** (within 3 months: ICCS, SBA, EAU) → cystography (ideally videourodynamics; or voiding cystourethrogram, VCUG) → ± baseline dimercaptosuccinic acid (DMSA) scan (EAU) → US at 6, 12, 18, 24 months then yearly.
  p.79
- UMPIRE (Centers for Disease Control and Prevention [CDC] Urologic and Renal Protocol for the Newborn and Young Child with SB) baseline: US normal 56%, mild hydronephrosis 40%, moderate–severe 4%; dilating VUR (grade III–V) **7%**; DMSA defects ≈8%. p.79
- TABLE **Hostile / high** · Detrusor leak point pressure (DLPP) or end-filling pressure **≥40 $cmH_2O$**; NDO with DSD (DSD removed in UMPIRE revision — poor agreement) p.79

### 3 · PROACTIVE (EARLY) MANAGEMENT

- **KEY POINT** Why proactive won: it treats the cause (pressure, failure to empty) before the consequence (reflux, dilatation, scarring, renal failure). Waiting for hydronephrosis means waiting for damage. p.80

### 4 · MANAGEMENT LADDER

- **ALGORITHM**
  CIC + antimuscarinic (± β3-agonist) → intradetrusor botulinum toxin → (vesicostomy as temporary bridge if CIC impossible) → augmentation cystoplasty ± continent catheterisable channel ± outlet procedure → Ch 48
  p.80
- Technique: general anaesthesia, rigid cystoscopy, **≈10 U/kg (max 360 U)**, ≈30 sites, **trigone spared**. Usual minimum age 2 y. p.80

### 8 · OTHER CAUSES

- **Autonomic dysreflexia** (lesion **≥T6**): bladder filling → unopposed sympathetic surge → severe hypertension, stroke, seizures. Monitor blood pressure and have α-blockers ready during urodynamics/cystography. p.81
- Upper motor neuron, coordinated: continence achieved but late; incontinence and urgency; NDO 30–61%, DSD ≈12%; one-third normal urodynamics. Upper-tract risk **<5%**; urodynamics only if conservative therapy fails or DSD symptoms/recurrent UTI. p.82
- **KEY POINT** Lesion level predicts pattern: **suprapontine** (CP) → coordinated overactivity, low renal risk; **suprasacral cord** (SCI, myelitis, tethering) → NDO + DSD, high risk; **sacral/peripheral** (cauda equina, pelvic surgery) → underactivity. Myelodysplasia is the exception — mixed and unpredictable. p.82

### 10 · MCQ ONE-LINERS

- **MCQ** CIC introduced by Lapides in 1971. p.82
- **MCQ** Commonest open NTD: myelomeningocele, usually lumbosacral. p.82
- **MCQ** Folic acid 400 µg/day; 4–5 mg/day if woman has SB. p.82
- **MCQ** Offspring risk with one affected parent: 3.7%. p.82
- **MCQ** Almost every open NTD has Chiari II malformation. p.82
- **MCQ** First urodynamics in myelomeningocele: within ≈3 months. p.82
- **MCQ** Dilating VUR in SB newborns: 7%. p.82
- **MCQ** DLPP ≥40 $cmH_2O$ = hostile bladder. p.82
- **MCQ** Cystatin C preferred to creatinine because of low muscle mass. p.82
- **MCQ** DSD diagnosed by EMG, not fluoroscopy. p.82
- **MCQ** Early intervention = CIC + antimuscarinic. p.82
- **MCQ** Expectant management: ESRD ≈30%; early intervention 0–1.6%. p.82
- **MCQ** No antibiotic prophylaxis for children on CIC. p.82
- **MCQ** Detrusor contraction mediated by M3 receptors. p.82
- **MCQ** Mirabegron approved from age 3; may cause hypertension. p.82
- **MCQ** Botulinum toxin dose ≈10 U/kg, max 360 U, trigone spared. p.82
- **MCQ** Vesicostomy is a temporary bridge. p.82
- **MCQ** Credé is harmful with a reactive external sphincter. p.82
- **MCQ** Reimplant grade ≥III VUR at augmentation. p.82
- **MCQ** Central precocious puberty affects ≈50% of girls with SB. p.82

- MCQ Normal male reproductive function usually if lesion at S1 or below. p.82
- MCQ Avoid intrauterine device and combined pill in women with SB. p.82
- MCQ Autonomic dysreflexia: lesions at or above T6. p.82
- MCQ Transverse myelitis: acute retention in >95%. p.82
- MCQ CP upper-tract deterioration <5%. p.82
- MCQ ARM bladder dysfunction stems from cord anomaly, not PSARP. p.83

## Ch 47 · Functional Disorders of the Bladder and Bowel in Children

### 1 · NORMAL BLADDER AND BOWEL FUNCTION

- **Infancy**: voiding pressures often **>100 cm $H_2O$** (boys > girls) and interrupted voiding are normal, disappearing by **~2 yr**; infants arouse before voiding. Dyscoordination beyond toddlerhood is therefore **dysfunction, not immaturity**. p.84
- KEY POINT Bladder and bowel are one functional unit: shared pelvic space ("zero-sum" volume), and afferents converging in the same dorsal root ganglia (cross-organ sensitisation). A loaded rectum provokes overactivity and poor emptying; treating constipation improves lower urinary tract symptoms (LUTS); anticholinergics given for the bladder can recreate the constipation that drove the symptoms. p.84

### 2 · DEFINITIONS — INTERNATIONAL CHILDREN'S CONTINENCE SOCIETY (ICCS)

- TABLE Storage · Increased frequency · **≥8 voids/day** p.84
- TABLE Decreased frequency · **≤3 voids/day** p.84
- **Daytime urinary incontinence (DUI)**: leakage while awake, age **≥5 yr**. **Enuresis**: intermittent leakage while asleep, ≥5 yr — **monosymptomatic (MNE)** = no other LUTS; **non-monosymptomatic (NMNE)** = with any LUTS. p.84
- KEY POINT **Trap:** dysfunctional voiding (neurologically intact) ≠ detrusor–sphincter dyssynergia (DSD; same discoordination with a neurological lesion). "Voiding dysfunction" is an old umbrella term — use "LUT dysfunction" (storage + voiding). Urgency incontinence does not by itself mean OAB: a voiding postponer also leaks with sudden urge, and treating him as OAB (anticholinergics) helps briefly then worsens emptying. p.85

### 4 · EVALUATION

- **Urinalysis** — false positives common. Specific gravity >1.005 makes diabetes insipidus unlikely. Persistent dipstick protein ≥1+ after urotherapy → first-morning protein:creatinine ratio (>**0.2 mg/mg** = abnormal). Pyuria/haematuria in girls may be vulvitis contamination. p.86
- ALGORITHM
  ≥2 of the following, ≥1/week for ≥1 month, IBS criteria not met: ≤2 defecations in toilet per week · ≥1 faecal incontinence episode per week · retentive posturing/volitional withholding · painful or hard stools · large rectal faecal mass · large-diameter stools that may block the toilet
  p.87
- No routine laboratory tests (thyroid, coeliac, calcium) or **plain abdominal radiograph** (poor reliability) — X-ray only if impaction suspected and examination impossible, or in refractory cases. Rectal diameter **≥30 mm** on transabdominal US suggests constipation (European Association of Urology/European Society for Paediatric Urology, EAU/ESPU), but gastroenterology guidelines advise against routine US. p.87

## 5 · MANAGEMENT OF DAYTIME INCONTINENCE — THE ALGORITHM

- ALGORITHM
  History + examination + urinalysis + PVR ± uroflow → exclude neurogenic (urodynamics, spinal MRI), diabetes, CKD, anatomical causes → screen and optimise neurodevelopmental disorder → **treat constipation first** → night-only symptoms? → treat as MNE → otherwise **standard urotherapy** (≥4 wk) → persisting → re-evaluate → **specific urotherapy** by diagnosis → refractory → third-line (botulinum toxin, neuromodulation)
  p.87

- TABLE **Oxybutynin** (non-selective; tertiary amine) · Immediate release (IR) **0.3–0.6 mg/kg/day** in 2–3 doses (max 15 mg); extended release (ER) 5–20 mg/day; patch 3.9 mg/day. Only United States-approved drug for paediatric non-neurogenic OAB (on observational data); liquid available. Dry mouth, flushing, constipation, cognitive effects. ER coating must not be crushed. p.88

- **Intradetrusor onabotulinumtoxinA** for refractory OAB: **5 units/kg, max 100 units** (higher doses no better, higher PVR); effect ~12 months; retention needing CIC is rare — counsel anyway. p.88

## 6 · NOCTURNAL ENURESIS

- Prevalence **5–10% at 7 yr**; ~15% resolve each year †; persists in ~2% of adults. Familial (loci on chromosomes 6 and 13). p.88

- **NMNE**: treat bothersome daytime LUTS (and constipation) first; minor daytime symptoms need not delay enuresis treatment. **MNE**: active treatment from **~6 yr**. p.89

- TABLE Use · Every night; parent must wake child; motivated family; continue if improving after a few weeks until **14 consecutive dry nights** · **0.2–0.4 mg** tablet or 120–240 µg lyophilisate ~1 h before bed; stop if no effect after ~1 week p.89

- ALGORITHM
  MNE ≥6 yr → alarm or desmopressin → failure → check correct use → switch or **combine** (alarm + desmopressin) → re-evaluate: hidden daytime LUTS, constipation (impaction may be silent), sleep-disordered breathing, neurodevelopmental disorder; uroflow + PVR → **add anticholinergic** to desmopressin (second line) → refractory: imipramine, β3-agonist, specialist psychoactive drugs
  p.89

## 7 · FUNCTIONAL CONSTIPATION — TREATMENT

- ALGORITHM
  Faecal impaction? → **disimpact** (oral polyethylene glycol (PEG) or enemas) → education, diary, toilet sitting, **maintenance PEG** → review at 2 wk → failing → reassess adherence, dose, drug, impaction, mental health → **≥2 months** maintenance → wean when ≥1 month symptom-free, ≥3 stools/week, Rome IV no longer met → relapse → reinstate / paediatric gastroenterology
  p.89

- TABLE PEG 3350 — disimpaction · **1–1.5 g/kg/day × 3–6 days** (>3 yr) p.89

- TABLE PEG 3350 — maintenance · **0.2–0.8 g/kg/day**, start 0.4 g/kg/day (>6 months) p.89

## 8 · RECURRENT UTI AND BBD

- In RIVUR/CUTIE, recurrence: **BBD + VUR 51% > BBD alone 35% > VUR alone 20%**. BBD outranks reflux as a risk factor. Wider caecal diameter (constipation) → more recurrence. p.90

## 10 · MCQ ONE-LINERS

- MCQ ICCS increased frequency: ≥8 voids/day; decreased: ≤3 voids/day. p.90

- MCQ Daytime incontinence and enuresis are diagnosed from age 5 years. p.90

- MCQ † EBC = (age + 1) × 30 mL. p.90

- MCQ Recurrent UTI is itself an ICCS symptom of BBD. p.90

- MCQ EAU/ESPU: treat constipation before specific treatment of LUT dysfunction. p.90

- MCQ Dysfunctional voiding: staccato flow + active pelvic floor EMG during voiding. p.90
- MCQ Hinman syndrome = non-neurogenic neurogenic bladder, end stage of dysfunctional voiding. p.90
- MCQ EMG lag time <2 s suggests detrusor overactivity. p.90
- MCQ First-line specific therapy for dysfunctional voiding: biofeedback. p.90
- MCQ Standard urotherapy trial: at least 4 weeks; rewards adherence, not dryness. p.90
- MCQ Oxybutynin IR dose 0.3–0.6 mg/kg/day, maximum 15 mg/day. p.90
- MCQ Trospium is a quaternary amine — does not cross the blood–brain barrier. p.90
- MCQ Mirabegron approved in children ≥3 yr only for neurogenic detrusor overactivity. p.90
- MCQ Paediatric refractory OAB botulinum dose: 5 units/kg, maximum 100 units. p.90
- MCQ PTNS needle ~5 cm above medial malleolus; great-toe flexion confirms placement. p.90
- MCQ Giggle incontinence: methylphenidate or biofeedback; biofeedback more durable. p.90
- MCQ Vaginal reflux: void with legs abducted; leak just after voiding. p.90
- MCQ Rome IV: ≥2 criteria, weekly, for ≥1 month; pain is not a criterion. p.91
- MCQ Plain abdominal radiograph not recommended to diagnose functional constipation. p.91
- MCQ Absent RAIR on anorectal manometry → rectal biopsy for Hirschsprung disease. p.91
- MCQ PEG disimpaction 1–1.5 g/kg/day for 3–6 days; maintenance start 0.4 g/kg/day. p.91
- MCQ Enuresis triad: nocturnal polyuria, small nocturnal capacity/overactivity, arousal failure. p.91
- MCQ Alarm is durable; desmopressin is immediate but not durable. p.91
- MCQ Desmopressin tablet 0.2–0.4 mg 1 h before bed; risk hyponatraemia with excess fluid. p.91
- MCQ Imipramine displaced from first line by cardiotoxicity. p.91
- MCQ Recurrent UTI risk: BBD + VUR (51%) > BBD alone > VUR alone. p.91

## Ch 48 · Lower Urinary Tract Reconstruction in Children

### 1 · GOALS AND PRINCIPLES

- KEY POINT Every reconstructive step is a storage–emptying trade. Raising capacity or outlet resistance converts a leaking but self-venting system into one that depends entirely on reliable emptying. A continent but uncatheterised reservoir causes hydronephrosis, sepsis or perforation — hence incontinence can be "protective", and CIC and motivation are prerequisites, not afterthoughts. p.92

### 2 · PATIENT SELECTION AND PREOPERATIVE EVALUATION

- **Storage pressure**: chronic filling pressure **>40 cm $H_2O$** impairs ureteral drainage → hydroureteronephrosis, falling glomerular filtration rate (GFR), acquired vesicoureteral reflux (VUR). First treat with anticholinergics + CIC; persistent poor compliance → **augmentation**. p.92
- Technique: fill at **≤10% of predicted capacity per minute** (fast filling falsely lowers compliance); body-temperature saline/contrast; catheter size affects leak pressure in infants. The Urologic Management to Preserve Initial Renal function (UMPIRE) protocol standardises paediatric neurogenic bladder care. p.92
- Serum electrolytes, blood urea nitrogen (BUN), creatinine; raised creatinine → nephrology and 24-h urine (creatinine clearance and **volume**). Impaired concentration → large urine volumes; the reservoir must hold **3–4 h** of output. p.93

### 4 · BLADDER AUGMENTATION (ENTEROCYSTOPLASTY)

- **Detubularise** on the antimesenteric border and **reconfigure into a sphere** — always. Intact bowel contracts at **60–100 cm $H_2O$**; detubularisation blunts contractions and a sphere gives maximal volume per surface area, so a shorter segment suffices. p.93
- Length: ileum **20–40 cm**, colon **~20 cm**; err on too large rather than too small. p.93
- **Ileocystoplasty**: segment at least **15–20 cm proximal to ileocaecal valve** (terminal ileum absorbs vitamin B12 †and bile salts) → re-anastomose, close mesenteric window → open antimesenterically → U, S or W fold or cup patch → anastomose posteriorly first. p.94
- KEY POINT Why is augmentation needed less now? Early anticholinergics, CIC, botulinum toxin and neuromodulation for the hostile neurogenic bladder lower augmentation rates (immediate vs expectant treatment: 17% vs 41%). Unproven by randomised trials, but the rationale for treating the bladder at diagnosis. p.94

### 5 · CONTINENT CATHETERISABLE CHANNELS

- Any supple, narrow tube tunnelled submucosally into a **low-pressure** reservoir is continent: rising reservoir pressure **compresses** the tunnelled segment. Appendix needs only a **~2 cm** tunnel. p.95
- ALGORITHM
  Mobilise caecum if high → detach appendix with a small caecal cuff (cuff to skin reduces stenosis; caecal flap extends a short appendix) → preserve mesoappendix → tunnel the **distal end** submucosally (posterolateral bladder; open or extravesical trough) → bring caecal end to skin without tension, twist or fascial compression → **skin flap** into stoma (V/rotational umbilical flap) → keep channel **short and straight**; fix bladder to anterior wall beneath fascia to prevent kinking → test catheterisation after every step and at different volumes → stent 6–12 French (Fr) for 10–14 days → surgeon does first catheterisation → CIC ≥4-hourly by day
  p.95

### 6 · BLADDER OUTLET PROCEDURES

- KEY POINT Choosing the outlet: a child who empties by voiding → AUS (the only option preserving voiding). A child on CIC who is being augmented → sling (girls) or flap-valve lengthening with a catheterisable channel. Failed repairs → bladder neck closure + Mitrofanoff. Always re-study the bladder afterwards. p.97

### 9 · COMPLICATIONS OF AUGMENTATION AND THEIR MANAGEMENT

- Counsel **~9%** risk; mostly myelodysplasia; **outlet procedures double the risk**; risk with missed CIC, substance abuse. p.98
- Ureterosigmoidostomy: adenocarcinoma at the anastomosis, mean latency **~26 yr**, risk vastly increased (urine + faeces → N-nitroso compounds). p.98
- Augmentation: adenocarcinoma reported from **4 yr**, highest after **gastrocystoplasty**; aggressive, often metastatic at diagnosis. Bladder cancer in spina bifida ~2–4%, mostly stage III–IV — part of the risk may be inherent to the underlying anomaly (exstrophy). Nephrogenic adenoma (benign) is the commonest tumour. p.98

### 12 · MCQ ONE-LINERS

- MCQ Pathogenic chronic filling pressure: >40 cm $H_2O$. p.99
- MCQ Fill paediatric urodynamics at ≤10% of predicted capacity per minute. p.99
- MCQ Reservoir must hold 3–4 h of urine output. p.99
- MCQ Bladder neck occlusion urodynamics unmask hostility before outlet surgery. p.99
- MCQ Ileum taken ≥15–20 cm proximal to the ileocaecal valve. p.99
- MCQ Typical augmentation: ileum 20–40 cm; sigmoid 15–20 cm. p.99
- MCQ Detubularisation always; intact bowel contracts at 60–100 cm $H_2O$. p.99

- MCQ Never permanent sutures/staples in the reservoir — stone nidus. p.99
- MCQ Ileum/colon cause hyperchloraemic acidosis; base deficit most sensitive marker. p.99
- MCQ Jejunum: hyponatraemic hyperkalaemic hypochloraemic acidosis. p.99
- MCQ Stomach: hypokalaemic hypochloraemic alkalosis and haematuria–dysuria syndrome. p.99
- MCQ Stomach preferred in renal insufficiency, short gut, irradiated bowel. p.99
- MCQ Mucus production: colon > ileum > stomach. p.99
- MCQ Augmentation stones are mostly struvite; daily irrigation ≥240 mL reduces them. p.99
- MCQ Delayed perforation ~9% (Indiana); outlet procedures double risk. p.99
- MCQ Perforation: cystogram with post-drain film; CT if negative; treat by surgery. p.99
- MCQ Ureterosigmoidostomy adenocarcinoma: mean latency ~26 years. p.99
- MCQ Earliest post-augmentation adenocarcinoma: 4 years. p.99
- MCQ Commonest tumour after cystoplasty: nephrogenic adenoma. p.99
- MCQ Mitrofanoff: appendix needs only a ~2 cm submucosal tunnel. p.99
- MCQ Flap valve tightens with pressure; nipple valve is effaced. p.99
- MCQ Commonest complication of continent channels: stomal stenosis. p.99
- MCQ Monti channels need about twice the revisions of appendicovesicostomy. p.99
- MCQ Monti: ileal ring retubularised transversely over 12 Fr. p.99
- MCQ Only outlet procedure preserving voiding: artificial urinary sphincter. p.99
- MCQ AUS cuff at bladder neck in girls and prepubertal boys; activate ~6 weeks. p.99
- MCQ Kropp: highest continence, difficult catheterisation; Pippi Salle: easier catheterisation. p.99
- MCQ Sling success predicted only by concomitant augmentation. p.99
- MCQ Bladder neck closure needs omentum/muscle interposition and a continent stoma. p.99
- MCQ Ureterocystoplasty needs a massively dilated ureter. p.99

## ★ Putting it together: common paediatric pathways

### A · ANTENATAL URINARY TRACT DILATION (UTD)

- ALGORITHM

**Prenatal:** UTD A1 (mild) before 28 weeks → repeat ultrasound (US) after 32 weeks. UTD A2–3 → US every 4 weeks to delivery. Male fetus with a thick-walled bladder and bilateral dilation → suspect posterior urethral valves (PUV).

**Postnatal:** first US **after 48 hours** (earlier scans under-read dilation because the newborn is relatively dehydrated), except suspected PUV or bilateral severe dilation, which are scanned before discharge.

**Then:** normal (anteroposterior diameter <10 mm, all parameters normal) → transient, no follow-up · mild → repeat US · moderate–severe or dilated ureter → voiding cystourethrogram (VCUG), mercaptoacetyltriglycine (MAG3) renogram and consider continuous antibiotic prophylaxis (CAP) (→ Ch 34).

p.100

### B · FIRST FEBRILE URINARY TRACT INFECTION (UTI)

- ALGORITHM

  **Diagnose properly:** catheter or suprapubic sample in the non-toilet-trained child; bag samples only to rule out.

  **Treat early:** antibiotics within 48 hours (delay beyond 72 hours doubles the odds of new scarring); oral therapy is fine if the child is not toxic.

  **Image:** US for all; VCUG if US is abnormal, infection is atypical or recurrent (American Academy of Pediatrics 2–24 months); dimercaptosuccinic acid (DMSA) scan at 4–6 months if scarring matters.

  **Prevent:** treat bladder and bowel dysfunction (BBD) and constipation; CAP for the high-risk child; circumcision lowers infant UTI risk (→ Ch 37, 41).

  p.100

### C · NEWBORN WITH POSTERIOR URETHRAL VALVES

- ALGORITHM

  Stabilise breathing → 5–7 Fr feeding tube per urethra (minimal or no balloon) → creatinine, electrolytes, fluid balance (watch for post-obstructive diuresis) → US then VCUG (dilated posterior urethra, hypertrophied bladder neck) → antibiotic prophylaxis → **endoscopic valve ablation** when stable; vesicostomy if the urethra is too small.

  Prognosis: nadir creatinine in the first year (<0.8 mg/dL good). Lifelong follow-up for valve bladder, CKD, blood pressure and continence; transplant → Ch 61 (book 1) (→ Ch 45).

  p.100

### E · DAYTIME WETTING AND BEDWETTING

- ALGORITHM

  History, examination, urinalysis, post-void residual ± uroflow → exclude neurogenic, anatomical and medical causes → **treat constipation first** → standard urotherapy for at least 4 weeks → specific therapy by diagnosis (antimuscarinic for overactive bladder, biofeedback for dysfunctional voiding) → refractory: botulinum toxin, neuromodulation.

  **Monosymptomatic bedwetting:** enuresis alarm or desmopressin (no fluids from 1 hour before until morning); combine for partial responders (→ Ch 47).

  p.101

### F · CHOOSING A SEGMENT FOR BLADDER AUGMENTATION

- KEY POINT Every reconstruction trades storage for emptying. A continent reservoir that is not reliably catheterised causes hydronephrosis, sepsis or perforation, so proven commitment to CIC comes before surgery. Spontaneous perforation of an augmented bladder presents with abdominal pain and sepsis and needs urgent diagnosis (→ Ch 48). p.101

# 6 Female Urology

467 core items · page numbers refer to the full book PDF

## Ch 101 · Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

### 1 · PERINEUM AND EXTERNAL GENITALIA

- **Vulva** = mons pubis, labia majora and minora, clitoris, vestibule. External urethral meatus lies **1 cm** anterior to the vaginal opening. p.2

### 4 · ENDOPELVIC FASCIA AND VISCERAL LIGAMENTS

- **Infundibulopelvic** (ovary → side wall, ovarian vessels; ureter just beneath) vs **ovarian ligament** (ovary → uterus) MCQ. p.3

- KEY POINT Support ligaments are connective tissue condensed around vessels and nerves — so the best anchors (uterosacral, cardinal) are exactly where suturing injures the ureter, pelvic plexus and sacral nerves. → Ch 108 p.3

### 5 · URETHRA AND CONTINENCE ANATOMY

- Length **4.7–5.1 cm**; behind symphysis; suspended by posterior pubourethral ligaments; rests on anterior vaginal wall. p.3

### 6 · BLADDER AND URETER

- ALGORITHM
  **Ureteric danger points**: pelvic brim (infundibulopelvic ligament/ovarian vessels) → under uterine artery (hysterectomy) → beside cardinal/uterosacral ligament near cervix → anterior vaginal fornix (vaginal surgery) → near sacral promontory (right ureter, sacrocolpopexy)
  p.4

### 7 · NERVES AT SURGICAL RISK

- KEY POINT Post-lithotomy foot drop is peripheral (fibular head), not pelvic. Post-op leg pain and swelling after long lithotomy → suspect **compartment syndrome**: surgical emergency († fasciotomy). p.4

### 8 · VESSELS AND LYMPHATICS

- **Presacral venous plexus**: valveless, high pressure (worse in lithotomy); a 1–2 mm vein can bleed > **1 L/min**. p.5

- ALGORITHM
  **Sacrocolpopexy**: expose ALL at promontory over L5/S1 disc → open peritoneum distally → suture into ALL over **S1** (strongest ALL; less discitis) → keep clear of left common iliac vein, middle sacral vessels, presacral plexus, right ureter (each ≈ 2.7 cm from mid-promontory)
  p.5

### 9 · ENDOSCOPIC ANATOMY

- ALGORITHM
  Distal urethra (glands, diverticular ostia, mesh, stones) → mid-urethral coaptation (external sphincter) → bladder neck coaptation → trigone and orifices (30° lens) → whole-mucosa survey
  p.5

### 11 · MCQ ONE-LINERS

- MCQ External urethral meatus lies 1 cm anterior to the vaginal opening. p.6
- MCQ Skene's glands are the female homologue of the prostate. p.6
- MCQ Compressor urethrae and urethrovaginal sphincter lie in the deep perineal space. p.6
- MCQ Coccygeus is not part of levator ani; it lies on the SSL. p.6
- MCQ Levator ani is supplied directly by S3–S5, mainly S4. p.6

- MCQ Martius flap works because of dual (femoral and internal iliac) labial supply. p.6
- MCQ Cooper's ligament: anchor for Burch colposuspension. p.6
- MCQ Obturator canal lies superolateral in the obturator foramen. p.6
- MCQ Pudendal nerve enters Alcock's canal via the lesser sciatic foramen. p.6
- MCQ Pudendal nerve is most vulnerable near the ischial spine during SSL fixation. p.6
- MCQ Sacral plexus exits immediately posterior to the SSL. p.6
- MCQ Post-lithotomy foot drop: common peroneal compression at fibular head. p.6
- MCQ Female urethra ≈ 5 cm; closure pressure mainly from oestrogen-dependent mucosa/submucosa. p.6
- MCQ Pubovaginal sling at bladder neck; synthetic MUS at mid-urethra. p.6
- MCQ Ureter passes under the uterine artery: "water under the bridge". p.6
- MCQ Sacrocolpopexy sutures go into ALL over S1. p.6
- MCQ Presacral vein of 1–2 mm can bleed more than 1 L/min. p.6
- MCQ Posterior division of internal iliac has no visceral branches. p.6
- MCQ Distal third urethra drains to inguinal nodes; proximal two-thirds to iliac/obturator. p.6
- MCQ T2-weighted MRI is the best test for urethral diverticulum. p.6

## Ch 108 · Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

### 1 · DEFINITIONS AND TERMINOLOGY

- **OAB syndrome**: urgency ± UUI, usually with frequency and nocturia; **defined by urgency**. Frequency ≥ **8 voids/24 h** is above normal but polyuria must be excluded. p.7

### 2 · EPIDEMIOLOGY OF FEMALE UI (HEADLINE)

- **Pregnancy**: commonest cause of incident UI in young women (≈ **40%**, mostly SUI). >30% still leak at 1 year; UI at 3 months usually persists long term. Pelvic floor muscle training (PFMT) during and after pregnancy reduces UI. p.7

### 3 · RISK FACTORS FOR FEMALE UI

- TABLE **Vaginal delivery** · Causes later **SUI, not UUI**. Caesarean section (CS) protects only if **every** delivery is CS; number needed to treat (NNT) **13** to prevent one moderate–severe UI. Elective CS < CS after labour. Baby ≥4 kg: odds ratio (OR) ≈1.5. Forceps, episiotomy, labour length: unproven p.8
- KEY POINT Two families of risk: obstetric injury and loss of support → **stress** UI; ageing, neurological, metabolic and lifestyle factors → **urgency**. Hence a woman's dominant UI type moves from stress to urge with age. p.8
- Much less common; **UUI > MUI > SUI**. Male SUI is almost always iatrogenic (radical prostatectomy (RP), radical pelvic surgery) or neurological. Remission up to **40%** (reversible urgency causes). p.8

### 4 · PHYSIOLOGY OF CONTINENCE

- **Female**: continence zone is the **proximal and mid-urethra**; bladder neck often incompetent even in continent women. Circular striated muscle compresses the mid-urethra under tonic pudendal drive; compressor urethrae and urethrovaginal sphincter distally. Oestrogen-sensitive spongy submucosa provides up to **30%** of closure. p.9

- KEY POINT The female outlet relies on a **single** mid-urethral sphincter compressed against a mobile backing — so female continence depends on support as much as on muscle. Continence needs support, intrinsic urethral properties and muscle together. p.9

### 6 · PATHOPHYSIOLOGY OF FEMALE SUI

- ISD was described with the autologous fascial pubovaginal sling for severe urethral dysfunction, usually after previous surgery; classified as **type III SUI** (Blaivas) MCQ. p.10
- **Valsalva leak point pressure (VLPP)**: abdominal pressure causing leakage (not valid during a detrusor contraction or poor compliance). **VLPP < 60 cm $H_2O$ = ISD**; historically directed to bulking or pubovaginal sling. p.10
- KEY POINT **Why slings work in hypermobility but Burch fails in ISD**: almost all female SUI includes some ISD. Sphincter-directed procedures (pubovaginal sling, MUS, even bulking) therefore work in hypermobile women; Burch colposuspension, which only corrects hypermobility, is less effective with severe ISD and a fixed urethra. p.10

### 11 · PATHOPHYSIOLOGY OF POP

- ALGORITHM
  **POP sequence**: levator trauma/denervation (labour) → loss of tone → widened urogenital hiatus → fascia/ligaments now carry the whole load → stretch and fail over time (ageing, straining, hysterectomy) → POP → hiatus widens further
  p.12
- Levator defects on MRI in up to **20%** of primiparas (none in nulliparas). **Levator avulsion** → advanced and multicompartment POP. p.12

### 12 · POP AND UI INTERACTION

- ALGORITHM
  Cystocele → urethral kink → obstruction masks SUI → POP repair relieves kink → SUI unmasked. Detect by UDS / stress test **with prolapse reduced** (pessary or speculum).
  p.12

### 14 · MCQ ONE-LINERS

- MCQ UUI does not require DO on urodynamics. p.13
- MCQ UD SUI requires leakage without a detrusor contraction. p.13
- MCQ OAB is defined by urgency. p.13
- MCQ Pregnancy is the commonest cause of incident UI in young women. p.13
- MCQ Vaginal delivery causes SUI, not UUI. p.13
- MCQ CS benefit is lost after even one vaginal delivery; NNT 13. p.13
- MCQ BMI ≥30 more than doubles UI risk; weight loss reverses it. p.13
- MCQ Systemic oestrogen increases UI; vaginal oestrogen does not. p.13
- MCQ Male UI: UUI > MUI > SUI; remission up to 40%. p.13
- MCQ Post-RP incontinence is mainly ISD (>70%). p.13
- MCQ Female continence zone: proximal and mid-urethra. p.13
- MCQ Submucosal vascular plexus contributes up to 30% of urethral closure. p.13
- MCQ Hammock hypothesis (DeLancey): repositioning the urethra is not essential. p.13
- MCQ Integral theory (Petros–Ulmsten): pubourethral ligament laxity; basis of tension-free MUS. p.13
- MCQ ISD = type III SUI; VLPP <60 cm $H_2O$. p.13
- MCQ Burch is less effective in ISD with a fixed urethra. p.13
- MCQ DeLancey level I loss → apical/vault prolapse. p.13

- MCQ DeLancey level III anterior loss → urethral hypermobility. p.13
- MCQ Commonest paravaginal defect: ATFP detachment from the ischial spine. p.13
- MCQ Describe POP by vaginal segment, not organ. p.13
- MCQ POP-Q reference point: hymen = 0. p.13
- MCQ Aa and Ap cannot exceed +3. p.13
- MCQ POP-Q stage 2: leading edge within 1 cm of hymen. p.13
- MCQ Strongest POP risk factor: vaginal childbirth (parity). p.13
- MCQ Elective CS ≈ nulliparous risk; CS after labour ≈ vaginal delivery. p.13
- MCQ Occult SUI: unmasked only when prolapse is reduced. p.13
- MCQ OAB improves after POP repair in 60–80%. p.13

## Ch 105 · Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

### 2 · HISTORY

- SUI coexists in ~**40%** with POP. Patients often expect prolapse repair to cure leakage → counsel. p.14

### 3 · PHYSICAL EXAMINATION

- **Urethral hypermobility**: **Q-tip test** — lubricated swab to the urethrovesical junction, Valsalva, angle change from horizontal **>30°** = hypermobility. Uncomfortable; now mostly teaching/trials. Vaginal swab test equivalent; **visual urethral mobility examination** (meatal rotation < or >30° on cough) is routine practice. p.15
- KEY POINT Stage II is the 2-cm window around the hymen. The III/IV boundary depends on tvl, not a fixed number. Worked example: tvl 8, Bp +5 → between +1 and +6 → **stage III**. p.16
- **Neurological** if suspected. **Bulbocavernosus reflex** (S2–S4) present in only ~**70%** of normal women → absence alone not pathological. Anal tone (S2–S4). p.16

### 5 · MANAGEMENT PATHWAY

- ALGORITHM
  History + examination + UA + PVR → exclude DIAPPERS, UTI, obstruction, tumour → classify and grade bother → red flags (pain, haematuria, recurrent UTI, pelvic mass, prior radiation/pelvic surgery, prolapse beyond introitus, fistula) → specialist work-up; otherwise initial management → reassess → extra tests only if they change management → surgery.
  p.17

### 7 · MCQ ONE-LINERS

- MCQ Commonest POP compartment: anterior. p.18
- MCQ Nocturnal enuresis = voiding, not leakage. p.18
- MCQ In MUI, treat the most bothersome component first. p.18
- MCQ Most specific POP symptom, most likely to resolve after repair: vaginal bulge. p.18
- MCQ Smoking: protective for primary POP, risk factor for UI. p.18
- MCQ Younger age is a risk factor for POP recurrence. p.18
- MCQ Occult SUI: revealed only after reduction of prolapse. p.18
- MCQ α-blockers worsen SUI in women; ACE inhibitors cause SUI via cough. p.18
- MCQ Cholinesterase inhibitors for dementia can precipitate UUI. p.18
- MCQ Negative supine CST → repeat standing. p.18

- MCQ Q-tip hypermobility: >30° from horizontal on Valsalva. p.18
- MCQ POP-Q reference point: hymen (0). p.18
- MCQ POP-Q stage II: leading edge −1 to +1 cm. p.18
- MCQ POP-Q stage IV: leading edge ≥ +(tvl − 2) cm. p.18
- MCQ Posterior speculum blade expelled on straining → level I (apical) support loss. p.18
- MCQ Bulbocavernosus reflex absent in ~30% of normal women — not pathological alone. p.18
- MCQ Required for index SUI surgical patient: UA, PVR, objective SUI demonstration. p.18
- MCQ 1-hour pad test: 500 mL drink; >50 g = severe. p.18
- MCQ Best imaging for mesh: translabial ultrasound. p.18
- MCQ Standard imaging for urethral diverticulum: MRI. p.18

## Ch 114 · Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

### 1 · DEFINITION AND PRINCIPLES

- KEY POINT Why first-line if surgery cures SUI more often? Because conservative therapy is risk free and at least as effective as drugs. This asymmetry of harm drives every recommendation below. p.19

### 3 · LIFESTYLE MODIFICATIONS

- TABLE Action · Reduce if 24-h output >~**2100 mL** · Increase to target p.20
- **Target**: six to eight 8-oz glasses, or **30 mL/kg/day**, minimum **1500 mL**, at set times. Institute of Medicine total water 2.7 L/day for women *includes food water*. p.20
- ~**250 mg/day** (≈2 cups of coffee) associated with moderate–severe UI in men. Coffee is not associated with nocturia. p.20

### 4 · PELVIC FLOOR MUSCLE TRAINING (PFMT)

- Cornerstone for SUI, UUI, MUI, urgency and frequency. Popularised by **Arnold Kegel (1948)** MCQ — SUI attributed to poor PFM awareness; repetitive contractions reduced it. p.21
- ALGORITHM
  Empty bladder → **lying**: 5 quick (1–2 s) + 5 sustained (5 s building to 10 s) → **sitting**: same → **standing**: same (= 30 contractions) → rest as long as each contraction (work:rest **1:1 to 1:2**) → twice daily = **60/day**.
  p.21

### 5 · BEHAVIOURAL TRAINING WITH URGE SUPPRESSION

- ALGORITHM
  Urge → do not rush, stay away from toilet → stand still or sit, relax → several quick PFM squeezes → concentrate on inhibiting, distract → urge subsides (it is a wave) → walk normally to void → once mastered, wait 5 min, then extend gradually.
  p.22
- Stand-alone efficacy for UUI: mean **60–80%** reduction in incontinence; also reduces urgency, frequency, nocturia in both sexes. p.22

### 7 · ELECTRICAL STIMULATION (E-STIM)

- TABLE UUI · **5–20 Hz** (low) p.23
- TABLE SUI (incl. post-prostatectomy) · **20–50 Hz** (higher) p.23

- KEY POINT E-stim is dosed by goal: sub-sensory, low-frequency, prolonged → detrusor inhibition; near-painful, higher-frequency, brief → sphincter contraction. Volitional PFMT achieves both, so E-stim earns its place only where volitional contraction is absent. p.23

### 8 · BLADDER TRAINING AND SCHEDULED VOIDING

- ALGORITHM
  Diary → initial interval ~1 h while awake (or longest comfortable; 30 min if needed) → void on waking, at each interval and at bedtime, not between → increase by **15–30 min/week** if fewer leaks and urgency controlled → target **3–4 h** → weekly review with positive reinforcement.
  p.24

### 12 · ADHERENCE, PREVENTION AND DELIVERY

- **Childbearing**: intensive supervised antenatal PFMT (12 weeks from 16–24 weeks' gestation) prevents late-pregnancy and early postpartum UI (grade A); postnatal programme after forceps or baby ≥ **4000 g**. Antenatal PFMT does **not** prevent postnatal prolapse. p.26

### 15 · MCQ ONE-LINERS

- MCQ Conservative therapy = neither surgery nor drugs. p.27
- MCQ Only grade A lifestyle recommendation: behavioural weight loss in overweight women. p.27
- MCQ 5–10% weight loss substantially reduces UI episodes. p.27
- MCQ Caffeine ~250 mg/day (two cups) associated with UI in men. p.27
- MCQ Fluid target: 30 mL/kg/day, minimum 1500 mL. p.27
- MCQ Stop fluids 3–4 h before bed for nocturia. p.27
- MCQ Commonest fluid problem in practice: over-restriction, not excess. p.27
- MCQ PFMT popularised by Kegel (1948); perineometer was the first BF device. p.27
- MCQ Correct PFM contraction: perineal elevation; incorrect: perineal descent. p.27
- MCQ Commonest PFMT error: Valsalva and abdominal/gluteal co-contraction. p.27
- MCQ Levator ani palpated at 4 and 8 o'clock. p.27
- MCQ PFMT work:rest 1:1–1:2; down-training 1:4. p.27
- MCQ Typical PFMT dose: 45–60 contractions/day. p.27
- MCQ "The knack": contraction just before and during a cough. p.27
- MCQ Supervised PFMT is grade A first-line at all ages. p.27
- MCQ BF adds no benefit to supervised PFMT; use for poor proprioception. p.27
- MCQ Vaginal cones cannot be retained with POP or wide introitus. p.27
- MCQ E-stim for UUI 5–20 Hz; for SUI 20–50 Hz. p.27
- MCQ E-stim niche: women unable to contract PFMs voluntarily. p.27
- MCQ Transcutaneous TNS = percutaneous TNS efficacy, fewer adverse effects. p.27
- MCQ Urge suppression: stop, squeeze quickly, let urge pass, then walk. p.27
- MCQ BT interval increased 15–30 min/week to 3–4 h. p.27
- MCQ BT dissociates voiding from urgency. p.27
- MCQ Prompted voiding: monitoring, prompting, praise. p.27
- MCQ Habit training = timed voiding individualised to the voiding pattern. p.27
- MCQ Only BT aims to lengthen the voiding interval. p.27
- MCQ Prompted voiding responders: 25–40% of nursing home residents. p.27
- MCQ AHRQ: behavioural therapy cures UUI more often than anticholinergics. p.27

- MCQ Grade D = no recommendation possible. p.27
- MCQ Continence pessary supports bladder neck and compresses urethra against pubis. p.27
- MCQ Serious pessary complications: erosion, impaction, vesicovaginal/rectovaginal fistula. p.27
- MCQ ATLAS: behavioural therapy ≥ pessary; combination no better than behavioural alone. p.27
- MCQ Antenatal PFMT prevents UI but not postnatal prolapse. p.27
- MCQ Indwelling catheter is the last resort for UI. p.27

## Ch 116 · Retropubic Suspension Surgery for Incontinence in Females

### 1 · DEFINITIONS AND MECHANISM

- KEY POINT Suspension corrects **hypermobility, not the sphincter**. It lifts and stabilises the bladder neck and proximal urethra retropubically, so it presupposes a mobile urethra. A fixed, ISD urethra needs a device that compresses (sling, artificial urinary sphincter (AUS)). p.28

### 2 · PATIENT SELECTION

- TABLE Immobile urethra (negative Q-tip ≈ **5-fold** failure) → sling or AUS · Obesity, menopause, age >60 p.28

### 4 · BURCH COLPOSUSPENSION — OPERATIVE STEPS

- ALGORITHM
  Low lithotomy, prep umbilicus to perineum incl. vagina, Foley → Pfannenstiel/low midline (± Cherney) → enter Retzius, sweep fat laterally → locate bladder neck by Foley balloon + vaginal finger → dissect bladder off vagina **laterally** → expose Cooper's ligaments → sutures → cystoscopy p.29
- Tie **distal first**, with vaginal elevation; tie **loosely** (Tanagho modification) — **≈ 2 fingerbreadths** left between bladder neck and pubis. Over-tightening → obstruction. p.29
- KEY POINT **Danger points**: never dissect in the midline over urethra/bladder neck (Santorini plexus bleeding, urethral/bladder injury, obstruction if sutures too medial). Passing through Cooper's may cause sudden heavy bleeding — pressure, clips, sutures. p.29
- TABLE Open (Pfannenstiel) · Cure **85–90% at 1 y, ≈70% at 5 y** p.29

### 5 · OTHER RETROPUBIC PROCEDURES

- KEY POINT AUS in women: second-line for ISD (recurrent or neurogenic SUI). Placed at bladder neck, preferably **abdominally**; the transvaginal route raises infection and cuff erosion risk. p.29

### 6 · PERIOPERATIVE CARE

- Incomplete emptying: teach clean intermittent catheterisation (CIC) (preferred over suprapubic tube) until PVR consistently **<150 mL**. p.29

### 11 · MCQ ONE-LINERS

- MCQ Burch anchors paravaginal tissue to Cooper's (iliopectineal) ligament. p.30
- MCQ MMK anchors to pubic symphyseal periosteum; complication: osteitis pubis. p.30
- MCQ Gap left between bladder neck and pubis: about two fingerbreadths. p.30
- MCQ Negative Q-tip test raises Burch failure about five-fold. p.30
- MCQ Q-tip hypermobility: angle change >30°. p.30
- MCQ Open Burch cure ≈85–90% at 1 y, ≈70% at 5 y. p.30
- MCQ Needle suspensions abandoned owing to suture pull-through. p.30
- MCQ Intraoperative cystoscopy is mandatory after Burch suture placement. p.30
- MCQ Burch-specific late complication: posterior compartment prolapse (enterocele). p.30

- MCQ SISTEr: autologous sling beat Burch but caused more voiding dysfunction. p.30
- MCQ CARE: prophylactic Burch at sacrocolpopexy reduces postoperative SUI. p.30
- MCQ ISD threshold: VLPP <60 or MUCP <20 cm $H_2O$. p.31
- MCQ AUS in women: second-line, preferably abdominal not transvaginal. p.31
- MCQ Burch sutures: first at mid-urethra level, most lateral. p.31
- MCQ Sutures must not enter the vaginal lumen. p.31
- MCQ Tanagho modification: sutures more lateral, tied loosely. p.31
- MCQ MMK and Burch obstruct more than paravaginal repair. p.31
- MCQ Paravaginal repair anchors to arcus tendineus fascia pelvis. p.31
- MCQ No ureteric efflux after Burch: release sutures. p.31
- MCQ Prefer CIC over suprapubic tube; target PVR <150 mL. p.31

## Ch 118 · Slings: Autologous, Biologic, Synthetic and Mid-Urethral

### 1 · HOW SLINGS WORK — THEORIES OF CONTINENCE

- Integral theory support: incontinent women have ≈ **40% less collagen** in skin/round ligament biopsies MCQ. p.32

### 2 · PREOPERATIVE ASSESSMENT AND PATIENT SELECTION

- **Examination**: pelvic organ prolapse (POP), atrophy, fistula, foreign material, urethral mobility. Q-tip (cotton-swab) test: hypermobility = **>30°** from rest on straining. p.33
- Abdominal/Valsalva leak point pressure (ALPP/VLPP) **<60 cm $H_2O$** = ISD; **>90** = little or no ISD. p.33

### 4 · PUBOVAGINAL SLING

- Width **<2 cm** — wide enough to compress without forming a narrow constricting band. p.34
- TABLE Synthetic at bladder neck · Strong inflammatory/foreign-body reaction; urethral perforation **15×**, vaginal exposure **14×** other materials (esp. woven polyester) · **No longer used at bladder neck** p.34
- KEY POINT **Fibrosis paradox**: more fibrosis should mean better support, but the inflammatory infiltrate that drives it also degrades the sling and destroys tissue → erosion. Autologous fascia remodels with minimal inflammation — the ideal balance. p.34

### 5 · AUTOLOGOUS PVS — OPERATIVE PROCEDURE

- ALGORITHM
  Pfannenstiel 6–7 cm, ≈2 cm above symphysis → mark **2 × 8 cm** graft (transverse or longitudinal) → excise → close fascia with running No. 1 polydioxanone (PDS) → graft cleaned of fat on side table → No. 1 PDS (or 1-0 polypropylene) stitched across each end, left long → keep in saline
  p.35
- Transverse incision: stay **≥2 cm above the symphysis** for tension-free closure. Undermining fascial edges helps closure but may weaken fascia. If closure impossible (rare) → bridge with mesh or biologic graft. p.35
- ALGORITHM
  Bladder empty → Stamey/Cobb-Ragde needles (or long clamp) passed **from above**, kept on the back of the pubis, out lateral to bladder into vaginal incision → **cystoscopy (70° lens)** while needles are moved → injury: withdraw and repass **more laterally**, re-scope after every pass → confirm bilateral ureteric efflux → sutures loaded, graft centred → needles withdrawn, ends tagged above
  p.36

- **Tension**: tie sutures over the rectus fascia leaving **two fingerbreadths** of slack — deliberately **loose**. Tighter only if lifelong CIC is planned. p.36

- No lifting >10 lb (≈4.5 kg) and no intercourse for **6 weeks** and until healing confirmed. p.36

### 7 · PVS COMPLICATIONS AND MANAGEMENT

- KEY POINT **The key diagnostic criterion is the temporal relationship** between surgery and onset of symptoms, not a urodynamic number. Preoperative UDS does not predict who will obstruct (SISTEr analysis). p.37

- ALGORITHM
  Transient retention (most void within **10 days**) → catheter/CIC, timed and double voiding, PFMT, biofeedback, anticholinergic → autologous PVS obstruction often improves: wait up to **3 months** → if complete retention, earlier: **<6 weeks** loosen autologous sling under anaesthesia (downward traction on urethra with cystoscope) → **>6 weeks** or failed conservative care → sling incision or urethrolysis
  p.37

- Commonest systemic: pulmonary, cardiovascular, neurological, gastrointestinal. Mortality ≈ **3 per 10,000** anti-incontinence procedures. p.38

### 8 · MID-URETHRAL SLING — MECHANISM AND MATERIALS

- KEY POINT **Why does a loose tape work?** At rest it does not touch the urethra, so voiding is unobstructed. On coughing, the mobile urethra descends against the tape: the posterior wall is arrested, the urethra kinks over it and the lumen closes. Hence a mobile urethra is needed — a fixed urethra never meets the tape, and tightening the tape to compensate creates obstruction. p.38

- Current standard: **type 1 (Amid) macroporous, monofilament, loosely woven polypropylene**, pore **>75 µm**. p.38

### 9 · MUS OPERATIVE PROCEDURES

- ALGORITHM
  Local infiltration (retropubic and paraurethral) → two suprapubic stabs ≈2 cm from midline → 1.5 cm midline vaginal incision starting 1.5 cm from meatus → minimal lateral tunnels to the endopelvic fascia (**not perforated**, unlike PVS) → rigid catheter guide deflects bladder to the opposite side → trocar through endopelvic fascia, Retzius, rectus, out suprapubically, **hugging the back of the pubis** → repeat → cystoscopy → tension → remove sheath → trim
  p.39

- TABLE Bladder perforation · **4.5%** · **0.6%** · Rare (but erosion higher with TVT-Secur) p.40

### 12 · MUS COMPLICATIONS AND MANAGEMENT

- KEY POINT **Concurrent vs staged sling at POP repair**. Concurrent: one operation and less postoperative SUI (risk ≈39% → 8–19% in meta-analysis), but more obstruction, storage symptoms and mesh events, and tension is hard to judge when SUI is only occult. Staged: avoids an unnecessary sling (some SUI resolves after POP repair) but commits a third of women to a second operation, usually within 12 months. Decide with the patient after testing for occult SUI with the prolapse reduced. p.41

- ALGORITHM
  Retention/high PVR → CIC or catheter (most resolve within 2 weeks) → persistent obstruction → **transvaginal midline sling incision at 2–4 weeks** (single small vertical incision, minimal dissection) → >3 months: tape fixed along its course → excise lateral portion (to pubis/ischiopubic ramus) → rarely formal urethrolysis
  p.42

- KEY POINT **Why the different timing from PVS?** Autologous fascia obstruction often settles, so one waits up to 3 months. Synthetic mesh becomes fixed by tissue ingrowth, rarely loosens, and late release needs wider dissection and causes more recurrent SUI — hence release within ≈4 weeks. p.42

### 14 · ASSESSING THE FAILED OR TROUBLESOME SLING

- ALGORITHM
  Symptom after sling → define: persistent/recurrent SUI vs storage symptoms vs voiding symptoms vs pain/discharge → history with **timeline** → examination (mesh exposure, POP, urethral mobility, tenderness over tape) → PVR, urinalysis/culture → **cystoscopy** (mesh in urethra/bladder, hypersuspension) → UDS selectively (stress leak, DO, obstruction) → tailored treatment
  p.43

### 17 · MCQ ONE-LINERS

- MCQ PVS sits at the bladder neck; MUS at the mid-urethra. p.44
- MCQ Integral theory: pubourethral ligaments, suburethral hammock, pubococcygeus. p.44
- MCQ Urethral hypermobility is a sign, not cause, of SUI. p.44
- MCQ Gold standard sling for all SUI forms: autologous fascial PVS. p.44
- MCQ Rectus fascia graft size: 2 × 8 cm. p.44
- MCQ Transverse rectus harvest stays ≥2 cm above symphysis. p.44
- MCQ Prior ventral hernia repair → use fascia lata. p.44
- MCQ Fascia lata is left unclosed at harvest. p.44
- MCQ PVS sutures tied over rectus with two-fingerbreadth slack. p.44
- MCQ Endopelvic fascia is perforated in PVS but not MUS. p.44
- MCQ Synthetic bladder-neck sling: 15× urethral perforation risk. p.44
- MCQ MUS mesh: type 1, macroporous (>75 µm) monofilament polypropylene. p.44
- MCQ ISD: MUCP <20 or ALPP <60 cm $H_2O$. p.44
- MCQ ValUE: UDS unnecessary before surgery for uncomplicated SUI. p.44
- MCQ SISTEr: PVS beats Burch; more voiding dysfunction. p.44
- MCQ TOMUS: retropubic = transobturator cure; more voiding dysfunction retropubic. p.44
- MCQ Bladder perforation: retropubic 4.5% vs transobturator 0.6%. p.44
- MCQ Groin pain: transobturator 6.4% vs retropubic 1.3%. p.44
- MCQ Bladder trocar injury: re-pass and continue. p.44
- MCQ Urethral injury at MUS: abandon mesh, stage repair. p.44
- MCQ Cystoscopy needs 70° lens and full bladder. p.44
- MCQ SIMS excess failure was due to withdrawn TVT-Secur. p.44
- MCQ SIMS needs tighter tension; restricted mobility predicts cure. p.44
- MCQ ISD or recurrent SUI: retropubic preferred over transobturator. p.44
- MCQ Fixed urethra predicts MUS failure better than low VLPP. p.44
- MCQ OPUS: prophylactic sling, number needed to treat 6.3. p.44
- MCQ Post-PVS obstruction: wait up to 3 months unless retention. p.44
- MCQ Post-MUS obstruction: incise sling at 2–4 weeks. p.44
- MCQ Transurethral bladder neck incision fails for sling obstruction (extraluminal sling). p.44

- MCQ Transvaginal urethrolysis least effective after PVS. p.44
- MCQ Urethral dilation after MUS is harmful. p.44
- MCQ Obstructing MUS usually lies too proximal. p.44
- MCQ Diagnosis of sling obstruction: temporal relationship of symptoms. p.44
- MCQ Commonest complication after MUS: urinary tract infection. p.44
- MCQ Remeex adjustable sling uses a suprapubic varitensor. p.44

## Ch 117 · Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

### 2 · CHOICE OF PROCEDURE BY COMPARTMENT

- KEY POINT Only paravaginal repair and uterosacral suspension are done by both routes. McCall = vaginal; Halban (linear sutures) and Moschcowitz (purse-string — may angulate the ureter) = abdominal. p.46

### 3 · ANTERIOR COMPARTMENT

- ALGORITHM
  Dorsal lithotomy, padded, hips/knees ~90° → catheter → hydrodissection ± vasoconstrictor → midline incision apex → bladder neck → dissect vaginal wall off pubocervical fascia (white, shiny surface) midline then laterally to ATFP → reduce bladder → interrupted 2-0/3-0 delayed-absorbable plication sutures bladder neck → apex, tied while assistant reduces → trim epithelium judiciously → running absorbable closure → IV dye + **cystoscopy** → pack.
  p.46
- Abandoned for SUI: American Urological Association (AUA) meta-analysis showed failure approaching **40%**; now a prolapse operation only. p.46
- TABLE De novo SUI · Rises with cystocele stage: stage 2 **41%**, early 3 **53%**, advanced 3–4 **66%**; point Ba predicts p.46
- KEY POINT **Confirming ureteric efflux**: indigo carmine (blue, IV), fluorescein 10% (green, IV), phenazopyridine (orange, oral pre-op). No efflux → ureteric catheter/retrograde pyelogram or remove plication sutures. A patent ureter may not efflux if under-hydrated — but never close without proving patency. p.47
- Lateral recurrence low (2–8%) but central recurrence ~**22%** regardless → treat as two independent failures. Abdominal route more durable at long follow-up. p.47

### 4 · OCCULT SUI AND THE CONCOMITANT SLING

- KEY POINT **Why the sling may also help the prolapse**: a suburethral sling adds distal anterior support, so it improves durability of the cystocele repair, not just continence. **Why not give everyone a sling**: OPUS showed more bladder perforation, UTI, bleeding and incomplete emptying, and patient-reported pelvic floor symptom scores did not differ — the benefit is real but must be weighed per patient. p.48
- ALGORITHM
  Advanced anterior POP → reduction stress test ± urodynamics → symptomatic, occult or urodynamic SUI → offer concomitant anti-incontinence procedure → continent and test negative → individual risk–benefit (OPUS calculator); **selective, not routine**.
  p.48

## 5 · APICAL COMPARTMENT

- **ALGORITHM**
  Intraperitoneal (after vaginal hysterectomy): stay sutures at 3 and 9 o'clock cuff dimples → excise enterocele sac → pack bowel → palpate ischial spines; ligaments posterior–medial → tent with Allis at 5 and 7 o'clock → **2–4** non-absorbable sutures per side, each proximal to the last → anterior repair/sling now → sutures through apical pubocervical and rectovaginal fascia (highest medially) → tie → **cystoscopy after tying, before trimming** → no efflux: remove most lateral ipsilateral suture first, then in sequence.
  p.48

- Results: apical success ~98%, anterior ~81% → recurrence is anterior by default; ureteric kinking/injury **1–11%**. Absorbable suture non-inferior to permanent. Premenopausal status raises recurrence. p.48

- **Safe zone**: **1.5–3.0 cm medial to the ischial spine** (1–2 fingerbreadths), **into** the ligament, limited depth, no entry above its upper border. Too lateral → pudendal bundle; too medial → thin fanned ligament (failure) + intraligamentous sacral nerves. p.49

- **ALGORITHM**
  Mark apex and reduce to intended ligament → midline incision → blunt entry to perirectal (or paravesical) space, rectum retracted medially → palpate spine and ligament → Breisky–Navratil/Heaney ± lighted retractors → 2 sutures (Capio or Miya hook/Deschamps) → other repairs → sutures out to apex; pulley stitch if permanent → close vagina halfway → tie apex flush to ligament (**no suture bridge**) → cystoscopy before trimming → close.
  p.49

- Complications: ipsilateral gluteal pain **15%** — usually resolves in 2–3 months (observe; local anaesthetic nerve injection if needed); pudendal vessel/nerve injury (too near spine); rectal injury (repair primarily); haemorrhage. p.49

- **KEY POINT** Both ligaments are safest in their **middle third** and dangerous at their bony ends: USLS intermediate portion; SSLF 1.5–3 cm medial to the spine. p.49

- **ALGORITHM**
  Low lithotomy; vaginal stent/end-to-end anastomosis (EEA) sizer → bowel packed, sigmoid to left → incise peritoneum over promontory, extend along **right** side of rectum (right ureter close) → expose anterior longitudinal ligament → (Halban culdoplasty if large enterocele) → dissect bladder off anterior vagina (~6–8 cm) and rectum off posterior vagina (to rectal reflection; to perineal body = colpoperineopexy) in the areolar plane above the fibromuscularis → Y-mesh sutured to both walls full-thickness **without entering the lumen** → tension → 2–3 permanent monofilament sutures to promontory at **S1** → trim → **retroperitonealise** → anti-incontinence procedure if indicated → cystoscopy → examine for ancillary vaginal repair.
  p.49

- **TABLE** Mesh exposure · Quote **~10% by 7 years** (extended CARE); mean onset 14 months — bleeding, discharge; more with microporous mesh; transvaginal excision ± partial colpocleisis, or abdominal excision → Ch 119 p.50

- **ALGORITHM**
  Elderly/comorbid, not sexually active → colpocleisis → fit, primary prolapse, wants vaginal route → USLS or SSLF ± vaginal hysterectomy or hysteropexy → young/active, recurrent, short vagina, high stage → SCP (minimally invasive preferred).
  p.50

## 7 · OBLITERATIVE SURGERY — COLPOCLEISIS

- **Urinary**: occult SUI unmasked and urethral traction → spare ≥**1.5 cm** of distal anterior wall; screen for SUI and offer concomitant sling if SUI/occult SUI. Frail women: retention risk and difficulty with CIC → consider suprapubic tube. p.51

- Outcome: success ~90–98%, satisfaction >90%, coital regret up to **13%**; body image and pelvic floor symptoms improve. These women are older and comorbid (43% >80 years), so shared decision-making and medical optimisation matter more than the operation itself. p.51

## 11 · MCQ ONE-LINERS

- MCQ Strongest determinant of satisfaction after POP surgery: absence of bulge symptoms, not anatomical cure. p.52
- MCQ Abdominal POP repair: more durable; vaginal: less morbidity, faster recovery. p.52
- MCQ Point D is recorded only when the cervix is present. p.52
- MCQ McCall culdoplasty is vaginal; Halban and Moschcowitz are abdominal. p.52
- MCQ Moschcowitz purse-string culdoplasty may angulate and obstruct the ureter. p.52
- MCQ Lateral (paravaginal) defect = pubocervical fascia detached from the arcus tendineus fasciae pelvis. p.53
- MCQ Anterior colporrhaphy and paravaginal repair do not treat SUI. p.53
- MCQ Paravaginal repair sutures: vaginal 5–7, abdominal 4–6, non-absorbable, 1 cm apart. p.53
- MCQ De novo SUI after cystocele repair rises with stage: 41% → 53% → 66%. p.53
- MCQ Concomitant sling reduces cystocele recurrence by about 55%. p.53
- MCQ OPUS: sling at vaginal POP repair cut 12-month incontinence from 43% to 27%. p.53
- MCQ CARE: Burch at sacrocolpopexy halved early de novo SUI (44% → 24%). p.53
- MCQ Uterosacral–ureter distance: 4 cm at sacrum, 0.9 cm at cervix. p.53
- MCQ Ureteric obstruction after USLS is due to kinking via the ureteric sheath. p.53
- MCQ Do cystoscopy after tying suspension sutures, before trimming them. p.53
- MCQ No efflux after USLS: remove the most lateral ipsilateral suture first. p.53
- MCQ SSLF sutures: 1.5–3 cm medial to the ischial spine, into the ligament. p.53
- MCQ Commonest recurrence after SSLF: anterior compartment (posterior-right axis deviation). p.53
- MCQ Gluteal pain after SSLF ≈15%, usually resolves within 3 months. p.53
- MCQ OPTIMAL: USLS and SSLF equivalent at 2 years. p.53
- MCQ Iliococcygeus fixation point: 1 cm distal to the ischial spine. p.53
- MCQ Sacrocolpopexy mesh: type I macroporous monofilament polypropylene, fixed at S1. p.53
- MCQ Vessel at risk above promontory: left common iliac vein; below S1: presacral plexus. p.53
- MCQ Mesh exposure after sacrocolpopexy ≈10% at 7 years (E-CARE). p.53
- MCQ Supracervical hysterectomy preferred with sacrocolpopexy to reduce mesh exposure. p.53
- MCQ Commonest failure site after sacrocolpopexy: posterior compartment. p.53
- MCQ Le Fort partial colpocleisis keeps lateral channels to drain the uterus. p.53
- MCQ Serious risk of obstructed Le Fort channels: pyometra. p.53
- MCQ Colpocleisis: spare ≥1.5 cm distal anterior wall to avoid urethral traction SUI. p.53
- MCQ Levator plication at rectocele repair abandoned because of dyspareunia. p.53
- MCQ Commonest rectovaginal fascial defect: transverse, detached from perineal body. p.53
- MCQ Constipation usually persists after anatomically successful rectocele repair. p.53
- MCQ Transvaginal synthetic POP mesh withdrawn by FDA in 2019. p.53

## Ch 119 · Genitourinary Complications of Mesh and Their Repair

### 1 · MESH, TERMINOLOGY AND REGULATION

- KEY POINT **Why pore size matters**: bacteria (~1 µm) enter small pores that macrophages and lymphocytes (~50 µm) cannot — infection is sealed from immunity. Large pores (~1500 µm knitted polypropylene) allow fibrocollagenous integration. Woven polypropylene (Marlex) is stiff and extrudes. No excess cancer or autoimmune disease with mesh. p.54

### 3 · EVALUATION OF SUSPECTED MESH COMPLICATION

- ALGORITHM
  History + **prior operative notes** (implant and excisions) → speculum and pelvic examination (exposure, bands, point tenderness, levator hypertonicity) → cystoscopy/urethroscopy if haematuria, recurrent urinary tract infection (UTI), storage lower urinary tract symptoms (LUTS) or voiding pain → urodynamics (UDS) if voiding dysfunction → imaging: translabial ultrasound locates sling; **MRI** shows sacrocolpopexy mesh (invisible on CT); CT to exclude haematoma.
  p.55

### 4 · MIDURETHRAL SLING COMPLICATIONS

- **Excision**: hydrodissect (1% lidocaine with adrenaline) → midline/horizontal incision, or **inverted-U flap** if not palpable → don't cut mesh on entry → locate (sound with metal instrument, ultrasound; may lie distal or beyond bladder neck) → expose full width lateral to urethra → pass instrument behind, divide ("divide and conquer") → excise ≥**1 cm** around defect → avoid blind lateral dissection (vulval haematoma, especially transobturator) → close. p.55
- After excision: SUI **42%** (reduced by simultaneous autologous fascial sling); urgency common. p.56
- KEY POINT **How much to remove**: retention — midline division; exposure — ≥1 cm each side; pain — maximal vaginal removal, rarely arms. Remove enough to fix the problem; **never remove asymptomatic mesh**; more removal = more bleeding, recurrence of POP and SUI. p.56

### 6 · SACROCOLPOPEXY MESH

- Exposure probability ~**10.5% at 7 years**. Risks: total hysterectomy (23% vs 5%), stage 3–4, multiple procedures. Cervix-sparing has low risk. p.56

### 8 · MCQ ONE-LINERS

- MCQ Only recommended pelvic implant mesh: type I macroporous monofilament knitted polypropylene. p.57
- MCQ Type I pore size: >75 µm. p.57
- MCQ "Erosion" should not be used for mesh complications. p.57
- MCQ Exposure = mesh visible through unhealed vaginal epithelium. p.57
- MCQ Perforation = mesh into a hollow viscus. p.57
- MCQ Transvaginal POP mesh: class III 2016; withdrawn 2019. p.57
- MCQ Slings and sacrocolpopexy mesh remain class II. p.57
- MCQ Urethral injury during sling dissection: abandon mesh placement. p.57
- MCQ Best imaging for sacrocolpopexy mesh: MRI. p.57
- MCQ Cystoscopy for trocar perforation requires a full bladder. p.57
- MCQ Small vaginal exposure: observation or topical oestrogen. p.57
- MCQ Exposure excision margin: at least 1 cm around defect. p.57
- MCQ Only predictor of conservative success: exposure size. p.57
- MCQ SUI after intraurethral mesh excision: 42%. p.57
- MCQ Sling division resolves retention in 95%. p.57

- MCQ Avoid urethral dilation for post-sling obstruction. p.57
- MCQ Infected sacrocolpopexy mesh: remove entirely. p.57
- MCQ Total hysterectomy at sacrocolpopexy raises exposure (23% vs 5%). p.57
- MCQ Exposed sacrocolpopexy suture: cut below knot in office. p.57
- MCQ Never remove asymptomatic mesh. p.57
- MCQ Complication unique to POP mesh: contraction (shrinkage), ~12%. p.57
- MCQ Groin pain after transobturator sling: excise via thigh through adductor brevis. p.57
- MCQ Male sling extrusion usually follows unrecognised urethral perforation without intraoperative cystoscopy. p.57
- MCQ Highest retention risk after male retrourethral sling: neobladder patients. p.57
- MCQ Mesh dissection most dangerous step: passing clamp behind mesh (blind). p.57

## Ch 121 · Urinary Tract Fistulae

### 1 · DEFINITION AND UNIVERSAL PRINCIPLES

- KEY POINT Trap: a fistula may be the only sign of recurrent cancer — biopsy the tract in anyone with prior malignancy or radiation. p.58

### 2 · VESICOVAGINAL FISTULA (VVF) — AETIOLOGY

- TABLE Cause · Obstructed labour, no access to operative obstetrics; young age, small pelvis, poor nutrition · Unrecognised surgical injury; **>75%** after hysterectomy p.58
- **Radiation**: obliterative endarteritis + lamina propria fibrosis → necrosis → ulceration → fistula. 0.6–2% after cervical cancer radiotherapy; typically **1.5–2 years** later (up to 30); dose-dependent, worse in smokers. p.59

### 3 · CLASSIFICATION OF VVF

- KEY POINT Trap: Goh type 1 is farthest from the meatus (best), type 4 closest (worst). Closer to the meatus = worse closure and worse continence. Goh predicts closure better. In practice always describe size, location (trigonal vs supratrigonal), tissue loss, ureteric and urethral involvement. p.59

### 4 · DIAGNOSIS OF VVF

- **Timing points to mechanism**: missed/failed cystotomy → first few days; thermal or crush injury → **7–10 days**; radiation → years. p.59
- **Upper tract imaging is mandatory** — ureteric injury coexists in up to **12%** of VVF. Computed tomography urogram (CTU) with delayed phase; MR urography if contrast allergy/renal failure. **Partial ureteric obstruction with a VVF = UVF until proven otherwise.** Unclear CTU → retrograde pyelogram + stent; fails → percutaneous nephrostomy (PCN). p.60
- KEY POINT The work-up answers three questions in order: is the fluid urine (creatinine), where does it come from (bladder, ureter, urethra, uterus — dye tests, endoscopy), and what else is damaged — the last is why upper tract imaging is compulsory even when the VVF is obvious. A PCN placed preoperatively also allows antegrade ureteroscopy during repair to locate the ureter in scar. p.60

### 5 · VVF — PROGNOSTIC FACTORS AND TIMING

- **Location**: most post-hysterectomy VVF are supratrigonal, midline. Within **1 cm** of a ureter → ureteric stent (identification, protection from oedema). Ureteric involvement → reimplant → abdominal route. Bladder neck involvement → reconstruction ± pubovaginal sling. Apical VVF (8–10 cm from introitus) is beyond Martius reach → peritoneal/omental flap. p.60
- TABLE Radiation-induced · ≥**12 months** after radiation ends · Late radiation injury still evolving p.60

### 6 · VVF — CONSERVATIVE AND ENDOSCOPIC TREATMENT

- **Catheter drainage** cures ~15%. Favourable: small (<1 cm), catheter within **3 weeks** of injury, fresh injury. Unfavourable: large, ≥6 weeks (tract epithelialised), radiation or malignancy (never heal — vascular damage). EAU: if attempted, trial ≥**12 weeks**. p.60

### 7 · VVF — SURGICAL REPAIR

- No randomised controlled trials (RCTs); first-attempt closure ≈ **90%** either route. Transvaginal: less blood loss, shorter stay and pain — some call it the gold standard unless contraindicated. Best route = the one the surgeon is most confident with. p.61

- **ALGORITHM**

  Lithotomy → cystoscopy (stent ureters if close; guidewire through tract) → ± suprapubic catheter + urethral catheter → episiotomy if narrow → Foley ≥8 Fr through fistula for traction → hydrodissect → **J- or U-incision** circumscribing tract with 1–3 mm cuff → raise anterior and posterior vaginal flaps widely → layer 1: 3-0/4-0 delayed absorbable (bladder wall + perivesical tissue) → layer 2: imbricating, **perpendicular** to layer 1 → watertight test 200–300 mL dyed fluid → ± Martius/peritoneal flap → advance opposite vaginal flap ≥2 cm beyond (layer 3) → cystoscopy for efflux

  p.61

- **ALGORITHM**

  Low lithotomy, ureteric catheters → low midline, extraperitoneal bladder mobilisation → sagittal cystotomy at dome → **bivalve** posteriorly down to and around the fistula → excise tract → dissect bladder off vagina ≥2–3 cm beyond → close vagina → **omentum/peritoneum** fixed beyond vaginal line → ancillary procedures → close bladder in 1–2 layers → suprapubic ± urethral catheter, fill test → drainage 10–14 days, cystogram

  p.62

### 10 · URETEROVAGINAL FISTULA (UVF)

- Usually unrecognised injury to the **distal ureter** at (often uneventful) abdominal hysterectomy — ureter passes under the uterine artery. Up to **25%** of VVF have a coexisting UVF. Delayed recognition raises fistula risk ~4-fold. p.63

- **ALGORITHM**

  Suspect UVF → exclude VVF + confirm with CTU/retrograde pyelogram → stent → remove 4–6 weeks → reimage → resolved, or persistent/stent fails → ureteroneocystostomy

  p.63

- **KEY POINT** UVF drains past a normally filling bladder (wet but voids normally); because the ureter heals over a stent if reached early, the interval from detection to stenting largely decides whether the patient needs an operation. p.63

### 14 · VESICOENTERIC FISTULA (VEF)

- Colovesical commonest; male : female **3 : 1** (uterus shields bladder; prior hysterectomy common in affected women). p.65

- **KEY POINT** VEF is a bowel disease presenting in the bladder — cure lies in removing the diseased bowel; the bladder usually heals itself. p.65

### 15 · UROCUTANEOUS FISTULAE

- **Urethrocutaneous fistula (UCF)**: commonest complication of hypospadias repair (up to 20%). Wait ≥**6 months**; exclude distal stricture. <1 cm → local closure; ≥1 cm or redo → vascularised regional flaps; **tunica vaginalis** cover better than dartos; massive loss → gracilis; neurogenic with pressure ulcers → ileal conduit. p.65

### 21 · MCQ ONE-LINERS

- **MCQ** Commonest acquired urinary fistula: VVF; developed world cause — hysterectomy. p.67

- **MCQ** Obstetric VVF mechanism: prolonged fetal head impaction → pressure necrosis. p.67

- MCQ Post-hysterectomy VVF typical site: vaginal cuff, supratrigonal. p.67
- MCQ Thermal/crush injury VVF presents at 7–10 days. p.67
- MCQ Double dye test: orange tampon = ureterovaginal; blue = VVF. p.67
- MCQ Phenazopyridine colours urine orange. p.67
- MCQ Coexisting ureteric injury in VVF: up to 12% — upper tract imaging mandatory. p.67
- MCQ Goh type 4: distal edge <1.5 cm from meatus — worst prognosis. p.67
- MCQ Waaldijk classification is based on closing mechanism involvement. p.67
- MCQ Catheter drainage cures ~15%; best if <1 cm and within 3 weeks. p.67
- MCQ EAU: conservative catheter trial ≥12 weeks; sealants not recommended. p.67
- MCQ Radiation VVF: delay repair ≥12 months after radiation. p.67
- MCQ Latzko = high partial colpocleisis; main drawback vaginal shortening. p.67
- MCQ O'Conor repair: transvesical bivalving to fistula, extraperitoneal. p.67
- MCQ Martius flap retained pedicle for VVF: superior (external pudendal). p.67
- MCQ Martius cannot reach apical VVF — use peritoneal or omental flap. p.67
- MCQ Omental flap supply: right and left gastroepiploic arteries. p.67
- MCQ Most important part of VVF repair: uninterrupted postoperative bladder drainage. p.67
- MCQ UVF: normal voiding with continuous leakage. p.67
- MCQ First-line UVF treatment: early ureteric stenting. p.67
- MCQ Youssef syndrome: menouria, amenorrhoea, continence — after caesarean section. p.67
- MCQ Commonest developed-world cause of urethrovaginal fistula: diverticulectomy. p.67
- MCQ Never place synthetic sling with urethrovaginal fistula repair. p.67
- MCQ York-Mason approach: posterior transsphincteric, prone jackknife. p.67
- MCQ Transperineal RUF repair uses gracilis interposition. p.67
- MCQ Martini grade 1–2 RUF: prior radiation/energy or salvage therapy. p.67
- MCQ Commonest cause of colovesical fistula: diverticulitis. p.67
- MCQ Most sensitive test for presence of VEF: poppy seed test. p.67
- MCQ Preferred imaging for VEF: CT (air in bladder). p.67
- MCQ Gouverneur syndrome: suprapubic pain, frequency, dysuria, tenesmus. p.67
- MCQ PVF: pubic pain on walking after radiation and outlet surgery; MRI best. p.67
- MCQ Ureteroarterial fistula commonest artery: common iliac; treat with stent graft. p.67
- MCQ Commonest complication of hypospadias repair: urethrocutaneous fistula. p.67
- MCQ Nephrocutaneous fistula: kidney non-functioning in ~90% — nephrectomy. p.67

## Ch 122 · Bladder and Female Urethral Diverticula

### 1 · BLADDER DIVERTICULA — DEFINITION AND PATHOLOGY

- ~90% in adults; male : female ≈ **9 : 1**. p.68
- Acquired: ~**70%** with benign prostatic hyperplasia (BPH); also bladder neck obstruction, stricture. In women look for dysfunctional voiding, prolapse, primary bladder neck obstruction, post-sling obstruction. p.68

- KEY POINT Congenital = weak wall at normal pressure; acquired = normal wall at high pressure — so an acquired diverticulum always sends the search back to the outlet. p.68

### 2 · BLADDER DIVERTICULA — EVALUATION

- KEY POINT The diagnostic work-up has two jobs: find the cause (outlet or neurogenic) and exclude cancer inside a sac that imaging, endoscopy and urodynamics each see imperfectly — so the tests are combined rather than any one relied upon. p.69

### 3 · MALIGNANCY IN BLADDER DIVERTICULA

- Histology: urothelial carcinoma **70–80%**, squamous cell carcinoma (SCC) 20–25%. p.69

- KEY POINT Treat a diverticular tumour as potentially invasive from the outset — the missing muscle layer explains early spread, perforation risk and inaccurate staging. p.69

### 4 · BLADDER DIVERTICULA — MANAGEMENT

- ALGORITHM

  Diverticulum → history, cystoscopy, cytology, VCUG → suspicious lesion? biopsy → cancer therapy → no lesion: upper tract imaging → HUN or symptoms/complications → urodynamics → treat lower urinary tract abnormality (BOO) → reassess → persistent HUN/symptoms due to diverticulum → good candidate: diverticulectomy; poor candidate: clean intermittent catheterisation (CIC) → asymptomatic, uncomplicated: surveillance

  p.69

### 7 · UD — PRESENTATION

- Third to seventh decades. **"Three Ds" — dysuria, dyspareunia, (postvoid) dribbling**: historical; all three in only ~**5%**. p.71

### 8 · UD — EVALUATION

- KEY POINT UD symptoms are those of common lower urinary tract complaints and do not track lesion size, so diagnosis depends on suspicion and examination; imaging must both confirm and **map** the lesion — especially extension towards the bladder neck, which raises the risk of bladder/ureteric injury and postoperative incontinence. p.71

### 9 · DIFFERENTIAL DIAGNOSIS OF PERIURETHRAL MASSES

- KEY POINT Two questions sort almost every periurethral mass: does it communicate with the urethra, and where does it sit relative to the meatus? Only UD communicate; lesions at or distorting the meatus are Skene, prolapse or caruncle; firm mobile solid masses are leiomyomas. p.72

### 11 · UD AND STRESS URINARY INCONTINENCE

- Significant de novo SUI after diverticulectomy alone: **7–16%**. Risk: size ≥30 mm, proximal, circumferential, short urethra. p.72

### 12 · SURGICAL REPAIR OF UD

- ALGORITHM

  Lithotomy, ring retractor, urethral Foley (suprapubic tube placed at the **start**, never at the end — it would cross the fresh urethral repair) → **inverted-U incision** (base distal, limbs to bladder neck, widening proximally) → raise vaginal flap off periurethral fascia → incise fascia **transversely**, raise proximal and distal fascial flaps → dissect sac to ostium, excise entire wall including inflamed urethral edge → close urethra watertight over catheter (as small as 12 Fr) → close fascia **perpendicular** to urethral line, obliterating dead space → ± **Martius flap** (poor tissue, attenuated fascia, inflammation) → advance vaginal flap (3 layers; 4 with Martius) → antibiotic vaginal pack

  p.73

## 13 · COMPLICATIONS

- TABLE Recurrent UD · **10–22%** (pooled ~10%) · Incomplete excision, poor closure, dead space; risk — circumferential, complex, proximal, multiple; redo harder p.73
- KEY POINT Every principle — complete sac removal, watertight closure, fascia closed off-axis, dead space obliterated — exists to prevent the three failures: recurrence, fistula, incontinence. p.73

## 14 · MCQ ONE-LINERS

- MCQ Bladder diverticulum lacks muscularis propria — cannot empty. p.74
- MCQ Male : female ratio of bladder diverticula ≈ 9 : 1. p.74
- MCQ Acquired diverticula: BOO in ~70% (BPH). p.74
- MCQ Congenital diverticulum site: posterolateral to ureteric orifice (paraureteric). p.74
- MCQ Hutch diverticulum: paraureteric in neuropathic bladder with VUR. p.74
- MCQ Syndromes: Menkes, Williams, Ehlers-Danlos. p.74
- MCQ Paradoxic enlargement of diverticulum seen on voiding VCUG. p.74
- MCQ Pressure-sink effect falsely lowers apparent detrusor contractility. p.74
- MCQ Commonest tumour in bladder diverticulum: urothelial carcinoma; second SCC. p.74
- MCQ Size alone is not an indication for diverticulectomy. p.74
- MCQ Treat outlet obstruction before or with diverticulectomy. p.74
- MCQ Endoscopic neck resection risk: retention by venting. p.74
- MCQ Commonest theory of UD: periurethral gland infection (Routh). p.74
- MCQ Periurethral glands drain into distal third of urethra. p.74
- MCQ UD ostium: posterolateral, 4 and 8 o'clock, mid-urethra. p.74
- MCQ All "three Ds" present in only ~5%. p.74
- MCQ Best imaging for UD: MRI (T2 hyperintense). p.74
- MCQ VCUG without voiding images is non-diagnostic. p.74
- MCQ Commonest carcinoma in UD: adenocarcinoma. p.74
- MCQ Stones in UD: 4–10%. p.74
- MCQ L/N/S/C3: location, number, size, configuration, communication, continence. p.74
- MCQ Absent periurethral fascia = pseudodiverticulum. p.74
- MCQ Diverticulectomy incision: inverted U. p.74
- MCQ Fascia closed perpendicular to urethral suture line. p.74
- MCQ Suprapubic tube placed at start of diverticulectomy, not the end. p.74
- MCQ No synthetic sling with UD repair — autologous fascial sling. p.74
- MCQ De novo significant SUI after diverticulectomy: 7–16%. p.74
- MCQ Recurrence after diverticulectomy: 10–22%; risk — circumferential. p.74
- MCQ Pericatheter VCUG at 14–21 days before catheter removal. p.74
- MCQ Skene gland cyst distorts meatus and does not communicate. p.74
- MCQ Gartner duct cyst may receive ectopic ureter — image upper tract. p.74

## Ch 77 · Sexual Function and Dysfunction in the Female

### 1 · FUNCTIONAL ANATOMY

- **Skene glands** are the prostate homologue (PSA-positive). **Bartholin glands** are the bulbourethral homologue. MCQ The G-spot has no agreed anatomical basis. p.75

### 2 · PHYSIOLOGY OF THE SEXUAL RESPONSE

- ALGORITHM
  Touch on the clitoral complex/vulva → pudendal (somatic) afferents → **S2–S4** → parasympathetic efferents (nervi erigentes → cavernous nerves) → arteriolar dilatation → engorgement + lubrication
  p.75

- KEY POINT Pudendal injury removes the **trigger**. Pelvic plexus or cavernous nerve injury removes the **response**. p.75

- ALGORITHM
  Neuronal nitric oxide synthase → nitric oxide (NO) → guanylate cyclase → cyclic guanosine monophosphate (cGMP) → ↓ $Ca^{2+}$ → smooth muscle relaxation → engorgement. Phosphodiesterase type 5 (PDE5) breaks down cGMP; arginase uses up the NOS substrate. Vasoactive intestinal polypeptide (VIP) works with NO.
  p.75

- TABLE Deficiency · Thin, dry vagina with fibrosis → dyspareunia. Risk rises when oestradiol is **≤50 pg/mL** · ↓ Libido, arousal, sensation, orgasm p.75

- KEY POINT Many women do not follow a linear sequence. Under Basson's model, lack of *spontaneous* desire (common after menopause) is not a dysfunction. The same idea led DSM-5 to merge desire and arousal into one diagnosis. p.76

### 3 · CLASSIFICATION

- TABLE Desire + arousal · Hypoactive sexual desire disorder (HSDD); female sexual arousal disorder (FSAD); sexual aversion disorder · **Female sexual interest/arousal disorder (FSIAD)**: **≥3 of 6** — ↓ interest; ↓ fantasies; ↓ initiation or receptivity; ↓ pleasure, ↓ response to cues, ↓ sensation (in 75–100% of encounters). Aversion dropped p.76

- TABLE Orgasm · Female orgasmic disorder (FOD) after normal excitement · FOD: delay, infrequency, absence **or ↓ intensity** in **75–100%** of encounters. Normal-excitement requirement removed p.76

- Female sexual dysfunction (FSD): ~**40%** of women have a lifetime sexual problem, but only ~**8–12%** have a distressing disorder. **HSDD is the most common**, FOD second. p.76

### 4 · CAUSES (BIOPSYCHOSOCIAL)

- **Drugs**: selective serotonin reuptake inhibitors (SSRIs), serotonin–noradrenaline reuptake inhibitors (SNRIs), antipsychotics (↑ prolactin); β-blockers, angiotensin-converting enzyme (ACE) inhibitors, calcium channel blockers, thiazides, spironolactone (antiandrogen); antiandrogens, gonadotrophin-releasing hormone (GnRH) agonists; opiates, alcohol. MCQ Combined hormonal contraceptives work through two mechanisms (↓ ovarian testosterone and ↑ SHBG); oral oestrogens, tamoxifen and thyroxine also ↑ SHBG. p.76

### 5 · EVALUATION

- ALGORITHM
  **PLISSIT**: **P**ermission (normalise) → **L**imited **I**nformation (response cycle, ageing, drugs) → **S**pecific **S**uggestions (lubricants, **vaginal oestrogen**, positions, vibrators) → **I**ntensive **T**herapy = **referral**
  p.77

## 6 · HYPOACTIVE SEXUAL DESIRE DISORDER

- ALGORITHM
  Screen (DSDS) → history + focused examination → **correct modifiable factors** (drugs, depression, GSM, pain, partner's ED) → education / sex or couples therapy → **generalised acquired** HSDD that persists → drug therapy → review at 8 weeks. **Refer**: situational HSDD not responding to counselling, psychiatric illness, sexual trauma
  p.78

- TABLE Dose · **100 mg at bedtime**, daily; about half respond by 8 weeks · **1.75 mg subcutaneous** ≥45 min before sex; ≤1 per 24 h, **≤8/month** p.78

## 8 · FEMALE ORGASMIC DISORDER

- Treatment: **education is the cornerstone**, plus directed masturbation, sensate focus and mindfulness. If arousal is impaired, treat the arousal problem. If arousal is intact, increase stimulation (vibrators; MCQ the EROS clitoral vacuum is the only FDA-cleared FSD device). Oestrogen or testosterone helps postmenopausal women with deficiency. Clitoral phimosis or adhesions can be corrected surgically. p.79

- ALGORITHM
  **SSRI-induced dysfunction**: was it present before treatment (depression itself causes dysfunction)? → wait for tolerance (up to ~4 months) → still present after 3–6 months: lower the dose or switch to or add **bupropion** or mirtazapine; consider sildenafil
  p.79

## 9 · GENITO-PELVIC PAIN / PENETRATION DISORDER

- **Vulvodynia**: vulval pain, usually burning, for **≥3 months** with no identifiable cause; localised or generalised, provoked or unprovoked. p.79

- **Provoked vestibulodynia (PVD)** is the commonest type (~**10–12%** of women; associated with irritable bowel syndrome and fibromyalgia). Primary PVD starts with the first attempt at penetration (tampon or intercourse). It is associated with nulliparity, family history, enuresis and dysmenorrhoea. Secondary PVD (~two-thirds) follows a period of pain-free sex; the vestibular area involved is larger and the clitoral hood is often the most painful site. p.79

- ALGORITHM
  Education + vulval care (avoid irritants) → topical lidocaine 5% or lidocaine–prilocaine (EMLA) **10 min before intercourse** → **multimodal pelvic floor physical therapy** (myofascial release, trigger points, stretching, down-training, biofeedback, graded **dilators**, electrical stimulation) + psychological/couples therapy → vestibular oestrogen if hypo-oestrogenic → **vestibulectomy** for refractory localised PVD
  p.79

- ALGORITHM
  Non-hormonal lubricants and moisturisers + regular sexual activity (first line according to the North American Menopause Society, NAMS) → **low-dose vaginal oestrogen** (†minimal absorption, no progestogen needed) or †vaginal dehydroepiandrosterone (DHEA, prasterone 6.5 mg nightly, converted locally to sex steroids) or †ospemifene 60 mg orally (a selective oestrogen receptor modulator that acts as an oestrogen agonist on the vagina; for women who cannot use vaginal products; risk of venous thromboembolism) → systemic hormone therapy if vasomotor symptoms are also present
  p.80

- KEY POINT †After **breast cancer**, use non-hormonal measures first. Vaginal oestrogen only after discussion with the oncologist (concern with aromatase inhibitors). p.80

- **Genital-sparing cystectomy** (when oncologically safe) allows sex to resume earlier (**~6 vs 14 weeks**). Type of urinary diversion does not matter. After cystectomy for benign disease, most women were not counselled beforehand and almost none were asked about sex at follow-up; a third of cancer patients wanted sex-therapy or partner referral. **Never trade oncological control for function.** → Ch 101 p.80



- MCQ Female sexual reflex: pudendal afferent, parasympathetic (S2–S4) efferent. p.81
- MCQ Orgasm and resolution are sympathetic (T12–L1). p.81
- MCQ The glans clitoris is non-erectile; dorsal nerves lie superficial to the tunica. p.81
- MCQ Skene glands are the prostate homologue (PSA-positive). p.81
- MCQ Basson model: responsive desire; spontaneous desire not required. p.81
- MCQ Masters–Johnson has no desire phase; Kaplan added it. p.81
- MCQ DSM-5 FSIAD needs ≥3 of 6 criteria, ≥6 months, with distress. p.81
- MCQ DSM-5 GPPPD merges dyspareunia and vaginismus. p.81
- MCQ DSM-5 FOD adds reduced intensity. p.81
- MCQ Commonest FSD is HSDD; second is FOD. p.81
- MCQ PLISSIT: intensive therapy means referral. p.81
- MCQ BSSC-W is not validated; DSDS is validated. p.81
- MCQ FSFI: 19 items, 6 domains. p.81
- MCQ PISQ is the only validated prolapse/incontinence sexual questionnaire. p.81
- MCQ Random testosterone measurement is useless in women. p.81
- MCQ Vaginal pH ≥4.5 without infection suggests atrophy. p.81
- MCQ Wickham striae are pathognomonic of lichen planus. p.81
- MCQ Flibanserin: 5-HT1A agonist/5-HT2A antagonist, 100 mg at bedtime; alcohol warning. p.82
- MCQ Bremelanotide: melanocortin agonist, 1.75 mg subcutaneously, ≤8/month; hyperpigmentation. p.82
- MCQ Both HSDD drugs are approved only before menopause. p.82
- MCQ Thyroxine and oral contraceptives raise SHBG. p.82
- MCQ Bupropion is the choice for SSRI-induced dysfunction. p.82
- MCQ Coital anorgasmia without clitoral stimulation is normal. p.82
- MCQ EROS clitoral vacuum is the only FDA-cleared FSD device. p.82
- MCQ PGAD: arousal not relieved by orgasm; associated with Tarlov cysts. p.82
- MCQ Vulvodynia requires ≥3 months of pain without a cause. p.82
- MCQ Multimodal pelvic floor physical therapy is standard for PVD. p.82
- MCQ Vestibulectomy extends to within 1–2 mm of the meatus. p.82
- MCQ Female cystectomy: preserve lateral vaginal walls and the distal urethra. p.82
- MCQ Combined anterior + posterior colporrhaphy increases dyspareunia. p.82
- MCQ TOT spares the dorsal clitoral nerve; TVT may denervate the vaginal wall. p.82
- MCQ Bladder pain syndrome has the worst sexual impact of any LUTD. p.82
- MCQ Genital-sparing cystectomy: sexual activity resumes at ~6 vs 14 weeks. p.82

- MCQ Coital leakage at orgasm suggests detrusor overactivity; at penetration, SUI. p.82
- MCQ Topical vestibular oestrogen works better than systemic because it reduces vulval nerve density. p.82

## ★ Putting it together: female pelvic medicine decisions

### A · WOMAN WITH URINARY INCONTINENCE: THE PATHWAY

- ALGORITHM

**1. Classify** by history and bladder diary: stress (SUI), urgency (UUI), mixed (MUI), overflow, functional. Examine: cough stress test, urethral mobility, pelvic organ prolapse (POP) staging, post-void residual (PVR), urinalysis (→ Ch 105).

**2. Urodynamics** are not needed before surgery for uncomplicated, demonstrable SUI (VALUE trial). Use them for mixed or unclear symptoms, voiding dysfunction, previous surgery, neurological disease or high-grade prolapse.

**3. Conservative first:** weight loss, fluid and caffeine adjustment, supervised pelvic floor muscle training (PFMT), bladder training; pessary or vaginal device; vaginal oestrogen in postmenopausal women (→ Ch 114).

**4. SUI surgery** if conservative care fails (→ Ch 116, 118); UUI → drugs, botulinum toxin, neuromodulation (voiding book).

p.83

### C · PELVIC ORGAN PROLAPSE: WHICH REPAIR?

- ALGORITHM

**Stage** with POP-Q; leading edge sets the stage; stage II = within 1 cm of the hymen (→ Ch 105).

**Asymptomatic** → observe. **Symptomatic** → pessary or surgery.

**Apex is the key:** anterior and posterior defects often follow apical loss, so support the apex — vaginal uterosacral or sacrospinous suspension, or abdominal/robotic sacrocolpopexy (most durable) (→ Ch 117).

**Not sexually active and frail** → colpocleisis (high success, low morbidity).

**Occult SUI:** discuss a concomitant sling (OPUS: fewer post-op leaks, more complications).

p.83

### D · REPAIRING A VESICOVAGINAL FISTULA

- KEY POINT Confirm the type (double-dye test), image the upper tracts, and biopsy any fistula after radiation or cancer. Most post-hysterectomy fistulae are supratrigonal and repaired vaginally (Latzko or layered); choose the abdominal route when the ureter is involved, the fistula is large or high, or vaginal repair has failed. Principles: excise or freshen the tract, tension-free watertight closure in layers, non-overlapping suture lines, a vascularised flap (Martius vaginally, omentum abdominally) when tissue is poor or irradiated, and continuous bladder drainage before a cystogram (→ Ch 121). p.84

### E · MESH COMPLICATIONS IN FOUR RULES

- ALGORITHM

1. Examine (speculum) and do cystoscopy for any suspicion; complications can appear years later.
2. Small vaginal exposure → vaginal oestrogen ± trimming; larger or symptomatic → excision with closure.
3. Mesh in the bladder or urethra → remove it (endoscopic or open/transvaginal) and repair; no new mesh at the same time.
4. Remove enough mesh to solve the problem; never remove asymptomatic mesh (→ Ch 119).

p.84

# 7 Voiding Dysfunction

335 core items · page numbers refer to the full book PDF

## Ch 102 · Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

### 1 · BONY PELVIS

- Pelvic brim at **50–60°** to horizontal. **Arcuate line** (promontory → pubis) separates false pelvis (iliac fossae) from true pelvis (urogenital organs). p.2

### 2 · LOWER ABDOMINAL WALL

- ~**4 cm**; transmits **spermatic cord and ilioinguinal nerve**. Deep ring (transversalis fascia) lies midway between anterior superior iliac spine (ASIS) and pubic tubercle, **lateral to inferior epigastric vessels**; superficial ring (external oblique) superolateral to pubic tubercle. p.2

### 4 · PERINEUM

- Perineal prostatectomy MCQ: **Young** divides the central tendon, leaving external sphincter on the rectum; **Belt** passes **under** the external sphincter on the anterior rectal wall. p.3

### 6 · PELVIC NERVES

- KEY POINT Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous). p.4

- Continues as **cavernosal nerves** posterolateral to prostate, **outside Denonvilliers fascia**; with prostatic vessels = **neurovascular bundle (NVB)** at **5 and 7 o'clock** at the apex. Injury at prostatectomy or rectal resection → ED. p.4

### 7 · PELVIC VISCERA

- Ureter in **Waldeyer's sheath** runs **1.5–2 cm** obliquely; narrow → stone impaction. p.5

### 8 · ENDOSCOPIC ANATOMY

- ALGORITHM
  Flexible cystoscopy (local) → bladder, orifices, trigone → retroflex at neck (intravesical prostatic protrusion, hidden tumours) → verumontanum with ejaculatory ducts → external sphincter coapts just distal → bulbar urethra
  p.5

- KEY POINT Verumontanum = sentinel at transurethral resection of the prostate (TURP): resection distal to it injures the external sphincter → incontinence. p.5

### 9 · RADIOGRAPHIC ANATOMY

- Malignant node: rounded, irregular, heterogeneous; size most specific at ≥**1 cm** short axis. p.5

- **CT urogram**: excretory images at **7–10 min**; iliac-crossing narrowing and peristaltic gaps are normal. p.5

- **Magnetic resonance imaging (MRI)**: best soft-tissue contrast. T1 — bone, gadolinium enhancement; T2 — organ anatomy, fluid, prostate, seminal vesicles, pelvic floor. Distended bladder wall >**3 mm** = thickening. External sphincter = T2-dark doughnut below the apex. p.6

### 10 · MCQ ONE-LINERS

- MCQ Sacrospinous ligament divides the sciatic foramen into greater and lesser. p.6

- MCQ Iliac crest corresponds to L4; umbilicus T10. p.6

- MCQ Below arcuate line, posterior rectus sheath is absent. p.6

- MCQ Inguinal canal: spermatic cord and ilioinguinal nerve. p.6

- MCQ Indirect hernia lies lateral to inferior epigastric vessels. p.6

- MCQ Medial umbilical ligament = obliterated umbilical artery. p.6

- MCQ Butterfly haematoma: anterior urethral injury confined by Colles fascia. p.6
- MCQ Puborectalis: key to faecal continence. p.6
- MCQ Belt approach passes under the external anal sphincter. p.6
- MCQ Internal pudendal artery runs in Alcock's canal. p.6
- MCQ Deep circumflex iliac arises from external iliac. p.6
- MCQ Superior vesical arises from patent umbilical artery. p.6
- MCQ Inferior vesical artery supplies the neurovascular bundle. p.6
- MCQ Node of Cloquet: distal limit of PLND. p.6
- MCQ Genitofemoral nerve: lateral limit of PLND. p.6
- MCQ Obturator nerve injury → weak thigh adduction. p.6
- MCQ Femoral nerve injury: psoas hitch, retractor blade. p.6
- MCQ NVB at 5 and 7 o'clock, outside Denonvilliers fascia. p.6
- MCQ Bladder neck α1 closure → antegrade ejaculation. p.6
- MCQ Intramural ureter: 1.5–2 cm. p.6
- MCQ Posterior urethra best imaged by VCUG. p.6
- MCQ Membranous urethra narrowing is normal. p.6
- MCQ CT urogram excretory phase: 7–10 minutes. p.6
- MCQ Phlebolith mimics a distal ureteric stone. p.6
- MCQ Distended bladder wall on MRI >3 mm = thickening. p.6
- MCQ Commonest rectal injury site at prostatectomy: distal rectum near apex. p.6
- MCQ Arcus tendineus fascia pelvis: puboprostatic ligament → ischial spine. p.6

## Ch 103 · Physiology and Pharmacology of the Bladder and Urethra

### 1 · OVERVIEW

- KEY POINT **Why single-organ therapy fails:** bladder and urethra act reciprocally, so boosting contractility in detrusor underactivity fails unless the urethra also relaxes. Afferent (sensory) modulation acts on both at once — reduce it for overactivity, augment it for underactivity — which makes it the most promising therapeutic lever. p.7

### 2 · FUNCTIONAL ANATOMY

- Wall ≈ **50% collagen, 2% elastin** besides muscle. Collagen mostly outside muscle bundles; types I, III, IV predominate. p.7
- TABLE Where · Intramural rhabdosphincter (predominant) — gives **~50% of static urethral resistance** · Periurethral pelvic floor (mixed) p.8
- TABLE Rhabdosphincter mix · Male **65% slow / 35% fast**; female **87% slow / 13% fast** p.8

### 3 · UROTHELIUM: BARRIER AND SENSOR

- KEY POINT A damaged urothelium produces symptoms twice over — by **solute leak** into the lamina propria and by **altered signalling** (more ATP, less NO) to afferents. p.8

### 4 · DETRUSOR SMOOTH MUSCLE PHYSIOLOGY

- ALGORITHM
  ACh → **M3** (Gq) → ↑ [$Ca^{2+}$]i (L-type $Ca^{2+}$ entry + phospholipase C → IP3 → SR release) → $Ca^{2+}$–calmodulin → myosin light chain kinase (MLCK) → phosphorylates MLC20 → actin–myosin cross-bridge cycling → contraction → myosin light chain phosphatase (MLCP) dephosphorylates as $Ca^{2+}$ falls → relaxation
  p.9

- KEY POINT **Motor–sensory coupling:** afferents lie beside myocytes; spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and hence sensory input — efferent and afferent functions are inseparable. p.10

### 6 · PERIPHERAL INNERVATION

- TABLE Origin · **S2–S4** sacral parasympathetic nucleus (intermediate grey) · Thoracolumbar intermediolateral cell column (†**T10–L2**) · **Onuf's nucleus**, S2–S4 ventral horn (lateral border) p.11

### 7 · CENTRAL CONTROL

- KEY POINT **Lesion localisation logic:** forebrain lesion → loss of tonic inhibition → DO with preserved coordination; pontine lesion → retention; suprasacral cord lesion → loss of PMC coordination → DO + **detrusor–sphincter dyssynergia (DSD)**; sacral/cauda lesion → decentralised, acontractile bladder → Ch 104. p.12

### 8 · STORAGE AND EMPTYING PHASES

- ALGORITHM
  Tension-receptor (Aδ) firing exceeds threshold → PAG/PMC switch flips (if cortex permits) → sympathetic and pudendal outflow off → **urethral sphincter relaxes first** (seconds earlier) + urethral smooth muscle relaxes (NO) → detrusor contracts (parasympathetic) → flow → urethral flow afferents reinforce contraction until empty
  p.13

- SUI women void at Pdet at Qmax ≈ **19 $cmH_2O$**; healthy men ≈ **60–70 $cmH_2O$**. Low voiding pressure in a woman ≠ impaired contractility; male obstruction nomograms must not be applied to women. p.13

### 9 · URETHRAL CLOSURE MECHANISMS

- **Contributors to urethral pressure**: smooth muscle, striated muscle and lamina propria/vascular plexus. Nicotinic blockade rarely lowers tone >40%; α-blockade lowers urethral pressure by about **one third** → smooth muscle (α1A) tone matters; little cholinergic contribution. p.13

- KEY POINT Continence = **active muscle tone + passive anatomical coaptation**. The integral theory led to the mid-urethral sling (MUS), the surgical standard for SUI, yet DeLancey's data suggest intrinsic sphincter function matters more than support → Ch 109. p.13

### 10 · COMPLIANCE

- **Compliance C = ΔV / ΔP**. Physiological filling keeps pressure **<10 $cmH_2O$**; women with SUI filled at 50 mL/min rose only a few $cmH_2O$ to capacity. p.13

- † A detrusor leak point pressure ≥ **40 $cmH_2O$** in low-compliance neurogenic bladders predicts upper tract damage — compliance is the main renal-risk parameter → Ch 104. p.14

### 11 · PATHOPHYSIOLOGY LINKS

- Post-void residual (PVR) ≥ **300 mL** warrants further evaluation (also if bothersome LUTS, hydronephrosis, renal insufficiency, culture-proven UTI); PVR <300 mL does not prove normal contractility. p.14

### 15 · MCQ ONE-LINERS

- MCQ Muscarinic subtype most numerous in detrusor: **M2**; subtype mediating contraction: **M3**. p.15

- MCQ Most highly expressed adrenoceptor in human detrusor: **β3**. p.15
- MCQ Dominant α-subtype in urethra and prostate: **α1A**; in human bladder: **α1D**. p.15
- MCQ Main $Ca^{2+}$ source triggering detrusor contraction: **L-type VDCC**. p.15
- MCQ Calcium sensitisation: **Rho kinase** inhibits MLCP. p.15
- MCQ Key $K^+$ channel braking detrusor excitability: **BK** (large-conductance $Ca^{2+}$-activated). p.15
- MCQ Uroplakins bind **type 1 fimbriae** of uropathogenic *E. coli*. p.15
- MCQ Urothelial transmitter exciting afferents via P2X3: **ATP**; inhibiting them: **NO**. p.15
- MCQ Parasympathetic transmitter relaxing urethra during voiding: **NO**. p.15
- MCQ Somatic EUS motoneurons: **Onuf's nucleus** (S2–S4 ventral horn). p.15
- MCQ PMC = **Barrington's nucleus** (dorsal pontine tegmentum); lesion → retention. p.15
- MCQ Pontine storage centre projecting to Onuf's nucleus: **L-region**. p.15
- MCQ Normal micturition reflex: **spinobulbospinal**, Aδ afferents. p.15
- MCQ Reflex emerging after SCI: **C-fibre spinal**, capsaicin/RTX-sensitive. p.15
- MCQ Dopamine tonically **inhibits** PMC — lost in Parkinson disease. p.15
- MCQ Loss of supraspinal suppression of guarding reflex → **DSD**. p.15
- MCQ Female rhabdosphincter: **87%** slow-twitch fibres. p.15
- MCQ Collagen type ↑ in poorly compliant bladder: **type III**. p.15
- MCQ α-blockade lowers urethral pressure by about **one third**. p.15
- MCQ Oxybutynin dry mouth mainly from metabolite **desethyloxybutynin**. p.15
- MCQ OBTX cleaves **SNAP-25**. p.15
- MCQ Duloxetine acts on **Onuf's nucleus** to ↑ sphincter tone. p.15
- MCQ Aged bladder: α-contraction ↑, β-relaxation ↓. p.15
- MCQ Neurotrophin central to DO plasticity: **NGF**. p.15
- MCQ RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin. p.15

## Ch 104 · Pathophysiology and Classification of Lower Urinary Tract Dysfunction

### 2 · PATHOPHYSIOLOGY

- KEY POINT UI can occur from ISD alone, **never from hypermobility alone**. **DeLancey's hammock**: a firm suburethral layer lets abdominal pressure compress the urethra. In half of continent women the continence point is the **mid-urethra**, not the bladder neck. Male sphincteric incontinence resembles ISD, not hypermobility. p.16

### 3 · FUNCTIONAL CLASSIFICATION (WEIN) — 2 × 2 TABLE

- KEY POINT
  - Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ
  - α-agonist tightens outlet (storage); α-blocker relaxes it (emptying).
  - OnabotulinumtoxinA: intradetrusor → storage; sphincteric → emptying.
  - Naming the failure names the treatment family — no exact diagnosis needed.

p.17

### 6 · NOCTURIA AND NEUROGENIC DYSFUNCTION

- **Polyuria >40 mL/kg/24 h**. **Nocturnal polyuria**: night volume ÷ 24-h volume >**20% young, >33% elderly**; from loss of nocturnal vasopressin rise or obstructive sleep apnoea. p.18
- KEY POINT DSD needs an intact sacral arc **disconnected from the pons** — hence suprasacral, never suprapontine. p.18

### 7 · MCQ ONE-LINERS

- MCQ Wein: failure to store or empty, due to bladder or outlet. p.18
- MCQ Storage failure outside bladder and outlet: fistula. p.18
- MCQ Micturition coordinating centre: pons. p.18
- MCQ Stress continence: equal pressure transmission to proximal urethra (Enhorning). p.18
- MCQ Collagen-driven low compliance is drug-unresponsive. p.18
- MCQ Flow and PVR cannot distinguish weak detrusor from obstruction. p.18
- MCQ DLPP assesses bladder; VLPP assesses outlet. p.18
- MCQ BOO: high detrusor pressure with low flow. p.18
- MCQ Incontinence never results from hypermobility alone. p.18
- MCQ Smooth sphincter dyssynergia: autonomic dysreflexia. p.18
- MCQ Dysfunctional voiding is non-neurogenic. p.18
- MCQ Polyuria: >40 mL/kg/24 h. p.18
- MCQ Nocturnal polyuria index >33% in elderly. p.18
- MCQ Terminal dribble: voiding symptom; post-micturition dribble: post-micturition. p.18
- MCQ Suprasacral lesion: DO + DSD. p.18
- MCQ Sacral lesion: acontractile detrusor. p.18
- MCQ Suprapontine lesion: synergic voiding. p.18
- MCQ Maximum cystometric capacity: can no longer delay voiding. p.18
- MCQ Normally silent bladder afferents: C fibres. p.18
- MCQ Bladder underactivity causes overflow incontinence too. p.19

## Ch 106 · Evaluation and Management of Males With Urinary Incontinence

### 2 · TYPES

- TABLE UUI — with urgency · Detrusor overactivity (DO) overcomes a normal outlet · ~**90%** have DO on urodynamics; = wet overactive bladder (OAB) p.20
- TABLE With chronic retention · Failure to empty; residual ≥**300 mL** · "Overflow" no longer ICS term; spinal cord injury (SCI) ≥T6 → autonomic dysreflexia risk p.20

### 3 · EVALUATION

- **Bladder diary**: only fully validated = International Consultation on Incontinence Questionnaire Bladder Diary (ICIQ-BD). **3 days = 7 days**; include a weekend. Gives frequency, volumes, nocturnal polyuria, leaks. p.21
- KEY POINT Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS — usually just before surgery. p.21

## 4 · URGENCY URINARY INCONTINENCE

- **ALGORITHM**
  Lifestyle + bladder training + pelvic floor muscle training (PFMT) ≥6 weeks (ideally 3 months) → antimuscarinic or β3-agonist (± combination) → UDS optional → onabotulinumtoxinA / sacral neuromodulation (SNM) / posterior tibial nerve stimulation (PTNS) → augmentation or ileal conduit
  p.21
- Conservative: cut fluid by **25%** (if intake >1 L/day); avoid caffeine and fizzy drinks; stop smoking; lose weight. Start with drugs for quicker relief. p.21
- **TABLE** Drawback · Retention ~**10%** → must accept self-catheterisation · Most invasive, costliest · Frequent visits; no maintenance protocol p.22
- SNM test: ≥**50%** improvement → permanent generator; technique → Ch 115. No head-to-head trials of the three. p.22

## 5 · STRESS URINARY INCONTINENCE

- **ALGORITHM**
  Supervised PFMT ≥3 months (± biofeedback, electrical stimulation; better started pre-op) → ± duloxetine → exclude anastomotic/urethral stricture and bladder dysfunction (UDS) → surgery by severity
  p.22
- After radical prostatectomy wait **6–12 months** — continence keeps improving. Meanwhile: penile clamp, condom catheter, pads; avoid long-term urethral catheter (traumatic hypospadias). p.22
- **AUS = gold standard** for moderate–severe SUI (~80% continence; Scott 1973 MCQ).
  †Components: urethral cuff, pressure-regulating balloon, scrotal pump; activated about 6 weeks after implantation; deactivate before urethral catheterisation or endoscopy to avoid erosion. p.22

## 8 · MCQ ONE-LINERS

- **MCQ** Commonest male UI type: UUI. p.23
- **MCQ** Most important male continence determinant: EUS. p.23
- **MCQ** Male SUI cause: prostatectomy. p.23
- **MCQ** New adult enuresis: exclude high-pressure chronic retention. p.23
- **MCQ** Significant residual in chronic retention: ≥300 mL. p.23
- **MCQ** Retention with SCI ≥T6 risks autonomic dysreflexia. p.23
- **MCQ** ~90% of men with UUI show DO. p.23
- **MCQ** Only fully validated diary: ICIQ-BD. p.23
- **MCQ** 3-day diary equals 7-day. p.23
- **MCQ** IPSS does not measure incontinence. p.23
- **MCQ** ICIQ-UI-SF score range: 0–21. p.23
- **MCQ** Severe leak: >4 pads or >400 g/day. p.23
- **MCQ** Flow cannot distinguish BOO from underactivity. p.23
- **MCQ** Fluid reduction: 25% if intake >1 L/day. p.23
- **MCQ** Cognitive impairment: prefer β3-agonist. p.23
- **MCQ** Mirabegron: monitor blood pressure. p.23
- **MCQ** OnabotulinumtoxinA OAB dose 100 U; neurogenic DO 200 U. p.23
- **MCQ** Retention after toxin in men: ~10%. p.23
- **MCQ** Wait 6–12 months after prostatectomy before SUI surgery. p.23

- MCQ AUS: gold standard, ~80% continence. p.23
- MCQ Bulking agents not for PPI. p.23
- MCQ ProACT contraindicated after radiotherapy. p.23

## Ch 107 · Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

### 1 · DEFINITION AND ROLE

- KEY POINT **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; detrusor–external sphincter dyssynergia (DESD); detrusor–internal sphincter dyssynergia (DISD); high-pressure detrusor overactivity throughout filling; detrusor leak point pressure **> 40 cmH₂O**; poor emptying with high storage pressure. Each is a route to sustained high intravesical pressure. p.24

### 2 · PREPARATION

- **ICS pre-invasive work-up**: validated symptom/bother score, **3-day frequency-volume chart**, free uroflowmetry and PVR. A normal free flow and low PVR beforehand let an abnormal invasive flow be recognised as artefact. p.24

### 3 · NON-INVASIVE COMPONENTS

- **Qmax depends on VV**: linear above 150 mL, hyperbolic below → minimum **VV 150 mL**. Alternatives: corrected Qmax (Qmax/√VV), Siroky nomogram (men), Liverpool nomogram (men and women). p.25

### 4 · EQUIPMENT, TRANSDUCERS AND QUALITY CONTROL

- KEY POINT **Quality control** † (ICS Good Urodynamic Practice): zero to atmospheric pressure; flush lines free of air; check initial resting Pves and Pabd are physiological and similar (Pdet near 0); **cough test** before filling, at intervals and before voiding — equal Pves and Pabd spikes confirm transmission (a biphasic Pdet spike means poor transmission → re-flush/reposition). Common artefacts: rectal contractions (fall in Pdet without cause), catheter displacement or loss of signal, kinked tubing, flow artefact from stream hitting the funnel edge or tapping. p.26

### 5 · FILLING CYSTOMETRY (CMG)

- Filling rates: standard adult **30–100 mL/min** (medium); † ICS physiological rate < body weight (kg)/4 mL/min; slow in neurogenic bladders. Neurogenic: physiological rate, body-warm saline (EAU). p.26
- Absence does not exclude it: **up to 50%** of women with urgency urinary incontinence (UUI) show no DO. DO occurs in a minority of asymptomatic volunteers, more on ambulatory UDS. p.26
- No agreed normal; < 20 mL/cmH₂O proposed as poor. But a small-capacity bladder gives a low number without danger → **absolute storage pressure matters more**: **> 40 cmH₂O** associated with upper-tract damage. p.26

### 6 · LEAK POINT PRESSURES

- TABLE **< 60** · Intrinsic sphincter deficiency (ISD) p.27
- McGuire: myelodysplastic children with **DLPP > 40 cmH₂O** → hydronephrosis or VUR in 85%. Not a validated safe cut-off (20 cmH₂O more sensitive in children) — target **as low as achievable**. p.27

### 7 · URETHRAL PRESSURE PROFILOMETRY (UPP)

- **MUCP ≤ 20 cmH₂O** → ISD ("type III" SUI), predicted sling/suspension failure. p.27

### 8 · PRESSURE–FLOW STUDIES (PFS)

- TABLE **Bladder outlet obstruction index (BOOI)** · PdetQmax − 2Qmax · **> 40 obstructed · 20–40 equivocal · < 20 unobstructed** p.28

- TABLE **Bladder contractility index (BCI)** · PdetQmax + 5Qmax · **> 150 strong · 100–150 normal · < 100 weak** p.28

- MCQ Bethanechol supersensitivity test (Lapides; Cannon's law of denervation): 2.5 mg SC; rise **≥ 15 cmH₂O** at 100 mL = positive (neurogenic). Unreliable, rarely changes management (CIC regardless). p.28

### 14 · MCQ ONE-LINERS

- MCQ Pdet = Pves − Pabd. p.30
- MCQ Water-filled transducer reference level: upper border of symphysis pubis. p.30
- MCQ Minimum voided volume for reliable Qmax: 150 mL. p.30
- MCQ Plateau flow curve: fixed obstruction such as stricture. p.30
- MCQ Standard adult filling rate: 30–100 mL/min. p.30
- MCQ Compliance = ΔV/ΔPdet; < 20 mL/cmH₂O proposed as poor. p.30
- MCQ Storage pressure > 40 cmH₂O associated with upper-tract damage. p.30
- MCQ DLPP > 40 cmH₂O in myelodysplasia: hydronephrosis/VUR in 85% (McGuire). p.30
- MCQ ALPP < 60 cmH₂O suggests ISD; ≥ 90 suggests hypermobility. p.30
- MCQ VLPP is lower than cough LPP in the same patient. p.30
- MCQ MUCP ≤ 20 cmH₂O defines ISD (type III SUI). p.30
- MCQ Up to 50% of women with UUI show no DO on CMG. p.30
- MCQ Single terminal IDC predicts storage-symptom resolution after TURP. p.30
- MCQ BOOI = PdetQmax − 2Qmax; > 40 obstructed. p.30
- MCQ BCI = PdetQmax + 5Qmax; < 100 weak. p.30
- MCQ BOOI cannot be used in women. p.30
- MCQ Blaivas–Groutz nomogram overdiagnoses female obstruction. p.30
- MCQ VUDS (Nitti) female BOO: radiographic narrowing with sustained contraction of any magnitude. p.30
- MCQ Guarding reflex is normal EMG rise during an IDC. p.30
- MCQ DESD requires a suprasacral lesion. p.30
- MCQ DISD and PBNO are diagnosed only by VUDS. p.30
- MCQ Asymptomatic bacteriuria: give prophylaxis before UDS. p.30
- MCQ Prophylaxis of choice: single oral trimethoprim–sulfamethoxazole within 1 h. p.30
- MCQ Impaired compliance: only absolute urodynamic indication to treat BOO. p.30
- MCQ EAU PFS trigger: Qmax > 10 mL/s (not < 10). p.30
- MCQ Bethanechol test positive: rise ≥ 15 cmH₂O (Cannon's law). p.30
- MCQ VUDS is gold standard invasive UDS in NLUTD. p.30

## Ch 109 · Neuromuscular Dysfunction of the Lower Urinary Tract

### 1 · PRINCIPLES AND CLASSIFICATION

- KEY POINT Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**; an open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**. p.31

### 3 · SPINAL CORD DISEASE

- **Spinal shock**: depressed cord excitability at/below lesion → areflexic, acontractile smooth bladder; bladder neck closed, some striated activity, no voluntary control → retention, overflow only. **Catheterise** (CIC preferred) keeping pressure low. Lasts usually **6–12 weeks** (up to 1–2 years); reflex contractions return with lower-limb deep tendon reflexes. p.33

- **Suprasacral SCI**: DO (C-fibre–mediated reflex emerges through afferent sprouting, NGF) + smooth synergia + striated DSD → storage *and* emptying failure at high pressure. Lesions at or above cord **T7–T8** (column T6) may add smooth sphincter dyssynergia. p.33

- ALGORITHM
  †Sit upright, loosen clothing → **remove stimulus** (drain bladder, unblock catheter, rectum) → if BP persists: nitroglycerin 2% paste above lesion (first line) → labetalol / intravenous nitroprusside → prophylaxis: terazosin or prazosin (α1) → intractable: sacral rhizotomy, sympathectomy, cordectomy
  p.34

### 4 · LESIONS DISTAL TO THE CORD

- **Cauda equina syndrome**: saddle anaesthesia, loss of voluntary anal/urethral sphincter control and sexual response; acontractile detrusor. **Decompression within 48 h** gives better urinary and rectal recovery. Sphincter recovers before detrusor. p.34

- KEY POINT Distal lesions hit the peripheral reflex arc → areflexia with a non-relaxing/fixed outlet; compliance falls only when preganglionic parasympathetic motor fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery. p.35

### 5 · SPECIFIC FUNCTIONAL AND MISCELLANEOUS CONDITIONS

- **Non-neurogenic chronic retention**: PVR ≥**300 mL** persisting ≥6 months on ≥2 occasions. High risk (hydronephrosis, chronic kidney disease, recurrent urinary tract infection [UTI], UI with skin sequelae) → drainage; suprapubic preferred to urethral long term. p.35

### 6 · EVALUATION AND RISK STRATIFICATION

- KEY POINT Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); DO in the high-risk row means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not by diagnosis. p.36

### 7 · MANAGEMENT

- ALGORITHM
  Low-pressure storage + regular emptying: CIC + antimuscarinic / β3 agonist → intradetrusor onabotulinumtoxinA → neuromodulation (selected) → augmentation cystoplasty / continent diversion → bladder neck closure + suprapubic catheter / ileovesicostomy / incontinent diversion (last resort)
  p.36

- **CIC is the gold standard**; keep volumes <**500 mL**. Credé/Valsalva not recommended (outlet resistance → high pressure). p.36

### 9 · MCQ ONE-LINERS

- MCQ Suprapontine lesion: DO with synergic sphincters; true DSD never occurs. p.37
- MCQ DSD requires a lesion between the PMC and the sacral cord. p.37
- MCQ Smooth sphincter dyssynergia added in SCI at or above cord T7–T8 (column T6). p.37
- MCQ Open bladder neck + denervated striated sphincter = MSA, not PD. p.37
- MCQ Post-TURP incontinence in "PD" historically reflected undiagnosed MSA. p.37
- MCQ Preferred OAB drug in dementia: β3 agonist. p.37
- MCQ Spinal shock usually lasts 6–12 weeks; bladder areflexic, bladder neck closed. p.37

- MCQ Autonomic dysreflexia: SCI above T6; hypertension with bradycardia. p.37
- MCQ First-line drug for acute AD: nitroglycerin 2% paste after removing stimulus. p.37
- MCQ Blaivas type 1 DSD: incomplete lesion, sphincter relaxes at peak detrusor pressure. p.37
- MCQ Untreated DSD: about half of men develop upper-tract complications. p.37
- MCQ Commonest UDS finding in lumbar disc disease: normally compliant areflexic bladder. p.37
- MCQ Cauda equina: decompression within 48 h improves urinary recovery. p.37
- MCQ Radical pelvic surgery: open bladder neck, fixed striated tone, ↓ compliance. p.37
- MCQ Elsberg syndrome: herpes simplex sacral radiculitis with retention and CSF pleocytosis. p.37
- MCQ Classic diabetic cystopathy: ↓ sensation, ↑ capacity, ↓ contractility, ↑ PVR. p.37
- MCQ Typical adult myelodysplastic bladder: areflexic with open bladder neck. p.37
- MCQ Fowler syndrome EMG: complex repetitive discharges and decelerating bursts; treat with SNM. p.37
- MCQ Non-neurogenic chronic retention: PVR ≥300 mL for ≥6 months on ≥2 occasions. p.37
- MCQ CIC volumes should stay below 500 mL. p.37
- MCQ No antibiotic prophylaxis for CIC or indwelling catheter users. p.37
- MCQ AUA/SUFU 2021: no surveillance cystoscopy in NLUTD. p.37
- MCQ Sacral deafferentation with Brindley stimulator abolishes AD and NDO. p.37
- MCQ Primary bladder neck obstruction: definitive treatment is bladder neck incision. p.37

## Ch 110 · Overactive Bladder

### 1 · DEFINITIONS AND TERMINOLOGY

- TABLE Increased daytime frequency · Patient considers they void too often by day; upper normal **7** voids p.38
- **Bother**: urgency and UUI impair quality of life (QoL) most; **UUI is the most bothersome lower urinary tract symptom (LUTS) in both sexes**. Frequency bothers at ≥**8** daytime voids; nocturia at ≥**2**. p.38

### 2 · EPIDEMIOLOGY AND BURDEN

- Adult OAB ≈ **12–29%**; spread reflects definition — loosest (EpiLUTS) highest, strict ICS (EPIC) lowest. p.38
- <Half ever tell a clinician (not asked, "must cope", unaware treatment exists). ~**92%** stop the first antimuscarinic within 2 years; half never refill. Reasons: **unmet expectations** > side effects > cost. p.38

### 3 · PATHOPHYSIOLOGY

- **DO is no longer the sine qua non**: only ~half with DO have urgency; only **44–69%** with urgency show DO. No true animal model (animals cannot report urgency). p.39
- KEY POINT Final common pathway: more afferent signal reaches awareness — the bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB. p.39

### 5 · CLINICAL ASSESSMENT

- **History**: symptoms and bother; SUI; voiding symptoms (BOO); haematuria; fluid/caffeine; occult neurology (new OAB with erectile dysfunction or tremor, spinal disease); pelvic surgery, radiotherapy, bowel (faecal incontinence ~**50%** of women with OAB); drugs; polyuria (diabetes, sleep apnoea, lithium). p.39

- **Patient-reported outcome measures**: International Consultation on Incontinence Questionnaire (ICIQ) modules pair symptoms (ICIQ-OAB) with QoL (ICIQ-OABqol); AUA/SUFU recommend Urogenital Distress Inventory (UDI-6), Incontinence Impact Questionnaire (IIQ) or OAB-q. MCQ OAB-V8 = screening tool, not severity score. p.40

### 6 · URODYNAMICS (UDS)

- **Always assess emptying**: each **10 mL** PVR raises catheterisation risk after onabotulinumtoxinA by **27%**; BOO relief does not guarantee cure — DO persists in up to **60%** after transurethral resection of prostate (TURP). p.40

### 7 · MANAGEMENT

- AUA/SUFU "lines" are **not a compulsory sequence**; often combined. Allow **4–12 weeks** before judging any therapy. p.40
- ALGORITHM
  Confirm OAB, exclude mimics → education → conservative → drugs → tibial nerve stimulation / sacral neuromodulation / onabotulinumtoxinA → augmentation or diversion; reassess at every step p.40
- Fluid reduction ~**25%** helps (avoid under-hydration); caffeine reduction improves urgency and frequency, not consistently UUI. p.41
- **Pelvic floor muscle therapy (PFMT) and bladder training**: leaks ↓ **50–80%**, negligible risk. p.41
- **Dementia**: antimuscarinic use >**3 months** → 10–65% higher risk (SUFU) → prefer β3 agonists or trospium, darifenacin, fesoterodine. p.41
- KEY POINT Escalate by invasiveness and reversibility, not severity alone: OAB is benign, so patient goals, frailty and PVR decide how far to climb. p.41

### 8 · MCQ ONE-LINERS

- MCQ Defining symptom of OAB: urgency. p.41
- MCQ Upper normal daytime voids (ICS): 7. p.41
- MCQ Most bothersome LUTS in both sexes: UUI. p.41
- MCQ OAB is a symptom diagnosis; DO is urodynamic. p.41
- MCQ SUI + OAB dry is not MUI. p.41
- MCQ Urinary + faecal incontinence: double incontinence. p.41
- MCQ OAB fear: leakage; BPS fear: pain. p.41
- MCQ Highest prevalence study: EpiLUTS; lowest: EPIC. p.41
- MCQ Antimuscarinic discontinuation by 2 years: ~92%. p.41
- MCQ Urgency patients showing DO: 44–69%. p.41
- MCQ Silent afferents awakened in pathology: C fibres. p.41
- MCQ Main regulatory level: PAG and PMC. p.41
- MCQ Bladder-centric theories: urothelium-based and myogenic. p.41
- MCQ Faecal incontinence in women with OAB: ~50%. p.41
- MCQ Most important examination component: pelvic/genital. p.41
- MCQ Diary grade 0: convenience void. p.41
- MCQ DO correlates best with UUI, worst with nocturia. p.41
- MCQ Each 10 mL PVR: 27% higher catheterisation risk after toxin. p.41
- MCQ DO persists after TURP in up to 60%. p.41

- MCQ PFMT cuts incontinence episodes 50–80%. p.41
- MCQ Antimuscarinic dementia risk: beyond 3 months. p.41
- MCQ Minimum therapeutic trial: 4–12 weeks. p.41
- MCQ Nocturia becomes bothersome at ≥2 episodes. p.41
- MCQ Sex gap in OAB lies mainly in UUI. p.41
- MCQ Each 5% weight lost after bariatric surgery: ~7–8% lower incontinence risk. p.41
- MCQ Most cost-effective third-line: onabotulinumtoxinA 100 U; costliest: SNS. p.41

## Ch 111 · Underactive Detrusor

### 1 · DEFINITIONS AND SYMPTOMS

- Clinically significant PVR is not agreed; one proposal: >**40%** of functional capacity (voided volume + PVR). p.42
- KEY POINT **Sensation decides the picture.** Absent sensation → infrequent voider; if emptying is poor, overflow incontinence worse at night (urethra relaxes in sleep). Preserved sensation → frequent voider with incomplete emptying and nocturia. DUA may also be asymptomatic. p.42

### 2 · EPIDEMIOLOGY

- In UDS series: men <50 **9–28%**; men >70 ~**48%**; women 12–45%; highest in the institutionalised elderly, often with **detrusor hyperactivity with impaired contractility (DHIC)** — involuntary contraction emptying <half the bladder. Criteria differ, so figures cannot be pooled. p.42

### 4 · DIAGNOSIS

- TABLE **Bladder contractility index (BCI)** (Abrams) · Pdet@Qmax + 5Qmax · >150 strong; 100–150 normal; **<100 weak**; male-derived p.44

### 5 · MANAGEMENT

- ALGORITHM
  Evaluate (diary, digital rectal examination, urinalysis, flow, PVR; neurological deficit → spinal MRI; stop offending drugs; treat constipation) → timed/double voiding → lower outlet resistance where relevant → ISC for high PVR (suprapubic catheter if ISC impossible) → selected: sacral neuromodulation, outlet surgery, diversion
  p.44
- KEY POINT Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** (bethanechol, toxin, electrotherapy). p.45
- Poor outcome in retention: voiding pressure <**45** cm $H_2O$, age >**80**, PVR >**1500 mL**. p.45

### 6 · MCQ ONE-LINERS

- MCQ DUA definition (ICS 2002): reduced strength and/or duration. p.45
- MCQ ICS definition of DUA has no numerical thresholds. p.45
- MCQ UAB has no pathognomonic symptom. p.45
- MCQ Failure to void only during UDS: bashful bladder. p.45
- MCQ DUA in men >70 undergoing UDS: ~48%. p.45
- MCQ Elderly with DO plus weak contraction: DHIC. p.45
- MCQ Untreated non-neurogenic male DUA at 10 years: minimal progression. p.45
- MCQ "Stiff balloon" hypothesis: Smith. p.45
- MCQ Degenerative condition with DUA and retention: multisystem atrophy. p.45

- MCQ Stroke: acute retention; chronic DO. p.45
- MCQ BCI = Pdet@Qmax + 5Qmax; weak <100. p.45
- MCQ Apparent acontractility contracting on ambulatory UDS: >80%. p.45
- MCQ Not recommended in DUA: muscarinic agonists, anticholinesterases, Credé. p.45
- MCQ Bethanechol: rare fatal cardiac arrest. p.46
- MCQ Significant PVR proposal: >40% of capacity. p.46
- MCQ Preferred drainage: ISC; long-term indwelling: suprapubic. p.46
- MCQ Strongest predictor of treatment response: preserved sensation. p.46
- MCQ Brindley stimulator needs intact efferents and detrusor. p.46
- MCQ Outlet surgery fails if voiding pressure <45, age >80, PVR >1500 mL. p.46
- MCQ TURP in low-PVR DUA: no benefit at 11 years. p.46

## Ch 113 · Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

### 2 · ANTIMUSCARINICS (OVERACTIVE BLADDER, OAB; DETRUSOR OVERACTIVITY, DO)

- ALGORITHM
  † OAB: behavioural therapy (bladder training, fluids, pelvic floor) → antimuscarinic *or* β3-agonist (β3 first in elderly/cognitive risk) → switch or combine → onabotA 100 U / sacral neuromodulation / percutaneous tibial nerve stimulation → augmentation cystoplasty
  p.47

### 3 · B3-ADRENOCEPTOR AGONISTS

- TABLE Mirabegron · 25 → 50 mg od · Avoid in **severe uncontrolled hypertension** (≥**180/110** mm Hg); monitor blood pressure; moderate CYP2D6 inhibitor (metoprolol, desipramine); monitor digoxin p.47

### 4 · COMBINATION THERAPY

- KEY POINT Combination beats solifenacin 5 mg and either monotherapy; in non-responders it **equals** solifenacin 10 mg but with less dry mouth — it gains **tolerability, not efficacy**, over dose escalation. No change in heart rate, blood pressure, QT or PVR; constipation roughly doubles. Benefit irrespective of age ≥65 or ≥75. p.48

### 5 · INTRADETRUSOR ONABOTULINUMTOXINA (ONABOTA)

- TABLE Dose · **100 U** † (≈20 sites) · **200 U** † (≈30 sites) p.48

### 6 · NOCTURIA — DESMOPRESSIN †

- **Hyponatraemia** is the key risk, rising with age ≥**65**: check sodium at baseline, within 1 week and at 1 month; stop if low. p.48

### 11 · MCQ ONE-LINERS

- MCQ Receptor mediating detrusor contraction: M3. p.50
- MCQ Most M3-selective antimuscarinic: darifenacin. p.50
- MCQ Quaternary amine, minimal CNS penetration: trospium. p.50
- MCQ Transdermal oxybutynin: less dry mouth — avoids N-desethyloxybutynin. p.50
- MCQ Fesoterodine shares tolterodine's active metabolite. p.50
- MCQ Absolute antimuscarinic contraindication: uncontrolled narrow-angle glaucoma. p.50
- MCQ Cognitive side effects: M1 blockade. p.50
- MCQ Mirabegron caution: severe uncontrolled hypertension. p.50

- MCQ Mirabegron inhibits CYP2D6. p.50
- MCQ Mirabegron add-on evidence exists only with solifenacin. p.50
- MCQ Combination vs solifenacin 10 mg: equal efficacy, less dry mouth. p.50
- MCQ OnabotA dose: 100 U idiopathic OAB, 200 U NDO. p.50
- MCQ Commonest onabotA AE: UTI. p.50
- MCQ Desmopressin key risk: hyponatraemia, especially age ≥65. p.50
- MCQ Duloxetine acts at Onuf's nucleus. p.50
- MCQ Commonest duloxetine AE: nausea. p.50
- MCQ Phenylpropanolamine banned: haemorrhagic stroke. p.50
- MCQ Bethanechol: no convincing evidence for detrusor underactivity. p.50
- MCQ Imipramine: anticholinergic plus outlet α-effect. p.50
- MCQ Intraprostatic toxins failed against sham injection. p.50
- MCQ RimabotulinumtoxinB limitation: short duration. p.50
- MCQ Flupirtine halted for hepatotoxicity. p.50

## Ch 115 · Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

### 1 · EVOLUTION AND MECHANISM

- Milestones MCQ: sacral neuromodulation (SNM) ~1985 (Tanagho, Schmidt); first tibial nerve stimulation 1983 (McGuire); percutaneous tibial nerve stimulation (PTNS) derived from acupuncture point **SP6**; first tibial implant 2006. p.51

### 2 · SACRAL NEUROMODULATION

- Every patient has a test. **Success = ≥50% improvement** in ≥1 diary parameter (frequency, voided volume, leaks) vs a 3–5-day baseline; retention — catheterisation reduced ≥50%, ideally none. p.51
- Battery: non-rechargeable ≥**10 years**, rechargeable ≥15 years. p.52
- **Magnetic resonance imaging (MRI)**: current devices conditional for whole-body MRI at **1.5 and 3 T**; at replacement remove all old components. p.52

### 3 · TIBIAL NERVE STIMULATION

- ALGORITHM
  Supine → 34-gauge needle ≈3–4 cm (three fingerbreadths) above medial malleolus, behind tibia, at 60–90° → ground pad same leg → 20 Hz, 200 µs, 0–10 mA → confirm: **great-toe flexion/toe fanning** (motor) or sole tingling (sensory) → **12 weekly sessions × 30 min** → taper → maintenance
  p.52

### 7 · MCQ ONE-LINERS

- MCQ SNM target root: S3 — bellows and great-toe flexion. p.53
- MCQ S2 response: calf/foot external rotation; S4: perineal response, no toe. p.53
- MCQ SNM test success: ≥50% improvement in a diary parameter. p.53
- MCQ Urodynamic DO does not predict SNM success. p.53
- MCQ FDA-approved SNM: UUI, urgency–frequency, NOUR. p.53
- MCQ Neurogenic bladder and pelvic pain: not FDA-approved for SNM. p.53
- MCQ Staged implant preferred for NOUR; PNE not used. p.53

- MCQ Bilateral SNM not superior to unilateral. p.53
- MCQ Modern SNM devices: conditional MRI at 1.5 and 3 T. p.53
- MCQ SNM in retention acts mainly by sphincter relaxation. p.53
- MCQ Fowler syndrome predicts good SNM response. p.53
- MCQ Current SNM theory: afferent modulation of forebrain awareness centres. p.53
- MCQ PTNS: 34-gauge needle, 3–4 cm above medial malleolus, 20 Hz. p.53
- MCQ PTNS protocol: 12 weekly 30-minute sessions, then maintenance. p.53
- MCQ PTNS derived from acupuncture point SP6. p.53
- MCQ First tibial nerve stimulation for bladder: McGuire, 1983. p.54
- MCQ SUmiT: first validated sham; 55% vs 21%. p.54
- MCQ PTNS inhibition of OAB-type reflex is naloxone-sensitive. p.54
- MCQ TTNS less effective than PTNS due to skin impedance. p.54
- MCQ Pudendal stimulation: salvage after failed SNM. p.54
- MCQ Brindley stimulator combined with posterior sacral rhizotomy. p.54
- MCQ ROSETTA used 200 U toxin — twice the registered OAB dose. p.54

## ★ Putting it together: voiding dysfunction decisions

### B · WHERE IS THE LESION? EXPECTED URODYNAMICS

- KEY POINT Dyssynergia appears only when the lesion lies between the pons and the sacral cord, because the pons coordinates detrusor contraction with sphincter relaxation. A detrusor leak point or storage pressure ≥40 $cmH_2O$, or poor compliance, threatens the kidneys and is the main reason to treat (→ Ch 107, 109). p.55

### C · OVERACTIVE BLADDER: TREATMENT LADDER

- ALGORITHM

**1.** Behavioural: fluid and caffeine adjustment, bladder training, pelvic floor muscle training; treat constipation; review drugs.
**2.** Drugs: antimuscarinic or β3-agonist (prefer β3-agonist or a low-CNS antimuscarinic in the elderly); combine if partial response; check post-void residual in men with obstruction (→ Ch 113).
**3.** Refractory: intradetrusor onabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn about CIC), percutaneous tibial nerve stimulation, or sacral neuromodulation (staged test; success = ≥50% improvement) (→ Ch 115).
**4.** Last resort: augmentation cystoplasty or urinary diversion.

p.55

### D · SPINAL CORD INJURY: MANAGING THE BLADDER

- ALGORITHM

**Spinal shock** (areflexic bladder): indwelling or intermittent catheter until reflexes return.
**Then** urodynamics; CIC + antimuscarinic for a high-pressure overactive bladder; onabotulinumtoxinA if needed; sphincterotomy or diversion if CIC is impossible; lifelong upper-tract surveillance.
**Autonomic dysreflexia** (lesion above T6): sudden hypertension, bradycardia, headache, flushing above the lesion → sit upright, loosen clothing, find and remove the stimulus (blocked catheter, full bladder, constipation), short-acting antihypertensive if pressure stays high (→ Ch 109).

p.56

# 8 Nocturia & Male Incont.

97 core items · page numbers refer to the full book PDF

## Ch 112 · Nocturia

### 2 · WHY NOCTURIA MATTERS

- **Common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate. Boston Area Community Health survey MCQ: higher in Black and Hispanic people, partly via socio-economic status. p.2
- **Bothersome**: most with ≥**2** voids are bothered, but one void can be very bothersome — **any nocturia that brings a patient to a urologist is clinically significant**. It may dominate the interview and mask other LUTS. p.2
- **Sleep**: urge-driven waking is the **commonest cause of adult sleep disruption**. Slow-wave sleep (N3), the deepest and most restorative stage, clusters in the early night and is not recovered later; most first voids occur within **2–3 h** of sleep onset, so they land in it. Non-rapid eye movement sleep is ≈75% of sleep, in 4–6 cycles nightly, but slow-wave sleep is only 10–25%. Hence daytime fatigue even when total sleep time is preserved, and a shorter time to first void means worse whole-night sleep. p.2
- KEY POINT Nocturia matters because of when it strikes and what it signals: the first void breaks restorative slow-wave sleep, and the symptom may be the visible edge of systemic disease. p.3

### 3 · PATHOPHYSIOLOGY — PRODUCTION, CAPACITY AND SLEEP

- TABLE NPi (ICS) · >**20%** young to >**33%** older adults · Most used; poor discrimination p.3
- TABLE NUP (ICS alternative) · >**90 mL/h** · More specific for production-driven nocturia p.3
- TABLE Global polyuria · >**40 mL/kg/24 h** (>2.8 L at 70 kg) · Uncommon, but serious differential — **always investigate** p.3
- **Krimpen study** MCQ: NPi >33% in 92% of older men with ≥2 voids but also 70% of men **without** nocturia. p.3
- True capacity = voided volume + post-void residual (PVR); MVV/NMVV are proxies. No consensus: normal usually >300–400 mL; **150–300 mL** suggests diminished capacity — either **24-h** (all voids small) or **night-specific** (small night voids, normal day voids; partly convenience voids). p.3
- ALGORITHM
  NUV 800 mL, MVV 200 mL → Ni = 4 → PNV = 3 → diary shows 6 night voids → NBCi = **3** (voiding well below capacity)
  p.3

### 4 · THE BLADDER DIARY

- ALGORITHM
  Measuring container → time and volume of every void → time, volume and type of every drink → mark bedtime (intending sleep) and rising → urgency grade and leakage per void
  p.4
- Duration: EAU **3 days** (Male: at least 3). Longer diaries are more reliable but adherence falls ("diary fatigue"); completed by fewer than half of urology patients. p.4
- KEY POINT Every cause works through one equation: production against capacity, gated by sleep. The diary measures both sides, and their ratio (Ni) predicts severity better than either alone. p.4

### 6 · SLEEP DISORDERS

- CPAP reduces nocturia by up to ≈**2 voids** in moderate-to-severe OSA — among the **largest effects of any intervention**. EAU Female (strong): suspected OSA → sleep clinic. p.5

- **STOP-BANG**: Snoring, Tired, Observed apnoea, Pressure (hypertension), BMI >**35**, Age >**50**, Neck >**40 cm**, Gender male; ≥**3** = high risk; highly sensitive for moderate-to-severe OSA. p.5

- KEY POINT OSA is the model for the whole chapter: one disorder raises production, lowers capacity and fragments sleep — which is why treating it gives one of the largest nocturia responses. p.5

### 7 · COMORBIDITIES

- **Hypertension**: blood pressure normally dips **10–20%** in sleep; **non-dipping** probably mediates the link via renin–angiotensin–aldosterone dysregulation → **nocturnal natriuresis**. Hydrochlorothiazide improved nocturia in men with α-blocker-refractory LUTS. p.5

- KEY POINT Comorbid disease reaches the bladder mostly through the kidney — third-spaced fluid, nocturnal natriuresis and failed antidiuresis all raise night production. Global polyuria and non-bothersome nocturia are both reasons to look beyond urology. p.5

### 8 · ACTIONS — LIFESTYLE AND BEHAVIOUR

- **Practice**: restrict fluid, caffeine and alcohol from **3–4 h** before sleep; older adults should drink more by day to avoid dehydration. Patients with high baseline intake gain most. p.6

### 9 · LOWER URINARY TRACT DYSFUNCTION

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), detrusor underactivity, neurogenic bladder and the genitourinary syndrome of menopause (GSM) → Voiding Dysfunction book, Ch 110. p.6

- TABLE 5ARI · Offer with LUTS and prostate >**40 mL** (weak) · — p.7

- KEY POINT Lower urinary tract drugs act on capacity and outflow, but most nocturia is driven by production — so effects are statistically significant but clinically small. Treat the mechanism, not the organ. p.7

### 11 · MANAGEMENT ALGORITHM AND REFRACTORY NOCTURIA

- ALGORITHM
  History + examination + diary + basic tests → identify Sleep CALM factors → optimise them (multidisciplinary) → conservative measures → antidiuretic or timed diuretic in selected refractory patients — diary at every step
  p.7

- TABLE Dose · 1.66 mcg if <65, no risk; ≥65 or at risk → **0.83 mcg** · **Sex-specific**: women **25 mcg**; men **50 mcg** p.7

- **Hyponatraemia** risk factors (sodium <**130 mmol/L**): **age ≥65**, baseline sodium <135, creatinine clearance <60 mL/min, doses above approved. Most occur within **3 weeks**. p.8

- KEY POINT Refractory nocturia is treated by moving urine production out of the night. Either way the price is sodium balance, so dose, age and monitoring define safe use. p.8

### 13 · MCQ ONE-LINERS

- MCQ ICS 2019 nocturia: voids during the main sleep period, any cause. p.8

- MCQ First morning void: volume in NUV, counted as daytime frequency. p.8

- MCQ Mnemonic for causes: Sleep CALM. p.8

- MCQ NP (ICS): NPi >33% in older adults, or NUP >90 mL/h. p.8

- MCQ Global polyuria: >40 mL/kg/24 h. p.8

- MCQ Nocturia index: NUV / MVV. p.8

- MCQ NBCi = ANV − PNV; >2 severe. p.8

- MCQ Most important diary sleep measure: FUSP. p.8

- MCQ CPAP can reduce nocturia by ≈2 voids. p.8

- MCQ Only US drug approved for nocturia: desmopressin. p.8
- MCQ ODT doses: women 25 mcg, men 50 mcg. p.8
- MCQ EAU sodium checks: baseline, day 3, day 7, 1 month. p.8
- MCQ Loop diuretics contraindicate desmopressin. p.8
- MCQ Most first voids occur within 2–3 h of sleep onset. p.8
- MCQ Commonest cause of adult sleep disruption: nocturia. p.9
- MCQ Gold standard severity measure: diary-recorded ANV. p.9
- MCQ Krimpen study showed NPi discriminates poorly. p.9
- MCQ Nocturnal-only diary cannot diagnose global polyuria. p.9
- MCQ OSA raises night output via atrial natriuretic peptide. p.9
- MCQ STOP-BANG cut-offs: BMI 35, age 50, neck 40 cm. p.9
- MCQ Normal nocturnal blood pressure dip: 10–20%. p.9
- MCQ Calcium channel blockers cause nocturia via oedema. p.9
- MCQ PDE5Is are not offered for nocturia. p.9
- MCQ Isolated nocturia is not a LUTS. p.9
- MCQ 5ARI recommendation needs prostate >40 mL. p.9
- MCQ V2 receptor gene lies on the X chromosome. p.9
- MCQ Main hyponatraemia risk factor: age ≥65. p.9
- MCQ Diuretic timing for NP: morning or early afternoon. p.9
- MCQ Investigate small night capacity when NBCi >1.3. p.9
- MCQ Copeptin testing is superseding water deprivation for DI. p.9

## Ch 123 · Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

### 1 · PATHOPHYSIOLOGY AND DEVICE MECHANISMS

- KEY POINT Every device either compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but act by compression — only the Advance depends absolutely on a coapting residual sphincter. p.10

### 2 · INDICATIONS AND DEVICE SELECTION

- Indication: **irreversible sphincter incompetence** with bothersome urinary incontinence (UI) after conservative care. Do not delay beyond **6 months** for severe/gravitational SUI, especially with a cystoscopic sphincter defect. p.10
- Severity (24-h pad weight): mild **<150 g**, moderate **<400 g**. AUS preferred if >250 g or after RT (MASTER trial → Voiding Dysfunction book, Ch 106). p.10
- Counsel all: expect ~1 thin pad/day; AUS lifespan **7–10 years**, replaceable; failed slings mostly go to AUS. p.11
- Radiation cystitis: hyperbaric oxygen and **6 months** free of bleeding before AUS. p.11
- KEY POINT Each device's contraindication mirrors the other's requirement. p.11

### 3 · THE HIGH-RISK ("FRAGILE") URETHRA

- Prior erosion: 5-year repeat erosion approaches **50%**. Urethroplasty: worse than RT — transection divides bulbourethral arteries (loses dual blood supply). p.11

- KEY POINT Erosion = spongiosal ischaemia under constant pressure. Every risk factor reduces perfusion; every countermeasure adds tissue (TC tunica) or lowers pressure — never squeeze harder. p.11

### 4 · OPERATIVE TECHNIQUE

- Intravenous antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg + first/second-generation cephalosporin or vancomycin. Sterile urine beforehand. p.11

- ALGORITHM
  Midline perineal incision → Colles fascia → bulbospongiosus divided → incise Buck fascia where it reflects off the urethra onto the diverging crura → 2-cm tunnel dorsal to Buck fascia over the urethral roof under vision → right-angle clamp → measure circumference → cuff (usually 4.0–4.5 cm) → tubing through bulbospongiosus deep to Colles fascia
  p.12

- **PRB**: abdominal (lower-quadrant incision → split muscle → preperitoneal pocket → close fascia round tubing) or via external ring (perforate inguinal floor, or ectopic submuscular above transversalis fascia). Avoid the blind ring route after mesh hernia repair or cystectomy. Fill **23 mL** saline or isotonic diluted contrast (not if iodine allergy). p.12

- TABLE **61–70** · Standard bulbar cuff p.12

- **Transcorporal cuff** (George Webster 2002 MCQ): parallel longitudinal tunical incisions on each corpus beside the spongiosum → dissect through intercorporal septum → ventral tunica stays on dorsal urethra → imbricate tunical edges → measure, place. For urethra <3.5 cm, revisions, prior erosion/urethroplasty (use a fresh site). Adds bulk to the thin dorsal spongiosum and avoids urethral injury; erections usually preserved; more transient retention; erosion benefit unproven. p.12

- ALGORITHM
  **Advance XP**: perineal incision → open bulbospongiosus → mark and divide perineal body 1–2 cm (releases bulb) → stab incision one fingerbreadth below adductor longus tendon → helical trocar outside-in (two "pops") onto fingertip at spongiosum–ramus angle → mesh tension-free, 4 absorbable sutures to spongiosum → tension arms equally under cystoscopy until mucosa coapts (bulb moves 3–4 cm proximally) → bury arms, layered closure
  p.12

### 5 · AUS COMPLICATIONS AND MANAGEMENT

- ALGORITHM
  Confirm cuff deactivated → small urethral catheter 24–48 h → fails second trial → **suprapubic cystostomy** (ultrasound/fluoroscopy-guided, avoid PRB) → retention for weeks = undersized cuff → replace cuff
  p.13

- **1–3%** primary, up to 10% high-risk; *Staphylococcus epidermidis*, *S. aureus*. Antibiotic-coated (InhibiZone) devices have not reduced AUS infection. p.13

- **2–15%**; up to 35% with several risk factors. Remove **all components** (assume infected); loose closure/secondary intention. Post-erosion stricture 8–35%. p.13

- Reimplant at **3–6 months** after confirming patency, at a **new site**; TC ± low-pressure PRB; nightly deactivation. Repeat erosion up to 35%. p.13

- KEY POINT Tempo points to cause: sudden = device, gradual = urethra. Every complication is managed to keep a salvageable urethra for the next cuff. p.13

### 9 · MCQ ONE-LINERS

- MCQ AUS = gold standard for all SUI severities. p.14

- MCQ Only Advance sling depends absolutely on residual sphincter. p.14

- MCQ Any prior RT → AUS first-line. p.14
- MCQ Detrusor hypocontractility → AUS, not sling. p.14
- MCQ Standard bulbar PRB: 61–70 cm $H_2O$. p.14
- MCQ Bladder neck PRB: 71–80 cm $H_2O$. p.14
- MCQ AUS activation delayed 4–6 weeks. p.14
- MCQ Transcorporal cuff introduced by Webster, 2002. p.14
- MCQ Bladder neck AUS contraindicated after RP. p.14
- MCQ Strong erosion risks: RT, prior erosion, urethroplasty. p.14
- MCQ Repeat erosion after prior erosion approaches 50% at 5 years. p.14
- MCQ Early AUS infection sign: scrotal pain. p.14
- MCQ Atrophy: downsize cuff, never raise PRB pressure. p.14
- MCQ Mechanical failure >3 years: replace whole device. p.14
- MCQ Retention after AUS: suprapubic cystostomy after second failed trial. p.14
- MCQ Commonest ProACT complication: perforation. p.14
- MCQ PRB fill volume: 23 mL. p.14
- MCQ Mild SUI: <150 g/24 h. p.14
- MCQ Erosion: explant all components. p.14
- MCQ Failed sling → AUS. p.14

## ★ Putting it together: nocturia and male stress incontinence

### A · NOCTURIA: THREE QUESTIONS FROM THE BLADDER DIARY

- ALGORITHM

**1. Global polyuria?** 24-h output >40 mL/kg → look for diabetes mellitus or insipidus, polydipsia — always investigate.

**2. Nocturnal polyuria?** Night fraction >20% (young) or >33% (older adults) → heart failure, sleep apnoea, oedema states, evening fluids; consider desmopressin with sodium monitoring.

**3. Reduced night capacity?** Nocturnal bladder capacity index >1.3 → overactive bladder, obstruction, incomplete emptying; treat the lower urinary tract. Screen sleep disorders in all ("Sleep CALM") (→ Ch 112).

p.15

### B · MALE STRESS INCONTINENCE AFTER PROSTATECTOMY

- KEY POINT Pelvic floor training first, but do not delay surgery beyond about 6 months for severe leakage. Grade by 24-h pad weight (mild <150 g, moderate <400 g). Mild–moderate leakage, contractile detrusor, no radiotherapy → male sling; heavy leakage (>250 g), any radiotherapy or a failed sling → artificial urinary sphincter (cuff, pressure-regulating balloon, scrotal pump). Erosion follows spongiosal ischaemia — risk rises after radiotherapy, previous erosion or urethroplasty; counter it by adding tissue (transcorporal cuff) or lowering pressure, never a tighter cuff. † An eroded or infected device is removed (→ Ch 123; Voiding Dysfunction book, Ch 106). p.15

# 9 Andrology

464 core items · page numbers refer to the full book PDF

## Ch 66 · Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

### 1 · SCROTUM

- KEY POINT Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum — †so scrotal violation in testis cancer adds inguinal nodes to the at-risk field. p.2

### 2 · TESTIS

- Volume **15–25 mL**, length 4.5–5.1 cm. Capsule: visceral tunica vaginalis → **tunica albuginea** (contains smooth muscle — possible contractility) → tunica vasculosa. p.2
- ~250 lobules; **600–1200 seminiferous tubules** (280–400 m) ending in rete testis. Interstitium **20–30%** of volume — **Leydig cells** (testosterone). **Sertoli cells** line tubules; their tight junctions form the blood–testis barrier. p.2
- KEY POINT **Surgical rules**: no traction suture through the lower pole; biopsy the **medial or lateral upper pole**; the midportion has the fewest vessels. After prior vasectomy the deferential collateral may be gone → preserve the testicular artery in later cord surgery and counsel about devascularisation. p.2
- Right vein enters IVC obliquely (valve-like); left enters left renal vein → **~90% of varicoceles left**. p.3

### 4 · VAS DEFERENS

- Length **30–35 cm**; lumen **0.2–0.7 mm**; **three muscle layers** (outer longitudinal, middle circular, inner longitudinal). p.4

### 5 · SEMINAL VESICLES AND EJACULATORY DUCTS

- **transrectal ultrasound (TRUS)**: post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** = obstruction; guides aspiration and vesiculography. **MRI** is modality of choice; CT for stones. Vesiculoscopy (4.5–6.9 Fr) before endoscopic treatment. p.4
- KEY POINT A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction. p.4

### 6 · PROSTATE

- TABLE Peripheral · **70%** · Most cancers; hyperechoic on TRUS p.4
- Normal **~25 mL**; enlarges with age, fastest in transition zone. Internal (preprostatic) sphincter — circular smooth muscle thickest at bladder neck — continence at rest. p.5
- TRUS volume = L × W × H × **0.52** (underestimates). **Multiparametric MRI**: T2, diffusion-weighted, dynamic contrast; Prostate Imaging Reporting and Data System (PI-RADS) v2.1; targeted biopsy reduces insignificant-cancer detection. p.5

### 7 · URETHRA

- TABLE Membranous (**2–2.5 cm**) · Striated sphincter; narrowing on retrograde urethrogram (RUG) p.5

### 8 · PENIS

- Sinusoidal smooth muscle **~45%** of volume; endothelial nitric oxide synthase sustains erection. p.6
- **Dorsal nerve** (pudendal, **S2–S4**) — sensation. **Cavernous nerves** (pelvic plexus, autonomic) — erection. p.6

- KEY POINT Erection is veno-occlusive: filled sinusoids compress subtunical and oblique emissary veins against the inelastic tunica; †tunical or smooth-muscle damage defeats this → venous leak. p.6
- TABLE Peak systolic velocity · **<30 cm/s** · Arterial insufficiency p.6
- TABLE End-diastolic velocity · **>5 cm/s** · Veno-occlusive dysfunction p.6
- TABLE Resistive index · **<0.80** · Veno-occlusive dysfunction p.6

### 9 · MCQ ONE-LINERS

- MCQ Cremaster: striated, from internal oblique. p.6
- MCQ Internal spermatic fascia continues transversalis fascia. p.6
- MCQ Scrotal lymph → superficial inguinal; testicular lymph → para-aortic/interaortocaval. p.6
- MCQ Testicular artery arises from aorta below renal artery. p.6
- MCQ Cremasteric artery from inferior epigastric; deferential from superior vesical. p.6
- MCQ Safest testis biopsy site: medial or lateral upper pole. p.6
- MCQ Countercurrent exchange: testicular arterial blood 2–4 °C below rectal. p.6
- MCQ ~90% of varicoceles are left-sided. p.6
- MCQ Hydrocele after varicocelectomy = lymphatic ligation. p.6
- MCQ Appendix testis müllerian; appendix epididymis wolffian. p.6
- MCQ Epididymal cauda lymph → external iliac nodes. p.6
- MCQ Vas crosses anterior to the ureter; lumen 0.2–0.7 mm. p.6
- MCQ Vasography only at reconstruction. p.6
- MCQ Seminal vesicle >16 mm or >6 mL post-ejaculation suggests obstruction. p.6
- MCQ Ejaculatory duct orifices: 5 and 7 o'clock. p.6
- MCQ Peripheral zone: 70% of gland, most cancers; BPH in transition zone. p.6
- MCQ TRUS ellipsoid coefficient 0.52. p.6
- MCQ Cavernous nerves travel in lateral prostatic fascia. p.6
- MCQ Striated sphincter motor neurons: Onuf nucleus. p.6
- MCQ Cowper glands open into bulbar urethra. p.6
- MCQ Common penile artery: bulbourethral, cavernosal, dorsal branches. p.7
- MCQ Deep dorsal vein lies beneath Buck fascia. p.7
- MCQ Glans is the distal expansion of corpus spongiosum. p.7
- MCQ Corporal smooth muscle ≈ 45% of volume. p.7
- MCQ Membranous urethra 2–2.5 cm long. p.7
- MCQ Penile duplex: peak systolic velocity (PSV) <30 cm/s arterial; end-diastolic velocity (EDV) >5 cm/s venous leak. p.7

## Ch 67 · Male Reproductive Physiology

### 1 · HYPOTHALAMIC–PITUITARY–GONADAL AXIS

- Gonadotrophin-releasing hormone (GnRH): 10-amino-acid peptide from preoptic and arcuate nuclei, reaching the anterior pituitary via the portal system; sole known function is LH/FSH release. Half-life **5–7 min**. p.8

- Rhythms: pulsatile (every **90–120 min**), circadian (testosterone highest in early morning), seasonal. p.8

- Testosterone ~**5 mg/day**, pulsatile. Metabolites: **dihydrotestosterone (DHT)** via 5α-reductase — more potent, needed in most peripheral tissues (not testis or muscle); **oestradiol** via aromatase — feedback especially on FSH. p.8

- KEY POINT The axis runs two loops: LH ↔ Leydig governed by testosterone, and FSH ↔ Sertoli governed by inhibin and oestradiol. Hence **LH reports Leydig function; FSH and inhibin B report the tubules** — raised FSH with low inhibin B signals spermatogenic failure. p.9

### 2 · TESTIS

- Volume 15–25 mL; **600–1200 tubules**, ~250 m; interstitium 20–30%; ~**700 million Leydig cells** in young adults. p.9

- **Countercurrent exchange** in the pampiniform plexus cools arterial blood **2–4 °C** below core and recycles testosterone from vein to artery. Loss of this gradient underlies dysfunction in varicocele and cryptorchidism. p.9

- Functions: unique adluminal milieu; ABP keeps intratubular androgen ~**50-fold** above serum; **inhibin B** (marker of Sertoli function in infertility); phagocytosis of residual bodies; fluid secretion; c-kit/kit ligand signalling with germ cells; spermiation. p.10

### 3 · SPERMATOGENESIS

- Duration classically **~64 days** (in vivo labelling 42–76 days). Output ~**123 million/day** (~1000 per heartbeat). p.10

### 4 · EPIDIDYMIS

- **Transport**: **2–12 days**; driven mainly by spontaneous rhythmic duct contraction (plus fluid resorption and efferent cilia). Recent ejaculation shortens only cauda transit. p.11

- KEY POINT Pregnancy occurs even after vasoepididymostomy at the efferent ductules (obstruction may let sperm mature proximally), but **the more distal the anastomosis, the better the fertility**. After vasectomy >15 years, motility falls after reversal — acquired epididymal dysfunction. p.12

### 5 · VAS DEFERENS, SEMINAL VESICLES, EJACULATORY DUCTS

- **Seminal vesicle**: 5–7 cm coiled tubule; supplies ~**80%** of seminal volume in the late ejaculate fractions; alkaline, **fructose**, prostaglandins; **semenogelin I** forms the coagulum. Semen also provides antioxidant protection (glutathione peroxidase, superoxide dismutase, catalase). p.12

### 7 · EJACULATION PHYSIOLOGY †

- ALGORITHM

  **Emission** (sympathetic, T10–L2 via hypogastric nerve; α1): vas, seminal vesicle and prostate contract → semen into posterior urethra; **bladder neck closes → Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions with relaxed external sphincter → antegrade ejaculate.

  p.13

### 8 · MCQ ONE-LINERS

- MCQ GnRH: decapeptide, half-life 5–7 min, pulses every 90–120 min. p.13

- MCQ Continuous GnRH agonist suppresses testosterone by abolishing pulsatility. p.13

- MCQ Kallmann syndrome: failed GnRH neurone migration. p.13

- MCQ Kisspeptin and leptin stimulate GnRH; opioids inhibit. p.13

- MCQ LH and FSH share the α subunit; β confers specificity. p.13

- MCQ Hyperprolactinaemia abolishes gonadotrophin pulsatility. p.13

- MCQ Testosterone production ~5 mg/day. p.13

- MCQ Testosterone feeds back mainly at hypothalamus; oestradiol at pituitary. p.13
- MCQ Inhibin B: Sertoli-derived, suppresses FSH. p.13
- MCQ SRY encodes an HMG-box protein; DAX1, WNT4 are antitestis genes. p.13
- MCQ INSL3 mediates transabdominal testicular descent. p.13
- MCQ StAR protein: rate-limiting cholesterol transport into mitochondria. p.13
- MCQ Young adult testis: ~700 million Leydig cells. p.13
- MCQ ABP keeps intratubular androgen ~50× serum. p.13
- MCQ Blood–testis barrier: Sertoli–Sertoli tight junctions. p.13
- MCQ Prepubertal testicular insult does not induce antisperm antibodies. p.13
- MCQ Spermatogenesis ~64 days; six stages in humans. p.13
- MCQ Helical waves make sperm production continuous. p.13
- MCQ Ap spermatogonia divide every 16 days. p.13
- MCQ Meiotic centromeres divide at anaphase II. p.13
- MCQ Acrosome from Golgi; flagellum from centriole. p.13
- MCQ AZFc deletion: sperm retrieval often possible; AZFa/b complete: unlikely. p.13
- MCQ Selfish spermatogonia (FGFR2/3) → spermatocytic seminoma. p.13
- MCQ Epididymal transit 2–12 days. p.13
- MCQ Fertilising ability acquired at late corpus/early cauda. p.13
- MCQ Vas: muscle-to-lumen ratio ~10:1. p.13
- MCQ Seminal vesicles supply ~80% of ejaculate; fructose source. p.13
- MCQ Ejaculate pH <7.2 suggests absent or obstructed seminal vesicles. p.13
- MCQ Semenogelin I forms the semen coagulum. p.13
- MCQ Paternal mtDNA is degraded after fertilisation. p.13
- MCQ Globozoospermia: PLCζ deficiency, ICSI may fail. p.13
- MCQ Inner dynein arm mutants: complete immotility. p.13

## Ch 68 · Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

### 2 · THE LONGEVITY GAP

- Life expectancy at birth: males **76.1** vs females 81.1 years (2016). p.14

### 4 · UROLOGIC DISORDERS OF FOCUS

- **Diagnosis**: low T confirmed on **more than one occasion** + well-documented symptoms. † American Urological Association (AUA): total T **<300 ng/dL** on **2 early-morning samples**. p.15
- KEY POINT † **Fertility caution**: exogenous T suppresses LH/follicle-stimulating hormone (FSH) and intratesticular T → oligo/azoospermia. Men wanting children: avoid TRT; use clomiphene, human chorionic gonadotropin (hCG) or aromatase inhibitors instead → Ch 69. p.15
- TABLE Monitoring · T, haematocrit and PSA at baseline, 3–6 months, then annually. Haematocrit **>54%** → hold/reduce dose, phlebotomy. PSA rise or abnormal digital rectal examination → urology evaluation p.15

### 5 · CARDIOVASCULAR PREVENTION

- TABLE LDL-C 70–189, 10-year risk ≥ **7.5%** · Moderate–high intensity statin p.16

### 7 · GASTROINTESTINAL, SKIN AND MENTAL HEALTH

- Iron deficiency anaemia: >30-fold risk of gastrointestinal (GI) malignancy. Colorectal screening **45–75** years. p.16

### 8 · MCQ ONE-LINERS

- MCQ Largest male:female mortality gap among leading causes — suicide (>3-fold). p.17
- MCQ Males have higher mortality in 12 of 15 leading causes of death. p.17
- MCQ Only US cancer with rising stage IV at diagnosis — prostate cancer. p.17
- MCQ ADAM questionnaire screens androgen deficiency; SHIM screens erectile function. p.17
- MCQ TD diagnosis — low T on more than one occasion plus symptoms. p.17
- MCQ † AUA total T cut-off — <300 ng/dL, two early-morning samples. p.17
- MCQ † Haematocrit on TRT — AUA withholds at >50%; Endocrine Society at >54%. p.17
- MCQ † TRT in man wanting fertility — contraindicated; suppresses spermatogenesis. p.17
- MCQ Chronic opioids — screen for testosterone deficiency (Endocrine Society). p.17
- MCQ TRT for major depression — not recommended. p.17
- MCQ Randomised TRT benefits — obesity, metabolic syndrome, diabetes, muscle, bone. p.17
- MCQ Any man with ED — considered at risk of CVD. p.17
- MCQ Highest-satisfaction ED treatment — penile prosthesis. p.17
- MCQ PDE5 inhibitor absolute contraindication — nitrates. p.17
- MCQ Only lipid target with primary-prevention benefit — LDL-C. p.17
- MCQ LDL-C ≥190 mg/dL — high-intensity statin. p.17
- MCQ ACC/AHA statin threshold — 10-year ASCVD risk ≥7.5%. p.17
- MCQ CAC most useful at — intermediate risk; zero score = <1% atherosclerosis. p.17
- MCQ Lipoprotein(a) — not lowered by statins. p.17
- MCQ Stage 1 hypertension (2017) — 130–139/80–89 mm Hg. p.17
- MCQ Weight gain ≥5 kg after 20 — raises coronary and T2DM risk. p.17
- MCQ Iron deficiency anaemia — >30-fold GI malignancy risk. p.17
- MCQ Colorectal screening starts at — 45 years. p.17
- MCQ UTI or retention in older man — evaluate for constipation. p.17
- MCQ Commonest skin cancer — basal cell carcinoma. p.17
- MCQ Psoriasis associated with — MI/CVD and psoriatic arthritis. p.17

## Ch 69 · Male Infertility

### 1 · DEFINITION AND OVERVIEW

- Infertility: no pregnancy after **1 year** of unprotected intercourse; ~15% of couples. Males and females contribute equally — always evaluate the man. p.18

### 2 · HISTORY

- KEY POINT **Exogenous testosterone (T) is a contraceptive**: it is aromatised to oestradiol (E2) and feeds back on the pituitary → ↓ luteinising hormone (LH) → ↓ intratesticular T (which must be far higher than serum) → ↓ spermatogenesis. Recovery usually ≥4 months after stopping (≤2 years' use); after longer use, unknown. p.18

- Orchidopexy within **18 months** (AUA); germ cells are lost progressively from 15–18 months. If missed, before 10 years improves later surgical sperm retrieval; age at orchidopexy is second only to T in predicting retrieval. p.19

### 3 · PHYSICAL EXAMINATION

- TABLE Seager caliper (long axis) · **≤4.6 cm** p.19

### 4 · ENDOCRINE EVALUATION

- ALGORITHM
  Morning sample: total T + sex hormone–binding globulin (SHBG) + albumin → calculated bioavailable T → LH, follicle-stimulating hormone (FSH) → E2; prolactin only if indicated
  p.19

- Adequate androgenisation: total T ≥ **280–300 ng/dL**. Low T is common even with density >20 million/mL → assess every infertile man (full endocrine work-up unnecessary above ~66 million/mL). p.19

- T (ng/dL) : E2 (pg/mL) ratio **<10:1** → aromatase excess. p.20

### 5 · SEMEN ANALYSIS

- TABLE Volume (mL) · **1.4** · 3.0 p.20

- TABLE Concentration (million/mL) · **16** · 66 p.20

- TABLE Normal forms, strict (%) · **4** · 14 p.20

### 6 · GENETIC TESTING

- ALGORITHM
  Non-obstructive azoospermia (NOA) or density **<5 million/mL** → karyotype + Y-microdeletion (ASRM/AUA 2021) · CBAVD (or idiopathic EDO) → CFTR testing of man **and partner**
  p.21

### 7 · IMAGING

- **TRUS** for EDO (azoospermia/oligozoospermia + low volume): seminal vesicle anteroposterior diameter **>1.5 cm** ± midline cyst; low specificity. p.21

### 8 · AZOOSPERMIA: OBSTRUCTIVE (OA) VS NON-OBSTRUCTIVE (NOA)

- TABLE FSH · **≤7.6 IU/L** · >7.6 IU/L p.21

- ALGORITHM
  Azoospermia → vasa absent? Yes → CFTR (both partners) · No → volume <1 mL? Yes → postejaculatory urinalysis: sperm → retrograde ejaculation; none → TRUS for EDO · Normal volume → FSH + testis size → OA vs NOA
  p.21

### 9 · SPECIFIC CAUSES AND TREATMENT

- Ejaculated sperm ~8%; testicular sperm retrieval ~**50%**. p.21

### 12 · MCQ ONE-LINERS

- MCQ Most important determinant of couple fertility — female partner's age. p.23

- MCQ Commonest identifiable correctable cause — varicocele. p.23

- MCQ Commonest genetic cause — Klinefelter (47,XXY). p.23

- MCQ Optimal abstinence before semen analysis — 1–2 days. p.23

- MCQ Sperm-safe lubricant — Pre-Seed. p.23

- MCQ Exogenous T effect on spermatogenesis — suppression via ↓ LH and intratesticular T. p.23

- MCQ Abnormal caliper testis long axis — ≤4.6 cm. p.23

- MCQ OA vs NOA FSH cut-off — 7.6 IU/L. p.23

- MCQ Adequate total T threshold — 280–300 ng/dL. p.23
- MCQ T:E2 ratio suggesting aromatase excess — <10:1. p.23
- MCQ WHO 6th 5th-centile concentration — 16 million/mL. p.23
- MCQ WHO 6th 5th-centile strict normal forms — 4%. p.23
- MCQ WHO 6th 5th-centile volume — 1.4 mL. p.23
- MCQ Only definitive semen result — azoospermia. p.23
- MCQ Low volume + azoospermia + no urinary sperm — TRUS for EDO. p.23
- MCQ TRUS sign of EDO — seminal vesicle >1.5 cm anteroposterior (AP). p.23
- MCQ Karyotype/Y-microdeletion indication — NOA or <5 million/mL. p.23
- MCQ AZFa or AZFb complete deletion — retrieval essentially nil. p.23
- MCQ Y deletion with retrievable sperm — AZFc. p.23
- MCQ CBAVD — test CFTR in man and partner. p.23
- MCQ Unilateral absent vas — renal ultrasound for agenesis. p.23
- MCQ Kallmann syndrome — HH + anosmia; KAL1. p.23
- MCQ Fertile eunuch — isolated LH deficiency. p.23
- MCQ First-line in HH — hCG ± FSH. p.23
- MCQ Klinefelter testicular sperm retrieval — ~50%. p.23
- MCQ Subclinical varicocele — do not repair. p.23
- MCQ Isolated right varicocele — exclude renal tumour. p.23
- MCQ Main varicocele injury mechanism — heat (lost countercurrent exchange). p.23
- MCQ Orchidopexy timing — by 18 months. p.23
- MCQ Cryptorchidism prognostic factor — laterality. p.23
- MCQ Prolactinoma drug of choice — cabergoline. p.23
- MCQ Globozoospermia — absent acrosome; ICSI. p.23
- MCQ Macrocephalic multiflagellate sperm — AURKC; ICSI not recommended. p.23
- MCQ Vitality test preserving sperm — hypo-osmotic swelling. p.23

## Ch 70 · Surgical Management of Male Infertility

### 2 · SURGICAL ANATOMY

- Epididymis may be dissected off the testis to the caput (inferior and medial epididymal arteries ligated) if the **superior epididymal artery is intact** → gains **4–6 cm**. p.24

### 3 · DIAGNOSTIC TESTIS BIOPSY

- ALGORITHM

  Local with cord block → larger testis (contralateral to a varicocele) → 2 cm scrotal incision → open tunica vaginalis → two 5-0 stay sutures parallel to albugineal vessels → 3–4 mm incision with 15° microknife, bipolar crossing vessels → pea-sized sample with iris scissors (no forceps) → touch prep → wet squash prep → negative: contralateral side → close tunica vaginalis continuous 5-0 (limits scarring); no antibiotics

  p.25

### 4 · VASOGRAPHY

- ALGORITHM

  High scrotal incision → vas at straight–convoluted junction → hemivasotomy at 25× → **vasal fluid examined first** → motile sperm: barbotage and cryopreserve **before any dye** → 24-G sheath into abdominal end, 1 mL saline: easy flow = patent → (rarely) dilute indigo carmine, blue urine in catheter confirms → formal contrast study only for obstruction proximal to the internal ring → close with 10-0 mucosa, 8-0 muscularis

  p.25

### 5 · VASOVASOSTOMY (VASECTOMY REVERSAL)

- **Length**, escalating: blunt dissection to external then internal ring → free convoluted vas from epididymal tunica (4–6 cm; **do not unravel**) → reroute beneath inguinal floor → dissect epididymis off testis (4–6 cm) → combined, gaps up to **10 cm**. p.26

- KEY POINT Why clear watery fluid still gets vasovasostomy: the epididymis is not yet blown out and sperm usually return. Toothpaste fluid means a secondary epididymal blow-out → open tunica vaginalis; a discrete level (dilated above, collapsed below) confirms. When in doubt, or inexperienced, do vasovasostomy. p.26

### 6 · VASOEPIDIDYMOSTOMY

- ALGORITHM

  Midline raphe → vas hemitransected at straight–convoluted junction, fluid sampled (no sperm), patency confirmed → full transection → open tunica vaginalis → site **above** obstruction/granuloma where tubules are dilated (if unclear: needle-puncture distal to proximal until sperm) → 3–4 mm tunic window → vas through separate tunica vaginalis opening, **fixed with 5-0 Prolene to tunica albuginea (tension relief)** → posterior tunic–vas 8-0 → intussusception sutures → outer tunic–vas adventitia

  p.27

### 7 · TRANSURETHRAL RESECTION OF THE EJACULATORY DUCTS (TURED)

- ALGORITHM

  24-Fr resectoscope; finger in rectum lifts posterior prostate → resect the **verumontanum** to open the dilated orifices/cyst (ducts exit at and lateral to the veru) → spare bladder neck, external sphincter and rectum → **blue efflux confirms**; minimal coagulation → catheter overnight, 7 days antibiotics

  p.28

### 8 · SPERM RETRIEVAL

- KEY POINT Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved. p.28

- ALGORITHM

  General/regional → testis delivered → 10× avascular plane on anterior albuginea → wide equatorial incision → bivalve (clam-shell) → systematic search, excise larger tubules → minced, examined at once by in-theatre andrologist → map positive areas → negative: contralateral → close albuginea 5-0, tunica vaginalis 4-0

  p.29

- TABLE **AZFa or AZFb deletion · 0% — do not operate** p.29

10 · VARICOCELECTOMY

- ALGORITHM
  Local (liposomal bupivacaine) + sedation → 2–3 cm incision over external ring → cord delivered on Penrose and backboard (exclude ilioinguinal and genital branch of genitofemoral nerves) → **micro-Doppler maps arteries before handling** (vasospasm) → open external then internal spermatic fascia → 10–25×, 1% papaverine → artery freed and looped; temporary occlusion confirms Doppler signal → **all veins except vasal doubly ligated/divided** (clips ≥1 mm, silk <2 mm, bipolar for tiny) → lymphatics preserved → deliver testis: ligate external spermatic and gubernacular veins → re-run cord → impulse test
  p.30

- KEY POINT Recurrence = a vein missed; hydrocele = a lymphatic tied; atrophy = an artery tied. Microsurgery with testicular delivery addresses all three. p.30

12 · MCQ ONE-LINERS

- MCQ Vas deferens receives no supply from cremasteric or cord vessels. p.30
- MCQ Never two simultaneous vasovasostomies on one vas — intervening segment necroses. p.30
- MCQ Epididymis off testis is safe if the superior epididymal artery is preserved; gains 4–6 cm. p.31
- MCQ Single puncture safe only at proximal caput; one stitch in corpus/cauda obstructs. p.31
- MCQ Diagnostic biopsy: azoospermia, normal testes, palpable vasa, normal FSH. p.31
- MCQ Biopsy fixative: Bouin's, Zenker's or glutaraldehyde — never formalin. p.31
- MCQ Most common biopsy complication: haematoma; most serious: epididymal biopsy. p.31
- MCQ Methylene blue kills sperm; use indigo carmine for patency. p.31
- MCQ Vasal granuloma at testicular end predicts better reversal outcome. p.31
- MCQ Toothpaste fluid without sperm → vasoepididymostomy. p.31
- MCQ Clear watery fluid without sperm → still vasovasostomy. p.31
- MCQ Grape-like clusters of sperm heads → epididymal obstruction. p.31
- MCQ Microdot method: six mucosal 10-0 sutures. p.31
- MCQ Anastomotic granuloma after vasovasostomy heralds obstruction. p.31
- MCQ LIVE: two longitudinal sutures placed before tubule opened. p.31
- MCQ No varicocelectomy simultaneous with vasoepididymostomy. p.31
- MCQ TURED: resect verumontanum; indigo carmine efflux confirms adequacy. p.31
- MCQ TURED complications: urine reflux, epididymitis, retrograde ejaculation. p.31
- MCQ MESA collection: capillary action, never negative pressure. p.31
- MCQ Best technique in NOA: microdissection TESE. p.31
- MCQ AZFa/AZFb complete deletion: zero retrieval — do not operate. p.31
- MCQ Sperm-containing tubules in NOA: larger, opaque; Sertoli cell–only thin and white. p.31
- MCQ Post-varicocelectomy hydrocele results from lymphatic ligation. p.31

## Ch 71 · Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · DEFINITION AND KEY CONCEPTS

- Historical milestones MCQ: oldest description of ED – Egyptian papyrus c. 2000 BCE; Leonardo da Vinci – erection is due to blood (hanged men); 1980s Doppler (Lue) – increased arterial inflow + decreased venous outflow; 1990s – nitric oxide (NO) as key mediator; 1998 – oral phosphodiesterase type 5 inhibitors (PDE5Is). p.32

## 2 · FUNCTIONAL ANATOMY

- Flaccid: sinusoidal blood is venous (partial pressure of oxygen, $Po_2$ **~35 mm Hg**); erect: arterial (**~90 mm Hg**). p.32

- KEY POINT Arterial and venous anatomy vary widely between men — which is why the same "nerve-sparing" or pelvic operation has different erectile outcomes in different patients. p.33

## 3 · HAEMODYNAMICS: PHASES OF ERECTION AND DETUMESCENCE

- TABLE 3 Full erection · Low (above flaccid) · Plateau **~100 mm Hg** p.33

## 4 · NEUROANATOMY

- TABLE Origin · **T11–L2** (often T10–T12) · **S2–S4** intermediolateral columns · S2–S4; motor from **Onuf nucleus** p.33

- KEY POINT nitric oxide synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections → NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally. p.34

## 5 · MOLECULAR MECHANISMS OF SMOOTH MUSCLE TONE

- Acetylcholine acts indirectly (endothelial NO release, presynaptic adrenergic inhibition); **atropine does not abolish** neurally induced erection MCQ. p.35

- KEY POINT Corporal tone is a tug-of-war over one phosphorylation site (MLC20): $Ca^{2+}$–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" = the phosphatase, not phosphorylated MLC. p.36

## 7 · RISK FACTORS AND ENDOTHELIAL DYSFUNCTION

- **Artery-size hypothesis**: penile arteries are smaller than coronary arteries, so the same endothelial disease becomes symptomatic in the penis first; ED preceded coronary artery disease (CAD) by a mean of **~2 years**. ED is a **sentinel** for cardiovascular disease and warrants cardiac risk stratification → Ch 72. p.36

## 8 · PATHOPHYSIOLOGY BY TYPE

- TABLE Upper cord (above sacral centre) · Preserved in **~95%**; brief, need continuous stimulation · Lost p.37

- TABLE Lower cord (sacral) · Preserved in only **~25%** · May occur (via thoracolumbar pathway); marginal tumescence p.37

- AUA: measure **morning, fasting** T in all men with ED (food acutely suppresses T); biochemical low T = **<300 ng/dL**; treat only with ≥2 low levels + attributable symptoms. Nocturnal erections need ~200 ng/dL. p.37

- Nocturnal (REM) erections are androgen-dependent; erections to visual stimuli are largely androgen-independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen. p.37

- KEY POINT Veno-occlusion is an emergent property of healthy tissue (elastic tunica + fully relaxed trabeculae), so most "venous leak" is really a cavernosal smooth muscle disease — the reason venous ligation surgery fails and is investigational. p.38

- **Chronic kidney disease**: up to **80%** of end-stage renal disease men have ED (hypogonadism, hyperprolactinaemia, atheroma, neuropathy); persists in ~half after transplant. p.38

- KEY POINT Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction, and TGF-β-driven fibrosis. p.39

## 9 · MCQ ONE-LINERS

- MCQ Erection = trabecular relaxation + arterial dilation + veno-occlusion. p.39

- MCQ Arterial inflow peaks in the latent (filling) phase, not at rigidity. p.39

- MCQ Full-erection intracavernous pressure ≈ 100 mm Hg; rigid phase suprasystolic via ischiocavernosus. p.39
- MCQ Emissary veins are compressed between inner circular and outer longitudinal tunica layers. p.39
- MCQ Ventral tunica thinnest (~0.8 mm); intracavernous pillars absent at 5–7 o'clock. p.39
- MCQ Glans and spongiosum lack veno-occlusion; pressure one-third to one-half of corporal. p.39
- MCQ Helicine arteries open directly into sinusoids without capillaries. p.39
- MCQ Accessory pudendal arteries in ~one-third; preserve at prostatectomy. p.39
- MCQ Deep dorsal vein drains into Santorini plexus. p.39
- MCQ Parasympathetic S2–S4 → erection; sympathetic T11–L2 → detumescence. p.39
- MCQ Onuf nucleus (S2–S4) innervates ischiocavernosus and bulbocavernosus via pudendal nerve. p.39
- MCQ Psychogenic erection absent with cord lesions above T9. p.39
- MCQ Complete upper cord lesion: reflexogenic erection preserved in ~95%. p.39
- MCQ PVN oxytocinergic neurons are the central pro-erectile output. p.39
- MCQ Apomorphine is a central D1/D2 agonist acting at PVN. p.39
- MCQ nNOS initiates, eNOS maintains erection. p.39
- MCQ NO → soluble guanylyl cyclase → cGMP → PKG-I → ↓ cytosolic $Ca^{2+}$. p.39
- MCQ PDE5 is the principal cGMP-hydrolysing enzyme in corpora; PDE6 is retinal. p.39
- MCQ Alprostadil acts via cAMP, independent of NO. p.39
- MCQ Rho-kinase inhibits myosin light chain phosphatase → sustained contraction. p.39
- MCQ Noradrenaline (α1) is the principal peripheral transmitter of flaccidity. p.39
- MCQ Atropine does not abolish neurally induced erection. p.39
- MCQ Gap junction protein connexin-43 synchronises corporal smooth muscle. p.39
- MCQ NOS needs cavernous $Po_2$ above ~50–60 mm Hg. p.39
- MCQ Prostacyclin (PGI2) contracts, PGE1 relaxes corporal tissue. p.39
- MCQ ED precedes coronary artery disease by ~2 years (artery-size hypothesis). p.39
- MCQ Nocturnal erections are androgen-dependent; visually evoked erections largely androgen-independent. p.39
- MCQ AUA low testosterone: <300 ng/dL on morning sample. p.39
- MCQ Commonest sexual symptom of low testosterone: reduced libido. p.39
- MCQ Thiazides and nonselective β-blockers carry highest antihypertensive ED risk. p.40
- MCQ Nebivolol and ARBs spare or improve erectile function. p.40
- MCQ 5α-reductase inhibitors: persistent ED despite minimal serum testosterone change. p.40
- MCQ Diabetes damages NO source but preserves response to exogenous NO. p.40
- MCQ Up to 80% of end-stage renal disease men have ED. p.40
- MCQ Venous leak after priapism shunt = persistent glans–cavernosum or cavernosum–spongiosum communication. p.40

## Ch 72 · Evaluation and Management of Erectile Dysfunction

### 1 · DEFINITION AND SIGNIFICANCE

- **Erectile dysfunction (ED)** (American Urological Association, AUA): consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction; minimum duration **3 months** (except trauma/surgery-induced ED). p.41
- Prevalence ~**20%** of adult men, rising with age (40–49 y up to 15%; 60–69 y up to 40%). One in four men presenting with new ED is <40 y, half of them severe. p.41

### 3 · BASIC EVALUATION

- KEY POINT Preserved nocturnal/morning erections prove an intact neurovascular axis — the single strongest historical pointer to psychogenic ED. p.42
- KEY POINT The basic work-up is designed to find the disease behind the ED — undiagnosed DM, dyslipidaemia, hypogonadism, CV risk — not to prove ED. p.43

### 4 · HORMONAL EVALUATION

- AUA threshold **300 ng/dL (10.4 nmol/L)**; sexual function correlates weakly with T above ~230 ng/dL. p.43
- Primary hypogonadism: low T, **high** LH/FSH (follicle-stimulating hormone) → testicular cause. Secondary: low T, low/inappropriately normal LH → prolactin; pituitary MRI if T **<150 ng/dL**, persistent hyperprolactinaemia or mass effect. p.43

### 5 · SPECIALISED TESTING

- TABLE PSV · **<25 cm/s** (25–35 borderline; >35 normal) · Persistently >25 cm/s p.44
- TABLE EDV · — · **>5 cm/s** p.44
- TABLE Resistive index (RI) = (PSV − EDV)/PSV · — · **<0.75** (normal >0.9) p.44
- KEY POINT Venous leak can be diagnosed only when arterial inflow is adequate and smooth muscle is fully relaxed — an anxious patient with incomplete relaxation will mimic veno-occlusive dysfunction, so redose before labelling a leak. Anatomical variants (dorsal–cavernous communications, early branching) can falsely suggest arteriogenic ED. p.44

### 6 · TREATMENT

- ALGORITHM
  Evaluate and treat reversible causes (lifestyle, drugs, hormones, psychosocial) + CV risk stratification → shared decision: phosphodiesterase type 5 inhibitor (PDE5I) | VED | IU alprostadil | ICI | (low-intensity extracorporeal shockwave therapy (Li-ESWT) investigational) → inadequate response: optimise dose/instructions, combine, switch → penile prosthesis → Ch 75
  p.45
- TABLE Half-life (h) · 3–5 · 4–5 · **17.5** · 3–5 p.45
- **Nitrates** in any form (glyceryl trinitrate, isosorbide, amyl nitrite "poppers") → profound hypotension; nitrate use >2 weeks earlier is not a contraindication. If angina occurs: no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). Hypotension: Trendelenburg → IV fluids → phenylephrine → intra-aortic balloon pump; no antidote. p.46

### 8 · MCQ ONE-LINERS

- MCQ Minimum duration for ED diagnosis: 3 months (except trauma/surgery). p.48
- MCQ ED precedes and predicts CV events; evaluate CV risk in all. p.48
- MCQ Strongest adjusted ED risk factor: antidepressant use; strongest disease: diabetes. p.48
- MCQ Princeton high-risk patient: defer sexual activity and ED therapy until cardiac stabilisation. p.48
- MCQ SHIM 5–7 severe, 8–11 moderate, 12–16 mild–moderate, 17–21 mild, 22–25 normal. p.48

- MCQ IIEF has 15 items across 5 domains. p.48
- MCQ Preserved morning erections suggest psychogenic ED. p.48
- MCQ Testosterone drawn 7–11 am; AUA cut-off 300 ng/dL. p.48
- MCQ About one-third of mildly low testosterone values normalise on repeat. p.48
- MCQ Pituitary MRI if testosterone <150 ng/dL with low/normal LH. p.48
- MCQ Bioavailable T = free + albumin-bound testosterone. p.48
- MCQ Ageing, cirrhosis, hyperthyroidism raise SHBG; obesity lowers it. p.48
- MCQ Duplex: PSV <25 cm/s = arterial insufficiency. p.48
- MCQ Duplex: EDV >5 cm/s with RI <0.75 = veno-occlusive dysfunction. p.48
- MCQ Side-to-side PSV difference >10 cm/s suggests unilateral arterial lesion. p.48
- MCQ Prolonged CIS/ICI erection reversed with diluted phenylephrine every 3–5 min. p.48
- MCQ RigiScan measures radial, not axial, rigidity. p.48
- MCQ NPTR false-negative in depression and ageing; false-positive in multiple sclerosis. p.48
- MCQ PDE5Is enhance but do not initiate erection. p.48
- MCQ Tadalafil: half-life 17.5 h, food-independent, back pain. p.48
- MCQ Avanafil: fastest Tmax (~30 min), least visual effects. p.48
- MCQ Vardenafil: avoid with class 1A/3 antiarrhythmics and congenital long QT. p.48
- MCQ Sildenafil blue vision due to PDE6 inhibition. p.48
- MCQ No nitrates for 24 h after sildenafil/vardenafil, 48 h after tadalafil. p.48
- MCQ Declare PDE5I failure only after up to 9–10 adequate attempts. p.48
- MCQ Alprostadil is the only FDA-approved intracavernosal agent; pain is commonest AE. p.48
- MCQ Papaverine: highest priapism and fibrosis risk; raises liver enzymes. p.48
- MCQ ICI contraindicated with monoamine oxidase (MAO) inhibitors (phenylephrine hypertensive crisis). p.48
- MCQ MUSE pellet inserted 3 cm into urethra; urethral pain in one-third. p.48
- MCQ VED constriction ring removed within 30 minutes. p.48
- MCQ Li-ESWT best evidence in mild–moderate vasculogenic ED; still investigational. p.48
- MCQ Revascularisation ideal: young nonsmoker, nondiabetic, isolated post-traumatic pudendal stenosis. p.48
- MCQ Commonest revascularisation complication: glans hyperaemia. p.48

## Ch 73 · Priapism

### 1 · DEFINITION AND CLASSIFICATION

- **Priapism**: full or partial erection lasting **≥4 h** beyond sexual stimulation/orgasm, or unrelated to stimulation. A urological emergency because of the risk of fibrosis, shortening and permanent erectile dysfunction (ED). p.49
- TABLE Share · **95%** · Subset of ischaemic · <5% p.49
- KEY POINT Trap: stuttering priapism is **ischaemic**, not a benign separate entity — episodes ≥4 h need the same first-line treatment. p.49

## 2 · AETIOLOGY

- **Recurrence after a shunt** may be shunt thrombosis (tunical collagen exposure → tissue factor/VIIa cascade). Regimen: **aspirin 325 mg + heparin 5000 U subcutaneously** pre-op → aspirin 81 mg daily × 2 weeks to keep the shunt patent. p.50
- Lifetime priapism risk **29–42%** of males with SCD; causes 23% of adult and **63% of paediatric** ischaemic priapism. Onset ~15 years; stuttering commoner than major episodes. p.50

## 4 · EVALUATION

- TABLE $pO_2$ (mm Hg) · **<30** · 40 · >90 p.50
- TABLE $pCO_2$ (mm Hg) · **>60** · 50 · <40 p.50
- TABLE pH · **<7.25** · 7.35 · 7.4 p.50
- ALGORITHM
  History + examination + gas → ischaemic → acute ischaemic algorithm | indeterminate → penile/perineal PDUS | non-ischaemic → expectant algorithm
  p.51

## 5 · MANAGEMENT OF ACUTE ISCHAEMIC PRIAPISM (AUA/SMSNA)

- TABLE ≥36 h · **100%** p.51
- ALGORITHM
  <36 h: aspiration ± irrigation + intracavernosal phenylephrine (blood pressure and cardiac monitoring) → distal shunt ± tunnelling → tunnelling if not done → proximal shunt / penoscrotal decompression → prosthesis if refractory
  >36 h: counsel high ED risk + analgesia/follow-up, OR early penile prosthesis
  p.51
- Dilute to **100–500 µg/mL**; **1-mL aliquots every 3–5 min**, continue up to **1 h** before declaring failure; †maximum ≈1 mg in 1 h (EAU). p.51
- Indicated for priapism **>36 h** or refractory to shunt ± tunnelling. Oedematous corpora dilate easily; delayed implantation meets dense fibrosis. p.52

## 6 · NON-ISCHAEMIC PRIAPISM

- ALGORITHM
  Observe, reassess at 4 weeks → persists → PDUS (fistula site, size) → counsel → selective internal pudendal/cavernosal artery embolisation → recurrence → repeat embolisation with non-resorbable agent → surgical ligation only after repeated failure
  p.52

## 9 · MCQ ONE-LINERS

- MCQ Priapism definition: erection ≥4 h unrelated to or beyond sexual stimulation. p.53
- MCQ Commonest type: ischaemic (95%). p.53
- MCQ Ischaemic gas: $pO_2$ <30, $pCO_2$ >60, pH <7.25. p.53
- MCQ Non-ischaemic gas resembles arterial blood: $pO_2$ >90, $pCO_2$ <40. p.53
- MCQ Glans and spongiosum remain flaccid in priapism. p.53
- MCQ Commonest cause in children: SCD (63%). p.53
- MCQ Classic antidepressant causing priapism: trazodone. p.53
- MCQ First-line ischaemic treatment: aspiration ± irrigation + intracavernosal phenylephrine. p.53
- MCQ Phenylephrine: 100–500 µg/mL, 1-mL aliquots every 3–5 min, up to 1 h. p.53
- MCQ Phenylephrine's key cardiac adverse effect: reflex bradycardia. p.53
- MCQ Aspiration site: 3 or 9 o'clock, avoiding dorsal neurovascular bundle. p.53

- MCQ Avoid chilled saline irrigation in SCD. p.53
- MCQ Winter shunt uses a Tru-Cut biopsy needle. p.53
- MCQ T-shunt: blade rotated 90° laterally after glans insertion. p.53
- MCQ Al-Ghorab: open excision of corporal tips via dorsal glans incision. p.53
- MCQ Corporal snake uses a 7/8 Hegar dilator. p.53
- MCQ Quackels shunt: cavernosum to bulbar spongiosum. p.53
- MCQ Grayhack shunt: saphenous vein to corpus cavernosum. p.53
- MCQ Bilateral proximal shunts are staggered ≥1 cm to avoid urethral stricture. p.53
- MCQ Priapism ≥36 h: ED essentially 100% — consider early prosthesis. p.53
- MCQ Early prosthesis: less shortening and higher satisfaction than delayed. p.53
- MCQ Non-ischaemic priapism onset is often delayed after trauma. p.53
- MCQ Non-ischaemic first-line: observation; ~60% resolve. p.53
- MCQ Failed resorbable embolisation → repeat with non-resorbable polyvinyl alcohol, not surgery. p.53
- MCQ Stuttering priapism was first described in SCD. p.53
- MCQ Priapism is a low-NO, low-PDE5 state. p.53
- MCQ PDE5Is cause priapism yet prevent stuttering episodes. p.53
- MCQ Penile duplex: ischaemic shows absent cavernosal flow. p.53
- MCQ Post-shunt recurrence may be thrombotic — aspirin + heparin prophylaxis. p.53
- MCQ Leukaemic priapism: leukapheresis plus cytotoxics. p.53
- MCQ First case report of priapism: 1824. p.53

## Ch 74 · Disorders of Male Orgasm and Ejaculation

### 2 · PHYSIOLOGY OF EJACULATION

- Reflex arc: genital sensory afferents → brain → spinal motor centres → efferents. Five groups of thoracolumbar and lumbosacral nuclei are coordinated by lumbar interneurons at **L3–L5** — the **spinal ejaculation generator**, modulated by genital and supraspinal input. p.54
- KEY POINT Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic. p.54

### 3 · PREMATURE EJACULATION — DEFINITION AND CLASSIFICATION

- All three required: (1) ejaculation always/nearly always before or within **about 1 min** of vaginal penetration (lifelong) **or** a clinically significant, bothersome reduction, often to **≤ about 3 min** (acquired); (2) inability to delay on all/nearly all penetrations; (3) **negative personal consequences** — distress, bother, frustration, avoidance of intimacy. p.55
- MCQ International Classification of Diseases (ICD-10) is the only older definition with a number (≤15 s); Masters and Johnson defined PE by partner satisfaction in ≥50% of coital episodes. p.55
- Community stopwatch **intravaginal ejaculatory latency time (IELT)**: positively skewed, median **5.4 min**; ≤1 min is statistically abnormal. p.55

## 6 · PREMATURE EJACULATION — TREATMENT

- **ALGORITHM**
  PE confirmed → secondary to ED/other dysfunction? **Yes** → treat primary cause → **No**: lifelong → selective serotonin reuptake inhibitor (SSRI) ± behavioural therapy (patient preference); acquired → treat cause ± behavioural/SSRI, attempt withdrawal after 6–8 weeks; variable/subjective → reassurance, education, psychotherapy
  p.56

- **KEY POINT** The real limiter in PE is adherence, not efficacy — most patients stop SSRIs and topicals (cost, loss of spontaneity, "antidepressant" stigma). p.57

## 7 · DELAYED EJACULATION, ANEJACULATION AND ANORGASMIA

- **ALGORITHM**
  Orgasm? **Never/sometimes** → inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique) → **Always** → ejaculate? No → **sperm in post-orgasm urine?** Yes = RE; No = aspermia → failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO)
  p.57

## 8 · RETROGRADE EJACULATION

- **Diagnosis**: post-masturbation urine; **>5–10 sperm/high-power field** confirms RE; more sperm in urine than ejaculate = significant RE. p.58

## 9 · SPERM RETRIEVAL IN SCI

- **ALGORITHM**
  **Penile vibratory stimulation (PVS)** — first line, success up to ~70% (needs intact T12–S1-level reflex) → **electroejaculation (EEJ)** — rectal probe stimulating hypogastric sympathetic efferents, effective and safe → prostate massage → surgical retrieval (TESE/TESA/MESA/PESA; testicular sperm extraction succeeds in most)
  p.59

## 11 · MCQ ONE-LINERS

- **MCQ** Most common male sexual disorder: ejaculatory dysfunction. p.59
- **MCQ** Spinal ejaculation generator: lumbar interneurons L3–L5. p.59
- **MCQ** Emission: sympathetic T10–L2; expulsion: somatic S2–S4 pudendal. p.59
- **MCQ** Bladder neck closes during expulsion, preventing retrograde flow. p.59
- **MCQ** Serotonin inhibits; dopamine (D2) promotes ejaculation. p.59
- **MCQ** Lifelong PE: 5-HT2C hyposensitivity / 5-HT1A hypersensitivity. p.59
- **MCQ** ISSM lifelong PE: IELT within about 1 min. p.59
- **MCQ** ISSM acquired PE: bothersome reduction, often ≤3 min. p.59
- **MCQ** ISSM three criteria: latency, inability to delay, negative consequences. p.59
- **MCQ** AUA/SMSNA lifelong PE latency: about 2 min. p.59
- **MCQ** DSM-5 severe PE: <15 s or before penetration. p.59
- **MCQ** Median community IELT: 5.4 min. p.59
- **MCQ** Only licensed oral PE drug: dapoxetine (not FDA). p.59
- **MCQ** Strongest ejaculatory delay among SSRIs: paroxetine. p.59
- **MCQ** Daily SSRI full effect: 2–3 weeks. p.59
- **MCQ** Hyperthyroidism → PE; hypothyroidism → DE. p.59
- **MCQ** Main sexual symptom in CP/CPPS: PE. p.59

- MCQ Normal IIEF-5 score: ≥22. p.59
- MCQ DE: ~25% lifelong, ~75% acquired. p.59
- MCQ Cabergoline for DE needs annual echocardiography. p.59
- MCQ Intranasal oxytocin for DE: RCTs negative. p.59
- MCQ Commonest causes of RE: TURP and diabetic neuropathy. p.59
- MCQ RE confirmed: >5–10 sperm/high-power field (HPF) in post-ejaculate urine. p.59
- MCQ Pseudoephedrine treats RE by closing bladder neck. p.59
- MCQ EDO semen: low volume, acidic, fructose-negative. p.59
- MCQ Dry ejaculation after RPLND: superior hypogastric plexus injury. p.60
- MCQ SCI first-line sperm retrieval: penile vibratory stimulation. p.60
- MCQ PVS/EEJ hazard: autonomic dysreflexia — nifedipine pretreatment. p.60
- MCQ Climacturia: urinary leakage at orgasm after radical prostatectomy. p.60
- MCQ Painful ejaculation with α-blockers: least with alfuzosin. p.60

## Ch 75 · Surgery for Erectile Dysfunction

### 2 · PROSTHESIS TYPES

- **Material**: Coloplast uses polyurethane (Bioflex) — cautery **<35 W**; Boston/Rigicon silicone–fabric–silicone trilaminate tolerates cautery. p.61

### 5 · OPERATIVE TECHNIQUE (THREE-PIECE)

- ALGORITHM

  Catheter → expose tunica → two 2-0 polydioxanone stay sutures per corpus → longitudinal corporotomy → dilate proximal then distal (Metzenbaum, Hegar/Brooks 7→13 mm) → goalpost test → measure (Furlow) → cylinders → test inflation 1 → close corporotomies → test inflation 2 → reservoir → pump → connect → test inflation 3 → irrigate, close

  p.62

### 6 · INTRA-OPERATIVE PROBLEMS

- KEY POINT Why no catheter after distal urethral perforation? Distal defects are not closed (poor exposure); the urinary stream draws tissue fluid into the urethra (Bernoulli effect), so urine does not enter the defect. Why wait 5–6 months? Solid scar must obliterate the false tract, or the dilator re-enters it. Almost every intra-operative error is **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley). p.63
- KEY POINT After robotic prostatectomy the peritoneal reflection is not replaced; bowel becomes fixed near the internal ring, so a blind inguinal Retzius reservoir can compress bowel on filling — choose an ectopic site or place under vision. p.63

### 9 · POST-OPERATIVE CARE

- Activation at **4–6 weeks**, partner present; daily cycling; intercourse thereafter. p.64

### 11 · MCQ ONE-LINERS

- MCQ First three-piece inflatable prosthesis: Scott, 1972. p.65
- MCQ Commonest complication of IPP: mechanical failure. p.65
- MCQ Commonest patient complaint after implant: shorter erection. p.65
- MCQ Shortening cause: pseudocapsule limiting cylinder expansion. p.65
- MCQ Poor dexterity → malleable rod. p.65
- MCQ Tactra core: nitinol; Genesis: silver; Rigi 10: stainless steel. p.65

- MCQ Titan cylinders are polyurethane (Bioflex); keep cautery <35 W. p.65
- MCQ Length-expanding cylinders: AMS 700 LGX and Infla 10 AX. p.65
- MCQ Best cylinders for curvature: AMS 700 CX, Titan, Infla 10 X. p.65
- MCQ Infrapubic approach risks the dorsal nerve; penoscrotal risks the urethra. p.65
- MCQ Day-of-surgery glucose <200 mg/dL matters more than HbA1c. p.65
- MCQ Prophylaxis: vancomycin or cephalosporin + gentamicin, ≤24 h. p.65
- MCQ InhibiZone coating: rifampin + minocycline. p.65
- MCQ Infected implant: remove all components. p.65
- MCQ Mulcahy salvage: explant, antiseptic washout, immediate replacement. p.65
- MCQ Salvage failure predictors: early onset, aggressive organism, cellulitis. p.65
- MCQ Hydrogen peroxide is dropped from salvage washes. p.65
- MCQ Revision washout lowers infection from ~10% to ~3%. p.65
- MCQ Lockout valve prevents autoinflation and allows ectopic reservoirs. p.65
- MCQ Commonest ectopic reservoir site: submuscular suprafascial. p.65
- MCQ Crural perforation repair: suture sling. p.65
- MCQ Distal urethral perforation: abandon ipsilateral cylinder; replace at 5–6 months. p.65
- MCQ SST deformity: ventral glans hypermobility; treat by glans fixation. p.65
- MCQ Extrusion: distal corporoplasty; erosion: explant. p.65
- MCQ Rods erode in insensate men — prefer inflatables. p.65
- MCQ Bladder reservoir erosion: exchange reservoir; bowel: remove entire device. p.65
- MCQ Pelvic radiation does not increase implant difficulty. p.65
- MCQ Post-priapism: early implant preserves length. p.65
- MCQ Activation 4–6 weeks after surgery. p.65
- MCQ Ohmmeter test detects leaks in retained parts. p.65

## Ch 76 · Diagnosis and Management of Peyronie Disease

### 1 · DEFINITION

- KEY POINT Misconceptions corrected: PD is **not rare** (3–20%; higher in diabetes and ED); **spontaneous resolution is rare** — up to half worsen untreated, so "wait 6–12 months" is a myth; occurs from teens to the 70s and in all races; flaccid trauma can trigger it; **calcification is not a marker of chronic disease**. p.66

### 2 · ANATOMY AND PATHOPHYSIOLOGY

- Longitudinal layer is thinnest at 3 and 9 o'clock and **absent between 5 and 7 o'clock** → dorsal buckling; **60–70% of plaques are dorsal** with septal involvement. p.66

### 3 · CLINICAL FEATURES AND PHASES

- TABLE Surgery · Not appropriate · When **≥12 months from onset and stable ≥6 months** p.67

### 5 · NON-SURGICAL MANAGEMENT

- **Protocol (IMPRESS)**: cycle = 2 injections of **0.58 mg** 24–72 h apart → **modelling** 24–72 h after the second; up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential — CCH without it is inferior. Most gain in the first 3 cycles. p.67
- **Criteria (AUA)**: stable disease, curvature **>30° and <90°**, intact erectile function (± medication). IMPRESS excluded ventral plaques, extensive calcification, disease <12 months. p.68

## 6 · SURGERY

- **ALGORITHM**

  Erection adequate for penetration (± phosphodiesterase type 5 inhibitor (PDE5I))? **No** → inflatable penile prosthesis + modelling → if curve persists, incision ± graft (graft if defect >2 cm)
  **Yes** → PDDUS → simple curve <60–70°, no hourglass/hinge, acceptable length loss → **plication** (if length loss unacceptable → grafting)
  **Yes** → complex/multiplanar >60–70°, severe indentation/hourglass, hinge, short penis (<9 cm stretched) → **plaque incision/partial excision + graft** if ED risk accepted (else prosthesis)

  p.69

## 8 · MCQ ONE-LINERS

- MCQ PD = fibrosis of the tunica albuginea. p.70
- MCQ Commonest plaque site: dorsal. p.70
- MCQ Longitudinal tunical layer absent between 5 and 7 o'clock. p.70
- MCQ Key cytokine: TGF-β1. p.70
- MCQ Associated: Dupuytren and Ledderhose disease. p.70
- MCQ Calcification does not indicate chronic disease. p.70
- MCQ Up to 50% worsen untreated; spontaneous resolution rare. p.70
- MCQ Surgery when ≥12 months from onset and stable ≥6 months. p.70
- MCQ Objective curvature: after ICI, goniometer, with PDDUS. p.70
- MCQ ICI erection test required before any invasive therapy (AUA). p.70
- MCQ Only AUA-endorsed oral therapy: NSAIDs for pain. p.70
- MCQ Vitamin E, tamoxifen, colchicine, carnitine: no benefit. p.70
- MCQ ESWT treats pain, not curvature. p.70
- MCQ Only FDA-approved PD drug: collagenase *Clostridium histolyticum*. p.70
- MCQ CCH dose: 0.58 mg, two injections per cycle, up to four cycles. p.70
- MCQ CCH indication: stable curvature 30–90°, intact erections. p.70
- MCQ Most serious CCH complication: corporal rupture. p.70
- MCQ CCH degrades collagen types I and III. p.70
- MCQ Modelling is essential after CCH. p.70
- MCQ Verapamil: calcium channel blocker; AUA "may offer". p.70
- MCQ Traction acts by mechanotransduction; needs ≥3 h/day. p.70
- MCQ Plication shortens the convex side. p.70
- MCQ 16-dot plication: four dots per stitch, extended Lembert pattern. p.70
- MCQ First incisionless plication: Essed–Schroeder. p.70
- MCQ Grafting for curves >60–70°, hourglass, hinge. p.70
- MCQ Strongest predictor of post-operative ED: pre-operative erection quality. p.70
- MCQ Ventral curves do poorly with grafting (veno-occlusive dysfunction). p.70
- MCQ Pericardium graft oversize ≤10%; small intestinal submucosa 25%. p.70
- MCQ PD + refractory ED → inflatable prosthesis + modelling. p.70
- MCQ Graft during prosthesis if tunical defect >2 cm. p.70
- MCQ Functionally straight: residual curvature ≤20°. p.70

## ★ Putting it together: andrology and sexual medicine decisions

### A · INFERTILE MAN: THE WORK-UP

- **ALGORITHM**

**1. History and examination** of both partners: duration, toxins, drugs (exogenous testosterone), mumps, cryptorchidism; testis volume, vasa (palpable?), varicocele, secondary sex characteristics (→ Ch 69).

**2. Two semen analyses** (abstinence 2–7 days). Low volume with acid pH → think ejaculatory duct obstruction or congenital bilateral absence of the vas deferens (CBAVD); check a post-ejaculate urine for retrograde ejaculation.

**3. Hormones** if the count is low or sexual function is impaired: follicle-stimulating hormone (FSH), testosterone, then luteinising hormone (LH) and prolactin if testosterone is low.

**4. Genetics** for azoospermia or severe oligozoospermia: karyotype and Y-chromosome microdeletions; cystic fibrosis transmembrane conductance regulator (CFTR) testing in CBAVD (test the partner too).

p.71

### B · AZOOSPERMIA: OBSTRUCTIVE OR NOT?

- **KEY POINT** † Vasectomy reversal or ICSI? Reconstruction suits a younger partner and the wish for more than one child; ICSI suits an older partner or a female factor. At reversal, inspect the fluid from the testicular vas: copious fluid — even clear and sperm-free — or sperm/sperm parts → vasovasostomy; thick toothpaste-like fluid or none → vasoepididymostomy (→ Ch 70). p.71

### C · ERECTILE DYSFUNCTION: THE LADDER

- **ALGORITHM**

**1. Assess:** history and a validated score (International Index of Erectile Function, IIEF-5, or Sexual Health Inventory for Men, SHIM), examination, glucose/glycated haemoglobin (HbA1c), lipids, morning testosterone. Erectile dysfunction (ED) is a cardiovascular sentinel — stratify cardiac risk before restarting sexual activity (→ Ch 72).

**2. Modify:** lifestyle, risk factors, offending drugs; treat hypogonadism (testosterone may rescue phosphodiesterase type 5 inhibitor [PDE5I] non-responders).

**3. Shared decision on any of:** PDE5I (never with nitrates; retry after correct-use counselling before calling it failure), vacuum erection device, intraurethral alprostadil, intracavernosal injection (alprostadil or trimix).

**4. Penile prosthesis** (inflatable, three-piece preferred) when the others fail or are unwanted; highest satisfaction. Infection: remove all components ± immediate salvage (→ Ch 75).

**Young man, focal arterial occlusion after pelvic or perineal trauma** → consider penile revascularisation.

p.72

### D · PRIAPISM: ISCHAEMIC VERSUS NON-ISCHAEMIC

- **KEY POINT** Sickle cell disease: treat the priapism exactly as above (no delay for transfusion), avoid chilled irrigation, and involve haematology. Stuttering priapism: daily PDE5I (paradoxical), pseudoephedrine at night, or hormonal suppression after puberty (→ Ch 73). p.72

### E · PEYRONIE DISEASE: WHICH TREATMENT?

- **ALGORITHM**

  **Active phase** (pain, changing curve): oral analgesia, penile traction; collagenase *Clostridium histolyticum* (CCH) for stable dorsal/lateral curves 30–90° with intact erections; intralesional verapamil or interferon α-2b as alternatives.
  **Surgery** only when stable — ≥12 months from onset and deformity unchanged ≥6 months. Document stretched penile length and erectile function first (→ Ch 76).
  **Erections adequate** + simple curve curvature <60°, no hourglasslt;60–70°, no hourglass or hinge, adequate length → tunical plication (shortens). **Erections adequate** + complex curve, hourglass or hinge, or short penis → plaque incision or partial excision + graft (more new ED).
  **Erections inadequate** for penetration despite PDE5I → inflatable prosthesis ± modelling.

  p.72

### F · PREMATURE EJACULATION (PE)

- **KEY POINT** Diagnose by latency (lifelong: about 1 min or less), inability to delay and distress — latency alone never suffices. Treat co-existing ED first. Options: behavioural (stop–start, squeeze), topical lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily selective serotonin reuptake inhibitor (paroxetine delays most), tramadol (off-label, addiction risk) (→ Ch 74). p.73

# 10 Testis, Penis & Urethra

383 core items · page numbers refer to the full book PDF

## Ch 78 · Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

### 1 · DEFINITION, BOUNDARIES AND ORGANS

- KEY POINT Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney. p.2

### 2 · POSTERIOR ABDOMINAL WALL AND SURFACE LANDMARKS

- TABLE Hilum · **L1** p.2

### 3 · RETROPERITONEAL FASCIAE AND SPACES

- **Renal fascia**: anterior lamina (Toldt, prerenal; thinner) and posterior lamina (Zuckerkandl, retrorenal; thicker, seen more on imaging). MCQ Gerota described only the anterior, yet his name covers both. Laminae merge laterally as the **lateroconal fascia** (separates anterior and posterior pararenal spaces). p.3

### 6 · LYMPHATICS

- Pelvic nodes → lumbar nodes → lumbar trunks → **cisterna chyli** (behind right side of aorta, in front of **L1–L2**) → thoracic duct, ascending behind aorta. p.4
- KEY POINT Why the hilum? The testis is embryologically retroperitoneal — it drains to where it began. Crossover is one-way (right → left), following global lymph flow; hence right-sided templates include interaortocaval and some para-aortic tissue. → Ch 80, 81 p.5

### 7 · NERVES

- **Parasympathetic**: craniosacral (III, VII, IX, X, S2–S4); ganglia in organ wall. **Sympathetic**: thoracolumbar **T1–L2**; trunk ganglia or splanchnics to aortic plexuses. p.5
- **Ejaculation**: † postganglionic fibres from the **T12–L3** sympathetic ganglia run anterior to the aorta/across the IVC and converge in the hypogastric plexus below the IMA; they mediate **seminal emission** and **bladder neck closure**. Disruption by extensive retroperitoneal dissection → loss of emission (anejaculation) or retrograde ejaculation. Erection (parasympathetic, pelvic) is unaffected. → Ch 81 p.5

### 9 · MCQ ONE-LINERS

- MCQ D1 duodenum and pancreatic tail are NOT retroperitoneal. p.6
- MCQ Renal hilum at L1; right kidney lower. p.6
- MCQ Perirenal space is open inferiorly — collections track into the pelvis. p.6
- MCQ Lateroconal fascia separates anterior and posterior pararenal spaces. p.6
- MCQ Costovertebral ligament tethers 12th rib to L1–L2 transverse processes. p.6
- MCQ Gonadal arteries arise anterolaterally from aorta below renal arteries. p.6
- MCQ Marginal artery of Drummond allows IMA ligation. p.6
- MCQ Abdominal ureter supplied medially, pelvic ureter laterally. p.6
- MCQ Left gonadal vein enters left renal vein at a right angle. p.6
- MCQ New right varicocele → image the retroperitoneum. p.6
- MCQ IMV can be mistaken for gonadal vein at reoperation. p.6
- MCQ Circumaortic left renal vein: posterior limb mimics a lumbar vein. † p.6
- MCQ Retrocaval ureter is a venous (IVC) developmental anomaly. † p.6

- MCQ Cisterna chyli: right of aorta, anterior to L1–L2. p.6
- MCQ Right testis primary landing zone: interaortocaval. p.6
- MCQ Left testis primary landing zone: left para-aortic. p.6
- MCQ Lymphatic crossover: right → left only. p.6
- MCQ Emission fibres from T12–L3 converge in hypogastric plexus below IMA. † p.6
- MCQ Genitofemoral nerve lies on anterior surface of psoas. p.6
- MCQ Genital branch of genitofemoral is motor to cremaster. p.6
- MCQ Obturator reflex during lateral-wall TURBT risks bladder perforation. p.6

## Ch 79 · Neoplasms of the Testis

### 1 · EPIDEMIOLOGY AND RISK FACTORS

- >95% of testicular neoplasms are **germ cell tumours (GCTs)**: seminoma vs non-seminomatous germ cell tumour (NSGCT). 95% testicular, 5% extragonadal. Commonest solid malignancy in men **20–40 y**; peak 25–35. p.7
- Cisplatin turned metastatic GCT from 5–10% to **80–90%** cure → the aim is now **minimum toxicity without losing curability**. Deaths come from platinum resistance and failure to clear residual disease early. p.7
- Bilateral **2%**, mostly metachronous; histology discordant in 30–50%. Highest incidence in White men, Scandinavia; lowest in Black men, Africa/Asia; incidence rising. p.7
- **GCNIS** (= intratubular germ cell neoplasia): arrested fetal gonocyte formed **before birth**; adjacent to invasive GCT in 80–90%; risk of invasion **50% at 5 y**, 70% at 7 y. Contralateral GCNIS 5–9%, ≈36% with atrophy/UDT/age <40. p.7
- KEY POINT GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes the risk. p.7

### 2 · PATHOLOGY AND CLASSIFICATION

- KEY POINT Management follows the least differentiated, most chemoresistant element: EC drives occult metastasis, choriocarcinoma drives haemorrhage, teratoma drives surgery. p.8

### 4 · DIAGNOSIS AND INITIAL MANAGEMENT

- TABLE Half-life · **5–7 days** · **24–36 h** · 24 h p.9

### 5 · STAGING

- Computed tomography (CT) abdomen–pelvis for all (also shows retroaortic renal vein, lower-pole artery before retroperitoneal lymph node dissection [RPLND]). False negative in **25–35% of CS I NSGCT** → landing-zone nodes 5–9 mm are suspicious; equivocal nodes with normal markers → repeat CT in 6–8 weeks. p.9
- KEY POINT IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is not valid at relapse. Seminoma risk depends on NPVM alone; mediastinal seminoma can be good risk, mediastinal NSGCT is always poor risk. p.10

### 6 · TREATMENT PRINCIPLES

- Options: orchidectomy, radiotherapy **≥20 Gy** (<20 Gy → recurrence), or observation. Radiotherapy kills germ cells but spares relatively radioresistant Leydig cells; still up to 40% later need testosterone. p.11

## 7 · NSGCT MANAGEMENT

- **ALGORITHM** CS I NSGCT → low risk (no LVI, no EC predominance) → surveillance → high risk → surveillance or BEP×1 or nerve-sparing RPLND (expert) → choose by patient preference and toxicity p.11
- **KEY POINT** Decision analysis: surveillance maximises quality-adjusted survival when relapse risk <≈1/3. All three options cure ≈100%; they differ in how many are treated needlessly vs how hard relapse is to treat. p.11
- Normal markers + residual mass **>1 cm** → **post-chemotherapy surgery (PCS)/RPLND**, full bilateral template. Histology: necrosis ≈40%, teratoma ≈45%, viable ≈15%. p.12

## 8 · SEMINOMA MANAGEMENT

- Risk factors: tumour **>4 cm** and **rete testis invasion** (relapse 12/16/32% with 0/1/2) — **LVI is not predictive** in seminoma; not validated, so surveillance remains preferred for all. p.13
- **ALGORITHM** Residual seminoma mass → <3 cm → observe → >3 cm → FDG-PET ≥6 weeks after chemotherapy (earlier = inflammatory false positives) → PET-negative → observe → PET-positive → template post-chemotherapy RPLND (or repeat PET/close observation); mass-only excision → 33% local relapse p.13

## 10 · TREATMENT SEQUELAE AND FOLLOW-UP

- **KEY POINT** Because cure is expected, late cardiovascular disease and second cancers — rising with each added modality — decide between equally curative options. p.14

## 13 · MCQ ONE-LINERS

- **MCQ** Commonest solid malignancy in men 20–40: GCT. p.15
- **MCQ** GCNIS: 50% invasive GCT at 5 years; forms before birth. p.15
- **MCQ** Only postpubertal GCT without GCNIS or i(12p): spermatocytic tumour — benign, sixth decade. p.15
- **MCQ** Raised AFP excludes pure seminoma and pure choriocarcinoma. p.15
- **MCQ** AFP half-life 5–7 days; hCG 24–36 hours. p.15
- **MCQ** Schiller–Duval bodies: yolk sac tumour. p.15
- **MCQ** Seminoma: CD117+, CD30−; EC: CD30+†, OCT3/4+. p.15
- **MCQ** Choriocarcinoma: very high hCG, haematogenous spread, brain haemorrhage. p.15
- **MCQ** hCG false positive in hypogonadism: LH cross-reactivity, corrected by testosterone. p.15
- **MCQ** Trans-scrotal orchidectomy alters drainage to inguinal nodes. p.15
- **MCQ** Right testis landing zone: interaortocaval; left: para-aortic/left renal hilar. p.15
- **MCQ** LVI upstages pT1 to pT2. p.15
- **MCQ** Stage IS: raised post-orchidectomy markers, no radiological disease → chemotherapy. p.15
- **MCQ** Any M1b = stage IIIC. p.15
- **MCQ** No poor-risk category for seminoma; mediastinal NSGCT always poor risk. p.15
- **MCQ** Brain imaging when hCG >10,000 IU/L. p.15
- **MCQ** CS I NSGCT occult metastasis predictors: LVI, EC predominance. p.15
- **MCQ** CS I seminoma relapse predictors: size >4 cm, rete testis invasion — not LVI. p.15
- **MCQ** Good-risk IGCCCG: BEP×3 or EP×4; intermediate/poor: BEP×4. p.15
- **MCQ** VIP replaces BEP when bleomycin lung toxicity must be avoided. p.15

- MCQ NSGCT residual mass >1 cm → RPLND; PET has no role. p.15
- MCQ Seminoma residual mass >3 cm → PET at ≥6 weeks. p.15
- MCQ Commonest histology in late relapse: yolk sac tumour. p.15
- MCQ Commonest somatic transformation in PCS specimens: rhabdomyosarcoma. p.15
- MCQ Growing teratoma syndrome: enlarging masses with normal markers → resect. p.15
- MCQ Carboplatin dose = 7 × (GFR + 25) using isotope GFR. p.15
- MCQ CS I seminoma standard: 20 Gy para-aortic if radiotherapy chosen. p.15
- MCQ GCNIS radiotherapy ≥20 Gy spares Leydig cells. p.15
- MCQ Commonest testicular tumour over 60 years: lymphoma†. p.15
- MCQ Reinke crystals: Leydig cell tumour†. p.15
- MCQ Large-cell calcifying Sertoli tumour: Carney complex, Peutz–Jeghers†. p.15

## Ch 80 · Surgery of Testicular Tumours

### 1 · EVALUATION OF THE TESTIS MASS

- History (growth, symptoms) → examination (supraclavicular nodes, breasts, abdomen, then scrotum) → scrotal ultrasound (not mandatory; key role is the **contralateral testis** — bilateral synchronous ≈ **1%**) → serum tumour markers (STMs). p.16

### 2 · RADICAL INGUINAL ORCHIDECTOMY

- ALGORITHM

  Supine; prep umbilicus to mid-thigh and genitalia; EUA locates external ring → 3–5 cm transverse incision along Langer lines over the canal (hockey-stick extension if needed) → open external oblique ≈ 4 cm → **preserve ilioinguinal nerve** → mobilise cord at pubic tubercle, **encircle with Penrose (early clamp)** → divide external spermatic fascia/cremaster → deliver testis by upward scrotal pressure → divide gubernaculum → dissect cord to peritoneal reflection at internal ring. p.16

- **High ligation**: vas and gonadal vessels ligated **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail — marks the gonadal stump so it can be found and excised at later retroperitoneal lymph node dissection (RPLND). Vas, being separate, need not be removed at RPLND. p.16
- KEY POINT † **Scrotal violation** (prior trans-scrotal orchidectomy/biopsy): complete with inguinal exploration, high cord ligation and excision of the scrotal scar; routine hemiscrotectomy is no longer needed (local recurrence low). Chemotherapy-era outcomes are not compromised. p.16
- **Indications (all three)**: polar tumour, **≤2 cm**, contralateral testis compromised/absent (synchronous bilateral, solitary testis). Justified only when anorchia risk outweighs local recurrence risk. **Never elective with a normal contralateral testis.** p.16
- Uncertain malignancy → inguinal exploration, early cord control, excisional biopsy ± frozen section. Very small masses (**<5 mm**), negative markers → low malignancy rate; partial orchidectomy or, selectively, surveillance. p.16
- Technique: as radical until delivery → isolate with towels → **intraoperative ultrasound** → cold ischaemia omissible if resection **<30 min** → incise tunica (vertical for ventral midline, horizontal for medial/lateral lesions — spares segmental arteries beneath tunica) → excise with rim of normal tubules → close tunica (absorbable) → three-point fixation. p.16
- Surrounding germ cell neoplasia in situ (GCNIS): completion orchidectomy vs adjuvant radiotherapy **18–20 Gy** (abolishes spermatogenesis, preserves Leydig/androgen function) vs observation (local recurrence ≈ 7.5%). p.16

### 3 · RPLND — TERMINOLOGY AND INDICATIONS

- TABLE Post-chemotherapy (PC-RPLND) · After induction chemotherapy · Residual mass **>1 cm**, normal markers p.17
- TABLE Late relapse · **>24 months** after complete response · Retroperitoneal relapse p.17
- CT within **6 weeks**, STMs within **7–10 days**; review vascular variants (duplicated renal vessels); no bowel prep; sperm banking offered. p.17

### 4 · TEMPLATES

- **PC-RPLND standard = residual mass + full bilateral infrahilar** (crossover with bulk; viable tissue unrecognisable intraoperatively). Modified PC template only if **all three**: lesion **≤5 cm** in landing zone, normal markers, International Germ Cell Cancer Collaborative Group (IGCCCG) good/intermediate risk. p.17

### 5 · OPEN RPLND TECHNIQUE

- KEY POINT Identify fibres **prospectively, before dividing anything**; sweep tissue off, loop them, then pass the packet beneath the nerve web. Avoid traction (neuropraxia caused transient anejaculation in early series). p.18
- No routine drain (if large dissection: remove when serous, **<100 mL/24 h** on diet); no nasogastric tube; clear fluids day 0, low-fat diet days 1–2; non-opioid analgesia; mobilise day 1; discharge day 3–5. p.18

### 6 · AUXILIARY PROCEDURES (PC-RPLND 23–45%)

- **Nephrectomy** (commonest): **left** (left hilar drainage); strongest predictor residual mass **>10 cm**; risk rises fibrosis < teratoma < viable and in salvage/desperation/redo. Concern = future cisplatin eligibility. p.18
- TABLE Repair · Narrowing **<25%** → primary/lateral venorrhaphy; >25% → patch or graft · Small → primary; significant → graft p.18
- **Liver**: histology discordant with retroperitoneum in about half; marker-negative residual lesions → resect, or open microwave ablation if **<3 cm** unresectable. p.19

### 8 · HISTOLOGY OF RESIDUAL MASS AFTER CHEMOTHERAPY

- TABLE Fibrosis/necrosis · **≈40%** · relapse-free survival (RFS)/CSS ≈95%; no adjuvant p.19
- TABLE Teratoma · **≈45%** · RFS 80–90%; must be resected (chemoresistant, grows, transforms); no adjuvant p.19
- TABLE Viable GCT · **≈15%** · Survival 50–70%; consider 2 adjuvant cycles p.19
- KEY POINT **Why resect every NSGCT residual >1 cm?** No imaging (including PET †) or marker separates necrosis from teratoma; teratoma is chemoresistant and can grow (growing teratoma syndrome), relapse late or undergo malignant transformation. Surgery is therefore both the staging test and the only treatment for teratoma; histology decides further therapy. p.19

### 10 · UNIQUE SITUATIONS

- **Seminoma residual mass**: dense desmoplasia, no teratoma, viable ≈10%, more complications. **<3 cm** → observe; **>3 cm** → FDG-PET (≥6 weeks after chemotherapy †); PET-avid → PC-RPLND (resectable, select), salvage chemotherapy or HDCT. p.20

### 11 · OUTCOMES, FERTILITY AND COMPLICATIONS

- **Ejaculation**: template alone or prospective nerve sparing → antegrade ejaculation **>90%** in primary RPLND; no oncological cost. † Anejaculation/retrograde ejaculation → sympathomimetics (pseudoephedrine, imipramine), post-ejaculatory urine sperm retrieval, electroejaculation or testicular sperm extraction. p.20
- TABLE Pulmonary (bleomycin) · PC 3–5%; acute respiratory distress syndrome · Low $FiO_2$, restrictive fluids; tachycardia alone is not a fluid trigger; accept $SpO_2$ **88–90%** p.20

- **ALGORITHM**
  Paracentesis → capped drain if reaccumulates → very-low-fat, high-protein diet + medium-chain triglycerides + octreotide → nil by mouth + parenteral nutrition → refractory: lymphangiography ± embolisation, peritoneovenous shunt, or exploration → continue low-fat diet ≥ **4 weeks** after resolution.
  p.21

## 13 · MCQ ONE-LINERS

- **MCQ** Suspected testis cancer → radical inguinal orchidectomy with early cord control. p.21
- **MCQ** Cord stump: non-absorbable suture, 1–2 cm tail, for retrieval at RPLND. p.21
- **MCQ** Bilateral synchronous testicular masses ≈ 1%. p.21
- **MCQ** Partial orchidectomy: polar, ≤2 cm, absent/compromised contralateral testis. p.21
- **MCQ** Cold ischaemia omissible if resection <30 minutes. p.21
- **MCQ** Adjuvant radiotherapy to GCNIS in remnant: 18–20 Gy. p.21
- **MCQ** Local recurrence after partial orchidectomy → completion radical orchidectomy. p.21
- **MCQ** Chemotherapy-first patients still need delayed orchidectomy (testis–metastasis discordance). p.21
- **MCQ** Primary RPLND requires normal post-orchidectomy markers. p.21
- **MCQ** Desperation RPLND = elevated markers after chemotherapy. p.21
- **MCQ** Late relapse: >24 months after complete response; treat surgically. p.21
- **MCQ** Right testis primary landing zone: interaortocaval. p.21
- **MCQ** Left testis primary landing zone: left para-aortic. p.21
- **MCQ** Second lumbar artery pair arises at the IMA level. p.21
- **MCQ** Right sympathetic fibres emerge beneath the IVC. p.21
- **MCQ** Postganglionic fibres L1–L4 converge on superior hypogastric plexus. p.21
- **MCQ** Commonest auxiliary procedure at PC-RPLND: left nephrectomy. p.21
- **MCQ** Strongest nephrectomy predictor: residual mass >10 cm. p.21
- **MCQ** IVC narrowing <25% → primary repair; aorta can never be ligated. p.21
- **MCQ** PC-RPLND histology: necrosis 40%, teratoma 45%, viable 15%. p.21
- **MCQ** Residual teratoma is chemoresistant; resect to prevent growing teratoma syndrome. p.21
- **MCQ** Chest residual: do RPLND first; retroperitoneal necrosis predicts thoracic necrosis ≈89%. p.21
- **MCQ** Commonest cause of redo RPLND: incomplete first surgery (retrocaval/retroaortic). p.21
- **MCQ** Seminoma residual <3 cm → observe; >3 cm → PET. p.21
- **MCQ** Seminoma residuals contain no teratoma. p.21
- **MCQ** Bleomycin patients: low $FiO_2$, restrictive fluids. p.21
- **MCQ** Commonest RPLND complication: ileus. p.21
- **MCQ** Chylous ascites triglyceride 2–8 times serum; risk ↑ with suprahilar dissection. p.21
- **MCQ** Low-fat diet ≥4 weeks after chylous ascites resolves. p.21
- **MCQ** Only predictor of OS after desperation RPLND: postoperative marker normalisation. p.21
- **MCQ** Leydig cell tumour: almost always stage I; Sertoli more aggressive. p.21

## Ch 81 · Minimally Invasive Retroperitoneal Lymph Node Dissection

### 1 · INDICATIONS AND RATIONALE

- **Primary (chemotherapy-naive)**: marker-negative clinical stage I nonseminomatous germ cell tumour (NSGCT); small-volume stage II NSGCT and seminoma. Diagnostic + therapeutic — stages and cures occult disease, and spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0. p.22
- **Postchemotherapy (PC-RPLND)**: NSGCT residual mass **>1 cm**; select seminomas. p.22
- First laparoscopic RPLND MCQ 1992; robotic 2006. Robot adds wristed instruments, **fourth-arm retraction**, easier suturing and **circumferential great-vessel dissection with lumbar control**. p.22
- KEY POINT Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach — the benefit of access alone is smaller than it looks. p.22

### 2 · GUIDELINES AND CONTROVERSY

- TABLE Intraoperative safety · Great-vessel surgery from a console; robot-trained surgeons may lack open skills; conversion **≈5%** p.22

### 3 · PATIENT SELECTION

- TABLE Small-volume II: 1–2 nodes in primary landing zone, none **>3 cm** · Large size change pre→post chemotherapy (desmoplasia) p.22

### 4 · UNCHANGING OPERATIVE PRINCIPLES

- Multidisciplinary decision; imaging within **≈4 weeks**, markers within **1–2 weeks**. p.23
- ALGORITHM
  Safe entry → retroperitoneal exposure → **split-and-roll** complete nodal removal → prospective nerve identification/sparing → **lumbar vessel division** → spermatic cord removal → closure
  p.23

### 7 · COMPLICATIONS AND MANAGEMENT

- ALGORITHM
  Pressure (instrument/assist port) + raise pneumoperitoneum → prepare: inform anaesthesia, blood, vascular help → if uncontrolled: keep pressure, undock, convert
  p.24
- Antegrade ejaculation preserved in **≥90%** after primary nerve-sparing MI-RPLND. p.24

### 8 · OUTCOMES VS OPEN

- Recent robotic node yields approach the open benchmark **≈30**; adjuvant chemotherapy falling but still too frequent. p.24
- Benchmark: recurrence-free ≈**85%** for limited pN+ without adjuvant chemotherapy. p.24

### 10 · MCQ ONE-LINERS

- MCQ PC-RPLND in NSGCT: residual mass >1 cm. p.25
- MCQ 20–30% with enlarged nodes are pN0 at RPLND. p.25
- MCQ First laparoscopic RPLND 1992; first robotic 2006. p.25
- MCQ NCCN is silent on MI-RPLND. p.25
- MCQ EAU: MI-RPLND only in specialist centres. p.25
- MCQ Ideal small-volume II for MIS: 1–2 landing-zone nodes, none >3 cm. p.25
- MCQ Large chemotherapy shrinkage predicts desmoplasia → favour open. p.25
- MCQ Post-chemotherapy standard: bilateral full template with nerve-sparing. p.25
- MCQ Lumbar vessels divided to clear retroaortic/retrocaval nodes. p.25

- MCQ Nerve-sparing preserves L1–L4 postganglionic fibres to hypogastric plexus. p.25
- MCQ Robotic MI-RPLND: steep Trendelenburg with peritoneal hammock. p.25
- MCQ Right template mobilisation includes Kocher manoeuvre. p.25
- MCQ High-fat meal pre-op highlights lymphatics. p.25
- MCQ Weck clips on great-vessel side of lumbar vessels. p.25
- MCQ Do not ligate right renal artery in interaortocaval space. p.25
- MCQ Left template: control lumbar vein into left renal vein. p.25
- MCQ Conversion rate ≈5%. p.25
- MCQ Antegrade ejaculation preserved ≥90% after primary MI-RPLND. p.25
- MCQ Open RPLND node-yield benchmark ≈30. p.25
- MCQ miR-371a-3p may select RPLND candidates. p.25

## Ch 82 · Tumours of the Penis

### 1 · OVERVIEW

- Squamous cell carcinoma (SCC) = **~95%** of penile malignancy. Rare in North America/Europe (**0.4–0.6%** of male cancers); up to **10%** in parts of Asia, Africa, South America. p.26

### 2 · PREMALIGNANT LESIONS — PENILE INTRAEPITHELIAL NEOPLASIA (PEIN)

- TABLE Erythroplasia of Queyrat (EQ) · Glans (± prepuce), uncircumcised; red, velvety, well-marginated; may ulcerate. Mimic: chronic balanitis · **10–33%** p.26
- TABLE Bowen disease (BD) · Shaft; sharply defined scaly erythematous plaque. Mimics: psoriasis, eczema, superficial basal cell carcinoma · **~5%** p.26

### 3 · AETIOLOGY AND RISK FACTORS OF SCC

- TABLE **Phimosis** · Present in **25–75%**; **7–10×** risk. The non-circumcision risk is mediated by phimosis. Cycle: poor hygiene → smegma → inflammation → fibrosis → phimosis p.27
- TABLE **HPV** · Low risk 6, 11 (warts, verrucous). High risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest** (also in metastases). Present in only **~50%** of SCC; tracks number of sexual partners; wives have ~3× cervical cancer risk p.27
- TABLE **Lichen sclerosus** (balanitis xerotica obliterans, BXO) · Cancer in **2.3–9%** on long follow-up; specifically HPV-**independent** SCC p.27
- KEY POINT Two routes converge on the preputial sac: HPV-driven (E6/E7, p16+, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none. Penile cancer is the only neoplasm with a simple prophylaxis that spares the organ. p.27

### 4 · NATURAL HISTORY AND SPREAD

- Distant metastases (lung, liver, bone, brain) uncommon (**1–10%**), late; distant without nodal disease unusual. Untreated: death within **2 years** (femoral erosion haemorrhage, sepsis, inanition); no spontaneous remission. p.27

### 6 · HISTOLOGY

- TABLE Usual (keratinising) **45–65%** · Independent · Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓ · 25–40% p.28

### 8 · STAGING — TUMOUR-NODE-METASTASIS (TNM) 8TH EDITION (AMERICAN JOINT COMMITTEE ON CANCER/UNION FOR INTERNATIONAL CANCER CONTROL, 2017)

- KEY POINT Traps: T1b and T2 share stage IIA; any N3 or T4 = stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery. p.29

### 9 · MANAGEMENT OF THE PRIMARY TUMOUR

- **Penectomy** = oncological gold standard (local recurrence **0–8%**) but harms sexual function and psyche; >half are ≤60 years → penile preservation where safe. p.29
- **Organ preservation suits**: PeIN, Ta, T1, grade I–II in **compliant** patients. **Against**: size **≥4 cm**, grade III, deep cavernosal invasion, unfit for salvage. p.30
- Microscopic proximal spread tracks grade: **~5 mm** (grade I–II), **~10 mm** (grade III); no skip lesions. p.30
- Only margins **<1 mm** raised recurrence → a **negative margin (>1 mm) on frozen section is enough; margins well under 1 cm are acceptable** for small low-grade tumours. p.30
- KEY POINT Margin width is a function of grade, not a fixed 2 cm. Low-grade tumours spread only millimetres and their recurrences are salvageable — so preservation is safe. In high-grade or cavernosal disease, local relapse itself becomes lethal, and amputation regains its place. p.30

### 10 · INGUINAL NODES — BRIEF ALGORITHM → CH 84

- Cure ~**80%** if ≤2 nodes, unilateral, no ENE, no pelvic nodes; pelvic nodes positive → ~**14%** 5-year survival with surgery alone. p.30
- Occult nodal risk by primary: PeIN/Ta/T1a **0–18%** (low risk); **T1b or above** one-third to >half (high risk). Corporal invasion ~60%. p.31
- ALGORITHM

  **cN0, low risk (PeIN, Ta, T1a)** → surveillance (compliant) → palpable node → FNAC/core → positive → ipsilateral inguinal lymph node dissection (ILND) + contralateral staging.
  **cN0, high risk (T1b+, LVI, PNI, grade III)** → bilateral **DSNB** (preferred; ultrasound ± FNAC first) or superficial / fascia-sparing / minimally invasive ILND → positive side → completion ILND.
  **cN1–cN2 mobile, <4 cm** → biopsy → fascia-sparing or radical ILND + contralateral staging; pelvic lymph node dissection (PLND) if >2 nodes or ENE.
  **Fixed / ≥4 cm / pelvic nodes** → cisplatin-eligible: neoadjuvant TIP → resect if response/stable; ineligible: chemoradiation → resect if response.
  **Adjuvant** if >2 nodes, bilateral, ENE or pelvic nodes.
  p.31

### 11 · RADIOTHERAPY

- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks**; penis vertical in a wax/Perspex block that acts as **bolus** (megavoltage beams spare skin, but tumour is cutaneous). ~55% recurrence-free at 5 years. Better control: dose >60 Gy, overall time ≤45 days, fraction ≥2 Gy; worse: T3, >4 cm, high grade. For unfit or locally advanced patients (± weekly cisplatin). p.31

### 16 · MCQ ONE-LINERS

- MCQ PeIN is defined by an intact basement membrane; WHO 2022 regards all PeIN as high grade. p.33
- MCQ Bowenoid papulosis: shaft of young men; does not progress to invasion. p.33
- MCQ Erythroplasia of Queyrat progresses in up to 33%; Bowen disease ~5%. p.33
- MCQ p16 immunohistochemistry is the surrogate marker for high-risk HPV. p.33
- MCQ HPV 16 is the commonest type in penile SCC; HPV is found in ~half. p.33
- MCQ Lichen sclerosus is linked to HPV-independent SCC. p.33
- MCQ Phimosis: 7–10-fold risk; adult circumcision gives little protection. p.33
- MCQ Neonatal circumcision protects against invasive cancer but not CIS. p.33
- MCQ Basaloid SCC: HPV-associated, >half node positive. p.33
- MCQ Verrucous carcinoma: pushing border, does not metastasise. p.33
- MCQ Sarcomatoid SCC is the most aggressive subtype. p.33

- MCQ Any proportion of anaplastic cells makes the tumour grade III. p.33
- MCQ pT1b = LVI, PNI, or high grade. p.33
- MCQ pT2 = corpus spongiosum; pT3 = corpora cavernosa. p.33
- MCQ Urethral invasion no longer upstages in TNM 8th edition. p.33
- MCQ pN1 = ≤2 unilateral inguinal nodes without ENE; pN3 = ENE or pelvic nodes. p.33
- MCQ Hypercalcaemia without bone metastases correlates with bulky inguinal disease. p.33
- MCQ No imaging reliably detects micrometastases in cN0 groins. p.33
- MCQ Negative surgical margin >1 mm suffices; the 2-cm rule is obsolete. p.33
- MCQ Nd:YAG preferred over $CO_2$ laser (6 mm vs 0.1 mm penetration). p.33
- MCQ Topical 5-FU/imiquimod is for Tis only. p.33
- MCQ Circumcision is mandatory before penile radiotherapy. p.33
- MCQ Brachytherapy suits T1–T2 glans tumours <4 cm. p.33
- MCQ Commonest late RT complication: meatal stenosis. p.33
- MCQ Most important prognostic factor: inguinal node status. p.33
- MCQ DSNB is the preferred staging for high-risk cN0 groins. p.33
- MCQ Antibiotic trial before FNAC of palpable nodes is no longer advised. p.33
- MCQ PLND if >2 positive inguinal nodes or ENE. p.33
- MCQ Bleomycin-containing regimens are discouraged for toxicity. p.33
- MCQ Penile BCC arises on the proximal shaft near the base. p.33
- MCQ Commonest sign of penile metastasis: priapism. p.33
- MCQ Commonest sources of penile metastases: bladder, prostate, rectum. p.33
- MCQ Penile lymphoma is treated systemically (R-CHOP). p.33

## Ch 84 · Inguinal Node Dissection

### 1 · WHY THE GROIN MATTERS IN PENILE CANCER

- Inguinal lymph node dissection (ILND) — first described by Basset (1912); Daseler (1948): radical ILND and five-zone anatomy MCQ. p.34
- Early vs delayed ILND (impalpable T2–3 nodes): ENE ≈20% vs ≈95%; 3-year disease-specific survival ≈84% vs ≈35%. → Manage groins within **3 months** of diagnosis. p.34
- Adequacy: ≥**8 nodes per side**. Guideline adherence is poor; omission predicts mortality — fear of morbidity must not cause omission. p.34

### 2 · INDICATIONS AND RISK GROUPS

- Unilateral palpable nodes → contralateral micrometastasis ≈**30%** → contralateral groin is staged. p.34

### 3 · STAGING WORK-UP

- KEY POINT Every groin test fails toward **false negatives**; a negative result never ends the pathway. p.34

6 · OPEN RADICAL ILND

- ALGORITHM
  Meticulous asepsis, frog-leg → oblique incision ≈2 cm below and parallel to inguinal ligament → flaps 3 cm above ligament to apex → sweep tissue off external oblique and cord → divide saphenous at apex (spare if minimal disease) → incise fascia lata over sartorius and adductor longus → dissect cephalad on anterior femoral vessels → divide saphenous at junction → clear deep nodes to femoral canal → **sartorius transposition** over vessels → close femoral canal (inguinal to Cooper's ligament) → suction drains, tack flaps.
  p.35

9 · COMPLICATIONS, PREVENTION AND POST-OP CARE

- TABLE Lymphocele / seroma · Unsealed lymphatics · Sealing devices, suction drain until **<30 mL/day**; aspirate/drain symptomatic or infected collections † p.36

11 · MCQ ONE-LINERS

- MCQ Strongest prognostic factor in penile cancer: nodal status. p.36
- MCQ Manage groins within 3 months of diagnosis. p.36
- MCQ Adequate ILND: ≥8 nodes per side. p.36
- MCQ ENE in a single node = pN3. p.36
- MCQ Commonest sentinel node zone: superomedial. p.36
- MCQ Femoral triangle: inguinal ligament, adductor longus medially, sartorius laterally. p.37
- MCQ Node of Cloquet: between femoral vein and lacunar ligament. p.37
- MCQ Modified ILND spares area lateral to femoral artery and below fossa ovalis. p.37
- MCQ Unilateral palpable node: contralateral micrometastasis ≈30%. p.37
- MCQ Penile lymphatic spread does not skip inguinal nodes. p.37
- MCQ Negative FNAC never ends the pathway. p.37
- MCQ Tumour-replaced node causes false-negative DSNB. p.37
- MCQ Sartorius transposition protects femoral vessels. p.37
- MCQ Flaps raised deep to Camper's fascia. p.37
- MCQ Commonest ILND complication: wound-healing problems. p.37
- MCQ Commonest complication after robotic ILND: lymphocele. p.37
- MCQ Fibrin sealant does not reduce lymphorrhoea. p.37
- MCQ cN3 nodes: neoadjuvant chemotherapy first. p.37
- MCQ Drain removal: output <30 mL/day. p.37
- MCQ Median time to inguinal recurrence ≈1 year. p.37
- MCQ Fossa ovalis, lateral to medial: nerve, artery, vein. p.37
- MCQ pT1b G2: intermediate risk (NCCN) but high risk (EAU-ASCO). p.37
- MCQ Glans tumours may drain directly to deep inguinal nodes. p.37
- MCQ VEIL suits only non-bulky, non-fixed nodes. p.37

## Ch 83 · Tumours of the Urethra

2 · ANATOMY → HISTOLOGY → LYMPHATICS

- KEY POINT Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women the line is also the **continence boundary**: the distal third can be excised without incontinence. p.38

### 5 · MALE POSTERIOR (PROSTATIC) URETHRAL CANCER

- KEY POINT The only question is "stroma breached?" TURP before BCG debulks, improves BCG contact and stages. Ductal disease is not T2 — do not jump to cystectomy. Repeat prostatic biopsies on follow-up. p.40

### 8 · URETHRAL RECURRENCE (UR) AFTER RADICAL CYSTECTOMY

- Risk **1–15%** (~5%); most within 2 years, ~11% after 5 years. p.41

### 9 · MCQ ONE-LINERS

- MCQ Commonest risk factor for male urethral cancer: urethral stricture. p.42
- MCQ Anterior male urethral cancer is mostly SCC; posterior mostly UC. p.42
- MCQ Female urethra: adenocarcinoma nearly as common as UC and SCC. p.42
- MCQ Urethral diverticulum carcinoma is usually adenocarcinoma. p.42
- MCQ Clear cell urethral adenocarcinoma may secrete PSA. p.42
- MCQ Palpable inguinal nodes in urethral cancer are almost always metastatic. p.42
- MCQ Prophylactic inguinal dissection has no proven benefit in cN0 urethral cancer. p.42
- MCQ Prostatic stromal invasion = T2 prostatic urethra. p.42
- MCQ Prostatic ductal involvement without stroma is Tis. p.42
- MCQ MRI is optimal for local staging of urethral cancer. p.42
- MCQ Distal male urethrectomy margin of 5 mm is acceptable. p.42
- MCQ Distal female urethra can be excised without incontinence. p.42
- MCQ Female urethral RT local control depends on length of urethra involved. p.42
- MCQ Chemoradiotherapy (5-FU + mitomycin C) preferred for advanced squamous PUC. p.42
- MCQ Every responder to urethral chemoradiotherapy develops stricture. p.42
- MCQ MVAC is active only against urothelial histology. p.42
- MCQ Male urethral recurrence after cystectomy is ~5%. p.42
- MCQ Prostatic urethral involvement triples male urethral recurrence risk. p.42
- MCQ Orthotopic neobladder lowers urethral recurrence versus cutaneous diversion. p.42
- MCQ Positive urethral margin: absolute indication for urethrectomy. p.42
- MCQ Female cystectomy: bladder neck involvement is main urethral risk. p.42
- MCQ Intraurethral BCG works for urethral CIS, not papillary disease. p.42

## Ch 85 · Benign Disorders of the Penis and Urethra

### 1 · RECONSTRUCTIVE SURGICAL TECHNIQUE

- **Sutures**: always **absorbable** (permanent suture = stone nidus); monofilament polydioxanone (PDS) preferred. Wound support needed **30–60 days**; chromic lasts only 14 days → avoid. **5-0** for most urethral work; **6-0** for single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA); small taper needle (RB-1). p.43

### 2 · TISSUE TRANSFER: GRAFTS AND FLAPS

- Buccal onlays are durable; failures occur within the **first 12 months**. p.44
- KEY POINT Graft trade-off: dense fine plexus and low mass favour take; carried dermis/lamina gives durability and resists contraction. This explains STSG vs FTSG, failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal. p.44

### 3 · SURGICAL ANATOMY (ESSENTIALS)

- KEY POINT Two envelopes, two vascular worlds: dartos carries skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery. p.44

### 4 · LICHEN SCLEROSUS (LS)

- Malignant transformation up to **8%** (mostly squamous cell carcinoma, SCC) → possibly premalignant. p.44

- KEY POINT LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, RUG before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy. p.45

### 6 · PREPUCE AND MEATUS

- ALGORITHM
  Penile block at base → foreskin retracted: mark cuff incision, straight across frenular base → incise to Buck's → foreskin advanced: mark coronal incision ± frenuloplasty → excise sleeve with **bipolar** haemostasis (adult vessels do not seal by pressure) → quadrant dartos sutures → skin closure
  p.46

- **Human immunodeficiency virus (HIV)**: foreskin rich in CD4+ cells, anaerobic microbiome, microtears; circumcision ↓ female-to-male HIV ~**60%**; also ↓ herpes simplex, human papillomavirus. Men who have sex with men: benefit mainly for insertive partners. p.46

### 7 · PENILE CURVATURE

- ALGORITHM
  Excise ventral dysgenetic tissue + mobilise spongiosum glans → perineum → artificial erection → ventral midline septotomy → still curved: dorsal tunical ellipse excision/plication under elevated neurovascular bundle (dorsal arteries may be dominant glans supply)
  p.46

### 9 · ADULT ACQUIRED BURIED PENIS (AABP)

- ALGORITHM
  Dorsal/ventral slit of phimotic ring → glans traction → circumcising incision, excise non-viable skin (keep shaft-base skin) → escutcheonectomy (harvest graft from it first; keep 2–3 cm skin at base) → tack Scarpa to rectus fascia/pubis → drains → cover shaft with STSG (0.012–0.020 inch), fibrin glue, bolster 5–7 days
  p.47

### 11 · MCQ ONE-LINERS

- MCQ Graft take: imbibition (~48 h) then inosculation (~48 h). p.47
- MCQ STSG: reliable take but contracts; FTSG: fastidious but durable. p.47
- MCQ Posterior auricular (Wolfe) FTSG behaves like an STSG. p.47
- MCQ OMG advantage: panlaminar plexus, thick lamina propria, wet-environment tolerance. p.47
- MCQ Z-plasty is a random transposition flap. p.47
- MCQ Bulbar urethra lies eccentric (dorsal) in the spongiosum. p.47
- MCQ McAninch flap relies on the superficial external pudendal subdermal plexus. p.47
- MCQ Commonest site of male genital LS: meatus. p.47
- MCQ LS never involves the posterior urethra. p.47
- MCQ LS malignant transformation: up to 8%, mainly SCC. p.47
- MCQ LS meatal stenosis: RUG before meatotomy. p.47
- MCQ Never use genital skin for LS urethroplasty. p.47
- MCQ Circinate balanitis: reactive arthritis. p.47

- MCQ Balanitis: commonest organism Candida. p.47
- MCQ Paraphimosis irreducible: emergency dorsal slit. p.47
- MCQ Circumcision reduces female-to-male HIV by ~60%. p.47
- MCQ Devine–Horton type IV = congenital curvature from tunical asymmetry. p.48
- MCQ Chordee without hypospadias: erection commensurate with flaccid size. p.48
- MCQ Meatus at glans tip: never divide urethra for ventral curvature. p.48
- MCQ Subclinical fracture curvature: excise and graft, not plicate. p.48
- MCQ Main risk factor for adult buried penis: obesity. p.48
- MCQ Classic infective cause of scrotal lymphoedema: Wuchereria bancrofti. p.48

## Ch 86 · Urethral Stricture Disease

### 2 · SURGICAL ANATOMY RELEVANT TO STRICTURE

- KEY POINT Retrograde glanular flow is what allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation becomes ischaemia — the thread linking vessel-sparing excision and PFUI repair failure. p.49

### 3 · AETIOLOGY

- KEY POINT Aetiology predicts biology: iatrogenic and straddle strictures are focal ischaemic/crush scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it. p.49

### 4 · PRESENTATION AND EVALUATION

- **Urethral rest**: suprapubic tube for **6–8 weeks** lets the hydrodilated urethra contract → true length. p.50

### 5 · DECISION MAKING

- ALGORITHM
  Short (<1.5 cm) bulbar, no dense spongiofibrosis, first treatment → dilation or internal urethrotomy (once) → recurrence → urethroplasty
  Bulbar 1–2 cm (up to 3–5 cm proximally) → excision and primary anastomosis (EPA)
  Longer bulbar → oral mucosa graft (OMG) onlay ± augmented anastomosis
  Penile → onlay (graft or flap); hostile bed/LS/failed hypospadias → staged OMG
  Panurethral/exhausted options/elderly → perineal urethrostomy
  p.50

### 7 · INTERNAL URETHROTOMY (DVIU)

- Direct vision internal urethrotomy (DVIU): incision through scar to healthy tissue so the lumen heals larger, by **secondary intention** — a race between epithelialisation and wound contraction; recurrence if contraction wins. Worse with greater luminal narrowing (>**74%**). p.50
- TABLE **<1.5 cm** · Longer p.51
- KEY POINT No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; repetition spends the patient's reconstructive capital. p.51

### 9 · SUBSTITUTION URETHROPLASTY

- KEY POINT A graft lives on its bed — it succeeds where the bed is thick and vascular (bulb, dorsal corpora) and is staged when the bed, or skin disease, cannot be trusted in one step. p.52
- Designs: ventral longitudinal (**Orandi**); dorsal transverse; hockey stick (7–9 cm); **circular island** — up to **15 cm**; scrotal island (not first choice). p.52

### 11 · PELVIC FRACTURE URETHRAL INJURY (PFUI)

- In ~**10%** of pelvic fractures. Commonest site: **bulbomembranous junction**, not prostatic apex ("apple off its stem" is wrong). Prepubertal: may extend into prostatic urethra. p.52
- Timing: **3–6 months** after trauma. Goal = primary anastomosis in all until proven impossible. Transpubic/abdominoperineal unnecessary; **endoscopic "core-through" not curative**. p.53
- **ALGORITHM**
  On-table endoscopy (exclude bladder stones) → exaggerated lithotomy → perineal incision → bulbospongiosus divided, bulb mobilised → Haygrove sound via suprapubic tract to proximal stump → excise fibrosis → spatulate both ends to **≥30 Fr** → 8–10 tension-free 4-0 polydioxanone (PDS) sutures over silicone stent
  p.53
- Success: high **90s %**. Failure is mostly **ischaemic stenosis** of the mobilised bulb (now a flap on retrograde flow), not anastomotic. p.53

### 15 · MCQ ONE-LINERS

- MCQ "Stricture" applies only to the anterior urethra. p.54
- MCQ Halving calibre cuts area ~75%. p.54
- MCQ Commonest causes: idiopathic, iatrogenic. p.54
- MCQ Bulbar urethra is dorsally eccentric in the spongiosum. p.54
- MCQ LS stricture mechanism: urine intravasation into glands of Littre. p.54
- MCQ Length assessment: dynamic RUG + VCUG. p.54
- MCQ Urethral rest for 6–8 weeks reveals true stricture length. p.54
- MCQ Bleeding during dilation means the scar is torn. p.54
- MCQ DVIU favourable: bulbar, <1.5 cm, first attempt. p.54
- MCQ Second urethrotomy adds virtually nothing. p.54
- MCQ EPA rules: total excision, spatulation, no tension. p.54
- MCQ Barbagli: dorsal graft spread-fixed onto corpora. p.54
- MCQ Grafts and flaps give equivalent results. p.54
- MCQ Onlay outperforms tubularised repair. p.54
- MCQ Avoid genital skin in LS strictures. p.54
- MCQ PFUI site: bulbomembranous junction. p.54
- MCQ PFUI repair timing: 3–6 months. p.54
- MCQ Endoscopic core-through is not curative for PFUI. p.54
- MCQ Progressive approach order: mobilisation, crural separation, infrapubectomy, rerouting. p.54
- MCQ PFUI repair failure: ischaemic bulbar stenosis. p.54
- MCQ Bilateral pudendal occlusion: revascularise before urethroplasty. p.54
- MCQ Bladder neck appearance does not predict continence after PFUI. p.54

## Ch 87 · Surgery of the Scrotum and Seminal Vesicles

### 1 · SURGICAL ANATOMY OF THE SCROTUM

- **Transdermal absorption** ~**40×** other skin; rich in 5α-reductase. MCQ p.55
- KEY POINT Three trunks (aorta, internal iliac, external iliac) anastomose in the cord (~10 cm above testis) and at the cauda. Hence the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal. p.55

### 3 · SCROTAL WALL: EXCISION, SCROTECTOMY, RECONSTRUCTION

- **Fournier gangrene (FG)**: commonest indication for partial scrotectomy. Fluids + broad-spectrum antibiotics + **urgent debridement** (spreads **2–3 cm/h**); mortality ~**20%**. Re-debride every **24–48 h** until no necrosis. Anorectal extension may need colostomy. p.55

- If >**50–60%** skin lost: no immediate reconstruction; testes in medial thigh pockets, saline gauze or within vacuum-assisted closure (VAC). p.55

- ALGORITHM
  Defect <50%, no tension → primary closure · <50% with tension → local scrotal advancement flap / secondary intention · >50% → VAC, then stretched scrotal advancement or delayed primary closure → no local tissue → fasciocutaneous thigh flap (superomedial/anterolateral) · meshed split-thickness skin graft (STSG) last.
  p.56

- KEY POINT Why grafts last: the scrotum works because it is lax; graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor. p.56

### 4 · HYDROCELE

- Fluid between parietal and visceral tunica vaginalis. **Congenital**: communicating via patent processus; ~**80%** resolve by 18 months. **Acquired**: production–absorption imbalance; men >40 y. p.56

- KEY POINT More tunica excised → lower recurrence but larger raw surface and haematoma risk; plication trades a little recurrence for a dry field. Recurrence after any successful repair is **0–6%**. p.56

- Single-session success modest; **retreatment** brings resolution to **85–96%**. Poor in multiloculated or >**750 mL** sacs. Fewer complications than surgery, lower satisfaction. p.56

### 6 · VARICOCELE REPAIR

- Present in ~**15%** of men. For orchialgia (dull ache on standing), **microsurgical repair** gives more complete relief. p.56

### 7 · VASECTOMY

- **Counsel**: not immediately sterile (contraception until post-vasectomy semen analysis, PVSA); repeat vasectomy <**1%**; pregnancy ~**1 in 2000** (0.05%) even after azoospermic PVSA; chronic pain ~**1–2%**. p.57

- TABLE Skin opening · 1.5–3 cm, median or bilateral · ≤**10 mm** · Puncture <10 mm p.57

- KEY POINT Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division of the vas is required; segment excision optional (**1 cm** adequate). p.57

- ALGORITHM
  PVSA at **8–16 weeks** (after ≥10–20 ejaculations) → azoospermia or rare non-motile sperm (RNMS, ≤**100,000/mL**) → stop other contraception · motile sperm → repeat at 6 months → still motile = failure → repeat vasectomy.
  p.57

- **Complications**: haematoma, infection, **sperm granuloma** (expected, not a complication; symptomatic up to 3 months → non-steroidal anti-inflammatory drugs (NSAIDs); persisting <5% → excise vasal ends/redo), post-vasectomy pain ~**2%** (cord block → denervation or reversal). p.58

### 8 · ORCHIDECTOMY AND PROSTHESIS

- **Prosthesis**: offer every man lacking a testis, before cancer orchidectomy. Silicone, saline-filled; pierce port ≤**5 times**, never post-op. Inguinal at radical orchidectomy/children; **high scrotal subdartos** pouch has least infection/erosion (implant away from suture line); trans-scrotal has higher extrusion. p.58

## 10 · CHRONIC SCROTAL PAIN

- **ALGORITHM**
  History, examination, duplex ultrasound → treat cause · none → exclude extrascrotal cause (cross-sectional imaging of abdomen, pelvis, spine, hip) → prostatitis/pelvic floor → treat · scrotal non-specific → support, NSAIDs, tricyclics, gabapentin ≥3 months → epididymal → epididymectomy · testis/post-vasectomy → spermatic cord block: >**50%** relief → microsurgical denervation (or reversal); <50% → pain clinic, inguinal orchidectomy last · post-hernia → ilioinguinal block >**90%** → neurolysis.
  p.58

## 12 · SEMINAL VESICLES: ANATOMY AND DISORDERS

- Seminal vesicles (SV): paired Wolffian (mesonephric) derivatives, bud at ~**12 weeks**; produce ~**60%** of seminal fluid. Join the ampulla to form the ejaculatory duct. p.59
- **Magnetic resonance imaging (MRI) is most accurate**; transrectal ultrasound (TRUS) suggests obstruction if SV width >**1.5 cm** or ejaculatory duct >**2.3 mm** — but confirm by **SV aspiration showing sperm**. p.59

## 14 · MCQ ONE-LINERS

- MCQ Commonest scrotal surgery complication: haematoma — prevented by dartos closure. p.60
- MCQ Dartos fascia limits Fournier gangrene spread to testes. p.60
- MCQ Injury to genital branch of genitofemoral: loss of cremasteric reflex. p.60
- MCQ Suspected testicular tumour: inguinal approach, cord clamp before handling testis. p.60
- MCQ Fournier gangrene spreads 2–3 cm/hour; re-debride every 24–48 h. p.60
- MCQ Lord plication: least haematoma, no drain, thin-walled hydrocele. p.60
- MCQ Jaboulay: partial excision and eversion of parietal tunica behind testis. p.60
- MCQ Sclerotherapy poor for multiloculated or >750 mL hydroceles. p.60
- MCQ Hydrocele recurrence after surgery: 0–6%, regardless of technique. p.60
- MCQ Spermatocele excision risk: obstructive azoospermia. p.60
- MCQ No-scalpel vasectomy: Li Shunqiang, China, 1974. p.60
- MCQ Recommended vas occlusion methods all use mucosal cautery. p.60
- MCQ Excised vas segment, if taken: 1 cm adequate. p.60
- MCQ PVSA at 8–16 weeks; RNMS ≤100,000 non-motile sperm/mL = success. p.60
- MCQ Pregnancy after successful vasectomy ~1 in 2000. p.60
- MCQ Sperm granuloma is not a complication. p.60
- MCQ Orchialgia: inguinal orchidectomy gives better pain outcome than scrotal. p.60
- MCQ Cord block >50% relief selects for microsurgical cord denervation. p.60
- MCQ Dartos pouch preferred over suture fixation for fertility. p.60
- MCQ Testicular prosthesis port: ≤5 punctures, never post-op. p.60
- MCQ Subinguinal microsurgical varicocelectomy: lowest recurrence and hydrocele †. p.60
- MCQ Seminal vesicles contribute ~60% of ejaculate volume. p.60
- MCQ Zinner syndrome: SV cyst + ipsilateral renal agenesis + ejaculatory duct obstruction. p.60
- MCQ TRUS obstruction: SV >1.5 cm, ejaculatory duct >2.3 mm; confirm by aspiration. p.60

## ★ Putting it together: testis, penis and urethra decisions

### A · TESTICULAR MASS: THE FIRST WEEK

- **ALGORITHM**

  Scrotal ultrasound → tumour markers (alpha-fetoprotein [AFP], human chorionic gonadotrophin [hCG], lactate dehydrogenase [LDH]) **before** orchidectomy → offer sperm banking → **radical inguinal orchidectomy** (never trans-scrotal) → repeat markers after orchidectomy (AFP half-life 5–7 days, hCG 1–1.5 days) → CT abdomen–pelvis and chest imaging → stage and risk group (→ Ch 79, 80).

  p.61

### C · PENILE CANCER: PRIMARY AND GROIN

- **ALGORITHM**

  **Primary:** penile intraepithelial neoplasia → topical 5-fluorouracil or imiquimod, laser, glans resurfacing. Invasive → organ-sparing excision, glansectomy, partial or total penectomy with perineal urethrostomy; margins under 1 cm are acceptable (→ Ch 82).

  **Groin, impalpable (cN0):** low risk (pTis/pTa/pT1a) → surveillance; intermediate/high risk → dynamic sentinel node biopsy or modified inguinal node dissection; a positive node → radical dissection.

  **Groin, palpable (cN+):** fine-needle aspiration → positive → bilateral radical inguinal dissection; fixed or bulky nodes (cN3) → neoadjuvant chemotherapy (paclitaxel–ifosfamide–cisplatin, TIP) first; pelvic dissection if ≥2 inguinal nodes or extranodal extension (→ Ch 84).

  p.61

# 11 Stones & Trauma

417 core items · page numbers refer to the full book PDF

## Ch 93 · Management of Upper Urinary Tract Obstruction

### 2 · DIAGNOSIS

- KEY POINT **Dilation is not obstruction.** Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows *where*; functional tests (renography, Whitaker) show *whether it matters*. Combine one of each to plan therapy. p.2

- **Renography**: hydrated → furosemide **0.3–0.5 mg/kg** at **20 min** (collecting system fills first). Technetium-99m mercaptoacetyltriglycine (MAG3) preferred over diethylenetriaminepentaacetic acid (DTPA) for image quality and dosimetry. **Obstructed: drainage half-time (T½) >40 min**; after successful pyeloplasty drainage occurs even before furosemide. p.2

- ALGORITHM

  **Whitaker test**: nephrostomy → bladder catheter (stops transmission of bladder pressure) → perfuse pelvis at **10 mL/min** → non-obstructed ≤12–15 cm $H_2O$; obstructed >**15-20 cm $H_2O$**. Invasive → only when other tests non-diagnostic. Errors: pelvic compliance, patient position

  p.2

### 3 · INDICATIONS AND CHOICE OF INTERVENTION

- **Neonates**: selective observation supported — operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults**, indeterminate → serial renography. p.2

- TABLE Crossing vessel present vs absent · **42% vs 86%** p.3

- **Nephrectomy**: thin parenchymal shell + normal contralateral kidney. Adults <**15%** differential function = non-salvageable; children — no agreed cut-off. Uncertain → temporary stent/nephrostomy, re-scan. Also extensive stones with chronic infection, or multiple failed repairs — only if the other kidney is adequate. p.3

### 4 · ENDOPYELOTOMY

- Pyeloplasty (≈**95%**) wins because it creates a wide, dependent funnel and reduces the pelvis; endourology wins only on recovery. p.3

- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy; marked vessel entanglement on three-dimensional (3D) CT. Such cases go robotic. p.3

- Stent: 14/7-Fr endopyelotomy stent (wide end across UPJ), or 8–10 Fr if passage difficult; traditionally **6 weeks** (2 vs 4 weeks equal). No strenuous activity 8–10 days. p.3

- ALGORITHM

  General anaesthesia → wire + safety wire → flexible ureteroscope via sheath → 200-µm holmium fibre → incise **laterally** into fat, extending into normal ureter → stent + Foley catheter → renogram 4 weeks after stent removal

  p.3

- ALGORITHM

  Percutaneous bleeding (thin atrophic parenchyma): bed rest, transfusion → **do not irrigate** nephrostomy (tamponade) → persistent → **selective embolisation** — low threshold, avoids exploration and nephrectomy

  p.4

- Follow-up: renogram 1 month after stent removal → 6 months → yearly for **2–3 years**. Durability gap widens: pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years. p.4

## 5 · PYELOPLASTY

- KEY POINT **Principles (open = minimally invasive)**: widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** pelvis-to-ureter transition. p.4

- MCQ Trendelenburg 1886 first attempt (patient died); Kuster 1891 first successful dismembered; **Anderson–Hynes 1949** = today's standard; Davis 1943 intubated ureterotomy; laparoscopic 1993; robotic 2002. p.4

- ALGORITHM
  Dissect ureter with periureteral tissue → lateral marking stitch → excise UPJ → spatulate **laterally** → apex to inferior pelvis → transpose anterior to vessel → watertight absorbable anastomosis (± reduction) → stent → drain
  p.4

- **Stent** out at **4–6 weeks** — reduces extravasation and fibrosis. Added nephrostomy in uncomplicated adults → longer stay, more infection; reserve for secondary UPJO. Suction drain near, not on, the suture line. p.4

- **Laparoscopic/robotic**: less morbidity, success >**90–95%** (= open). Robot's advantage is **ergonomic** (wristed suturing, short learning curve), not outcome. Transperitoneal most used (space, familiar anatomy); retroperitoneal smaller space; transmesenteric needs a visible dilated pelvis (pre-stenting hides it). p.4

- ALGORITHM
  Foley out 24–36 h → drain out if negligible; output rises after Foley removal → **replace Foley 7 days** (reflux along stent) → stent out 4–6 weeks → renography
  p.4

## 6 · OTHER UPJ RECONSTRUCTIONS

- KEY POINT Every UPJ repair asks: **what tissue bridges the gap while keeping blood supply?** Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small. p.5

- TABLE Culp–DeWeerd spiral flap · Large pelvis, long proximal narrowing · Longest flap; length:width ≤**3:1** p.5

## 8 · URETERAL STRICTURE — EVALUATION AND ENDOUROLOGY

- Stricture after ureteroscopy (URS) ≈**3%** vs 1.5% after shock-wave lithotripsy (SWL); impacted stone >2 months → ≈1 in 4. Thulium laser linked with more strictures than holmium (higher energy). p.5

- Endourology needs ipsilateral function ≥**25%** (algorithm 20%). p.5

- **Balloon dilation**: retrograde pyelogram → wire → 4-cm, 5–8-mm high-pressure balloon inflated until the **waist disappears**, 10 min → stent 2–4 weeks; antegrade if retrograde impossible. Works by longitudinal splits, like an incision. Contraindicated: infection, stricture >**2 cm**. Favourable: short, recent (≤3 months), non-anastomotic, well vascularised (≈85%); anastomotic ≈50%; transplant strictures respond well. p.6

- ALGORITHM
  <2 cm + function >20% → endoscopic incision | >2 cm + >20% → reconstruction | <20% → nephrectomy (children may still merit repair). Transplant → consider balloon. Renography for 2 years
  p.6

- KEY POINT Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across. p.6

## 9 · URETERAL STRICTURE — RECONSTRUCTION LADDER

- TABLE Ureteroureterostomy · **2–3 cm** · Upper/mid p.6

- TABLE Psoas hitch · **6–10 cm** · Below pelvic brim p.6
- TABLE Boari flap · **10–15 cm** · Lower/mid; spiral may reach pelvis p.6
- ALGORITHM
  Persistent leak: X-ray stent position → drain off anastomosis → stop suction → Foley + anticholinergic → nephrostomy
  p.6
- **Ileal ureter**: isoperistaltic; contraindicated by creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Distal ileum ≥15 cm from ileocaecal valve; segments >15 cm prevent reflux to the pelvis. No proven benefit of tapering. Complications: leak, mucus obstruction, ischaemic necrosis. Progressive dilation + worsening acidosis → seek outlet obstruction; endoscopic surveillance from year 3. p.7
- KEY POINT Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung. p.7

## 10 · URETEROENTERIC ANASTOMOTIC STRICTURE

- ALGORITHM
  <2 cm + >20% → right: endoscopic incision; **left: formal repair** (sigmoid mesentery bleeding) | >2 cm → formal repair | <20% → nephrectomy or observation
  p.7

## 11 · RETROPERITONEAL FIBROSIS

- Age 40–60, male:female 2–3:1. Mass centres on distal aorta at **L4–L5**, encasing ureters; usually bilateral. Back/flank pain commonest; weight loss, fever, hypertension; half present uraemic. IVC involvement → leg oedema, deep vein thrombosis. p.7
- ALGORITHM
  Uraemia → **nephrostomy** or stents → manage **post-obstructive diuresis** → stop culprit drug, biopsy → steroids → ureterolysis
  p.8

## 12 · MCQ ONE-LINERS

- MCQ Obstructed renogram: T½ >40 minutes despite furosemide. p.8
- MCQ Whitaker: 10 mL/min; obstructed >15–20 cm $H_2O$. p.8
- MCQ Endopyelotomy with crossing vessel: 42% vs 86% without. p.8
- MCQ Endopyelotomy incision: lateral. p.8
- MCQ Avoid lower-pole access for antegrade endopyelotomy. p.8
- MCQ Adult kidney <15% function: non-salvageable. p.8
- MCQ Standard pyeloplasty: Anderson–Hynes (1949). p.8
- MCQ Foley Y-V-plasty: high insertion only. p.8
- MCQ Spiral flap length:width ≤3:1. p.8
- MCQ Retrocaval ureter: always right-sided. p.8
- MCQ Endoureterotomy: distal anteromedial, proximal lateral. p.8
- MCQ Commonest Boari complication: recurrent stricture. p.8
- MCQ TUU: donor ureter passes above the IMA. p.8
- MCQ Ileal ureter contraindicated with creatinine >2 mg/dL. p.8
- MCQ Ureteroenteric stricture commoner on left. p.8
- MCQ RPF: aorta at L4–L5; medial ureteric deviation. p.8

- MCQ RPF first line: prednisone. p.8
- MCQ Ureterolysis: always bilateral. p.8
- MCQ PET: most sensitive test of RPF activity. p.8
- MCQ Commonest identifiable RPF cause: drugs (methysergide); commonest malignant: lymphoma. p.8
- MCQ Furosemide in renography: 0.3–0.5 mg/kg at 20 minutes. p.8
- MCQ Preferred renography isotope: MAG3. p.8
- MCQ Crossing vessel: prognostic factor, not contraindication, for endopyelotomy. p.8
- MCQ Endourological contraindication: obstructed segment >2 cm. p.8
- MCQ Only dismembered pyeloplasty excises the abnormal UPJ. p.8
- MCQ Ureterocalicostomy: resect lower-pole parenchyma, not simple nephrotomy. p.8
- MCQ Stricture after URS ≈3% vs SWL 1.5%. p.8
- MCQ Psoas hitch bridges 6–10 cm; Boari flap 10–15 cm. p.8
- MCQ Left ureteroenteric stricture: formal repair (sigmoid mesentery bleeding). p.8
- MCQ Non-refluxing ureteroenteric anastomosis strictures more often. p.8
- MCQ Ureterolysis: start distally; omental wrap most definitive. p.8

## Ch 94 · Upper Urinary Tract Trauma

### 2 · HAEMATURIA

- Significant injury suggested by **gross haematuria**, or **microhaematuria (>5 red blood cells per high-power field [RBC/hpf]) with shock** (systolic blood pressure [SBP] **<90 mm Hg**). p.9
- KEY POINT Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight. p.9

### 3 · CLASSIFICATION — AMERICAN ASSOCIATION FOR THE SURGERY OF TRAUMA (AAST) KIDNEY INJURY SCALE (2018)

- TABLE II · Perirenal haematoma confined to Gerota's fascia; laceration **≤1 cm**, no extravasation p.9
- Grades I–III defined by **depth (1 cm)**; IV–V by collecting system and vessels. MCQ Trap: main-vessel injury with contained bleeding was grade IV in 1989, now V; segmental vascular injury (pseudoaneurysm, arteriovenous fistula) now IV. p.9

### 5 · IMAGING STUDIES

- **Contrast CT with arterial/nephrographic and delayed excretory phases** is gold standard: arterial phase shows bleeding; delayed phase (~**10 min**) shows urinary extravasation, which is missed without it. CT cannot reliably show **renal vein injury** — perfusion looks normal; medial haematoma with normal arterial findings suggests it. p.10
- **Intraoperative one-shot intravenous pyelogram (IVP)**: unstable patient at laparotomy without CT, mainly to confirm a **functioning contralateral kidney**: **2 mL/kg contrast → single film at 10 min**. Alternatives: palpate the other kidney, or loop the injured ureter and give intravenous (IV) methylene blue. p.10

### 6 · NONOPERATIVE MANAGEMENT

- **Failure triggers**: instability; transfusion of **2–3 units**; new flank pain or guarding; enlarging urinoma with fever or rising white count. Most failures need only stent, nephrostomy or AE. p.11

### 7 · URINARY EXTRAVASATION

- KEY POINT Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained. p.11

## 8 · INTERVENTION

- **ALGORITHM**
  Eviscerate small bowel → incise peritoneum over aorta above inferior mesenteric artery → extend to ligament of Treitz → follow aorta to left renal vein → loop vein, retract cephalad → loop both renal arteries → reflect colon → open Gerota's fascia → occlude vessels if bleeding
  p.11

- **ALGORITHM**
  Exposure → vascular control → limited debridement → suture-ligate vessels → **watertight collecting-system closure** → approximate parenchyma over bolster → cover with Gerota's fascia or **pedicled omentum** → drain
  p.12

- **Absolute indication for revascularisation**: solitary kidney or bilateral injury, warm ischaemia ideally **<120 min**. Unilateral with normal other kidney → observe (thrombosis) or nephrectomy; revascularisation salvages only ~⅓. p.12

## 10 · URETERAL INJURY — AETIOLOGY AND GRADING

- **TABLE** II · Laceration **<50%** transection p.12
- **TABLE** IV · Complete transection with **<2 cm** devascularisation p.12

## 11 · URETERAL INJURY — DIAGNOSIS

- **KEY POINT** Ureteral injury is diagnosed by absent contrast beyond it — so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative. p.13

## 12 · URETERAL INJURY — MANAGEMENT

- **ALGORITHM**
  Stable at laparotomy → repair now · unstable → damage control, delayed repair · contusion → stent or excise and repair · late or partial → stent → fails → nephrostomy, delayed repair · ureterovaginal fistula → stent first
  p.13

- **Boari flap**: gaps up to **12 cm**; needs normal bladder; length:width ≤3:1; rarely acute. p.14

## 13 · MCQ ONE-LINERS

- **MCQ** Ipsilateral rib 9–12 fractures: strongest association with renal injury. p.14
- **MCQ** Blunt microhaematuria without shock or deceleration: no imaging. p.14
- **MCQ** About half of UPJ injuries have no haematuria. p.14
- **MCQ** Children: hypotension only after ~50% blood loss. p.14
- **MCQ** AAST 2018: segmental vascular injury is grade IV. p.14
- **MCQ** AAST 2018: main renal vessel laceration is grade V. p.14
- **MCQ** Cortical rim sign: main renal artery occlusion. p.14
- **MCQ** Largest MiGUTS contributor: haematoma rim distance. p.14
- **MCQ** One-shot IVP: 2 mL/kg, film at 10 minutes. p.14
- **MCQ** Isolate renal vessels before opening Gerota's fascia. p.14
- **MCQ** Left renal vein can be ligated; right cannot. p.14
- **MCQ** Urinoma: 0–20 HU. p.14
- **MCQ** Page kidney: compression → renin-mediated hypertension. p.14
- **MCQ** Most sensitive test for ureteral injury: retrograde ureterography. p.14
- **MCQ** AAST ureter IV vs V: devascularisation <2 vs >2 cm. p.14
- **MCQ** Trauma ureteroneocystostomy: non-tunnelled. p.14

- MCQ Boari flap bridges up to 12 cm. p.14
- MCQ Methylene blue: serotonin syndrome risk. p.14
- MCQ Ureteroarterial fistula: endovascular covered stent. p.14
- MCQ Commonest site of iatrogenic ureteral injury: distal ureter. p.14
- MCQ Commonest site of external ureteral injury: proximal ureter. p.14
- MCQ Revascularisation absolute indication: solitary kidney or bilateral injury. p.14
- MCQ Only absolute intraoperative exploration indication: pulsatile or expanding haematoma. p.14
- MCQ AE failure predictors: instability, grade V, extravasation, gross haematuria. p.14
- MCQ Penetrating renal injury: image with any haematuria. p.14
- MCQ Delayed renal bleeding: usually within 21 days; treat by AE. p.14
- MCQ No bowel interposition or autotransplantation in acute ureteral repair. p.14
- MCQ Psoas hitch: lower-third mainstay, 95–100% success. p.14

## Ch 95 · Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

### 1 · OVERVIEW AND EPIDEMIOLOGY

- Lifetime prevalence **1–15%**; United States (US) prevalence ≈ **10%** and rising — partly a detection artefact of more computed tomography (CT). p.15

### 3 · NUCLEATION, GROWTH, AGGREGATION AND RETENTION

- Normal urine holds calcium oxalate at ≈ 4× its aqueous solubility; with inhibitors it precipitates only at **7–11×**. p.16
- Subepithelial papillary plaque (Randall, 1937 MCQ) — present in most stone formers on endoscopy. p.16
- KEY POINT **Plaque vs plug**: plaque is the idiopathic route (an interstitial apatite anchor); plug is intratubular crystal obstruction with tubular injury — the route of secondary systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Insults such as infection or shock-wave lithotripsy (SWL) may convert a calcium oxalate former into a brushite former. p.16

### 4 · INHIBITORS, PROMOTERS AND MATRIX

- **Matrix** = non-crystalline component, ≈ **2.5%** of stone weight; contains **matrix substance A** (in urine of stone formers, not healthy people) and mostly inflammatory proteins. Role undetermined. p.17

### 5 · MINERAL METABOLISM

- TABLE Proximal tubule · **60–65%** · Paracellular; claudin-2 p.17

### 6 · CLASSIFICATION OF NEPHROLITHIASIS

- TABLE Calcium oxalate (pure or mixed with apatite) · **60%** p.18
- KEY POINT Classification is by the **urinary abnormality, not the stone**: calcium stones are a final common product of many derangements, so composition starts the work-up rather than ending it. Fewer than 3% of formers have no identifiable abnormality — evaluation almost always finds a target. Work-up and treatment → Ch 99. p.18

### 7 · HYPERCALCIURIA

- **Definition**: > **200 mg/day** after 1 week of 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day; or > 7 mmol/day (men), > 6 mmol/day (women). Any cut-off is artificial — risk is graded. p.18

- KEY POINT **Fasting urine + PTH separate gut from kidney**: renal leak → fasting urinary calcium high, PTH high, serum calcium normal; absorptive → fasting urine normal, PTH suppressed. The exact mirror images. p.19

- **PHPT** causes ≈ 5% of nephrolithiasis; stones occur in < 5% of PHPT. Suspect when serum calcium > **10.1 mg/dL**; repeat with intact PTH; ionised calcium helps when total calcium is normal (normocalcaemic PHPT). p.19

### 8 · HYPEROXALURIA

- **Definition**: urinary oxalate > **40 mg/day**. Raises calcium oxalate saturation. p.20

- Autosomal recessive hepatic defects of glyoxylate metabolism: glyoxylate cannot become glycine, so it is oxidised to oxalate. Urinary oxalate ≥ **100 mg/day** → stones, marked nephrocalcinosis, renal injury → falling glomerular filtration rate (GFR) → **systemic oxalosis** (vicious circle). p.20

- ALGORITHM

  Fat malabsorption → luminal fatty acids saponify calcium and magnesium → less calcium to bind oxalate → more free oxalate → **colonic hyperabsorption** (bile salts and fatty acids also ↑ colonic permeability)

  p.20

### 9 · HYPERURICOSURIA, HYPOCITRATURIA AND HYPOMAGNESIURIA

- **Definition**: urinary uric acid > **600 mg/day**; commonest cause dietary purine excess. Others: gout, myeloproliferative disease, hypoxanthine-guanine phosphoribosyltransferase (HGPRT) deficiency. p.21

- **Definition**: urinary citrate < **320 mg/day**. p.21

- KEY POINT Citrate excretion reads proximal-tubular **intracellular pH**: anything acidifying the cell (acidosis, hypokalaemia, acid diet) makes the tubule reclaim citrate — which is why alkali (potassium citrate) is the treatment → Ch 99. p.21

### 10 · RENAL TUBULAR ACIDOSIS

- KEY POINT Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria — the recipe for **calcium phosphate** stones. p.22

### 11 · URIC ACID STONES

- Uric acid pKa **5.35**; sodium urate ≈ 20× more soluble than free acid. Solubility of undissociated uric acid ≈ 96–100 mg/L — easily exceeded below pH 6; at pH 6.5 > 1200 mg/L stays soluble. †Pure uric acid stones are radiolucent on plain kidney–ureter–bladder (KUB) radiograph but visible on non-contrast CT (NCCT). p.22

- ALGORITHM

  Insulin resistance / renal lipotoxicity (tubule burns fatty acids not glutamine; ↓ sodium–hydrogen exchanger 3 [NHE3]) → ↓ proximal ammoniagenesis → $H^+$ buffered by titratable acid instead of ammonium → **persistently low urine pH** → undissociated uric acid → stone

  p.22

- KEY POINT Uric acid stones are a disease of **urine pH, not uric acid** — a renal manifestation of metabolic syndrome. Because there is no inhibitor, raising urine pH above the pKa removes the main driver; †alkalinisation can dissolve existing uric acid stones → Ch 99. p.23

### 13 · INFECTION (STRUVITE) STONES

- ALGORITHM

  Urea → (urease) $2NH_3 + CO_2 \rightarrow NH_3 + H_2O \rightarrow NH_4^+ + OH^-$ → urine pH 7.2–8.0 (urease keeps producing ammonia, so pH keeps rising) → $CO_2$ → carbonate; phosphate → $PO_4^{3-}$ → with magnesium: **struvite**; with calcium: carbonate apatite

  p.23

### 15 · ANATOMIC PREDISPOSITION AND PREGNANCY

- KEY POINT Anatomy sets the stage but metabolism writes the stone: patients with anatomical stones still need full **metabolic evaluation**, and surgical correction alone (e.g. pyeloplasty) does not prevent recurrence — treating the metabolic defect substantially cuts it → Ch 99. p.24

### 17 · MCQ ONE-LINERS

- MCQ Commonest stone: calcium oxalate (≈ 60%); calcium stones ≈ 80% overall. p.25
- MCQ Most urinary salts sit in the metastable zone — where inhibitors and therapy act. p.25
- MCQ Urine pH changes affect phosphate, uric acid and cystine — not oxalate. p.25
- MCQ Randall plaque: calcium apatite, originating in thin-limb basement membrane. p.25
- MCQ Ductal plugs suggest a secondary cause (enteric hyperoxaluria, distal RTA, cystinuria). p.25
- MCQ No known urinary inhibitor of uric acid or cystine. p.25
- MCQ Calcitriol: most potent stimulator of gut calcium absorption; PTH has no direct gut effect. p.25
- MCQ Hypercalciuria: > 200 mg/day on 400 mg calcium diet, or > 4 mg/kg/day. p.25
- MCQ Absorptive: normal fasting urinary calcium, suppressed PTH; renal: high fasting calcium, high PTH. p.25
- MCQ Thiazide challenge unmasks normocalcaemic PHPT. p.25
- MCQ Sarcoid hypercalciuria: extrarenal calcitriol, responds to corticosteroids. p.25
- MCQ Dent disease: ClC-5, low-molecular-weight proteinuria, X-linked. p.25
- MCQ Hyperoxaluria > 40 mg/day; primary hyperoxaluria ≥ 100 mg/day. p.25
- MCQ PH1: AGT deficiency, glycolate in urine; liver–kidney transplant. p.25
- MCQ Enteric hyperoxaluria needs an intact colon; ileostomy gives uric acid stones instead. p.25
- MCQ Hyperuricosuria > 600 mg/day; hypocitraturia < 320 mg/day. p.25
- MCQ Distal RTA: urine pH > 6, hypocitraturia, calcium phosphate stones, up to 70%. p.25
- MCQ Recessive distal RTA ($H^+$-ATPase B1) has sensorineural deafness. p.25
- MCQ Uric acid pKa 5.35; low urine pH is the dominant cause of uric acid stones. p.25
- MCQ Cystinuria: SLC3A1 (chromosome 2, type A), SLC7A9 (chromosome 19, type B). p.25
- MCQ Cystine solubility rises meaningfully only at alkaline pH (≈ 1000 mg/L at pH 9). p.25
- MCQ Struvite needs urease; Proteus mirabilis commonest; E. coli rarely urease-positive. p.25
- MCQ Most staghorn calculi are metabolic (calcium phosphate), not struvite. p.25
- MCQ Topiramate and acetazolamide cause calcium phosphate stones via distal RTA-like state. p.25

## Ch 96 · Acute Care of Urinary Lithiasis

### 1 · PRESENTATION

- Haematuria ~**90%**; absent in >10%. p.26

### 2 · DIAGNOSIS

- TABLE Low-dose NCCT (≤**3 mSv**) · Equal for stones ≥3 mm · Weak <3 mm, obesity p.26
- KEY POINT Unknown diagnosis → NCCT answers everything at once. Known stone → US showing hydronephrosis suffices. p.26

### 4 · MEDICAL EXPULSIVE THERAPY (MET)

- Trial of passage ± MET for **4–6 weeks**, then intervene. p.27

5 · URGENT INTERVENTION

- **Stent failure predictors**: pericalcular ureteric thickness (P-CUT) ≥**7.5 mm** (strongest); transverse stone diameter ≥9.5 mm; multiple/large stones → prefer PCN. p.27

- ALGORITHM

  Colic → analgesia + NCCT → infection/AUA criterion? Yes → drainage only (stent/PCN) → definitive later. No → stone <10 mm → trial ± MET 4–6 weeks → fails → URS/SWL

  p.27

6 · SPECIAL POPULATIONS

- Imaging: **US → MRI → low-dose NCCT** (American College of Obstetricians and Gynecologists [ACOG]). Physiological hydronephrosis ~90%, rarely below iliac vessels; resistive index difference ≥**0.06** vs contralateral = obstruction. Fetal risk not significant <50 mSv. p.27
- Conservative first; ~30% need intervention. Stent/PCN encrust → change every **4–6 weeks**; early gestation favours PCN (each stent exchange needs anaesthesia). Preterm delivery highest with PCN. p.27

8 · MCQ ONE-LINERS

- MCQ Gold standard for acute colic: NCCT. p.28
- MCQ NCCT misses indinavir and matrix stones. p.28
- MCQ KUB unacceptable for acute diagnosis. p.28
- MCQ First-line imaging in pregnancy/children: US. p.28
- MCQ Haematuria absent in >10% of stones. p.28
- MCQ First-line analgesia: NSAID (ketorolac 30 mg). p.28
- MCQ Antispasmodics and forced fluids: no benefit. p.28
- MCQ MET benefit: distal stones 5–10 mm. p.28
- MCQ Silodosin: better expulsion, more retrograde ejaculation. p.28
- MCQ Trial of passage limit: 4–6 weeks. p.28
- MCQ Passage predicted by transverse width. p.28
- MCQ Stone >6.5 mm wide: ~9% pass. p.28
- MCQ Infection + obstruction: drain only. p.28
- MCQ Strongest stent-failure predictor: P-CUT ≥7.5 mm. p.28
- MCQ Pregnancy imaging order: US → MRI → low-dose CT. p.28
- MCQ Pregnancy hydronephrosis distal to iliac vessels suggests stone. p.28
- MCQ Resistive index difference ≥0.06 indicates obstruction. p.28
- MCQ URS permissible in any trimester. p.28
- MCQ First stone after 60: exclude aortic and cardiac causes. p.28

## Ch 97 · Surgical Management of Ureteral Calculi

1 · BACKGROUND AND DIAGNOSIS

- Medical expulsive therapy (MET) with α-blockers remains controversial after SUSPEND → Ch 96. Passage becomes less likely with time → intervene after **4–6 weeks**. p.29

2 · INDICATIONS AND PREPARATION

- KEY POINT Why only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen and uncontrollable; laser URS works intraluminally under vision. p.29

### 3 · CHOOSING TREATMENT

- TABLE SWL · Historical limit 2 cm; SFR falls above **10–15 mm** p.30

- KEY POINT Constant trade-off: SWL = fewer complications, more sessions; URS = one session, slightly more morbidity. p.30

### 4 · URETEROSCOPES

- TABLE Cost · Cheap · Repair ~every **15** cases · Suits low volume p.30

### 5 · CONSENT AND ACCESS

- **Difficult ureter**: manoeuvre now, or stent for passive dilation and return later (extra anaesthetic, but among the safest). Second-look after pre-stenting: **2–4 weeks** (≥14 days for SFR; ≤4 weeks for lower sepsis). p.30

- ALGORITHM
  Open-ended catheter to 1–2 mm short of stone → RPG outlines obstruction → contrast hydrodissection + hydrophilic floppy-tip wire (catheter directs force) → advance catheter past stone → exchange for stiff wire (straightens tortuous ureter)
  p.31

### 6 · LITHOTRIPSY

- Tissue damage above **43 °C**; ureter holds a small volume that heats fast. **Heat tracks power, not laser type.** p.32

- Cap power at **≤10 W in ureter**, ≤25 W in kidney; sheath for outflow; continuous room-temperature or cold irrigant. Stopping irrigation to limit retropulsion → thermal injury and stricture. p.32

### 8 · SWL FOR URETERAL CALCULI

- Technique (detail → Ch 98): GA/sedation (10–15 mm movement impairs fragmentation); upper ureter supine, mid-ureter over sacrum **prone**, below sacroiliac joint supine; bubble-free coupling; **60–90 shocks/min** beats 120; ramp energy (protective vasoconstriction). p.32

### 9 · SPECIAL SITUATIONS

- **Pregnancy**: conservative first (if no infection, solitary kidney, anuria, fetal distress); **SWL contraindicated**; URS safe under any anaesthesia with shielded limited fluoroscopy. Stents encrust faster → change **every 4 weeks**; retrograde stent before nephrostomy. p.33

### 10 · COMPLICATIONS

- KEY POINT Nearly every serious URS complication is force without vision — blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage. p.33

### 13 · MCQ ONE-LINERS

- MCQ US misses mid-ureteral stones. p.34

- MCQ Obstructed infected kidney: drain only. p.34

- MCQ Laser URS: anticoagulation may continue. p.34

- MCQ SWL SFR falls above 10–15 mm. p.34

- MCQ Pneumatic lithotripsy: never via flexible scope. p.34

- MCQ Hydrophilic wire: best to bypass, easily displaced. p.34

- MCQ PTFE wire damages flexible-scope channel on backloading. p.34

- MCQ Sheath tip placed above iliac vessels. p.34

- MCQ Pre-stenting unnecessary for ureteral stones. p.34

- MCQ URS after pre-stenting: 2–4 weeks. p.34

- MCQ Thermal injury threshold: 43 °C. p.34
- MCQ Ureteral laser power cap: 10 W. p.34
- MCQ Laser heat depends on power, not laser type. p.34
- MCQ TFL drilling melts stone — paint instead. p.34
- MCQ Single perioperative antibiotic dose suffices. p.34
- MCQ SWL rate 60–90 shocks/min beats 120. p.34
- MCQ Mid-ureteral stone over sacrum: SWL prone. p.34
- MCQ Strongest SWL failure predictor: impaction (ring sign). p.34
- MCQ Forced diuresis after SWL: no benefit. p.34
- MCQ Pregnancy: SWL absolutely contraindicated; stents changed 4-weekly. p.34
- MCQ Avulsion: commonest proximal third; nephrostomy, delayed repair. p.34
- MCQ Post-URS stricture: 3–6%; image every patient. p.34

## Ch 98 · Surgical Management of Upper Urinary Tract Calculi

### 1 · CHOOSING THE APPROACH

- KEY POINT Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude *a* modality, not surgery. p.35
- Truly asymptomatic = no symptoms and no sequelae (including no recurrent urinary tract infection [UTI]). Natural history not benign: **~50%** have a stone event within **5 years**. p.35
- Residual fragments **>4 mm** after ureteroscopy (URS) → treat (more growth, events, reintervention). p.35
- SWL predictors on CT: stone-free rate (SFR) best with density **<900 HU** (Hounsfield units) and skin-to-stone distance (SSD) **<10 cm**. p.35

### 2 · SHOCK WAVE LITHOTRIPSY

- First human SWL 1980 (Dornier); the electrohydraulic HM3 remains the efficacy benchmark MCQ. p.36
- Shock wave: peak positive **30–100 MPa**, then negative tail ≈ −10 MPa; generated at F1, focused on the stone at **F2**. A focal zone **wider than the stone** breaks better and tolerates respiratory movement. p.36
- **Absolute contraindications**: pregnancy; uncorrected coagulopathy or current antiplatelet/anticoagulant use; untreated UTI; severe untreated hypertension; aortic/renal artery aneurysm (aorta >**4 cm**); obstruction distal to the stone; inability to target (skeletal deformity). p.37
- **Rate**: **60–90 shocks/min** beats 100–120 — fast rates build a cavitation cloud that shields the stone and injures tissue. p.37
- **Refuted**: diabetes, hypertension, long-term renal decline, infertility MCQ. p.37

### 3 · URETEROSCOPY (RETROGRADE INTRARENAL SURGERY, RIRS)

- Landmarks MCQ: Bozzini's Lichtleiter 1806 (first endoscope); Takayasu 1974 first ureteral access sheath (UAS). p.37

### 4 · PERCUTANEOUS NEPHROLITHOTOMY

- Landmarks MCQ: Fernström and Johansson 1976 first percutaneous stone removal; Wickham 1984 first tubeless; Valdivia-Uría 1987 first supine; Jackman 1998 first mini-perc (13 Fr). p.38

- **ALGORITHM**
  Confirm system and ureter clear → withdraw sheath under vision, safety wire last out → compress tract → persistent bleeding: re-inflate balloon or **Kaye tamponade catheter** → alert anaesthetist → **selective angioembolisation**
  p.39

### 7 · MCQ ONE-LINERS

- MCQ Asymptomatic renal stones: ~50% have an event within 5 years. p.40
- MCQ Stone size predicts intervention, not symptoms. p.40
- MCQ Treat residual fragments >4 mm after URS. p.40
- MCQ SWL best: density <900 HU, SSD <10 cm. p.40
- MCQ Patient who must stay on anticoagulant: URS. p.40
- MCQ Lower pole >1 cm: SWL not first-line. p.40
- MCQ EAU 10–20 mm: SWL and endourology in equipoise. p.40
- MCQ Triple D score predicts SWL, not PCNL success. p.40
- MCQ Modern lithotripters: electromagnetic source. p.40
- MCQ Piezoelectric: smallest focal volume, lowest energy. p.40
- MCQ Most powerful initial comminution mechanism: dynamic squeezing. p.40
- MCQ No routine stent with SWL. p.40
- MCQ First ureteral access sheath: Takayasu, 1974. p.41
- MCQ UAS lowers intrarenal pressure, not proven to cut infection. p.41
- MCQ Stent omission: normal anatomy, function, no injury, no second stage. p.41
- MCQ Ureteral avulsion: immediate nephrostomy, delayed reconstruction. p.41
- MCQ First percutaneous stone removal: Fernström and Johansson, 1976. p.41
- MCQ PCNL safe on low-dose aspirin. p.41
- MCQ Uncontrolled tract bleeding: Kaye catheter, then angioembolisation. p.41
- MCQ Horseshoe kidney: upper pole access usually subcostal. p.41
- MCQ Transplant stones present without colic (denervated graft). p.41

## Ch 99 · Prevention of Urinary Stone Disease

### 1 · STONE DISEASE AS A CHRONIC CONDITION

- Recurrence up to **50%** within 5–10 years. **ROKS** (Recurrence of Kidney Stone) nomogram predicts a second stone; ROKS 2 predicts multiple recurrences. Risks: young age, male sex, high body mass index (BMI), family history, prior asymptomatic stone, brushite/struvite/uric acid stone; low urine volume and low citrate. p.42

### 3 · GENERAL MEASURES (ALL FORMERS)

- **Fluid** — the cornerstone for every stone type (lowers supersaturation). Target urine ≥**2.5 L/day** (≈3 L intake). Adherence is poor (only ~half reach target); smart bottles and apps may help. p.42
- KEY POINT Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption. p.42

### 5 · HYPERCALCIURIA — THIAZIDES

- **Preferred drug** (AUA): hypercalciuria with recurrent calcium stones, **and** recurrent calcium stones with normal evaluation. First used by Yendt (1966) MCQ. p.43

### 6 · HYPEROXALURIA

- TABLE Urine oxalate · >40, often 60–70+ mg/day; low calcium · Often >**100 mg/day**; high plasma oxalate p.43

### 7 · HYPOCITRATURIA

- Juices: **alkali citrate, not total citrate, matters**. Orange juice (potassium citrate) raises pH and citrate but also oxalate; lemonade (citric acid, little alkali) has most evidence and controllable sugar. Over-the-counter supplements with high total citrate may be acidic (Moonstone) MCQ. p.44

### 8 · URIC ACID STONES — A PH DISEASE

- pKa ≈ **5.5**; >90% soluble at pH 6.5. Urine pH <5.5 is the commonest cause (high acid excretion, low ammonium excretion). p.44
- **Potassium citrate** to pH **6.0–6.5**; avoid >7.0. Sodium alkali risks calcium stones. p.44

### 9 · CYSTINE STONES

- ALGORITHM

  **Dilute** (urine ≥**3 L/day** including at night; sodium ≤2300 mg/day; moderate protein — not in children) → cystine <**250 mg/L** → **alkalinise** with potassium citrate to pH **7.0–7.5** → **thiol** (tiopronin)

  p.44

### 10 · INFECTION (STRUVITE) STONES

- **Acetohydroxamic acid (AHA)**: urease inhibitor; prevents new stones and growth. AUA: only for residual/recurrent struvite after surgical options exhausted. Toxicity in ~30%: haemolytic anaemia, **deep venous thrombosis**, tremor, headache. Contraindicated with creatinine >**2.5 mg/dL** and pregnancy. p.45

### 11 · MONITORING

- ALGORITHM

  Start therapy → 24-hour urine within **6 months** → then at least yearly urine + blood for side effects → new stone: **re-analyse** → periodic imaging

  p.45

### 13 · MCQ ONE-LINERS

- MCQ Most important single risk factor: low urine volume. p.45
- MCQ Target urine output ≥2.5 L/day. p.45
- MCQ Calcium restriction increases stone risk. p.45
- MCQ 24-hour urine is not part of first-stone screening. p.45
- MCQ Thiazide hypocitraturia arises via hypokalaemia. p.45
- MCQ Amiloride plus thiazide: never add potassium. p.45
- MCQ Enteric hyperoxaluria: calcium with meals. p.45
- MCQ Enteric hyperoxaluria: liquid potassium citrate. p.45
- MCQ PH1 gene: AGXT; commonest type. p.45
- MCQ Lumasiran targets glycolate oxidase. p.45
- MCQ Sodium citrate does not lower urinary calcium. p.45
- MCQ Urine pH ≥7 favours calcium phosphate stones. p.45
- MCQ Orange juice raises urinary oxalate. p.45
- MCQ Uric acid pKa 5.5. p.45
- MCQ Uric acid target pH 6.0–6.5. p.45
- MCQ Allopurinol: not first-line for uric acid stones. p.45

- MCQ Cystine target <250 mg/L; pH 7.0–7.5. p.45
- MCQ Thiol of choice: tiopronin. p.45
- MCQ AHA contraindicated if creatinine >2.5 mg/dL. p.45
- MCQ AHA's most severe toxicity: deep venous thrombosis. p.45
- MCQ Repeat 24-hour urine within 6 months of therapy. p.45

## Ch 100 · Lower Urinary Tract Calculi

### 1 · BLADDER CALCULI: CLASSIFICATION AND AETIOLOGY

- TABLE Patient · Children; peak **2–4 years**; male ≫ female · Men >60 with outlet obstruction p.46
- KEY POINT Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child. p.46

### 4 · SPECIAL SITUATIONS

- Risk peaks **3 months** after injury; 15–30% within 10 years; quadruples after first stone. Acute phase: immobilisation hypercalciuria (calcium stones); later: infection. p.47

### 7 · MCQ ONE-LINERS

- MCQ Endemic bladder stone peak: 2–4 years. p.48
- MCQ Endemic stone composition: ammonium acid urate. p.48
- MCQ Secondary bladder stone composition: struvite. p.48
- MCQ Commonest bladder stone presentation: terminal haematuria. p.48
- MCQ Foreign-body stones: heterogeneous nucleation. p.48
- MCQ Bladder stone from BOO: absolute indication for BPH surgery. p.48
- MCQ Laser: modality of choice for transurethral cystolithotripsy. p.48
- MCQ Percutaneous cystolithotomy spares the urethra. p.48
- MCQ ESWL preferred with artificial sphincter or penile prosthesis. p.48
- MCQ Continent pouch stone: percutaneous approach. p.48
- MCQ SCI stone risk peaks 3 months post-injury. p.48
- MCQ CIC lowers stone risk versus indwelling catheter. p.48
- MCQ Transplant bladder stone: remove exposed suture. p.48
- MCQ Urethral calculi: 0.3–1% of stones. p.48
- MCQ Commonest urethral stone site: posterior urethra. p.48
- MCQ Urethral stone presentation: acute painful retention. p.48
- MCQ Anterior urethral stone: never push back. p.48
- MCQ Primary urethral stone: struvite, from stricture or diverticulum. p.48
- MCQ Prostatic calculi: no link with adenocarcinoma. p.48
- MCQ Preputial calculi: send prepuce for histology. p.48
- MCQ Seminal vesicle stones: TRUS most accurate. p.48

## Ch 54 · Management of Paediatric Kidney Stone Disease

### 1 · EPIDEMIOLOGY: THE ADULT DISEASE STARTED EARLY

- Early-onset disease = onset < **24 years**. Incidence rising fastest in **adolescent girls**. About 50% recur within 3–5 years (≥ adult rate). p.49

- TABLE Calcium oxalate (CaOx) · ≈ **80%** — mostly idiopathic, on Randall's plaques, not genetic or infective · 75–80% p.49

- KEY POINT Stone formers have more CKD, hypertension and fracture; childhood is the bone-accrual window, so seek low BMD early. p.49

## 2 · DIAGNOSIS AND IMAGING

- ALGORITHM
  Suspected stone → **ultrasound (US)** grey-scale + colour Doppler → positive: treat as indicated → negative + high suspicion or hydronephrosis → **low-dose non-contrast computed tomography (NCCT)** → negative + low suspicion: stop.
  p.49

## 3 · METABOLIC EVALUATION — EVERY CHILD

- TABLE Calcium · < **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y) p.50

- KEY POINT Never read a child's urine against adult absolute limits: infants physiologically excrete more calcium, so the school-age 0.21 ratio overcalls hypercalciuria. Citrate is an inhibitor — its abnormality is too *little*. p.50

## 5 · MANAGEMENT PRINCIPLES

- Residual fragment threshold: **4 mm on US** (2 mm CT threshold + ≈ 2 mm US overestimate). p.51

- TABLE Uncomplicated ureteric ≤ **10 mm** · Observation ± α-blocker p.51

- TABLE Renal ≤ **20 mm** · SWL or URS p.51

## 6 · URETEROSCOPY

- ALGORITHM
  General anaesthesia + sterile culture → padded lithotomy (infants: gel rolls) → cystoscopy → retrograde pyelogram + safety wire → calibrate orifice with 8/10 Fr coaxial dilator → **if it will not pass: 4.6–4.8 Fr stent ≥ 14 days, return later** → URS → stent if long case, many passes, trauma/oedema.
  p.51

## 8 · PERCUTANEOUS NEPHROLITHOTOMY

- ALGORITHM
  Hypothermia prevention → ureteric catheter → prone (latex precautions in myelomeningocele) → US/fluoroscopic puncture of posterior calyx → serial dilators → 18 Fr sheath (24 Fr staghorn; trim in small children) → nephroscope + ultrasonic/laser → flexible check → nephrostomy (7 Fr infant to 16 Fr adolescent).
  p.52

## 11 · MCQ ONE-LINERS

- MCQ Early-onset stone disease: onset before 24 years; female predominant. p.53

- MCQ Commonest paediatric stone: calcium oxalate (≈ 80%). p.53

- MCQ Alkaline paediatric urine: more CaP, fewer uric acid stones. p.53

- MCQ First imaging: US; twinkle artefact confirms stone. p.53

- MCQ Commonest metabolic abnormality: idiopathic hypercalciuria (30–50%). p.53

- MCQ Hexagonal crystals = cystine. p.53

- MCQ PH1 gene *AGXT*; urinary glycolate raised. p.53

- MCQ Dent disease: X-linked *CLCN5*, low-molecular-weight proteinuria. p.53

- MCQ Bartter = alkalosis; dRTA = acidosis; both hypercalciuric. p.53

- MCQ CYP24A1 loss mimics vitamin D intoxication. p.53

- MCQ α-Blocker MET for ureteric stones ≤ 10 mm. p.53
- MCQ No routine pre-stenting before paediatric URS. p.53
- MCQ Tight ureter: stent ≥ 14 days, no balloon dilatation. p.53
- MCQ Renal ≤ 20 mm: SWL or URS; > 20 mm: PCNL. p.53
- MCQ Low-dose CT before PCNL is a strong recommendation. p.53
- MCQ Significant residual fragment on US: 4 mm. p.53
- MCQ Slower SWL rate (80/min) clears better than 120/min. p.53
- MCQ Calcium is not restricted — it binds gut oxalate. p.53
- MCQ Thiols and lumasiran: only drugs approved for paediatric prevention. p.53
- MCQ Pyridoxine: trial in every PH1 patient. p.53

## Ch 125 · Genital and Lower Urinary Tract Trauma

### 2 · PENILE FRACTURE

- ALGORITHM
  Prompt exploration → ventral penoscrotal (or small lateral) incision → evacuate haematoma → identify tear (artificial erection with saline/dye) → irrigate → close tunica with interrupted **2-0/3-0 absorbable** → repair urethra if injured
  p.54

### 3 · PENETRATING, BITE, AMPUTATION, ZIPPER AND STRANGULATION INJURIES

- **Gunshot**: explore immediately (do not delay for imaging) → irrigate → remove foreign matter → antibiotics → close tunica; artificial erection confirms closure. Almost never isolated. Urethral injury in **15–50%** → RUG or intraoperative dye injection. Viable tissue → primary urethral repair; high-velocity/shotgun tissue loss → suprapubic diversion and staged repair. p.55

- ALGORITHM
  Suprapubic cystostomy → debride → tunica with 2-0 absorbable → urethra over catheter 4-0 → minimal neurovascular dissection → dorsal artery **11-0**, dorsal vein **9-0**, dorsal nerve **10-0** nylon → skin cover
  p.55

### 4 · TESTICULAR INJURY

- ALGORITHM
  Explore (transverse or vertical scrotal incision) → remove necrotic/extruded tubules → close tunica with small absorbable sutures → large defect: tunica vaginalis flap/graft (no synthetic graft) → drain
  p.56

- Salvage **>90%** if repaired within **72 h**; non-operative care → infection, atrophy, pain, late orchidectomy. Solitary testis: protective cup, sport not restricted. p.56

### 6 · BLADDER INJURY

- Retrograde (stress) cystography ~100% accurate: fill to discomfort or **350 mL**; scout → full → **drainage film** (else posterior leak missed). Underfilling gives false negatives. p.57

- KEY POINT Indications for immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, and laparotomy for other reasons. p.57

### 7 · POSTERIOR URETHRAL INJURY (PFUI)

- ALGORITHM
  Blood at meatus → RUG before catheter: 16-Fr catheter in fossa navicularis, balloon 3 mL → penile traction → oblique position → inject 20 mL contrast → image during injection
  p.58
- Defect fills with scar — a **stenosis**, not a stricture. Repair at **2–3 months** once ambulant. p.58

### 10 · MCQ ONE-LINERS

- MCQ Tunica albuginea thinnest ventrolaterally. p.59
- MCQ Penile fracture deviates away from the tear. p.59
- MCQ Eggplant deformity: Buck's fascia intact. p.59
- MCQ Penile fracture urethral injury: ~20% West, ~3% Middle East. p.59
- MCQ Urethral check at fracture repair: flexible cystoscopy. p.59
- MCQ Penile fracture repair delay up to 7 days: outcome unchanged. p.59
- MCQ Penile gunshot: urethral injury 15–50%. p.59
- MCQ Human bites: leave open; dog bites: close primarily. p.59
- MCQ Penile replantation: dorsal artery with 11-0 nylon. p.59
- MCQ Amputation: double-bag; psychiatric referral always. p.59
- MCQ Testicular salvage >90% if explored within 72 hours. p.59
- MCQ Haematocele: up to 80% hide rupture. p.59
- MCQ Dislocated testis: superficial inguinal pouch. p.59
- MCQ Penile STSG non-meshed; scrotal STSG meshed. p.59
- MCQ Bladder injury: pelvic fracture in 83–95%. p.59
- MCQ Pelvic fracture: bladder injury in 5–10%. p.59
- MCQ Cystography: gross haematuria + pelvic fracture. p.59
- MCQ Cystography fill volume: 350 mL. p.59
- MCQ Plain cystography needs drainage film; CT cystography does not. p.59
- MCQ Extraperitoneal extravasation: flame-shaped. p.59
- MCQ Extraperitoneal rupture: 22-Fr catheter, cystogram at 2–3 weeks. p.59
- MCQ Intraperitoneal rupture: immediate repair. p.59
- MCQ PFUI disrupts the bulbomembranous junction. p.59
- MCQ PFUI acute: suprapubic tube. p.59
- MCQ PFUI urethroplasty at 2–3 months. p.59
- MCQ PFUI gap is stenosis, not stricture. p.59
- MCQ Straddle injury: bulbar urethra; anastomotic urethroplasty. p.59

## Ch 62 · Paediatric Genitourinary Trauma

### 1 · RENAL INJURY: HOW THE CHILD DIFFERS

- TABLE Blood pressure after haemorrhage · Maintained until late (≈ **25–30% volume loss**) — hypotension is a late sign · Falls earlier; reliable p.60

## 2 · INDICATIONS FOR IMAGING

- Paediatric mechanism triggers: fall > **10 feet (3 m)**, high-speed motor vehicle collision, blow to abdomen or flank. Associated predictors: thoracic or intra-abdominal organ injury; rib, spine, femur or pelvic fracture. p.60

- Haematuria + hypotension + associated injury + mechanism together give > **98% sensitivity** for significant GU injury in children. European Association of Urology (EAU) agrees, adding associated abdominal injury. p.60

- KEY POINT Trap: "no haematuria, so no imaging". Haematuria is only one of four routes into imaging; in children the two adult markers fail precisely, so mechanism must drive the decision. p.60

## 4 · GRADING

- KEY POINT Grades 1–3 differ only in depth (1 cm) and never involve the collecting system — entry into it makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011). p.61

- TABLE Perirenal haematoma > 3.5 cm adult, > **2.2 cm child** · Bleeding needing angioembolisation p.61

## 5 · NON-OPERATIVE MANAGEMENT (NOM)

- ALGORITHM
  Confirm stability → grade injury → bed rest → serial haemoglobin/haematocrit + abdominal examination → mobilise when gross haematuria resolves and counts are stable.
  p.61

- Complications: grade 1–3 without devitalised segment ≈ zero. Grade 3 (devitalised) to 5: ≈ **25%** bleeding needing embolisation, ≈ **15%** urinoma needing intervention, ≈ 5% open exploration. p.62

## 6 · COMPLICATIONS DURING NOM

- Only grade 4–5 leak; usual source is the parenchymal laceration (also pelvis, ureter, UPJ). Up to **85%** resolve spontaneously → managed non-operatively first (AUA, EAU). p.62

- TABLE Timing · From injury · Typically **1–2 weeks** (up to 1 month) p.62

## 7 · OPERATIVE MANAGEMENT

- ALGORITHM
  Contralateral kidney confirmed → open Gerota fascia → control main vessels → expose whole injury → debride → oversew vessels, close collecting system (absorbable) → close parenchyma over bolster/omentum (sliding-clip renorrhaphy) → drain. Penetrating: inspect ipsilateral ureter.
  p.63

## 9 · URETERAL TRAUMA

- TABLE II / III · < 50% / ≥ **50%** transection p.64

- TABLE V · Avulsion, > **2 cm** devascularisation p.64

## 10 · BLADDER INJURY

- KEY POINT Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor — many need secondary procedures, even closure and diversion. p.64

## 11 · URETHRAL INJURY

- ALGORITHM
  Suprapubic diversion → wait several months (haematoma resolves, bladder descends, gap shrinks) → cystogram + RUG ± endoscopy via both routes → transperineal excision and primary anastomosis → tension: partial pubectomy, bladder mobilisation → catheter 2–3 weeks → voiding trial, cystogram → suprapubic tube out.
  p.65

### 12 · GENITAL INJURY

- **KEY POINT** Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Examine anus and rectum; involve social services before discharge. p.65

### 13 · MCQ ONE-LINERS

- **MCQ** Commonest organ in paediatric GU trauma: kidney. p.66
- **MCQ** Children maintain blood pressure until ≈ 25–30% blood loss. p.66
- **MCQ** Up to two-thirds of grade ≥ 2 injuries have normal urinalysis. p.66
- **MCQ** Paediatric imaging is mechanism-led: fall > 10 feet, high-speed collision. p.66
- **MCQ** Penetrating flank injury needs imaging even without haematuria. p.66
- **MCQ** Delayed CT phase detects urinary extravasation. p.66
- **MCQ** Paediatric high-risk perirenal haematoma: > 2.2 cm. p.66
- **MCQ** Non-operative management is the default at every renal grade. p.66
- **MCQ** Up to 85% of urine leaks resolve spontaneously. p.66
- **MCQ** Urinoma triad: flank pain, ileus, low-grade fever. p.66
- **MCQ** First-line for symptomatic urinoma: ureteral stent. p.66
- **MCQ** Delayed renal bleeding: usually 1–2 weeks, pseudoaneurysm or fistula. p.66
- **MCQ** One-shot pyelogram: 2 mL/kg, film at 10–15 minutes. p.66
- **MCQ** Stable non-pulsatile haematoma at laparotomy: do not explore. p.66
- **MCQ** UPJ disruption: medial extravasation, empty distal ureter. p.66
- **MCQ** Post-traumatic hypertension is renin-mediated: block the renin–angiotensin system. p.66
- **MCQ** Ureteral injury: haematuria absent in 25–50%. p.66
- **MCQ** Child's bladder more intraperitoneal: isolated bladder injury commoner. p.66
- **MCQ** Intraperitoneal bladder rupture: always repair. p.66
- **MCQ** Paediatric posterior urethral injury: often prostatic or bladder neck. p.66
- **MCQ** Pelvic fracture urethral injury: suprapubic tube, delayed perineal anastomosis. p.66
- **MCQ** Mogen clamp: highest risk of glans amputation. p.66
- **MCQ** Normal genital examination does not exclude abuse. p.66

## ★ Putting it together: stones, obstruction and trauma decisions

### A · RENAL COLIC IN THE EMERGENCY DEPARTMENT

- **ALGORITHM**

  **1.** Analgesia first: a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless renal failure, bleeding or pregnancy. Non-contrast computed tomography (NCCT) confirms; ultrasound first in pregnancy and children (→ Ch 96).

  **2. Decompress now** (retrograde stent or percutaneous nephrostomy, equally effective) if obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, impending renal deterioration, refractory pain or intractable vomiting. Treat the stone later — never definitive lithotripsy in an infected obstructed kidney.

  **3. Otherwise** stone <10 mm → trial of passage ± medical expulsive therapy (α-blocker; benefit confined to distal stones 5–10 mm) for 4–6 weeks → if it fails, ureteroscopy (URS) or shock wave lithotripsy (SWL) (→ Ch 97).

  p.67

### B · CHOOSING THE STONE OPERATION

- KEY POINT SWL absolute contraindications: pregnancy, uncorrected coagulopathy or antithrombotic drugs, untreated urinary tract infection, uncontrolled hypertension, aortic or renal artery aneurysm, and obstruction distal to the stone. Do not stent routinely after SWL or uncomplicated URS (→ Ch 98). p.67

### C · PREVENTING THE NEXT STONE

- ALGORITHM

  **Everyone:** fluids for urine output ≥2.5 L/day †, normal calcium intake (restriction increases stones), low sodium, less non-dairy animal protein, more fruit and vegetables (→ Ch 99).

  **High risk or recurrent** → stone analysis, serum chemistry and one or two 24-hour urines → targeted therapy → repeat 24-hour urine within 6 months.

  p.67

### D · RENAL TRAUMA

- ALGORITHM

  **Image** (contrast CT with delayed phase) for gross haematuria, microhaematuria with shock, deceleration injury or penetrating flank/abdominal trauma; blunt microhaematuria without shock → observe (→ Ch 94).

  **Stable, staged grade I–IV and selected V** → non-operative management (>90% success in blunt injury). Contrast blush → angioembolisation. Symptomatic urine leak → retrograde stent + urethral catheter.

  **Explore** for instability despite resuscitation or an expanding/pulsatile haematoma. Unstable at laparotomy without CT → one-shot intravenous pyelogram (2 mL/kg, film at 10 min) to confirm a working contralateral kidney.

  p.68

### G · CHILDREN ARE DIFFERENT

- KEY POINT **Stones:** ultrasound first, low-dose NCCT only if ultrasound is negative and suspicion stays high; every child gets a full metabolic evaluation after the first stone (monogenic disease is common). Renal ≤20 mm → SWL or URS; >20 mm → PCNL (→ Ch 54). **Trauma:** imaging is driven by mechanism (rapid deceleration, fall >3 m, flank blow, rib or spine fracture), not by haematuria or hypotension, because children hold their blood pressure until late; a perirenal haematoma >2.2 cm predicts bleeding needing angioembolisation (→ Ch 62). p.68

# 12 Bladder & Diversion

357 core items · page numbers refer to the full book PDF

## Ch 129 · Tumours of the Urinary Bladder

### 1 · EPIDEMIOLOGY (HEADLINE)

- Median age at diagnosis **73**; ~9 in 10 after 55. **Male:female ≈ 3:1** – smoking, sex differences in carcinogen metabolism and sex steroids. p.2

### 2 · RISK FACTORS

- **Smoking**: **30–40%** of UC; risk **2–3×**; pipes and cigars too; carcinogens found in e-cigarette users. **N-acetyltransferase 2 (NAT2) slow acetylators** are at highest risk, because NAT2 acetylation detoxifies aromatic amines. Risk falls progressively after quitting, approaching never-smokers after **>15 years**. ~20% still smoke at diagnosis; active smoking raises recurrence in non-muscle-invasive bladder cancer (NMIBC) and cancer death in muscle-invasive bladder cancer (MIBC). p.2

### 3 · PATHOLOGY

- TABLE Dysplasia · Normal thickness, loss of polarity, umbrella cells kept; CK20, p53 · 14–19% progress; **60%** → CIS if prior cancer p.3
- KEY POINT Small cell is treated as systemic disease: primary cystectomy gives poor disease-free survival not rescued by adjuvant chemotherapy, whereas neoadjuvant chemotherapy markedly improves cancer-specific survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation – proof that chronic inflammation drives it. p.4

### 4 · MOLECULAR BIOLOGY

- TCGA: luminal-infiltrated responds poorly to cisplatin NAC (immunotherapy candidate); neuronal → etoposide–cisplatin; APOBEC high-mutational-load subset has better survival. MCQ Micropapillary maps to a luminal class. p.5

### 5 · CLINICAL PRESENTATION AND EVALUATION

- **Painless gross haematuria in 85%**, intermittent; microscopic haematuria in nearly all. Irritative symptoms suggest CIS. p.5
- TABLE Suspicious for HGUC · N:C ≥**0.7** + ≥2 features, few cells (<5–10) p.6
- ALGORITHM
  Positive high-grade cytology + negative cystoscopy → random bladder and prostatic urethral biopsies → selective upper-tract washings for cytology ± ureteroscopy
  p.6

### 6 · STAGING AND RISK GROUPS

- **2017 change**: regional nodes without distant metastasis = stage III (previously IV). MCQ Common iliac = N3, not M1a. p.7

### 7 · MCQ ONE-LINERS

- MCQ Commonest presentation: painless gross haematuria (85%). p.7
- MCQ Smoking accounts for 30–40% of UC; risk near never-smokers after >15 years. p.7
- MCQ Slow NAT2 acetylators who smoke have highest risk. p.7
- MCQ Cyclophosphamide carcinogen: phosphoramide mustard; risk above 20 g cumulative. p.7
- MCQ Only S. haematobium is linked to bladder SCC. p.7
- MCQ Radiation-associated bladder cancer latency: 15–30 years; IMRT carries little risk. p.7
- MCQ Leukoplakia of bladder is not premalignant. p.7

- MCQ Cystitis glandularis may progress to adenocarcinoma. p.7
- MCQ Dysplasia with prior bladder cancer progresses to CIS in 60%. p.7
- MCQ CIS lacks FGFR3 mutations; carries TP53, RB, PTEN alterations. p.7
- MCQ RAS mutations are mutually exclusive with FGFR3. p.7
- MCQ Basal/squamous is the commonest consensus class (35%); neuroendocrine-like has worst survival. p.7
- MCQ Luminal markers: GATA3, KRT20, PPARG, FOXA1; basal: CK5/6, CK14. p.7
- MCQ Any micropapillary component → manage as micropapillary; early cystectomy even for NMIBC. p.7
- MCQ Plasmacytoid variant: CDH1 loss, intraperitoneal spread, chemosensitive. p.7
- MCQ Small cell bladder cancer: chromogranin/synaptophysin positive; cisplatin–etoposide first. p.7
- MCQ Preoperative radiation worsened survival in bladder SCC. p.7
- MCQ Bladder adenocarcinoma: exclude secondary source – colonoscopy. p.7
- MCQ Urachal carcinoma is staged by Sheldon; treated by en bloc dome, urachus and umbilicus. p.7
- MCQ Paris System HGUC: N:C ratio ≥0.7 with ≥2 features. p.7
- MCQ UroVysion FISH: chromosomes 3, 7, 17 and 9p21. p.7
- MCQ BTA false positive with haematuria (complement factor H in blood). p.7
- MCQ Highest-sensitivity urine marker: ImmunoCyt. p.7
- MCQ AJCC 2017: N1–N3 without M1 is stage III. p.7
- MCQ Only prostatic stromal invasion makes T4a. p.7

## Ch 130 · Management Strategies for Non–Muscle-Invasive Bladder Cancer

### 1 · DEFINITION AND NATURAL HISTORY

- **Non–muscle-invasive bladder cancer (NMIBC)**: urothelial carcinoma (UC) not invading detrusor: Ta, T1 and carcinoma in situ (CIS). Forms **70–80%** of new bladder cancers: Ta 60–70%, T1 20–30%, CIS ~10%. Pathology, grading and staging → Ch 129. p.8
- Recurrence **40–60%**; progression from ~6% (low-grade [LG] Ta) to 17% (high-grade [HG] T1). p.8
- **Grade, not stage, is the strongest predictor of progression**; CIS is second. HG Ta and HG T1 progress at similar rates. Primary CIS progresses in **>50%**; untreated CIS reaches muscle in 40–83%. p.8
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in roughly **one-third to one-half**; clinical T1 is unreliable. p.8
- KEY POINT NMIBC is two diseases under one label. LG disease is a **recurrence** problem, and HG disease is a **progression** problem. Every decision follows from which one the patient has: chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) immunotherapy treats progression, and cystectomy is for progression that cannot be controlled. p.8

### 2 · RISK STRATIFICATION

- TABLE Low · LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP) · Single post-op chemotherapy instillation p.8
- TABLE High · HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement · Induction BCG + maintenance to **3 yr**. BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene, gemcitabine–docetaxel p.8

- EORTC **overestimates** risk because its cohorts had little maintenance BCG, perioperative mitomycin C (MMC) or repeat transurethral resection of bladder tumour (TURBT). CUETO was derived in BCG-treated patients, so it fits current practice better. MCQ Both use **WHO 1973** grading. p.8

### 3 · ENDOSCOPIC MANAGEMENT

- Assess the upper tracts at diagnosis with computed tomography (CT) urography, or retrograde pyelography at TURBT. Upper tract tumours occur in **<5%**; risk factors are trigonal tumour, CIS and high-risk disease. p.9

- Rationale: residual tumour at the resection site in **≥40%**. In HG T1, re-resection upstages to muscle invasion in **25–30%**, and in up to ~50% if the first specimen had no muscle. It changes treatment in one-third and lowers recurrence and progression in T1 (randomised data). p.9

- TABLE Recommended (within **6 weeks**, include muscularis propria) · Consider p.9

- Small (**≤1 cm**) LG recurrences: office fulguration or laser ablation. No tissue is obtained, so a **prior LG diagnosis and negative cytology are mandatory**. p.9

- TABLE Gain · ~**one-third more CIS**; also smaller papillary tumours · More papillary tumours and CIS p.9

### 4 · SINGLE POST-OPERATIVE INSTILLATION

- MMC given **within 6 h** reduces recurrence; given at 24 h it does not. Level 1 evidence, with the greatest benefit in an **initial solitary LG papillary tumour**. Incremental benefit in recurrent or multiple tumours is limited. p.9

### 6 · BACILLE CALMETTE-GUÉRIN (BCG)

- ALGORITHM
  Wait **2–4 weeks** after TURBT (re-epithelialisation) → urinalysis before each dose (exclude infection or gross bleeding) → 1 vial in 50 mL saline by catheter → retain 1–2 h, restricting fluids, diuretics and caffeine beforehand. If catheterisation is traumatic, delay several days.
  p.10

- CIS initial CR up to **84%**; ~half are durable. Only ~30% are free of recurrence or progression at 10 yr, so close follow-up is mandatory. p.11

- ALGORITHM
  **SWOG (Lamm 2000)**: induction weekly ×6 → 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**. High risk: full dose for 3 yr. Intermediate risk: 1 yr (EORTC 30962).
  p.11

- MCQ BCG is resistant to **pyrazinamide**. Avoid cycloserine (psychiatric effects). Asymptomatic granulomatous prostatitis needs no treatment. p.11

- ALGORITHM
  Prioritise high risk → reduce dose, especially in maintenance → split vials → chemotherapy for intermediate risk (single agent) and high risk (Gem/Doce) → early RC for the highest risk.
  p.11

### 7 · RECURRENCE AFTER BCG

- TABLE **BCG-unresponsive** · Persistent/recurrent CIS or HG Ta/T1 despite adequate BCG, or HG recurrence within **6 months (Ta/T1)** or **12 months (CIS)** of last dose · Unlikely to benefit from more BCG → strongly consider RC p.12

- KEY POINT **Trap:** HG T1 at 3 months after induction alone already counts as unresponsive. CIS or HG Ta at 3 months does not; it is "resistant" and gets more BCG. p.12

## 8 · EARLY RADICAL CYSTECTOMY IN HIGH-RISK NMIBC

- Rationale: T1 failing BCG at 3 months progresses in **82%**; CIS failing BCG progresses in ~50%. With prolonged intravesical therapy, only ~¼ are alive with an intact bladder at 15 yr. Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%**, falling to 64% if the disease was understaged. Even a 12-week delay worsens survival. p.12

- ALGORITHM

  **Strongly consider RC** for: BCG-unresponsive HG disease · persistent HG T1 on re-resection · T1 with CIS, LVI or variant histology · deep lamina propria invasion · diffuse CIS · diverticular HG tumour · distal ureteric or prostatic ductal involvement · tumour too large or inaccessible to resect completely · patient preference.

  p.12

## 9 · SURVEILLANCE

- TABLE Low · 3 months → annually from 12 months; consider stopping at **≥5 yr** · Consider · Only if haematuria p.13

- TABLE High · **3–4-monthly for 2 yr** → 6-monthly for 2 yr → annually **for life** · Same schedule · Annually for 2 yr, then lengthen p.13

## 12 · MCQ ONE-LINERS

- MCQ Strongest predictor of NMIBC progression: grade; second: CIS. p.13
- MCQ LG T1 and HG Ta ≤3 cm are AUA intermediate risk. p.13
- MCQ Any CIS, LVI or variant histology is high risk. p.13
- MCQ EORTC and CUETO tables use WHO 1973 grading. p.13
- MCQ Re-TURBT within 6 weeks for all T1 and for incomplete resection. p.13
- MCQ HG T1 without muscle in the specimen: ~50% upstaged at re-resection. p.13
- MCQ Residual T1 at re-TURBT: progression up to 80%. p.13
- MCQ Diverticular tumour beyond the lamina propria = T3a. p.13
- MCQ Blue-light cystoscopy: greatest gain is CIS detection (~one-third more). p.13
- MCQ NBI wavelengths: 415 and 540 nm; false positives ~60%. p.13
- MCQ Single instillation: within 6 h; ineffective at 24 h. p.13
- MCQ US standard single instillation: gemcitabine 2 g in 100 mL (SWOG S0337). p.13
- MCQ Never give BCG immediately after TURBT: risk of septic death. p.13
- MCQ BCG binds fibronectin; an effective response is Th1. p.14
- MCQ BCG is the only intravesical agent that reduces progression. p.14
- MCQ SWOG maintenance: 3 weekly doses at 3 and 6 months, then 6-monthly to 3 years. p.14
- MCQ BCG is resistant to pyrazinamide. p.14
- MCQ Prednisone for BCG sepsis only with antimycobacterial cover. p.14
- MCQ Valrubicin: FDA-approved for BCG-refractory CIS; CR ~20%. p.14
- MCQ Adequate BCG: ≥5/6 induction + ≥2/3 maintenance (T1: induction alone). p.14
- MCQ BCG-unresponsive window: 6 months for Ta/T1, 12 months for CIS. p.14
- MCQ CIS at 3 months after induction = BCG-resistant → reinduce. p.14
- MCQ Pembrolizumab in BCG-unresponsive CIS: CR 41% at 3 months, 19% at 12 months. p.14
- MCQ Nadofaragene: adenovirus carrying interferon-α2b gene, every 3 months. p.14
- MCQ N-803 is an IL-15 superagonist given with BCG. p.14

- MCQ T1 failing BCG at 3 months: 82% progression. p.14
- MCQ Progression under surveillance does worse than de novo muscle-invasive disease. p.14
- MCQ High-risk cystoscopy: 3–4-monthly for 2 years, then lifelong. p.14
- MCQ UroVysion FISH targets chromosomes 3, 7, 17 and 9p21. p.14

## Ch 131 · Management of Muscle-Invasive and Metastatic Bladder Cancer

### 1 · NATURAL HISTORY AND HISTOLOGY

- Of new bladder cancers, **20–30%** are muscle-invasive bladder cancer (MIBC) and **~12%** node-positive or metastatic. p.15
- **Primary MIBC** = invasive at diagnosis; **secondary** (**~20%**) = progressed from non-muscle-invasive bladder cancer (NMIBC). Secondary MIBC does worse: more understaging and lower pathological response to neoadjuvant chemotherapy (NAC), so it is biologically different. p.15
- Most deaths after adequate local therapy occur within **2 years** and are systemic. They come from micrometastases already present at diagnosis, which is why every curative pathway wraps local treatment in systemic therapy. p.15

### 2 · CLINICAL STAGING

- High-grade T1 or no muscle in the specimen → repeat TUR within **6 weeks**. Without muscle, >40% are upstaged. p.15
- **Prostatic urethral biopsy** (risk: trigone or bladder-neck tumour, carcinoma in situ [CIS], multifocal disease). Take a full loop from mid-prostate to the verumontanum at **5 and 7 o'clock**, where prostatic ducts are densest and CIS is commonest. Stromal involvement affects the need for NAC, urethrectomy and the diversion choice. Full thickness shows mucosa, ducts and stroma for accurate staging. p.15
- **Image before TUR**: post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**. Hydronephrosis strongly suggests extravesical disease. p.16
- **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the low-signal muscularis line on MRI: intact in 1–3, interrupted in 4, breached into fat in 5. A score of **≥3** is suspicious for MIBC, so "equivocal" counts as positive. p.16

### 4 · RADICAL CYSTECTOMY: PRINCIPLES AND EXTENT

- Delaying RC more than **12 weeks** after diagnosis leads to more extravesical and nodal disease and worse survival. p.16
- KEY POINT An organ is spared only where it is oncologically silent. In men the gatekeepers are the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall. p.17

### 5 · PELVIC LYMPH NODE DISSECTION: EXTENT AND EVIDENCE

- About **25%** are node-positive at RC, and nodal status is the strongest predictor of survival. Recurrence is **70–80%** if node-positive vs ~30% if node-negative. p.17

### 7 · OUTCOMES AFTER RC

- **~50%** of RC patients ultimately die of bladder cancer, usually from metastases within 2 years. p.18
- **Pelvic recurrence risk**: low (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**. p.18

### 8 · NEOADJUVANT THERAPY

- TABLE Renal · Glomerular filtration rate (GFR) **<60 mL/min** · Creatinine clearance <30 mL/min p.18

## 10 · BLADDER PRESERVATION

- **Dose**: continuous course to **60–66 Gy** with cystoscopic restaging at 3–4 months (commonest). Split course (≈40 Gy → restage → consolidate) is historical (MGH/RTOG). **55 Gy in 20 fractions** is non-inferior. p.19

- KEY POINT TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopic surveillance decide success. Complete TUR cuts the need for cystectomy (~11% vs 42%). p.19

## 11 · METASTATIC UROTHELIAL CARCINOMA

- ALGORITHM
  Platinum-eligible → GC/ddMVAC (or gemcitabine-carboplatin; or EV + pembrolizumab) → response or stable disease → avelumab maintenance → progression → pembrolizumab, or erdafitinib if FGFR-altered → EV → sacituzumab govitecan. Platinum-ineligible → pembrolizumab → EV.
  p.20

## 12 · FOLLOW-UP

- **After bladder preservation**: cystoscopy every 3–4 months in years 1–2, every 6 months in years 3–4, then yearly to **10 years**. CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5. Biopsy any recurrence. p.20

## 14 · MCQ ONE-LINERS

- MCQ Secondary MIBC: about 20% of MIBC; worse outcome and lower NAC response. p.21
- MCQ Small cell bladder cancer → NAC with etoposide + cisplatin before RC. p.21
- MCQ Prostatic urethral biopsy sites: 5 and 7 o'clock beside the verumontanum. p.21
- MCQ VI-RADS ≥3 is suspicious for muscle invasion. p.21
- MCQ Common iliac nodes = N3; nodes beyond = M1a. p.21
- MCQ Prostatic urethral CIS does not make pT4a; stromal invasion does. p.21
- MCQ Basal subtype gains most from NAC; p53-like is chemoresistant. p.21
- MCQ Standard PLND cranial limit: common iliac bifurcation or ureteric crossing. p.21
- MCQ About 25 nodes gives ~75% chance of detecting nodal disease. p.21
- MCQ SWOG S1011: extended PLND gives no survival gain and more toxicity. p.21
- MCQ Male urethral recurrence: highest risk with prostatic stromal invasion. p.21
- MCQ Intraoperative frozen urethral margin decides orthotopic diversion. p.21
- MCQ Cisplatin-ineligible: GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, NYHA III–IV. p.21
- MCQ ABC NAC meta-analysis: 5% absolute 5-year survival gain. p.21
- MCQ Adjuvant nivolumab (CheckMate 274) for one year in pT3–4 or N+ disease. p.21
- MCQ BC2001 sensitiser: mitomycin C + 5-fluorouracil. p.21
- MCQ TMT salvage cystectomy needed in 10–30%. p.21
- MCQ CIS and hydronephrosis are contraindications to TMT. p.21
- MCQ Partial cystectomy is the treatment of choice for urachal adenocarcinoma. p.21
- MCQ Avelumab maintenance after platinum response: JAVELIN Bladder 100. p.21
- MCQ Second-line pembrolizumab beat chemotherapy (KEYNOTE-045); atezolizumab did not (IMvigor211). p.21
- MCQ Enfortumab vedotin targets Nectin-4; sacituzumab govitecan targets Trop-2. p.21
- MCQ Erdafitinib signature toxicity: hyperphosphataemia. p.21

- MCQ Positive prostatic urethral biopsy does not preclude a neobladder if the frozen margin is negative. p.21
- MCQ Upper tract is the commonest site of late recurrence after RC. p.21
- MCQ Low-grade papillary urethral tumour: resect pre-RC; urethrectomy not needed. p.21
- MCQ Platinum rechallenge if last platinum was at least 12 months earlier. p.21

## Ch 132 · Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

### 2 · TRANSURETHRAL RESECTION OF BLADDER TUMOUR (TURBT)

- Upper tracts: computed tomography (CT) urogram, magnetic resonance imaging (MRI), or ultrasound + retrograde pyelogram. Iodinated contrast if glomerular filtration rate (GFR) **≥60** mL/min; gadolinium if GFR **>30**. p.22
- ALGORITHM
  Lithotomy → antibiotic within 1 h (avoid fluoroquinolones) → 30° lens with visual obturator (urethra, orifices) → 70° lens (bladder neck, dome, anterior wall) → 120° if still incomplete → record map → blue-light / narrow-band imaging where available (more tumours found, less recurrence; useful for carcinoma in situ (CIS))
  p.22
- Tumour extends beyond its visible edge, so take a **~2 cm** margin of normal-looking tissue where possible. p.22
- KEY POINT Perforation can follow an adequate resection of an advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak extravesically. p.23
- TABLE Single-dose mitomycin C / epirubicin · NMIBC; **within 24 h** · Meta-analysis: ~**13%** absolute fall in recurrence p.23
- TABLE Bacille Calmette–Guérin (BCG) · High-grade; most active agent · **Never immediately.** Wait **2–4 weeks** and confirm no gross haematuria († risk of systemic BCG absorption). p.23

### 3 · RADICAL CYSTECTOMY: PREPARATION

- **Venous thromboembolism (VTE) prophylaxis**: pneumatic compression + pharmacological prophylaxis started **before induction**. Extend for **4 weeks** (28 vs 8 days → major VTE reduced, no extra bleeding). Enoxaparin (low-molecular-weight heparin, LMWH) is standard. A DOAC (apixaban) is as effective with better compliance, so consider it for the extended course. p.23

### 4 · OPEN RADICAL CYSTECTOMY: TECHNIQUE

- ALGORITHM
  Lower midline incision; fascia opened exactly in the midline → Retzius space; bladder freed from sidewalls → urachus divided; peritoneum incised lateral to the medial umbilical ligaments → vas / round ligament divided → caecum and small-bowel root mobilised; left Toldt line incised → **retrosigmoid window** (route for the left ureter) → self-retaining retractor (check the vena cava is not compressed) → ureters freed with adventitia; obliterated umbilical artery divided first for length; ureters ligated and divided → PLND → vesical pedicles stapled, clipped or sealed
  p.24
- ALGORITHM
  Peritoneum over seminal vesicles incised (keep a margin if tumour is at the base) → rectum separated; Denonvilliers fascia incised → posterior pedicles controlled → endopelvic fascia opened → dorsal venous complex ligated → urethra divided
  p.24

- **ALGORITHM**
  Ovarian vessels ligated → posterior cervical fornix incised → lateral/posterior pedicles controlled → urethra excised from above or the introitus → **clamshell vaginal closure** (keeps girth, loses length), watertight, 2-0 polyglactin → vaginal pack out day 2
  p.24

### 5 · PELVIC LYMPHADENECTOMY

- Node yield reflects surgical quality: **≥10 nodes** → better survival. In NMIBC at RC, yield >20 → better cancer-specific survival. p.25

### 6 · PARTIAL CYSTECTOMY

- **ALGORITHM**
  PLND → cystotomy away from tumour → full-thickness excision with perivesical fat, 1–2 cm margin on frozen section → reimplant ureter if needed → 2–3-layer closure, test watertight → warm water irrigation (against seeding) → drain → cystogram day 7, then remove catheter
  p.25

### 7 · ROBOTIC RADICAL CYSTECTOMY (RARC)

- **KEY POINT** Bladder cancer has no effective salvage. Urine spillage, a poor PLND or a positive soft-tissue margin leads to almost uniformly fatal recurrence, so robotic technique is judged on oncology first. p.25

- **ALGORITHM**
  Sigmoid released → posterior plane over presacral packet → cul-de-sac / prostate apex → Retzius laterally → anterior pedicle clipped → ureters freed (**never grasp the ureter**) → **extended PLND with ureters intact** → ureters clipped; margin sent; left ureter passed under sigmoid → posterior pedicles → dorsal venous complex → urethra (clip proximally only if neobladder)
  p.25

### 10 · MCQ ONE-LINERS

- **MCQ** Bipolar TURBT uses 0.9% saline; monopolar uses water or 1.5% glycine. p.26
- **MCQ** Obturator jerk: lateral wall tumour; prevent with low distension, bipolar energy, muscle relaxant. p.26
- **MCQ** Ureteric orifice resection: pure cutting current; routine stent unnecessary. p.26
- **MCQ** Palpable mass after complete TURBT = cT3b. p.26
- **MCQ** Extraperitoneal perforation → catheter drainage; intraperitoneal → laparotomy and repair. p.26
- **MCQ** Single post-TUR chemotherapy within 24 h cuts recurrence ~13%; not progression. p.26
- **MCQ** BCG: never immediately; wait 2–4 weeks without gross haematuria. p.26
- **MCQ** Maximal TURBT before cystectomy does not improve survival. p.26
- **MCQ** Iodinated contrast needs GFR ≥60; gadolinium contraindicated if GFR <30. p.26
- **MCQ** Routine mechanical bowel preparation not advised before RC. p.26
- **MCQ** Alvimopan: peripheral μ-opioid antagonist reducing post-cystectomy ileus. p.26
- **MCQ** Extended VTE prophylaxis for 4 weeks after RC; DOAC acceptable. p.26
- **MCQ** Standard PLND lateral limit: genitofemoral nerve; distal: Cooper ligament. p.26
- **MCQ** ≥10 nodes removed is associated with better survival. p.26
- **MCQ** Positive urethral frozen section contraindicates orthotopic neobladder. p.26
- **MCQ** Prostate-sparing RC risk: occult prostate cancer. p.26
- **MCQ** Female RC standard: anterior exenteration; female sex is an independent risk for death. p.26
- **MCQ** Postmenopausal ovaries still produce testosterone; consider sparing. p.26

- MCQ Female neobladder needs ≤cT2 and uninvolved bladder neck. p.26
- MCQ Partial cystectomy: solitary MIBC without CIS; cystogram day 7. p.26
- MCQ Urachal adenocarcinoma: en bloc umbilicus, urachus and dome. p.26
- MCQ RAZOR: robotic non-inferior to open for 2-year progression-free survival. p.26
- MCQ Obturator nerve is not the posterior limit of obturator dissection. p.26

## Ch 133 · Use of Intestinal Segments in Urinary Diversion

### 1 · SURGICAL ANATOMY

- Do not rely on more than **8 cm** of bowel surviving away from a straight vessel → clear ≤8 cm of mesentery at a segment end. Base the pedicle on a palpable artery. p.27
- KEY POINT Watershed zones are fed by the smallest terminal branches of two arteries, so they become ischaemic first in hypotension or embolism → site anastomoses to one side of them. p.27

### 2 · SEGMENT SELECTION

- **B12**: liver stores last **3–5 yr**; low levels common, clinical deficiency rare. Sensitive markers: homocysteine, methylmalonic acid. Many give empirical parenteral B12 after 5–7 yr of ileal diversion. p.28

### 3 · BOWEL PREPARATION AND ENHANCED RECOVERY

- Bacterial load: stomach low; descending colon **$10^{10\text{-}12}$/g**. p.28

### 4 · INTESTINAL ANASTOMOSES

- **Ogilvie syndrome**: dilated caecum within 3 days; water-soluble enema excludes obstruction; rupture risk with caecum **>12 cm** or rising white count → neostigmine/colonoscopic decompression; surgery only for ischaemia, peritonitis or refractory cases. p.29

### 5 · URETEROINTESTINAL ANASTOMOSES

- KEY POINT Conduit pressure is damped by ureteric peristalsis and not transmitted to the renal pelvis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop pelvic colonisation, so a stented refluxing anastomosis is the default. Reflux also allows loopogram imaging of the upper tracts in urothelial cancer. p.30
- Open excision and reanastomosis most durable; endoincision ~50–60% at 2 yr; balloon not durable. Poor endourological outcome: stricture <1 yr post-op, **≥1.5 cm**, left-sided. Metal stents only for limited life expectancy. p.30

### 6 · CONDUIT DIVERSION

- ALGORITHM
  Segment **10–15 cm**, 10–15 cm from ileocaecal valve → transilluminate, keep a major arcade, clear 2 cm of mesentery at ends (windows ~5 cm, not more) → ileoileostomy; segment caudal to it → close mesenteric window → flush until clear → right ureter found by mobilising caecum, left via line of Toldt and passed behind sigmoid mesentery → stented anastomoses (Yankauer sucker as passer) → fix base retroperitoneally in right lower quadrant → stoma
  p.30

### 8 · RENAL FUNCTION AND DETERIORATION

- TABLE Screen · Creatinine + urine protein · Creatinine **<2.0 mg/dL**, no proteinuria → proceed p.31
- TABLE GFR · Inulin clearance · **>35 mL/min** (average tolerance >40) p.31
- TABLE Acidification · Ammonium chloride load · Urine pH **≤5.8** p.31
- TABLE Concentration · Water deprivation · Osmolality **≥600 mOsm/kg** p.31

## 9 · METABOLIC COMPLICATIONS

- KEY POINT Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na, Cl; high K); ileum/colon absorb $NH_4Cl$ (high Cl). Potassium is low with stomach and ileum/colon – **only jejunum is hyperkalaemic**. p.32
- **Gastrin** correlates with bicarbonate; above **120 ng/L** small gastrin rises cause large bicarbonate rises. Distension stimulates gastrin → warn high-gastrin patients about dehydration and overdistension. p.32
- **Ureterosigmoidostomy**: 6–29% (mean 11%) after **10–20 yr** latency; mostly adenocarcinoma at the ureterocolic junction where urothelium meets colon bathed in faeces. p.33

## 12 · MCQ ONE-LINERS

- MCQ Griffiths point: splenic flexure; Sudeck's point: rectosigmoid. p.34
- MCQ Do not clear more than 8 cm of mesentery from a small-bowel segment end. p.34
- MCQ Only jejunum causes hyperkalaemia. p.34
- MCQ Stomach: hypochloraemic hypokalaemic metabolic alkalosis. p.34
- MCQ Ileum/colon: hyperchloraemic hypokalaemic metabolic acidosis via ammonium chloride absorption. p.34
- MCQ Haematuria–dysuria syndrome: gastric segments; treat with H2 blockers. p.34
- MCQ Gastrin above 120 ng/L sharply raises bicarbonate. p.34
- MCQ Life-threatening alkalosis: arginine hydrochloride. p.34
- MCQ Jejunal syndrome: salt, bicarbonate, thiazide. p.34
- MCQ Chlorpromazine and nicotinic acid block chloride transport via cyclic AMP. p.34
- MCQ Correct acidosis alone → hypokalaemic paralysis; give potassium. p.34
- MCQ Commonest cause of altered sensorium: hyperammonaemia. p.34
- MCQ Methotrexate toxicity: drain continent pouch during chemotherapy. p.34
- MCQ Continent diversion: GFR >35, urine pH ≤5.8, osmolality ≥600. p.34
- MCQ Bowel invalidates urine pH, osmolality, urea and creatinine. p.34
- MCQ Osteomalacia prevented by correcting base deficit >2.5 mEq/L. p.34
- MCQ B12 liver stores last 3–5 years. p.34
- MCQ Wallace anastomosis: lowest stricture rate. p.34
- MCQ Left ureter stricture: under the IMA. p.34
- MCQ Most important prophylaxis: intravenous anaerobic-cover antibiotic before incision. p.34
- MCQ Alvimopan: peripheral μ-opioid antagonist, max 15 doses. p.34
- MCQ Loop (Turnbull) stoma: obese; more hernia, less stenosis. p.34
- MCQ Transverse conduit: irradiated pelvis. p.34
- MCQ Treat pure Proteus or Pseudomonas in conduit urine. p.34
- MCQ Ogilvie syndrome: caecum >12 cm → neostigmine. p.34
- MCQ Never place a drain on an intraperitoneal anastomosis. p.34
- MCQ Stoma aperture: keep subcutaneous fat; admit two fingers. p.34
- MCQ Ureterosigmoidostomy cancer: 10–20 years; excise old anastomosis. p.34

## Ch 134 · Continent Urinary Diversion

### 1 · DEFINITION AND BASIC PRINCIPLES

- **Principle 3 – capacity:** mature volume **≥500 mL**. 44 cm ileum double-folded holds <200 mL initially and stretches to 500–600 mL at low pressure within a year — a large initial volume is unnecessary. Ileum stretches more than colon. p.35

- KEY POINT Continence is a pressure balance: low reservoir pressure against adequate outlet resistance. Detubularisation lowers the first; the sphincter or efferent valve supplies the second. Failure of either causes leakage. The original Camey I (tubular ileum) failed because its pressure was too high. p.35

### 2 · PATIENT SELECTION

- ALGORITHM

  Estimated glomerular filtration rate (eGFR) >40 → intact external sphincter → adequate bowel → not frail/no severe liver, cardiac or psychological disease → men: no cancer at distal prostatic urethra; women: no palpable anterior vaginal tumour or cervical invasion → willing to self-catheterise (both sexes) → offer orthotopic

  p.35

- Warn that intraoperative findings force a change in up to **~10%**; agree the back-up diversion. Mark a conduit site in everyone; authors prefer the umbilicus for a continent cutaneous stoma. p.35

- **Renal function** — the key contraindication. Bowel absorbs urea, $K^+$, $Cl^-$ and secretes $Na^+$, $HCO_3^-$ → acid load → hyperchloraemic metabolic acidosis if kidneys are poor. Threshold: creatinine < **1.7–2.2 mg/dL** or eGFR > **35–40 mL/min**. Moderate impairment (eGFR 40–59) is safe: decline is no faster than with conduit. Obstructive uraemia often reverses after cystectomy. All patients: hydration, avoid nephrotoxins, oral sodium bicarbonate. Metabolic detail → Ch 133. p.36

### 3 · CONTINENCE MECHANISM AND ITS PRESERVATION AT CYSTECTOMY

- ALGORITHM

  Divide anterior attachments → incise endopelvic fascia, sweep levators → ligate superficial dorsal vein → slightly incise puboprostatic ligaments → ligate DVC, divide close to apex, suspend to pubic periosteum → incise anterior urethra → place 2-0 sutures → divide catheter → posterior sutures through rectourethralis/Denonvilliers → divide posterior urethra → tag sutures until after ureteroenteric anastomosis

  p.37

- ALGORITHM

  Posterior pedicles → sharp plane off anterior vagina → Foley balloon (~20 mL) marks junction → sweep fat off anterior urethra → free bladder neck and first 0.5 cm urethra → clamp bladder neck (prevents tumour spill) → divide just distal → frozen section

  p.37

### 4 · CHOICE OF BOWEL AND URETERIC REFLUX

- **Antireflux is unnecessary** in a neobladder: Valsalva voiding raises pressure equally throughout, and every antireflux valve can obstruct. Direct (Wallace/Leadbetter) anastomosis has the lowest stricture rate (**3–6%**) vs Le Duc 7–29%. **Obstruction, not reflux, destroys kidneys.** p.38

### 5 · ORTHOTOPIC NEOBLADDER DESIGNS

- ALGORITHM

  **Studer:** isolate ileum 15–20 cm proximal to ileocaecal valve (distal division in avascular plane; short proximal mesenteric division) → discard ~5 cm proximally for mobility → 40–44 cm reservoir (two 20–22 cm limbs) + afferent limb (longer if ureters short) → open near mesentery → oversew back wall (running 3-0 polyglycolic acid) → fold opposite way, close → stented end-to-side ureteroileal anastomoses → urethral anastomosis at dependent point

  p.38

- KEY POINT Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later. p.39

### 7 · PERIOPERATIVE CARE

- **Drainage** (authors): 7 Fr single-J stents tied to a 24 Fr urethral (or caecostomy) catheter, irrigated routinely to clear mucus; drain out when low; review ~day 10; catheter out at **3 weeks**; no routine pouchogram unless drain output is significant. Then frequent timed voids, lengthened progressively, plus pelvic floor exercises. p.39

### 8 · RESULTS AND COMPLICATIONS

- Daytime continence develops over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2; persistent nocturnal incontinence **20–50%**. Risk: age >65, colon, diabetes, no nerve sparing, hysterectomy. p.40
- KEY POINT Men fail by **leaking** (sphincter without its reflexes) — augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks) — support the pouch. p.41

### 11 · MCQ ONE-LINERS

- MCQ Mature neobladder target capacity: ≥500 mL at low pressure. p.41
- MCQ Detubularisation abolishes coordinated peristaltic contractions; a sphere maximises volume per surface area. p.41
- MCQ Most distensible bowel muscle: ileal circular; least: colonic longitudinal. p.41
- MCQ Renal threshold for continent diversion: eGFR >35–40 mL/min or creatinine <1.7–2.2 mg/dL. p.41
- MCQ Strongest male urethral recurrence risk: prostatic stromal invasion. p.41
- MCQ Strongest female urethral risk: bladder-neck tumour; no skip lesions to urethra. p.41
- MCQ Bladder CIS does not predict urethral recurrence. p.41
- MCQ Best test for urethral involvement: intraoperative frozen section of urethral margin. p.42
- MCQ Orthotopic neobladder protects against urethral recurrence (HR 0.44). p.42
- MCQ Commonest technical reason to abandon orthotopic diversion: short/fatty mesentery. p.42
- MCQ Self-catheterisation after neobladder: 10–20% of men, ~30% of women. p.42
- MCQ Rhabdosphincter lies anterolateral to urethra, supplied by pudendal nerve. p.42
- MCQ Omental flap between vagina and pouch prevents pouch–vaginal fistula. p.42
- MCQ Hautmann W uses ~70 cm ileum; Studer 40–44 cm plus afferent limb. p.42
- MCQ Kock nipple late problems: stenosis, extussusception, staple stones. p.42
- MCQ Lowest ureteric stricture: direct Wallace/Leadbetter refluxing anastomosis, ~3–6%. p.42
- MCQ Studer vs T pouch randomised controlled trial (RCT): no renal difference; T pouch more reoperations. p.42
- MCQ Mitrofanoff appendix channel needs ≥5–6 cm length and ≥12 Fr calibre. p.42
- MCQ Indiana pouch continence: imbricated ileocaecal valve plus tapered terminal ileum. p.42
- MCQ Undrained urine leak: percutaneous drainage/nephrostomy, not early open repair. p.42
- MCQ Alvimopan is a peripheral μ-opioid antagonist reducing postoperative ileus. p.42
- MCQ Night-time incontinence persists in 20–50%; bowel water secretion contributes. p.42
- MCQ Retention: women 20–30% vs men 4–10%; female cause is mechanical angulation. p.42
- MCQ Treat retention with intermittent self-catheterisation; drugs are ineffective. p.42
- MCQ Treat bacteriuria only if symptomatic or Proteus/Pseudomonas. p.42

- MCQ Persistent cutaneous stomal incontinence requires surgical revision of efferent mechanism. p.42
- MCQ Female urethral recurrence essentially never occurs without bladder-neck involvement. p.42
- MCQ Indiana valve imbricated over a 16 Fr catheter with non-absorbable Lembert sutures. p.42
- MCQ Prior bowel resection: reuse the old anastomosis as one reservoir end. p.42
- MCQ Pouch–vaginal fistula beyond first weeks needs surgical repair, not drainage alone. p.42
- MCQ Annual vitamin B12 and lifelong upper-tract imaging after continent diversion. p.42

## Ch 135 · Minimally Invasive Urinary Diversion

### 1 · CONTEXT

- International Robotic Cystectomy Consortium (IRCC): ICUD rose ≈**11%/year**, driven by **ileal conduit** (commonest robotic diversion). p.43

### 2 · PATIENT SELECTION AND CONTRAINDICATIONS

- TABLE Neobladder (absolute) · Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function — same as open p.43

- ALGORITHM
  Pasadena Consensus learning pathway: RARC + extracorporeal diversion → intracorporeal conduit (≈30–100 cases) → continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.
  p.43

### 3 · PREOPERATIVE CARE AND PERIOPERATIVE BUNDLE

- TABLE Thromboprophylaxis · Pneumatic compression + stockings; weight-based low-molecular-weight heparin before and up to **4 weeks** after p.43
- TABLE Antibiotics · Broad-spectrum intravenous within **1 h** p.43

### 5 · ANATOMIC CONSIDERATIONS (NERVE SPARING)

- TABLE Sphincter · One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex · Mainly mid–distal third; continence survives loss of bladder neck and cranial third p.44

### 6 · INTRACORPOREAL ILEAL CONDUIT

- ALGORITHM
  Left ureter under sigmoid mesentery to right → 12 cm ileum, 15–20 cm from ileocaecal valve → marionette stitch → isolate with 45-mm stapler → ureteroileal anastomoses (left first) + stents → bowel reanastomosis → stoma.
  p.44

### 7 · INTRACORPOREAL NEOBLADDER

- ALGORITHM
  Urethral anastomosis (Van Velthoven barbed suture) → detubularise distal 40 cm (cold scissors, over 24-Fr chest tube) → posterior plate → ureters to afferent limb/chimney, stented → close anterior wall after 22-Fr catheter confirmed.
  p.45

- **ALGORITHM**
  Mobilise right colon to mid-transverse → divide ileum 15 cm from ileocaecal junction and transverse colon proximal to middle colic → restore ileocolic continuity immediately (prevents malrotation) → resect appendix base → detubularise along anterior taenia, fold vertically → ureters through posterior plate, double-J; 24-Fr caecostomy via appendiceal stump → taper ileum over 14-Fr catheter, imbricate ileocaecal valve → umbilical stoma.
  p.46

- **KEY POINT** Capacity comes from the detubularised right colon; continence from the tapered ileum and imbricated ileocaecal valve. p.46

### 8 · POSTOPERATIVE CARE: DRAINS, CATHETERS, STENTS

- **TABLE** Neobladder catheter · **3–4 weeks**, after leak-free pouchogram p.46

### 9 · OUTCOMES

- **Operative time** is a quality measure (prolonged time → more complications, mortality). Strongest determinant: diversion type (conduit ≈70 min shorter). ICUD ≈ ECUD time (redocking cancels saving). Proficiency ≥**66 RARCs**. p.46

- **KEY POINT** Why is night continence worse? Daytime continence relies on the urethrosphincteric guarding reflex (urine in posterior urethra → sphincter contraction). In deep sleep this reflex is inactive, so continence depends on reservoir capacity — hence detubularisation, nerve and sphincter preservation, night alarm and limiting nocturnal output. Assess at 6–12 months, when capacity is maximal. p.46

### 11 · MCQ ONE-LINERS

- **MCQ** Commonest robotic diversion: ileal conduit. p.47
- **MCQ** Annual rise in ICUD (IRCC): ≈11%. p.47
- **MCQ** Creatinine contraindicating neobladder: >2 mg/dL. p.47
- **MCQ** Main contraindication to robotic route: poor cardiopulmonary tolerance of steep Trendelenburg. p.47
- **MCQ** Bowel preparation before ICUD: none, no oral antibiotics. p.47
- **MCQ** Low-molecular-weight heparin duration: up to 4 weeks. p.47
- **MCQ** Stay <5 days: ≈twofold 90-day major complications. p.47
- **MCQ** Target time: ≤4 h conduit, ≤5 h neobladder. p.47
- **MCQ** Marionette stitch: untied, held by instrument. p.47
- **MCQ** Conduit: 12 cm ileum, 15–20 cm from ileocaecal valve. p.47
- **MCQ** Stricture-reducing enterotomy: athermal buttonhole with serosal removal. p.47
- **MCQ** Guidewire removed only after stent anchoring. p.47
- **MCQ** Karolinska innovation: early neobladder–urethral anastomosis. p.47
- **MCQ** Only stapled intracorporeal pouch: Padua. p.47
- **MCQ** Longest ileum: USC (65 cm); shortest: Saint-Augustin Y (40 cm). p.47
- **MCQ** Neobladder catheter out: 3–4 weeks after pouchogram. p.47
- **MCQ** Strongest determinant of operative time: diversion type. p.47
- **MCQ** IRCC proficiency: ≥66 RARCs. p.47
- **MCQ** Parastomal hernia: ≈one-third by 3 years. p.47
- **MCQ** Most reoperations: diversion-related, after 90 days. p.47
- **MCQ** Night incontinence: guarding reflex inactive in sleep. p.47

- MCQ iROC: QoL benefit lost after 12 weeks. p.47
- MCQ Indiana pouch continence: tapered ileum + imbricated ileocaecal valve. p.47
- MCQ Reoperation predictors: neoadjuvant chemotherapy, high-grade complication. p.47
- MCQ ICG aids robotic uretero-enteric stricture revision. p.47
- MCQ Good continence predictors: ECOG status, retained filling sensation. p.47

## Ch 124 · Bladder Surgery for Benign Disease

### 1 · SURGICAL ANATOMY ESSENTIALS

- KEY POINT Three landmarks organise every benign bladder operation: the space of Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety — knowing where each lies decides whether the operation can stay extraperitoneal. p.48
- KEY POINT Common aftercare for almost every operation here: watertight **two-layer 2-0 polyglactin** closure (mucosa first, then seromuscular), leak test by filling, large-bore Foley ≈ **1 week**, pelvic drain on self-suction out at ≈ **48 h** when output is low, **cystogram before catheter removal**. High drain output → compare drain-fluid creatinine with serum creatinine (raised = urine leak). p.48

### 2 · BLADDER DIVERTICULA

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**; augmentation for neurogenic bladder. **Endoscopic outlet surgery goes first** — irrigation afterwards would distend the bladder and disrupt fresh suture lines. p.49
- ALGORITHM
  Supine, anterior superior iliac spine (ASIS) at table break, slight flexion → 18 Fr Foley, fill ≈ 200 mL, plug → lower midline incision, split recti, incise transversalis fascia (distension pushes the peritoneal reflection up) → sponge-stick develops Retzius and gutters, sweeping peritoneum off the dome → vertical midline cystotomy dome to above bladder neck, 2-0 stay sutures at apices → inspect mucosa, identify orifices (5 Fr ureteral catheter if concern) → excise sac → close defect → 20–22 Fr Foley, two-layer cystotomy closure, leak test → optional suprapubic catheter through a separate punch cystotomy → pelvic drain.
  p.49
- TABLE Delivery · Allis clamp through the neck everts sac into bladder; or (not evertible) sac dissected out of pseudocapsule from inside with Allis countertraction on the neck; mucosal incision ≈ **0.5 cm** round neck · Same mucosal incision; finger or suction tip in sac pushes it out; Kittner clears fat; extravesical neck incised circumferentially p.50
- Lithotomy, steep Trendelenburg; cystoscopy first (retrograde pyelogram, selective catheterisation or transillumination). Pneumoperitoneum **15 mm Hg**; robotic assistant port contralateral to the diverticulum (iliac bone restricts an ipsilateral port). p.50
- ALGORITHM
  Asymptomatic leak → keep Foley and drain → symptomatic collection → percutaneous drain → drain study and cystogram before removal → persistent → nephrostomy tubes vs exploration and closure.
  p.50
- KEY POINT Diverticulectomy is a dissection problem, not a reconstruction problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely — and fix the outlet first. p.50

### 3 · BLADDER STONES, FOREIGN BODIES AND FISTULAE

- **Percutaneous**: distended bladder → 1-cm incision two fingerbreadths above symphysis → 18-gauge needle under ultrasound or cystoscopic vision (prior surgery: imaging to exclude interposed bowel) → stiff guidewire coiled in bladder → dilate to **30 Fr** sheath → 24–26 Fr nephroscope; stones <1 cm grasped intact, larger fragmented (pneumatic/ultrasonic); entrapment-bag option. Suprapubic catheter over the wire, clamped at 24–48 h for trial of voiding (or urethral catheter out at 24–48 h). p.51

- **Transvesical laparoscopy**: trocars into bladder under cystoscopic vision, pneumovesicum **8–12 mm Hg**; avoids the peritoneum; suited to eroded mesh. p.51

### 4 · PARTIAL CYSTECTOMY FOR BENIGN DISEASE AND URACHAL SURGERY

- Full-thickness excision preserving overlying perivesical fat, with ≈ **2-cm margin** if capacity allows; biopsies/frozen section of margins; reimplant if orifice involved; 22 Fr Foley, two-layer closure, drain. p.52

- KEY POINT Why margins for benign lesions? Leiomyoma, paraganglioma and endometriosis are infiltrative and recur after incomplete excision — so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, margin). Urachal surgery is the exception: the whole tract matters, a wide margin does not. p.52

### 6 · URETERAL REIMPLANTATION IN ADULTS

- KEY POINT Why refluxing is acceptable in adults: VUR is rarely the indication, and refluxing anastomoses show no faster renal decline, more stenosis or (on long-term data) more infection than tunnelled ones, while needing less ureteral length and less bladder mobilisation. p.53

- TABLE Tunnel · **3–5 × ureteric diameter** (5:1 robotically) · None p.53

- Sutures: full-thickness through ureter; mucosa plus some detrusor on bladder (mucosal apposition); knots outside the lumen. Stent: **6 Fr × 26–28 cm double-J**, placed before completion. p.54

- Primary reimplant for defects up to **4 cm**; longer → psoas hitch/Boari; renal descent adds up to **5 cm**. More bladder reach: divide contralateral medial umbilical ligament, free peritoneum off the dome, ligate the contralateral pedicle. p.54

- Foley 1 week with cystogram; drain at 48 h; stent **≈6 weeks**, removed cystoscopically. p.54

- KEY POINT Reimplantation fails for one reason in two forms — ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other. The robot helps mainly by easing intracorporeal suturing. p.55

### 7 · BRIDGING DISTAL URETERAL DEFECTS

- TABLE **Boari flap** · Tubularised anterior wall flap · Up to **15 cm** (mid-ureter) · Base ≥**4 cm**, apex ≥**3 cm**, length:base ≤**3:1**, measured on empty bladder p.55

- KEY POINT The bladder is the donor tissue: hitch first, flap second — every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and the bladder's capacity. p.55

### 8 · ENTEROCYSTOPLASTY (AUGMENTATION)

- TABLE **Ileum** (commonest) · Abundant, mobile, long mesentery, robust supply · 25 cm (15–20 cm minimally invasive) taken **≈20 cm** proximal to ileocaecal valve (spares terminal ileum: B12/bile salts); acidosis risk p.56

- **ALGORITHM**
  Midline incision, peritoneum opened → isolate ileum, check tension-free reach to dome → transilluminate mesentery (arcade reaches ends) → staple bowel; divide mesentery keeping a broad base (no torsion) → side-to-side stapled anastomosis; close mesenteric window (internal hernia) → excise staple lines, irrigate, **detubularise** on antimesenteric border, fold into U → **bivalve bladder widely** along the dome, anterior to posterior → wide-mouthed transmural anastomosis (robotically: start posteriorly, U apex at bladder neck) → suprapubic tube through separate cystotomy ± Foley → leak test, drain.
  p.56

- Stones (mucus, bacteria, stasis) — commonest long-term complication; irrigation may reduce mucus. Neoplasia (adenocarcinoma and urothelial) → surveillance endoscopy from **10 years**. Routine blood tests for metabolic and vitamin deficiency. p.56

- **KEY POINT** Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem: detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, blood tests and endoscopic surveillance manage the price. p.56

## 9 · CONTINENT CATHETERISABLE CHANNELS

- **ALGORITHM**
  Minimally invasive appendicovesicostomy: lithotomy, 10–15° Trendelenburg → stay suture on appendix → detach from caecum with cuff, close caecal defect → spatulate distal end → ≈4-cm submucosal detrusor tunnel at dome or anterior wall → button-hole mucosa, anastomose with 4-0 polydioxanone → catheter in channel, stoma matured in right lower quadrant or umbilicus, pelvic drain.
  p.57

- **KEY POINT** Use the appendix when present and long enough; reconfigure ileum only when it is not. Either way, continence comes from tunnel length through detrusor, and failure to catheterise regularly risks the bladder, not just the channel. p.57

## 11 · MCQ ONE-LINERS

- **MCQ** Bladder diverticulum wall lacks functional muscularis propria. p.58
- **MCQ** Hutch diverticulum: congenital, posterolateral, beside ureteric orifice, with VUR. p.58
- **MCQ** Cancer within a diverticulum → partial or radical cystectomy, not diverticulectomy. p.58
- **MCQ** Endoscopic prostate surgery precedes diverticulectomy at the same sitting. p.58
- **MCQ** Cleavage plane in diverticulectomy: sac–pseudocapsule plane. p.58
- **MCQ** Combined approach hazard: incisions not meeting → excess bladder wall excised. p.58
- **MCQ** Commonest complication of diverticulectomy and partial cystectomy: urinary extravasation. p.58
- **MCQ** Urine leak confirmed by drain-fluid creatinine exceeding serum creatinine. p.58
- **MCQ** Best imaging for bladder stone size: non-contrast CT. p.58
- **MCQ** Percutaneous cystolithotripsy avoided after prior abdominal surgery or augmentation (bowel injury). p.58
- **MCQ** Prolonged transurethral cystolitholapaxy risks urethral stricture. p.58
- **MCQ** Percutaneous cystolithotripsy sheath: 30 Fr; stones <1 cm removed intact. p.58
- **MCQ** Bladder paraganglioma: syncope on micturition; α-block before instrumentation. p.58
- **MCQ** Paraganglioma cystoscopy: yellow raised mucosa; MIBG confirms function. p.58
- **MCQ** Bladder endometriosis is mainly extravesical; treated by partial cystectomy. p.58
- **MCQ** Urachus = median umbilical ligament; excise with dome cuff, no wide margin. p.58

- MCQ IC/BPS cystectomy success: small fibrotic bladder, Hunner lesions. p.58
- MCQ Simple cystectomy preserves prostate; no lymphadenectomy. p.58
- MCQ Poorly functioning kidney with distal stricture → nephrectomy, not reimplant. p.58
- MCQ Nonrefluxing tunnel length: 3–5 times ureteric diameter. p.58
- MCQ Neo-hiatus superior-medial, toward bladder neck; lateral sites kink. p.58
- MCQ Reimplant stent: 6 Fr double-J for about 6 weeks. p.58
- MCQ Primary reimplant ≤4 cm; psoas hitch gains ≈5 cm; Boari up to 15 cm. p.58
- MCQ Psoas hitch sutures parallel to fibres: spare genitofemoral and femoral nerves. p.58
- MCQ Reimplant the ureter before tying psoas hitch sutures. p.58
- MCQ Boari flap: base ≥4 cm, length:base ≤3:1, measured empty. p.58
- MCQ Psoas hitch/Boari contraindicated with small, non-compliant bladder. p.58
- MCQ Ileal augment: take segment ≈20 cm proximal to ileocaecal valve. p.58
- MCQ Detubularisation abolishes peristalsis; narrow anastomosis forms pseudodiverticulum. p.58
- MCQ Augment perforation: missed catheterisation; emergency exploration. p.58
- MCQ Augmented bladder: surveillance cystoscopy from 10 years. p.58
- MCQ Channel continence depends on submucosal detrusor tunnel. p.58
- MCQ Yang-Monti: 2–4 cm ileum retubularised transversely. p.58
- MCQ Commonest channel complication: stomal stenosis. p.58

## ★ Putting it together: the big treatment choices

### C · CHOOSING THE URINARY DIVERSION

- TABLE Must have · Usable ileum (not irradiated or diseased) · eGFR **>35–40**; intact sphincter; negative urethral frozen section; willing to self-catheterise · eGFR >35–40; dexterity; reliable efferent limb (appendix, tapered ileum) p.60

### E · METASTATIC UROTHELIAL CARCINOMA IN FOUR STEPS

- ALGORITHM
  **1. Fitness**: cisplatin-eligible, cisplatin-ineligible but platinum-eligible, or platinum-ineligible.
  **2. First line**: enfortumab vedotin (EV) + pembrolizumab (EV-302); otherwise gemcitabine–cisplatin (or gemcitabine–carboplatin if cisplatin-ineligible).
  **3. After platinum response or stable disease**: avelumab maintenance (JAVELIN Bladder 100).
  **4. Later lines**: pembrolizumab; erdafitinib if FGFR-altered; EV; sacituzumab govitecan. Platinum-ineligible: pembrolizumab first. → Ch 131
  p.60

# 13 Kidney & UTUC

319 core items · page numbers refer to the full book PDF

## Ch 136 · Benign Renal Tumours

### 1 · THE BENIGN SMALL RENAL MASS

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign. p.2

### 2 · RENAL CYSTS AND THE BOSNIAK CLASSIFICATION

- **Simple cyst**: any size, homogeneous, **≤20 Hounsfield units (HU)** → no further imaging. Heterogeneity, >20 HU, septa, wall thickening, nodules or calcification → Bosniak staging. p.2
- TABLE I · Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification · Benign · No follow-up p.2
- TABLE II · Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity · Highly likely benign · No follow-up p.2
- TABLE IIF · Minimally thickened (**3 mm**) enhancing wall/septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI · Likely benign · Active surveillance up to **5 yr** p.2
- TABLE III · Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall/septum · **~50%** · ≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible p.2
- TABLE IV · Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion · **~90%** p.2
- KEY POINT Memory ladder: wall ≤2 → 3 → ≥4 mm → irregularity → nodule. Traps: any calcification is compatible with II; a Bosniak I wall may enhance; large (≥3 cm) hyperdense cysts are no longer IIF. Bosniak I/II growth is expected and not a sign of malignancy. p.2
- KEY POINT **Why surveil small III/IV?** Bosniak III has a low positive predictive value, and cystic RCC is indolent: much of its volume is fluid, so tumour burden is low and size predicts outcome poorly (proposed stage **pT1c**). The number needed to treat to prevent one metastasis is therefore very high. p.2
- TABLE Cause · PKD1 (chr 16, polycystin-1); PKD2 (chr 4, polycystin-2) — ciliary calcium signalling · End-stage renal disease (ESRD) on dialysis; cysts from proximal tubule; **80%** by 10 yr p.3
- TABLE RCC · Risk not raised, but earlier, more sarcomatoid, harder to diagnose · Risk **≥10×**; multifocal, bilateral, male; papillary commoner; distinct World Health Organization (WHO) 2022 entity p.3
- ADPKD: hypertension (renin–aldosterone, ischaemia); extrarenal cysts (liver, pancreas, spleen, ovary, seminal vesicle, arachnoid). US diagnosis in PKD1 relatives aged 15–29: **≥3 cysts**; a normal scan excludes disease reliably only from age 40. p.3

### 3 · ONCOCYTOMA

- KEY POINT **Why can a core not clear oncocytoma?** Sampling error, and chRCC contains oncocytoma-like areas — a substantial minority of biopsy "oncocytomas" are RCC at surgery. EAU: definitive diagnosis is not possible on a needle core; adequate tissue with CK7 and cytogenetics helps. p.3

### 4 · ANGIOMYOLIPOMA

- Perivascular epithelioid cell tumour: dysmorphic vessels, smooth muscle, fat. Sporadic (women, 4th–5th decade) or tuberous sclerosis complex (TSC; AML in **55–90%**) or lymphangioleiomyomatosis (LAM). p.3
- TABLE Treatment · If triggers met (below) · Everolimus if **>3 cm** · As RCC p.4

- Macroscopic fat (**−15 to −20 HU** on non-contrast CT) is diagnostic. US unreliable (some RCCs hyperechoic). MRI: signal loss on fat suppression, India-ink artefact. Fat-poor AML mimics RCC → MRI/biopsy. p.4

- ALGORITHM

  **Spontaneous retroperitoneal haemorrhage** (Wunderlich†) → resuscitate → **SAE** → nephrectomy only if SAE unavailable or unsuitable

  **Symptomatic** (pain, haematuria, barriers to care) → SAE → nephron-sparing surgery (NSS) or RN → ablation in selected cases

  **Asymptomatic sporadic**: >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, childbearing age/pregnancy → treat; otherwise surveil

  **TSC**: >3 cm or growing → everolimus first line; SAE or kidney-sparing resection second

  p.4

- KEY POINT **Why abandon 4 cm?** Bleeding risk rises steeply with size but the 4 cm cut-off was poorly specific, so most size-treated patients were overtreated. AML <4 cm → surveillance; observed AML ≥4 cm → thin-slice CT angiography to detect aneurysm. "Barriers to care" is an indication because surveillance presumes access to imaging and emergency care. Pregnancy raises bleeding risk and stakes → consider prophylactic treatment before conception. p.4

## 5 · OTHER BENIGN EPITHELIAL AND MESENCHYMAL TUMOURS

- **Papillary adenoma**: unencapsulated, **≤15 mm** (≤5 mm until 2015); shares +7, +17, −Y and KMT2C with papillary RCC; no metastatic capacity; acceptable in a donor kidney. p.4

## 6 · LANDMARK TRIAL

- TABLE EXIST-2 · Phase III randomised, everolimus vs placebo, TSC/LAM AML · ≥50% volume reduction **42% vs 0%**; no bleeding requiring embolisation · Approved; mTOR inhibition first line in TSC p.5

## 7 · MCQ ONE-LINERS

- MCQ Simple cyst: ≤20 HU. p.5
- MCQ Bosniak III ~50%, IV ~90% malignant. p.5
- MCQ ≤3 thin septa = II; ≥4 = IIF. p.5
- MCQ IIF: surveillance up to 5 years. p.5
- MCQ Proposed cystic RCC stage: pT1c. p.5
- MCQ PKD1 chr 16; PKD2 chr 4. p.5
- MCQ ACKD cysts arise from proximal tubule. p.5
- MCQ Everolimus: stomatitis, acne, proteinuria, hypercholesterolaemia. p.5
- MCQ Commonest benign enhancing SRM: oncocytoma. p.5
- MCQ Birt–Hogg–Dubé: folliculin, chr 17. p.5
- MCQ Oncocytoma: KIT positive, CK7 single-cell. p.5
- MCQ Segmental enhancement inversion favours oncocytoma. p.5
- MCQ AML: HMB-45, Melan-A positive; keratin negative. p.5
- MCQ TSC2 tuberin 16p13; TSC1 hamartin 9q34. p.5
- MCQ AML intralesional aneurysm ≥5 mm predicts bleeding. p.5
- MCQ TSC AML >3 cm: everolimus. p.5
- MCQ Papillary adenoma ≤15 mm. p.5
- MCQ Metanephric adenoma: WT1, CD57 positive; polycythaemia. p.5
- MCQ DICER1: paediatric cystic nephroma. p.5

- MCQ Leiomyoma: commonest origin renal capsule. p.5
- MCQ Renal haemangioblastoma: VHL. p.5
- MCQ Reninoma: hypertension plus hypokalaemia in young woman. p.5

## Ch 137 · Malignant Renal Tumours

### 2 · CLASSIFICATION & PATHOLOGY

- KEY POINT Trap: epithelioid angiomyolipoma is malignant, whereas classic angiomyolipoma (AML), papillary adenoma and oncocytoma are benign. The papillary type 1/2 split is **abandoned** — "type 2" proved to be several diseases (MET-altered, MiT translocation, FH-deficient) with very different prognoses, which is why WHO moved to molecular definitions. p.6
- KEY POINT Why does localised clear cell RCC do worse than papillary/chromophobe, yet metastatic clear cell does better? It is intrinsically more aggressive, but its VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs). p.7

### 3 · MOLECULAR BIOLOGY & HEREDITARY SYNDROMES

- ALGORITHM
  Normoxia: VHL protein + elongin B/C + cullin-2 + RBX1 = E3 ubiquitin ligase → HIF-1α/2α degraded
  VHL loss or hypoxia → **HIF-2α** accumulates → dimerises with ARNT → ↑ VEGF (angiogenesis), PDGF, TGF-α, GLUT1, erythropoietin, CA-IX → hypervascular clear cell RCC
  p.8

### 5 · DIAGNOSIS

- ALGORITHM
  Simple cyst → observe
  Complex/solid: estimated glomerular filtration rate (eGFR) >45 → renal CT or MRI with/without contrast · eGFR 30–45 → hydrate, same · eGFR <30 → non-contrast CT or gadolinium MRI · contrast allergy → USG/contrast USG or MRI
  Fat → AML · complex cyst → Bosniak · solid enhancing → management options ± renal mass biopsy
  p.11

### 6 · STAGING & PROGNOSIS

- KEY POINT TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus supradiaphragmatic thrombus) are lost — record all, and use integrated nomograms. p.12

### 10 · OTHER RENAL MALIGNANCIES

- KEY POINT Almost all non-RCC malignancies are **hypovascular**; the key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis — biopsy, don't resect). RCC nodes land interaortocaval (right) or para-aortic (left); nodes centred elsewhere suggest lymphoma. A solitary enhancing mass years after organ-confined non-renal cancer is more likely primary RCC. p.17

### 12 · MCQ ONE-LINERS

- MCQ Most accepted environmental risk factor for RCC: tobacco. p.17
- MCQ VHL gene 3p25; with elongin B/C targets HIF-α for degradation. p.17
- MCQ Main HIF isoform in clear cell RCC: HIF-2α; inhibitor belzutifan. p.17
- MCQ VHL type 2C: phaeochromocytoma only; type 1: no phaeochromocytoma. p.17
- MCQ Only oncogene familial RCC: MET (hereditary papillary, 7q31). p.17
- MCQ BHD: FLCN 17p11.2 — chromophobe/hybrid oncocytic tumours, pneumothorax. p.17
- MCQ HLRCC: FH 1q42; leiomyomas; excise at any size. p.17

- MCQ Papillary RCC: trisomy 7 and 17; clear cell: 3p loss, 5q gain. p.17
- MCQ Cup-like CA-IX, CK7+: clear cell papillary tumour (benign). p.18
- MCQ Medullary carcinoma: sickle cell trait, SMARCB1 loss. p.18
- MCQ Commonest paediatric RCC: Xp11 TFE3 translocation. p.18
- MCQ ISUP grade 4: sarcomatoid or rhabdoid differentiation. p.18
- MCQ Stauffer syndrome: interleukin-6; persistence after nephrectomy = residual tumour. p.18
- MCQ CT fat <−20 HU = AML; enhancement >15–20 HU = RCC-suspicious. p.18
- MCQ TNM 8th: pelvicalyceal invasion = T3a. p.18
- MCQ Caval wall invasion = T3c at any level. p.18
- MCQ VHL abdominal MRI from age 15, every 2 years. p.18
- MCQ Hypothermia for PN if ischaemia ≥25–30 min. p.18
- MCQ 30% of recurrences occur after 5 years. p.18
- MCQ Commonest renal sarcoma: leiomyosarcoma. p.18
- MCQ Commonest malignant renal neoplasm overall: metastasis (lung primary). p.18
- MCQ Renal carcinoid: horseshoe kidney, 82-fold risk. p.18
- MCQ Cryoablation lethal temperature: −20 °C or colder; iceball ~1 cm beyond tumour. p.18
- MCQ Renal leiomyosarcoma > liposarcoma; retroperitoneum: liposarcoma commonest. p.18
- MCQ Genetic counselling for all renal malignancy at age ≤46. p.18

## Ch 138 · Urothelial Tumours of the Upper Urinary Tract and Ureter

### 1 · EPIDEMIOLOGY

- Upper tract urothelial carcinoma (UTUC) = **5–10%** of urothelial cancers; incidence rising (ureteric tumours). p.19
- Peak age 70–90; male:female ≈ **2:1**. In the Balkans UTUC is **40%** of renal neoplasms. p.19
- Bilateral ≤ 5%; multifocal **10–20%**; concomitant carcinoma in situ (CIS) 11–36%; concurrent bladder cancer **17%**. p.19
- After UTUC treatment, the bladder recurs in **22–47%** and the contralateral upper tract in 2–6%. p.19
- KEY POINT **Why UTUC presents invasive:** the upper tract urothelium looks like bladder urothelium but sits on a much thinner (mesodermal) muscle layer, so tumours reach muscle, renal parenchyma or periureteric fat earlier than in the bladder. p.19

### 2 · RISK FACTORS

- **Smoking** – the most important modifiable factor; relative risk (RR) **2.5–7**, dose-dependent, ureter > pelvis. Stopping for > 10 yr cuts the risk by 60–70%. p.19
- KEY POINT All four behave as a **field disease**, with more multiple and bilateral recurrences. That matters for treatment: kidney-sparing options become more attractive, but BEN patients often also have renal insufficiency. An analgesic abuse history is present in 22% of pelvic and 11% of ureteric tumours. p.19
- Caused by mutations in the MMR genes MLH1, MSH2, MSH6 and PMS2. UTUC risk is ≈ **14×** that of the general population and ≈ **75×** in MSH2 carriers. A germline mutation is found in **5–9%** of screened UTUC patients. p.20

### 3 · PATHOLOGY & MOLECULAR BIOLOGY

- > 90% of tumours are urothelial carcinoma (UC). Variant histology occurs in ≈ **25%** (squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial) and marks high-grade, aggressive disease. **Micropapillary** is an independent predictor of worse outcome. p.20

- Ureteric inverted papillomas harbour malignancy in ≈ **18%**, so follow them for ≥ 2 yr. p.20

### 4 · CLINICAL FEATURES & DIAGNOSIS

- TABLE **Computed tomography urography (CTU)** – first line; the EAU prefers it to MR urography · Sensitivity ≈ **92%**, specificity ≈ 95%. Signs: filling defect (50–75%), obstruction, non-visualisation. Misses flat lesions and invasion of small tumours. UC ≈ **46 Hounsfield units (HU)** vs ≈ 100 HU for a uric acid stone. p.20

- KEY POINT **Why URS biopsy cannot stage:** samples are small and rarely include deep layers, so a Ta biopsy is often upstaged to T3 at resection. Stage is therefore **inferred** from grade, appearance, hydronephrosis and cytology, and patients treated with kidney-sparing surgery need intensive follow-up. p.21

### 6 · PROGNOSTIC FACTORS & RISK STRATIFICATION

- **Preoperative factors:** age (predictive, but never a reason by itself to deny curative RNU); smoking; high American Society of Anesthesiologists (ASA) score and poor Eastern Cooperative Oncology Group performance status (ECOG-PS); obesity; adverse blood markers (low albumin, high neutrophil-to-lymphocyte ratio, high C-reactive protein); **surgical delay – do RNU within 12 wk** for invasive disease. p.21

- Size: **2 cm** is the best cut-off for predicting ≥ pT2. p.21

- KEY POINT Because true stage is known only after RNU, every preoperative factor is a stand-in for invasion. The rule is asymmetric: a single adverse feature makes the tumour high risk. **Trap:** a small, low-grade unifocal tumour **with hydronephrosis is high risk**. p.22

### 7 · RECURRENCE PATTERNS

- **Upper tract after bladder cancer:** 2–4%, late (median > 3 yr); 0.75–6.4% after cystectomy. About **30%** of patients with bladder CIS treated with bacillus Calmette–Guérin (BCG), or cystectomised for BCG-refractory disease, develop UTUC. Risk factors: distal ureter/orifice tumour, multifocal or recurrent non-muscle-invasive disease, CIS, reflux. p.22

- KEY POINT **Why recurrence is asymmetric:** urine flows downhill. Tumour cells seed antegrade, so bladder recurrences are common, early and monoclonal. The field effect (independent clones in carcinogen-exposed urothelium) explains multifocal tumours of different clonal origin; both operate. p.22

### 8 · MANAGEMENT OVERVIEW & FOLLOW-UP

- ALGORITHM
  Cystoscopy + CTU ± URS biopsy/cytology + contralateral function → LS screening if indicated → risk-stratify
  **Low risk** → kidney-sparing treatment (endoscopic or segmental) with intensive surveillance
  **High risk** → RNU with bladder cuff ± lymph node dissection (LND), within 12 wk
  **Clinically node-positive (cN+)** → systemic chemotherapy first → surgery if there is a response
  After RNU → single postoperative intravesical chemotherapy dose → cystoscopic follow-up for everyone
  p.22

### 9 · MCQ ONE-LINERS

- MCQ UTUC = 5–10% of urothelial cancers; Balkans: 40% of renal neoplasms. p.23

- MCQ Smoking RR 2.5–7; ureter at greater risk than pelvis. p.23

- MCQ Lynch: UTUC risk 14×; MSH2 carriers 75×. p.23

- MCQ RAS-mutated molecular subtype has the worst prognosis. p.23
- MCQ Aristolochic acid signature: A:T→T:A transversion, TP53 codon 139. p.23
- MCQ BEN: familial, not inherited; bladder cancer not increased in affected families. p.23
- MCQ Arsenic UTUC: blackfoot disease; women 2:1. p.23
- MCQ Phenacetin latency ≈ 22 yr; capillarosclerosis is pathognomonic. p.23
- MCQ Ureteric inverted papilloma: malignancy ≈ 18%; follow ≥ 2 yr. p.23
- MCQ CTU: sensitivity 92%, specificity 95%. p.23
- MCQ UC ≈ 46 HU; uric acid stone ≈ 100 HU. p.23
- MCQ Cytology sensitivity: G1 20%, G2 45%, G3 75%. p.23
- MCQ Pelvic tumour invading renal parenchyma = T3, not T4. p.23
- MCQ N1 = single node ≤ 2 cm; any N+ = AJCC stage IV. p.23
- MCQ Necrosis > 10% is an independent adverse predictor. p.23
- MCQ Lymph node density ≥ 30% predicts poor outcome. p.23
- MCQ Uroplakin III and p27 expression are favourable. p.23

## Ch 139 · Surgical Management of Upper Urinary Tract Urothelial Tumours

### 1 · PRINCIPLES & PATIENT SELECTION

- KEY POINT Low grade does not automatically mean ablation: obstruction or multifocality moves an LG tumour into the unfavourable group, where RNU retains a role. p.24
- KEY POINT The algorithm balances two axes — tumour risk and renal reserve. High risk with adequate reserve goes to extirpation, low risk to preservation; imperative indications are where the two conflict. p.24
- ALGORITHM
  Imaging suspicion → CT/MR urogram → endoscopy + biopsy + cytology → risk group + shared decision
  High risk + adequate glomerular filtration rate (GFR) → RNU ± neoadjuvant chemotherapy · Low risk or poor GFR → kidney-sparing
  p.24

### 2 · ENDOSCOPIC (KIDNEY-SPARING) MANAGEMENT

- AUA/SUO: endoscopic treatment for tumours **<1.5 cm**; caution ≥1.5 cm (invasion risk). p.25
- KEY POINT Does URS spread tumour? Prior ureteroscopic manipulation increases bladder recurrence (downstream seeding), but pre-RNU URS has not increased metastasis. Mitigate with a single postoperative intravesical instillation. p.26

### 3 · SEGMENTAL URETERECTOMY

- Ipsilateral recurrence after conservative ureteric surgery **33–55%**, mostly distal to the original site, and lifelong. p.26

### 4 · POSITIVE CYTOLOGY & CARCINOMA IN SITU (CIS)

- ALGORITHM
  Positive voided cytology → repeat → CT urogram + bladder biopsies → bladder positive: treat bladder
  Persistent positive with negative bladder (or after bacillus Calmette–Guérin (BCG) "6+1") → selective upper-tract cytology + retrograde pyelography (RPG) + prostatic urethral biopsy (men) + URS
  p.27

### 6 · RADICAL NEPHROURETERECTOMY

- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%**, cannot be practically surveyed, and worsens survival. After bowel diversion, excise the ureteroenteric anastomosis with a bowel cuff. p.28

- KEY POINT Approach matters less than principle: port-site seeding and peritoneal carcinomatosis follow spillage, not the approach. p.29

- Removing **≥8 nodes** improves detection and disease-specific survival above pT1. Surgeon, not approach, determines yield. Outcome: pN0 > pNx > pN+. Specific complication: chylous leak. p.29

### 7 · PERIOPERATIVE ADJUNCTS

- **Neoadjuvant**: AUA/SUO standard option for high-risk UTUC, especially if post-op creatinine clearance (CrCl) expected **<50** mL/min. EA8141: CrCl >50 → accelerated MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×4; 30–49 → gemcitabine–carboplatin; pathological complete response 14%, <ypT2N0 60%. p.30

### 10 · LANDMARK TRIALS

- TABLE POUT · Phase III adjuvant platinum vs surveillance, pT2–4/N+ · Stopped early for efficacy; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46% · Adjuvant platinum standard after RNU for ≥pT2/N+ p.31

### 11 · MCQ ONE-LINERS

- MCQ Retained intramural stump recurrence: 30–75%. p.31
- MCQ AUA/SUO endoscopic size cut-off: <1.5 cm. p.31
- MCQ Most important predictor of recurrence after endoscopic treatment: grade. p.31
- MCQ Distal-third ureteric blood supply is lateral. p.31
- MCQ Ho:YAG penetration <0.5 mm; Nd:YAG 5–6 mm; thulium <0.2 mm. p.31
- MCQ Percutaneous tract dilated to 30 Fr; tract seeding ≈0.3%, all high grade. p.31
- MCQ Psoas hitch ≈5 cm; Boari flap further 10–15 cm. p.31
- MCQ Boari: width ≥3× ureteric diameter; base ≥2 cm wider than apex. p.31
- MCQ Ileal ureter: 20–25 cm ileum, ≥15 cm from ileocaecal valve. p.31
- MCQ ≥8 nodes removed improves disease-specific survival above pT1. p.31
- MCQ Right renal pelvis template includes interaortocaval nodes. p.31
- MCQ Renal pelvis template caudal limit: inferior mesenteric artery. p.31
- MCQ Bladder recurrence after RNU: 30–50%, mostly within 2 years. p.31
- MCQ UGN-101 (Jelmyto): FDA 2020, low-grade UTUC; stricture ≈44%. p.31
- MCQ Commonest BCG upper-tract complication: renal granulomas. p.31
- MCQ Upper-tract CIS: 6-week BCG induction, preferably via nephrostomy. p.31
- MCQ Antegrade instillation pressure column ≤25 cm $H_2O$. p.31
- MCQ Neoadjuvant chemotherapy favoured when post-RNU CrCl expected <50. p.31
- MCQ First ureteroscopic tumour treatment: Goodman 1984; antegrade: Tomera 1982. p.31
- MCQ First laparoscopic nephroureterectomy: Clayman 1991. p.31
- MCQ Renal descensus gains up to 8–10 cm, more on the left. p.31
- MCQ Upper-tract CIS diagnosed by persistently positive selective cytology with normal imaging and URS. p.31
- MCQ Bipolar resection avoids hypo-osmotic irrigant absorption in percutaneous resection. p.31
- MCQ Ureteroscopic resection never goes beyond lamina propria. p.31

- MCQ Ileal ureter: Foley ≥1 week; cystogram + nephrostogram before tube removal. p.31

## Ch 141 · Contemporary Open Surgery of the Kidney

### 1 · ROLE & PREOPERATIVE WORK-UP

- GFR **<60** or significant proteinuria → perioperative nephrology consult; poorly controlled diabetes or hypertension needs optimising even when GFR ≥60. p.32

- Imaging maps kidney, vessels and neighbours (colon, pleura, pancreas, spleen, liver): computed tomography (CT)/magnetic resonance imaging (MRI); contrast generally safe if GFR **>30**. Suspected thrombus → MR or CT venography, **≤1 month old** (rapid propagation). Venography shows cranial and caudal extent and suggests caval wall invasion, which decide liver mobilisation, level of caval control and need for graft or resection. p.32

### 2 · SURGICAL ANATOMY & APPROACHES

- **SMA**: anterior aorta **1–2 cm above LRV**, veers left → mistaken for left renal artery. p.32

### 4 · RADICAL NEPHRECTOMY

- ALGORITHM

  **Right:** survey abdomen → Toldt, iliac → hepatic flexure (protect right colic/ileocolic) → Kocher → IVC → loop and palpate renal vein → artery behind vein → **artery then vein** → kidney outside Gerota's.

  **Difficult hilum:** interaortocaval artery ligation → kidney decompresses → flaccid vein checked, double-ligated (0 silk + 3-0 Prolene) → lateral artery now easy to reach, divided. Or: lower pole → ureter and gonadal vein (spare gonadal if safe; large tumours risk avulsion from IVC/LRV) → divide ureter → lift kidney posterior to anterior to show artery.

  **Left:** Toldt splenic flexure → iliac → renocolic ligament (pad pancreas) → aorta leads to LRV; gonadal vein leads up to it → artery above/behind LRV → 0 silk + 3-0 Prolene, confirm occlusion → vein.

  p.34

- KEY POINT **Why artery first?** Tying the vein first leaves the kidney engorged with arterial inflow and collaterals bleed. Arterial ligation shrinks the kidney and softens the vein so it can be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or that the SMA was tied. p.34

- **Adrenalectomy** only for upper-pole tumour with lost plane, adrenal mass, diffuse involvement. Metastasis **<5%**; CT/MRI miss 20–25%, so use clinical indicators. Contralateral metastasis on follow-up is as common as ipsilateral → prophylactic removal risks adrenal insufficiency without benefit. p.34

- **LND**: nodes positive ≈5%. European Organisation for Research and Treatment of Cancer (EORTC) randomised trial (772 patients): no overall survival (OS) benefit, but many small low-stage tumours enrolled. LND is prognostic, may reduce retroperitoneal recurrence. **Do if nodes enlarged** (short axis **≥1.5 cm ≈ 50% positive**); consider in high risk (cytoreduction, >10 cm, grade ≥3, sarcomatoid, thrombus, T4). p.34

### 5 · IVC TUMOUR THROMBECTOMY

- RCC = commonest cause; **5–10%** of RCC. Untreated → most die within 1 yr, so surgery (± systemic therapy) is the backbone, with RN + LND. p.35

- TABLE I · IVC **≤2 cm** above ostium · Milk back, Satinsky partial clamp, ostial cavotomy p.35

- ALGORITHM
  **Level I (right):** usually partially occlusive, nonadherent. Friable or embolised thrombus → control IVC before handling tumour. Midline/subcostal → minimal handling of renal vein and IVC → right artery ligated interaortocavally → loop LRV, suprarenal and infrarenal IVC → **Rummel tourniquets** (18-Fr red rubber tube on vessel loop) → pinch IVC above thrombus, milk thrombus toward ostium → Satinsky around ostium (confirm thrombus inside jaws) → circumferential ostial cavotomy, keeping ≥half IVC width → extract with traction on renal vein → tie gauze over vein stump.
  **Level I–II:** early renal artery ligation → tourniquets **infrarenal IVC → contralateral renal vein → suprarenal IVC** → cavotomy → extract → heparinised saline flush, inspect wall (biopsy/resect suspicious intima) → 4-0 Prolene → release **caudal clamp first**, flush 5–10 mL on held breath → cranial clamp.
  p.35

- **Pringle** needed if clamping above hepatic veins (else liver engorges and bleeds): ≤60 min, ideally **≤20 min** (liver ischaemia, portal thrombosis, splenic congestion and rupture). Pringle + clamps above and below liver = total hepatic vascular occlusion. Not needed if clamping below the major hepatic veins with accessory veins tied. p.36

- KEY POINT **"Shortcuts" that become "longcuts":** opening IVC before all inflow controlled; side-biting a level I; clips on IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping test clamp. p.37

## 6 · PARTIAL NEPHRECTOMY

- Multifocality: clear cell ≈2%, papillary ≈10% → inspect whole kidney. Hereditary: observe to **3 cm**, except **HLRCC** (hereditary leiomyomatosis and RCC) → early wide excision; use intraoperative ultrasound (IOUS). Preserve perirenal fat and Gerota's, see whole surface, resect all visible tumours; reapproximate fat and fascia so later surgery is easier. p.37

- TABLE Warm ischaemia · Aim **<30 min**; within limits, preserved parenchyma dominates p.37

- KEY POINT PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes, so ischaemia is individualised; hypothermia and mannitol are optional (role in solitary kidney/CKD unclear). p.37

## 8 · MCQ ONE-LINERS

- MCQ CKD-EPI 2021 is race-free, preferred over MDRD. p.38
- MCQ Thrombus imaging must be ≤1 month old. p.38
- MCQ Postinfarction syndrome lasts up to 72 h. p.38
- MCQ Subcostal flank: poor hilar access, not for PN. p.38
- MCQ Turner-Warwick popularised supracostal incision. MCQ p.38
- MCQ Right renal artery lies behind renal vein and IVC. p.39
- MCQ SMA arises 1–2 cm above LRV. p.39
- MCQ Left gonadal and adrenal veins drain into LRV. p.39
- MCQ Robson (1969) defined radical nephrectomy. p.39
- MCQ Adrenal metastasis <5%; imaging misses 20–25%. p.39
- MCQ Node short axis ≥1.5 cm ≈ 50% positive. p.39
- MCQ EORTC LND trial: no OS benefit. p.39
- MCQ Venous thrombus in 5–10% RCC. p.39
- MCQ Level I: ≤2 cm above ostium. p.39

- MCQ Level III defined by major, not accessory, hepatic veins. p.39
- MCQ Budd–Chiari contraindicates thrombectomy. p.39
- MCQ Pringle ≤60 min, ideally ≤20 min. p.39
- MCQ Abort clamp trial: CO ↓>50% or MAP ↓>30%. p.39
- MCQ DHCA core 16–18 °C. p.39
- MCQ Patch width = diameter × fraction resected × 3. p.39
- MCQ IVC graft: PTFE 16–20 mm; lifelong aspirin. p.39
- MCQ Bland group D → infrarenal IVC resection. p.39
- MCQ Multifocality: papillary ≈10%, clear cell ≈2%. p.39
- MCQ Hereditary RCC: operate at 3 cm, except HLRCC. p.39
- MCQ Hyperfiltration if nephron mass ↓>80%. p.39
- MCQ Resection margin 5–10 mm. p.39
- MCQ Hypothermia, mannitol, off-clamp: no RCT benefit. p.39
- MCQ Urine leak: drain creatinine > serum. p.39
- MCQ Pseudoaneurysm: angiography over CT angiography. p.39
- MCQ SMV: never ligate. p.39
- MCQ No cell saver in thrombectomy. p.39
- MCQ Supracostal intercostal incision hugs the upper rib border (spares bundle). p.39
- MCQ Accessory hepatic veins drain caudate lobe; safely ligated. p.39
- MCQ Pringle = clamping portal vein, hepatic artery, bile duct. p.39
- MCQ Lymphatics above renal vein clipped to prevent chylous ascites. p.39

## Ch 142 · Laparoscopic and Robotic Surgery of the Kidney

### 1 · PRINCIPLES & PATIENT SELECTION

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ. Minimally invasive surgery (MIS) vs open: less pain and blood loss, shorter convalescence, better cosmesis, **equivalent long-term outcomes**. Indications same as open. p.40
- KEY POINT The patient must tolerate the **pneumoperitoneum**, not the renal operation — so cardiopulmonary reserve, not age or BMI, decides fitness. p.40

### 2 · ACCESS & PORT PLACEMENT

- **Robotic TP:** 3D view, wristed instruments, tremor elimination, motion scaling. Arm spacing **8 cm (Si) / 6 cm (Xi)**; 12 mm low-midline assistant port for sutures, clamps, staplers, suction (5 mm if no stapler needed, e.g. pyeloplasty). p.40
- KEY POINT RP pitfalls: balloon inflated within muscle → flank hernia; too anterior → peritoneal/colon injury; too posterior → quadratus/psoas bleeding. Bail-out: open the peritoneum and continue TP. p.41

### 3 · NEPHRECTOMY — OPERATIVE STEPS

- ALGORITHM
  Reflect colon → find ureter/gonadal vessels on psoas → lift kidney, expose hilum → artery then vein → upper pole (adrenal spared or taken) → ureter and lateral attachments → bag → extract
  p.41

## 4 · SIMPLE VS RADICAL NEPHRECTOMY

- KEY POINT Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery unchanged. p.42

## 5 · PARTIAL NEPHRECTOMY

- ALGORITHM
  Intraoperative ultrasound (size, depth) → open Gerota's away from lesion → score capsule → clamp hilum (Satinsky en bloc or bulldogs) → cold-scissor excision → close vessels/collecting system → renorrhaphy → unclamp → drain
  p.43

- **Trifecta:** negative margin + no major complication within 30 days + **WIT ≤25 min**. p.43

- **Collecting system** entered in **~60%** (large, endophytic tumours). Repair with 2-0/3-0 polyglactin; **always leave a drain** (urinoma). A routine ureteric catheter does not prevent fistula. Check integrity with intravenous indigo carmine or retrograde instillation; argon beam can destroy the repair suture. p.43

- **Ischaemia:** historical warm limit **30 min**. **Preserved parenchymal volume predicts function better than WIT.** Consider cold ischaemia (ice slush, cold perfusion) for solitary kidney, CKD or long expected clamp time. p.43

- KEY POINT Why no clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped. Precise excision saves nephrons; minutes of ischaemia matter less. p.43

## 7 · COMPLICATIONS & MANAGEMENT

- Overall **~15%**, mostly postoperative; PN > RN > simple. Learning curve ~20 laparoscopic nephrectomies. Robotic PN halves open conversion vs laparoscopic. p.44

- ALGORITHM
  Major bleed: locate → temporary control (grasper/pressure) → inform anaesthetist (blood, open set) → clear field, raise pneumoperitoneum or add trocar → definitive clip/suture → else convert and call vascular surgeon
  p.45

- KEY POINT MIS complications are the open ones in altered form and timing: the lethal ones — access vessel injury, thermal bowel injury — happen out of view and present late or atypically. p.45

## 9 · MCQ ONE-LINERS

- MCQ First laparoscopic nephrectomy: Clayman 1991; first laparoscopic PN: Winfield 1993. p.46
- MCQ Better predictor of difficulty than BMI: visceral adipose tissue. p.46
- MCQ Robot arm spacing: 8 cm Si, 6 cm Xi. p.46
- MCQ RP vs TP radical nephrectomy: consistent RP advantage is shorter operative time. p.46
- MCQ En bloc hilar stapling: no arteriovenous fistula; saves ~45 min. p.46
- MCQ Trifecta WIT ≤25 min; historical warm ischaemia limit 30 min. p.46
- MCQ Collecting system entered in ~60% of laparoscopic PN. p.46
- MCQ Positive margins 0–7% of MIS PN; matter only in high-risk tumours. p.46
- MCQ Port-site recurrence ~0.1% of laparoscopic cancer operations. p.46
- MCQ >5 cm tumour upstaged to pT3a: more recurrence after PN than RN. p.46
- MCQ Cryo iceball 0.5–1 cm beyond tumour; two freeze–thaw cycles. p.46
- MCQ RFA 60–100°C; >105°C causes vaporisation; no real-time imaging. p.46
- MCQ Device failure: staplers > non-locking clips > locking clips. p.46
- MCQ Bowel injury mortality ~3%; thermal causes about half. p.46

- MCQ Unrecognised pneumothorax after laparoscopic nephrectomy ~0.3%. p.46
- MCQ Neuromuscular injury in 2.7% of laparoscopic urological surgery. p.46
- MCQ Learning curve ~20 laparoscopic nephrectomies. p.46
- MCQ Commonest MIS complication: vascular injury. p.46
- MCQ Robotic PN morbidity plateaus at 18–20 hospital cases/year. p.46
- MCQ Retroperitoneal access of choice: peritoneal dialysis, dense adhesions. p.46
- MCQ Ureter lies just posterior to gonadal vessels; peristalsis identifies it. p.46
- MCQ Haematuria ~day 12 after PN: pseudoaneurysm → selective embolisation. p.46
- MCQ Balloon inflated within flank muscle → postoperative flank hernia. p.46
- MCQ Nephroptosis: >5 cm or 2 vertebral bodies; asymptomatic in 20% of women. p.46

## Ch 143 · Focal Therapy for Renal Tumours

### 1 · RATIONALE & INDICATIONS

- KEY POINT **Trap:** TA is never the default for a fit patient. p.47
- TABLE American Urological Association (AUA) 2021 · Consider for cT1a **≤3 cm**, esp. comorbid; PN prioritised; percutaneous preferred p.47
- TABLE European Association of Urology (EAU) 2022 · TA not routinely **>3 cm**; CA not **>4 cm**; for frail/comorbid p.47

### 2 · SELECTION & WORK-UP

- ALGORITHM
  Counsel (age, comorbidity, CKD risk, size/location) → biopsy → re-counsel with histology → PN / surveillance / percutaneous ablation
  p.47

### 3 · MECHANISMS

- Kidney dies at −19.4 °C, tumour may need −50 °C → target **≤−40 °C**. Iceball edge is 0 °C; <−20 °C only 3 mm inside it → iceball **5–10 mm beyond tumour**. Every cryo rule (margin, multiple probes, double cycle, duration) exists to push the −40 °C isotherm beyond the tumour edge. p.47
- Heat: 40–45 °C sublethal; **≥60 °C** coagulative necrosis; ≥100 °C vaporisation/charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**. p.47

### 4 · CHOICE OF ENERGY

- KEY POINT **Why RFA is self-limiting:** charring raises impedance and halts current, so temperature is capped at 105 °C and electrodes are cooled. MWA *radiates* energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink. p.48

### 5 · TECHNIQUE

- ALGORITHM
  IV antibiotics → prone/flank on computed tomography (CT) gantry → ultrasound (US), CT or magnetic resonance imaging (MRI)-guided placement → non-contrast CT confirms position → ablate → contrast CT (completeness, haematoma)
  p.49
- **CA:** multiple probes **1–2 cm apart**; remove only after full thaw, without resistance – early removal fractures tumour and bleeds. Size the iceball on CT, where it is a distinct hypodense zone, or on real-time US. Tumours >4 cm can be ablated, but bleeding rises with size and probe number. p.49

### 6 · FOLLOW-UP, SUCCESS & SALVAGE

- **AUA:** CT/MRI with and without contrast + chest imaging at **6, 12, 24, 36, 48, 60 months**; after 5 years, shared decision. Consider risk-adjusted follow-up for larger or high-grade tumours. US is not routine unless a contrast-enhanced US (CEUS) protocol exists. p.49

### 7 · ONCOLOGICAL OUTCOMES VS PN

- KEY POINT **Why comparisons are biased:** benign tumours and under-graded biopsies flatter TA; occult pT3a and growth-selected tumours penalise it. Older guidelines restricted TA to non-surgical candidates, so many patients die of other causes before progression is detected. European Organisation for Research and Treatment of Cancer (EORTC) 30904 (PN vs RN, ≤5 cm) had only 12 RCC deaths at 9.3 years – trials need huge cohorts. p.49

### 8 · COMPLICATIONS & MANAGEMENT

- Major (Clavien ≥3) **≤5%**, fewer than PN; percutaneous < laparoscopic. Laparoscopic complications fall with centre experience. Pain is minimal: modern cryoprobe shafts are insulated, and MWA has shaft cooling and no grounding pads. p.50

### 10 · MCQ ONE-LINERS

- MCQ AUA: TA for cT1a ≤3 cm; only NCCN allows cT1b. p.50
- MCQ Argon cools, helium warms (Joule–Thomson). p.50
- MCQ Cryo target ≤−40 °C; iceball 5–10 mm beyond tumour. p.50
- MCQ MWA works by dielectric hysteresis; RFA by ionic friction. p.51
- MCQ >70 °C for 1 min lethal to RCC. p.51
- MCQ Only RFA needs grounding pads and is limited by charring. p.51
- MCQ MWA least susceptible to heat sink. p.51
- MCQ Only CA shows a visible real-time zone. p.51
- MCQ After RFA: fibrotic halo; after CA: >50% shrinkage in year 1. p.51
- MCQ Pyeloperfusion temperature opposes the ablation. p.51
- MCQ CA preferred for central tumours. p.51
- MCQ Commonest complication: haemorrhage; major ≤5%. p.51
- MCQ Lowest complication rate: MWA (network meta-analysis). p.51
- MCQ CSS and MFS equal TA vs PN. p.51
- MCQ Histotripsy is non-thermal; HIFU thermal + cavitation. p.51
- MCQ IRE preserves extracellular matrix and vessels. p.51
- MCQ Normal renal parenchyma dies at −19.4 °C. p.51
- MCQ <−20 °C reached only 3 mm inside iceball edge. p.51
- MCQ RFA 450–1200 kHz; MWA 915 MHz or 2.45 GHz. p.51
- MCQ D5W, not saline, for hydrodisplacement with RFA. p.51
- MCQ Success = complete loss of enhancement on CT/MRI. p.51
- MCQ AUA follow-up imaging: 6, 12, 24, 36, 48, 60 months. p.51
- MCQ Biopsy before ablation, ideally at a separate session. p.51
- MCQ Laparoscopic ablation: prior scarring or failed hydrodisplacement. p.51
- MCQ 12-tine umbrella RFA electrode gives the most spherical zone; cool-tip the largest. p.51
- MCQ HIFU cavitation threshold >3500 W/cm². p.51

## Ch 144 · Treatment of Advanced Renal Cell Carcinoma

### 1 · OVERVIEW

- Renal cell carcinoma (RCC) is a family of distinct cancers. About **1/3** present with synchronous metastases; a further **20–40%** of localised tumours later metastasise. Metastatic RCC (mRCC) 5-yr survival ≈ **12%**; 10-yr <5%. p.52

### 2 · PROGNOSTIC MODELS

- TABLE Corrected calcium · **>10 mg/dL** · > upper limit of normal (ULN) p.52
- TABLE Lactate dehydrogenase (LDH) · **>1.5 × ULN** · — p.52
- TABLE Time factor · No prior nephrectomy · **<1 yr** diagnosis → systemic therapy p.52
- KEY POINT Both models score host reserve (PS, anaemia), tumour-driven systemic effects (calcium, LDH, neutrophils, platelets) and pace of disease. Three factors are shared; IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and a short interval. Poor risk = ≥3 factors in both. The IMDC risk group now **decides the first-line regimen**. p.52

### 3 · BIOLOGY AND DRUG TARGETS

- TABLE Sunitinib · VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off** · Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism** p.53

### 4 · SURGERY IN METASTATIC DISEASE

- ALGORITHM
  mRCC → histology + IMDC risk + PS + metastatic burden
  Fit, low burden, favourable features, resectable primary → consider upfront CN → systemic therapy
  Intermediate/poor risk or high burden → systemic therapy first → deferred CN in responders
  p.54
- KEY POINT CN helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy. When systemic therapy is active, operating first on poor-risk patients mainly delays it – hence selection and deferral. p.54
- No randomised trial; retrospective series. Isolated **pulmonary** metastases are most often resected with curative intent; 5-yr survival **35–50%** after complete resection of limited disease. p.54

### 5 · SYSTEMIC THERAPY FOR CCRCC

- TABLE Durable complete response (CR) · <2% · **7–9%**; >60% of CRs stay disease-free p.54
- TABLE Regimen · Schedule/route immaterial · **600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses p.54
- KEY POINT **Choosing**: IO/TKI gives higher response and PFS with better tolerance; IO/IO (ipilimumab + nivolumab) gives deeper, durable responses (CR 9%) at the cost of autoimmune toxicity – they have never been compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk because favourable-risk patients did better on sunitinib in CheckMate 214. Avelumab + axitinib is approved but not NCCN-preferred (no OS benefit). PD-L1 testing does not select patients for any regimen. Single-agent TKI or temsirolimus is the fallback when IO is contraindicated. p.55
- **Ipilimumab + nivolumab**: grade 3–4 events **46%**; immune-related events needing high-dose glucocorticoids 35%; permanent discontinuation 22%. The chosen dose (nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4) was the least toxic in CheckMate 016. Nivolumab alone is better tolerated than everolimus (grade 3–4: 19% vs 37%). p.55

### 6 · NON-CLEAR CELL RCC

- KEY POINT In nccRCC the hereditary syndrome names the driver, and the driver names the drug: VHL → VEGF/mTOR/HIF-2α; MET → MET inhibitors; FH → glucose restriction. p.56

- MCQ MSKCC: LDH >1.5 × ULN, calcium >10 mg/dL, no prior nephrectomy. p.57
- MCQ IMDC adds neutrophilia, thrombocytosis; drops LDH. p.57
- MCQ Poor risk = ≥3 factors; IMDC poor-risk median OS 7.8 mo. p.57
- MCQ VHL: chromosome 3p; complexes with elongin B/C and CUL2. p.57
- MCQ HPRC: MET, chromosome 7, type 1 papillary. p.57
- MCQ Sporadic papillary RCC: trisomy 7 in >2/3. p.57
- MCQ HLRCC: fumarate hydratase; type 2 papillary; bevacizumab + erlotinib. p.57
- MCQ Most selective VEGFR TKI: axitinib. p.57
- MCQ Only anti-PD-L1 used in RCC combinations: avelumab. p.57
- MCQ Lenvatinib uniquely inhibits FGFR1–4; cabozantinib inhibits MET and AXL. p.57
- MCQ Sunitinib: 50 mg, 4 weeks on, 2 off; causes hypothyroidism. p.57
- MCQ Pazopanib: raised ALT; COMPARZ noninferior to sunitinib. p.57
- MCQ High-dose IL-2: 600,000–720,000 IU/kg 8-hourly, max 15 doses. p.57
- MCQ IL-2 dose-limiting toxicity: vascular leak syndrome. p.57
- MCQ IL-2 durable CR 7–9%; only agent approved on phase II response. p.57
- MCQ Temsirolimus: IV weekly; only mTOR inhibitor with OS benefit. p.57
- MCQ Temsirolimus is listed for intermediate/poor risk only; active surveillance for favourable only. p.57
- MCQ CheckMate 025 primary endpoint OS, HR 0.73. p.57
- MCQ CheckMate 214: CR 9% vs 1%. p.57
- MCQ CLEAR PFS HR 0.39 – largest among IO combinations. p.57
- MCQ JAVELIN Renal 101: only combination without OS benefit. p.57
- MCQ First-line regimens limited to intermediate/poor risk: ipilimumab + nivolumab, cabozantinib, temsirolimus. p.57
- MCQ PD-L1 status does not select first-line therapy. p.57
- MCQ Belzutifan targets HIF-2α; approved after IO and TKI, and for VHL disease. p.57
- MCQ Chemotherapy response in ccRCC: 5.5–6%. p.57
- MCQ Collecting duct carcinoma: gemcitabine + carboplatin. p.57
- MCQ Spontaneous regression after nephrectomy: <1%. p.57
- MCQ Metastasectomy 5-yr survival after complete resection: 35–50%. p.57
- MCQ SURTIME: deferred CN OS 32.4 vs 15.0 mo. p.57
- MCQ CARMENA: ~45% MSKCC poor risk; sunitinib alone noninferior. p.57
- MCQ Haematuria from unresectable primary: angioembolisation. p.57
- MCQ Autoimmune events with ipilimumab predicted response. p.57

## ★ Putting it together: the big treatment choices

### A · SMALL RENAL MASS (SRM): WHICH TREATMENT?

- TABLE Best for · Elderly, frail, life expectancy <5 y; solid <2 cm or mainly cystic · cT1a **≤3 cm**, comorbid; biopsy first · cT1a standard of care; imperative in solitary kidney, bilateral, hereditary tumours or chronic kidney disease (CKD) · Only when all three: high complexity, no CKD/proteinuria, expected estimated glomerular filtration rate (eGFR) **>45** p.58

### E · METASTATIC RCC IN FOUR STEPS

- ALGORITHM

**1. Risk group (IMDC, International Metastatic RCC Database Consortium; 6 factors)**: Karnofsky <80%, haemoglobin < lower limit of normal, calcium, neutrophils or platelets above upper limit of normal, <1 year from diagnosis to systemic therapy → 0 favourable · 1–2 intermediate · ≥3 poor.

**2. First line**: immunotherapy (IO) + tyrosine kinase inhibitor (TKI) (pembrolizumab + axitinib, nivolumab + cabozantinib, pembrolizumab + lenvatinib) for any risk; ipilimumab + nivolumab for intermediate/poor risk.

**3. Cytoreductive nephrectomy**: upfront only if fit, low tumour burden and favourable risk; otherwise systemic therapy first and deferred nephrectomy in responders (CARMENA, SURTIME).

**4. Later lines**: switch mechanism; belzutifan (HIF-2α inhibitor) after both IO and TKI.

p.59

# 14 Adrenal

139 core items · page numbers refer to the full book PDF

## Ch 145 · Surgical and Radiographic Anatomy of the Adrenals

### 1 · EMBRYOLOGY

- At birth gland is **2×** adult weight; **fetal zone involutes in year 1**, replaced by definitive zone (ZG, ZF) under adrenocorticotropic hormone (ACTH). Zona reticularis (ZR) reaches adult volume at 6–8 y. p.2

### 2 · ZONES AND HORMONES

- TABLE Fasciculata · **~80%** · Lipid-laden "clear cells" in straight cords · Cortisol p.2

### 4 · BLOOD SUPPLY

- KEY POINT Right side is riskier: high, retrocaval, short vein — little working length; avulsion tears the cava. p.2

### 5 · IMAGING APPEARANCE

- **Computed tomography (CT)**: best spatial resolution; inverted **V/Y**, limbs 3–6 mm (thinner than crus). Right posterior to IVC; left anteromedial to upper pole, behind splenic vessels. Unenhanced **≤10 HU** (Hounsfield units) = lipid-rich tissue. Morphology only. p.3

### 6 · MCQ ONE-LINERS

- MCQ Cortex from coelomic mesoderm; medulla from neural crest. p.3
- MCQ Adrenal and gonad share the adrenogonadal ridge. p.3
- MCQ Renal agenesis: adrenal orthotopic, often discoid. p.3
- MCQ Neonatal adrenal is twice adult weight; fetal zone involutes in year 1. p.3
- MCQ Largest zone: fasciculata (~80%) — cortisol. p.3
- MCQ Glomerulosa → aldosterone; reticularis → DHEAS. p.3
- MCQ Testicular adrenal rests present in CAH. p.3
- MCQ Preganglionic fibres reach medulla without synapsing. p.3
- MCQ Arteries: superior (inferior phrenic), middle (aorta), inferior (renal). p.3
- MCQ Right adrenal vein: short, into posterior IVC. p.3
- MCQ Left adrenal vein: into left renal vein, joined by inferior phrenic. p.3
- MCQ Cortical venous cortisol aids medullary epinephrine synthesis. p.3
- MCQ Normal CT: inverted V/Y, limbs thinner than crus. p.3
- MCQ ≤10 HU unenhanced = lipid-rich adrenal tissue. p.3
- MCQ On T2 MRI the normal gland resembles fat. p.3
- MCQ US principal role: solid vs cystic mass. p.3
- MCQ Only angiographic test giving function: adrenal venous sampling. p.3
- MCQ Left gland lies behind pancreas and splenic vessels. p.3
- MCQ Adrenal weight ≈5 g, equal in both sexes. p.3
- MCQ Zona reticularis expands at 6–8 years. p.3

## Ch 146 · Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

### 1 · APPLIED ANATOMY AND PHYSIOLOGY

- **Veins**: right short, straight into inferior vena cava (IVC); left long, joined by inferior phrenic vein → left renal vein. Anomalous venous drainage in up to **9%** – matters for adrenal vein sampling (AVS). Arterial supply triple (inferior phrenic, aorta, renal) and overlapping → partial adrenalectomy rarely infarcts the remnant. p.4

### 2 · EVALUATION OF AN ADRENAL MASS – PRINCIPLES

- KEY POINT Two questions: **(1) Is it malignant?** – answered by imaging. **(2) Is it hormonally active?** – answered by biochemistry. Either = treat (usually resect). No imaging feature separates functional from non-functional adenoma, so biochemistry is never omitted. p.4

### 3 · IMAGING

- **Unenhanced computed tomography (CT)** – single best test. **≤10 Hounsfield units (HU)** = intracellular lipid = adenoma (in a patient without known malignancy); diagnoses >70% of adenomas. ≤0 HU is 100% specific but insensitive. Measure density yourself if the report omits it. p.5
- Limits: up to **7%** of metastases are <10 HU; rare low-density phaeochromocytomas exist → biochemistry for every lesion. p.5
- TABLE Adenoma if · **>60%** · **>40%** p.5
- ALGORITHM
  Mass → unenhanced CT ≤10 HU or MRI drop-out → adenoma | >10 HU → 15-min washout CT → APW >60% / RPW >40% = adenoma; below = adenoma very unlikely → manage as indeterminate
  p.5

### 4 · HORMONAL WORK-UP (ALL LESIONS >1 CM)

- KEY POINT Every incidentaloma: **1-mg dexamethasone suppression test (DST) + metanephrines**. Add **aldosterone-to-renin ratio (ARR)** if hypertension or unexplained hypokalaemia. >10% of incidentalomas are functional, yet 75–80% are inadequately evaluated. **Never biopsy, sample or excise before excluding phaeochromocytoma** – manipulation can provoke a fatal catecholamine surge. Canadian Urological Association (CUA) guidance allows omitting metanephrines in asymptomatic <10 HU lesions. p.6
- TABLE **Overnight 1-mg low-dose DST** (preferred for adrenal masses) · 1 mg at 11 pm–midnight, cortisol 8–9 am. **≤1.8 µg/dL (50 nmol/L)** excludes; 1.8–5.0 = possible autonomous secretion; **>5.0 µg/dL (140 nmol/L)** >95% specific · Oral oestrogens (↑ cortisol-binding globulin) → false positive; CYP3A4 inducers (phenytoin, carbamazepine, rifampicin) → false positive; inhibitors (itraconazole, ritonavir, diltiazem) → false negative; unaffected by renal function p.6

### 5 · BIOPSY AND INCIDENTALOMA MANAGEMENT

- ALGORITHM
  Imaging characterises it? → no biopsy | functional? → no biopsy | ACC suspected? → no biopsy | will result change management (e.g., known primary) and justify risk? → biopsy, **after excluding phaeochromocytoma**
  p.7

### 6 · PRIMARY ALDOSTERONISM (PA)

- Morning (8–10 am), seated: plasma aldosterone concentration (PAC, ng/dL) ÷ plasma renin activity (PRA, ng/mL/h). Positive: **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 to avoid spuriously high ratios; high ARR with low PAC suggests false positive. p.8
- **Lateralisation**: (A/C dominant) ÷ (A/C non-dominant) **>2–4:1** → ipsilateral adrenalectomy. Cortisol correction adjusts for dilution. p.8

- **ALGORITHM**
  Indication → ARR + → confirm → (family history/onset <20 → FH-I test) → surgical candidate? No → MRA | Yes → CT → age <40 + unilateral >1 cm nodule → adrenalectomy; otherwise → AVS → lateralised → laparoscopic adrenalectomy; not lateralised → MRA
  p.9

### 7 · CUSHING SYNDROME AND CORTISOL EXCESS

- **ALGORITHM**
  Exclude exogenous steroid → screen (1-mg DST / LNSC ×2 / UFC ×2) → abnormal → endocrinologist confirms → **morning ACTH**: low → adrenal (CT) | normal/high → pituitary MRI + **inferior petrosal sinus sampling after CRH** (central:peripheral gradient = Cushing disease; none = ectopic)
  p.9

### 8 · PHAEOCHROMOCYTOMA AND PARAGANGLIOMA (PPGL)

- **KEY POINT** The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014). p.10
- **KEY POINT** **Why never β before α?** β-blockade removes β2-mediated arteriolar dilatation; circulating catecholamines then act unopposed on α1 receptors → hypertensive crisis. Hence use selective β1 agents. p.11

### 11 · ADRENAL METASTASES

- With known malignancy, **>50%** of new adrenal lesions are metastases. p.13

### 12 · ADRENAL INSUFFICIENCY

- **KEY POINT** † **Adrenal crisis management**: do not wait for results – draw cortisol/ACTH, then **hydrocortisone 100 mg IV bolus**, followed by 200 mg/24 h (infusion or 50 mg 6-hourly); rapid 0.9% saline (1 L in the first hour) with dextrose for hypoglycaemia; treat precipitant; taper to maintenance over 1–3 days. Fludrocortisone unnecessary while hydrocortisone >50 mg/day. Teach sick-day rules (double oral dose; injectable hydrocortisone) and a steroid alert card. p.14

### 15 · MCQ ONE-LINERS

- **MCQ** Unenhanced CT ≤10 HU in adrenal mass = lipid-rich adenoma. p.15
- **MCQ** Adenoma on 15-min washout: APW >60%, RPW >40%. p.15
- **MCQ** Only adrenal lesion exempt from the 4-cm resection rule: myelolipoma. p.15
- **MCQ** Every incidentaloma: 1-mg DST plus metanephrines; ARR if hypertensive or hypokalaemic. p.15
- **MCQ** 1-mg DST cortisol ≤1.8 µg/dL (50 nmol/L) excludes autonomous cortisol secretion. p.15
- **MCQ** Oral contraceptives cause false-positive DST by raising cortisol-binding globulin. p.15
- **MCQ** Most sensitive test for phaeochromocytoma: plasma free metanephrines (supine). p.15
- **MCQ** β-blockers cause false-positive ARR; ACE inhibitors/ARBs false-negative. p.15
- **MCQ** Commonest PA subtype: bilateral idiopathic hyperplasia (≈60%). p.15
- **MCQ** FH-I: CYP11B1/CYP11B2 chimeric gene, treated with glucocorticoid. p.15
- **MCQ** AVS may be skipped: age <40, unilateral adenoma >1 cm, normal contralateral gland. p.15
- **MCQ** AVS lateralisation ratio >2–4:1 (cortisol-corrected aldosterone). p.15
- **MCQ** Gold standard pituitary vs ectopic ACTH: inferior petrosal sinus sampling with CRH. p.15
- **MCQ** Commonest ectopic ACTH source: bronchial carcinoid. p.15
- **MCQ** Paradoxical cortisol rise after dexamethasone: PPNAD (Carney complex). p.15
- **MCQ** Nelson syndrome follows bilateral adrenalectomy for Cushing disease. p.15

- MCQ PPGL gene most associated with malignancy: SDHB. p.15
- MCQ SDHD shows maternal imprinting (disease when inherited from father). p.15
- MCQ VHL phaeochromocytoma secretes normetanephrine, not metanephrine. p.15
- MCQ FDG PET poor in MEN-2; $^{123}$I-MIBG best there. p.15
- MCQ Never start β-blocker before α-blockade in phaeochromocytoma. p.15
- MCQ Extrapyramidal side effects: metyrosine, not phenoxybenzamine. p.15
- MCQ Malignant phaeochromocytoma is defined only by metastases. p.15
- MCQ Weiss criteria: ≥3 of 9 = adrenocortical carcinoma. p.15
- MCQ ENSAT stage IV ACC = distant metastasis only. p.15
- MCQ Primary adrenal insufficiency alone causes hyperkalaemia and hyperpigmentation. p.15

## Ch 147 · Surgery of the Adrenal Glands

### 1 · SURGICAL ANATOMY (ESSENTIALS)

- **Right**: pyramidal, behind bare area of liver (fused in ~10%), inferior vena cava (IVC) medial, duodenum anteromedial. Vein **<1 cm**, into **posterior IVC at 45°**; variants into right hepatic or right renal vein. p.16

### 2 · INDICATIONS

- TABLE Size · **≥4 cm**; >2 cm growing ≥1 cm in 12 months p.16
- TABLE Imaging · Non-contrast CT (computed tomography) **>10 HU** (Hounsfield units); washout <50% with delayed >35 HU; chemical-shift MRI (magnetic resonance imaging) not adenoma; fluorodeoxyglucose positron emission tomography (FDG-PET) uptake ≥ liver p.16
- KEY POINT Trap: bilateral hyperplasia causing aldosteronism is **not** surgical — hence adrenal venous sampling (AVS) before adrenalectomy for aldosteronism. p.16
- Inert, ≤4 cm, ≤10 HU, homogeneous → repeat CT at 12 months. Myelolipoma (≤ −20 HU fat): operate only if **>7 cm**, symptomatic (haemorrhage) or atypical; still exclude PCC. p.16

### 4 · PRE-OPERATIVE PREPARATION BY PATHOLOGY

- TABLE **PCC** · α-blockade **2 weeks** (phenoxybenzamine preferred; doxazosin, calcium-channel blocker, metyrosine alternatives) → β-blocker **only after α**, for tachycardia → volume expansion (fluids, high-salt diet) · Unopposed β-blockade removes β2 vasodilatation → hypertensive crisis. Chronically constricted, contracted patients collapse once vein tied. p.17

### 6 · OPERATIVE STEPS

- ALGORITHM
  Elevate liver (no capsular tears) → Kocher (duodenum medial) → expose IVC → up from renal vein insertion (inferior limit) → find short adrenal vein → **warn anaesthesia**, clip/seal early → superior and lateral dissection → psoas (posterior limit) → bag, intact.
  p.17
- ALGORITHM
  Reflect colon at line of Toldt, outside Gerota → mobilise spleen/pancreas up → follow left renal vein (or gonadal vein) to adrenal vein → clip after warning → medial dissection, protect pancreas and splenic artery → divide ganglion nexus superomedially → bag.
  p.17
- KEY POINT Why ligate the vein early in PCC? Limits catecholamine surge during handling; but hypotension follows — tell anaesthesia first. p.17

### 10 · OUTCOMES AND FOLLOW-UP

- PCC: lifelong recurrence up to **16%** → annual blood pressure and metanephrines. p.18

- ACC: advanced-stage 5-year survival **<15%** — multidisciplinary follow-up. p.18

11 · MCQ ONE-LINERS

- MCQ First laparoscopic adrenalectomy: Gagner, 1992, transperitoneal. p.19
- MCQ Steroid cover for Cushing was the first major safety advance. p.19
- MCQ Functional tumours excised regardless of size. p.19
- MCQ Non-functional incidentaloma ≥4 cm → adrenalectomy. p.19
- MCQ Non-contrast CT ≤10 HU → lipid-rich adenoma. p.19
- MCQ Best cortisol screen: 1-mg overnight DST. p.19
- MCQ Best PCC screen: plasma free or urinary metanephrines. p.19
- MCQ Up to 40% of primary aldosteronism is normokalaemic. p.19
- MCQ AVS distinguishes unilateral adenoma from bilateral hyperplasia. p.19
- MCQ Preferred α-blocker for PCC: phenoxybenzamine, ~2 weeks. p.19
- MCQ Never start β-blocker before α-blockade. p.19
- MCQ Right adrenal vein enters posterior IVC at 45°. p.19
- MCQ Left adrenal vein drains to left renal vein. p.19
- MCQ Kocher manoeuvre exposes IVC for right adrenalectomy. p.19
- MCQ Retroperitoneoscopy: renal artery met before vein. p.19
- MCQ Posterior limit of dissection: psoas. p.19
- MCQ Suspected ACC → open, intact, en bloc resection. p.19
- MCQ Commonest adrenal metastasis source: lung, then kidney. p.19
- MCQ Myelolipoma: operate if >7 cm or symptomatic. p.19
- MCQ Partial adrenalectomy: lateral dissection is safe. p.19
- MCQ In partial adrenalectomy main vein can be sacrificed. p.19
- MCQ Hypotension after PCC vein ligation → volume first. p.19
- MCQ PCC recurrence lifelong up to 16%. p.19
- MCQ Ablation of PCC or ACC discouraged outside trials. p.19
- MCQ Peritonitis contraindicates transperitoneal, not retroperitoneal, access. p.19

## Ch 140 · Retroperitoneal Tumours

1 · ANATOMY OF THE RETROPERITONEUM

- KEY POINT The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement. p.20

4 · PATHOLOGY — MESENCHYMAL TUMOURS (SARCOMA)

- Soft-tissue sarcoma <1% of adult cancers; **15–20%** arise in the RP. Diagnosis needs integrated IHC, molecular and clinical data. p.21
- TABLE **Well-/de-differentiated liposarcoma** (WD/DD; commonest, **~60%**) · Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+ · Radical, function-preserving compartmental resection p.21
- TABLE **Leiomyosarcoma** (second, **~20%**) · From vessel media (IVC), collecting-system muscle; no fat; SMA, desmin, h-caldesmon+ · Negative margins + adjuvant therapy p.21

- Behaviour: WD — slow, local recurrence up to 40%, rarely metastasises; DD — rapid, distant metastasis up to 20%, 5-year cancer-specific mortality ~30%. Mean time to recurrence ~**2 years** → lifelong surveillance. p.22

- KEY POINT Liposarcoma family sorts on MDM2: amplified (WD/DD) = local disease won or lost at the margin; absent (pleomorphic) = systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable. p.22

### 6 · GERM CELL, HAEMATOLOGICAL, CYSTIC AND UNKNOWN PRIMARY

- KEY POINT Histology decides first treatment more than anatomy: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning a sarcoma resection. p.23

### 7 · SURGERY FOR RP SARCOMA — PRINCIPLES

- Nephrectomy is the commonest auxiliary procedure: up to **55%** in primary RP sarcoma; renal function falls but rarely needs dialysis — † confirm contralateral function (split renogram) beforehand. p.23

### 8 · RPLND IN GERM CELL TUMOURS (ESSENTIALS)

- **Post-chemotherapy RPLND (PC-RPLND)**: all residual disease + full bilateral infrahilar template (standard). Modified unilateral only if: lesion **≤5 cm** in primary landing zone before and after chemotherapy, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good/intermediate risk. p.24

- **Nerve-sparing**: postganglionic fibres L1–L4 converge on the superior hypogastric plexus below the IMA; identify prospectively, loop, dissect packets around them → antegrade ejaculation **90–100%**. Temporary anejaculation = neuropraxia. p.24

### 9 · COMPLICATIONS AND MANAGEMENT

- ALGORITHM
  **Chylous ascites**: paracentesis + low-fat medium-chain triglyceride (MCT) diet ± octreotide → reaccumulation: indwelling drain → nil by mouth + total parenteral nutrition → persistent >100 mL/day: peritoneovenous (LeVeen) shunt or surgical ligation (last resort) → continue MCT diet 1–3 months after resolution.
  p.25

### 13 · MCQ ONE-LINERS

- MCQ Commonest primary RP sarcoma: liposarcoma (~60%); second: leiomyosarcoma (~20%). p.25
- MCQ Negative HU fat on CT: liposarcoma, angiomyolipoma, myelolipoma. p.25
- MCQ WD/DD liposarcoma: MDM2/CDK4 amplification on 12q; pleomorphic liposarcoma is MDM2-negative. p.25
- MCQ Dense nodule within a fatty RP mass suggests de-differentiation. p.25
- MCQ SFT: NAB2-STAT6 fusion; Doege-Potter syndrome = IGF-mediated hypoglycaemia. p.25
- MCQ Synovial sarcoma: t(X;18), TLE1, calcification. p.25
- MCQ Ewing and DSRCT both t(11;22) EWSR1 — partners FLI1 vs WT1. p.26
- MCQ Lymphoma in RP: biopsy, then chemotherapy — never resect. p.26
- MCQ Young man with RP mass: AFP and hCG first. p.26
- MCQ Extragonadal RP GCT is often a burnt-out testicular primary. p.26
- MCQ Isochromosome 12p marks GCT, including unknown-primary midline tumours. p.26
- MCQ Commonest extra-adrenal paraganglioma site: organ of Zuckerkandl. p.26
- MCQ MPNST: H3K27me3 loss; half in NF-1; parent nerve must be sacrificed. p.26
- MCQ Desmoids never metastasise but recur locally; nuclear β-catenin. p.26

- MCQ PEComa unresectable: mTOR inhibitors. p.26
- MCQ FNCLCC grade: differentiation + mitoses + necrosis; score 6–8 = grade 3. p.26
- MCQ Cystic lymphangioma: complete excision; aspiration/marsupialisation recur. p.26
- MCQ Aorta always reconstructed; chronically occluded IVC can be ligated. p.26
- MCQ Chylous ascites first line: paracentesis + MCT diet ± octreotide. p.26
- MCQ Right testis drains interaortocaval; left drains para-aortic. p.26
- MCQ Unilateral PC-RPLND only if mass ≤5 cm in landing zone, normal markers, good/intermediate risk. p.26
- MCQ Only Ewing sarcoma has standard neoadjuvant chemotherapy among RP sarcomas. p.26

## ★ Putting it together: adrenal and retroperitoneal decisions

### A · ADRENAL INCIDENTALOMA: TWO QUESTIONS

- ALGORITHM

**1. Is it malignant?** Unenhanced CT ≤10 Hounsfield units (HU), or signal drop-out on chemical-shift MRI → lipid-rich adenoma. If >10 HU → washout CT (absolute washout >60% or relative >40% = adenoma). Size >4 cm, suspicious imaging or growth >1 cm → consider resection (→ Ch 146).

**2. Is it functioning?** Test everyone: 1 mg overnight dexamethasone suppression test (cortisol), plasma or urine metanephrines (phaeochromocytoma), and aldosterone–renin ratio if hypertensive or hypokalaemic.

**Silent, <4 cm, benign imaging** → no further follow-up needed.

p.27

### C · ADRENALECTOMY: WHICH APPROACH?

- KEY POINT Laparoscopic (transperitoneal lateral or posterior retroperitoneoscopic) is standard for benign tumours. Choose open surgery for large suspected adrenocortical carcinoma or local invasion, because capsular rupture causes tumour spill and recurrence. Right side: the short adrenal vein drains straight into the inferior vena cava, so control it carefully. Partial adrenalectomy for bilateral or hereditary disease preserves function (→ Ch 147). p.27

### D · RETROPERITONEAL MASS

- ALGORITHM

**Image** (CT/MRI) → **exclude** paraganglioma (metanephrines), germ cell tumour (testis exam, markers) and lymphoma before biopsy.

**Biopsy** by core needle via a retroperitoneal route when the diagnosis would change treatment.

**Sarcoma:** complete en bloc resection with negative margins at the first operation (multivisceral if needed) in a specialist centre; never enucleate. Lifelong CT follow-up, especially for liposarcoma (→ Ch 140).

p.27

# 15 Renal & Transplant

157 core items · page numbers refer to the full book PDF

## Ch 88 · Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

### 1 · KIDNEY: POSITION AND RELATIONS

- Retroperitoneal; length ≈**11 cm**, weight 125–170 g (male). Lies T12–L3; **right 1–2 cm lower** (liver); descends ≈3 cm on inspiration and standing. p.2

### 3 · PARENCHYMA AND ITS IMAGING

- Glomerular filtration rate (GFR) ≈**125 mL/min**; ≈99% of filtrate reabsorbed. Urothelium lines calyces to urethra → urothelial carcinoma at any level. p.2
- TABLE **Nephrographic · 80–120 s · Best for renal masses** p.3

### 4 · RENAL VESSELS, LYMPHATICS AND NERVES

- TABLE Length · 2–4 cm · **6–10 cm** p.3

### 5 · PERCUTANEOUS PUNCTURE AND VESSELS

- KEY POINT Why the fornix? Arteries run on the anterior and posterior faces of the infundibula; a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Consequences of vascular injury: haemorrhage, arteriovenous fistula, pseudoaneurysm, loss of renal function. p.4

### 8 · RADIOLOGIC ANATOMY OF THE URETER

- KEY POINT **Weigert–Meyer rule**: upper-moiety orifice **inferior and medial** → ureterocele, obstruction; lower moiety → vesicoureteral reflux (VUR). p.5

### 10 · ENDOSCOPIC ANATOMY

- Lower-pole infundibulum needs ≈**140°** deflection. Papillae: pale discs ringed by fornices; ducts of Bellini dilate with obstruction. p.5
- ALGORITHM
  Infundibulotomy: superior → inferior → posterior quadrants; **never anterior**
  p.5

### 11 · MCQ ONE-LINERS

- MCQ Hilum: vein, artery, pelvis, posterior segmental artery. p.5
- MCQ Gerota fascia open inferiorly. p.5
- MCQ Supracostal access above 11th rib risks pneumothorax. p.5
- MCQ Best CT phase for renal mass: nephrographic. p.5
- MCQ Horseshoe ascent arrested by IMA. p.5
- MCQ Five segmental end arteries. p.5
- MCQ Accessory renal arteries: 25–28%. p.5
- MCQ Commonest left renal vein anomaly: circumaortic. p.5
- MCQ Safest puncture: calyceal fornix. p.5
- MCQ Sampaio A-II: crossed calyces. p.6
- MCQ Weigert–Meyer: upper moiety inferomedial. p.6
- MCQ Distal ureteric endoureterotomy: medial. p.6
- MCQ Peristalsis is myogenic. p.6
- MCQ Lyon golf-hole: grade 3. p.6

- MCQ Infundibulotomy: never anterior. p.6
- MCQ Narrowest ureter: intramural. p.6
- MCQ Right kidney 1–2 cm lower than left. p.6
- MCQ Anterior pararenal space crosses the midline. p.6
- MCQ Interlobar arteries run in columns of Bertin. p.6
- MCQ Posterior segmental artery anterior to ureter causes UPJO. p.6
- MCQ Left renal vein: 6–10 cm, three tributaries. p.6
- MCQ Nutcracker: left renal vein between SMA and aorta. p.6
- MCQ Most dangerous puncture: upper-pole infundibulum. p.6
- MCQ Ureteric narrowings: UPJ, iliac crossing, UVJ. p.6

## Ch 90 · Fundamentals of Nephrology

### 1 · WHY A UROLOGIST NEEDS NEPHROLOGY

- **Nephron**: ≈**1 million** per kidney; glomerulus + tubule (proximal tubule → loop of Henle → distal tubule → collecting duct). Number fixed at birth; reduced nephron mass (nephrectomy, agenesis) → **hyperfiltration** of the remainder. p.7
- KEY POINT Whoever operates on, obstructs, relieves or doses drugs through the kidney is changing volume, electrolytes and acid–base balance. The glomerulus is a pressure-driven filter set by arteriolar tone; the tubule is an energy-hungry line working at the edge of hypoxia — most disorders below fail one or the other. p.7

### 2 · MEASURING KIDNEY FUNCTION

- KEY POINT Trap: after cystectomy or bladder repair, rising SCr with ascites may be **intraperitoneal urine leak** reabsorbed across the peritoneum — not kidney injury. p.7
- Normal excretion <**150 mg/day** (barrier limits passage; proximal tubule reclaims). More = barrier/tubular damage or overproduction (myeloma paraproteins). †Nephrotic range >3.5 g/day. p.7

### 3 · ACUTE KIDNEY INJURY

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32. GFR depends on holding glomerular hydrostatic pressure by adjusting afferent and efferent tone. p.8
- TABLE 1 · Rise ≥**0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†) · <0.5 mL/kg/h for 6 h p.8
- TABLE FeNa · <**1%** · >1% p.9
- TABLE Fractional excretion of urea (FeUrea) — on diuretics/alkalosis · <**35%** · >35% p.9
- **Management**: hypovolaemia → isotonic fluid; heart failure → diuretics; postoperative hypotension → hold antihypertensives. MAP >**65 mm Hg** suffices in healthy adults, but elderly vasculopaths autoregulate at higher pressures. p.9
- SCr rise within **48 h** of intravenous iodinated contrast (usually computed tomography [CT]): vasoconstriction + direct tubulotoxicity. Risk: age, diabetes, CKD, anaemia, volume, arterial route. p.10
- **Gadolinium** (magnetic resonance imaging [MRI]): not nephrotoxic at usual dose but groups I and III linked to **nephrogenic systemic fibrosis (NSF)** — avoid in AKI, eGFR <**30**, dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) may be used. **Group III (gadoxetate) is treated as group I**, not intermediate. p.10

- Onset days–weeks after the drug. Consider in unexplained hospital AKI and **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ. p.10

- Non-recovery persisting ≥**90 days** = CKD or end-stage kidney disease (ESKD). p.11

### 5 · CHRONIC KIDNEY DISEASE

- **Definition (KDIGO)**: abnormal kidney structure or function for ≥**3 months**. Structure alone qualifies — a congenitally atrophic kidney with normal GFR is CKD. Classify by **cause, GFR category, albuminuria category**. p.11

- TABLE Anaemia (normocytic) · Erythropoietin falls, especially GFR <30 · Replete iron, exclude other causes → erythropoiesis-stimulating agent; target haemoglobin **10–11.5 g/dL**; >13 harmful p.11

- TABLE Acidaemia · Low ammoniagenesis, bicarbonate loss · Alkali to keep bicarbonate ≥**22** — slows progression, helps bone p.12

- Glycated haemoglobin (HbA1c) <7.5%; blood pressure <**130/80**. Agent matters only in proteinuric disease → RAAS blockade. p.12

### 6 · HAEMATURIA AND PROTEINURIA — WHEN TO INVOLVE A NEPHROLOGIST

- †Microhaematuria = ≥**3** red cells/high-power field on microscopy; a positive dipstick needs microscopic confirmation. p.12

- ALGORITHM
  Haematuria ± proteinuria → urine microscopy + ACR + SCr/eGFR → red cell casts, dysmorphic cells, AKI or systemic features → **urgent nephrology**; isolated non-glomerular haematuria → urological evaluation; isolated albuminuria → stage CKD, refer if criteria below.
  p.12

### 7 · ELECTROLYTE AND ACID–BASE DISORDERS

- Total body water = 0.6 × weight; ⅔ intracellular; extracellular ¾ interstitial, ¼ intravascular. Of 1 L: D5 → **83 mL** intravascular; 0.9% saline → 250 mL; 5% albumin → 1000 mL. p.13

- **Rate**: chronic/unknown onset → rise ≤**8–12 mEq/L per 24 h**, else **osmotic demyelination**. Highest risk: fluid-resuscitated hypovolaemia removes the ADH stimulus → brisk water diuresis; overcorrection → DDAVP + free water. Severe symptoms (seizure, obtundation) → **3% saline to raise 4–6 mEq/L** over hours, then slow. p.13

- Free water; chronic → fall ≤**0.5 mEq/L/h** (else cerebral oedema). Hypo-correction shrinks brain cells; hyper-correction swells them. p.13

- Electrocardiogram (ECG): peaked T → long PR, wide QRS → P loss → sine wave → ventricular fibrillation. >**6.0** generally urgent. p.13

- ALGORITHM
  **Stabilise** (any ECG change): calcium gluconate 10% 10 mL intravenously → **shift**: insulin 10 U + 50 mL 50% dextrose; nebulised salbutamol 20 mg → **remove**: haemodialysis if oliguric/ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; potassium binder → stop culprit drugs.
  p.13

- Treat if symptomatic or ≥**14 mg/dL**: **isotonic saline** 200 mL/h, titrated to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin (rapid, tachyphylaxis 48 h); intravenous bisphosphonate (sustained); denosumab. p.14

- MCQ Laxative abuse → negative urine AG, urine pH >5.8 → **ammonium acid urate stones**. p.14

### 8 · URINARY TRACT OBSTRUCTION AND POST-OBSTRUCTIVE DIURESIS

- Physiological after relief of prolonged **bilateral** obstruction (typically catheterising chronic retention). Up to **300 mL/h** common; starts within hours, lasts ≈2 days; usually self-limited. p.14

- **ALGORITHM**
  Relieve obstruction → free oral fluids, no routine IV → output persistently high, hypovolaemia or obtunded/nil by mouth → metabolic panel every **4–6 h** (Na, K, phosphate, calcium, magnesium) → replace ≈ **75%** of urine output with **0.45% saline** + electrolytes.
  p.15

- **KEY POINT** Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement keeps the patient expanded and sustains it. p.15

- Small ureteric stone with low-grade obstruction: observation for **2–4 weeks** is functionally safe. Beyond 4 weeks outcome is unpredictable (recovery reported up to 5 months). p.15

### 10 · MCQ ONE-LINERS

- **MCQ** Most accurate estimate: combined creatinine–cystatin C equation. p.15
- **MCQ** Trimethoprim and cimetidine raise SCr by blocking secretion, not GFR. p.15
- **MCQ** Rising SCr with ascites after bladder surgery: suspect intraperitoneal urine leak. p.15
- **MCQ** Normal protein excretion <150 mg/day; first-line test: spot ACR. p.15
- **MCQ** KDIGO AKI stage 1: SCr rise ≥0.3 mg/dL or urine <0.5 mL/kg/h for 6 h. p.15
- **MCQ** Efferent constriction lowers flow but raises GFR. p.15
- **MCQ** NSAIDs block afferent dilatation; RAAS blockers block efferent constriction. p.15
- **MCQ** Prerenal: FeNa <1%; FeUrea <35% on diuretics. p.15
- **MCQ** Muddy-brown granular casts: ATN; red cell casts: glomerulonephritis; white cell casts: AIN. p.15
- **MCQ** Diuretics treat overload in ATN, not ATN itself. p.15
- **MCQ** Contrast AKI prevention: intravenous fluid before contrast in at-risk patients. p.15
- **MCQ** Gadolinium group III handled as group I; group II allowed in advanced CKD. p.16
- **MCQ** Commonest biopsy lesion in checkpoint-inhibitor AKI: AIN. p.16
- **MCQ** Uraemic bleeding: DDAVP 0.3 µg/kg; tachyphylaxis after 3 doses. p.16
- **MCQ** CKD anaemia target haemoglobin 10–11.5 g/dL. p.16
- **MCQ** Chronic hyponatraemia: correct ≤8–12 mEq/L/24 h — osmotic demyelination. p.16
- **MCQ** Chronic hypernatraemia: lower ≤0.5 mEq/L/h — cerebral oedema. p.16
- **MCQ** Sodium polystyrene sulfonate: only emergency-approved binder; risks colonic necrosis. p.16
- **MCQ** Vomiting causes hypokalaemia through kaliuresis, not gastric loss. p.16
- **MCQ** Topiramate: proximal-type RTA, high urine pH, calcium phosphate stones. p.16
- **MCQ** Negative urine anion gap: gastrointestinal bicarbonate loss. p.16
- **MCQ** Post-obstructive diuresis: replace ≈75% of output with 0.45% saline if unable to drink. p.16
- **MCQ** Complete UUO: full recovery at 1 week, none after 6 weeks (canine). p.16

## Ch 91 · Renal Transplantation

### 1 · END-STAGE DISEASE AND WHY TRANSPLANT

- Chronic kidney disease (CKD) is defined by reduced estimated glomerular filtration rate (eGFR), albuminuria, or both. Kidney Disease: Improving Global Outcomes (KDIGO) stage **G5 (eGFR < 15 mL/min/1.73 m²)** = end-stage renal disease (ESRD), needing dialysis or transplant. CKD staging, dialysis modalities and their complications → Ch 90. p.17

- **MCQ** First successful kidney transplant: identical twins, 1954. p.17

## 2 · RECIPIENT EVALUATION

- KEY POINT Two questions drive every test: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)? p.17
- Evaluate before dialysis; deceased-donor listing allowed at eGFR ≤**20**. Elderly may gain from a faster older-donor kidney. p.17

## 4 · DECEASED DONATION

- Paediatric donors <**15 kg** → **en bloc** kidneys on aortic and caval conduits into an adult with body mass index (BMI) <30 and low immunological risk; thrombosis 10–15%, but long-term survival equals living donation. p.18
- ALGORITHM
  Procurement: midline incision + sternotomy → mobilise bowel off retroperitoneum → control aorta and inferior vena cava (IVC) → identify aberrant/iliac accessory arteries **before cannulation** → perfuse; ice via opened Gerota fascia → kidneys with perinephric fat, **periureteric tissue preserved** → en bloc removal → back-table separation, cold storage
  p.18

## 5 · LIVING DONATION

- TABLE Mortality · **3/10,000** p.19

## 6 · THE OPERATION

- ALGORITHM
  **Vein first** (shortens arterial ischaemia): end-to-side to external iliac vein, valves excised, 5-0/6-0 polypropylene → artery end-to-side to external iliac (punch arteriotomy, avoid calcified clamp sites) → bulldogs, release iliac clamps, haemostasis → reperfuse → position without kink or stretch
  p.20
- ALGORITHM
  **Lich–Gregoir** extravesical: fill bladder → myotomy to mucosa → open mucosa distally → spatulate ureter → mucosa-to-mucosa 5-0/6-0 absorbable ± stent, distal anchoring stitch → close detrusor loosely as short tunnel
  p.20

## 10 · MCQ ONE-LINERS

- MCQ ESRD: KDIGO G5, eGFR <15. p.21
- MCQ Deceased-donor listing at eGFR ≤20. p.21
- MCQ Genetic FSGS does not recur; primary does. p.21
- MCQ Primary hyperoxaluria: liver–kidney transplant. p.21
- MCQ Live vaccines contraindicated after transplant. p.21
- MCQ Incidental renal tumour <1 cm: no treatment needed first. p.21
- MCQ O universal donor; AB universal recipient. p.21
- MCQ Class II HLA: DR, DQ, DP. p.21
- MCQ Hyperacute rejection: preformed DSA. p.21
- MCQ En bloc for donors <15 kg. p.21
- MCQ Machine perfusion reduces DGF only. p.21
- MCQ Donor mortality 3/10,000. p.21
- MCQ First laparoscopic donor nephrectomy: Ratner, 1995. p.21
- MCQ Venous anastomosis before arterial. p.21
- MCQ Lower pole artery supplies ureter. p.21

- MCQ Commonest ureteric anastomosis: Lich–Gregoir. p.21
- MCQ Venous thrombosis Doppler: reversed diastolic flow. p.21
- MCQ Stenosis: parvus-tardus; angioplasty first. p.21
- MCQ Best lymphocele treatment: laparoscopic fenestration. p.21
- MCQ Late transplant nephrectomy: subcapsular. p.21
- MCQ TCMR first line: intravenous steroids. p.21
- MCQ dd-cfDNA negative predictive value >90%. p.21
- MCQ Gibson incision: lower hernia rate than hockey stick. p.21
- MCQ Torsion occurs only with intraperitoneal grafts. p.21
- MCQ Hume test: renal vein occlusion → pulsatile swelling. p.21

## Ch 92 / 61 · Urologic Complications of Renal Transplantation and Paediatric Transplant Considerations

### 1 · OVERVIEW

- Urologic complications are the **commonest surgical adverse event** (**8–9.5%**). Most arise from the **transplant ureter and ureterovesical anastomosis** — this organises the whole subject. p.22

### 2 · HAEMATURIA AND URETERAL STENTS

- Meta-analyses: removal before week 3 reduces UTI without more leak or stenosis; optimum **14–21 days**. Removal **before 14 days increases leak** — hence a window, not a date. p.22

- ALGORITHM
  Computed tomography (CT) in every case (burden, distribution → approach) → **percutaneous nephrostomy first** (decompression, function recovery, antegrade access) → clearance by combined percutaneous nephrolithotomy (PCNL), retrograde intrarenal surgery (RIRS) and extracorporeal shock wave lithotripsy (ESWL)
  p.22

### 4 · URETERAL OBSTRUCTION AND STRICTURE

- KEY POINT Lymphocele spans both periods. Trap: a late obstruction is not automatically a stricture — exclude stones and lymphocele. p.23

- Stricture: **0.5–9%**; primary cause **ischaemia**. Risks are perfusion or healing insults: cold ischaemia >435 min, acute rejection, urinoma/abscess, delayed graft function (DGF), multiple arteries, older donor, and transplant-specific **BK virus and cytomegalovirus (CMV)**. p.23

### 6 · LYMPHOCELE

- **<15%** need treatment. Efficacy: aspiration alone worst < drain ± sclerotherapy < open surgery < **laparoscopic peritoneal window** (best). †The window drains lymph into the peritoneum, where it is absorbed. p.24

### 7 · STONES AND UTI

- Cystitis: fluoroquinolone, amoxicillin-clavulanate, third-generation cephalosporin or nitrofurantoin (creatinine clearance >30 mL/min) for **5–10 days**. Pyelonephritis/complicated: **14–21 days**, and confirm the graft drains. p.24

### 10 · PAEDIATRIC PRETRANSPLANT PREPARATION

- Target: age-expected capacity at detrusor pressure **<40 cm $H_2O$**. First line antimuscarinics (oxybutynin; then solifenacin, mirabegron, intradetrusor botulinum toxin) with clean intermittent catheterisation (CIC). Prove emptying by repeated low residuals. p.25

- **ALGORITHM**
  Catheterise → instil increasing saline volumes → set dwell → drain → step up volumes (± medication) → target capacity for age → **confirm by urodynamics** → clear for transplant; inadequate → augment
  p.26

- **KEY POINT** A renal allograft must never be used to test whether a bladder works. p.26

- Indications: capacity **<75%** of expected, pressures **>40 cm $H_2O$**, CIC needed every **≤3-4 h** despite maximal therapy. Augmentation does not clearly raise transplant risk. p.26

### 11 · PAEDIATRIC RECONSTRUCTION AND NATIVE NEPHRECTOMY

- Major reconstruction **before transplant**; allow **6–12 weeks** after enterocystoplasty. Exception: transplant in first 2 years, before continence matters. Document with a diagram. p.26

### 12 · PAEDIATRIC TRANSPLANT URETER

- **ALGORITHM**
  Partially fill bladder → traction sutures → size ureter → detrusor incision **3–4 cm** → excise distal mucosal disc → drain bladder → spatulate ureter 4–5 mm → interrupted mucosa-to-mucosa fine absorbable → anchor adventitia (prevents eversion) → close detrusor loosely at entry
  p.26

### 13 · PAEDIATRIC COMPLICATIONS

- **ALGORITHM**
  Rising creatinine + hydronephrosis → rejection signs → biopsy, treat rejection → no rejection and unimproved by bladder drainage → stent/nephrostomy → failed retrograde → percutaneous antegrade study and stent → focal stenosis: balloon + stent 4–6 weeks → open repair without delay
  p.27

### 14 · MCQ ONE-LINERS

- **MCQ** Commonest early urologic complication: urinary leak. p.27
- **MCQ** Predominant late (>3 months) obstruction cause: ureteral stricture. p.27
- **MCQ** Primary cause of transplant stricture: ischaemia. p.27
- **MCQ** Optimal stent removal: 14–21 days; earlier increases leak. p.27
- **MCQ** Persistent haematuria after biopsy: arteriovenous fistula. p.27
- **MCQ** Urine leak proof: fluid creatinine far above serum. p.27
- **MCQ** Lymphocele peak: 6 weeks. p.27
- **MCQ** Best lymphocele treatment: laparoscopic peritoneal window. p.27
- **MCQ** mTOR inhibitors associated with lymphocele. p.27
- **MCQ** Reflux into adult graft: 50–86% radiologically, <1% significant. p.27
- **MCQ** TURP safe ≥3 weeks post-transplant, not with stent. p.27
- **MCQ** Post-transplant RCC: 90% in native kidneys. p.27
- **MCQ** Bladder cancer risk ~3-fold, more aggressive. p.27
- **MCQ** Paediatric target pressure: <40 cm $H_2O$. p.27
- **MCQ** Augment if capacity <75% expected for age. p.27
- **MCQ** Gastric augmentation: haematuria–dysuria syndrome. p.27
- **MCQ** Allow 6–12 weeks after enterocystoplasty before transplant. p.27
- **MCQ** Preferred paediatric anastomosis: extravesical non-refluxing ureteroneocystostomy. p.27
- **MCQ** Commonest paediatric urologic complication: ureteral stenosis (6%). p.27

- MCQ Use largest urethral catheter at transplantation. p.27
- MCQ >50% paediatric obstructions occur within 100 days. p.27

## ★ Putting it together: kidney function and transplant decisions

### A · RISING CREATININE AFTER UROLOGICAL SURGERY

- ALGORITHM

**1. Postrenal first:** check the catheter, scan the bladder and kidneys (hydronephrosis?). After cystectomy or bladder repair, rising creatinine with ascites may be an intraperitoneal urine leak reabsorbed across the peritoneum — drain fluid creatinine far above serum confirms it (→ Ch 90).
**2. Prerenal or intrinsic:** fractional excretion of sodium (FeNa) <1% and bland urine → volume depletion, recovers with fluids; >1% with granular casts → acute tubular necrosis. Review nephrotoxins and contrast; stop renin–angiotensin blockers and metformin while unwell.
**3. Dialysis** only for "AEIOU" that fails medical treatment: refractory acidosis, electrolytes (potassium), intoxication, overload, uraemic complications.

p.28

### B · POST-OBSTRUCTIVE DIURESIS

- KEY POINT After relieving bilateral obstruction (or obstruction of a solitary kidney), monitor urine output, electrolytes and weight. Most of the diuresis is physiological, clearing retained salt and water; † it becomes pathological when it continues after the patient is back to normal volume. † Do not replace every millilitre — that keeps the diuresis going. If the patient cannot drink, replace about 75% of output with 0.45% saline and adjust to electrolytes (→ Ch 90). p.28

### E · CHILD GOING FOR TRANSPLANT

- ALGORITHM

Most have congenital anomalies of the kidney and urinary tract, chiefly posterior urethral valves. Before listing, make the bladder safe: capacity appropriate for age at detrusor pressure <40 cm $H_2O$, using antimuscarinics and clean intermittent catheterisation, and augmentation if needed. Remove dilated refluxing native ureters or kidneys causing infection, polyuria or hypertension. A poor bladder left untreated destroys the graft (→ Ch 92/61).

p.29

# 16 BPH

251 core items · page numbers refer to the full book PDF

## Ch 148 · Anatomy and Development of the Prostate

### 1 · DEVELOPMENT (EMBRYOLOGY)

- TABLE **10** · Solid epithelial buds invade mesenchyme; bud pattern fixes adult ductal pattern p.2
- KEY POINT McNeal: UGS mesenchyme loses inductive capacity with age; **re-awakening of embryonic induction** in ageing TZ stroma may initiate new glands → BPH. p.2

### 2 · ZONAL ANATOMY (MCNEAL)

- Base = bladder neck; apex = junction with membranous urethra. Prostatic urethra bends **≈35°** at the verumontanum; **13–15** ducts open dorsolaterally. p.2
- TABLE Peripheral (PZ) · Longest, distal · ≈70 · **70%** of cancers; †palpable on digital rectal examination p.3
- KEY POINT Trap: CZ (around ejaculatory ducts) ≠ TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). Enlarging TZ compresses PZ into the **surgical (pseudo)capsule** — the enucleation plane → Ch 152. p.3

### 5 · SECRETIONS AND CLINICAL MARKERS

- Prostatic fluid: **37–44%** of semen, acidic (pH ≈6.3); seminal vesicle fluid (majority, alkaline) supplies semenogelins. p.3
- Coagulation → liquefaction by kallikreins within **20–30 min**; nonliquefaction if incomplete by 60 min. p.3
- KEY POINT Benign cells: high zinc blocks aconitase → citrate exported (inefficient metabolism). Cancer: **zinc lost**, citrate oxidised → low citrate (†basis of magnetic resonance spectroscopy). p.4

### 6 · MCQ ONE-LINERS

- MCQ Prostate derives from urogenital sinus (endoderm). p.4
- MCQ Budding begins at week 10. p.4
- MCQ Initiation requires AR in mesenchyme, not epithelium. p.4
- MCQ Gate to budding: AR gene demethylation. p.4
- MCQ UGS mesenchyme is an instructive inducer (Cunha). p.4
- MCQ SRD5A2 deficiency → incomplete prostate. p.4
- MCQ Andromedin required for development: FGF10. p.4
- MCQ Epithelial specification marker: Nkx3-1. p.4
- MCQ BPH arises in transition zone. p.4
- MCQ 70% of cancers arise in peripheral zone. p.4
- MCQ Central zone surrounds ejaculatory ducts. p.4
- MCQ Anterior fibromuscular stroma is non-glandular. p.4
- MCQ Prostatic urethra angulates ≈35° at verumontanum. p.4
- MCQ Luminal keratins 8/18; basal 5/14. p.4
- MCQ Hillock cell marker: KRT13; club: SCGB1A1. p.4
- MCQ Most neuroendocrine cells are neural crest–derived. p.4
- MCQ Mouse has no PSA gene. p.4
- MCQ Liquefaction normal within 20–30 minutes. p.4

- MCQ Highest tissue zinc: prostate. p.4
- MCQ Zinc inhibits mitochondrial aconitase. p.4
- MCQ Cancer: low zinc, low citrate. p.4
- MCQ PSMA is a folate hydrolase. p.4
- MCQ PAP: first prostate cancer marker. p.4
- MCQ Gestational DES → squamous metaplasia. p.4
- MCQ TCDD → prostatic hyperinnervation. p.4
- MCQ †Neurovascular bundles: posterolateral, outside capsule. p.4
- MCQ †Adenoma arterial supply enters at 5 and 7 o'clock. p.4

## Ch 149 · Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

### 1 · DEFINITIONS AND THE FOUR OVERLAPPING ENTITIES

- "**Prostatism**" is obsolete and wrong: it made the prostate the sole source of LUTS. Paul Abrams coined "LUTS" to replace it MCQ. p.5
- † Urodynamic BOO is graded by the bladder outlet obstruction index (BOOI = PdetQmax − 2Qmax): **>40 obstructed, 20–40 equivocal, <20 unobstructed**. p.5
- KEY POINT Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are four separate dimensions; obstruction can only be diagnosed by measuring it (pressure–flow). p.5

### 2 · EPIDEMIOLOGY (HEADLINE)

- Pooled autopsy data: **no BPH under 30**; prevalence rises from the fourth decade, ≈ **50% in the 50s**†, peaking at **88% in the 80s** (approaches 100% in the ninth decade). p.5
- Proposed cut-offs: volume **>30 mL**; common community composite: IPSS >7, maximum flow rate (Qmax) <15 mL/s, volume >20 mL. p.6
- TABLE Mild · **0–7** p.6
- TABLE Moderate · **8–19** p.6
- TABLE Severe · **20–35** p.6
- Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**; transition zone volume (TZV) ≈ 15 → 25 mL. p.6

### 3 · AETIOLOGY

- KEY POINT Core concept: BPH is an **accumulation** disease. Volume = proliferation − death. Established BPH shows equal or reduced replication with ↑ anti-apoptotic genes (bcl-2); androgens act chiefly by **suppressing cell death**. That is why androgen withdrawal shrinks the gland. Early BPH may involve rapid proliferation. p.6
- Serum testosterone falls with age, yet **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. Intraprostatic DHT is **maintained, not elevated**, in BPH (earlier "raised DHT" reports used cadaveric controls in whom DHT falls after death) MCQ. p.7
- ALGORITHM
  Testosterone → enters stromal cell → **type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) → DHT → acts on stroma (autocrine) or diffuses to epithelium (paracrine) → DHT–AR complex → DNA → transcription of growth factors → epithelial survival/growth
  p.7
- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% adrenal (negligible role). p.7

- SRD5A2 gene on **chromosome 2p23** MCQ: V89L ↓ activity, A49T ↑ activity, long TA repeats ↓ activity; links to BPH inconsistent. Promoter methylation in up to 30% of BPH specimens. p.7
- **Waist circumference** (central adiposity) predicts prostatectomy and frequent LUTS independently of BMI. BMI >35 vs <25: odds ratio **3.5** for prostate ≥40 mL. Biases: DRE under-detects enlargement in the obese; obese men steered away from surgery. p.9
- Case–control study of men with large resected glands (>37 g) at young age: first-degree male relatives had ≈ **4-fold** risk of surgically treated BPH (brothers highest, ≈6–8-fold). p.9
- Segregation analysis: **autosomal dominant**. ≈ **50%** of men operated on before 60 may have heritable disease vs ≈9% after 60. p.10

## 4 · PATHOLOGY

- **Middle lobe** must be **periurethral** in origin (no TZ tissue at the bladder neck) MCQ. p.10
- KEY POINT **Why does the capsule matter?** The capsule (unique to the human prostate) transmits the pressure of tissue expansion inward onto the urethra. The dog gets BPH but rarely obstruction because it has no capsule. Clinical proof: transurethral incision of the prostate (TUIP) incises the capsule and relieves obstruction with no change in volume. Hence **size does not correlate with obstruction**; geometry, capsule and tone matter more. p.10

## 5 · PATHOPHYSIOLOGY OF OBSTRUCTION

- ALGORITHM
  Prostatic hyperplasia → ↑ urethral resistance (static + dynamic) → detrusor compensation (↑ pressure to maintain flow) → cost paid in storage function → storage LUTS ± impaired emptying → (minority) decompensation
  p.11
- Response is largely adaptive, but **many LUTS arise from obstruction-induced bladder change**, not the obstruction itself. ≈ **One-third** keep significant voiding dysfunction (mostly storage) after surgical relief of obstruction. p.11
- ALGORITHM
  **Three-stage remodelling**: smooth muscle hypertrophy (adaptive) → compensation → decompensation (detrusor underactivity, underactive bladder)
  p.11
- KEY POINT Why persisting storage LUTS after transurethral resection of the prostate (TURP)? Obstruction leaves a collagen-rich, dedifferentiated, overactive detrusor that removing the adenoma cannot reverse. Counsel patients with DO preoperatively. Acute urinary retention (AUR) is **not** the inevitable end of this sequence: many men in AUR have adequate detrusor function and a precipitant. p.11

## 6 · NATURAL HISTORY

- ≈ **40%** improve symptomatically without active treatment, whether on watchful waiting or placebo (placebo does not beat watching). Placebo response rises with baseline severity; active drug gives ≈ twice the improvement in each stratum. p.12
- Qmax falls ≈ **2.1%/year**; rapid decline predicted only by age ≥70 and Qmax <10 mL/s. p.12
- Volume grows ≈ **0.6 mL/year** (≈1.9%/year), faster with higher baseline PSA and larger volume. p.12
- Placebo arm of **Medical Therapy of Prostatic Symptoms (MTOPS)**: overall progression **4.5 per 100 patient-years** (17% at 4 years); mostly **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**. p.12
- KEY POINT Why use PSA? It is a cheap surrogate for volume, and volume is the main driver of AUR and surgery. Among >100 candidate predictors, a multi-variable model was only slightly better than PSA alone. p.13

## 7 · COMPLICATIONS

- KEY POINT Serious complications (death, renal failure, stones, infection) have become rare. What remains common is **AUR and surgery**, both predicted by the same factors: age, symptoms, low flow and a large gland (PSA). p.13
- TABLE Later surgery · **75%** · **26%** p.13
- Incidence rises with selection: community ≈ **4.5–6.8/1000 person-years** (Physicians' Health Study, Olmsted County); watchful waiting arm of the Veterans Affairs (VA) trial 9.6; PLESS placebo 18; MTOPS placebo 6. Hazard is constant over time. p.13
- Autopsy: ≈ **8×** commoner with BPH, with **no** increase in renal/ureteric stones (stasis, not metabolic). Clinical risk small; screen if haematuria or stuttering stream. p.14

## 9 · MCQ ONE-LINERS

- MCQ BPH is hyperplasia (↑ cell number), not hypertrophy. p.15
- MCQ All BPH nodules arise in the transition zone or periurethral region. p.15
- MCQ The middle lobe is periurethral in origin. p.15
- MCQ Earliest periurethral nodules are stromal, resembling embryonic mesenchyme. p.15
- MCQ No histological BPH below age 30; autopsy prevalence 88% in the 80s. p.15
- MCQ Autopsy prevalence is similar across ethnic and geographic groups. p.15
- MCQ Prepubertal castrates never develop BPH; androgens are permissive. p.15
- MCQ Intraprostatic DHT is maintained, not raised, in BPH. p.15
- MCQ Type 2 5α-reductase sits in stromal interstitial fibroblasts; SRD5A2 on chromosome 2p23. p.15
- MCQ Finasteride inhibits type 2; dutasteride types 1 and 2. p.15
- MCQ About 90% of prostatic androgen is testicular-derived DHT. p.15
- MCQ ERα is stromal and proliferative; ERβ is epithelial and antiproliferative. p.15
- MCQ FGF-7 (KGF): stromal source, receptor only on epithelium. p.15
- MCQ TGF-β inhibits epithelium but stimulates fibroblasts. p.15
- MCQ First prostatic growth factor identified: basic FGF (FGF-2). p.15
- MCQ Familial BPH: autosomal dominant, early onset, large gland, normal 5-ARI response. p.15
- MCQ Only humans and dogs develop BPH; the dog rarely obstructs (no capsule). p.15
- MCQ TUIP relieves obstruction without reducing volume — evidence for the capsule's role. p.15
- MCQ Dynamic component is mediated by α1a-adrenoceptors. p.15
- MCQ Trabeculation reflects increased detrusor collagen. p.15
- MCQ About one-third keep storage symptoms after relief of obstruction. p.15
- MCQ IPSS bands: mild 0–7, moderate 8–19, severe 20–35; lacks incontinence and pain items. p.15
- MCQ DRE underestimates prostate volume, more so in large glands. p.15
- MCQ Olmsted: prostate grows ≈0.6 mL/year; Qmax falls ≈2.1%/year. p.15
- MCQ About 40% improve on watchful waiting or placebo. p.15
- MCQ Trial "placebo effect" partly reflects regression to the mean. p.15
- MCQ MTOPS placebo progression 4.5/100 patient-years, mostly symptomatic. p.15
- MCQ No BPH-related renal insufficiency occurred in MTOPS. p.15
- MCQ Spontaneous AUR leads to surgery in 75%; precipitated AUR in 26%. p.15
- MCQ PSA >1.4 ng/mL and volume >40 mL predict AUR and surgery. p.15

- MCQ Bladder stones are 8× commoner with BPH; renal stones are not. p.15
- MCQ Finasteride is first-line for BPH-related haematuria. p.15
- MCQ Vasectomy is not a risk factor for BPH. p.15

## Ch 150 · Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

### 1 · DEFINITIONS AND THE DIAGNOSTIC PROBLEM

- KEY POINT A man with bothersome LUTS cannot be assumed to have BPH-related BOO until fully assessed. The work-up has two aims: **grade severity and bother**, and **identify the cause**, because the cause dictates treatment. Every test is judged by one question: can it separate outlet obstruction from bladder dysfunction? Symptoms, flow, postvoid residual (PVR), prostate volume and cystoscopy all fail; only the **pressure–flow study** passes, so it is the gold standard, reserved for men in whom the answer changes surgery. p.16

### 2 · HISTORY AND SYMPTOM ASSESSMENT

- **AUA Symptom Index (AUASI)**: 7 questions — frequency, nocturia, weak stream, hesitancy, intermittency, incomplete emptying, urgency — each 0–5, total **0–35**. p.16
- TABLE Mild · **0–7** p.17
- TABLE Moderate · **8–19** p.17
- TABLE Severe · **20–35** p.17
- Minimum change the patient perceives: **3 points**. Men with severe baseline symptoms need proportionately smaller change, so **percentage** reduction is more reliable. p.17

### 3 · PHYSICAL EXAMINATION AND DIGITAL RECTAL EXAMINATION

- PV does not correlate with symptom severity or degree of obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression. p.17

### 4 · FREQUENCY–VOLUME CHARTS AND NOCTURIA

- **Duration**: reference **3 days** — adherence falls with 7 days, reliability rises with more days. Keeping a diary has a bladder-training effect; tell the patient not to change his routine. p.17
- **Nocturnal polyuria** (ICS): nocturnal output **>20%** of 24-h output in younger men, **>33%** over 65 years. p.17
- ALGORITHM

  **Nocturia work-up**: FVC → classify: †global polyuria (>40 mL/kg/24 h → look for diabetes mellitus/insipidus, excess intake) / nocturnal polyuria (heart failure, sleep apnoea, evening fluids, diuretic timing) / reduced bladder capacity (OAB, BOO) → treat cause → if persistent, look for other causes such as **sleep apnoea** (AUA).

  p.18

### 5 · LABORATORY TESTS

- **Indication**: only if a PCa diagnosis would change management — i.e. life expectancy **>10 years**. Counsel on false positives/negatives and biopsy complications. p.18
- **Volume surrogate**: area under curve ≈0.76–0.78. Age-specific PSA cut-offs predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Predicts transrectal ultrasound (TRUS) volume within ±20% in most men. p.18

### 6 · INSTRUMENTAL INVESTIGATIONS

- Thresholds: **<30 mL** insignificant; persistently **>50 mL** important; **200–300 mL** may indicate DUA and predicts poorer response; **>300 mL** = significant chronic retention. p.18

- Errors: inaccurate when voided volume **<150 mL** (125–150) → repeat; single flow underestimates achievable Qmax; void in preferred position, interpret only voids the patient considers representative. p.19

- Cut-offs: Qmax **10 mL/s** — higher specificity; **15 mL/s** — higher sensitivity. Qmax <15 mL/s is the working definition of obstruction but does not exclude decompensation. p.19

- TABLE Cannot void **>150 mL** p.19

- TABLE PVR **>300 mL** p.19

- TABLE Age **>80** or **<50 years** with predominantly voiding LUTS p.19

- TABLE **Intravesical prostatic protrusion (IPP)** · Tip of protrusion to base, sagittal suprapubic view, bladder 100–200 mL; **≥10 mm** = high grade · Reasonable accuracy; predicts need for medical therapy and **failed trial without catheter (TWOC)**. Not a substitute for PFS p.19

## 7 · EVALUATION ALGORITHMS

- ALGORITHM
  **EAU (men ≥40 years)**: history (incl. sexual function) → symptom score → examination + DRE → urinalysis → PVR → PSA (if life expectancy >10 years and result changes management).
  Abnormal DRE, high PSA, abnormal urinalysis, suspected neurological disease → manage under relevant guideline.
  Not bothersome + normal tests → no further work-up. Significant PVR → renal ultrasound ± renal function.
  Bothersome → FVC (storage/nocturia) → prostate ultrasound + uroflow if treatment contemplated → medical algorithm. Surgery contemplated → endoscopy if it alters choice; PFS for specific indications.
  p.20

- ALGORITHM
  **AUA**: Step 1 history, examination, IPSS, urinalysis ± PVR/flow; **PVR >300 mL → manage as non-neurogenic chronic urinary retention** regardless of symptoms → Step 2 counsel: behavioural/lifestyle, medical therapy, procedural referral → Step 3 reassess at **4–12 weeks**: improved → routine follow-up; nocturia predominant → FVC + drug trial; incomplete response → FVC, PVR/flow, urodynamics, cystoscopy → OAB pathway, surgical algorithm (BOO), or both (mixed).
  p.20

## 8 · CONSERVATIVE MANAGEMENT — WATCHFUL WAITING (WW)

- TABLE Fluids · **1.5–2 L/day** adjusted for climate/activity (guided by FVC); restrict before journeys; none for 2 h before bed p.21

- KEY POINT Decision to treat is driven by **bother and complicating features**, not by symptom score or gland size alone. p.21

## 9 · MEDICAL THERAPY — INDICATIONS AND CHOICE

- ALGORITHM
  **EAU medical algorithm**: not bothersome → WW ± lifestyle | nocturnal polyuria → lifestyle ± vasopressin analogue | storage predominant → lifestyle ± antimuscarinic or β3-agonist | **PV >40 mL and long-term treatment intended** → 5-ARI ± α1-blocker or PDE5I | otherwise → α1-blocker or PDE5I → residual storage symptoms → add antimuscarinic or β3-agonist.
  **AUA**: α-blocker first (or PDE5I first if erectile dysfunction [ED] coexists) → **prostate >30 mL** → consider adding 5-ARI → inadequate response → trial of PDE5I → still inadequate → discuss surgery.
  p.21

- KEY POINT **Reading drug trials**: placebo improves both scores and flow (ideal: 4-week placebo run-in); efficacy is judged by the 3-point/percentage threshold; Qmax gains should be expressed as percentages. Treatment goals: relieve symptoms, reduce BOO, improve emptying, and **prevent progression**. p.21

### 10 · A1-ADRENERGIC BLOCKERS

- KEY POINT **Why titrate terazosin/doxazosin?** α1b blockade causes **first-dose hypotension** and syncope; bedtime starting dose and titration avoid it. Uroselective agents are fixed-dose. Trap: doxazosin GITS starts at 4 mg without titration. p.22

- KEY POINT **IFIS** — triad: **flaccid billowing iris, iris prolapse toward incisions, progressive miosis** despite dilation during cataract surgery. Cause: α1a blockade of iris dilator → dilator atrophy, so it occurs **even years after stopping**; stopping the drug pre-operatively does **not** reduce risk. Highest with **tamsulosin**, then alfuzosin, doxazosin, naftopidil. Tell the patient to inform the ophthalmologist. †In a man awaiting cataract surgery, defer starting an α1-blocker until after the operation. p.23

- KEY POINT One efficacy, three toxicity profiles: **α1b → vascular** (terazosin, doxazosin); **α1a → ejaculation and IFIS** (silodosin, tamsulosin); **α1d → storage benefit** (naftopidil). Choosing an α-blocker is choosing the side effect the patient tolerates best. p.23

### 11 · 5A-REDUCTASE INHIBITORS (ANDROGEN MANIPULATION)

- **Slow onset**: maximum PV reduction at **6 months**; perceptible benefit needs ≥6 months (trials of 6–12 months can miss benefit). p.23

- **Volume dependent**: best when prostate **>40 mL** (PLESS PV >55 mL → 70% risk reduction). Guideline indication: enlarged prostate (EAU >40 mL; AUA >30 mL) with moderate-to-severe LUTS. p.23

- KEY POINT 5-ARIs treat the gland, not the symptom: slow, volume-dependent, modest symptomatically, but they prevent AUR and surgery — and they halve PSA, which is both a trap and a tool. p.24

### 12 · ANTIMUSCARINICS FOR STORAGE LUTS

- KEY POINT Antimuscarinics treat the bladder, not the outlet, so every recommendation is anchored to a PVR ceiling (200 mL) — the question is how well the detrusor empties. p.24

### 13 · B3-AGONISTS

- **Mirabegron**: start **25 mg** → **50 mg** daily. Urodynamic safety trial in BOO: Qmax unchanged; PVR rose only at 100 mg. Lacks anticholinergic burden (dry mouth less than solifenacin/tolterodine). p.24

- AEs: nasopharyngitis, headache, hypertension (similar to placebo; BP rise ≤1 mm Hg). **Contraindicated in severe uncontrolled hypertension** (**≥180/110 mm Hg**); measure BP before and during treatment; QT prolongation at 200 mg. Moderate CYP2D6 inhibitor †. p.24

### 15 · COMBINATION THERAPY

- **Indication**: moderate-to-severe LUTS at high risk of progression — PV **>30–40 mL**, PSA **>1.5 ng/mL**; monotherapy doses once daily. p.25

- KEY POINT Most progression events are symptomatic. **α-blockers prevent symptomatic progression only; 5-ARIs prevent the hard events (AUR, surgery).** The α-blocker gives rapid dynamic relief; the 5-ARI gives slow static shrinkage. p.25

### 17 · PROGNOSIS AND FOLLOW-UP ON MEDICAL THERAPY

- ALGORITHM

  **Follow-up**: **4–12 weeks** — response (α-blocker) and AEs → **6 months** (5-ARI response) → **annually**: symptom score, flow, PVR; PSA on 5-ARI if life expectancy >10 years. WW: yearly review.

  p.26

- ALGORITHM
  **AUR**: urethral catheter decompression (†suprapubic if urethral fails) → start **α1-blocker** (e.g. alfuzosin 10 mg) → **TWOC after ~3–5 days** (median ~5) → voids: continue medical therapy (add 5-ARI if large gland), monitor → fails: re-catheterise; up to **3 TWOCs** may be tried → recurrent AUR or failure → PFS and surgery → Ch 151.
  p.26
- Immediate surgery resolves BOO but emergency surgery has higher morbidity → **TWOC preferred** for first episodes; succeeds in about **60%**. Recurrence after successful TWOC is common, mostly within **6 months**. p.26
- **TWOC failure predictors**: age, severe LUTS, drained volume **>1 L**, spontaneous AUR, high-grade IPP. p.26
- KEY POINT AUR follows the two-component model: the **α-blocker relieves the dynamic component and rescues the TWOC**; the **5-ARI shrinks the static component and prevents the next episode**. p.27



- MCQ IPSS = 7 AUASI questions + 1 QoL (bother) question; total symptom score 0–35. p.27
- MCQ IPSS bands: mild 0–7, moderate 8–19, severe 20–35. p.27
- MCQ Minimum patient-perceptible IPSS change: 3 points. p.27
- MCQ PSA >1.6 / 2.0 / 2.3 ng/mL (50s/60s/70s) predicts PV >40 mL. p.27
- MCQ 5-ARIs lower PSA by ~50% after 6–12 months; obtain baseline PSA. p.27
- MCQ Nocturnal polyuria: >33% of 24-h output at night in men over 65. p.27
- MCQ Uroflow unreliable when voided volume <150 mL. p.27
- MCQ Uroflowmetry cannot distinguish BOO from detrusor underactivity. p.27
- MCQ PFS indicated for age >80 or <50 with voiding LUTS, PVR >300 mL, voided volume <150 mL. p.27
- MCQ AUA: PVR >300 mL → manage as chronic urinary retention. p.27
- MCQ IPP ≥10 mm is high grade and predicts TWOC failure. p.27
- MCQ Exclude median lobe by cystoscopy before prostatic urethral lift. p.27
- MCQ α1a constitutes ~70% of prostatic α1-receptors. p.27
- MCQ Most α1a-selective agent: silodosin; only α1d-selective: naftopidil. p.27
- MCQ Highest IFIS risk: tamsulosin; stopping pre-operatively does not prevent IFIS. p.27
- MCQ Doxazosin GITS starts at 4 mg without titration. p.27
- MCQ Highest anejaculation: silodosin; lowest among uroselectives: alfuzosin. p.27
- MCQ ALLHAT doxazosin arm stopped for excess heart failure. p.28
- MCQ Finasteride inhibits type 2; dutasteride types 1 and 2 (DHT ↓90%). p.28
- MCQ PLESS: finasteride halved AUR and surgery over 4 years. p.28
- MCQ MTOPS combination reduced progression 66%; doxazosin did not reduce AUR. p.28
- MCQ CombAT entry: PV ≥30 mL and PSA ≥1.5 ng/mL. p.28
- MCQ Antimuscarinics in BPH: only if PVR <200 mL. p.28
- MCQ Mirabegron contraindicated in uncontrolled BP ≥180/110 mm Hg. p.28
- MCQ Tadalafil 5 mg daily improves IPSS without increasing Qmax. p.28
- MCQ Only phytotherapy endorsed by EAU: hexane-extracted Serenoa repens 320 mg. p.28

- MCQ TWOC succeeds in ~60%; α-blocker raises success (ALFAUR 62% vs 48%). p.28
- MCQ TWOC failure predictors: older age, drained volume >1 L, spontaneous AUR. p.28
- MCQ Only 5-ARIs, not α-blockers, prevent AUR. p.28

## Ch 151 · Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

### 1 · SCOPE AND PRINCIPLES

- KEY POINT Prostate volume predicts progression (acute urinary retention (AUR), need for surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County). p.29

### 2 · WORK-UP AND INDICATIONS FOR SURGERY

- History, examination, IPSS and urinalysis at baseline (St 1); repeat IPSS after treatment (St 3). IPSS bands: **0–7 mild, 8–19 moderate, 20–35 severe**. p.29

### 3 · CHOOSING THE OPERATION: SIZE, SHAPE, BLEEDING RISK, SEXUAL FUNCTION

- KEY POINT Only enucleation spans every size band; only simple prostatectomy joins it above 80 cc. TUIP is the only option confined to small glands. MISTs and RWT live in the 30–80 cc band — so cystoscopic or imaging assessment of volume and median lobe precedes every recommendation. p.30

### 4 · MONOPOLAR TURP (M-TURP) — THE BENCHMARK

- Monopolar current passes loop → patient → grounding pad, so irrigant must be non-ionic (sterile water, **1.5% glycine**†, sorbitol) and is **hypo-osmolar** — the root of TUR syndrome. p.30
- ALGORITHM

  Inspect bladder (tumour → resect it first, defer TURP; diverticula) → identify ureteric orifices, bladder neck, trigone, verumontanum, external sphincter (indigo carmine/methylene blue if orifices hidden) → resect median lobe first → lateral lobes → anterior tissue and apex last → evacuate chips (Ellik/Toomey) → final haemostasis → large-bore three-way catheter

  p.30
- Even an uncomplicated TURP absorbs **800–1000 mL** irrigant — account for it in fluid prescription. Bladder spasm or rectal pain → check catheter for clot obstruction. p.31
- Warn: dysuria while re-epithelialisation occurs (weeks); eschar passage with delayed bleeding at **1–4 weeks**; storage symptoms may persist — if emptying confirmed, antimuscarinic or β3-agonist; detrusor overactivity may take months to settle. p.31

### 5 · TURP COMPLICATIONS AND THEIR MANAGEMENT

- **Mechanism**: prostatic venous pressure ≈ **10 mm Hg**; irrigation at higher pressure enters opened sinuses → dilutional hyponatraemia, hypervolaemia. Glycine adds ammonia toxicity (encephalopathy) and transient visual disturbance/blindness. p.31
- **Risk factors**: gland **>45 cc**, resection **>90 min**, capsular perforation, high irrigant height. Keep bag ≈ **60 cm** above patient; +10 cm more than doubles absorption. p.31
- **Diagnosis**: neurological status + serum $Na^+$ (check after long/large resections). $Na^+$ **<120 mEq/L** = severe (coma, seizures, herniation). p.31
- ALGORITHM

  Suspect TUR syndrome → stop/expedite surgery, achieve haemostasis, stop irrigation → send $Na^+$ → †loop diuretic if fluid-overloaded → severe central nervous system (CNS) symptoms: **3% saline 100 mL bolus** (raises $Na^+$ 2–3 mEq/L), repeat up to ×2 at 10-min intervals (max 300 mL); target rise 4–6 mEq/L; ≤12 mEq/L in first 24 h acceptable in acute post-TURP hyponatraemia (≤8–10 mEq/L/24 h if presentation >48 h) → high-dependency unit (HDU) monitoring

  p.31

- KEY POINT Why is acute post-TURP hyponatraemia safer to correct than chronic? The brain has not yet shed its organic osmolytes, so osmotic demyelination (central pontine myelinolysis) is less likely. Bipolar TURP with saline eliminates TUR syndrome. p.31

- TABLE **Retrograde ejaculation** · Bladder neck resected — about **two-thirds** · Counsel pre-operatively; erection, orgasm, desire unaffected p.32

### 7 · TRANSURETHRAL INCISION OF THE PROSTATE (TUIP)

- ALGORITHM
  Collins knife / thin loop / holmium–thulium laser → incision at 5 and/or 7 o'clock, starting distal to ureteric orifice → through bladder neck → end proximal to verumontanum → deepen to capsule (± fat) → cautery → catheter out, discharge day 0–1
  p.33

### 8 · BIPOLAR TRANSURETHRAL VAPORISATION (TUVP)

- KEY POINT Vaporisation trades haemostasis for a coagulated, specimen-free bed: the coagulum explains dysuria and retention; the lack of a clean anatomical plane explains weaker durability. The same trade recurs with PVP. p.33

### 9 · LASER PHYSICS

- TABLE Thulium:YAG / thulium fibre laser (TFL) · 2013 / 1940 nm · Water (TFL at absorption peak; ~4× holmium) · **0.2 mm** (shallowest) · ThuLEP, ThuVEP, ThuVARP p.33

### 11 · HOLMIUM LASER ENUCLEATION OF THE PROSTATE (HOLEP)

- ALGORITHM
  550 µm end-firing fibre, 24–28 Fr continuous-flow resectoscope, saline, ~2 J × 40–50 Hz (coag 1–2 J × 20 Hz) → 5 and 7 o'clock bladder-neck incisions to capsule (6 o'clock single incision if no median lobe) → carry to just proximal to verumontanum, join, enucleate median lobe into bladder → perimontanal inverted-U incision, develop lateral lobe–capsule plane at apex → sweep anteriorly to 11–12 and 12–1 o'clock → incise apical mucosal strip (avoids sphincter traction) → anterior plane to bladder neck ("Baleen" sign heralds bladder) → release lobes into bladder → haemostasis (defocused beam) → morcellation via nephroscope → 22 Fr three-way catheter, ~60 mL balloon
  p.34

- Retrograde ejaculation ≈ **three-quarters** (whole TZ removed); median-lobe-only HoLEP preserves it in ~90%. Erectile function unchanged or improved. p.35

### 13 · MINIMALLY INVASIVE SURGICAL THERAPIES (MISTS)

- Local anaesthesia; typically **4–6 implants** anterolaterally (between dorsal veins and neurovascular bundles), bladder neck to apex. Gland tissue is more compliant than capsule, so tension opens the lumen. p.35

- KEY POINT PUL moves tissue; WVTT ablates it. That is why PUL cannot treat a median lobe, needs no catheter and causes minimal dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritative symptoms. Both preserve the bladder neck and hence antegrade ejaculation. p.36

### 15 · TRUE MISTS AND TEMPORARY DEVICES

- **iTind (TIPD)**: three nitinol struts (12, 5, 7 o'clock) + 6 o'clock anchoring leaflet; left **5–7 days** → ischaemic pressure necrosis remodels bladder neck and prostatic urethra; removed by collapsing into an 18 Fr catheter. AUA St 41: 25–75 cc, no obstructing median lobe. Sham RCT: responder end point met, absolute IPSS difference not significant; sexual function preserved. p.36

### 16 · PROSTATE ARTERY EMBOLISATION (PAE)

- **ALGORITHM** Femoral access under local anaesthesia → pelvic angiography → catheter balloon with contrast as landmark → 1.9–2.4 Fr microcatheter into prostatic arteries (most often from internal pudendal; often duplicated) → microspheres until stasis — bilateral better than unilateral p.36

### 17 · OVERALL COMPARISON

- **KEY POINT** The "gold standard" for a given man is the operation that meets his goals safely — chosen by shared decision-making around his most bothersome symptoms, prostate size/shape, bleeding risk, wish to preserve ejaculation, and the surgeon's own expertise. p.37

### 19 · MCQ ONE-LINERS

- **MCQ** Only procedure associated with TUR syndrome: monopolar TURP. p.38
- **MCQ** TUR syndrome risk factors: gland >45 cc, resection >90 min. p.38
- **MCQ** Prostatic venous pressure ≈ 10 mm Hg; irrigant height ≈ 60 cm. p.38
- **MCQ** 3% saline 100 mL raises sodium by 2–3 mEq/L; maximum 300 mL. p.38
- **MCQ** Retrograde ejaculation after TURP ≈ two-thirds; after HoLEP ≈ three-quarters. p.38
- **MCQ** Distal landmark protecting the external sphincter at TURP: verumontanum. p.38
- **MCQ** Recurrent BNC: transurethral incision plus mitomycin C; intractable: Y-V plasty. p.38
- **MCQ** TUIP: only glands <30 cc without median lobe. p.38
- **MCQ** Only endoscopic procedure with efficacy superior to TURP: HoLEP. p.38
- **MCQ** HoLEP and ThuLEP are size-independent (AUA St 38). p.38
- **MCQ** Higher bleeding risk: HoLEP, ThuLEP or PVP (St 43). p.38
- **MCQ** PVP wavelength 532 nm, chromophore haemoglobin. p.38
- **MCQ** Nd:YAG abandoned: ~1 cm penetration causing sloughing and retention. p.38
- **MCQ** Highest dysuria among transurethral techniques: PVP. p.38
- **MCQ** PUL contraindication: obstructing median lobe. p.38
- **MCQ** Avoid PUL encrustation: implants >1.5 cm distal to bladder neck. p.38
- **MCQ** WVTT energy: convection, 103 °C steam for 9 s, ~4 kJ total. p.38
- **MCQ** WVTT median lobe needle: 45° off midline, never straight posterior. p.38
- **MCQ** Aquablation needs general/spinal anaesthesia, so is not a MIST. p.38
- **MCQ** iTind left in situ 5–7 days; AUA range 25–75 cc. p.38
- **MCQ** BNC after HoLEP is commoner in small glands (<40 cc). p.38

## Ch 152 · Simple Prostatectomy: Open, Laparoscopic and Robotic

### 1 · DEFINITION AND SURGICAL ANATOMY

- **Simple prostatectomy** = enucleation of the hyperplastic adenoma only, for benign prostatic hyperplasia (BPH); radical prostatectomy removes the whole gland for cancer. Gold standard for bladder outlet obstruction with glands **>80 g**. p.39
- Eponyms MCQ: **Freyer** popularised suprapubic transvesical; **Millin (1945)** retropubic; first laparoscopic 2002, first robotic 2008. p.39

### 2 · INDICATIONS

- BPH, gland **>80–100 g**, bothersome lower urinary tract symptoms (LUTS) refractory to medical or prior endoscopic therapy. p.39

### 3 · PRE-OPERATIVE EVALUATION

- KEY POINT Work-up answers three questions: is the gland big enough (TRUS)? is there cancer that makes a simple operation wrong (DRE, PSA, MRI)? will the bladder empty once unobstructed (urodynamics)? p.39

### 4 · OPEN TECHNIQUES

- KEY POINT All open variants = four moves: enter → pseudocapsular plane → divide urethra at apex **proximal to the verumontanum**, no traction on sphincter → 5/7 o'clock haemostasis + bladder neck mucosal advancement. p.40

### 5 · HAEMOSTASIS

- ALGORITHM
  Postoperative bleeding → catheter traction (50 mL balloon) ± continuous bladder irrigation (CBI) → exclude coagulopathy → theatre: endoscopic clot evacuation + fulguration → reopen fossa if severe
  p.40

### 7 · LAPAROSCOPIC AND ROBOTIC SIMPLE PROSTATECTOMY

- ALGORITHM
  Vertical cystotomy, stay sutures → identify orifices → bladder neck incision 3→9 o'clock posteriorly under median lobe → develop **posterior plane** (most distinct) to apex → circumferential lobes → divide anterior commissure, see verumontanum → transect urethra **just proximal to veru** → 5/7 o'clock barbed sutures → **vesicourethral mucosal advancement** (double-armed suture from 6 o'clock) → two-layer bladder closure
  p.41

### 8 · OUTCOMES AND COMPARISONS

- KEY POINT Minimally invasive routes improve perioperative morbidity only; function is identical because the enucleation is identical. Open transfusion now **<5%**. p.41

### 9 · COMPLICATIONS AND MANAGEMENT

- TABLE Bladder neck contracture · Open **2–6%**; minimally invasive <1% · Dilation or Collings knife incision to 22 Fr p.41
- TABLE Erectile dysfunction · **3–5%** · — p.41
- TABLE **Retrograde ejaculation** · **80–90%** · Counsel pre-op p.41
- KEY POINT Why? The adenoma lies inside the pseudocapsule and proximal to the verumontanum: removing it destroys the bladder neck (retrograde ejaculation) but spares the external sphincter and extracapsular neurovascular bundles (incontinence, erectile dysfunction rare). p.42

### 10 · MCQ ONE-LINERS

- MCQ Simple prostatectomy threshold: gland >80 g. p.42
- MCQ Enucleation plane: surgical (pseudo)capsule = compressed peripheral zone. p.42
- MCQ Adenoma arteries enter posterolaterally at bladder neck: 5 and 7 o'clock. p.42
- MCQ Retropubic approach without opening bladder: Millin, 1945. p.42
- MCQ Suprapubic transvesical prostatectomy popularised by Freyer. p.42
- MCQ Bladder stones, diverticulum or large median lobe → suprapubic. p.42
- MCQ Millin capsulotomy: transverse, ≈2 cm distal to bladder neck. p.42
- MCQ Suprapubic cystotomy: 1–2 cm proximal to bladder neck. p.42
- MCQ Transverse incision avoids distal extension into sphincter. p.42
- MCQ Main Millin disadvantage: DVC bleeding. p.42
- MCQ Urethra divided proximal to verumontanum. p.42

- MCQ Catheter balloon 50 mL prevents migration into fossa. p.42
- MCQ Malament: purse-string partitioning fossa from bladder. p.42
- MCQ O'Conor: posterior capsular plication. p.42
- MCQ Suspected leak: check drain creatinine. p.42
- MCQ First laparoscopic simple prostatectomy: Mariano, 2002. p.42
- MCQ First robotic simple prostatectomy: Sotelo, 2008. p.42
- MCQ Robotic step removing need for CBI: vesicourethral mucosal advancement. p.42
- MCQ STEP uses pneumovesicum at 8 mm Hg. p.42
- MCQ Functional outcomes equal across open and robotic. p.42
- MCQ Retrograde ejaculation after enucleation: 80–90%. p.42
- MCQ Bladder neck contracture: incise with Collings knife to 22 Fr. p.42
- MCQ No TUR syndrome with simple prostatectomy. p.42

## ★ Putting it together: the BPH pathway

### A · MAN WITH LOWER URINARY TRACT SYMPTOMS (LUTS)

- ALGORITHM

  **1. Assess:** history and International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), digital rectal examination, urinalysis, prostate-specific antigen (PSA) if it changes management, frequency–volume chart if nocturia. Uroflow and post-void residual (PVR) before invasive therapy (→ Ch 150).

  **2. Look for complications** that make surgery the answer: refractory retention, recurrent urinary tract infection, bladder stones, recurrent haematuria from the prostate, renal impairment from obstruction.

  **3. Mild or not bothered** → watchful waiting with lifestyle advice.

  **4. Bothersome** → medical therapy (table B); if it fails or the patient prefers → procedure (table C).

  p.43

### D · TURP COMPLICATIONS AT A GLANCE

- KEY POINT TUR syndrome (monopolar, hypotonic glycine absorption): confusion, visual change, hyponatraemia → stop, check sodium, diuretic for overload, 3% saline for severe neurological symptoms, correct slowly. Bipolar saline resection prevents it. Late problems: retrograde ejaculation (about two-thirds), urethral stricture (meatus, bulb), bladder neck contracture (small glands, over-resection; ~6 months), incontinence (pelvic floor training, then sling or artificial sphincter) (→ Ch 151). p.44

# 17 Prostate Cancer

570 core items · page numbers refer to the full book PDF

## Ch 153 · Epidemiology, Aetiology and Prevention of Prostate Cancer

### 1 · EPIDEMIOLOGY — THE HEADLINE

- Worldwide: **second commonest** non-skin cancer and **fifth** cause of cancer death in men; 2020 ≈ **1.4 million cases, 375,000 deaths**. First for incidence in 112/185 countries. p.2
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest); United States (US) lifetime risk ≈ **1 in 8**; 5-year survival 97%, yet second cause of cancer death in US men. p.2
- **Incidence and mortality are decoupled.** Incidence tracks screening intensity and wealth (Northern/Western Europe, North America, Australia); mortality peaks where African ancestry meets poor access — **Caribbean has the highest mortality in the world** MCQ. Asia is lowest on both, but rising (Western diet, obesity, screening). p.2
- **Latent (autopsy) cancer** rises steeply with age: ~5% <30 y → 29% at 60–69 → **59% >79 y**; higher in Black men at every age. p.2
- KEY POINT Why is aetiology so hard to study? Natural history spans decades, and in a prostate-specific antigen (PSA)-screened population "a diagnosis of prostate cancer signifies, in most cases, that a man happened to have a PSA test at that time." Risk-factor studies may therefore identify determinants of *being tested*. The preferred end point is **aggressive disease** (advanced stage, metastasis, death). **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits spuriously appear to raise incidence while lowering mortality. p.2

### 2 · ESTABLISHED RISK FACTORS

- Black men: highest incidence (172.6/100,000 in US) and **twice** the cancer-specific mortality of White men, with younger age and more metastatic disease at presentation. p.2
- TABLE Both father and brother · **9×** p.2

### 5 · POSSIBLE RISK FACTORS — BODY, METABOLISM, LIFESTYLE

- **Hormones**: adult serum testosterone does *not* predict risk MCQ; genetically higher free testosterone, IGF-1, insulin and early puberty modestly do. p.4
- **Composite healthy lifestyle** (no smoking, activity, healthy weight, diet) → **30–50%** lower lethal cancer; even in the highest polygenic-risk quartile lifetime lethal risk falls from 5.3% to 1.6% — genes are not destiny. p.4
- Transgender women keep the prostate; on oestrogen median PSA ≈ **0.02 ng/mL** (50-fold lower) → normal-looking PSA is falsely reassuring; cut-points must be lowered. p.4

### 6 · SUPPLEMENTS

- KEY POINT Why did vitamin E work in the Alpha-Tocopherol, Beta-Carotene (ATBC) trial but fail in SELECT? ATBC used a tenfold lower dose in **heavy smokers** with minimal PSA screening (aggressive, clinically detected cancers); SELECT used 400 international units (IU) in non-smokers with widespread screening (indolent cancers). Different disease, different answer. p.4

### 7 · CHEMOPREVENTION (PRIMARY PREVENTION)

- KEY POINT Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does — lifestyle advice benefits both. p.5

### 8 · PSA SCREENING

- Screening reduces prostate cancer death by ~**20–30%**; the incidence excess in the screened arm shrinks with time while the mortality benefit builds then plateaus. p.5

9 · LANDMARK TRIALS

- TABLE **PCPT** (Prostate Cancer Prevention Trial) · 18,882 average-risk men, PSA ≤3; finasteride 5 mg vs placebo 7 y; end-of-study biopsy · Cancer ↓ **24.8%** (18.4 vs 24.4%); Gleason 7–10 ↑ 6.4 vs 5.1%; 18-y cancer death ↓ 25% (non-significant) · Fewer cancers, more high-grade; not approved p.6
- TABLE **REDUCE** (REduction by DUtasteride of prostate Cancer Events) · High risk: PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg vs placebo 4 y · Cancer ↓ **22.8%**; Gleason 8–10 ↑ (0.9 vs 0.6%; years 3–4: 12 vs 1) · Same answer as PCPT; FDA warning p.6

10 · MCQ ONE-LINERS

- MCQ Highest prostate cancer mortality region: Caribbean. p.6
- MCQ Latent cancer at autopsy >79 years ≈ 59%. p.6
- MCQ Three established risk factors: age, race, family history. p.6
- MCQ Father and brother both affected: ~9-fold risk. p.6
- MCQ Twin-study heritability: up to 58%. p.6
- MCQ Black men: twice the cancer-specific mortality of White men. p.6
- MCQ Ancestry-related risk locus: 8q24. p.6
- MCQ HOXB13 mutation: hereditary prostate cancer, prostate-specific homeobox gene. p.6
- MCQ BRCA2 carriers: aggressive disease; caution with active surveillance. p.6
- MCQ MMR-deficient (Lynch) disease: responds to immune checkpoint inhibitors. p.6
- MCQ Polygenic risk scores predict any cancer, not aggressive cancer. p.6
- MCQ Smoking: no increase in incidence but ~40% higher mortality. p.6
- MCQ Vasectomy is not a prostate cancer risk factor. p.6
- MCQ Transgender women on oestrogen: median PSA 0.02 ng/mL. p.6
- MCQ Adult serum testosterone does not predict prostate cancer risk. p.6
- MCQ High-dose zinc supplements: increased lethal cancer. p.6
- MCQ SELECT: selenium/vitamin E no benefit; vitamin E possibly harmful. p.6
- MCQ PCPT: finasteride reduced cancer by 24.8%. p.6
- MCQ PCPT and REDUCE: more high-grade cancer on 5ARI. p.6
- MCQ No drug is FDA-approved for prostate cancer prevention. p.6
- MCQ 5ARIs lower PSA ~50% — double measured value. p.6
- MCQ ERSPC: ~20% reduction in prostate cancer death. p.6
- MCQ PLCO negative because the control arm was contaminated by screening. p.6
- MCQ USPSTF: Grade D (2012) → Grade C for age 55–69 (2018). p.6
- MCQ Composite healthy lifestyle: 30–50% fewer lethal cancers. p.6

## Ch 154 · Prostate Cancer Biomarkers

2 · PSA BIOLOGY

- PSA = human kallikrein 3 (**hK3**), gene KLK3 on **chromosome 19q13**; androgen-regulated serine protease from luminal epithelium that **liquefies semen** (cleaves semenogelin). Half-life **2–3 days**. p.7
- Biopsy: rise lasts ≥**4 weeks**; cycling >55 km ~10% rise; ejaculation transient in men ≥50 (back to baseline in 24 h — ask, and repeat after abstinence). p.7

- **5α-reductase inhibitors (5ARIs)**: PSA falls ~**50%** by 12 months → **double** it (overestimates in first 6 months, underestimates after years). 5ARI improves PSA's screening performance (Prostate Cancer Prevention Trial, PCPT). p.7

### 3 · PSA DERIVATIVES

- TABLE **PSA density** (PSAD) · PSA ÷ prostate volume; cut-off **0.15** (range 0.08–0.15) · Better than PSA; used after negative/equivocal magnetic resonance imaging (MRI) (data for 0.15 there are lacking); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification p.7

- KEY POINT Limit of %fPSA: validated against *any* cancer, not GG ≥2, so it overestimates clinically significant risk. Risk calculators (PCPT, Prostate Biopsy Collaborative Group) incorporate it for shared decisions. p.8

### 4 · BLOOD AND URINE TESTS (NCCN 2023 PRE-BIOPSY BIOMARKERS)

- TABLE **4Kscore** · Blood: total PSA, fPSA, iPSA, **hK2** + age, DRE, prior biopsy · **7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis/death risk p.8

- **TMPRSS2:ERG**: fusion of androgen-regulated TMPRSS2 with the E26 transformation-specific (ETS) transcription factor ERG; **~100% specific** but in only ~half of tumours → combined with PCA3 (doubles specificity at 95% sensitivity). p.8

- KEY POINT Why must urine tests follow a DRE? Urine carries cancer RNA only from shed prostate cells, and an attentive DRE massages cells into the urethra. ExoDx avoids this because exosomes (small vesicles carrying RNA) are shed into urine spontaneously. Urinary markers are highly cancer-specific; their weakness is coverage, so they are combined into panels. p.8

### 7 · CLINICAL ALGORITHM

- ALGORITHM
  Raised PSA → repeat (exclude infection, instrumentation, ejaculation, 5ARI) → MRI and/or biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density) → biopsy if risk high
  Negative biopsy, persistent suspicion → MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx → targeted re-biopsy
  Newly diagnosed favourable-risk → PSA density, Prolaris/Oncotype/Decipher → AS vs treatment
  After RP → † PSA should be undetectable; ≥0.2 ng/mL confirmed = biochemical recurrence → Decipher/PORTOS for adjuvant/salvage radiotherapy
  p.10

### 8 · MCQ ONE-LINERS

- MCQ PSA is human kallikrein 3; gene on chromosome 19q13. p.10
- MCQ PSA liquefies semen by cleaving semenogelin. p.10
- MCQ Serum PSA half-life: 2–3 days. p.10
- MCQ hK2 activates proPSA by cleaving its 7-aa leader. p.10
- MCQ Major PSA complex: α1-antichymotrypsin. p.10
- MCQ PSA–α2-macroglobulin is not detected by standard assays. p.10
- MCQ Cancer: lower % free PSA. p.10
- MCQ Per cell, cancer produces less PSA than benign tissue. p.10
- MCQ 5ARI halves PSA; % free PSA unchanged. p.10
- MCQ PSA rise after biopsy persists ≥4 weeks. p.10
- MCQ PSA density cut-off: 0.15. p.10
- MCQ PSA velocity should never alone trigger biopsy. p.10
- MCQ [–2]proPSA: peripheral zone; BPSA: transition zone. p.10
- MCQ PHI = (p2PSA ÷ fPSA) × √tPSA. p.10

- MCQ 4Kscore unique component: hK2; cut-off 7.5%. p.10
- MCQ IsoPSA unaffected by 5ARI. p.10
- MCQ PCA3: long non-coding RNA, independent of prostate volume. p.10
- MCQ PCA3 requires post-DRE urine; FDA-approved after negative biopsy. p.10
- MCQ ExoDx: urine exosomes, no DRE needed. p.10
- MCQ TMPRSS2:ERG: ~100% specific, present in half of tumours. p.10
- MCQ SelectMDx genes: DLX1 and HOXC6; needs prostate volume. p.10
- MCQ ConfirmMDx: methylation of GSTP1, APC, RASSF1 — field effect. p.10
- MCQ Prolaris: cell-cycle progression genes. p.10
- MCQ Oncotype DX predicts adverse pathology at prostatectomy. p.10
- MCQ Decipher ≥0.4 post-RP may favour radiotherapy. p.10
- MCQ BRCA2: highest-risk, most aggressive germline mutation. p.10
- MCQ HOXB13 G84E raises risk but not aggressiveness. p.10

## Ch 155 · Prostate Biopsy: Techniques and Imaging

### 1 · INDICATIONS AND CONTRAINDICATIONS

- **Repeat biopsy**: rising/persistent PSA; atypical small acinar proliferation (ASAP; **≈50%** cancer risk); atypical intraductal proliferation; positive urinary prostate cancer gene 3 (PCA3)/ConfirmMDx with persistent suspicion; suspicious magnetic resonance imaging (MRI) lesion; recurrence after ablation. **Follow-up**: active surveillance (AS). p.11
- PSA density (PSAD) = PSA ÷ volume; **>0.15 ng/mL/cc** suggests cancer. Volume (ellipsoid) = π/6 × transverse × anteroposterior × length; 1 $cm^3$ ≈ 1 g. p.11

### 2 · PRE-BIOPSY MULTIPARAMETRIC MRI (MPMRI)

- KEY POINT Trap: DCE never creates or downgrades a PI-RADS 4–5, and never upgrades a TZ lesion. mpMRI still misses a minority of significant cancers (mostly 3+4; a third primary pattern 4): MRI-invisible lesions, reader discordance, registration or needle error, small lesion in a large gland. Hence systematic cores remain relevant. p.11

### 6 · ANTIBIOTIC PROPHYLAXIS

- Transrectal: every patient, regardless of risk. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin ± aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before; duration **≤24 h**. Oral route only for quinolones. p.13
- Infection risk factors: antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI/prostatitis/biopsy infection, many prior biopsies (AS), travel (India, South-East Asia), health-care worker, rectal fluoroquinolone-resistant *E. coli*. Response: augmented (>1 agent) or rectal-swab-targeted prophylaxis, or go transperineal. p.13

### 8 · COMPLICATIONS AND MANAGEMENT

- TABLE Sepsis · **0.3–3.1%** · Carbapenem until cultures; up to 2 weeks if admitted p.13
- ALGORITHM

  **Post-biopsy fever** → assess for sepsis (fever/hypothermia, tachycardia, tachypnoea, confusion, hypotension) → cultures → **no empirical fluoroquinolone** (resistant *E. coli*; also resistant to co-trimoxazole, ampicillin) → carbapenem (or amikacin/3rd-generation cephalosporin per sensitivity) → resuscitate.

  p.14

- ALGORITHM
  **Persistent rectal bleeding** → pressure (probe/finger) → tamponade (inflated sheath or Foley balloon) → endoscopy with adrenaline/sclerosant → angioembolisation → transrectal suture. p.14

- KEY POINT "Commonest complication": bleeding overall; infection among serious complications/admissions. Admissions for infection rose from 1.0% to 4.1% in the resistance era. The asymmetry (bleeding frequent but trivial, infection rare but dangerous) drives universal prophylaxis and the move to the perineum. p.14

### 9 · REPEAT AND SATURATION BIOPSY

- Yield falls each round: European Randomized Study of Screening for Prostate Cancer (ERSPC) **22% → 10% → 5% → 4%** on biopsies 2–4. p.14

### 11 · MCQ ONE-LINERS

- MCQ Percentage of cancers isoechoic on TRUS: 39%. p.14
- MCQ Commonest false-positive hypoechoic lesion: benign prostatic hyperplasia foci and prostatitis. p.14
- MCQ PSAD suggesting malignancy: >0.15 ng/mL/cc. p.14
- MCQ Sextant biopsy described by Hodge, 1989. p.14
- MCQ Current standard systematic biopsy: 12 cores (AUA). p.14
- MCQ Needle gauge for core biopsy: 18 G; throw 15–25 mm. p.14
- MCQ Maximum cores per formalin jar: 2. p.15
- MCQ Transrectal biopsy wound class: III (contaminated). p.15
- MCQ AUA prophylaxis duration: ≤24 h. p.15
- MCQ Most consistent infection risk factor: antibiotic exposure within 6 months. p.15
- MCQ Dominant organism in post-biopsy sepsis: fluoroquinolone-resistant *E. coli*. p.15
- MCQ Empirical drug for post-biopsy sepsis: carbapenem. p.15
- MCQ Aminoglycoside substitute in renal insufficiency: aztreonam. p.15
- MCQ Aspirin before biopsy: continue low dose. p.15
- MCQ INR threshold for biopsy: <1.5. p.15
- MCQ Commonest complication overall: bleeding (haematuria). p.15
- MCQ Haematospermia lasts ≈1 month. p.15
- MCQ Dominant PI-RADS sequence in PZ: DWI; in TZ: T2WI. p.15
- MCQ DCE role: upgrades PZ score 3 to 4 only. p.15
- MCQ Lesion size giving automatic PI-RADS 5: >1.5 cm. p.15
- MCQ Significant cancer definition in PROMIS: Grade Group ≥3 or core length ≥6 mm. p.15
- MCQ PROMIS mpMRI sensitivity 93% vs TRUS 48%. p.15
- MCQ PRECISION: targeted 38% vs systematic 26% significant cancer. p.15
- MCQ Cores per MRI target: 2–4. p.15
- MCQ Cognitive vs fusion vs in-bore: no significant detection difference. p.15
- MCQ EAU 2023 preferred route: transperineal. p.15
- MCQ Commonest transperineal block: periapical triangle. p.15
- MCQ Repeat-biopsy yield ERSPC biopsy 2: 22%. p.15

- MCQ ASAP on biopsy: ≈50% cancer on repeat. p.15
- MCQ Absent seminal vesicle: ipsilateral renal agenesis in 79%. p.15
- MCQ Müllerian cysts midline; wolffian cysts lateral. p.15

## Ch 156 · Pathology of Prostatic Neoplasia

### 1 · PRECURSOR AND INTRADUCTAL LESIONS

- **HGPIN is a precursor**: carcinoma seen budding off it; ~**20%** of HGPIN carries the transmembrane protease serine 2 : ETS-related gene (**TMPRSS2:ERG**) fusion; further hits needed for invasion. p.16
- KEY POINT Intraductal lesions are graded by what they signify: HGPIN is a weak risk marker, IDC-P usually marks high-grade cancer already present. Traps: re-biopsy for HGPIN; surveillance for grade group 1 with IDC-P. p.16

### 2 · ADENOCARCINOMA: LOCATION, SPREAD, STAGING

- TABLE Peripheral zone (PZ) · †~70% · †~70% · Main mass in clinical T2 and **85%** of T1c; †posterolateral → palpable on DRE, well sampled by systematic biopsy p.16
- Volume vs stage: EPE uncommon <**0.5 cm³**; nodes/SVI uncommon <**4 cm³**. p.17

### 3 · GRADING: GLEASON AND GRADE GROUPS

- TABLE 1 · 3+3=6 · 0% · **96%** p.18
- TABLE 2 · 3+4=7 · 0.6% · **88%** p.18
- TABLE 3 · 4+3=7 · 4.3% · **63%** p.18
- TABLE 4 · 4+4, 3+5, 5+3 = 8 · — · **48%** p.18
- TABLE 5 · 4+5, 5+4, 5+5 · — · **26%** p.18
- KEY POINT The steep drop is between GG2 and GG3, so 3+4 and 4+3 are never lumped as "7". Trap: 3+5 and 5+3 are GG4; GG5 starts at 9. Because pure GG1 cannot metastasise, whether to call it "cancer" is debated. p.18
- KEY POINT †Why it matters: cribriform pattern and IDC-P mark aggressive disease and argue against active surveillance even in GG2, so they must appear in the report. Aggregate grading of a targeted lesion avoids both under-grading (one core alone) and over-grading (highest core alone). p.18
- TABLE Biopsy GG1 → RP · ~**⅓** upgraded p.18

### 4 · NEEDLE BIOPSY ASSESSMENT

- **"Atypical, suspicious for carcinoma" (ASAP)**: ~5% of biopsies; cancer on follow-up ~**40%**, PSA does not stratify; ~80% of these GG1, ~5% GG4–5. **Seek expert review before re-biopsy** (most often changed diagnosis); then markers, imaging or re-biopsy sampling the atypical site. p.19
- KEY POINT Sampling error is one-directional: a bad biopsy is believed, a good biopsy is provisional, an atypical biopsy is re-examined before it is repeated. p.19

### 5 · TUR AND RADICAL PROSTATECTOMY SPECIMENS

- **EPE**: focal (few glands) vs nonfocal — extent predicts progression. **SVI far more ominous**: **65%** 5-yr progression. p.19
- **Margins**: ~15% positive in robotic series; only ~50% progress. Distance to ink irrelevant. Report **extent (focal vs nonfocal, cut-off 3 mm) and grade at margin** — guide adjuvant radiotherapy. p.19

### 8 · MCQ ONE-LINERS

- MCQ LGPIN — not reported. p.20

- MCQ Isolated HGPIN, even multifocal — no repeat biopsy. p.20
- MCQ IDC-P — basal cells retained; usually spread of high-grade invasive cancer. p.20
- MCQ GG1 + IDC-P on biopsy — upgraded at RP in >80%; not for AS. p.20
- MCQ Commonest tumour site — posterior peripheral zone; multifocal >85%. p.20
- MCQ Microscopic bladder neck invasion — pT3a. p.20
- MCQ 2005 biopsy Gleason — commonest + highest pattern. p.20
- MCQ Cribriform and glomeruloid glands — pattern 4. p.20
- MCQ 3+5=8 — GG4. p.20
- MCQ 5-yr BCR-free GG1→5 — 96, 88, 63, 48, 26%. p.20
- MCQ Biopsy GG1 upgraded at RP — about one-third. p.20
- MCQ Tertiary pattern 5 <5% — stays in its GG. p.20
- MCQ Basal cell markers — p63, HMWCK; absence supports cancer. p.20
- MCQ ASAP — ~40% cancer on follow-up; PSA does not stratify. p.20
- MCQ Cowper gland mimics cancer at apex; SV/central zone mimic HGPIN at base. p.20
- MCQ Perineural invasion — report on biopsy, omit at RP. p.20
- MCQ Focal vs nonfocal positive margin — 3 mm. p.20
- MCQ SVI — 65% 5-yr progression. p.20
- MCQ Hormone-treated cancer — not graded; post-ablation residual cancer — graded. p.20
- MCQ Ductal adenocarcinoma — graded pattern 4; mimics papillary urothelial cancer at verumontanum. p.20
- MCQ Urothelial vs prostatic — GATA3 vs PSA/NKX3.1/prostein. p.20
- MCQ †Largest zone and commonest site of cancer — peripheral zone (~70%). p.20
- MCQ †BPH arises from — transition zone; central zone cancers are rare but aggressive. p.20
- MCQ †IDC-P in Gleason grade — included (ISUP 2019); excluded, reported separately (GUPS 2019). p.20
- MCQ †Several MRI-targeted cores from one lesion — aggregate grade (ISUP 2019). p.20

## Ch 157 · Diagnosis and Staging of Prostate Cancer

### 2 · SCREENING IN BRIEF

- TABLE Cancer-specific mortality · **20% relative reduction** (16 years) · No difference (17 years) p.21

### 3 · DIGITAL RECTAL EXAMINATION AND PSA

- Lower PSA: hypogonadism, obesity, statins, 5α-reductase inhibitors (5-ARIs), BPH surgery. On 5-ARI (finasteride 1 or 5 mg, dutasteride): **double PSA** after ≥12 months, or treat the nadir as new baseline and investigate any rise. p.21

### 5 · TNM (AJCC 8TH EDITION, 2017)

- KEY POINT T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b, not T4; non-regional nodes are M1a, not N1; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is assigned by DRE, not by MRI. p.22

### 6 · PREDICTING TUMOUR EXTENT

- **PSA** correlates with stage but cannot stage an individual (overlap, BPH contribution, high-grade tumours make less PSA per gram). Organ-confined: ≈80% at PSA <4, 66% at 4–10, <50% at >10. Nodes: ≈20% at PSA >20, **75% at >50**. p.22

### 8 · STAGING IMAGING

- KEY POINT Why not image low-risk men? With a low pre-test probability, most positive scans are false positives, which cause further tests, anxiety and wrong denial of curative treatment. p.23

### 11 · MCQ ONE-LINERS

- MCQ Early prostate cancer: usually asymptomatic. p.24
- MCQ ERSPC mortality reduction: 20% relative. p.24
- MCQ PLCO null result explained by control-arm contamination. p.24
- MCQ Current USPSTF grade (55–69): C. p.24
- MCQ PSA is organ-specific, not cancer-specific. p.24
- MCQ PSA rises ≈4% per mL of prostate volume. p.24
- MCQ Correction for 5-ARI after 12 months: double PSA. p.24
- MCQ Finasteride 1 mg lowers PSA as much as 5 mg. p.24
- MCQ Main source of serum PSA without cancer: transition zone. p.24
- MCQ Free PSA cut-off suggesting cancer: ≤25%. p.24
- MCQ PSA velocity threshold (PSA 4–10): >0.75 ng/mL/year. p.24
- MCQ Incidental cancer in >5% of resected tissue: T1b. p.24
- MCQ Impalpable cancer found on needle biopsy: T1c. p.24
- MCQ Tumour >half of one lobe, not both: T2b. p.24
- MCQ Seminal vesicle invasion: T3b. p.24
- MCQ Rectal or bladder invasion: T4. p.24
- MCQ Non-regional node metastasis: M1a. p.24
- MCQ Bone metastasis: M1b. p.24
- MCQ Gleason 4+3: Grade Group 3. p.24
- MCQ Most important biopsy prognostic finding: grade. p.24
- MCQ D'Amico high risk: PSA >20, Gleason ≥8 or T2c. p.24
- MCQ D'Amico low risk needs all three criteria. p.24
- MCQ Nodal involvement at PSA >50: ≈75%. p.24
- MCQ Organ-confined at PSA <4: ≈80%. p.24
- MCQ No routine bone scan in asymptomatic low/intermediate risk. p.24
- MCQ FDA-approved PSMA agents: $^{68}$Ga-PSMA-11, piflufolastat F-18. p.24
- MCQ Decipher: 22-RNA tissue classifier. p.24
- MCQ Oncotype DX Prostate: 17 genes. p.24
- MCQ Prolaris measures cell-cycle progression genes. p.24
- MCQ PLND value: staging, not proven therapy. p.24
- MCQ Pathological stage beats clinical stage for prognosis. p.24

## Ch 158 · Active Management Strategies for Localised Prostate Cancer

### 1 · THE MODERN QUESTION: TIMING AS WELL AS MODALITY

- KEY POINT Cancer mortality risk is always weighed against competing mortality. A competing risk argues against treatment only when it will act *before* the cancer does. p.25

### 2 · HOST EVALUATION

- Prostate Cancer Outcomes Study (PCOS): 10-year prostate cancer mortality **3–18%** by risk group vs **≥33%** other-cause mortality with any comorbidity. p.25
- **Charlson Comorbidity Index (CCI)**: most used; designed for 1-year inpatient mortality. **Prostate cancer-specific CCI (PCCI)** reweights for 10-year mortality. **Care Assessment Needs (CAN) score** (automated, Veterans Health Administration): best predictor of 5-year survival MCQ. p.25
- TABLE Prior transurethral resection (TUR) defect · Surgery · Brachytherapy dosimetry difficult; ↑ incontinence. Do transurethral resection of prostate (TURP) *before* radiation, delay EBRT **6–8 weeks** p.26

### 3 · RISK STRATIFICATION

- TABLE Very low · **All**: T1c; Gleason ≤6/grade group (GG) 1; PSA <10 ng/mL; <3 positive cores, ≤50% each; PSA density <**0.15** · Not indicated p.26
- KEY POINT Favourable groups need ALL criteria; adverse groups ANY one. Traps: PSA density appears only in very low risk; GG3 alone makes intermediate risk unfavourable. American Urological Association (AUA)/American Society for Radiation Oncology (ASTRO)/Society of Urologic Oncology (SUO) use very low/low/intermediate/high without substratification. p.26
- **proPSMA** (randomised, 302 high-risk): gallium-68 PSMA PET-CT vs conventional imaging — accuracy **92% vs 65%**, sensitivity 85% vs 38%. PSMA PET's advantage is specificity/positive predictive value; a negative scan cannot exclude small nodal disease. p.26

### 4 · OBSERVATION VS ACTIVE SURVEILLANCE

- Limitations: **25–45%** occult high grade on systematic biopsy; declining adherence; no agreed triggers. Reclassification predictors: PSA density >0.15, core cancer density. p.27

### 6 · LANDMARK TRIALS

- TABLE **SPCG-4** (Scandinavian Prostate Cancer Group-4) · RP vs WW; 695 men; pre-PSA era, ~90% palpable · 23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute difference 11%; NNT **8**. Greatest benefit age <65 (RR 0.45) and intermediate risk · RP saves lives in palpable, intermediate-risk disease in younger men; not representative of screen-detected cancer p.27
- TABLE **ProtecT** (Prostate Testing for Cancer and Treatment) · 1643 screen-detected men aged 50–69; active monitoring vs RP vs radiotherapy + 3–6 months ADT; 77% low risk · 15 yr: cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); metastasis higher with monitoring (half of all progression); 60% of monitored men treated by 15 yr; RP = radiotherapy (RT) for mortality and metastasis · Initial AS is safe in favourable risk (incl. much favourable intermediate); surgery and radiation are oncologically equivalent; age <65 → lower cancer mortality with surgery than RT p.27
- KEY POINT Why do randomised trials show so little benefit? Competing mortality and low-risk enrolment dilute the effect, so benefit appears only in younger, intermediate-risk men with long follow-up. The consistent observational advantage of RP over radiation is mostly selection bias. ProtecT expected 15% mortality and observed ~1% at 10 years — hence low power. p.27

### 7 · FUNCTIONAL OUTCOMES COMPARED

- KEY POINT Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow cumulative decline. Curves converge by 15 years — the choice is *when and where* the patient accepts harm. p.28

## 9 · STEPWISE DECISION ALGORITHM

- ALGORITHM

  **Decision 1 — treat or not?** Disease (grade, stage, genomics, family history) + host (age, comorbidity, longevity, preference) → no treatment = WW (short longevity) or AS (very low/low, selected favourable intermediate).
  **Decision 2 — partial or whole gland?** Focality, size, grade, potency, LUTS → partial = focal ablation (limited-focality intermediate risk, ideally in a trial) → Ch 163.
  **Decision 3 — surgery vs radiation vs whole-gland ablation** → expected outcomes, toxicities, baseline function, risk of secondary treatment.
  **Decision 4 — delivery**: open vs robotic RP, lymph node dissection, surgeon; EBRT vs brachytherapy, fractionation, ADT → Ch 162.

  p.29

- **Whole-gland ablation**: little long-term data, frequent retreatment, salvage surgery hampered by fibrosis. Cryoablation → ED almost always; high-intensity focused ultrasound (HIFU) → better function but retreatment **29–38%**. p.29

## 10 · MCQ ONE-LINERS

- MCQ SPCG-4: RP vs watchful waiting, pre-PSA era, ~90% palpable disease. p.29
- MCQ SPCG-4 at 23 years: cancer-death RR 0.56, NNT 8. p.29
- MCQ SPCG-4 benefit greatest in men <65 years and intermediate risk. p.29
- MCQ PIVOT: RP no mortality benefit in low risk; underpowered (731 of 2000 planned). p.29
- MCQ PIVOT 19.5 years: significant all-cause benefit only in intermediate risk. p.29
- MCQ ProtecT 15-year cancer mortality: ~2–3% in all three arms (P = 0.53). p.29
- MCQ ProtecT: monitoring → more metastasis but no mortality difference. p.29
- MCQ ProtecT radiotherapy arm received 3–6 months neoadjuvant ADT. p.29
- MCQ ProtecT: 60% of active monitoring arm treated by 15 years. p.29
- MCQ Early divergence of survival curves in observational RP vs RT studies indicates selection bias. p.30
- MCQ Brachytherapy, not EBRT, causes the worst acute obstructive/irritative symptoms. p.30
- MCQ EBRT signature toxicity: bowel problems persisting at 24 months. p.30
- MCQ RP improves obstructive/irritative scores beyond AS by 1–2 years. p.30
- MCQ PCOS: functional differences between RP and RT disappear by 15 years. p.30
- MCQ CEASAR: RP erectile disadvantage seen only in the highest baseline-function quartile. p.30
- MCQ Previous pelvic radiation is the absolute contraindication to radiotherapy. p.30
- MCQ Brachytherapy technically difficult with gland >60 $cm^3$ (pubic arch interference). p.30
- MCQ Delay EBRT 6–8 weeks after TURP. p.30
- MCQ Radiation planned in retention: change to intermittent catheterisation to cycle the bladder. p.30
- MCQ Charlson index originally predicted 1-year inpatient mortality. p.30
- MCQ CAN score outperforms CCI and PCCI for 5-year survival. p.30
- MCQ PSA density criterion appears only in NCCN very-low-risk group. p.30
- MCQ GG3 alone makes intermediate risk unfavourable. p.30
- MCQ Primary Gleason pattern 5 = very high risk. p.30
- MCQ Staging imaging begins at unfavourable intermediate risk. p.30

- MCQ Partin tables predict pathological stage; accurate 72.4%. p.30
- MCQ Epstein: PSA density <0.15, Gleason ≤6, <3 cores, ≤50% any core. p.30
- MCQ Most important counselling element: longevity assessment. p.30
- MCQ Most influential factor in patient choice: urologist's recommendation. p.30
- MCQ Whole-gland HIFU retreatment 29–38%; cryoablation causes ED almost always. p.30

## Ch 159 · Active Surveillance of Prostate Cancer

### 1 · RATIONALE: OVERDIAGNOSIS

- **Latent reservoir**: autopsy microfocal cancer ≈ age as % (30% in 30s, ≥80% over 80). Prostate-specific antigen (PSA) + biopsy detects cancer in 15–25%, but lifetime risk of prostate cancer death is only **3%**. Harbouring : death ≈ 20:1; diagnosis : death ≈ 6:1. Estimated prostate overdiagnosis 87–94%. p.31

### 2 · AS VERSUS WATCHFUL WAITING

- KEY POINT AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them. p.31

### 3 · BIOLOGY OF GRADE GROUP 1

- True grade progression ~**1–2% per year**, mostly high-volume Gleason 6; follow-up must be long term. p.31
- Deaths in low-risk series = **occult higher grade missed at biopsy**, present in ~**30%** after 10–12-core systematic biopsy (85% of it 3+4). p.31
- KEY POINT The main hazard of AS is **sampling error at entry**, not biology over time ("wolf in sheep's clothing"). The protocol's first job is to get the entry grade right. p.32

### 4 · PATIENT SELECTION

- TABLE 15-yr cancer mortality · **5%** actuarial (1.5% actual) · **0.5%** p.32

### 5 · MONITORING PROTOCOL

- ALGORITHM
  Diagnosis → mpMRI (if not already done) → **confirmation biopsy** (systematic + targeted) within 1–3 yr → PSA 6-monthly, DRE every 1–2 yr, MRI ≤ every 18–24 months → repeat biopsy every 3–5 yr while stable → trigger → definitive therapy, or switch to WW when life expectancy falls
  p.33
- KEY POINT Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger. p.33

### 10 · MCQ ONE-LINERS

- MCQ Lifetime risk of prostate cancer death (North America) — 3%. p.35
- MCQ AS intent — curative, deferred; WW — palliative. p.35
- MCQ Switch AS to WW — life expectancy <5–10 years. p.35
- MCQ Epstein very low risk — ≤2 cores, no core >50%, PSA density <0.15. p.35
- MCQ Pure Gleason 6 metastatic potential — effectively zero. p.35
- MCQ Occult higher grade after systematic biopsy — ~30%. p.35
- MCQ True grade progression on AS — ~1–2% per year. p.35
- MCQ Main hazard of AS — misattribution of grade at entry. p.35
- MCQ Pattern 3 vs 4 — PTEN deleted in 70–90% of pattern 4. p.35
- MCQ Cyclin D2 — lost in pattern 4. p.35

- MCQ Intervention on AS at 5 years — ~25%, regardless of cohort. p.35
- MCQ 15-yr cancer mortality — Toronto 5% actuarial vs Hopkins 0.5%. p.35
- MCQ Commonest cause of death on AS — cardiovascular disease. p.35
- MCQ Highest upgrading risk germline mutation on AS — BRCA2. p.35
- MCQ Family history inflation — screening artefact, not biology. p.35
- MCQ Confirmation biopsy — within 1–3 years; systematic + targeted. p.35
- MCQ PSA on AS — 6-monthly. p.35
- MCQ Repeat biopsy while stable — every 3–5 years. p.35
- MCQ Definite trigger — GG ≥3. p.35
- MCQ PSA rise alone — prompts MRI/biopsy, not treatment. p.35
- MCQ DWI signal — inversely related to Gleason score. p.35
- MCQ Serial MRI alone — insufficient to exclude progression. p.35
- MCQ Decipher — 22-marker RNA classifier. p.35
- MCQ Genomic test on AS — only higher-risk candidates; never sole basis. p.35
- MCQ ProtecT active monitoring — more metastases, no mortality difference at 15 years. p.35
- MCQ SPCG-4 benefit — age <65 and GG ≥3. p.35
- MCQ REDEEM — dutasteride reduced volume progression, not upgrading. p.35
- MCQ Selenium and vitamin E — contraindicated on AS. p.35
- MCQ USPSTF PSA screening — Grade D (2012) → Grade C (2018). p.35
- MCQ Neuroendocrine progression without PSA rise — check testosterone, image. p.35

## Ch 160 · Open Radical Prostatectomy

### 1 · PRINCIPLES

- KEY POINT The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins. p.36

### 2 · SURGICAL ANATOMY

- **Inferior vesical artery** ends as **urethral** branches (bladder neck, **5 and 7 o'clock** pedicles) and **capsular** branches (wrapped in nerves → visible marker of the cavernous nerves). p.36
- Pelvic plexus = parasympathetic **S2–S4** + sympathetic (hypogastric nerve); beside the rectum, **5–11 cm** from the anal verge, midpoint at the seminal vesicle tip. Ligating the "lateral pedicle" midway cuts both vessels and nerves. p.36
- KEY POINT Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist. p.36

### 3 · PATIENT SELECTION AND PREOPERATIVE CARE

- **Defer 6 weeks after biopsy, 12 weeks after TURP** so haematoma and adhesions settle — crucial for NVB preservation and avoiding rectal injury. p.37

### 4 · RETROPUBIC RP — OPERATIVE STEPS

- ALGORITHM

  Lower midline extraperitoneal incision → pelvic lymph node dissection (PLND) → endopelvic fascia → puboprostatic ligaments → DVC → urethra → posterior sphincter → NVB release → pedicles → bladder neck and seminal vesicles → reconstruction → anastomosis

  p.37

- KEY POINT Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB. p.38

### 8 · POSTOPERATIVE COMPLICATIONS

- Mortality **0.2%**. p.39
- **Thromboembolism** = major cause of death; highest with PLND, peaks **14–28 days**. Compression, early ambulation, patient education; treat with low-molecular-weight heparin. p.39
- KEY POINT Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs — hence early re-exploration, clips near the bundles and a peritoneal window. p.39

### 11 · MCQ ONE-LINERS

- MCQ RP goals in order: cancer, complications, continence, potency. p.40
- MCQ Defer RP 6 weeks after biopsy, 12 weeks after TURP. p.40
- MCQ Most useful open RP instrument: fibreoptic headlight. p.40
- MCQ DVC common trunk lies directly over the urethra. p.40
- MCQ Pelvic plexus: 5–11 cm from anal verge, centred on seminal vesicle tip. p.40
- MCQ Striated sphincter: vertical tubular sheath, not flat diaphragm. p.40
- MCQ Correct nerve-sparing plane: interfascial. p.40
- MCQ Denonvilliers fascia is excised completely with the specimen. p.40
- MCQ Commonest positive margin site: apex. p.40
- MCQ Pubourethral ligament is spared to support the sphincter. p.40
- MCQ Stapled DVC control saves 200–400 mL blood. p.40
- MCQ Urethral sutures: mucosa and submucosa only. p.40
- MCQ Near the NVB: clips, never thermal energy. p.40
- MCQ Induration or fixed bundle → excise that NVB. p.40
- MCQ Tsuzuki ≥2 criteria → no high anterior release on that side. p.40
- MCQ Complete seminal vesicle excision is essential. p.40
- MCQ "Turtle sign": intussuscepted bladder neck. p.40
- MCQ Rectal injury: two-layer repair plus omental interposition. p.40
- MCQ Delayed haemorrhage needing transfusion → explore and evacuate. p.40
- MCQ Major cause of death after RP: thromboembolism, peak 14–28 days. p.40
- MCQ Recalcitrant contracture: incision plus triamcinolone. p.40
- MCQ Post-RP incontinence: intrinsic sphincter deficiency. p.40
- MCQ Perineal RP spares the DVC → less bleeding. p.40
- MCQ Pelvic plexus: parasympathetic S2–S4 plus hypogastric sympathetic fibres. p.40
- MCQ Hernia after open RP: up to 20% — repair preperitoneally at RP. p.40
- MCQ Wide excision clamp passes medial → lateral to protect the rectum. p.40
- MCQ Continence definition: 0–1 pad per 24 hours. p.40
- MCQ ED recovery predictors: age <65, potent, no diabetes, bilateral sparing. p.40

## Ch 161 · Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

### 1 · BACKGROUND

- KEY POINT The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery → Ch 160. p.41

### 2 · PATIENT SELECTION

- Multiparametric MRI (mpMRI): specificity **0.88**, sensitivity **0.61** for extracapsular extension (ECE); guides extent of nerve sparing and wider resection. p.41

### 4 · ACCESS

- Veress or Hasson entry; insufflation 12–15 mm Hg. Camera port at umbilicus, **15–17 cm** from pubic symphysis; three 8-mm robotic ports; assistant 12-mm and 5-mm ports on the patient's right. p.42

### 5 · OPERATIVE STEPS (TRANSPERITONEAL ANTERIOR RALRP)

- ALGORITHM
  Drop bladder (Retzius) → endopelvic fascia → DVC ligation → bladder neck → seminal vesicles and vasa → posterior plane (Denonvilliers) → pedicles and NVB release → apex: DVC and urethra → inspect → pelvic lymph node dissection (PLND) → entrapment → posterior reconstruction → anastomosis → extraction
  p.42

- Divide anteriorly in the midline → deflate balloon, lift catheter → inspect for median lobe and orifices → divide posterior neck angled **45°** downward. p.42

### 7 · PELVIC LYMPH NODE DISSECTION

- TABLE **Extended (ePLND)** · + internal iliac · ~**63%** p.44

- **Indications**: low risk (nodes <5%) → may omit. American Urological Association (AUA) 2018: unfavourable intermediate or high risk. European Association of Urology (EAU) 2021: nomogram nodal risk **>5%**. AUA/American Society for Radiation Oncology (ASTRO) 2022: risk-stratified by nomogram. p.44

- **Lymphocele — commonest**: radiological up to 60%, clinically significant **2–5%**. **Transperitoneal access is not protective** (bladder adheres and walls off the fossa) → **peritoneal interposition flap** reduces it. Presents with irritative symptoms and pelvic pressure, classically ipsilateral deep vein thrombosis (DVT); may get infected. Asymptomatic → leave. Symptomatic → percutaneous drain → recurrence: sclerosant or laparoscopic marsupialisation. p.44

### 9 · MIS-SPECIFIC COMPLICATIONS

- KEY POINT Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated. p.45

### 12 · MCQ ONE-LINERS

- MCQ Only absolute contraindications: uncorrectable bleeding and unfitness for general anaesthesia. p.46

- MCQ Prior extraperitoneal mesh hernia repair → transperitoneal approach. p.46

- MCQ mpMRI for ECE: specificity 0.88, sensitivity 0.61. p.46

- MCQ Retrovesical approach suits a median lobe. p.46

- MCQ Posterior bladder neck divided angled 45° downward. p.46

- MCQ When in doubt, divide bladder neck more proximally. p.46

- MCQ Default nerve-sparing plane: interfascial. p.46

- MCQ Posterolateral ECE risk → excise Denonvilliers fascia. p.46
- MCQ Cavernous nerves are highly heat-sensitive — athermal release. p.46
- MCQ Commonest positive margin site: apex. p.46
- MCQ Agreed benefit of Rocco stitch: tension-free anastomosis. p.46
- MCQ Retzius-sparing: faster early continence, higher positive margins. p.46
- MCQ Extended PLND captures ~63% of prostatic drainage. p.46
- MCQ EAU threshold for PLND: nomogram nodal risk >5%. p.46
- MCQ Extended PLND lateral limit: genitofemoral nerve. p.46
- MCQ Commonest PLND complication: lymphocele. p.46
- MCQ Transperitoneal approach does not prevent lymphocele; peritoneal flap does reduce it. p.46
- MCQ Do not remove catheter before day 5 (retention). p.46
- MCQ Bladder neck stricture: <2% robotic vs up to 16% open. p.46
- MCQ Critical robotic failure: <0.5%. p.46
- MCQ Yaxley RCT: equal function, less blood loss robotic. p.46
- MCQ Obesity and gland size do not raise PSM. p.46
- MCQ Cavernous nerves lie at 3 and 9 o'clock at mid-urethra. p.46
- MCQ Salvage RALRP continence: 33–80%. p.46

## Ch 162 · Radiation Therapy for Prostate Cancer

### 1 · RADIOBIOLOGY AND FRACTIONATION

- Conventional fractionation = **1.8–2 gray (Gy)** daily (1 Gy = 1 J/kg). The **linear-quadratic model** predicts a tissue's response to fraction size from its **α/β ratio**. p.47
- Prostate cancer has few cycling cells, so it behaves like a late-responding tissue: α/β **≈1.5 Gy** (external beam radiation therapy [EBRT] cohorts 1.0–1.7), **lower than the rectum (~3 Gy)**. p.47
- TABLE Moderate hypofractionation · 2.25–4 Gy · ~4 wk (**60 Gy/20**) p.47
- TABLE Ultrahypofractionation = stereotactic body RT (SBRT) · ≥5 Gy · ≤5–7 fractions (**36.25 Gy/5**) p.47

### 2 · EBRT TECHNIQUES

- TABLE Planning target volume (PTV) · CTV + margin for motion and set-up: **3–7 mm** with image guidance p.48
- **Rectal spacer**: biodegradable hydrogel or hyaluronic acid injected transperineally into Denonvilliers' fascia separates prostate from anterior rectal wall. Randomised: late rectal toxicity **2% vs 7%** (hydrogel, 79.2 Gy); grade ≥2 rectal 2.9% vs 13.8% (hyaluronic acid, 60 Gy/20). Safe with EBRT, SBRT and brachytherapy. p.48

### 4 · HYPOFRACTIONATION

- **Dose ceiling**: no oncological gain above **40 Gy/5**; higher doses buy toxicity, not control. p.48

### 5 · BRACHYTHERAPY (BT)

- TABLE $^{125}$I · **59.6 d** · 140–160 Gy · 108–110 Gy p.49
- **Post-implant dosimetry** (CT ± MRI at 14–30 days): **D90** (dose covering 90% of gland) >90% of prescription; **V100** (% of gland receiving prescription) >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. Low D90 predicts biochemical failure. MCQ D90 is a dose, V100 a volume. p.49

- **ASCENDE-RT** (intermediate–high risk; 12 mo ADT + pelvic EBRT 46 Gy): **LDR boost** vs EBRT boost to 78 Gy → EBRT boost had **twice the biochemical failure** (HR 2.04); no OS difference; price = 5-yr grade 3 GU toxicity **18.4% vs 5.2%**. p.49
- KEY POINT Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant. p.49

### 6 · ADT WITH RT

- TABLE Unfavourable intermediate · **4–6 mo** · — · RTOG 94-08, DFCI 95-096, TROG 96.01 (6 > 3 mo); RTOG 99-10: 9 mo no better than 4 p.50
- TABLE High / very high · **24–36 mo** (≥18) · 12–36 mo · European Organisation for Research and Treatment of Cancer (EORTC) 22863, RTOG 92-02, EORTC 22961 (36 > 6 mo); PCS IV: 18 = 36 mo p.50

### 9 · FOLLOW-UP AND BIOCHEMICAL FAILURE AFTER RT

- TABLE Failure · 3 consecutive rises · PSA **≥ nadir + 2 ng/mL** p.51
- **Biopsy**: optimal at **30–36 mo** (post-mitotic cell death). Do not grade tumours with marked radiation effect; marked effect behaves as negative. Basal cell marker (high-molecular-weight cytokeratin) separates radiation atypia from cancer. mpMRI-targeted biopsy; recurrence arises at the original dominant nodule. p.51

### 13 · MCQ ONE-LINERS

- MCQ α/β of prostate cancer ≈ 1.5 Gy — lower than rectum (~3 Gy). p.52
- MCQ Low α/β is the radiobiological rationale for hypofractionation. p.52
- MCQ Moderate hypofractionation standard: 60 Gy in 20 fractions (CHHiP, PROFIT). p.52
- MCQ CHHiP 57 Gy/19 was not shown noninferior. p.52
- MCQ HYPO-RT-PC: 42.7 Gy in 7 fractions; PACE-B: 36.25 Gy in 5. p.52
- MCQ SBRT dose ceiling: no benefit above 40 Gy in 5 fractions. p.52
- MCQ Dose escalation improved biochemical control in every trial but never OS. p.52
- MCQ IMRT reduces GI but not GU toxicity versus 3D-CRT. p.52
- MCQ Rectal spacer sits in Denonvilliers' fascia; halves-to-thirds late rectal toxicity. p.52
- MCQ Half-life: $^{125}$I 59.6 d, $^{103}$Pd 17 d, $^{131}$Cs 9.7 d. p.52
- MCQ LDR $^{125}$I monotherapy dose: 140–160 Gy; boost ~110 Gy. p.52
- MCQ Single-fraction 19 Gy HDR monotherapy fails — multifraction is standard. p.52
- MCQ D90 goal >90% of prescription; V100 >85% of gland. p.52
- MCQ BT relative contraindications: IPSS >20, Qmax <10 mL/s, recent TURP, pubic arch interference. p.52
- MCQ ASCENDE-RT: EBRT boost doubled biochemical failure versus LDR boost. p.52
- MCQ RTOG 0232: adding EBRT to BT in intermediate risk adds toxicity, not control. p.52
- MCQ ADT radiosensitises by reducing Ku70-dependent DNA repair. p.52
- MCQ Unfavourable intermediate risk: 4–6 months ADT with RT. p.52
- MCQ High risk: 18–36 months ADT with EBRT (PCS IV: 18 = 36). p.52
- MCQ HDR boost made extra ADT redundant in TROG RADAR. p.52
- MCQ Nodal trials likely negative because L5/S1 border misses common iliac nodes. p.52
- MCQ Phoenix definition: PSA nadir + 2 ng/mL. p.52

- MCQ Benign PSA bounce: ~1/3 after LDR; associated with better outcome. p.52
- MCQ Optimal post-RT biopsy timing: 30–36 months. p.52
- MCQ PSADT <3 months after RT predicts prostate cancer death. p.52
- MCQ Radiation proctitis bleeding: 6–18 months; argon plasma coagulation effective. p.53
- MCQ SPPORT: pelvic nodal RT + short-term ADT best for salvage. p.53
- MCQ Early salvage preferred over adjuvant RT (RADICALS-RT, RAVES, GETUG-AFU 17). p.53

## Ch 163 · Focal Therapy of Prostate Cancer

### 1 · DEFINITION AND RATIONALE

- Prostate cancer is multifocal in **50–91%** of prostatectomy specimens. **Index lesion (IL)** = largest/highest-grade focus; hypothesis: the IL drives the natural history and metastasis. p.54
- **For**: most extraprostatic extension (EPE) arises from the IL; IL holds **80–90%** of tumour volume; satellites are mostly **<0.5 mL** (Stamey's "insignificant" cancer) — discounting them makes 52–93% of glands functionally unifocal; metastases are monoclonal. p.54
- KEY POINT "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance. p.54

### 2 · PATIENT SELECTION

- **PROMIS** (mpMRI vs TMB) for clinically significant prostate cancer (csPCa): sensitivity **93%**, specificity 41%, NPV 89% → lesions need histological confirmation and MRI-negative areas still need systematic biopsy. p.54
- mpMRI **underestimates tumour volume** (5–30%) → margin needed: **5 mm** ablates >95%, 6 mm 98.6% of csPCa → often hemi- or zonal ablation. p.54
- ~**90%** of outfield csPCa lies within **1 cm** of the target (the "penumbra") → biopsy in and around the lesion. p.54
- KEY POINT Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland — every consensus has kept all three. p.55

### 4 · ENERGY SOURCES COMPARED

- KEY POINT Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants (cryo-enhancers, SBRT boost) all aim to make that edge lethal. Hence laser's precise but narrow zone gives the most infield recurrence. p.56

### 5 · OUTCOMES

- HEAT (HIFU): 7-year failure-free survival **69%** (68% intermediate, 65% high risk). p.56

### 7 · FOLLOW-UP AND DEFINITION OF FAILURE

- **PSA** falls only **50–80%** (residual epithelium) — no "zero" target; Phoenix/American Society for Radiation Oncology (ASTRO) definitions are not validated after FT. Proposed: nadir + **1.0** ng/mL at 12 months or nadir + 1.5 at 24–36 months predicts csPCa. p.57
- ALGORITHM
  mpMRI + targeted biopsy (4–6 cores) of treated zone at **3–6 months** → mpMRI at 1–2 and 5 years (targeted biopsy for new lesion) → systematic biopsy at 1–2 and 5 years for the untreated gland p.57

## 9 · LANDMARK TRIALS

- TABLE **PCM301** (Azzouzi 2017) · 415 low-risk men; padeliporfin VTP vs AS · Progression 28% vs 58%; more negative biopsies at 2 yr; conversion to radical therapy 4 yr **24% vs 53%**; continence preserved · First level 1 evidence for FT — but Gleason 6 only, surrogate end points (not survival), possible reclassification p.57
- KEY POINT Open problems: can insignificant satellites be safely left; cost of repeated treatment; adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy — prostate FT relies on surveillance and retreatment instead. p.58

## 10 · MCQ ONE-LINERS

- MCQ FT rests on the index lesion hypothesis. p.58
- MCQ Stamey insignificant cancer: volume <0.5 mL. p.58
- MCQ Index lesion holds ~80–90% of tumour volume. p.58
- MCQ FT "sweet spot": completely ablatable Gleason 3+4. p.58
- MCQ Maximum grade for FT by 2016 consensus: 4+3; exclude cribriform/ductal. p.58
- MCQ Erectile dysfunction is not a contraindication to FT. p.58
- MCQ PROMIS: mpMRI sensitivity 93%, specificity 41% for csPCa. p.58
- MCQ ~90% of outfield csPCa lies within 1 cm of the MRI target. p.58
- MCQ A 5-mm margin ablates >95% of csPCa. p.58
- MCQ Hockey-stick ablation is partial-gland ablation, not FT. p.58
- MCQ Transrectal HIFU suits posterior lesions; anterior lesions → transperineal or transurethral. p.58
- MCQ HIFU lethal temperature ≥55 °C; laser ≥60 °C. p.58
- MCQ Cryotherapy uses Joule–Thomson effect; urethral warmer reduces urethral injury. p.58
- MCQ Commonest complication of focal cryotherapy: urinary retention. p.58
- MCQ IRE is non-thermal; field strength >1600 V/cm. p.58
- MCQ IRE needs deep anaesthesia to prevent muscle contraction. p.58
- MCQ VTP drug: padeliporfin (WST11), activated by 753-nm light. p.58
- MCQ PCM301: VTP vs AS in low risk; only randomised FT trial. p.58
- MCQ PCM301 end points were progression and conversion, not survival. p.58
- MCQ Laser: most predictable zone but highest infield failure. p.58
- MCQ Focal HDR brachytherapy: urethral stricture up to 8.2%. p.58
- MCQ PSA falls only 50–80% after FT. p.58
- MCQ Outfield cancer early after FT = selection failure. p.58
- MCQ First follow-up biopsy of treated zone at 3–6 months. p.58
- MCQ PI-FAB reads DCE first. p.58
- MCQ COLD registry: partial cryo ~2× better erectile recovery than whole-gland. p.58
- MCQ Most feared salvage complication: rectourethral fistula. p.58
- MCQ FORECAST compares focal salvage cryotherapy and HIFU. p.58

## Ch 164 · Treatment of Locally Advanced Prostate Cancer

### 3 · STAGING AND RISK ASSESSMENT

- TABLE PSMA PET · Nodal specificity ≥95%; **proPSMA**: accuracy **92% vs 65%** for conventional imaging · Nodal sensitivity only ~40–66% p.59

### 4 · RADICAL PROSTATECTOMY (RP) IN HIGH-RISK DISEASE

- Outcomes, cT3: about half recurrence-free at 8–10 yr on surgery alone; disease-specific survival (DSS) **~80–90% at 10 yr** although biochemical disease-free survival is often only ~50%. p.60
- KEY POINT Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37) — higher than ECE, SVI or nodes — so remove all prostatic tissue regardless of stage. p.60
- TABLE Positive margin · — · **4.37** (highest) p.60
- Dose: **≥64 Gy** (dose response as in primary RT). **GETUG-AFU 16**: **3 months** of AD with salvage RT improves 10-yr progression-free survival (PFS) 64% vs 49% (HR 0.54). Pelvic nodal salvage RT (SPPORT) and technique → Ch 162. p.61
- ALGORITHM
  **pN1 after RP**: favourable (single or few nodes, Gleason ≤8) → observation with PSA monitoring is reasonable → adverse features (>3 nodes, Gleason >8) → AD + RT (best OS) → avoid AD alone where fit.
  p.61

### 5 · RT + ADT

- **Duration**: intermediate risk 4–6 mo; high risk prolonged — NCCN **18–36 mo** with EBRT, 12–36 mo with brachytherapy boost. p.61
- **SPCG-7/SFUO-3** (78% cT3; total AD → continuous flutamide ± RT): cancer death relative risk (RR) **0.44**, overall mortality RR 0.68; PSA recurrence 75% vs 26%. p.62

### 7 · OTHER OPTIONS AND COMPARISONS

- ALGORITHM
  **Locally advanced / high-risk M0**: stage (PSMA PET or CT + bone scan) → fit, resectable → RP + ePLND → adverse pathology: early salvage RT (adjuvant if pT3b/Gleason 8–10/margins+); pN1: observe or AD + RT. Alternative → dose-escalated RT (± brachytherapy boost, ± pelvic nodes) + 18–36 mo ADT → very high risk or N1: + 2 yr abiraterone → unfit for local therapy: ADT alone.
  p.62
- KEY POINT Viva: "Why not ADT alone for T3 disease?" — SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity. p.62

### 9 · MCQ ONE-LINERS

- MCQ Locally advanced = T3–4, N0/N1, M0. p.63
- MCQ Clinical overstaging of cT3 occurs in 7–27%. p.63
- MCQ Strongest pathological predictor of progression after RP: positive surgical margin. p.63
- MCQ Kattan RP nomogram does not apply to cT3b or node-positive disease. p.63
- MCQ CAPRA: each 2-point rise doubles recurrence risk. p.63
- MCQ PSMA PET nodal specificity ≥95%, sensitivity only ~40–66%. p.63
- MCQ proPSMA: PSMA PET accuracy 92% vs 65% for conventional imaging. p.63
- MCQ NAD before RP: clinical but rarely pathological downstaging; no survival gain. p.63
- MCQ NAD does not reduce SVI or nodal positivity. p.63
- MCQ EORTC 22911: adjuvant RT benefit only with positive margins. p.63

- MCQ SWOG 8794: only adjuvant RT trial showing OS benefit. p.63
- MCQ Postoperative RT dose ≥64 Gy. p.63
- MCQ GETUG-AFU 16: 3 months AD with salvage RT improves PFS. p.63
- MCQ Adjuvant AD in pT3N0: no survival benefit. p.63
- MCQ pN1: AD + RT beats AD alone; AD alone raises other-cause mortality. p.63
- MCQ RTOG 86-10: 4 months AD reduced disease-specific mortality, not OS. p.63
- MCQ TROG 96.01: 6 months, not 3, reduced all-cause mortality. p.63
- MCQ EORTC 22863: 3 years goserelin improved OS. p.63
- MCQ RTOG 94-13: whole pelvis + adjuvant AD arm had worst OS. p.63
- MCQ NCCN ADT with EBRT for high risk: 18–36 months. p.63
- MCQ SPCG-7: adding RT to ADT more than halves cancer death. p.63
- MCQ RTOG 0521: docetaxel added to RT + ADT, OS HR 0.69. p.63
- MCQ STAMPEDE high-risk M0: N1, or two of Gleason ≥8, cT3–4, PSA ≥40. p.63
- MCQ Bicalutamide 150 mg increased mortality in localised disease. p.63
- MCQ Cryoablation vs RT: equal survival, worse biochemical control. p.63
- MCQ SPCG-15 compares RP with RT + ADT in cT3N0. p.63

## Ch 165 · Management of Recurrent and Newly Metastatic Prostate Cancer

### 1 · BIOCHEMICAL RECURRENCE: CONCEPT AND DEFINITIONS

- After radical prostatectomy (RP): **¼–⅖** develop BCR within 10 years; only about half of intermediate/high-risk men with BCR ever fail clinically. p.64
- Memorial Sloan Kettering (MSK): PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis and secondary therapy. p.64
- TABLE Post-RP — American Urological Association (AUA) · PSA **≥0.2 ng/mL** · No gland → any PSA suspect p.64
- TABLE After external beam radiotherapy (EBRT) ± androgen deprivation therapy (ADT) · **Phoenix**: **nadir + 2 ng/mL** · Remnant gland keeps making PSA; validated against metastasis and death p.64
- TABLE After low-dose-rate (LDR) brachytherapy · ≥0.2 may suit better; **<0.2 at 4 years ≈ cure** · ≥2× EBRT dose → very low nadir; Phoenix may flatter brachytherapy p.64
- TABLE **Low risk** (needs BOTH) · PSA doubling time (PSADT) **>1 year** AND GG <4 · Interval to failure **>18 months** AND GG <4 p.64
- KEY POINT **PSA bounce** after RT: transient rise of 0.1–0.5 ng/mL at a mean ≈ **9 months** (post-radiation prostatitis, delayed cell death), then a fall. It is **favourable** (better BCR-free survival) → monitor closely; never salvage on a bounce. p.64

### 2 · RISK PREDICTION AND NATURAL HISTORY

- **Nomograms**: biopsy-based before treatment, pathology-based after. MSK pre-op (PSA, stage, Gleason, biopsy cores; concordance 0.76–0.79) and post-op (adds extracapsular extension, seminal vesicle, nodes, margins, adjuvant RT; 10-year BCR-free survival, concordance 0.79–0.81) — only MSK tools reach 15-year cancer-specific survival. **CAPRA** (Cancer of the Prostate Risk Assessment) = pre-op; **CAPRA-S** = post-op ("surgery") MCQ. **Salvage RT nomogram** (PSA, pathological Gleason, PSADT, margins, nodes, ADT): concordance 0.69; in GG 3–5, a positive margin >3 mm roughly halves survival. p.64

- **Pound series** (untreated post-RP BCR): median **8 years** BCR → metastasis, then **5 years** metastasis → death; predictors: time to BCR <2 years, PSADT <10 months, Gleason 8–10. Most men with BCR do not die of prostate cancer. p.65

### 3 · IMAGING AT RECURRENCE

- Bone scan and computed tomography (CT): rarely positive below PSA **10 ng/mL**. p.65
- Positron emission tomography (PET)/CT tracers: choline; $^{18}$F-fluciclovine (**EMPIRE-1**: fluciclovine-guided salvage improved 3-year control); **prostate-specific membrane antigen (PSMA)** PET ($^{68}$Ga-PSMA-11, $^{18}$F-piflufolastat) — highest detection: ≈½ positive at PSA **<0.5**, ≈100% above 2. Detection rises with PSA for every tracer. p.65
- KEY POINT Two curves run in opposite directions: salvage RT works best at the lowest PSA, but imaging detection and certainty of true recurrence improve only as PSA rises. Risk features (grade, margins, kinetics) let us act early without overtreating. p.65

### 4 · SALVAGE AFTER RADICAL PROSTATECTOMY

- **Dose**: **≥64–65 Gy** to the fossa (guidelines); 70 Gy adds nothing (SAKK 09/10). Pelvic nodes 40–45 Gy. Intensity-modulated RT (IMRT) standard. p.65

### 5 · SALVAGE AFTER RADIOTHERAPY

- TABLE Salvage RP (open or robotic) · EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, no nodes/metastases on imaging, initial cT1–T2; gives pathological staging. Poor if seminal vesicle or nodal involvement · Incontinence, bladder neck contracture, rectal injury, near-universal impotence (robotic: fewer) p.66
- TABLE Salvage cryotherapy · PSA <10 (ideally <4), PSADT **≥16 months**; focal or whole-gland; post-cryo nadir <0.2 predicts success · **Urethrorectal fistula** (now 1–3%), erectile dysfunction; incontinence now single-digit thanks to **urethral warming** p.66

### 6 · NON-CURATIVE APPROACH TO BCR: OBSERVATION VS ADT

- AUA: observation (or trial) over hormones after maximal local therapy; ADT only for higher metastatic risk — PSADT **<12 months** or high grade; intermittent preferred. EAU: observe EAU low-risk BCR, limited life expectancy or men declining local therapy. Surveillance PET/CT, especially if PSADT <12 months. p.66
- **Messing (ECOG) trial**: immediate ADT after RP for node-positive (pN+) disease improved OS and progression-free survival (PFS) — but it was small, pre-dated modern imaging, had bulky nodes (up to 20), a fifth had persistent PSA, and the delayed arm was treated only at clinical progression. Low-volume pN+ with undetectable PSA: **20–40%** cured by surgery alone. p.66
- **EMBARK** — high-risk BCR (PSADT **≤9 months**, PSA ≥1 after RP or ≥2 above nadir after RT): leuprolide vs leuprolide + enzalutamide vs enzalutamide; metastasis-free survival (MFS) 71.4% / 87.3% / 80.0% (combination HR 0.42) → enzalutamide approved for **high-risk non-metastatic hormone-sensitive prostate cancer (nmHSPC)**. p.66

### 7 · ANDROGEN DEPRIVATION THERAPY: BIOLOGY AND AGENTS

- KEY POINT **Flare** matters almost only with spinal or weight-bearing bone metastases: give an antiandrogen with (or before) the agonist for **21–28 days**; simultaneous start is as good as a week's lead-in. For **impending cord compression** or severe bone pain choose an **antagonist or orchiectomy** (or ketoconazole).
  **Antiandrogen withdrawal**: stopping bicalutamide on CAB at progression gives >50% PSA fall in 15–30% (mutant AR now activated by the drug); 0–10% with ARTAs; no OS gain. p.67

## 8 · ADT SIDE EFFECTS AND THEIR MANAGEMENT

- TABLE **Osteoporosis / fracture** · Loss of peripheral oestrogen; fracture risk ↑20–45%, rises with duration · Baseline hip dual-energy x-ray absorptiometry (DEXA) + FRAX; all: calcium 1000–1200 mg + vitamin D 400–1000 international units (IU)/day, weight-bearing exercise, stop smoking. Drug if T-score −1 to −2.5 with 10-year hip fracture **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly; repeat DEXA at 1 year p.67

## 9 · METASTATIC HORMONE-SENSITIVE PROSTATE CANCER (MHSPC)

- Prognosis: metachronous low volume best (median OS ≈92 vs ≈43 months synchronous/high volume); nodes only > bone > visceral > bone + visceral. **SWOG 9346 PSA nadir at 7 months**: ≤0.2 → median OS **75** months; 0.2–4 → 44; >4 → 13. p.68
- TABLE **CHAARTED high volume** · **≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion). Low volume = not high volume · Docetaxel benefit subgroup; RT to primary in low volume p.68
- TABLE **LATITUDE high risk** · **≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral metastasis · Entry for abiraterone p.69
- Docetaxel **75 mg/m² 3-weekly × 6**; meta-analysis HR 0.77; neutropenia; benefit greatest in high volume. p.69

## 10 · LANDMARK TRIALS

- TABLE Radiation Therapy Oncology Group (RTOG) 0534 (SPPORT) · Bed vs bed + 4–6 months ADT vs pelvis + ADT · 5-year freedom from progression 70.9 / 81.3 / **87.4%** · Add ADT and pelvic nodes (endpoint not OS) p.69
- TABLE CHAARTED (ECOG 3805) · ADT ± docetaxel · OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months · Benefit in high volume only p.70

## 11 · MCQ ONE-LINERS

- MCQ AUA BCR after RP: PSA ≥0.2 ng/mL; NCCN uses ≥0.1. p.70
- MCQ Phoenix definition after EBRT: PSA nadir + 2 ng/mL. p.70
- MCQ PSA <0.2 ng/mL 4 years after LDR brachytherapy ≈ cure. p.70
- MCQ EAU low-risk BCR after RP: PSADT >1 year AND GG <4. p.70
- MCQ EAU low-risk BCR after RT: interval to failure >18 months AND GG <4. p.70
- MCQ Only modifiable predictor of BCR after RP: positive surgical margin. p.70
- MCQ Pound: median 8 years BCR → metastasis, 5 years metastasis → death. p.70
- MCQ PSMA PET detects ≈half of recurrences at PSA <0.5 ng/mL. p.70
- MCQ Minimum salvage RT dose to prostate bed: 64–65 Gy. p.70
- MCQ RTOG 9601: 2 years bicalutamide 150 mg with SRT improved OS. p.70
- MCQ Salvage cryotherapy selection: PSA <10, PSADT ≥16 months. p.70
- MCQ EAU salvage RP: PSA <10, GG ≤3, life expectancy ≥10 years, initial cT1–T2. p.70
- MCQ EMBARK high-risk BCR: PSADT ≤9 months. p.70
- MCQ Castrate testosterone level: <50 ng/dL. p.70
- MCQ LHRH agonist raises FSH; antagonist lowers FSH and causes no flare. p.70
- MCQ Flare prophylaxis: antiandrogen for 21–28 days. p.70
- MCQ Impending cord compression in hormone-naive man: antagonist or orchiectomy. p.70
- MCQ Apalutamide: rash and hypothyroidism; enzalutamide: seizures. p.70
- MCQ Abiraterone needs prednisone to suppress ACTH-driven mineralocorticoid excess. p.70

- MCQ Commonest ADT side effect: hot flushes; most effective treatment: oestrogens. p.70
- MCQ SWOG 9346: intermittent ADT non-inferiority unproven in metastatic disease. p.70
- MCQ CHAARTED high volume: ≥4 bone metastases (≥1 appendicular) or visceral. p.70
- MCQ LATITUDE high risk: ≥2 of Gleason ≥8, ≥3 bone lesions, visceral. p.70
- MCQ Negative docetaxel mHSPC trial: GETUG-AFU 15. p.70
- MCQ Messing trial: immediate ADT for pN+ after RP improved OS, but nodes were bulky and imaging dated. p.70
- MCQ HORRAD: prostate RT in mHSPC — no overall OS benefit; low-volume signal later confirmed by STAMPEDE. p.70
- MCQ Cyproterone acetate: steroidal antiandrogen; now used 50–100 mg/day for hot flushes. p.71
- MCQ SWOG 9346: PSA nadir ≤0.2 ng/mL at 7 months → median OS 75 months. p.71
- MCQ Darolutamide in mHSPC is approved only with docetaxel (ARASENS). p.71
- MCQ STOMP and ORIOLE: MDT delays ADT/progression in oligometastases; no OS data. p.71

## Ch 166 · Management of Castration-Resistant Prostate Cancer

### 2 · CLINICAL STATES AND PROGRESSION CRITERIA

- **PSA doubling time (PSADT)** predicts the pace: PSADT **<3 months** → rapid course → consider aggressive therapy. p.72
- TABLE Bone scan · New lesions early may be **flare** of healing disease; rarely shows response · Rescan at **12–16 weeks** as new baseline; progression = **≥2 new lesions** ("2 + 2" rule) p.72
- KEY POINT Why the rules exist: every marker (PSA, bone scan, PSMA PET) can flare early, so progression is called on confirmed, combined evidence — otherwise effective therapy is abandoned too soon. p.72

### 3 · GERMLINE AND SOMATIC TESTING

- Somatic homologous recombination repair (HRR) mutations in up to **20–25%** of mCRPC; mismatch repair deficiency (dMMR)/microsatellite instability-high (MSI-H) in **2–5%** → if found, genetic counselling for Lynch syndrome. p.73

### 5 · MCRPC: ANDROGEN-SIGNALLING INHIBITORS

- KEY POINT **Why prednisone with abiraterone?** CYP17 blockade lowers cortisol → adrenocorticotropic hormone (ACTH) rises → excess mineralocorticoid precursors. Low-dose prednisone suppresses ACTH. Enzalutamide needs no steroid (prednisone may even activate a promiscuous AR). Class effects of ARPIs: fatigue, falls and fractures — caution in men >75 years. p.74

### 6 · CYTOTOXIC CHEMOTHERAPY

- TABLE Docetaxel · **75 mg/m² 3-weekly** + prednisone 5 mg twice daily; first-line chemotherapy · Neutropenia, fatigue, oedema, **neuropathy**, hyperlacrimation, nail dystrophy p.74
- TABLE Cabazitaxel · **20 mg/m²** (25 if low neutropenic risk) 3-weekly + prednisone + granulocyte colony-stimulating factor (G-CSF); post-docetaxel · **Neutropenia** (grade 3–4 82%), **diarrhoea**; little neuropathy p.74

### 7 · RADIOPHARMACEUTICALS

- TABLE Emitter / targeting · **Alpha** (path **<100 µm** → spares marrow); calcium mimetic, bone-seeking · Beta; PSMA-directed — needs PSMA PET-positive disease p.74

## 9 · IMMUNOTHERAPY

- **Pembrolizumab** (anti-programmed cell death protein 1 [PD-1], 200 mg IV 3-weekly): tumour-agnostic approval for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase**, after ≥1 prior line. Immune toxicity: colitis, hepatitis, endocrinopathy, pneumonitis. p.75

## 10 · NEUROENDOCRINE/AGGRESSIVE-VARIANT CRPC

- TABLE Survival · Years · **<12 months** p.75

## 12 · COMPLICATIONS AND PALLIATIVE CARE

- ALGORITHM
  Back pain ± motor/sensory deficit or bladder/bowel change → **high-dose IV dexamethasone** (load 10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day) → **urgent whole-spine magnetic resonance imaging (MRI)** → definitive RT ± decompressive surgery → taper steroids over 2–3 weeks
  p.76

## 13 · BONE-TARGETED THERAPY

- TABLE Dose · **4 mg IV 4-weekly**, adjusted for creatinine clearance · **120 mg subcutaneous (SC) 4-weekly**; no renal adjustment p.76
- TABLE Toxicity · Flu-like fever, myalgia, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ) · Hypocalcaemia (grade ≥3 5%), hypophosphataemia, ONJ **2–4%** p.77
- KEY POINT **ONJ** occurs mostly with dental extraction or poor dentition → dental assessment first; avoid in these men. Give calcium 1000 mg + vitamin D 800 international units (IU) daily with both agents to prevent hypocalcaemia. Bone agents are strongly recommended alongside radium-223. p.77

## 15 · MCQ ONE-LINERS

- MCQ CRPC requires castrate testosterone <50 ng/dL with progression. p.78
- MCQ PCWG3 bone progression: ≥2 new lesions, confirmed on a scan after 12–16 weeks. p.78
- MCQ LHRH suppression is continued throughout CRPC therapy. p.78
- MCQ nmCRPC treatment threshold in SPARTAN, PROSPER, ARAMIS: PSADT ≤10 months. p.78
- MCQ Abiraterone is active in nmCRPC but not approved there. p.78
- MCQ Abiraterone must be taken fasting; food raises absorption up to tenfold. p.78
- MCQ Enzalutamide: seizures; apalutamide: rash and hypothyroidism. p.78
- MCQ Darolutamide and apalutamide do not cross the blood–brain barrier. p.78
- MCQ TAX 327: 3-weekly docetaxel 75 mg/m² beat mitoxantrone. p.78
- MCQ Cabazitaxel escapes P-glycoprotein efflux; toxicity neutropenia and diarrhoea. p.78
- MCQ Mitoxantrone palliates but does not prolong survival. p.78
- MCQ Radium-223: alpha emitter, path <100 µm, bone-only symptomatic disease. p.78
- MCQ Radium-223 must not be combined with abiraterone/prednisone. p.78
- MCQ $^{177}$Lu-PSMA-617 typical toxicity: dry mouth. p.78
- MCQ Sipuleucel-T antigen: prostatic acid phosphatase fused to GM-CSF. p.78
- MCQ Sipuleucel-T improves OS without lowering PSA. p.78
- MCQ Pembrolizumab indication: MSI-H/dMMR or tumour mutational burden ≥10/megabase. p.78
- MCQ dMMR prevalence in mCRPC: 2–5%. p.78
- MCQ PARPI benefit greatest in BRCA2; rucaparib not approved for ATM. p.78
- MCQ CHIP can cause false-positive DNA-repair calls on liquid biopsy. p.78

- MCQ Neuroendocrine CRPC: low PSA, lytic lesions, visceral disease — treat with platinum–etoposide. p.78
- MCQ Small cell prostate cancer signature: ≥2 of PTEN, TP53, RB1 loss. p.78
- MCQ Sequential ARPI benefit only 15–30% (cross-resistance). p.78
- MCQ CARD: cabazitaxel beats a second ARPI after docetaxel and one ARPI. p.78
- MCQ First step in cord compression: high-dose dexamethasone, then MRI. p.78
- MCQ Patchell: decompressive surgery + RT superior to RT alone. p.78
- MCQ Denosumab dose: 120 mg SC 4-weekly, no renal adjustment. p.78
- MCQ Zoledronic acid: 4 mg IV 4-weekly, renal dose adjustment. p.78
- MCQ Bone agents reduce SREs but do not improve survival. p.78
- MCQ ONJ risk: dental extraction, poor dentition — dental check first. p.78

## ★ Putting it together: the prostate cancer pathway

### A · FROM RAISED PSA TO DIAGNOSIS

- ALGORITHM

  **1. Raised prostate-specific antigen (PSA) or abnormal digital rectal examination (DRE)** → repeat PSA; exclude infection, instrumentation, 5α-reductase inhibitors (double the PSA). Consider PSA density and biomarkers (→ Ch 154).

  **2. Multiparametric magnetic resonance imaging (mpMRI) before biopsy** → Prostate Imaging Reporting and Data System (PI-RADS) score (→ Ch 155).

  **3. Biopsy**: targeted biopsy of PI-RADS 3–5 lesions plus systematic cores; transperineal route lowers infection risk.

  **4. Grade** with the International Society of Urological Pathology (ISUP) Grade Group (GG 1–5) (→ Ch 156). **Stage and risk-group** (→ Ch 157); staging imaging (prostate-specific membrane antigen positron emission tomography [PSMA PET], or computed tomography [CT] + bone scan) from unfavourable intermediate risk upwards.

  p.79

### B · LOCALISED DISEASE: RISK GROUP → TREATMENT

- KEY POINT Trials comparing treatment with observation (SPCG-4, PIVOT, ProtecT) show the benefit of radical treatment is confined to younger men with intermediate-risk or higher disease and long life expectancy. ProtecT showed RP and RT give equal cancer survival at 15 years, with different side-effect profiles. p.79

ALGORITHM

**1. Biochemical recurrence** (Ch 165): PSMA PET to localise. After RP: salvage RT. After RT: biopsy-proven local recurrence → local salvage. High-risk recurrence (PSA doubling time ≤9 months) → ADT + enzalutamide (EMBARK).

**2. Metastatic hormone-sensitive (mHSPC)**: never ADT alone. ADT + an androgen-receptor pathway inhibitor (ARPI: abiraterone, enzalutamide, apalutamide, darolutamide); triplet with docetaxel for fit, high-volume disease. RT to the prostate for low-volume disease (STAMPEDE arm H). Offer germline and somatic genetic testing to all.

**3. Non-metastatic castration-resistant (nmCRPC)** with PSA doubling time ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).

**4. Metastatic castration-resistant (mCRPC)** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); $^{177}$Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); poly(ADP-ribose) polymerase (PARP) inhibitor if BRCA/homologous-recombination mutation (PROfound); pembrolizumab if microsatellite instability-high. Bone protection (denosumab or zoledronic acid) throughout.

p.80